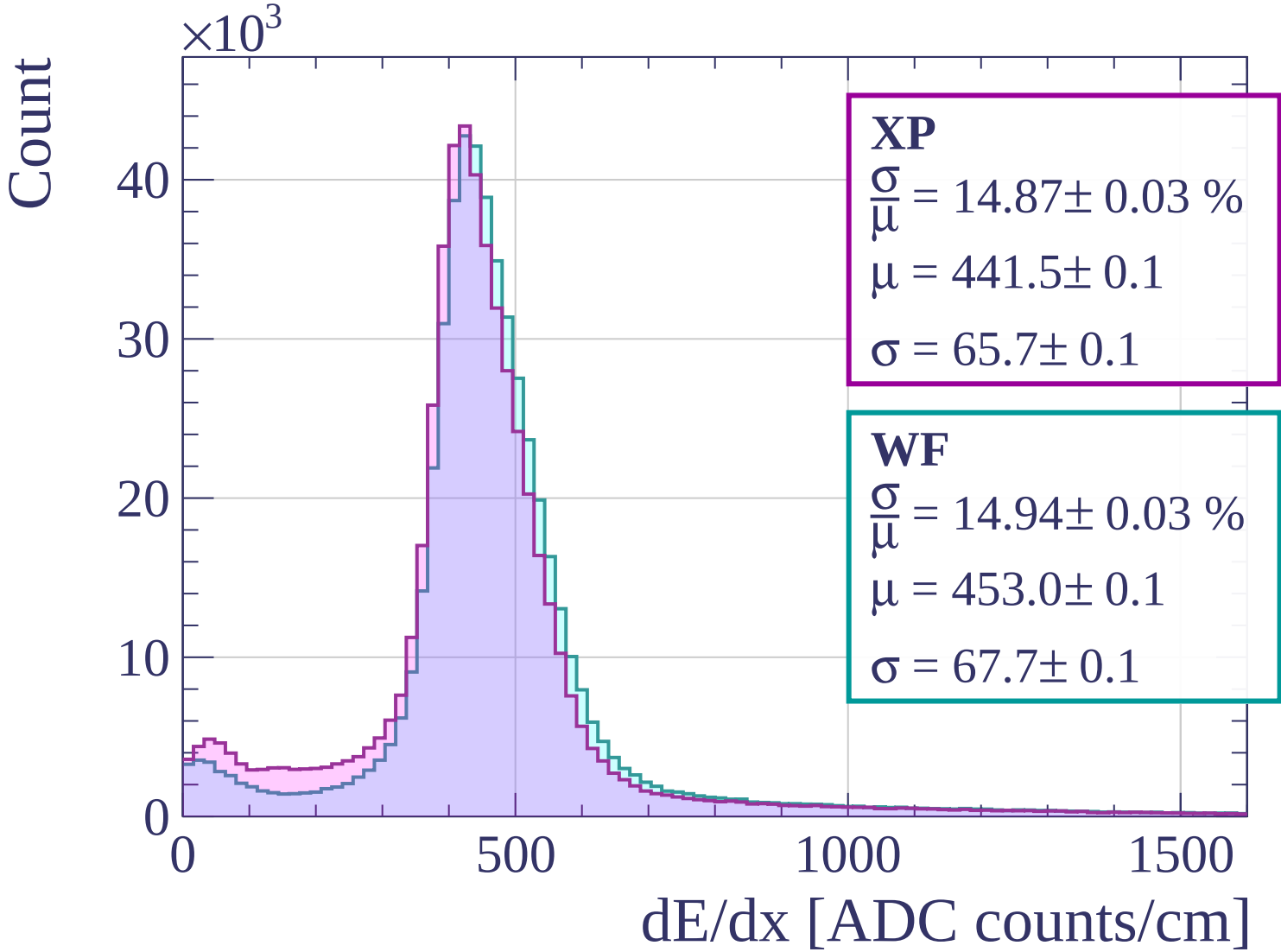
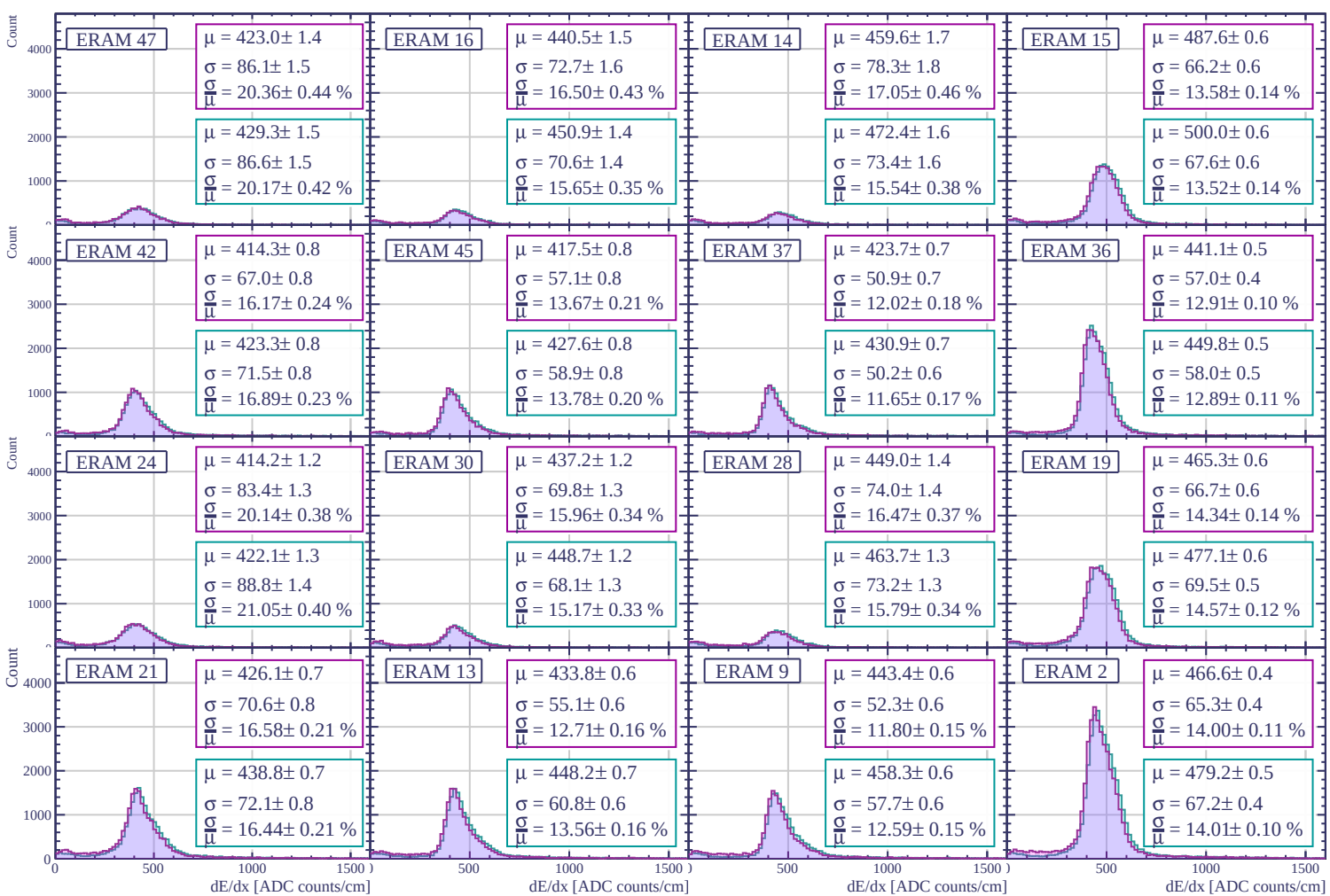
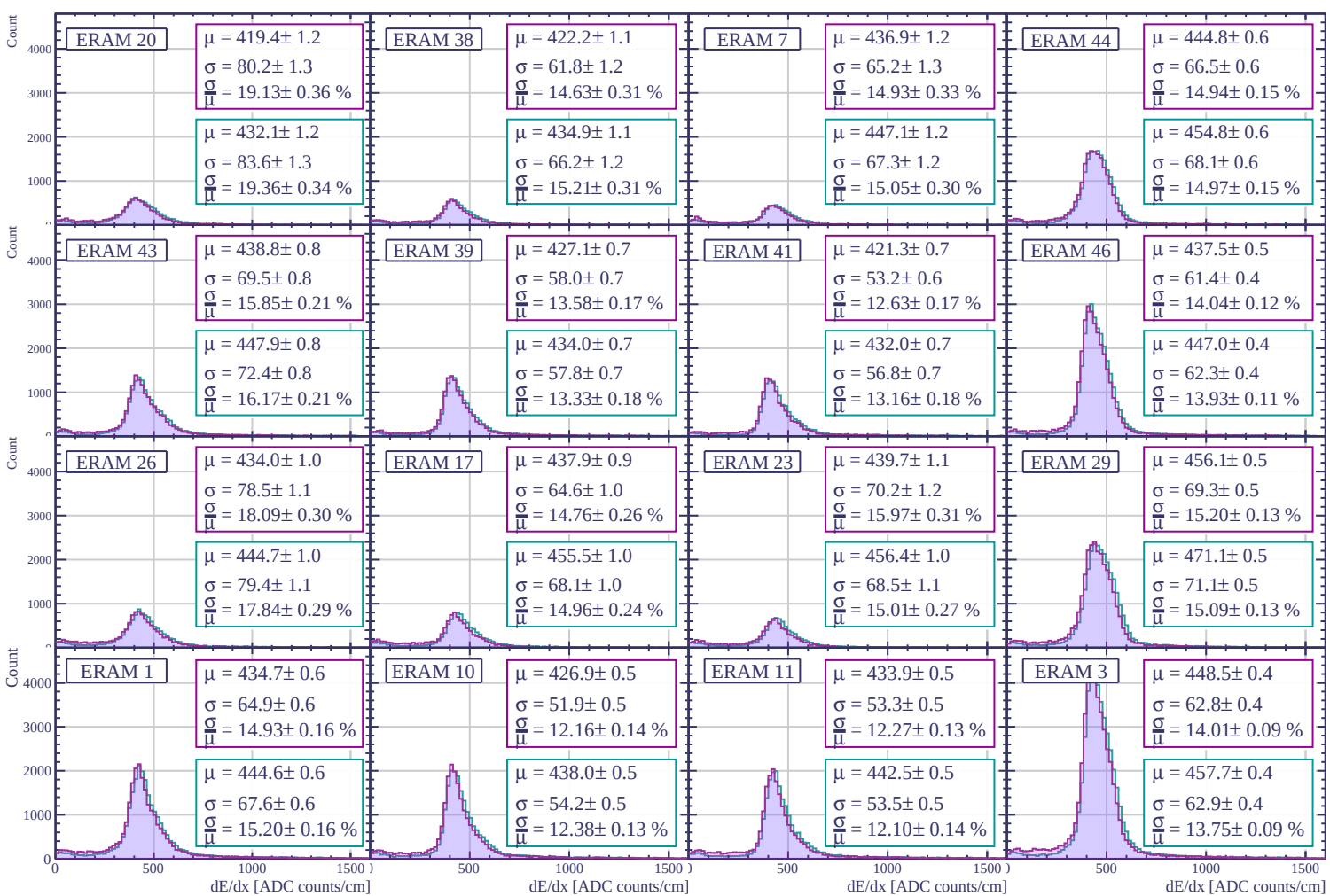
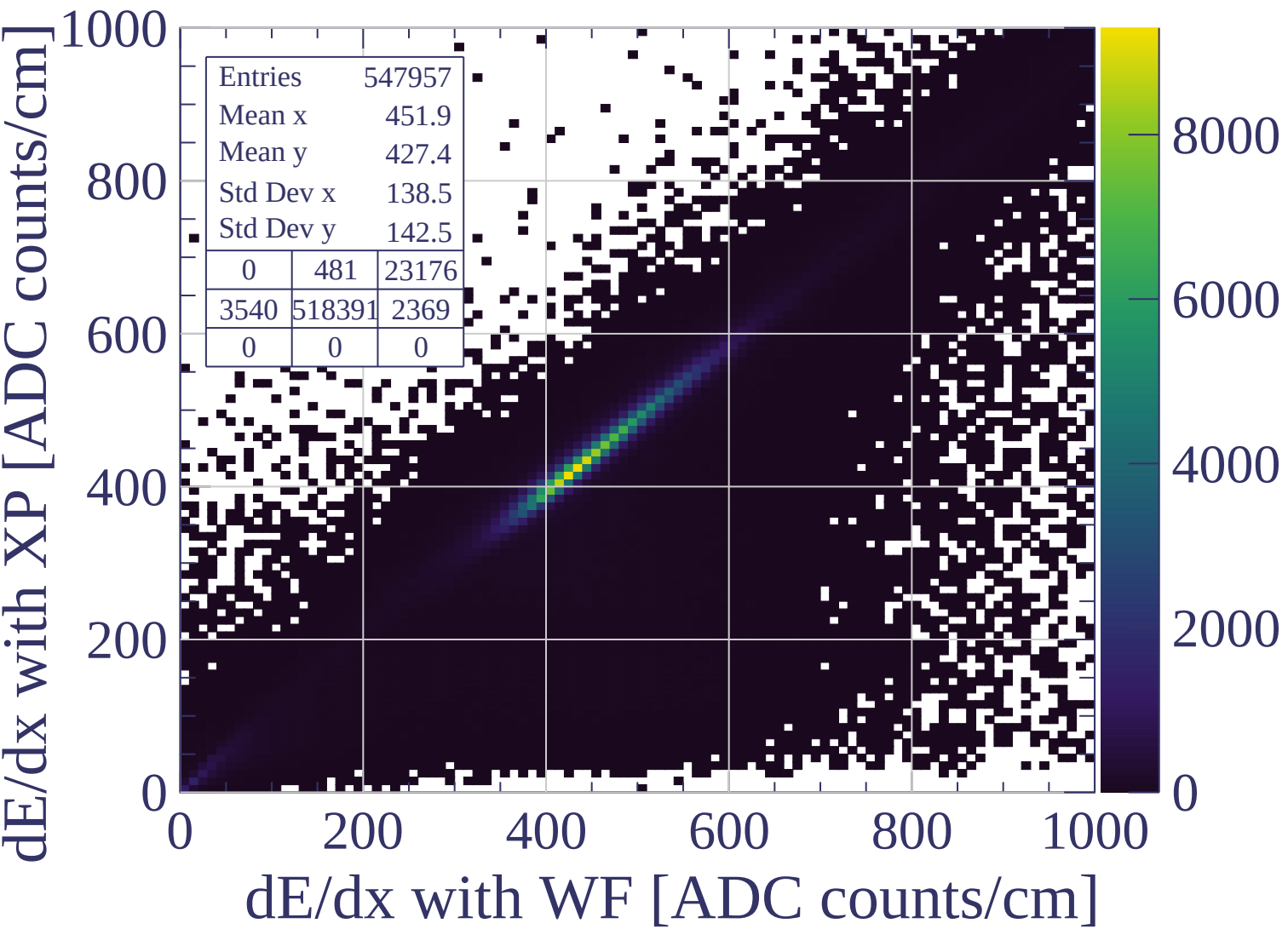
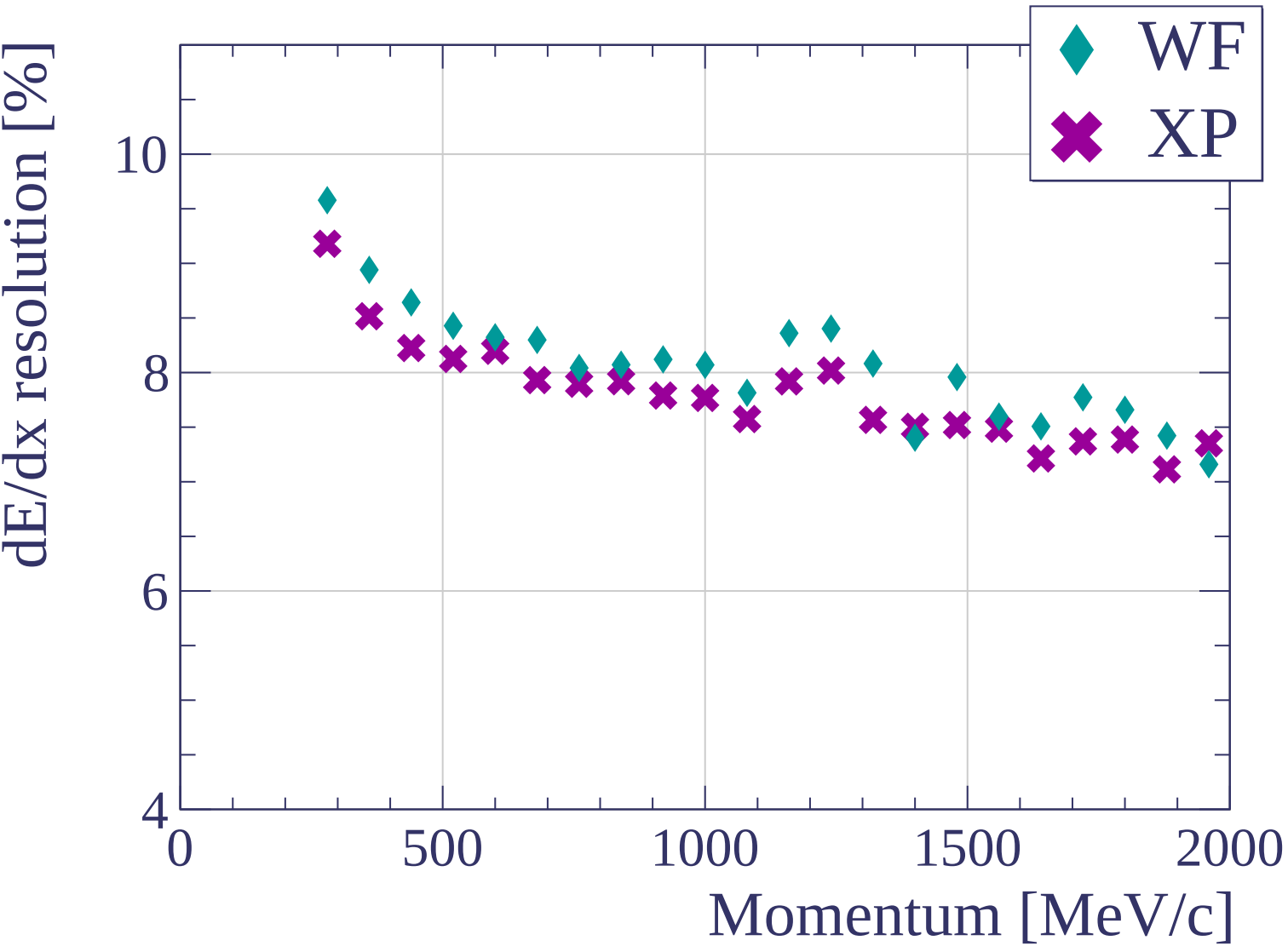


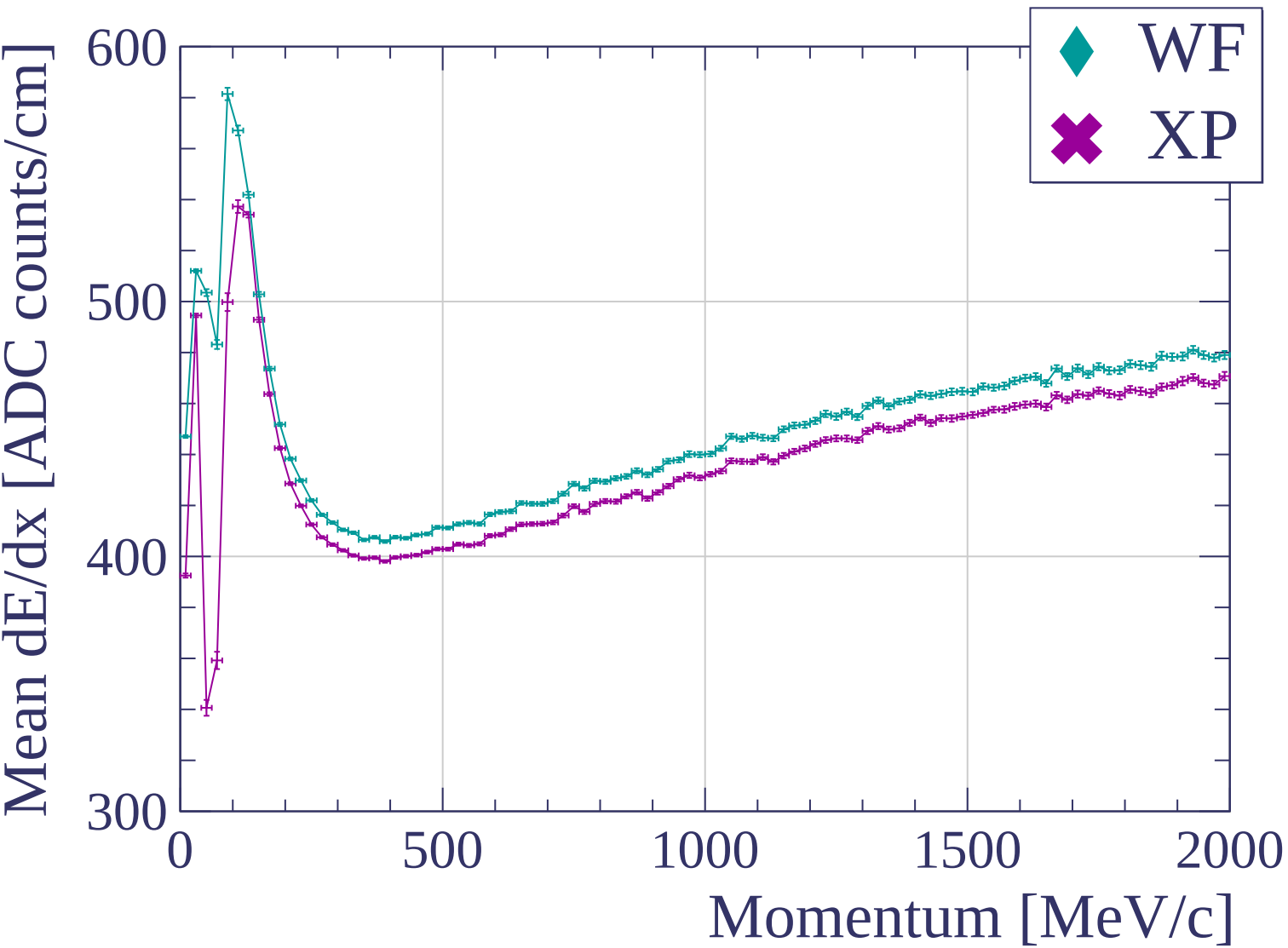


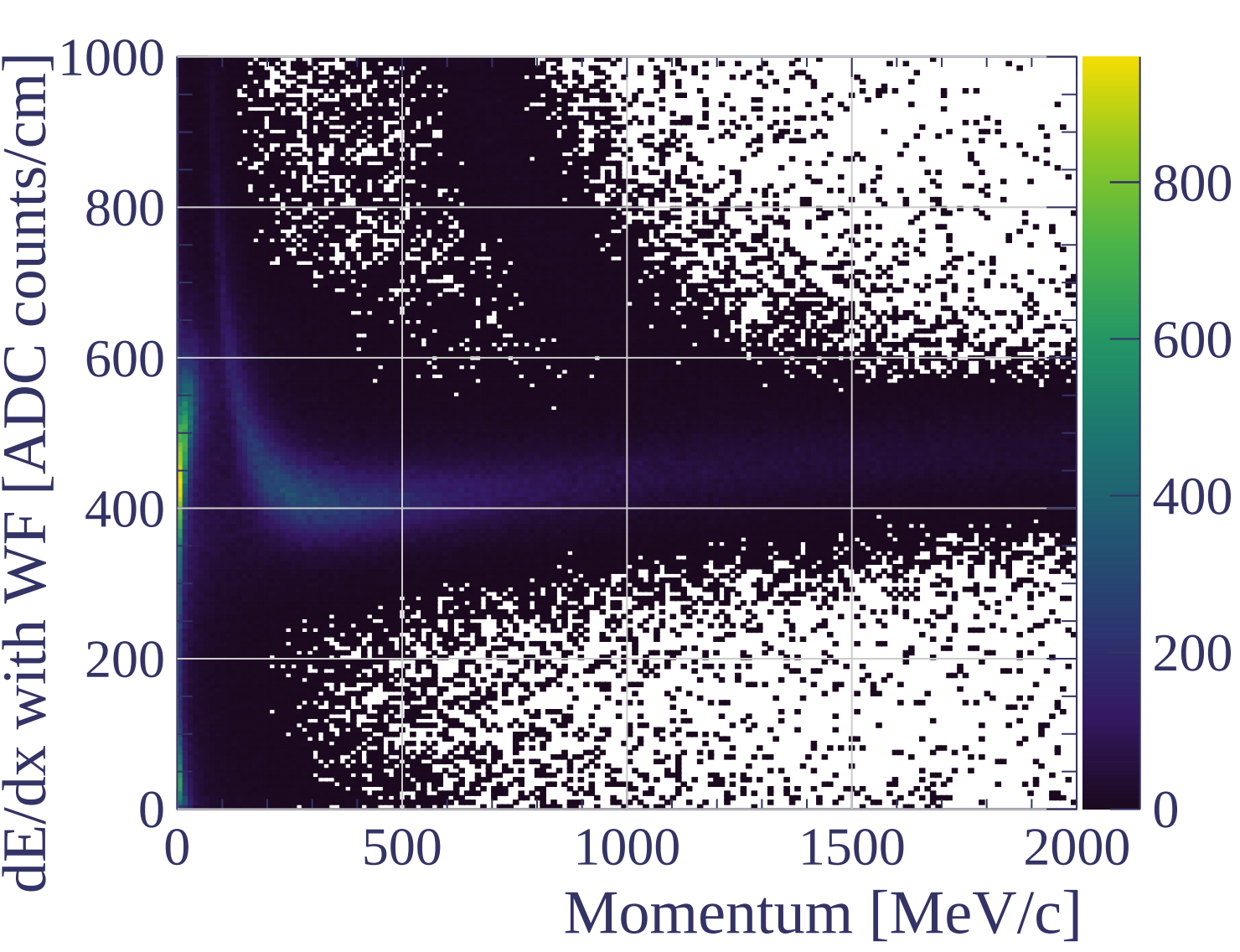


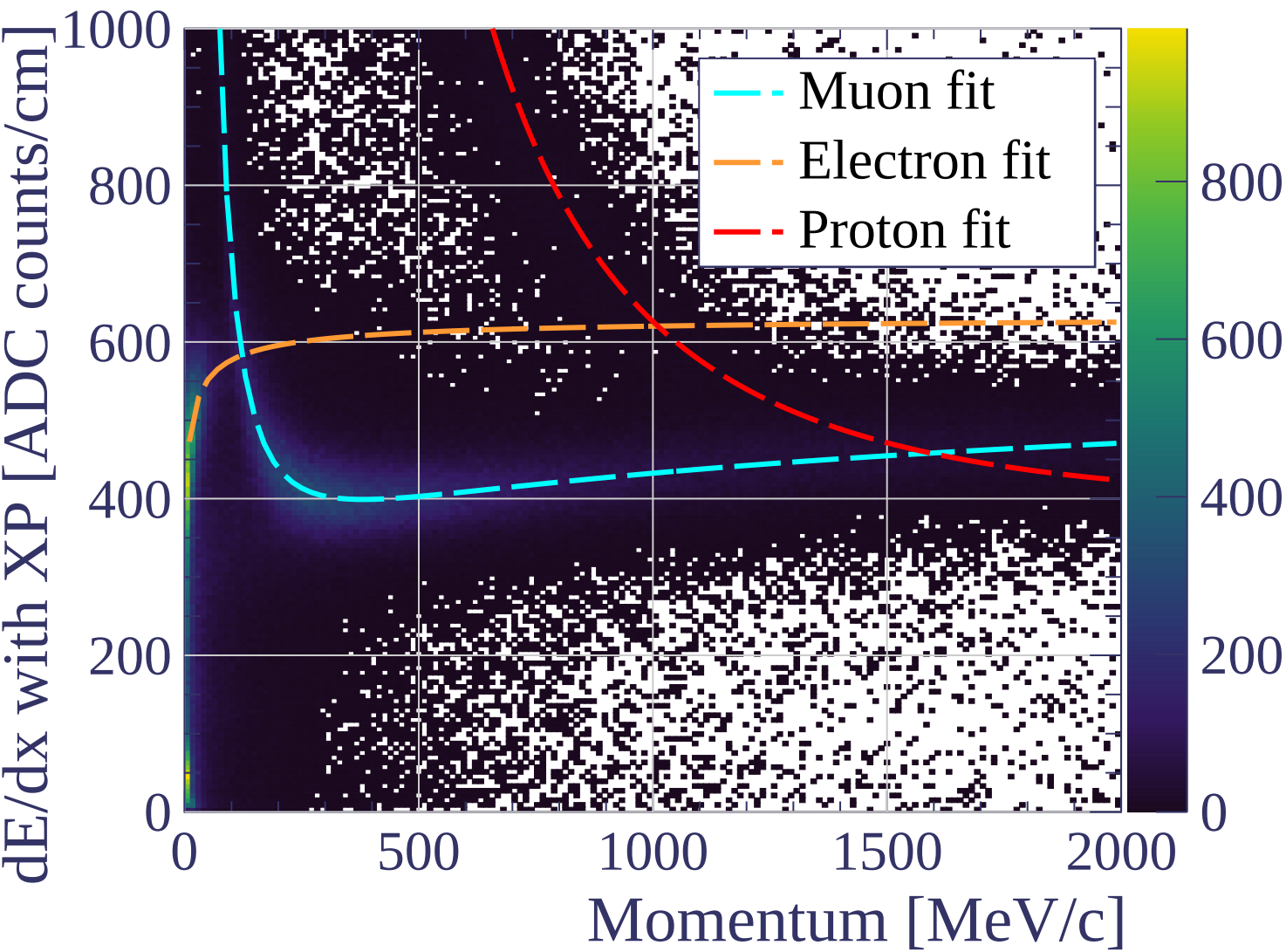




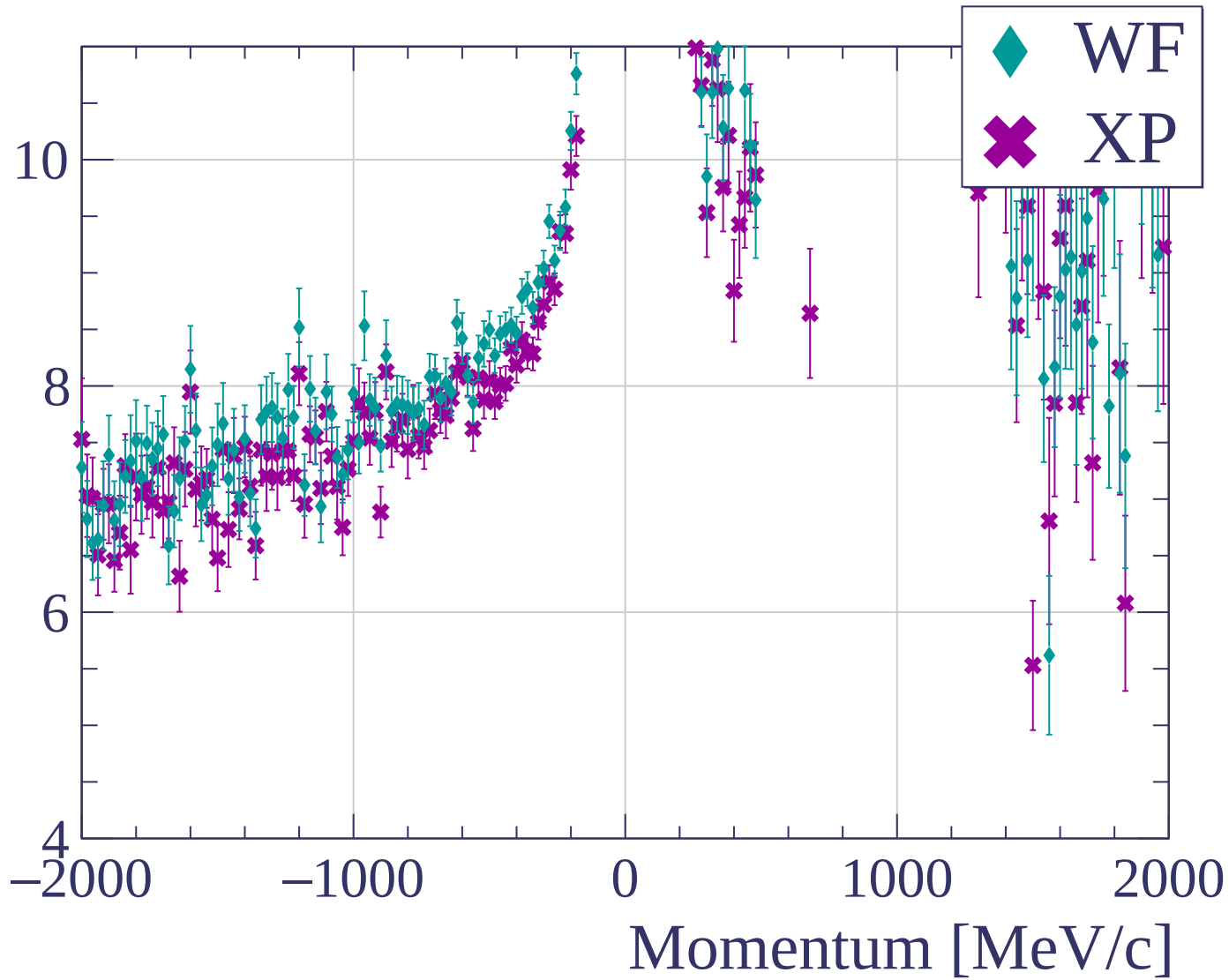


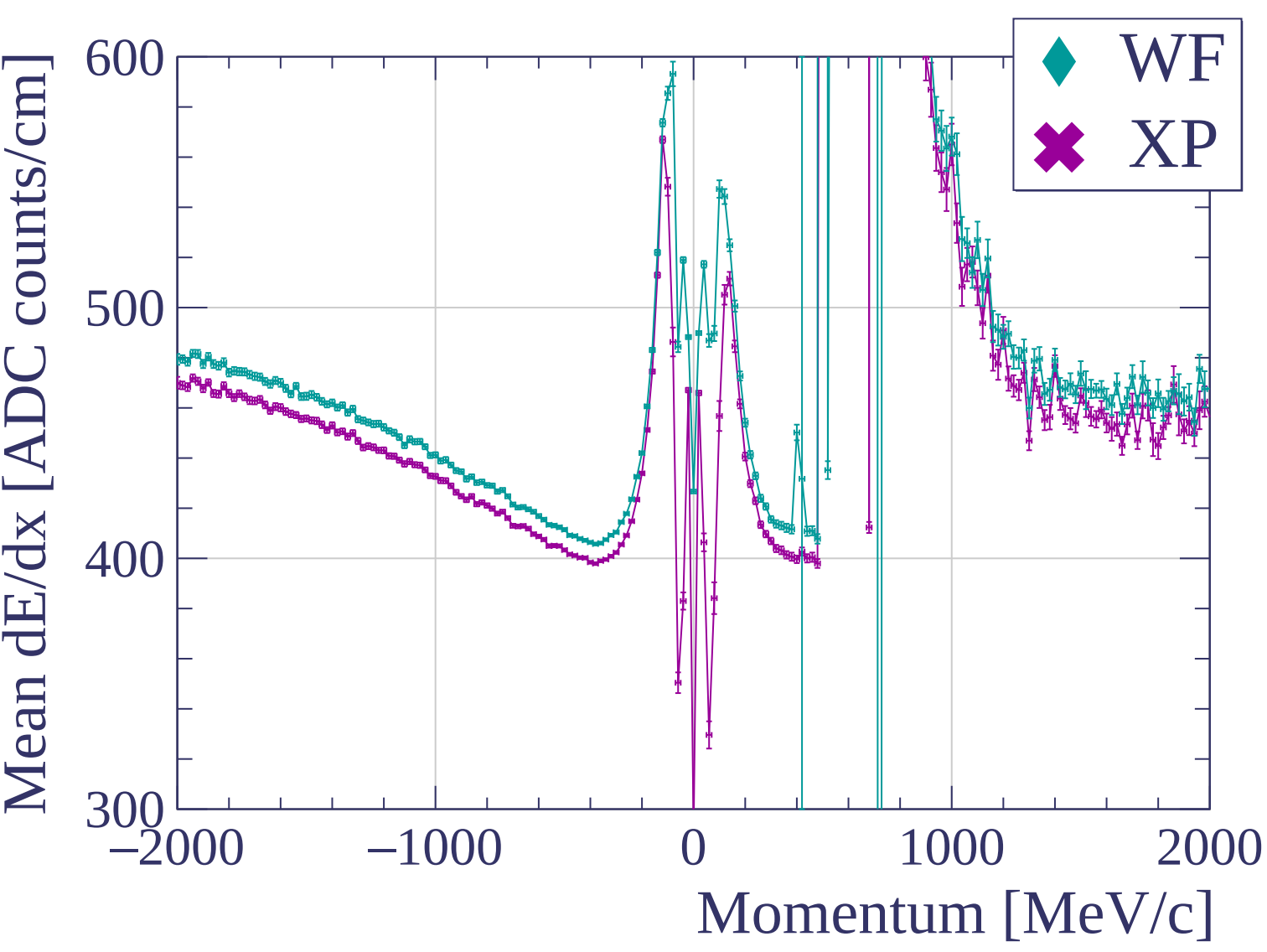


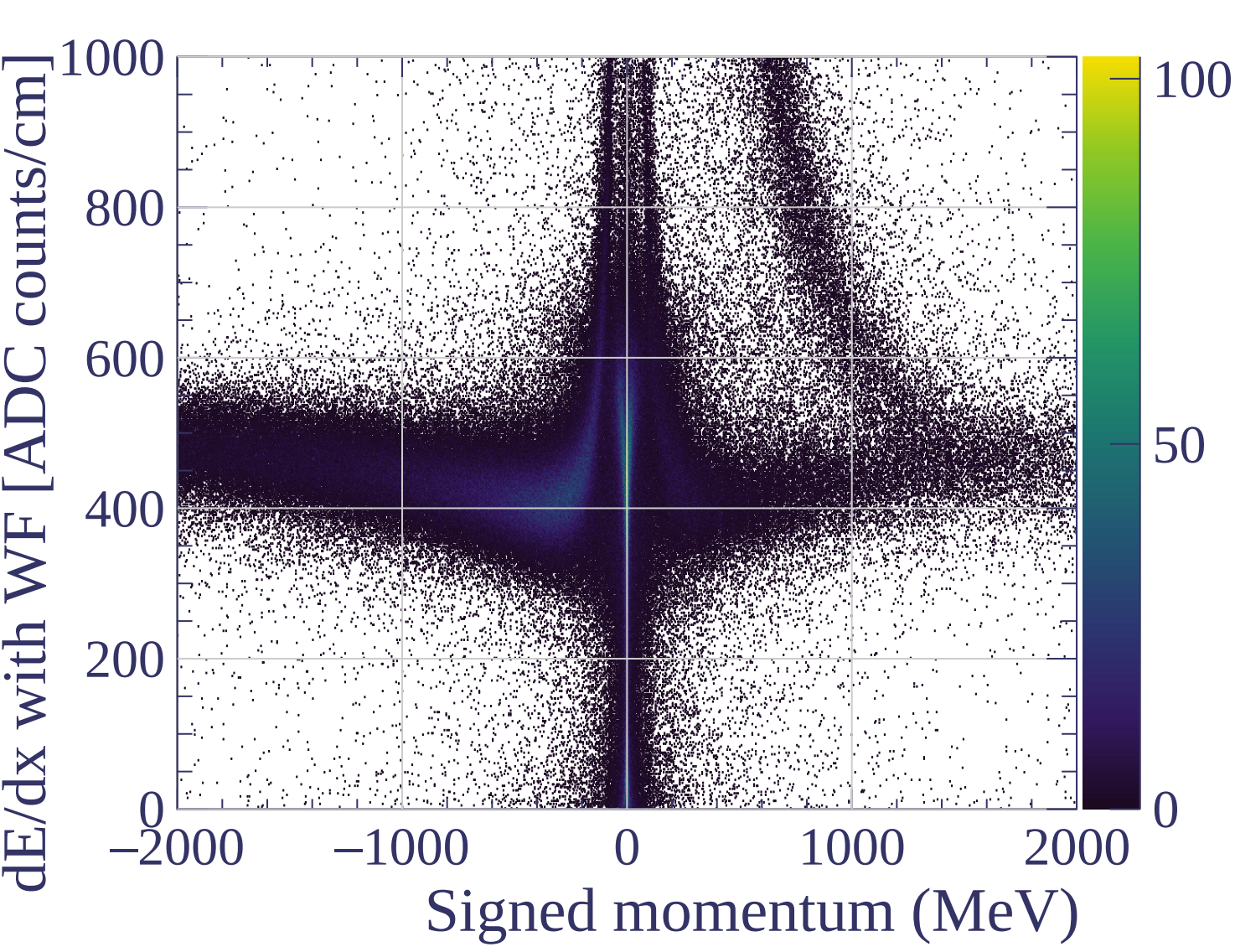


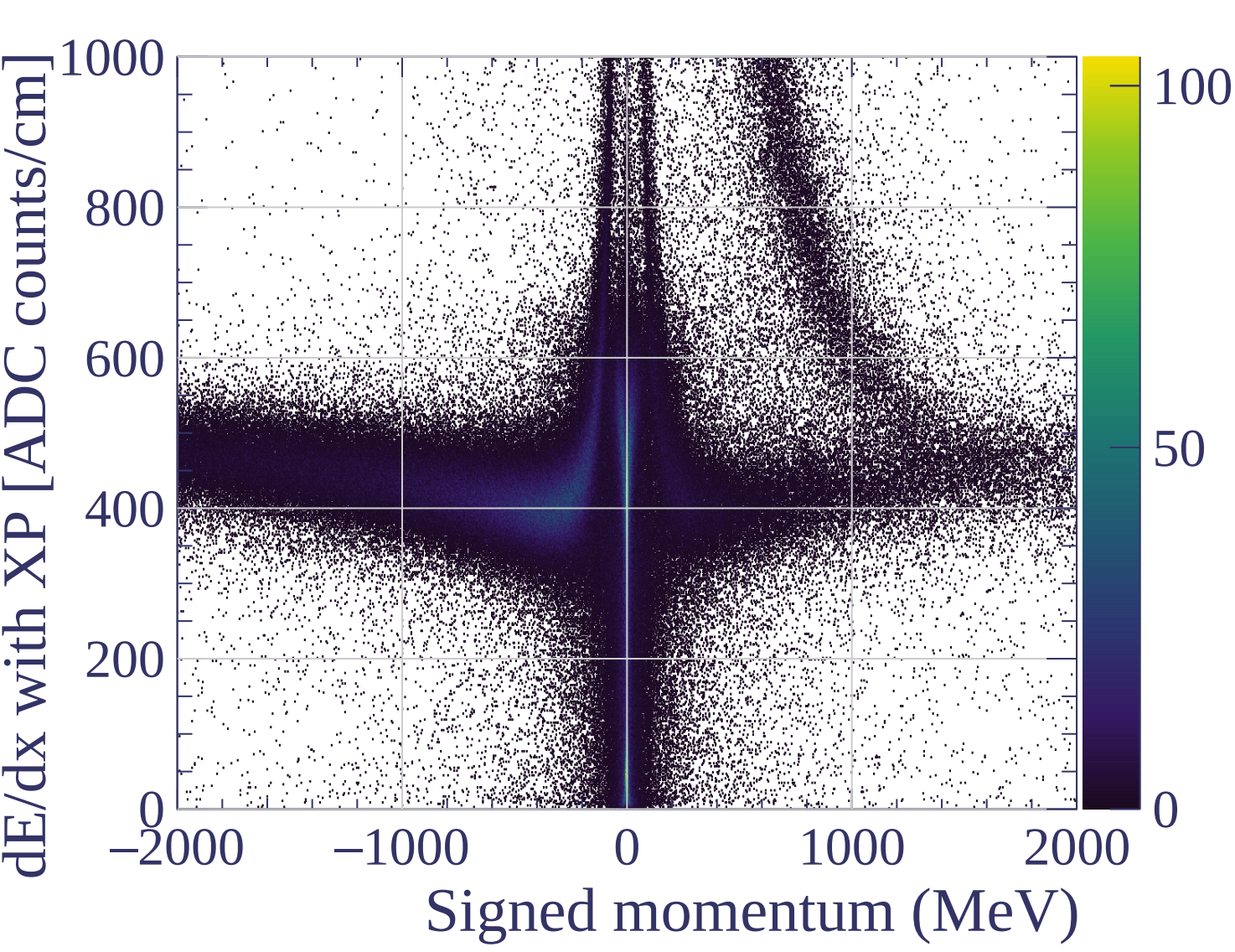


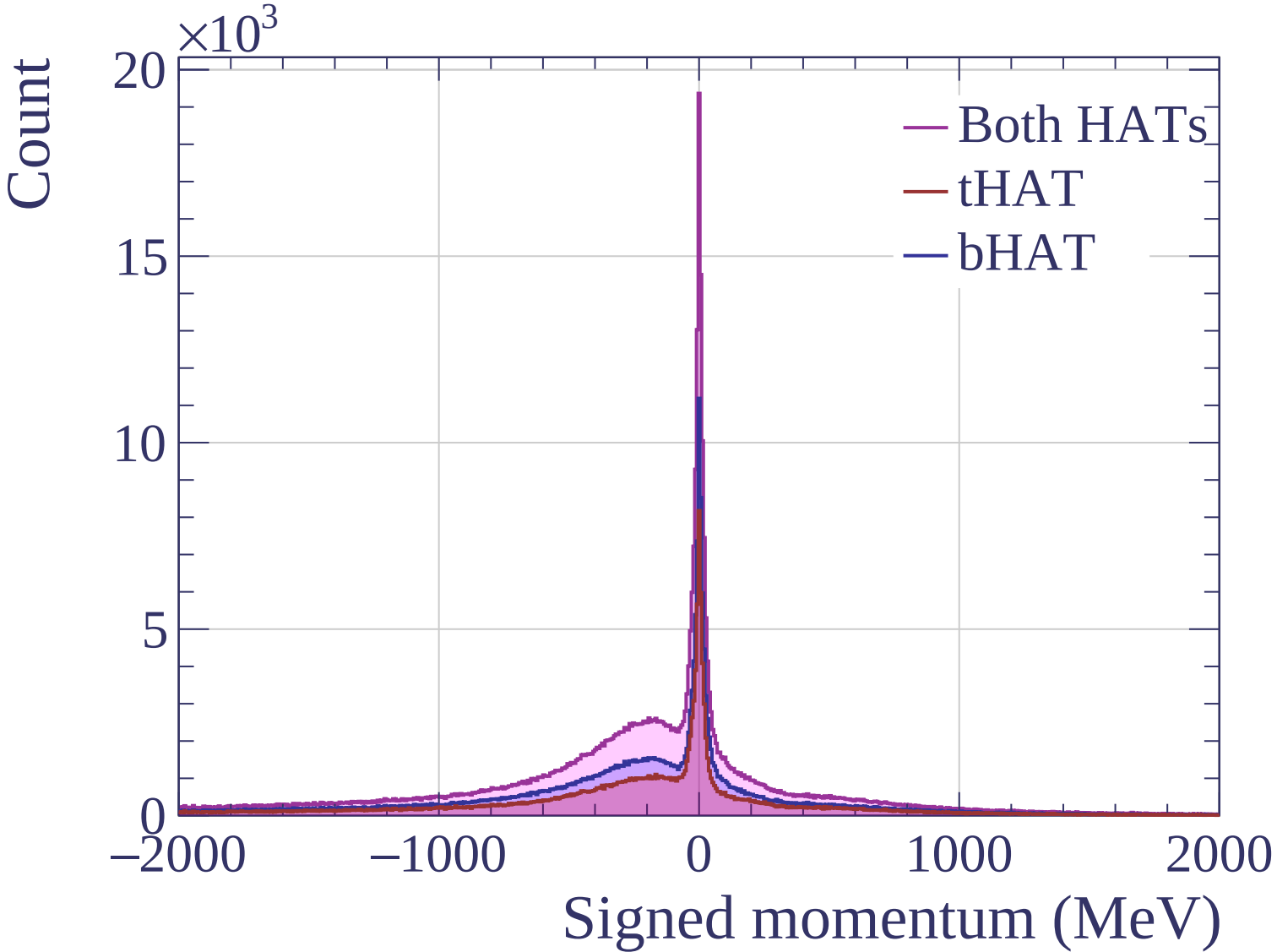
dE/dx resolution [%]

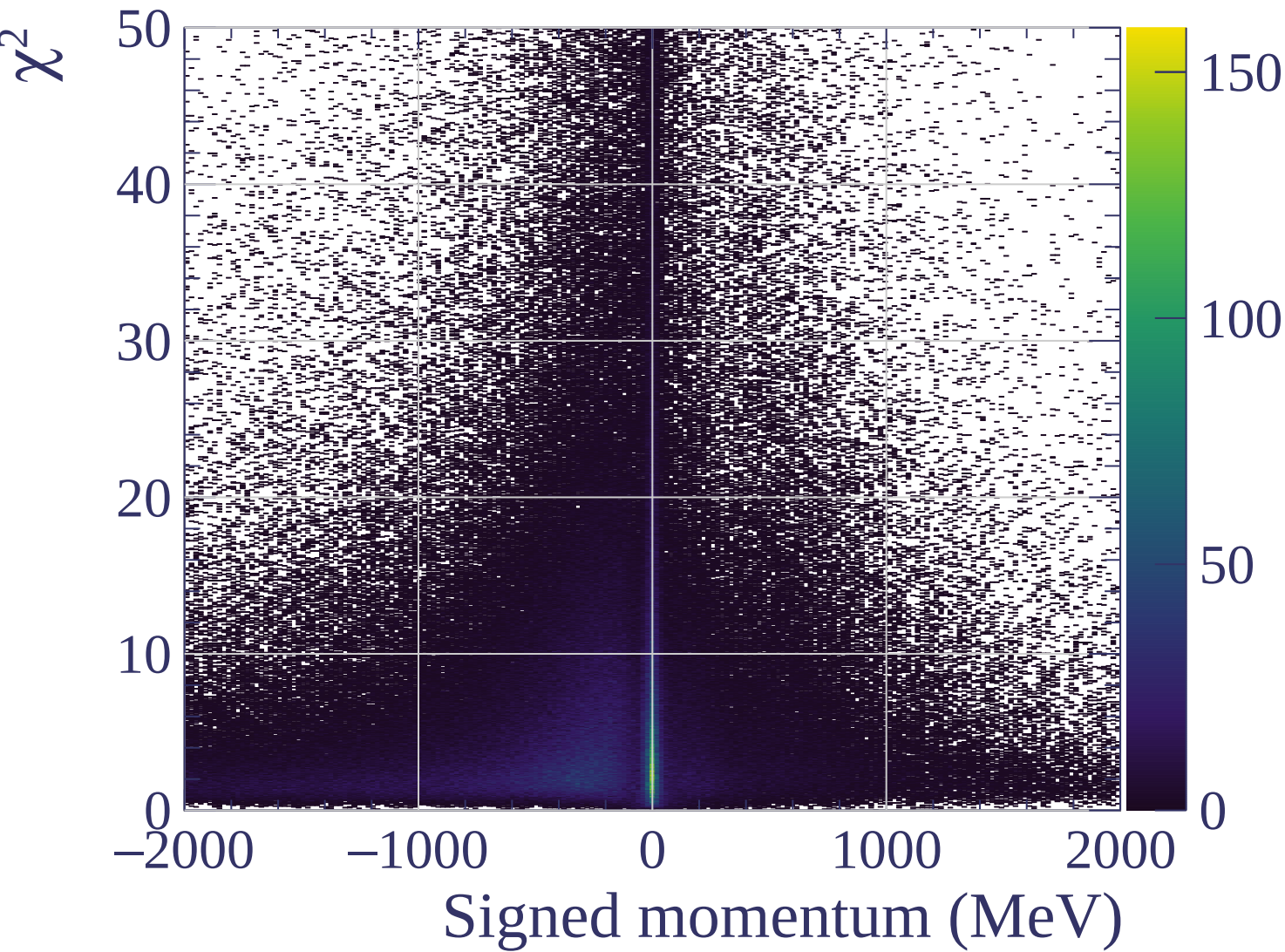


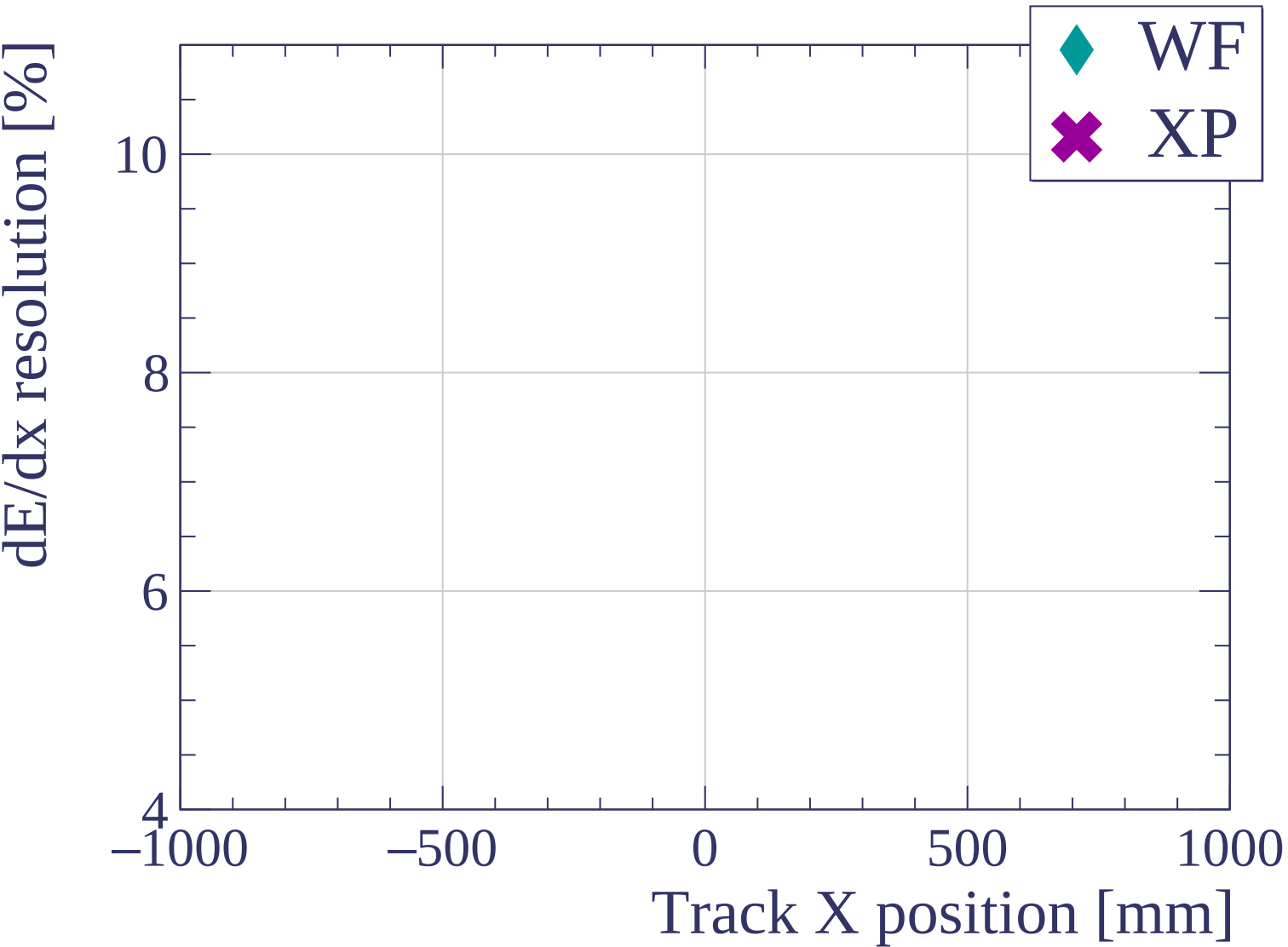


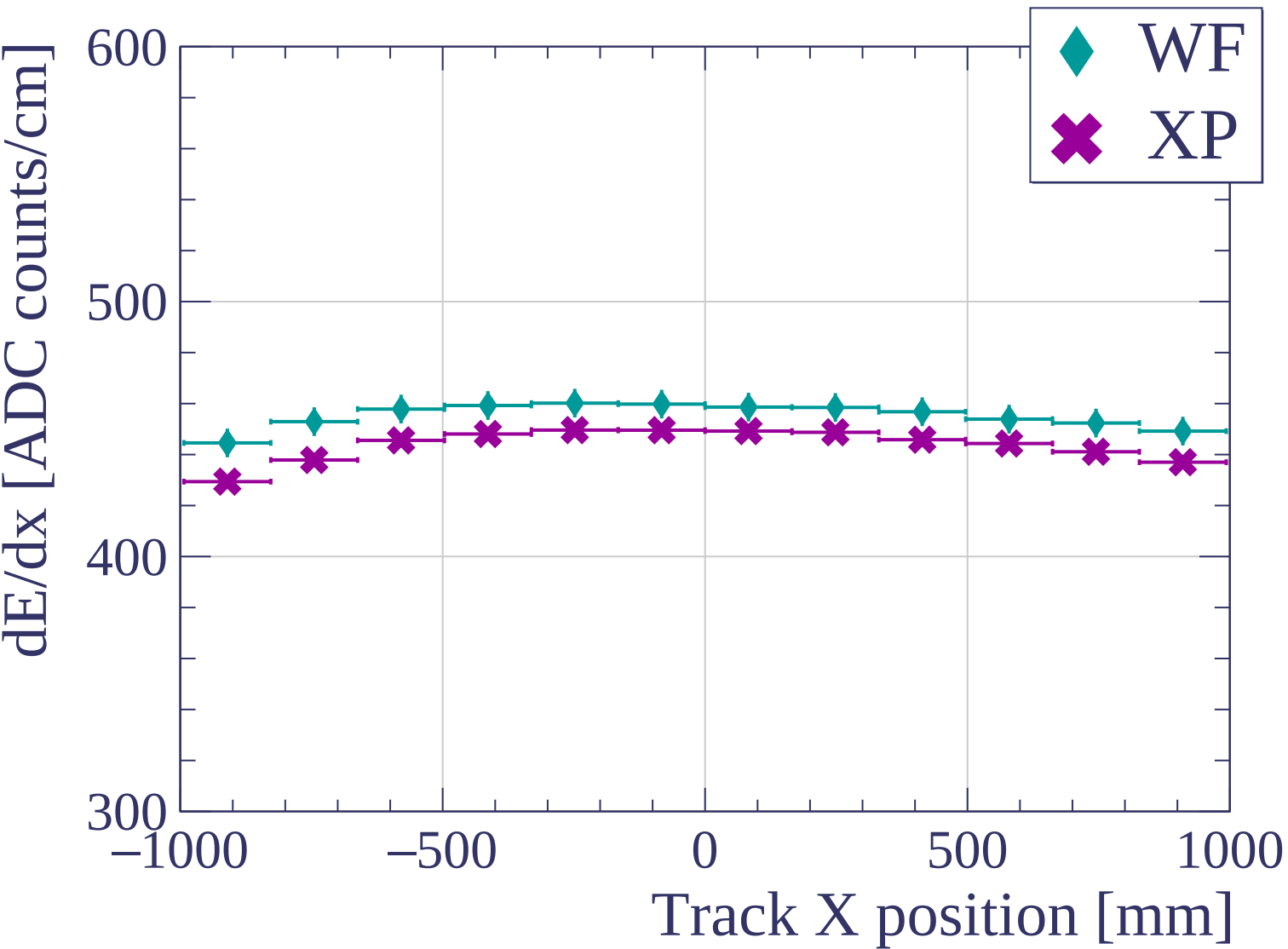


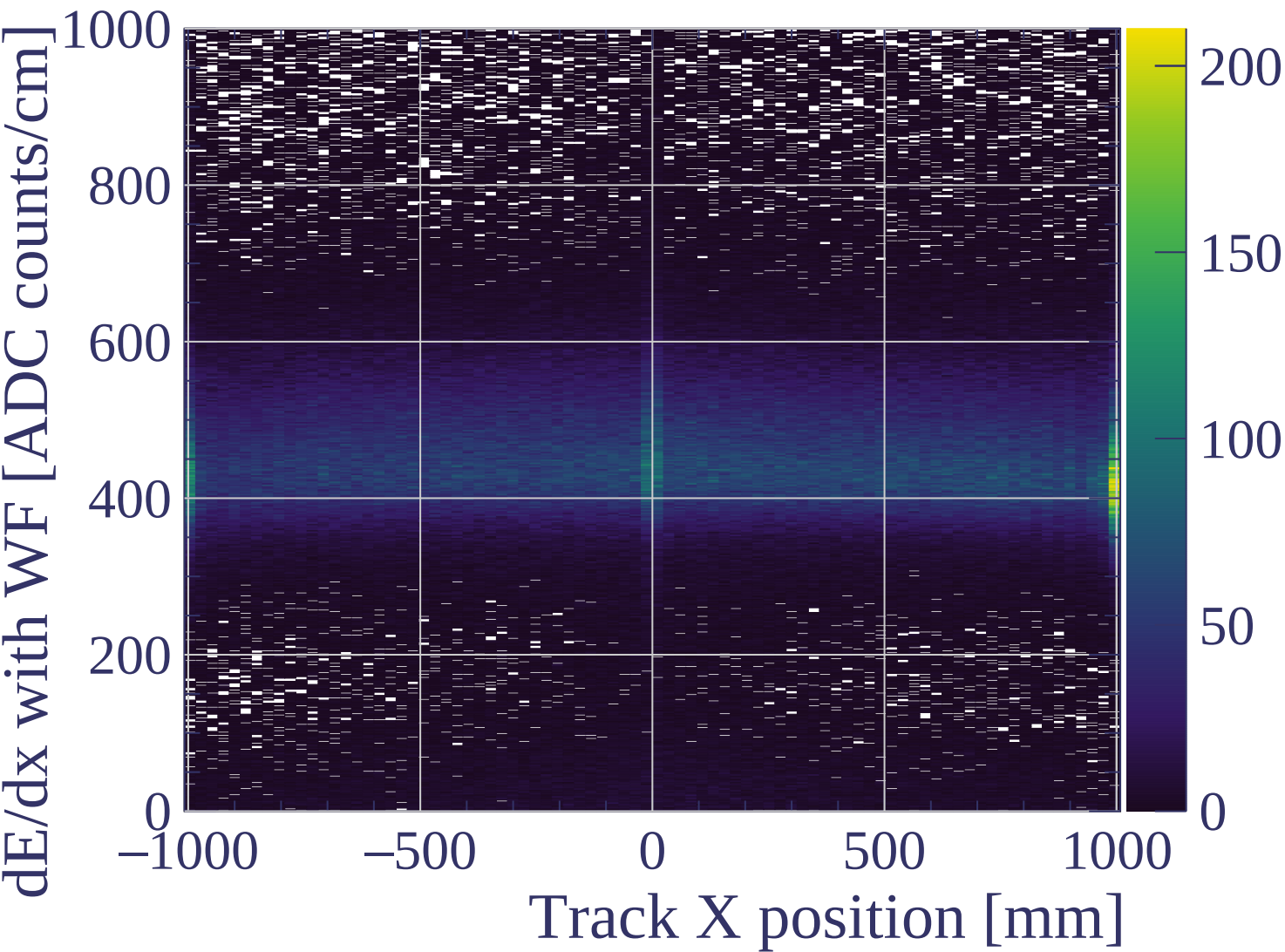


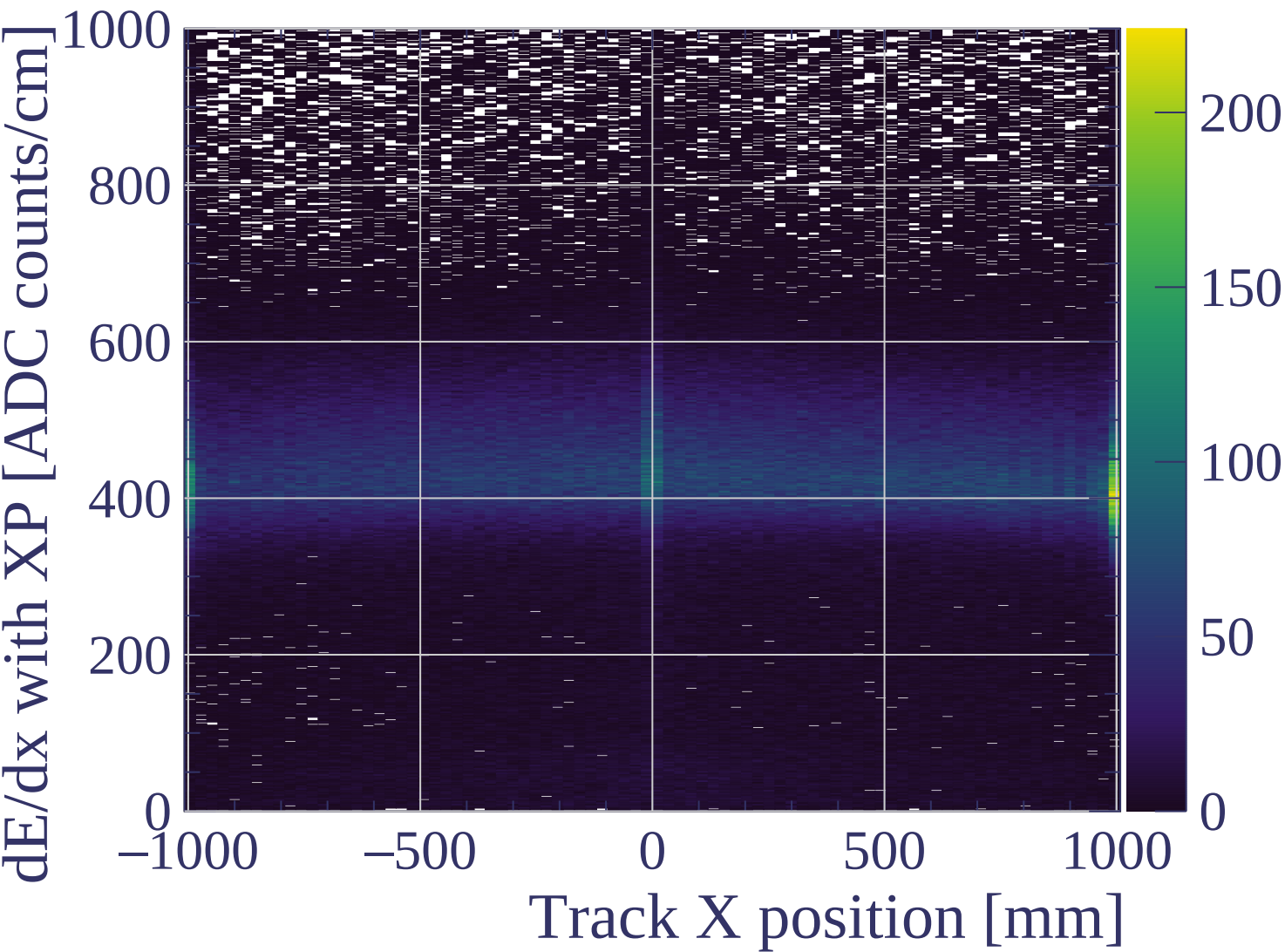




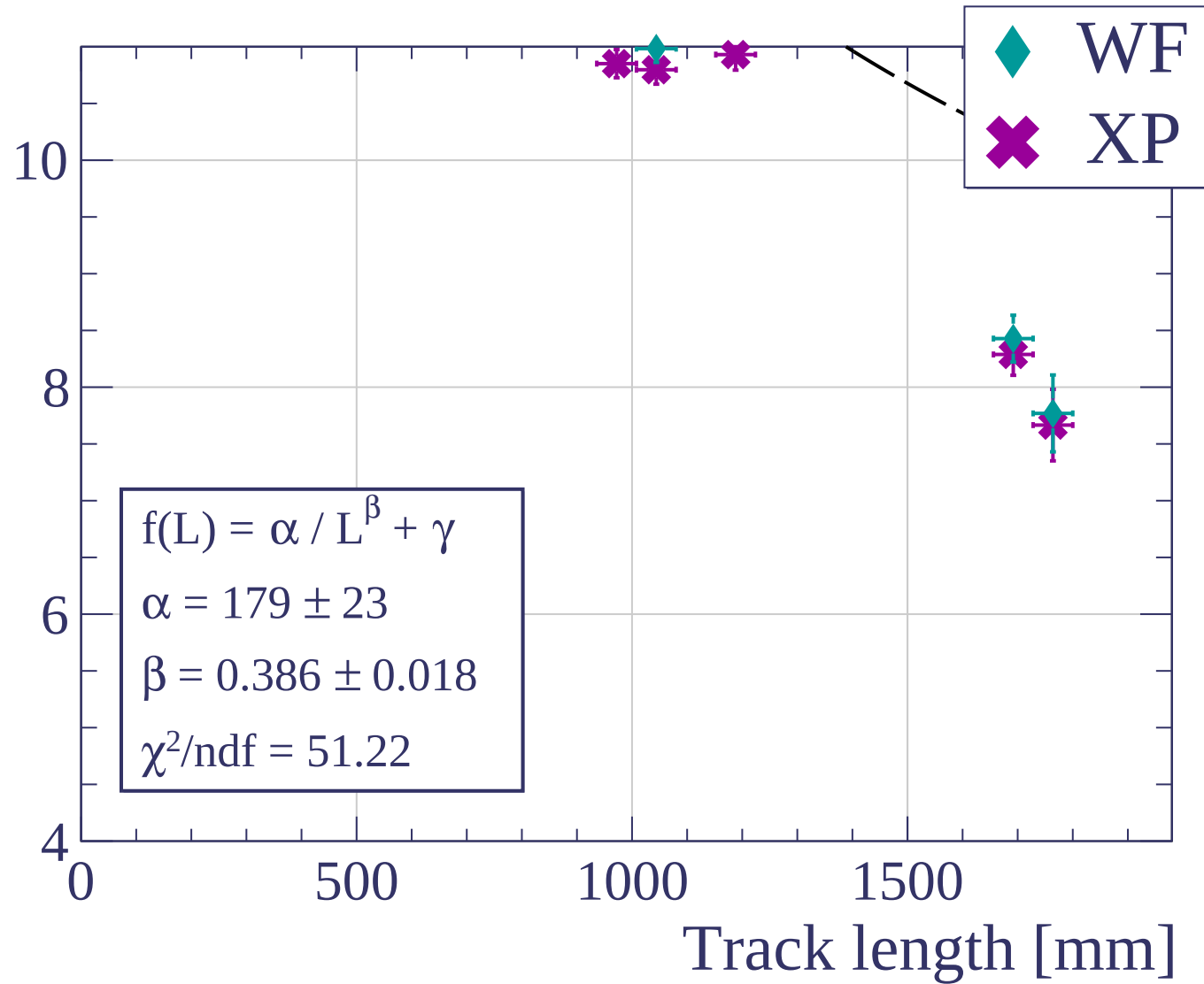


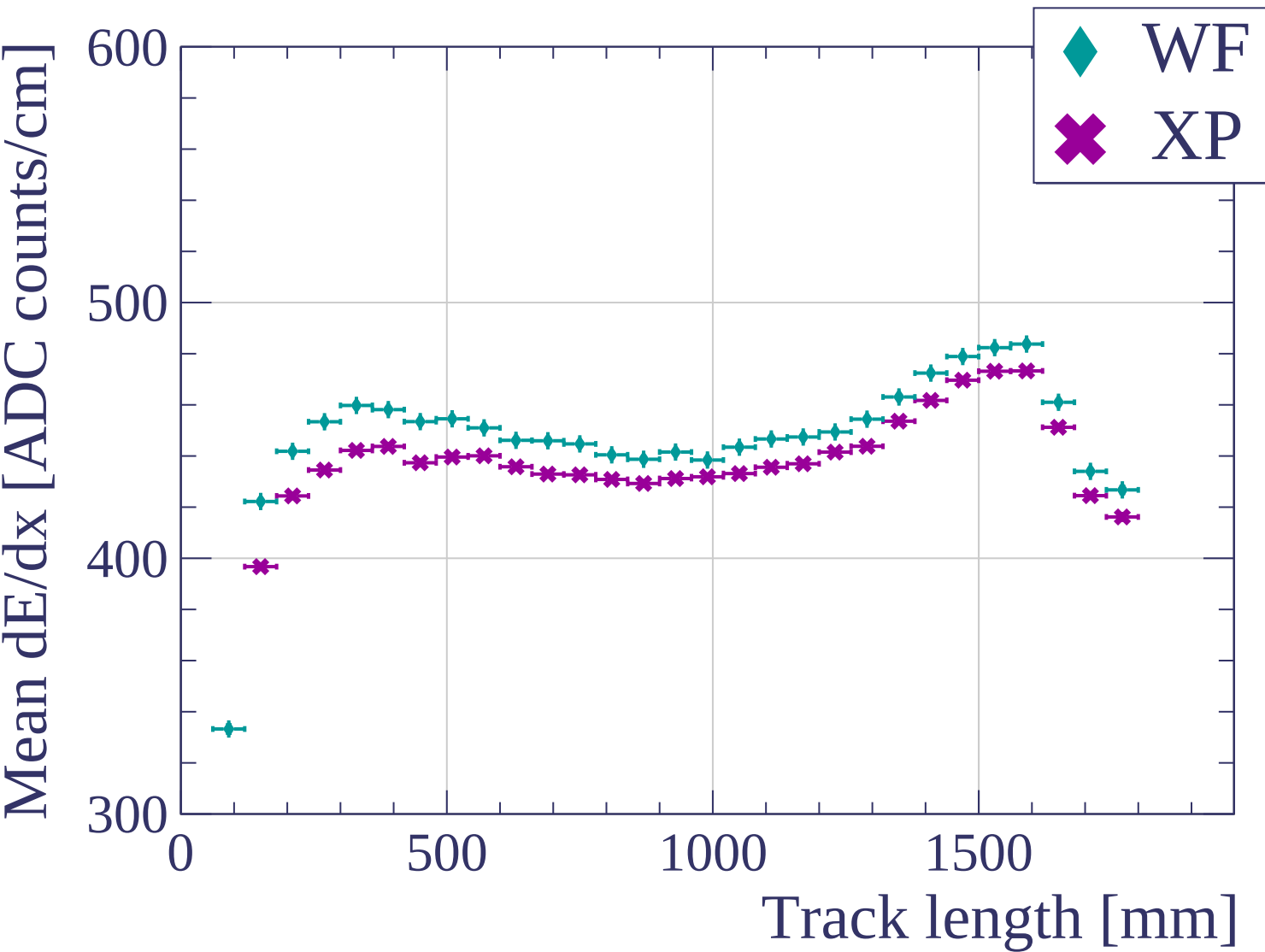


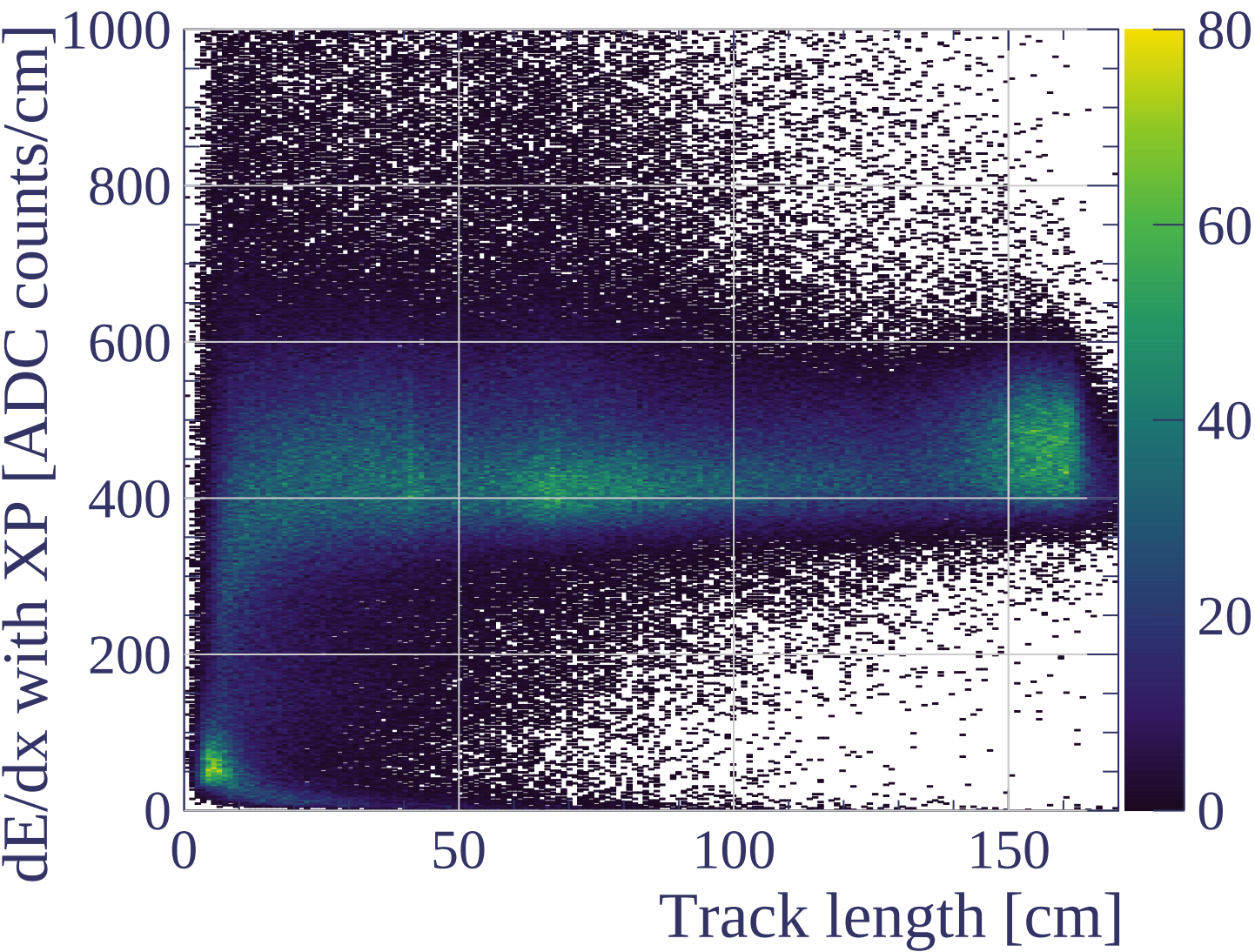


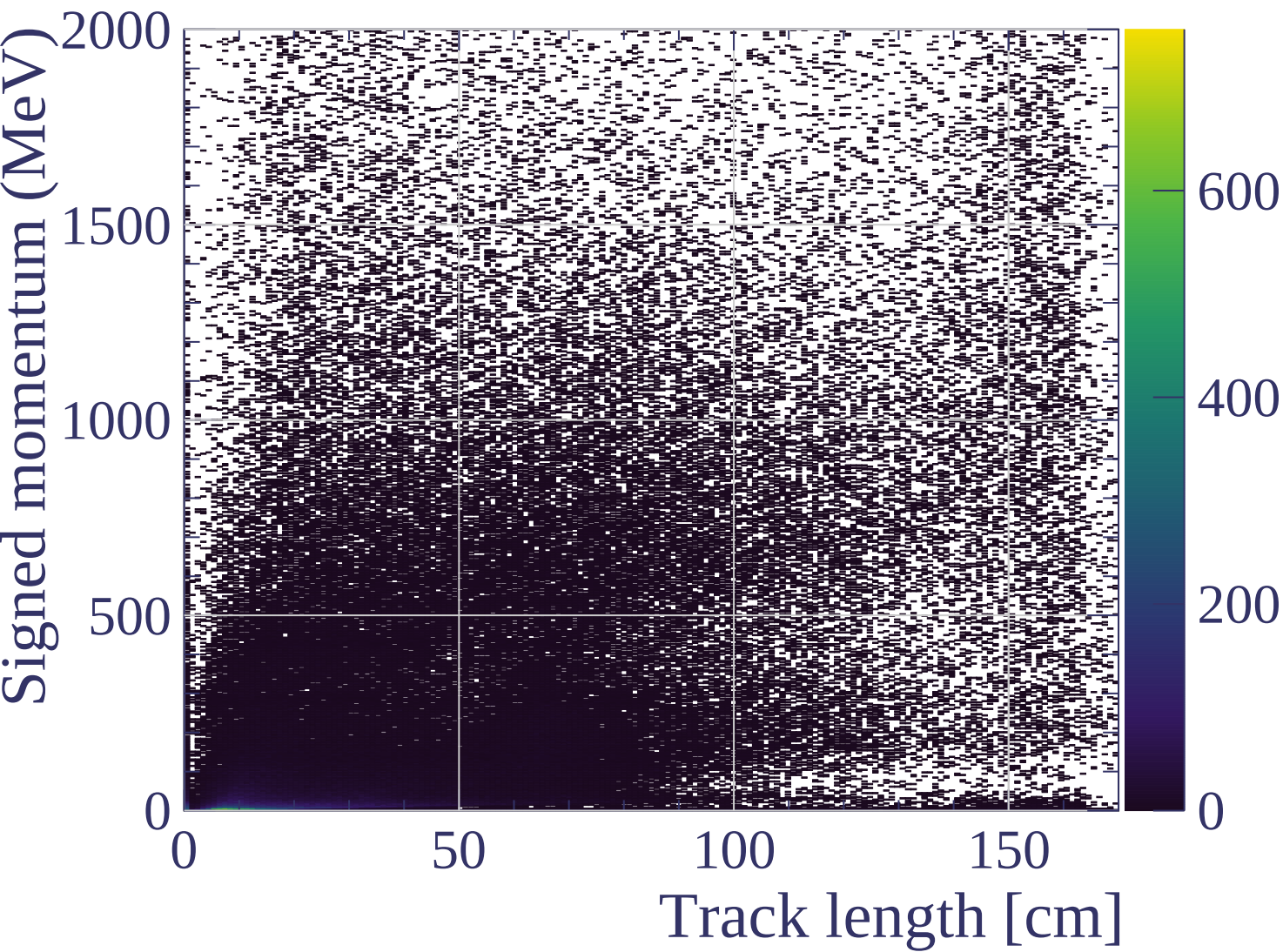


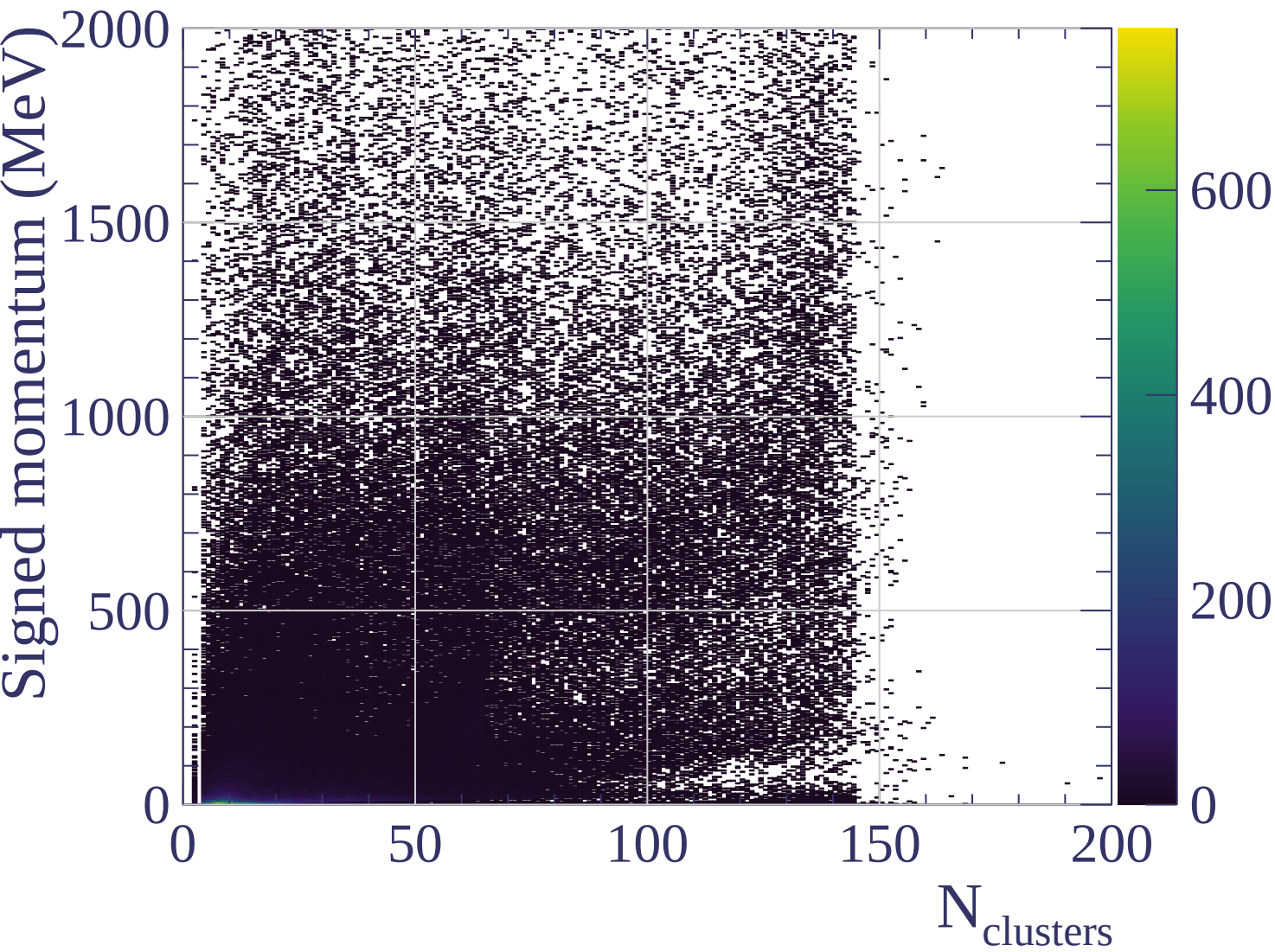
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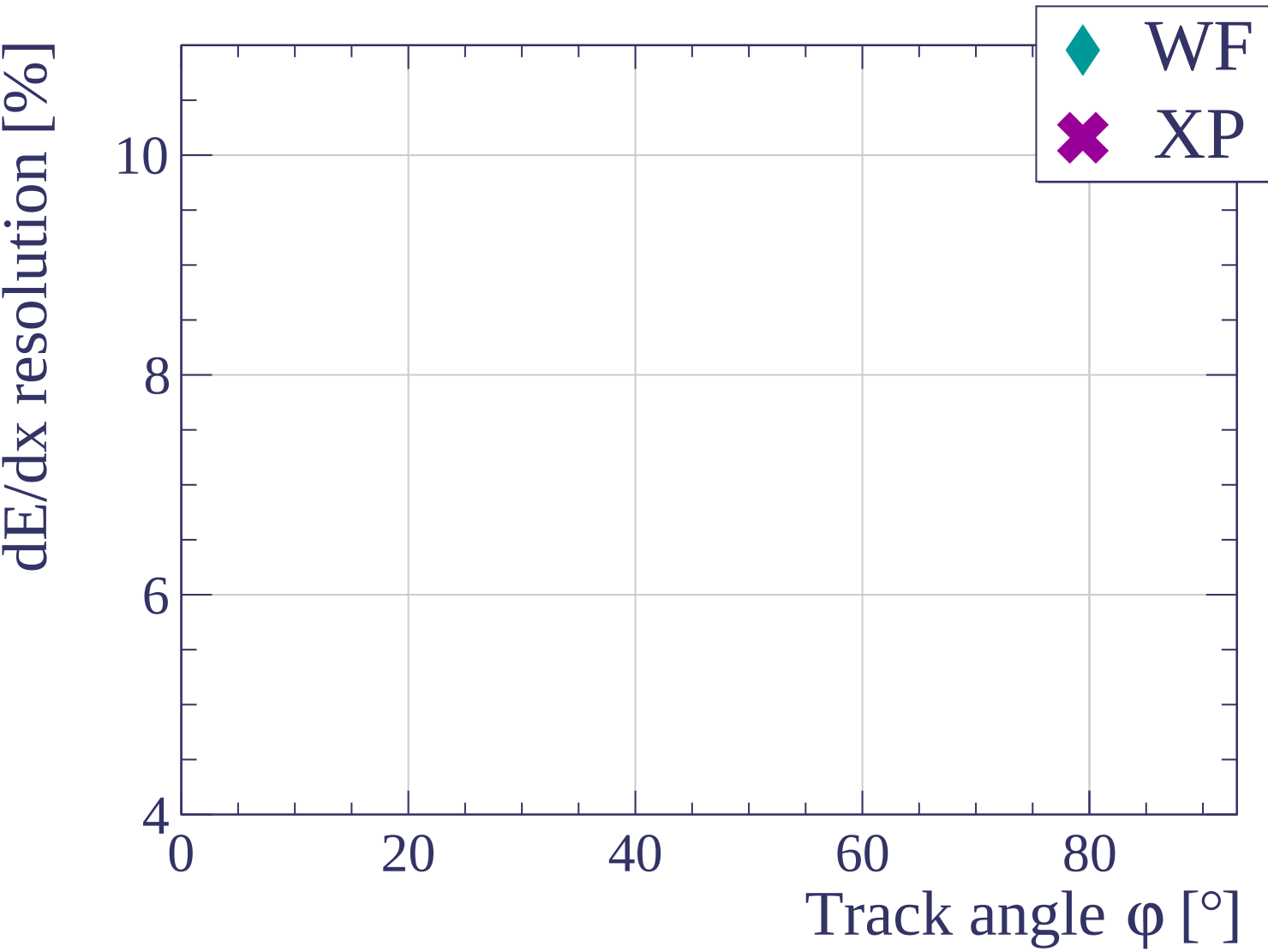


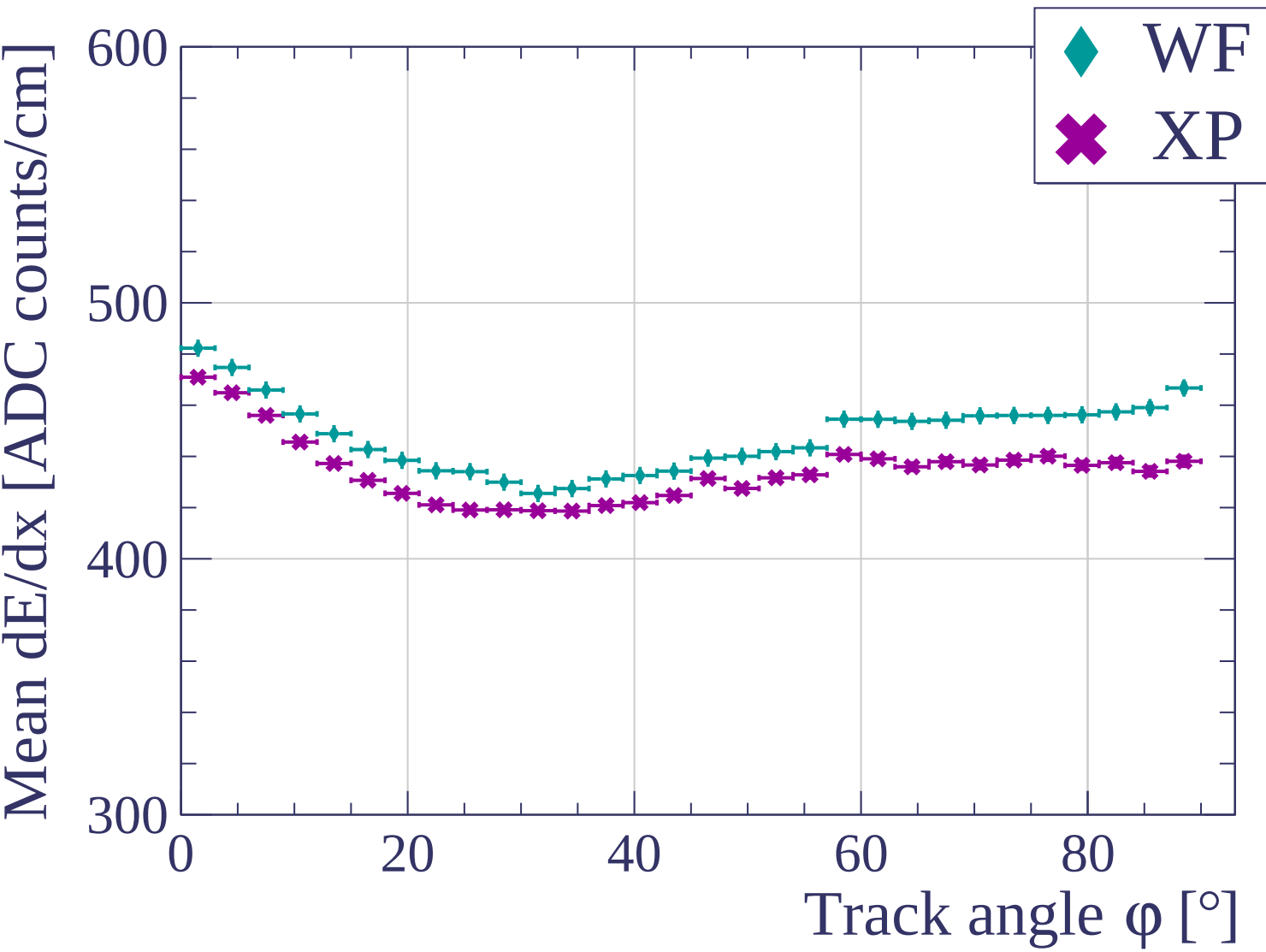


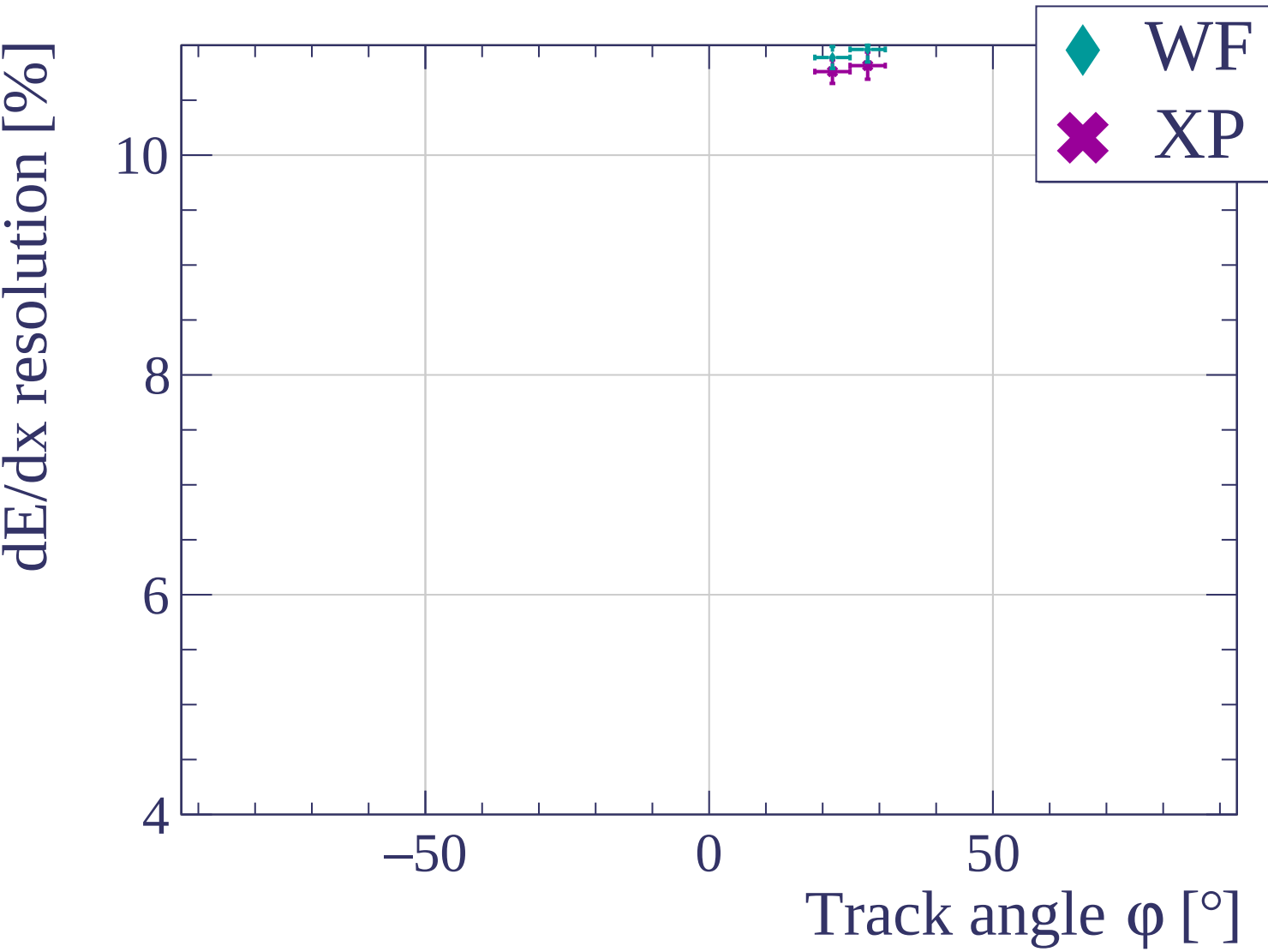


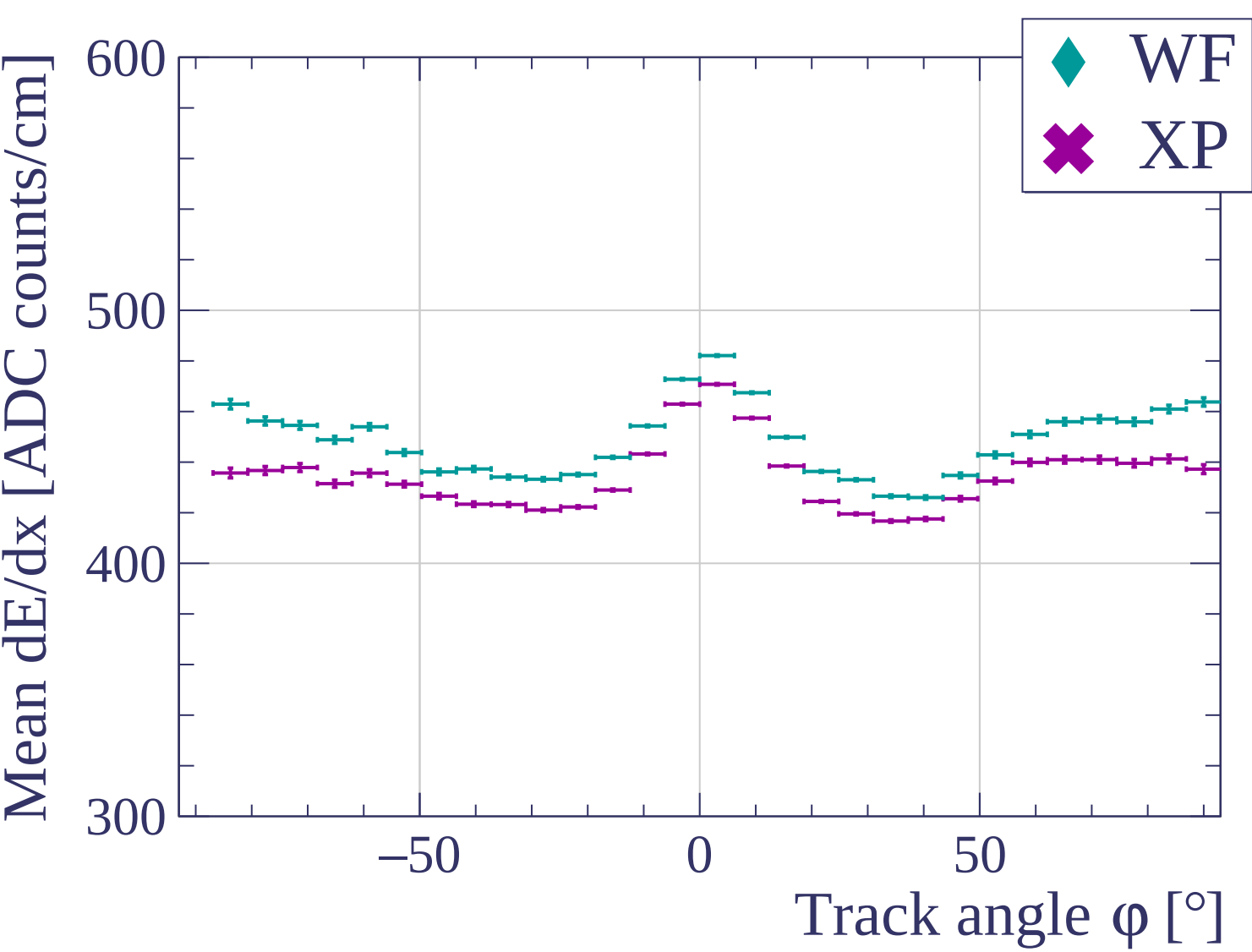


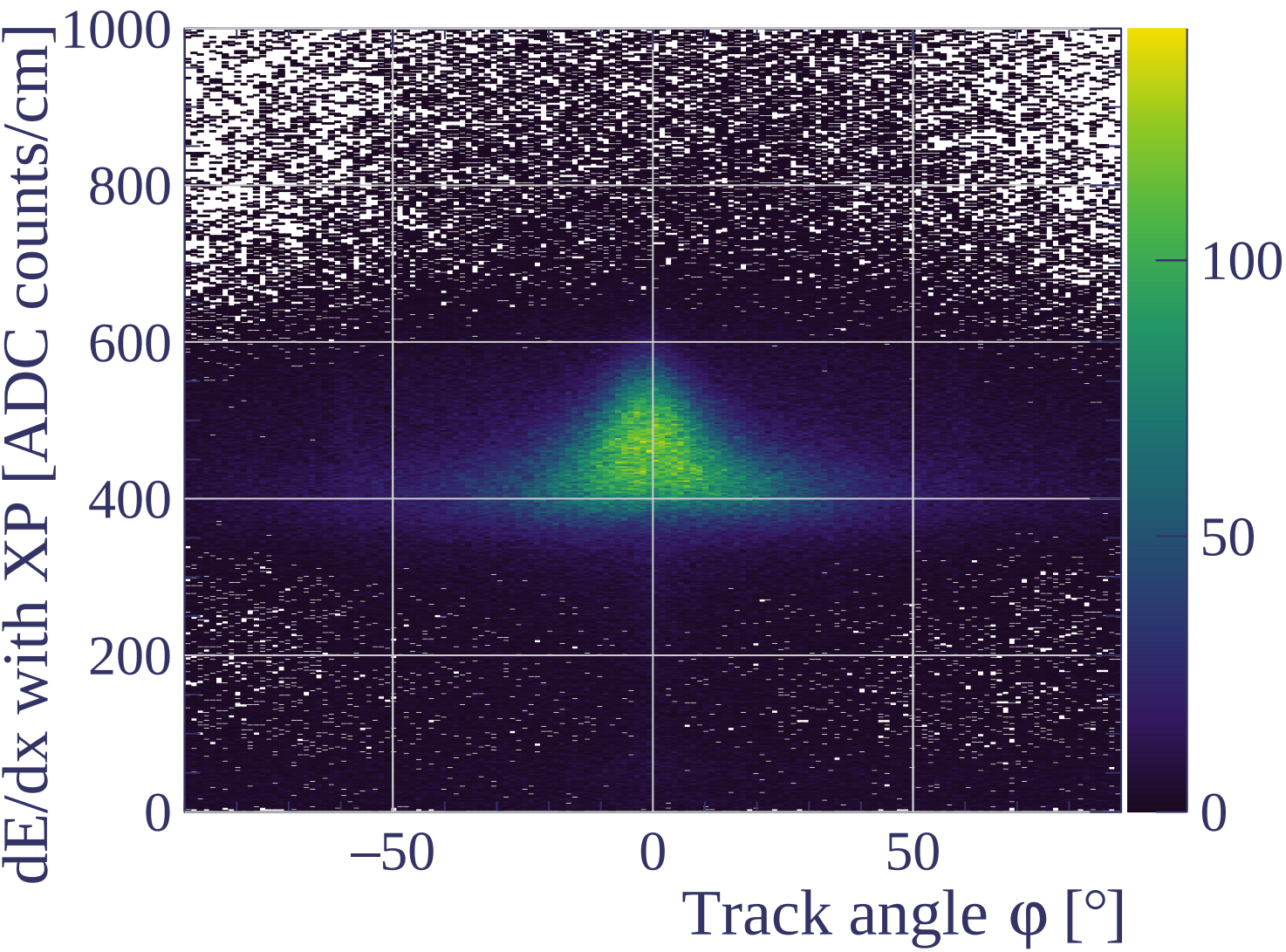


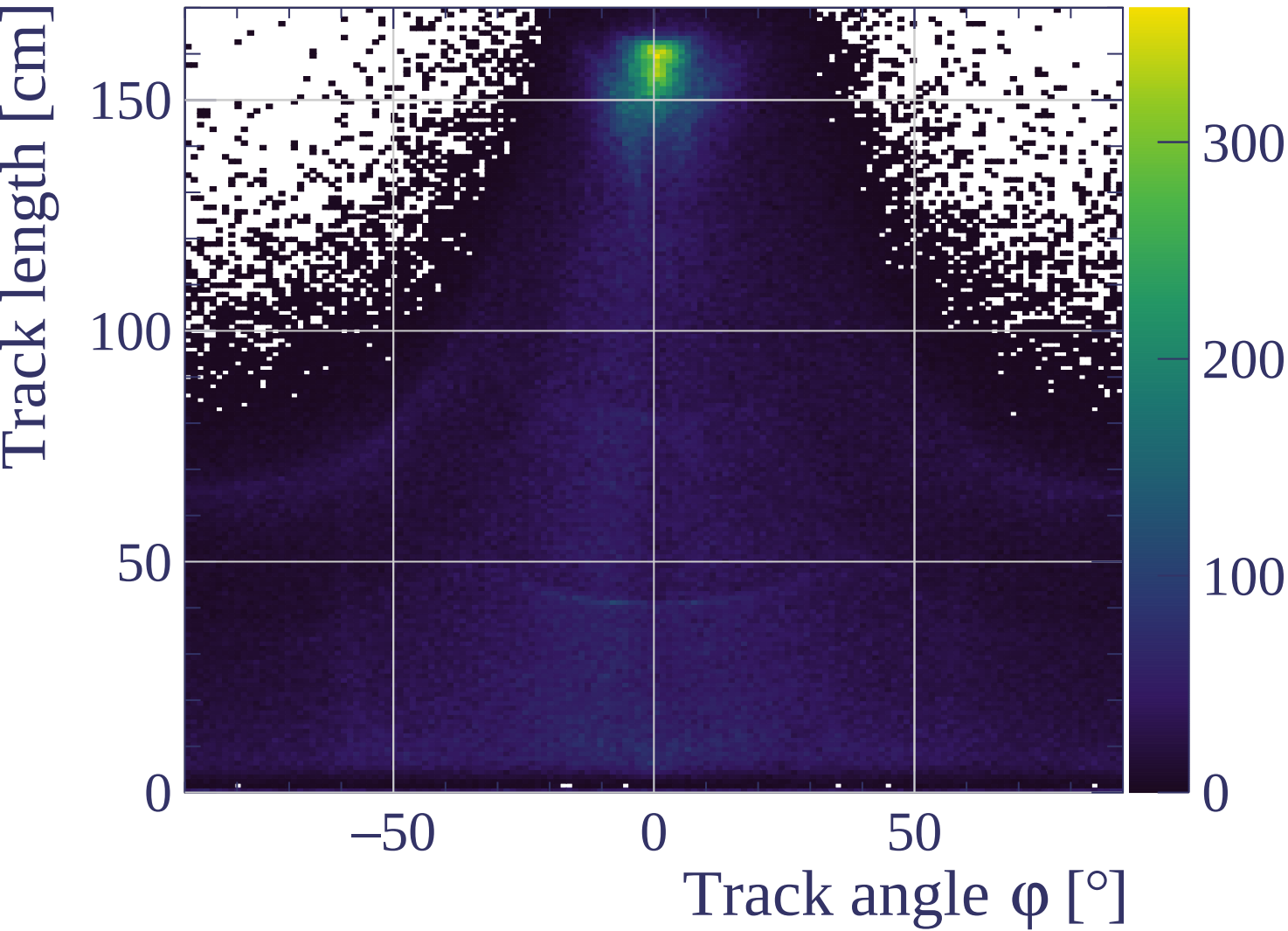


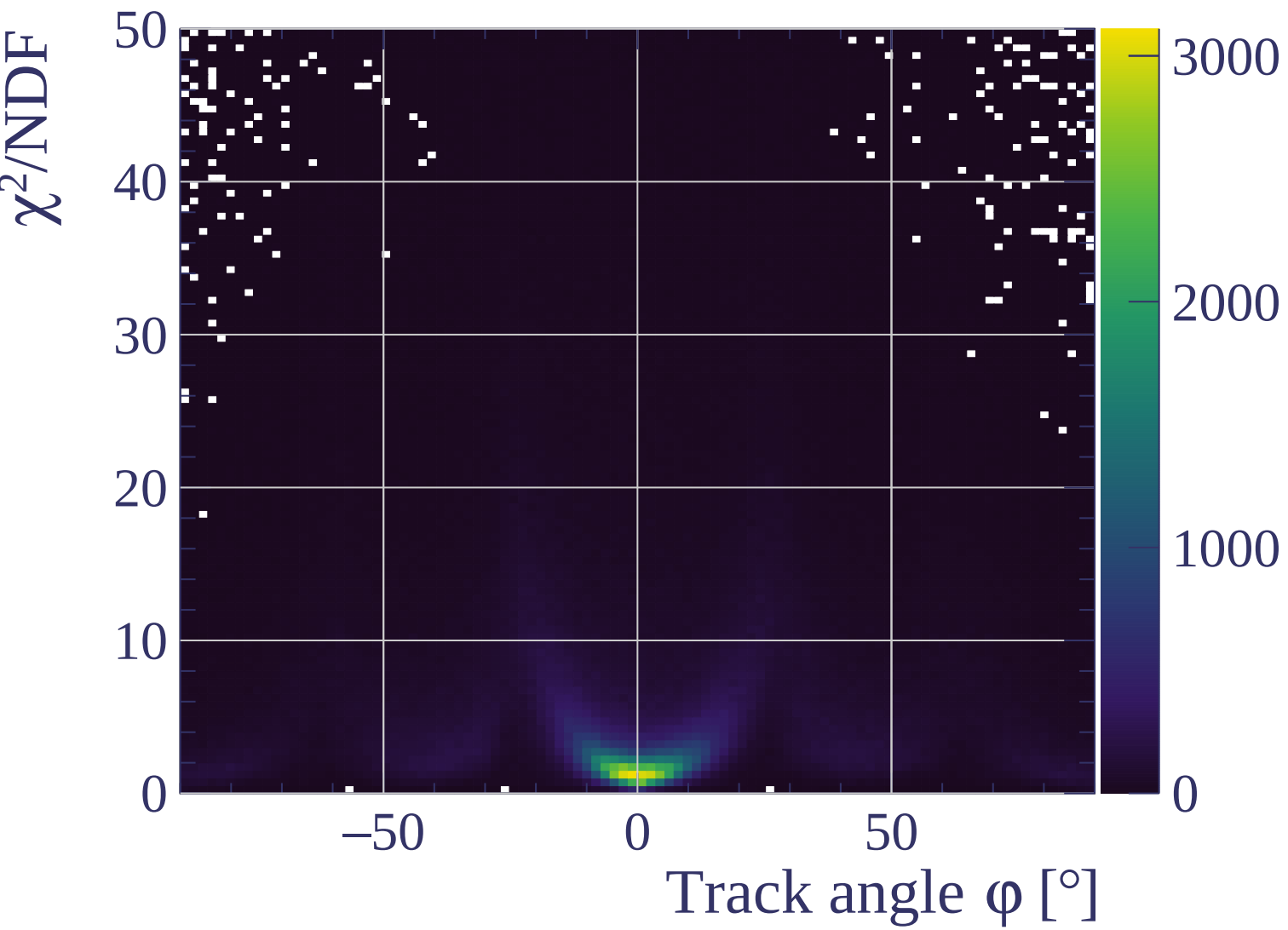


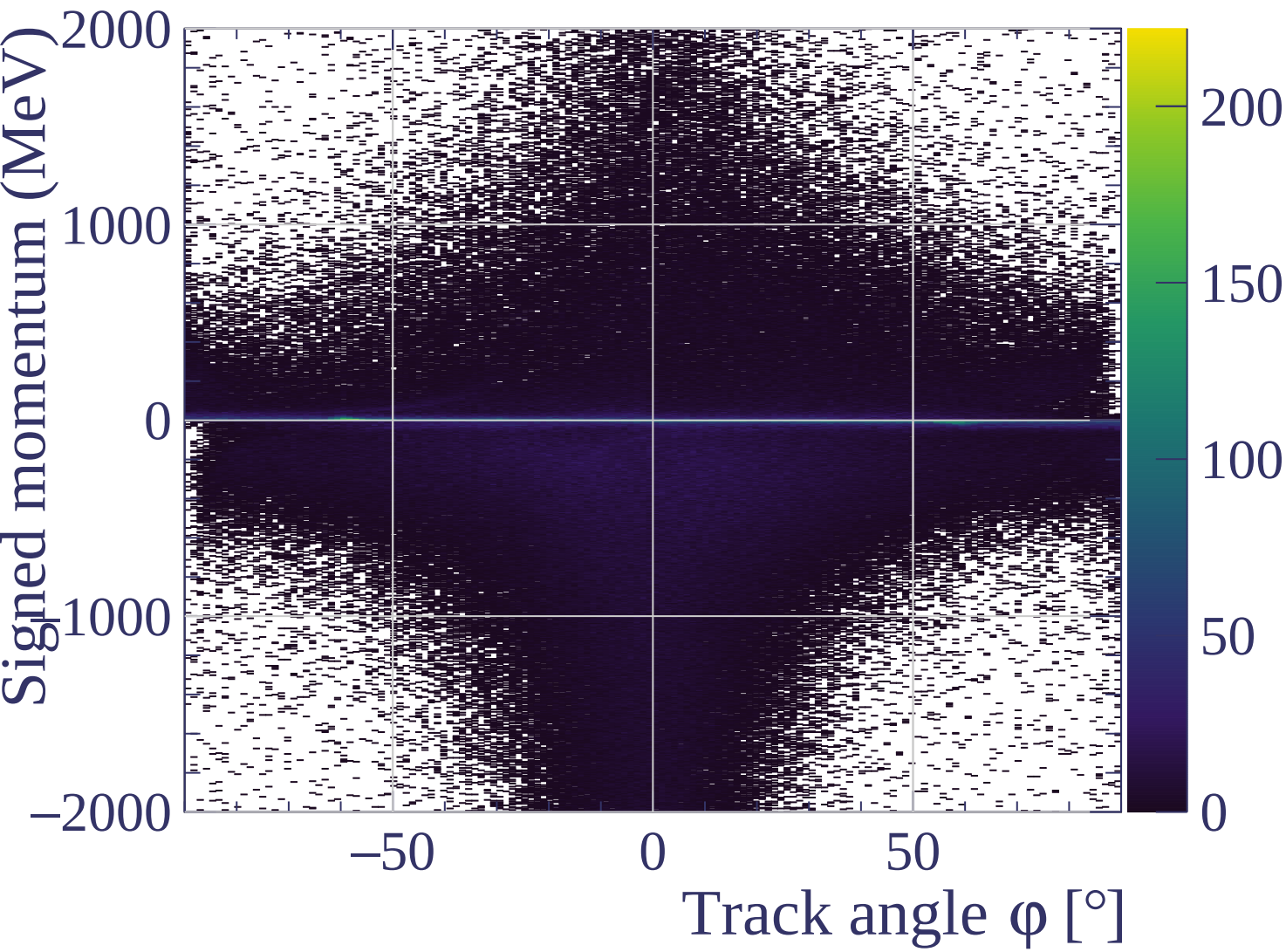


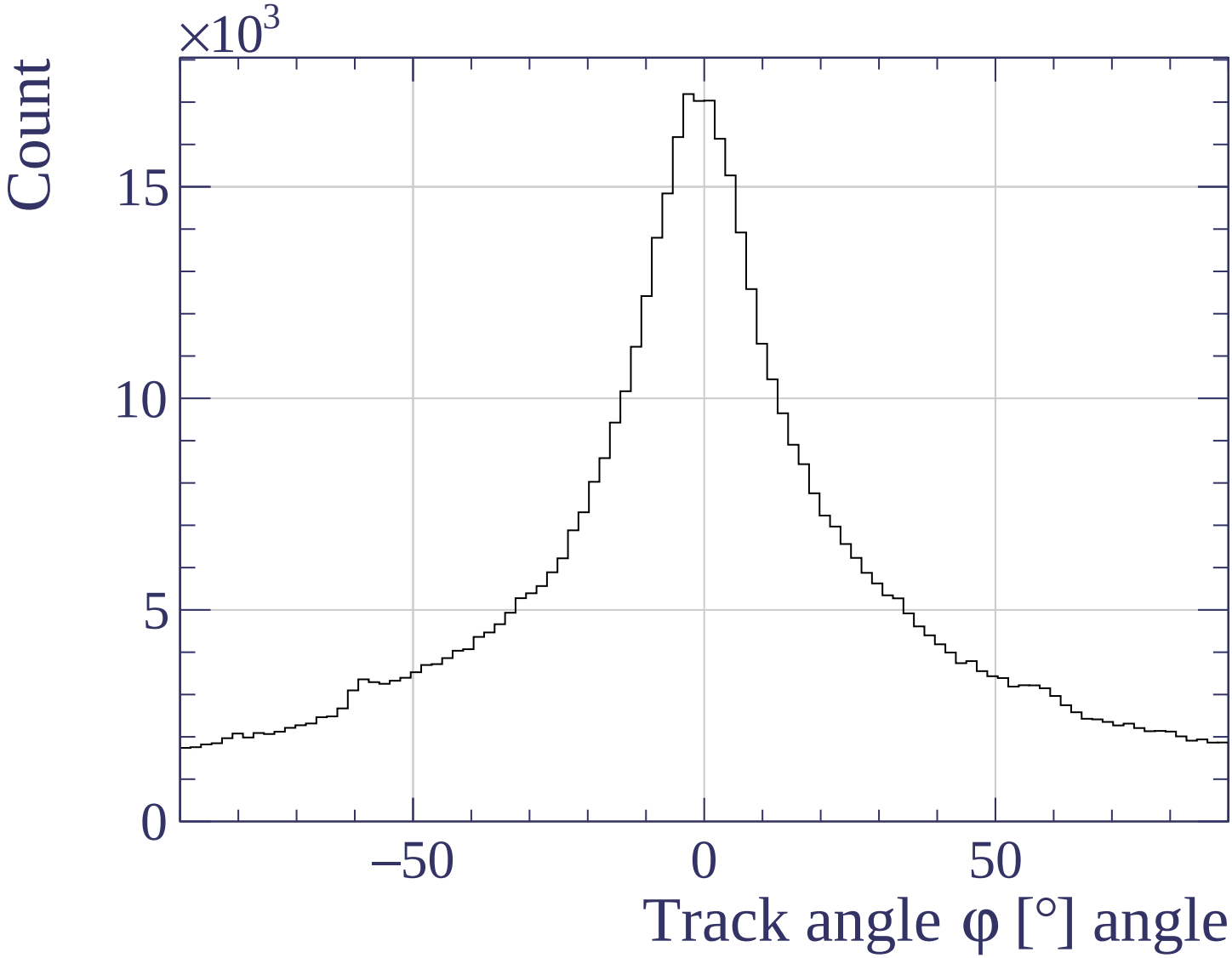


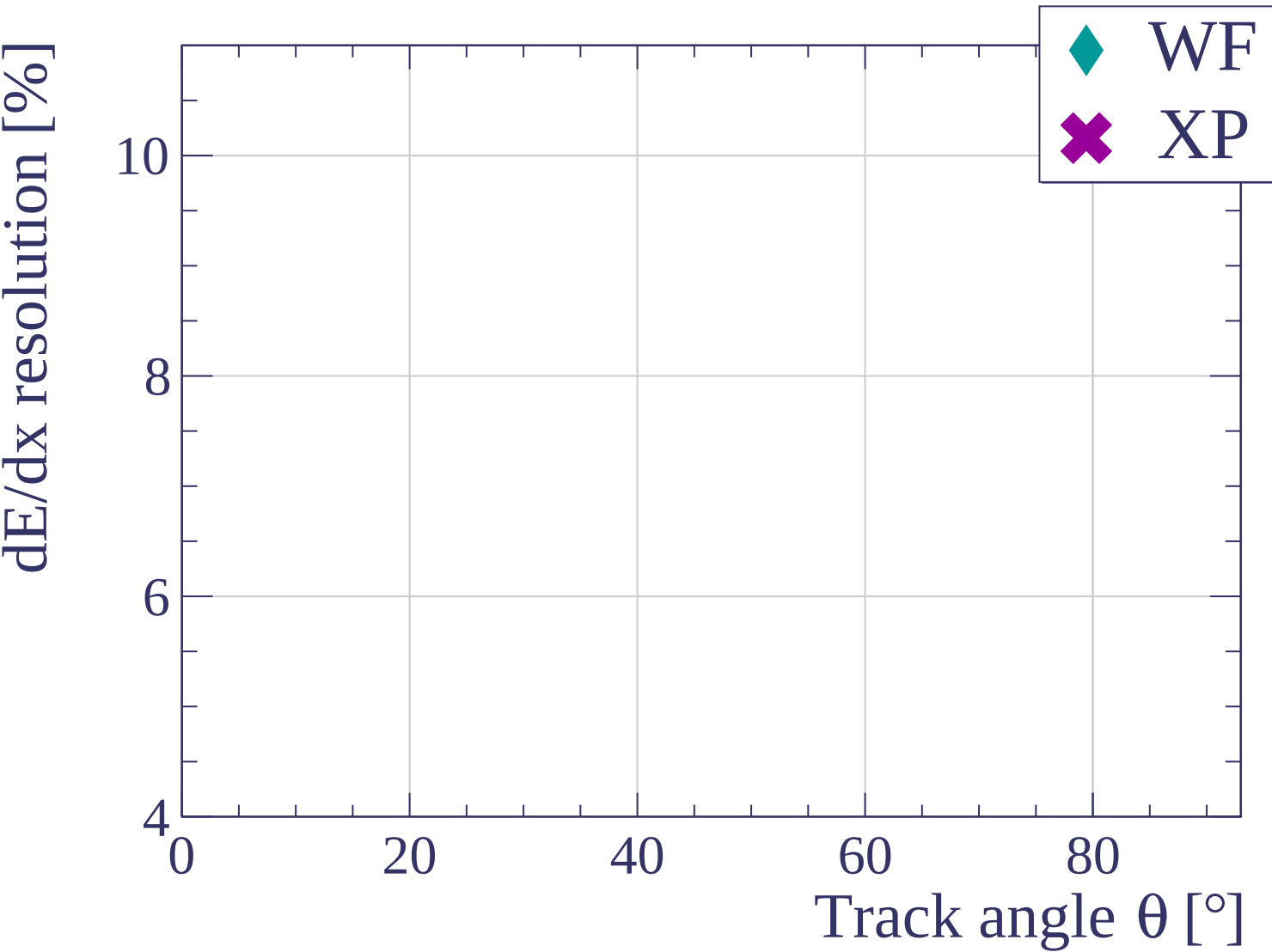


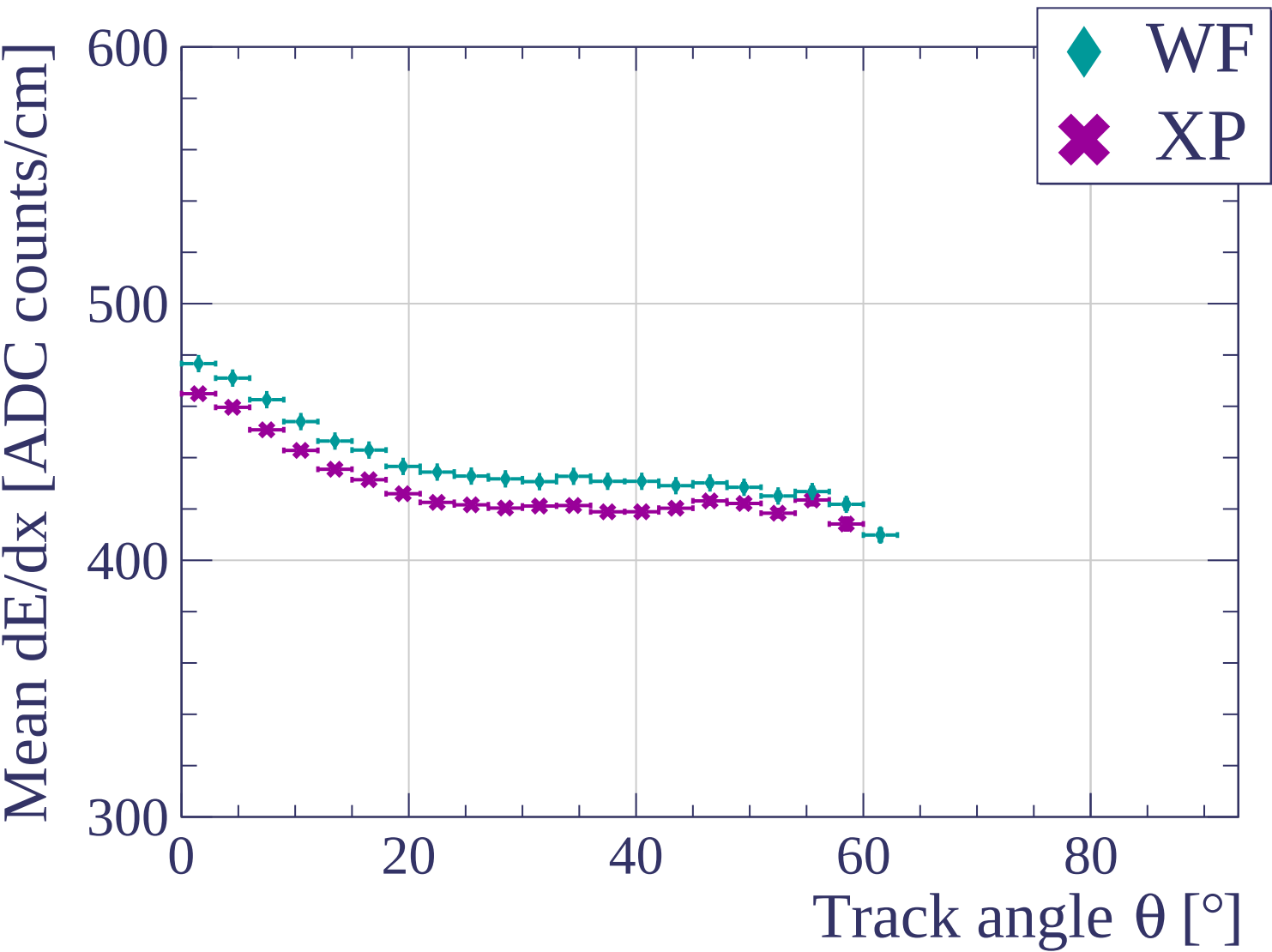


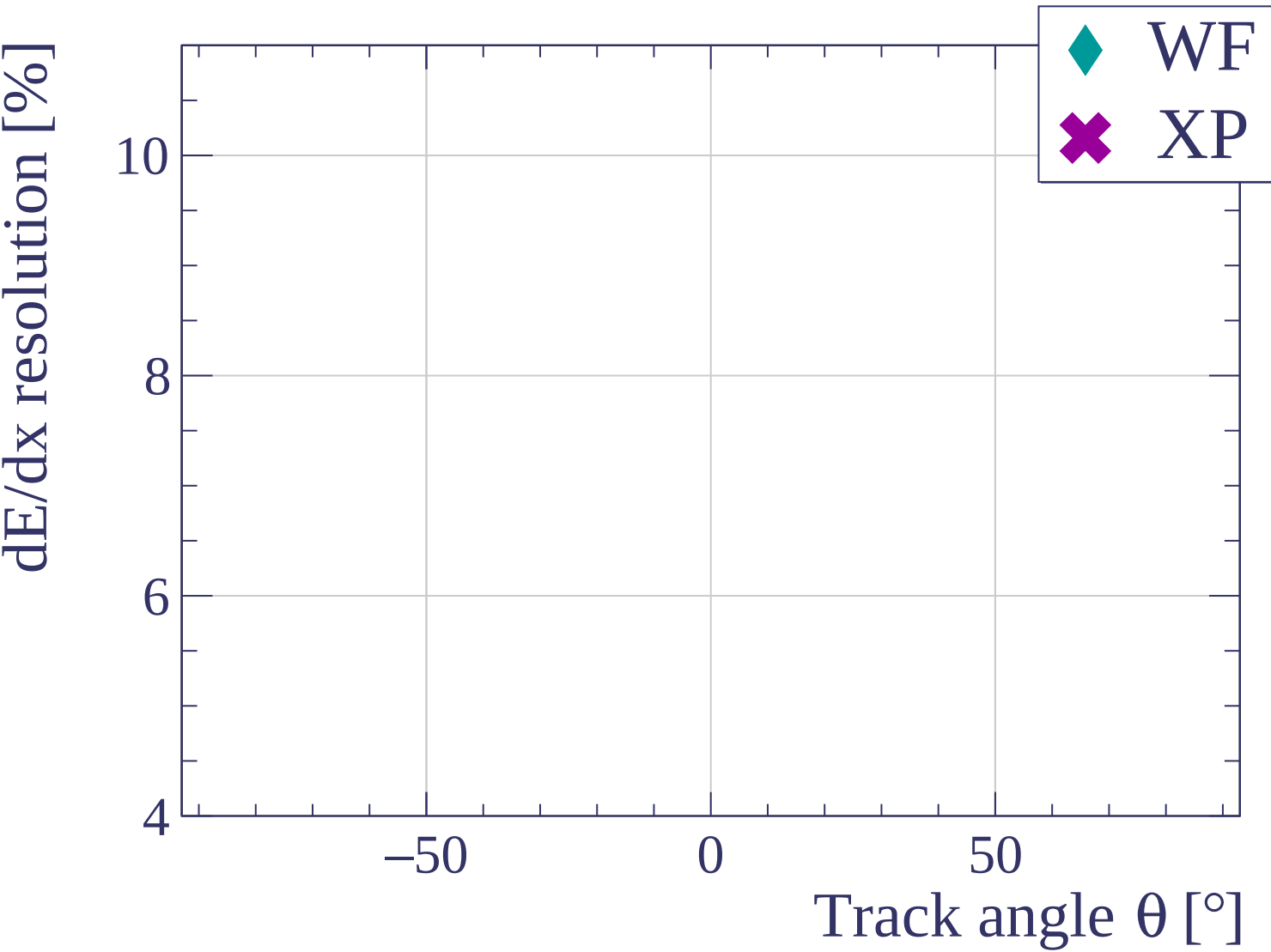


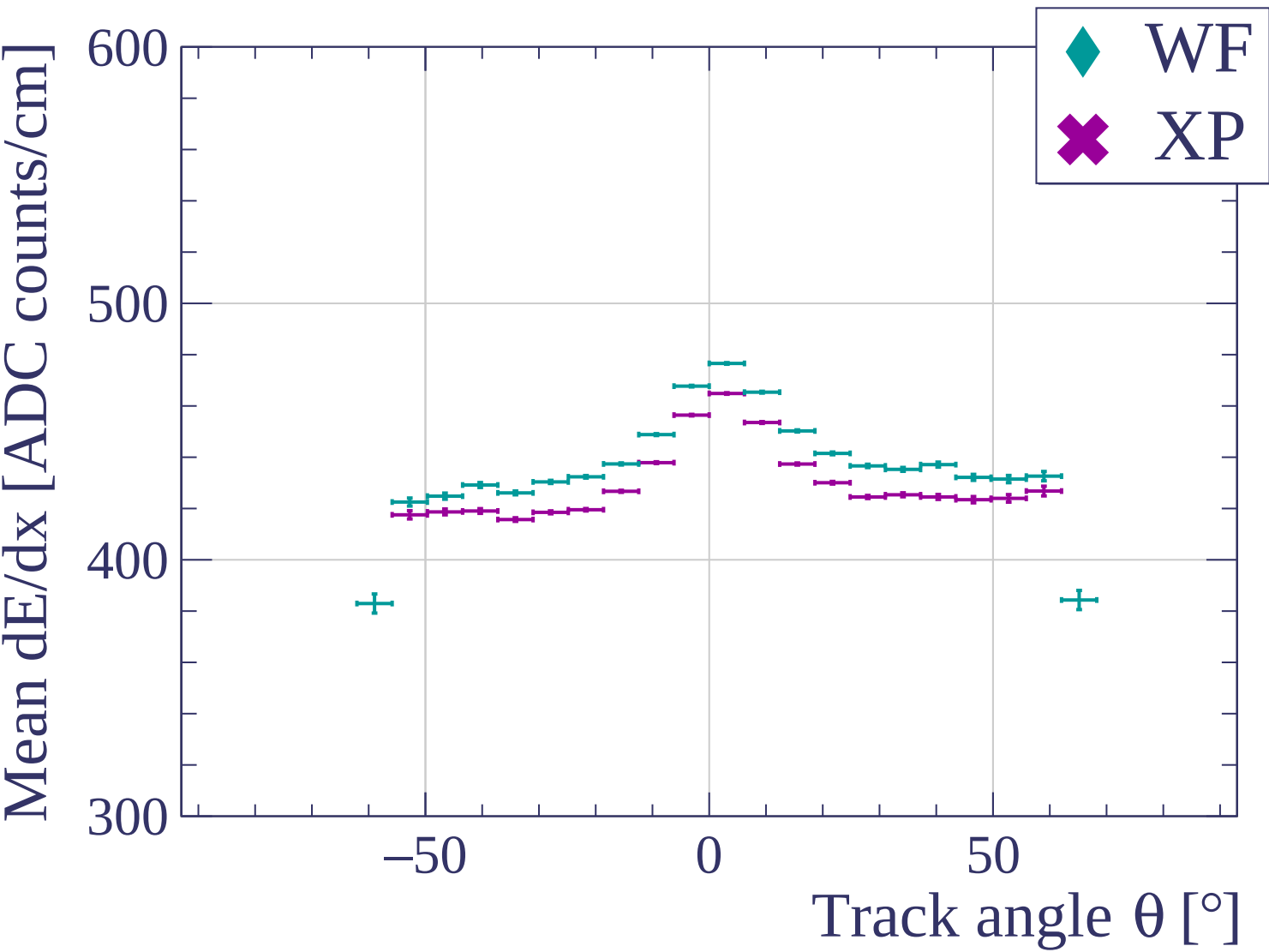


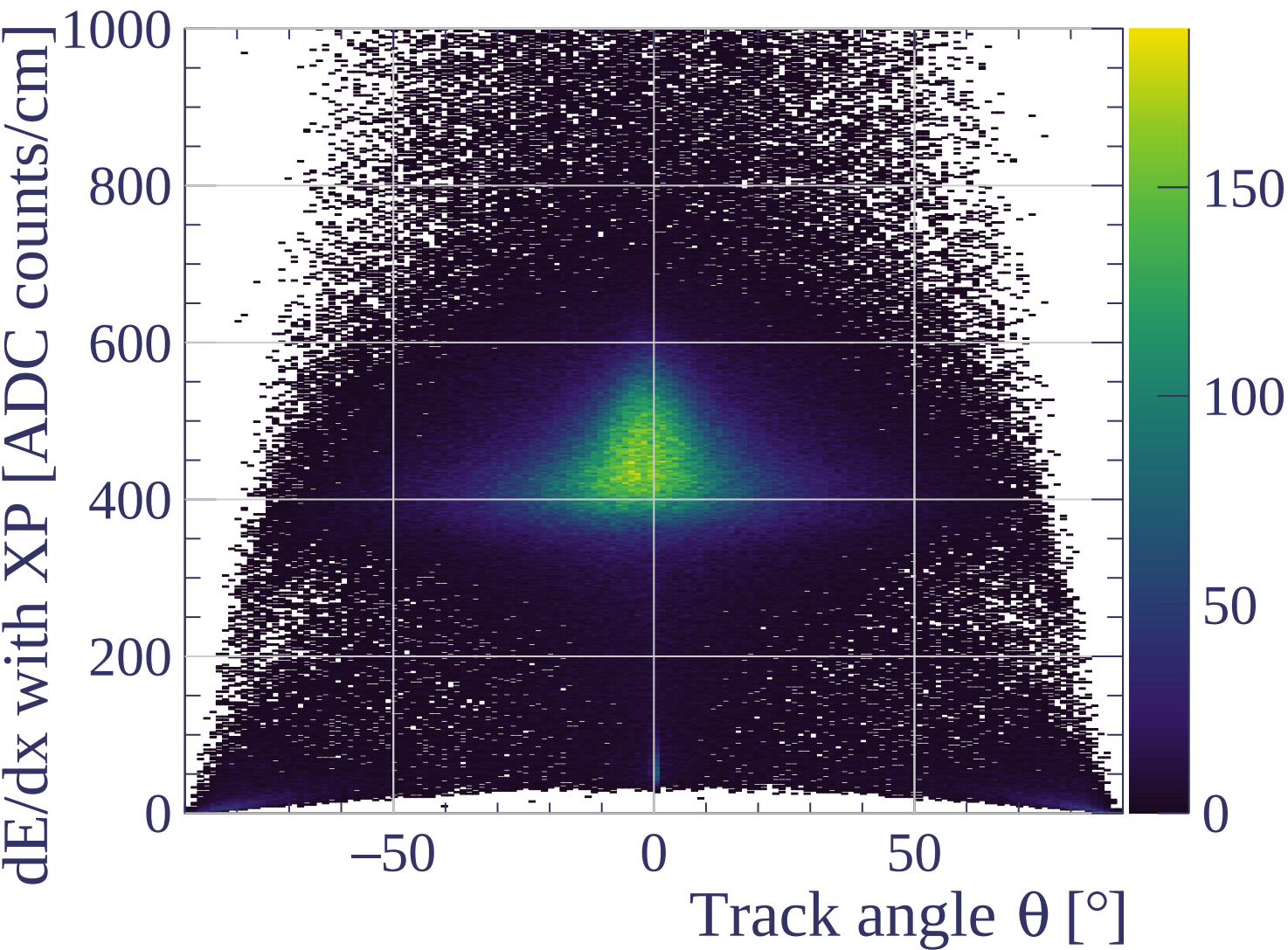


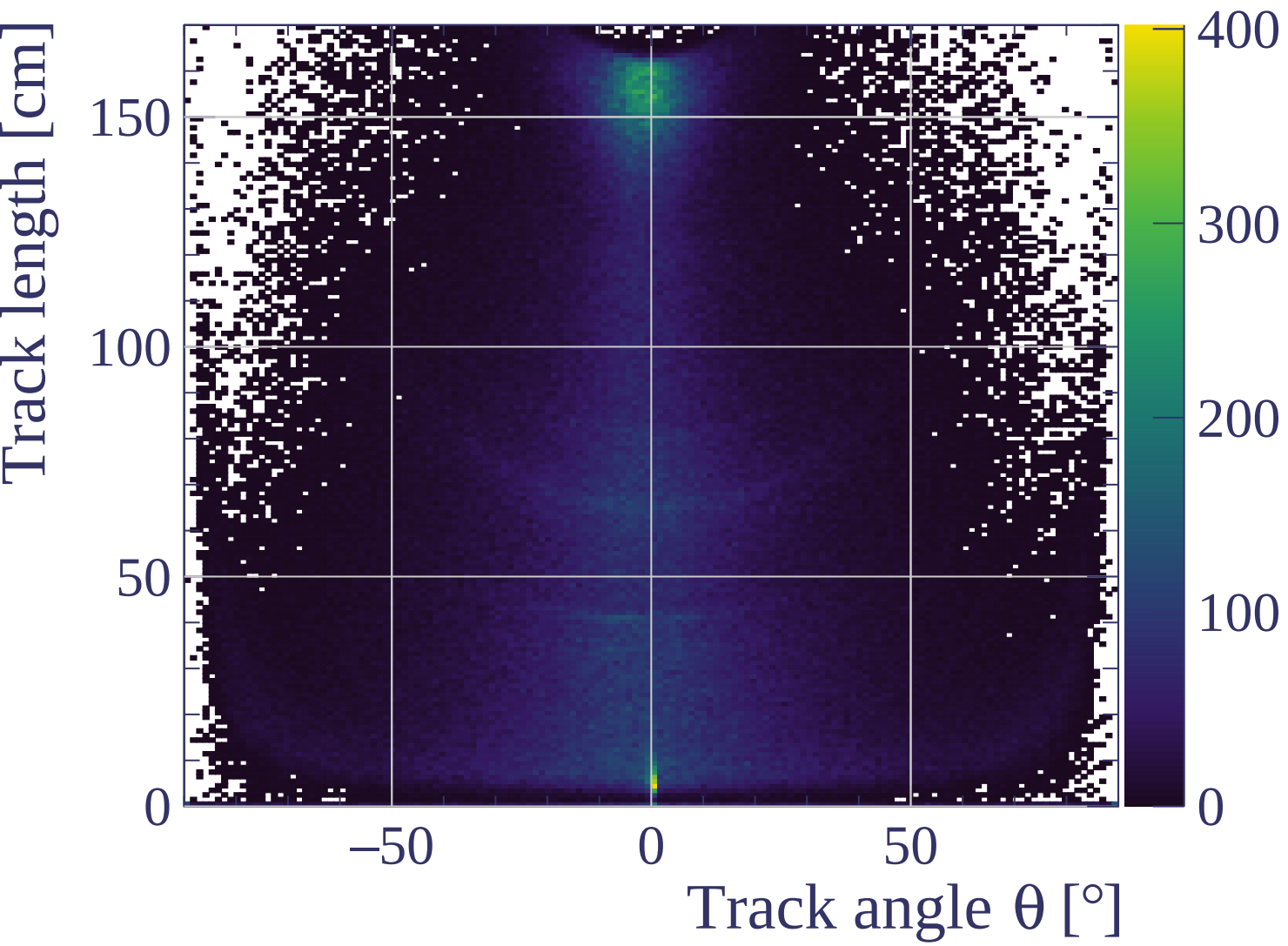


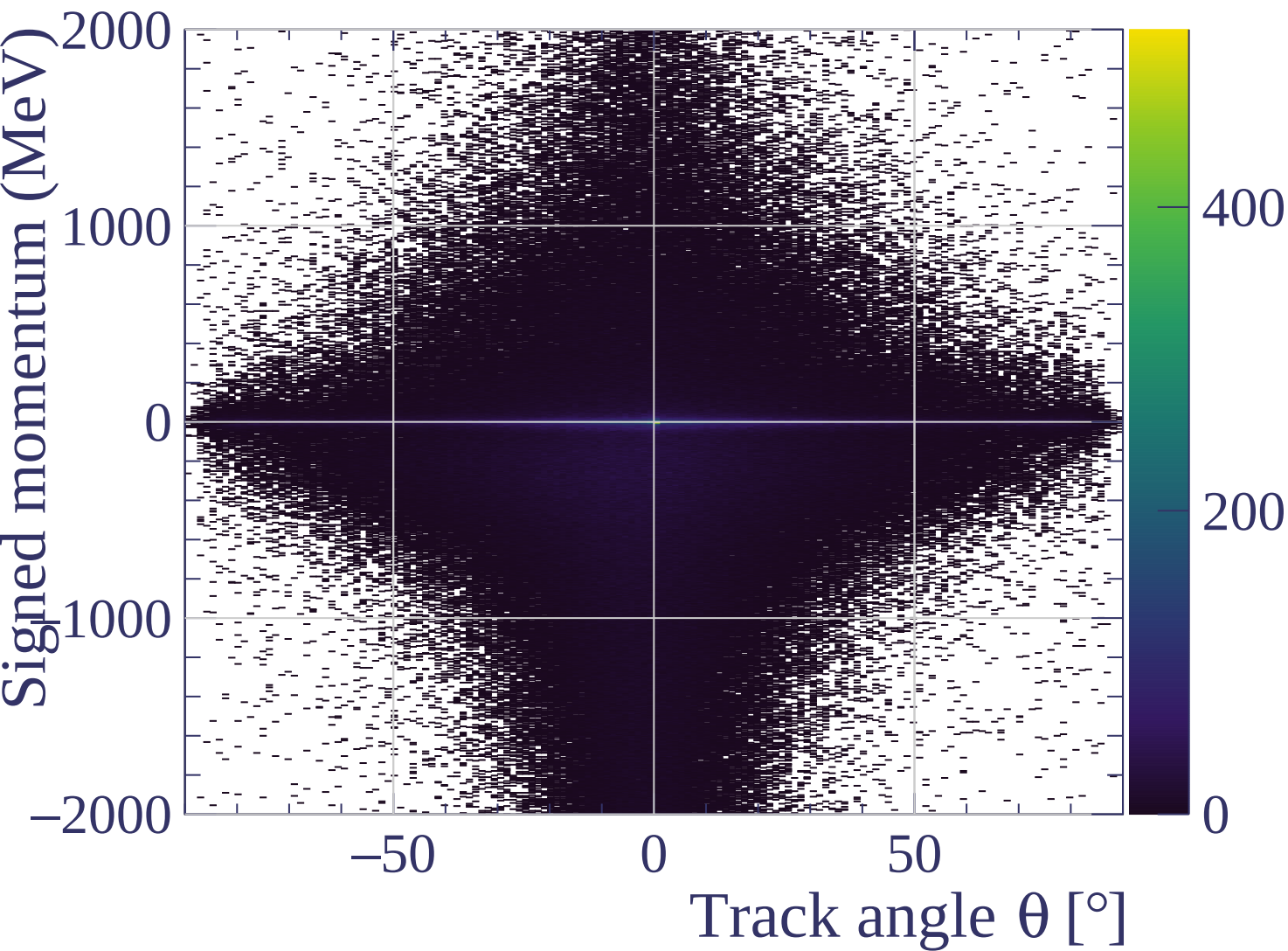


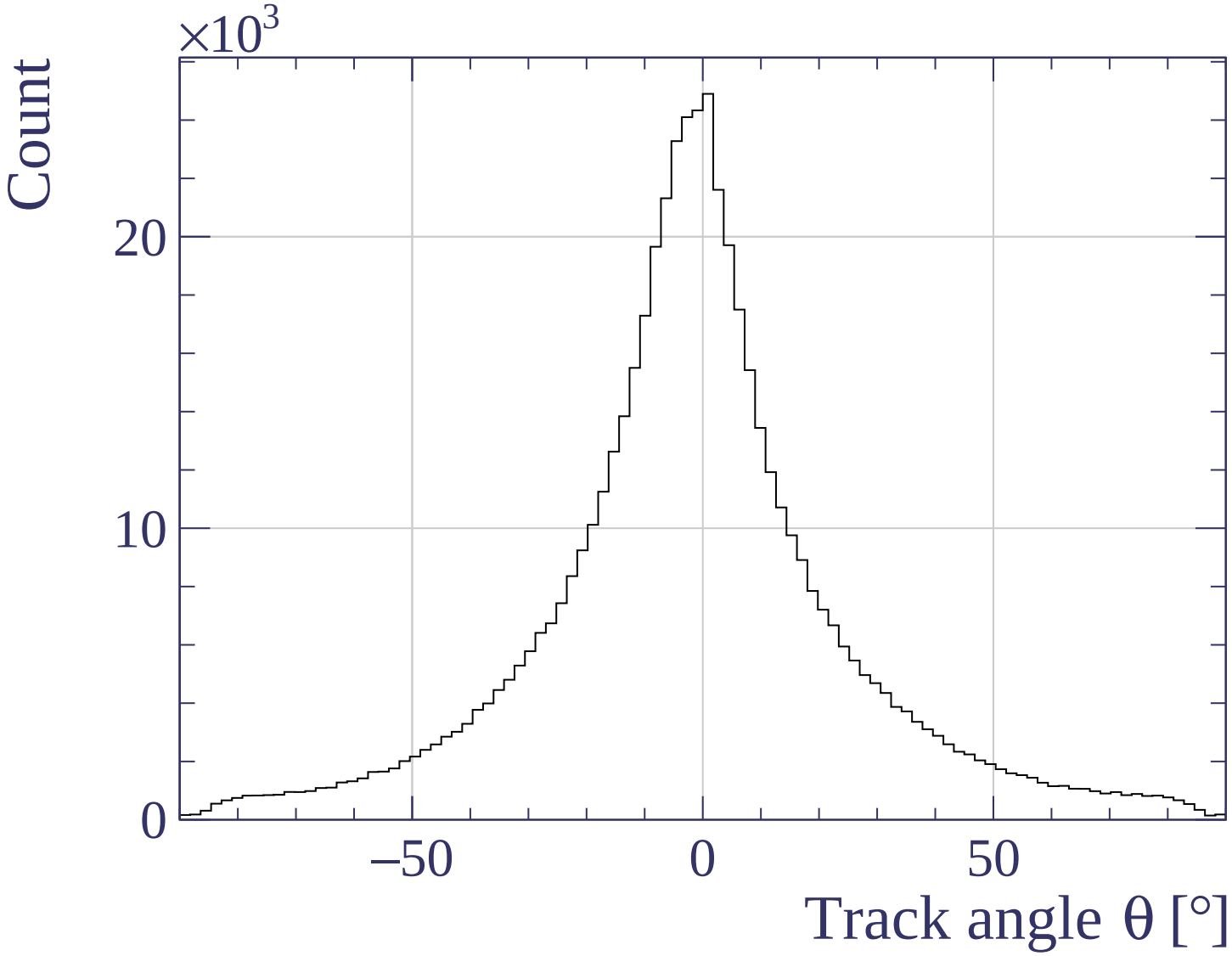




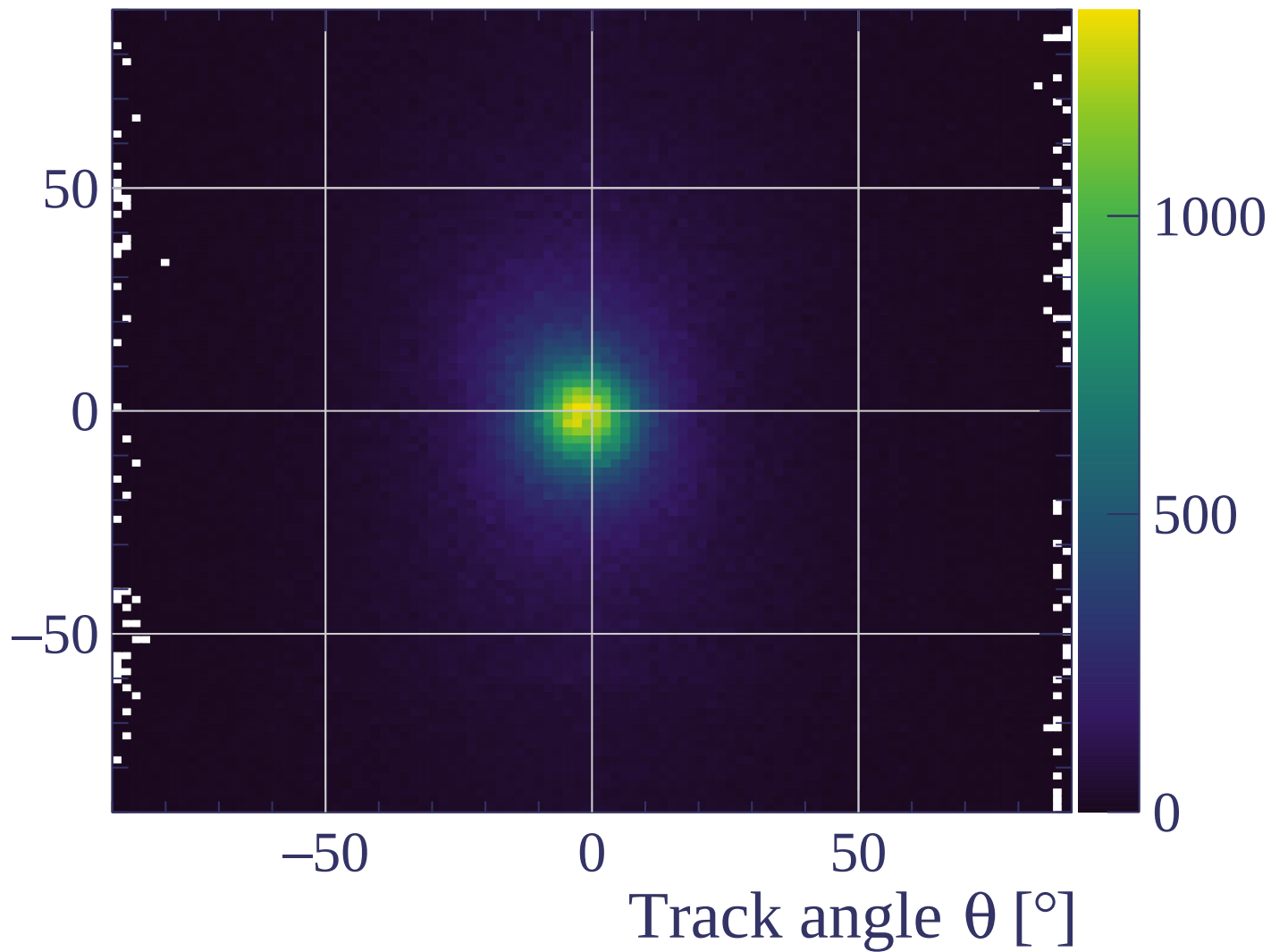




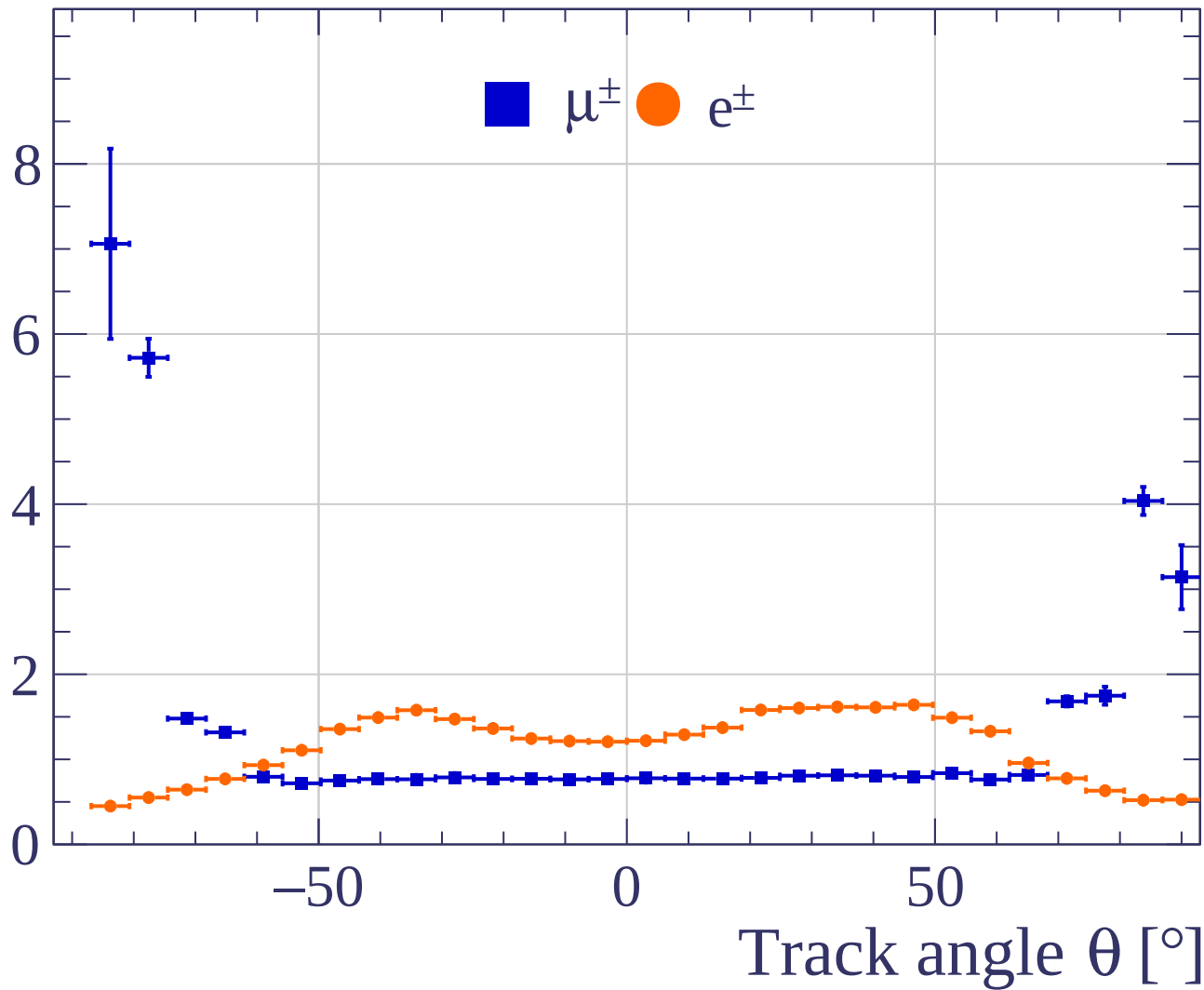




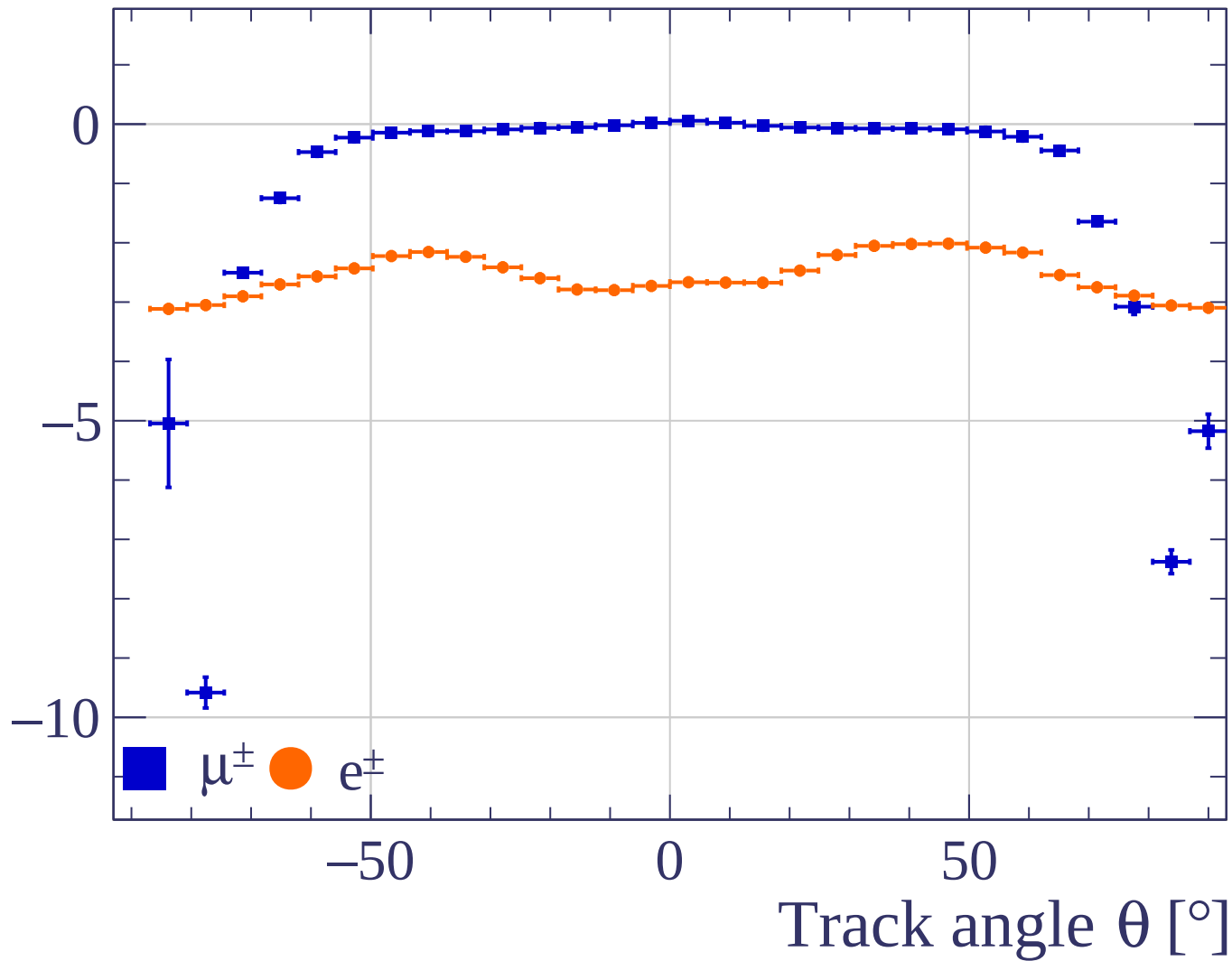
Track angle ϕ [°]



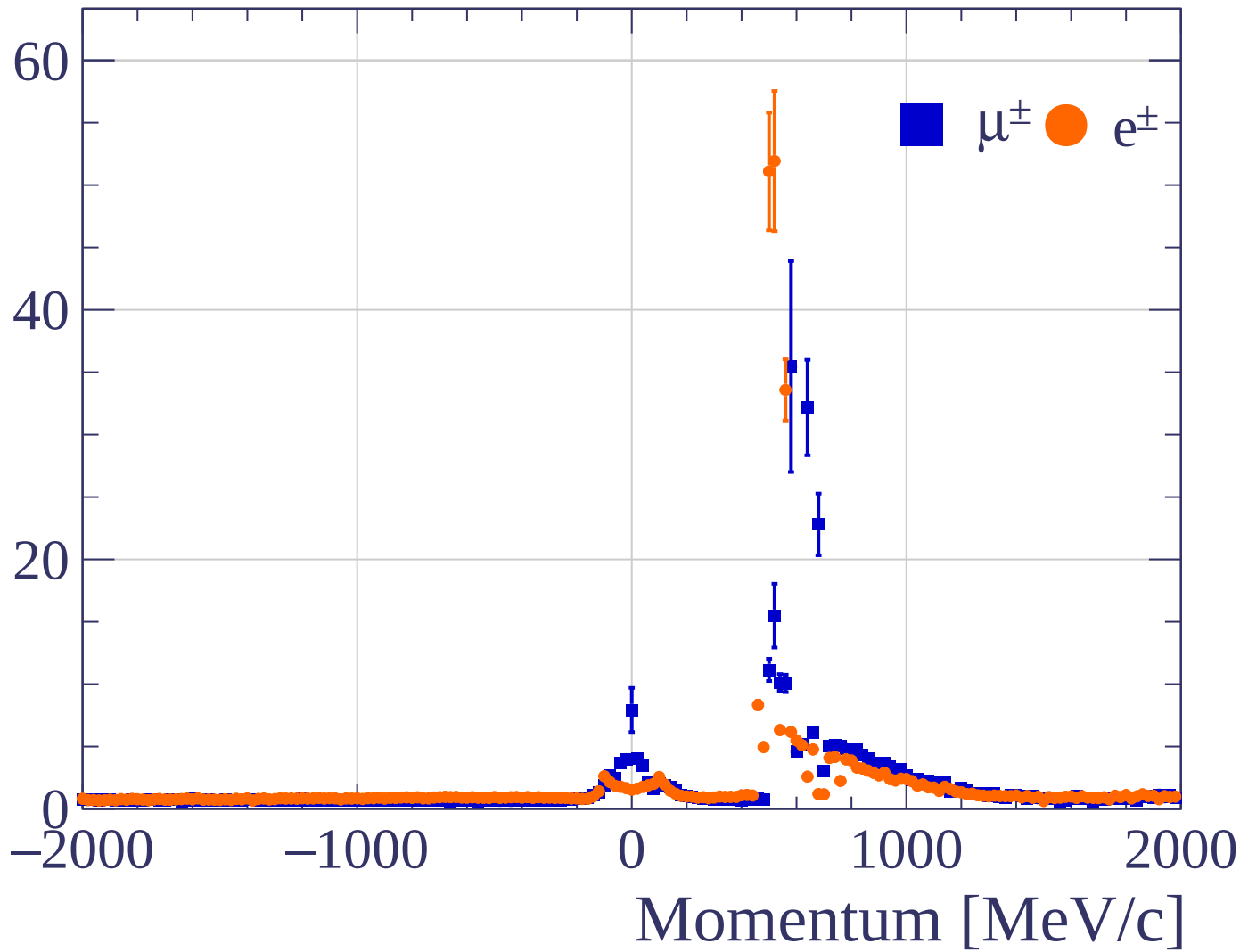
Pulls standard deviation



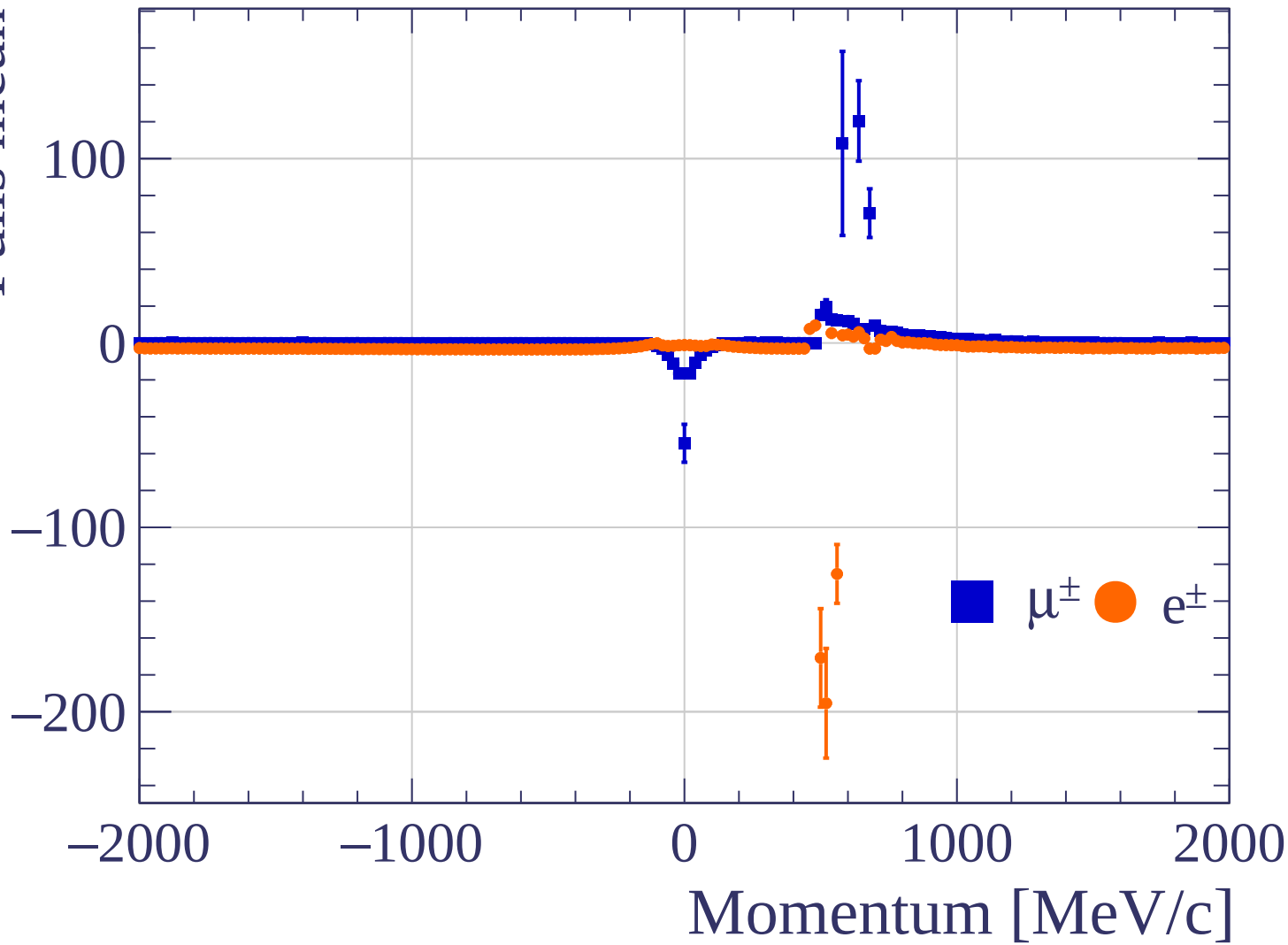
Pulls mean

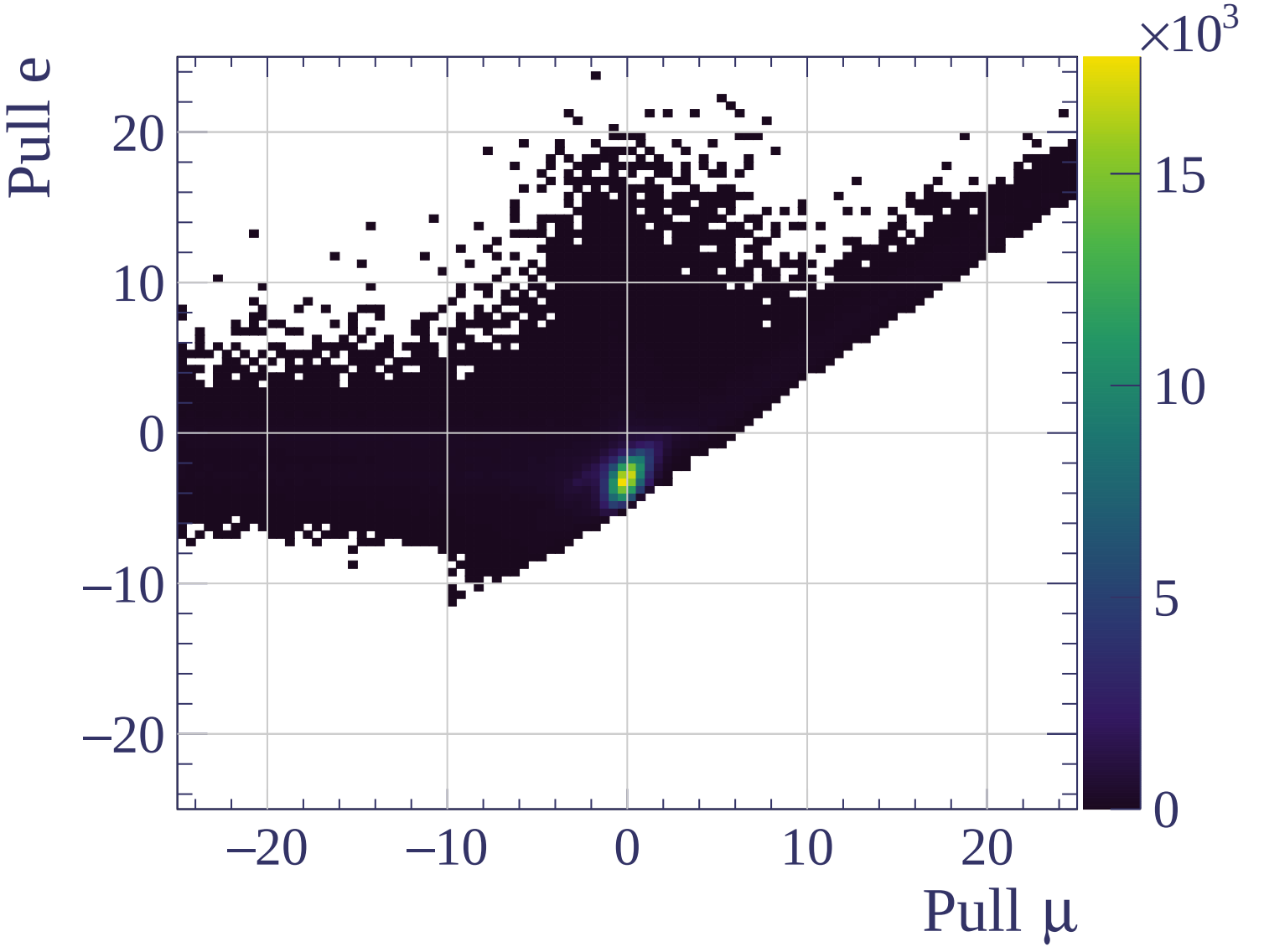


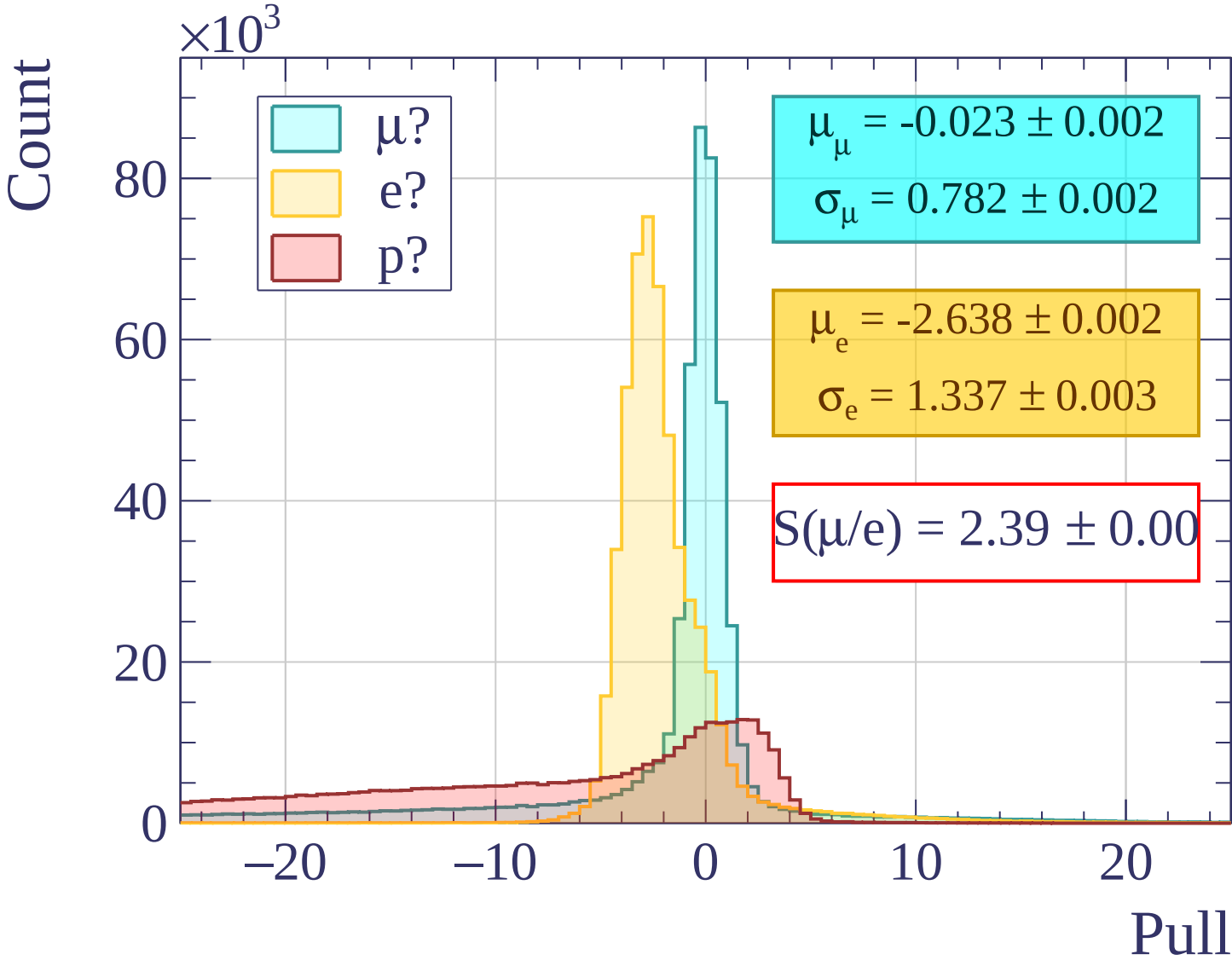
Pulls standard deviation

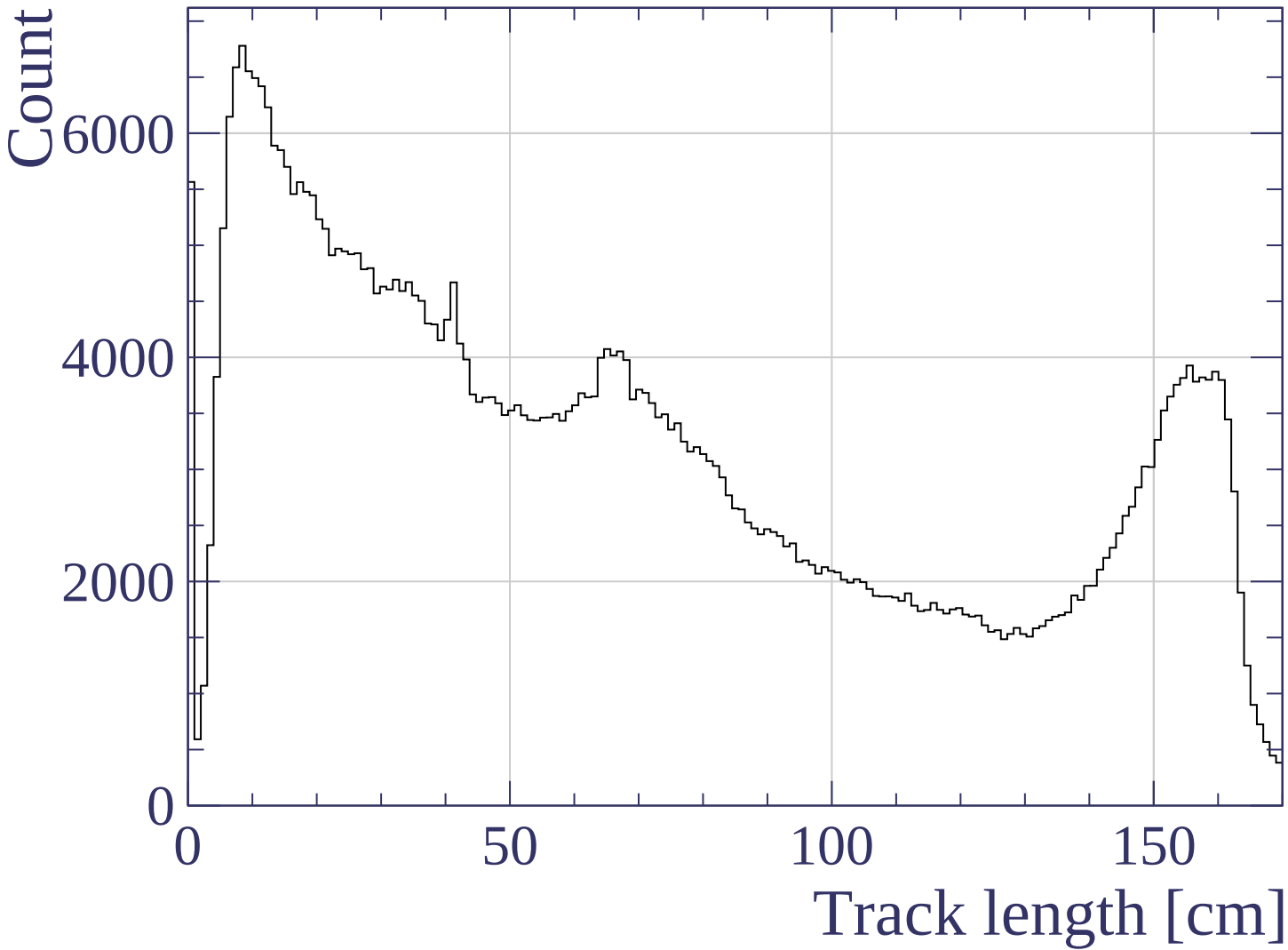


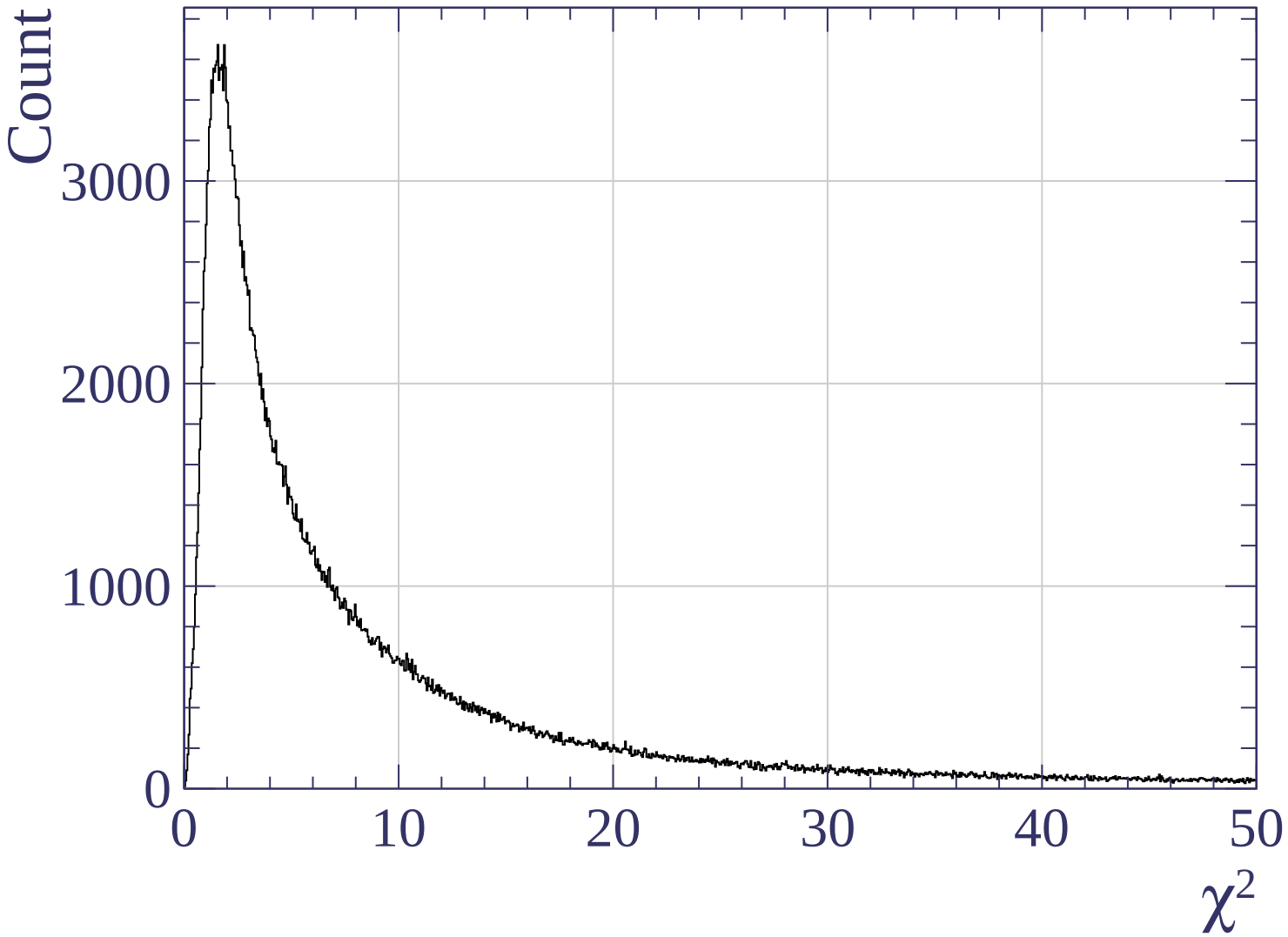
Pulls mean

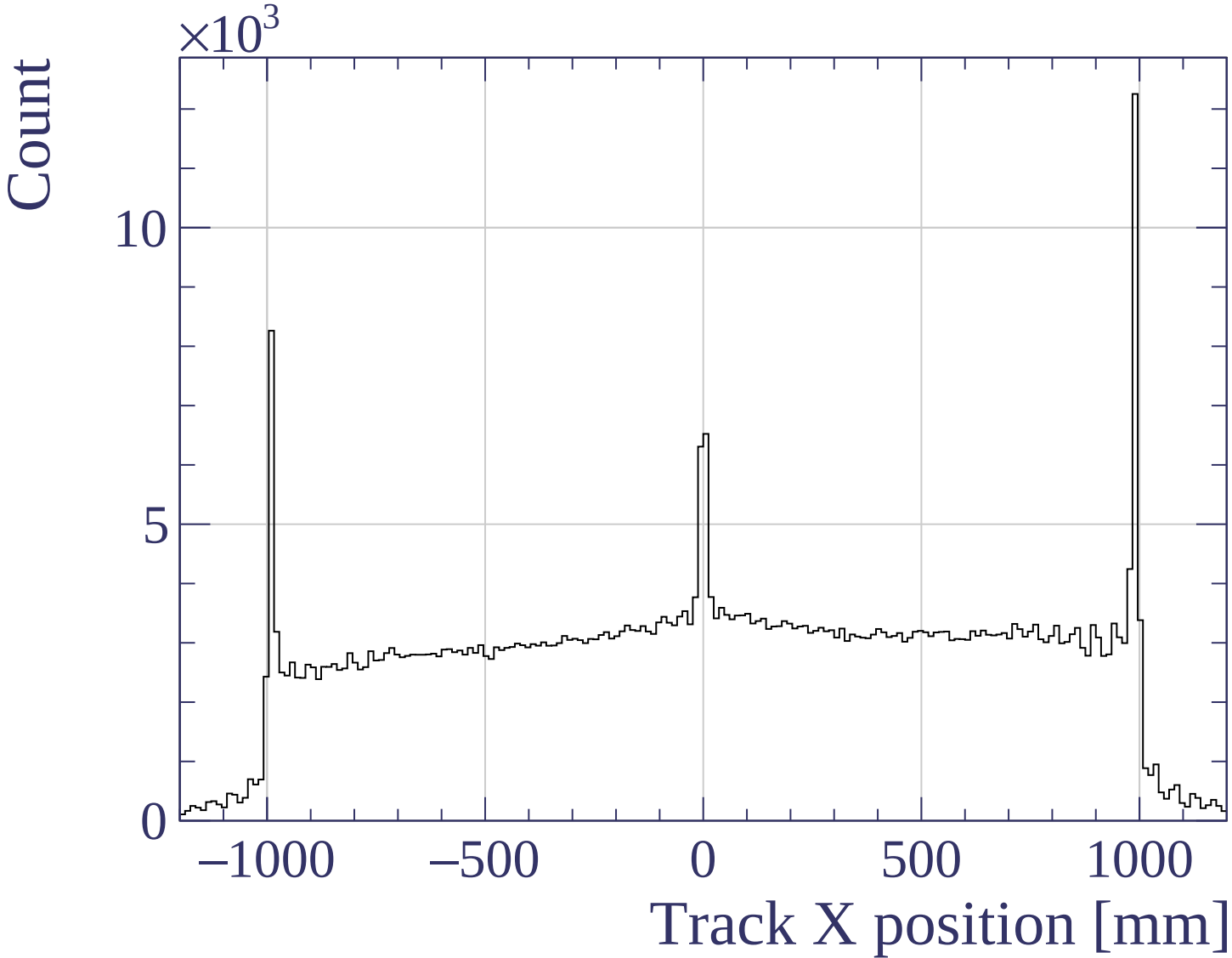


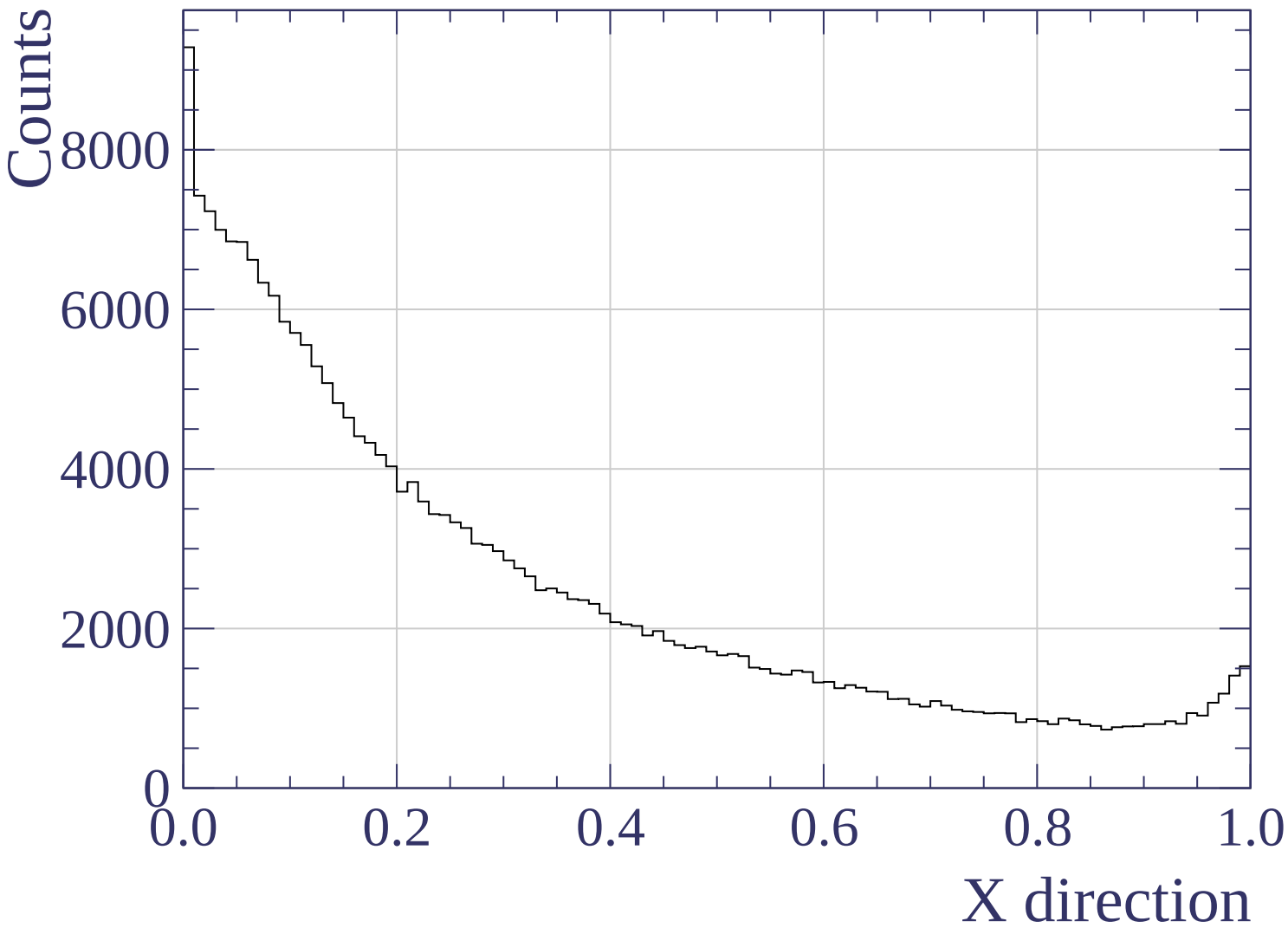


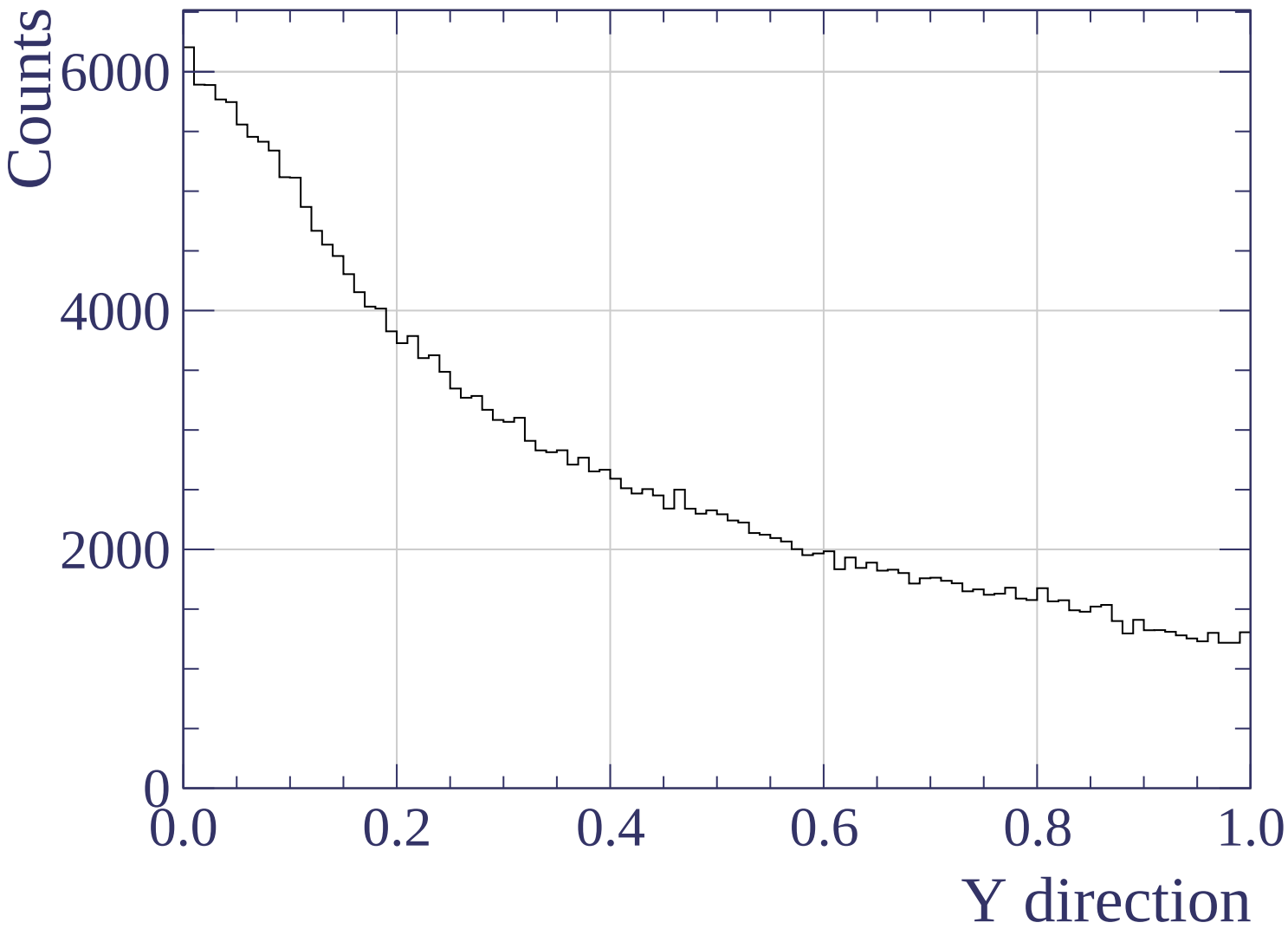


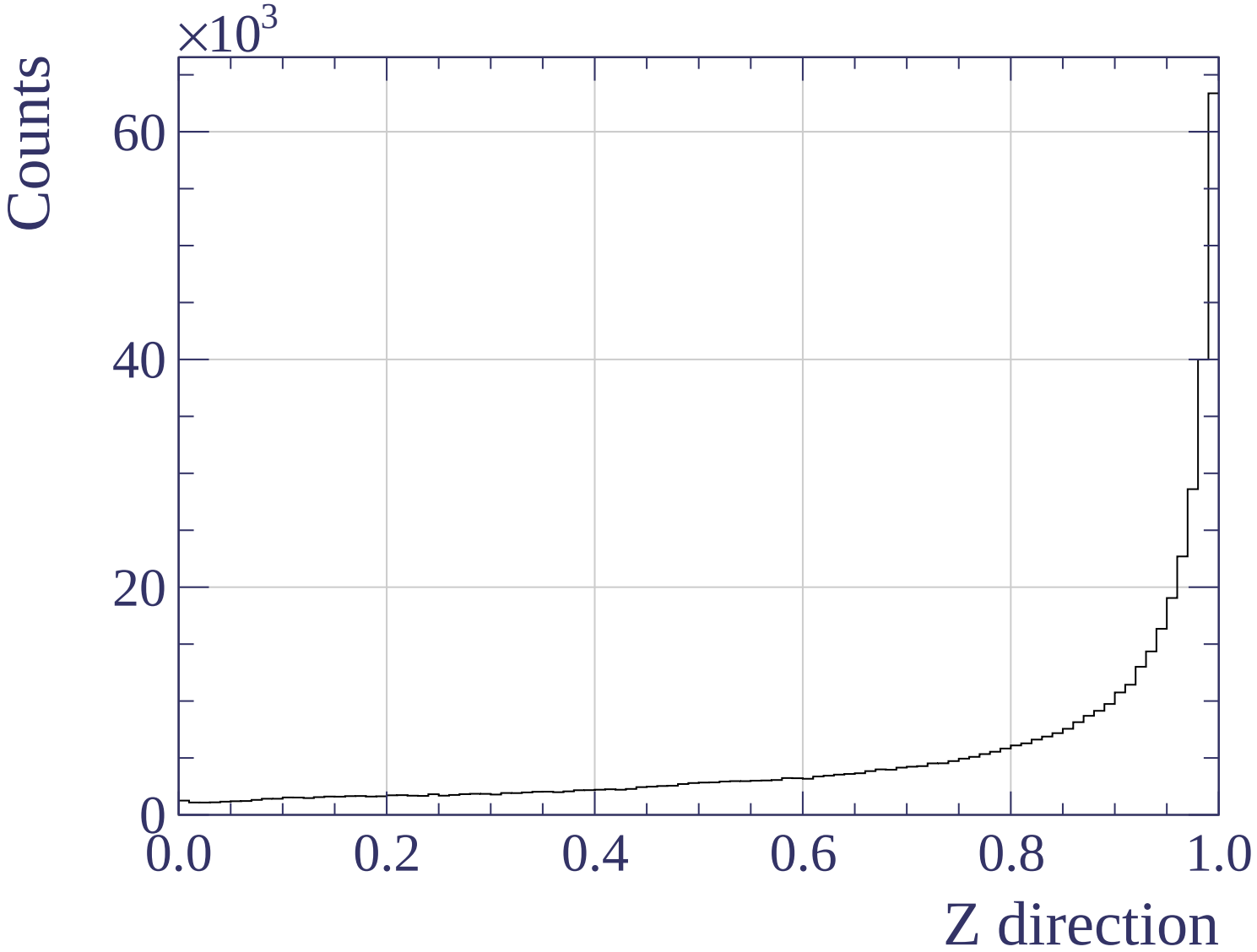


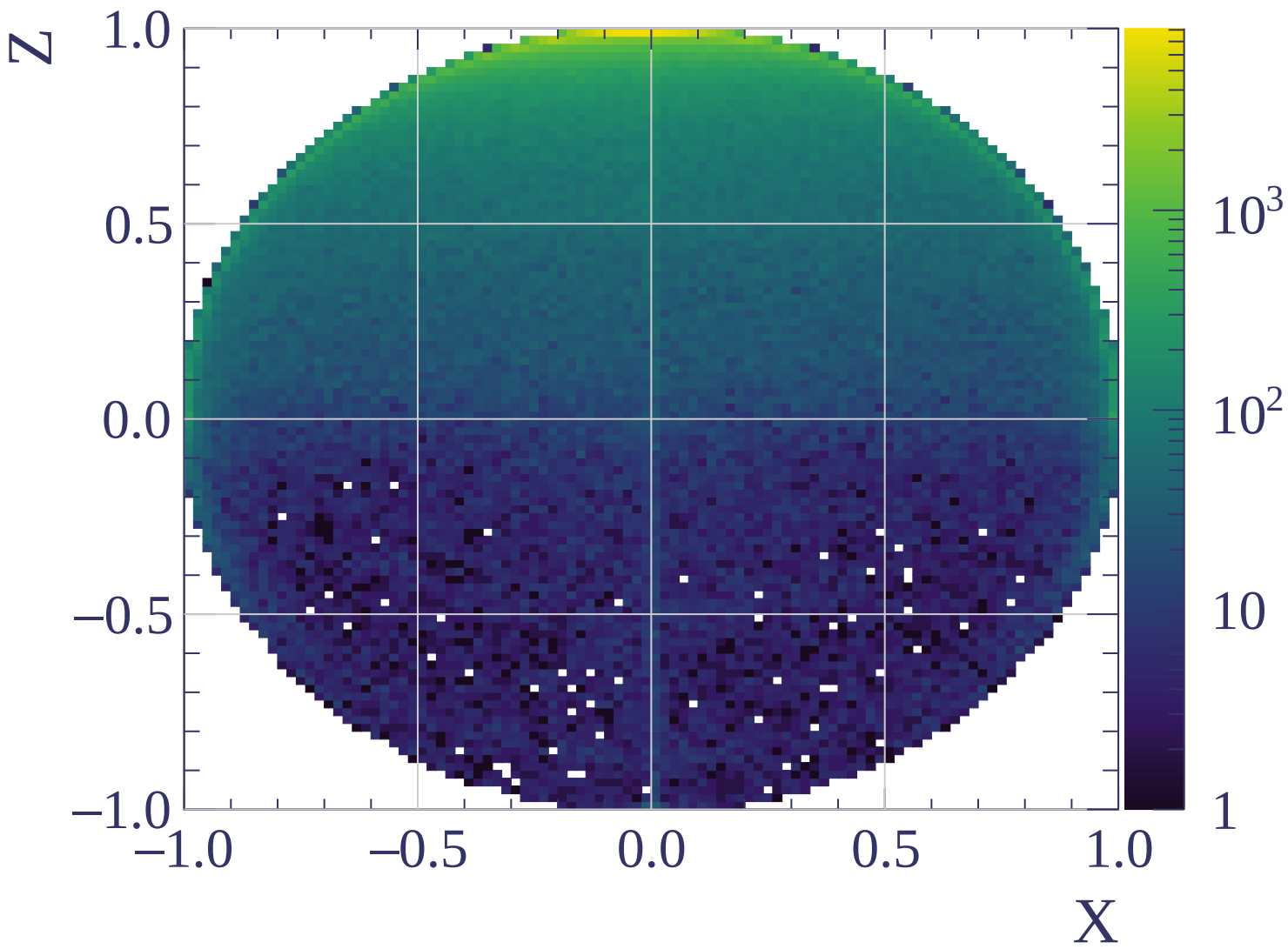


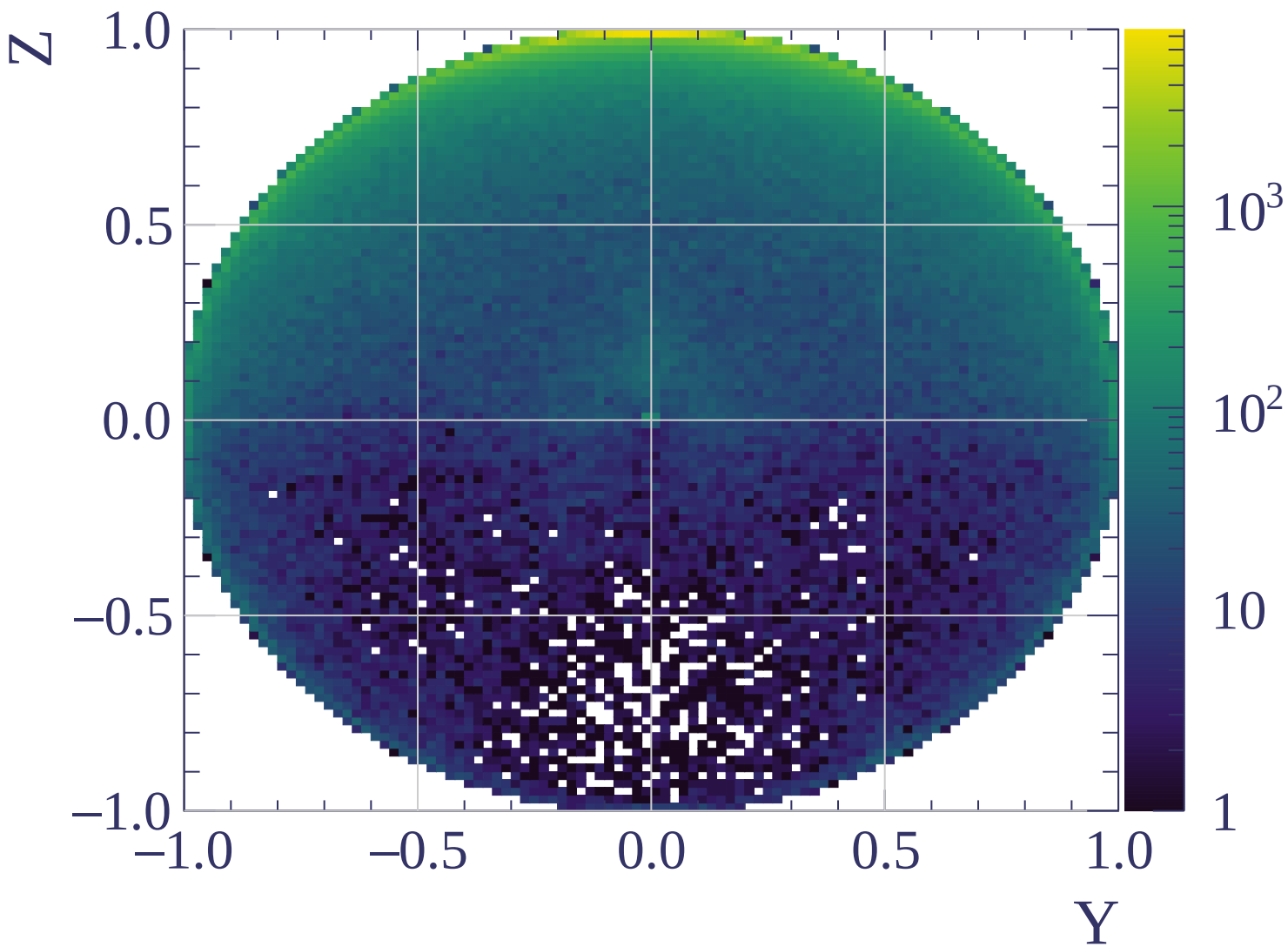


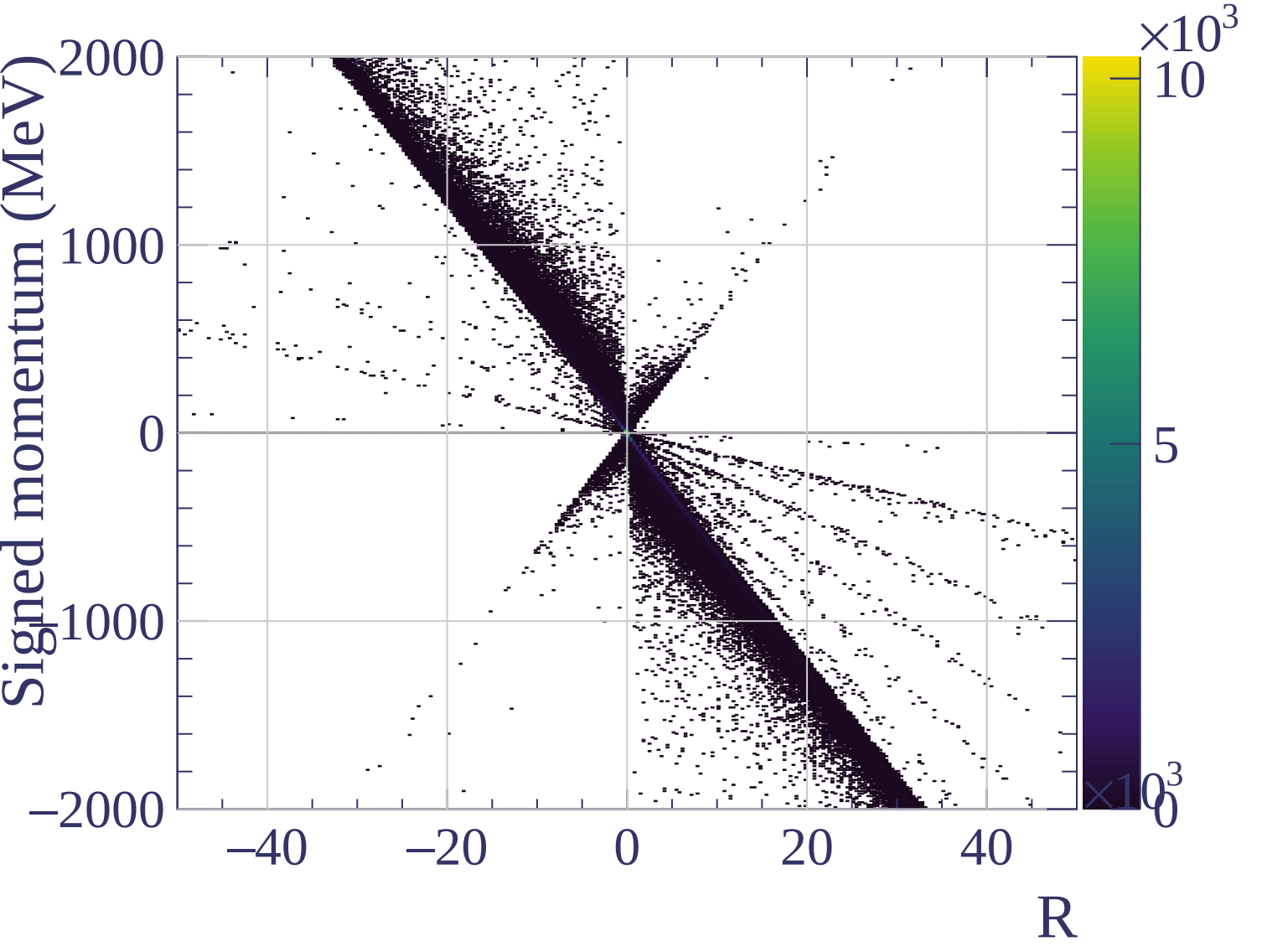


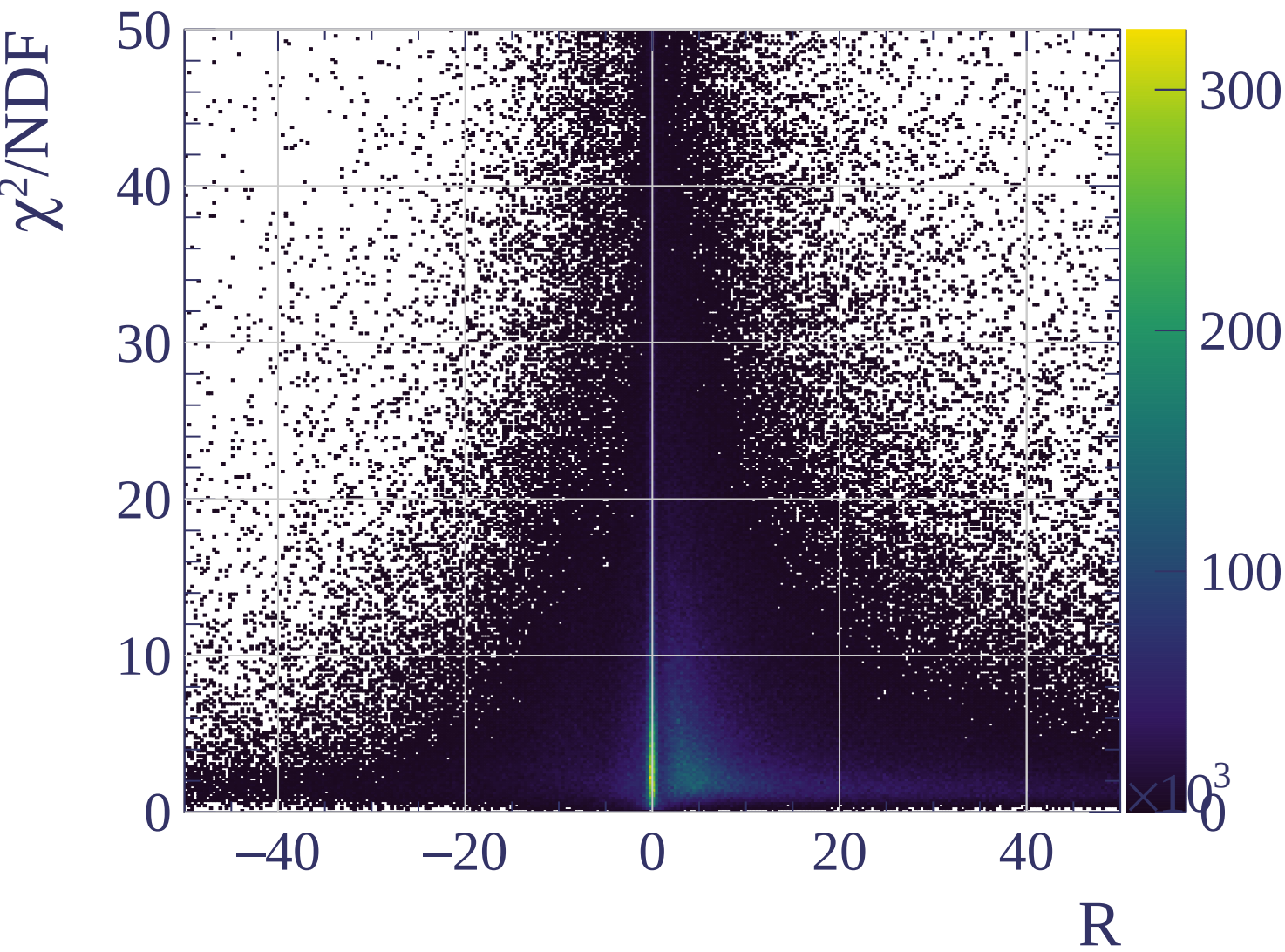


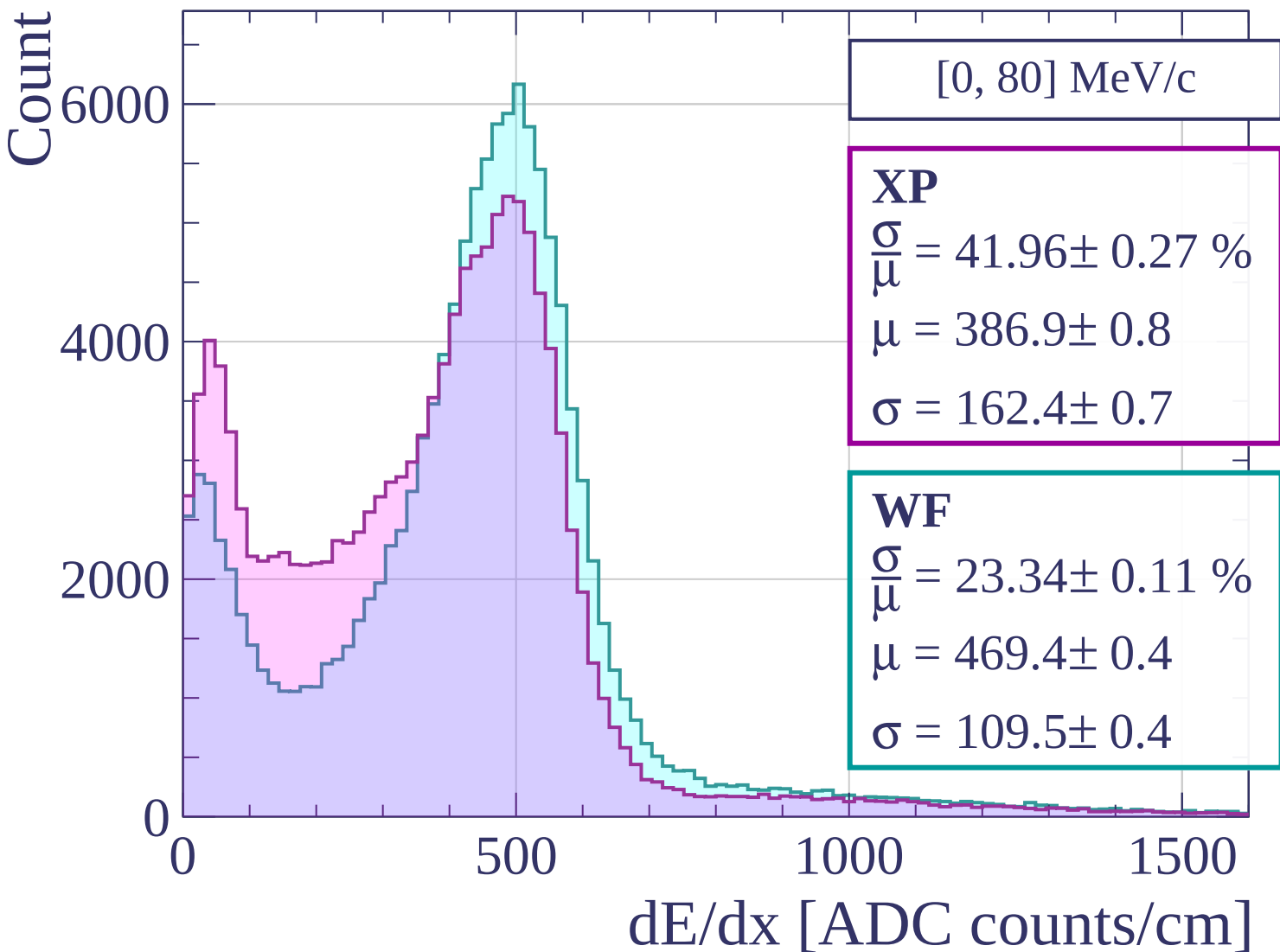


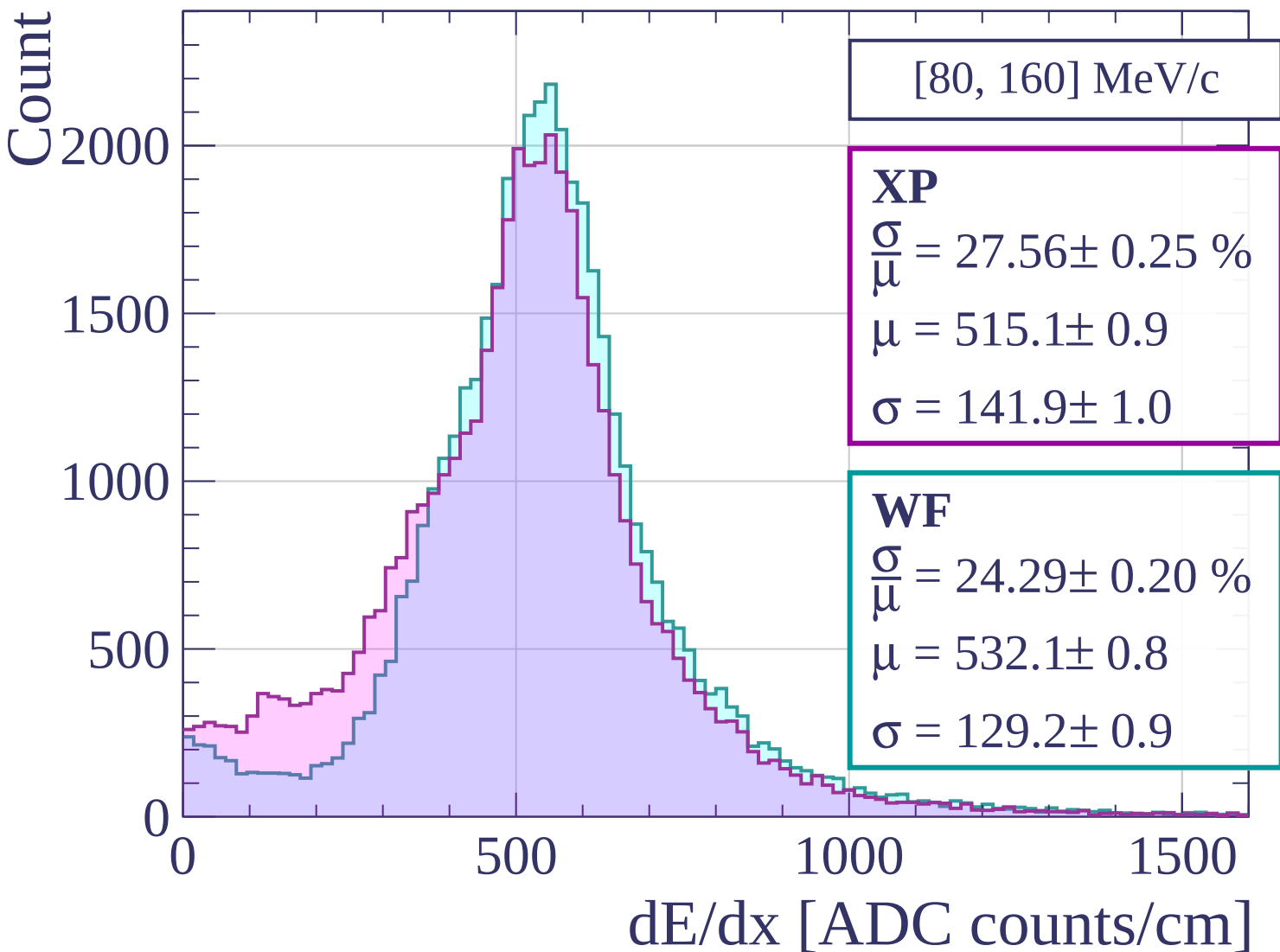


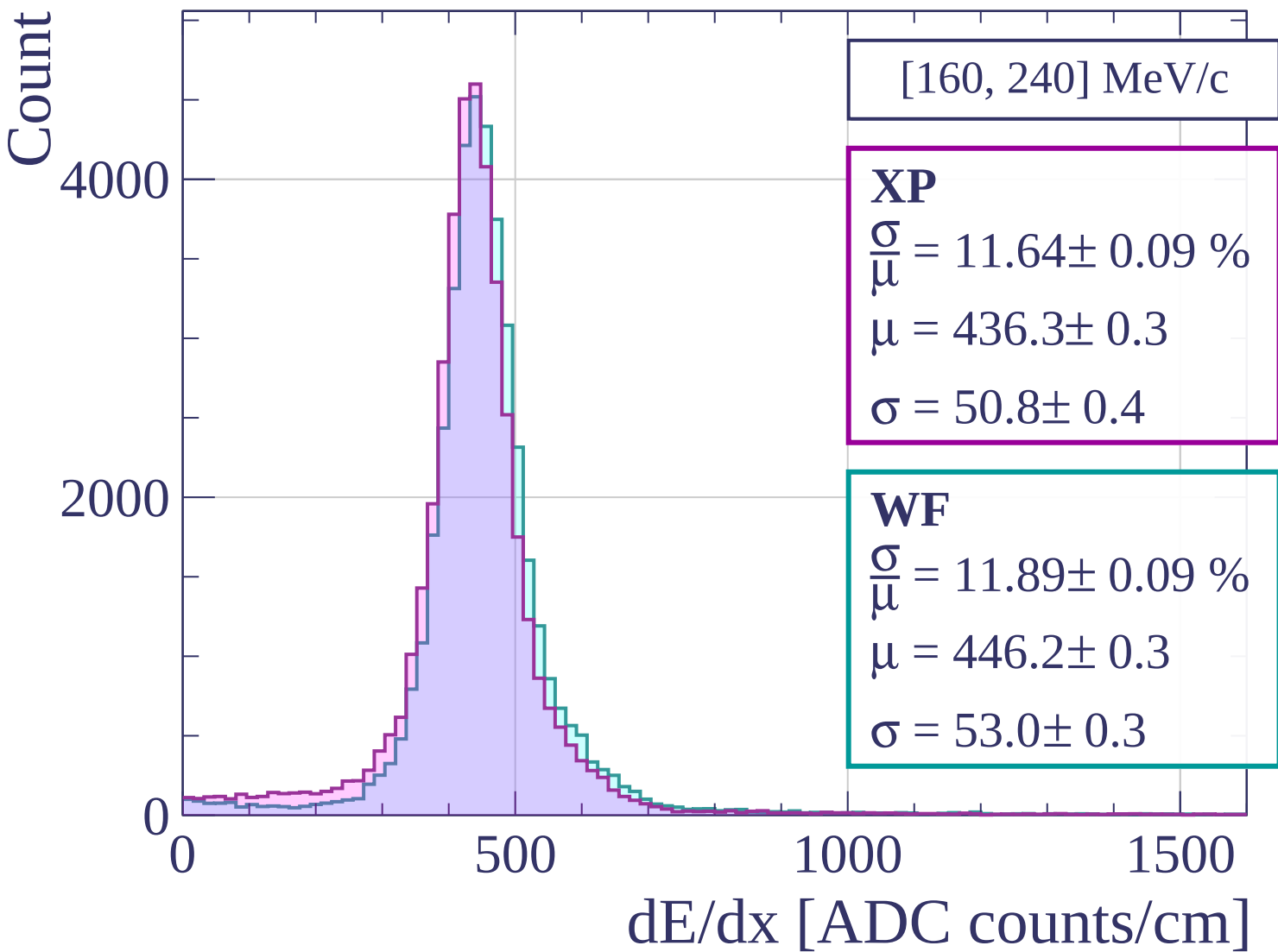


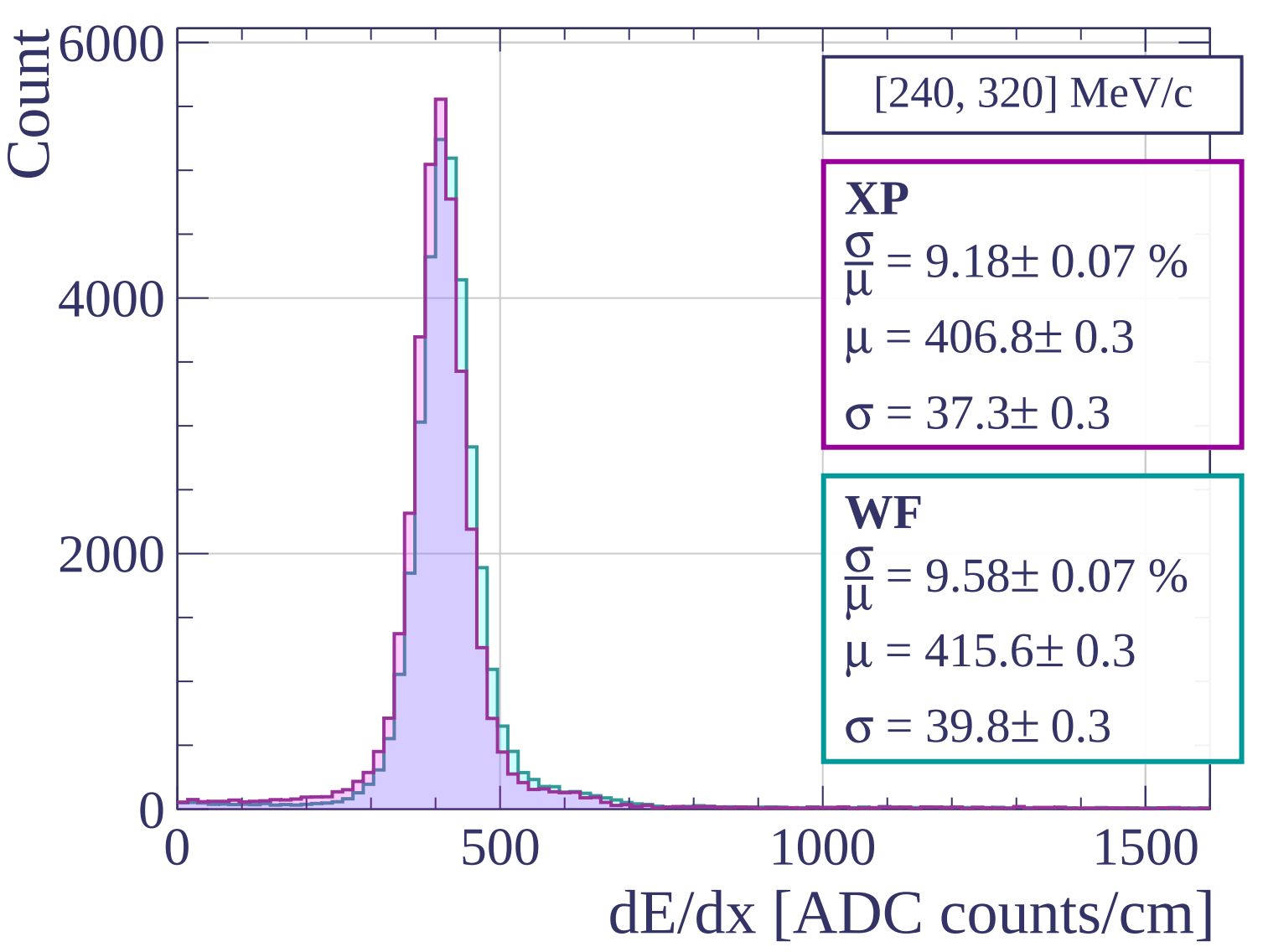


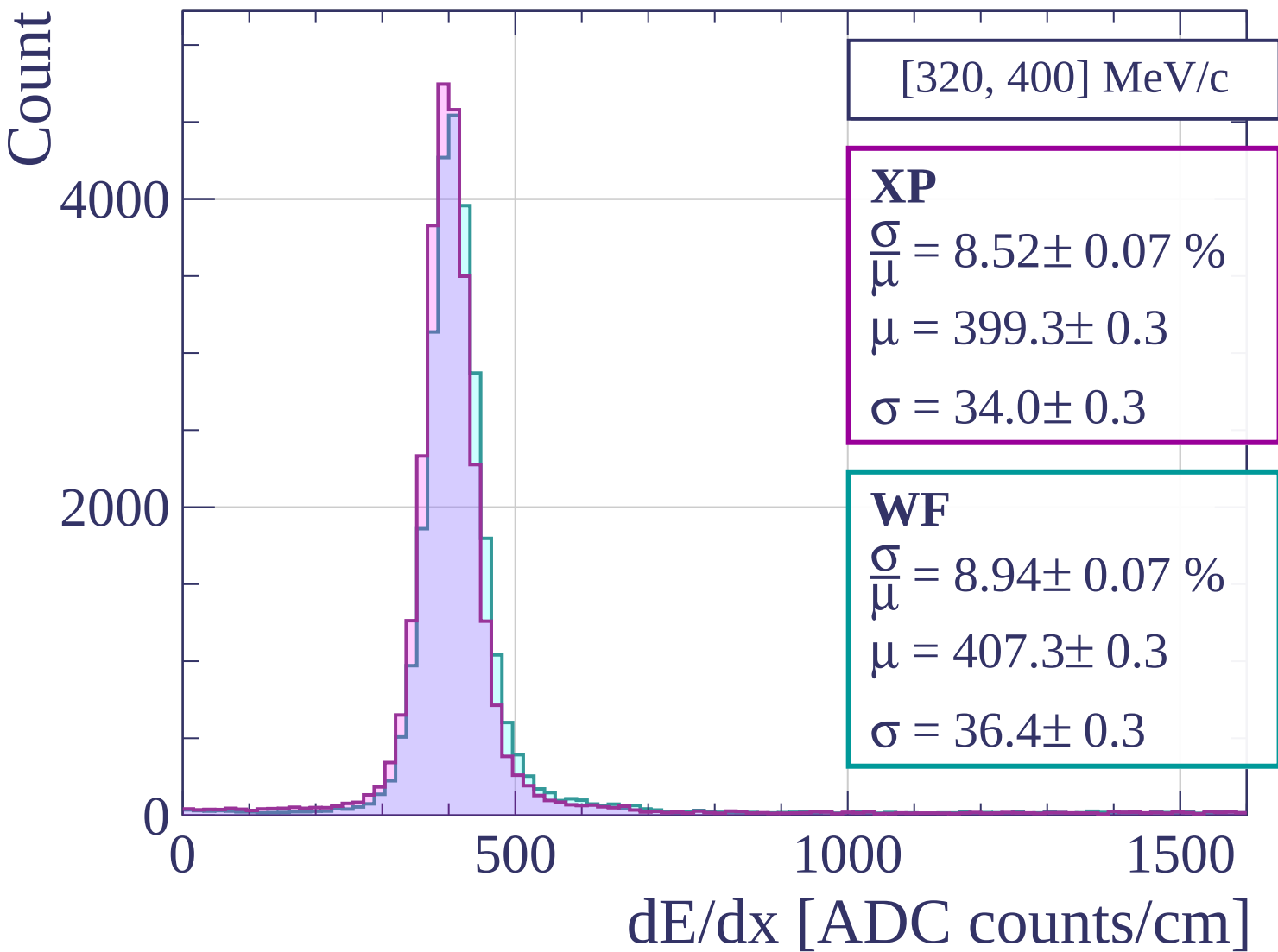


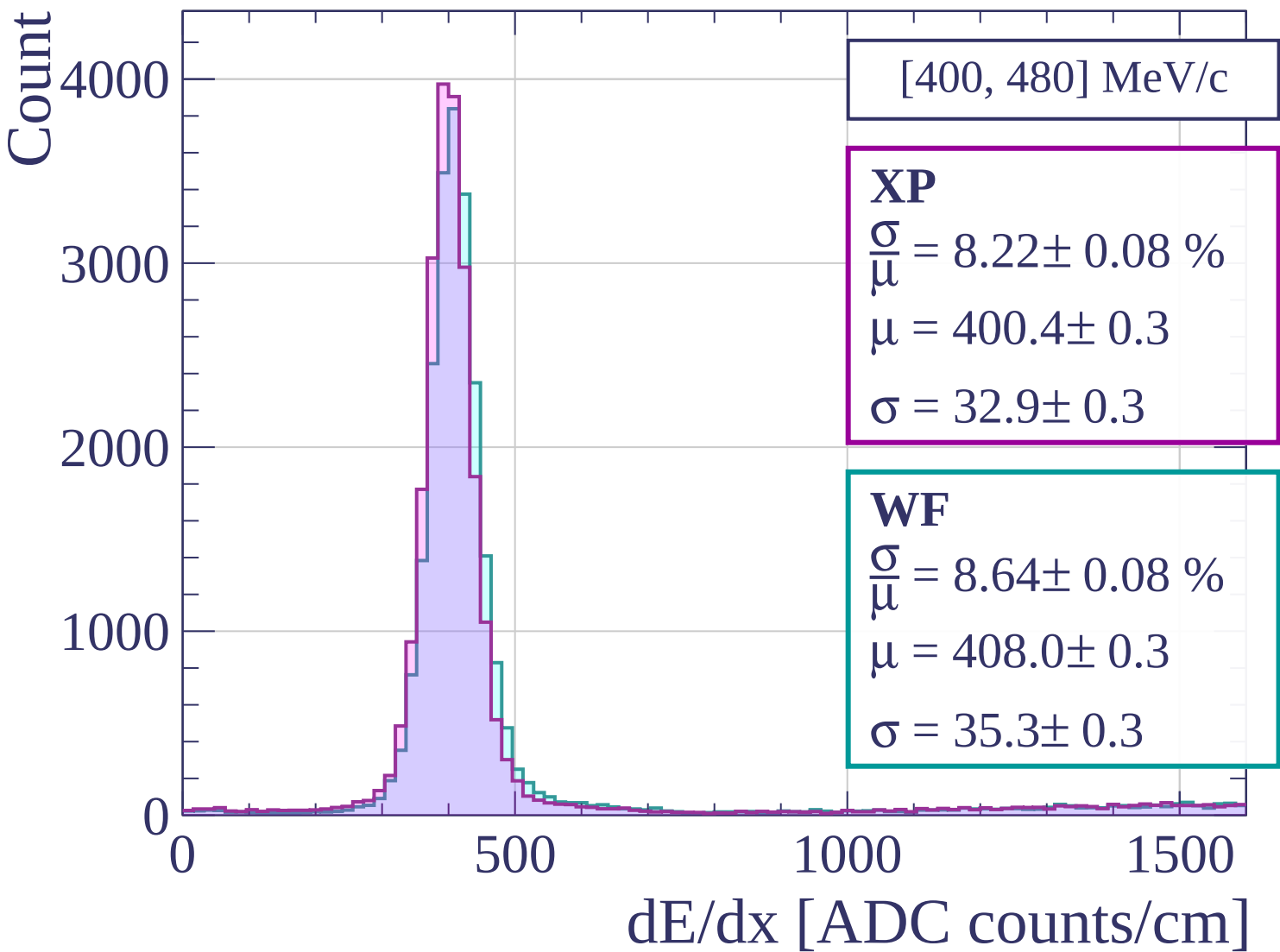


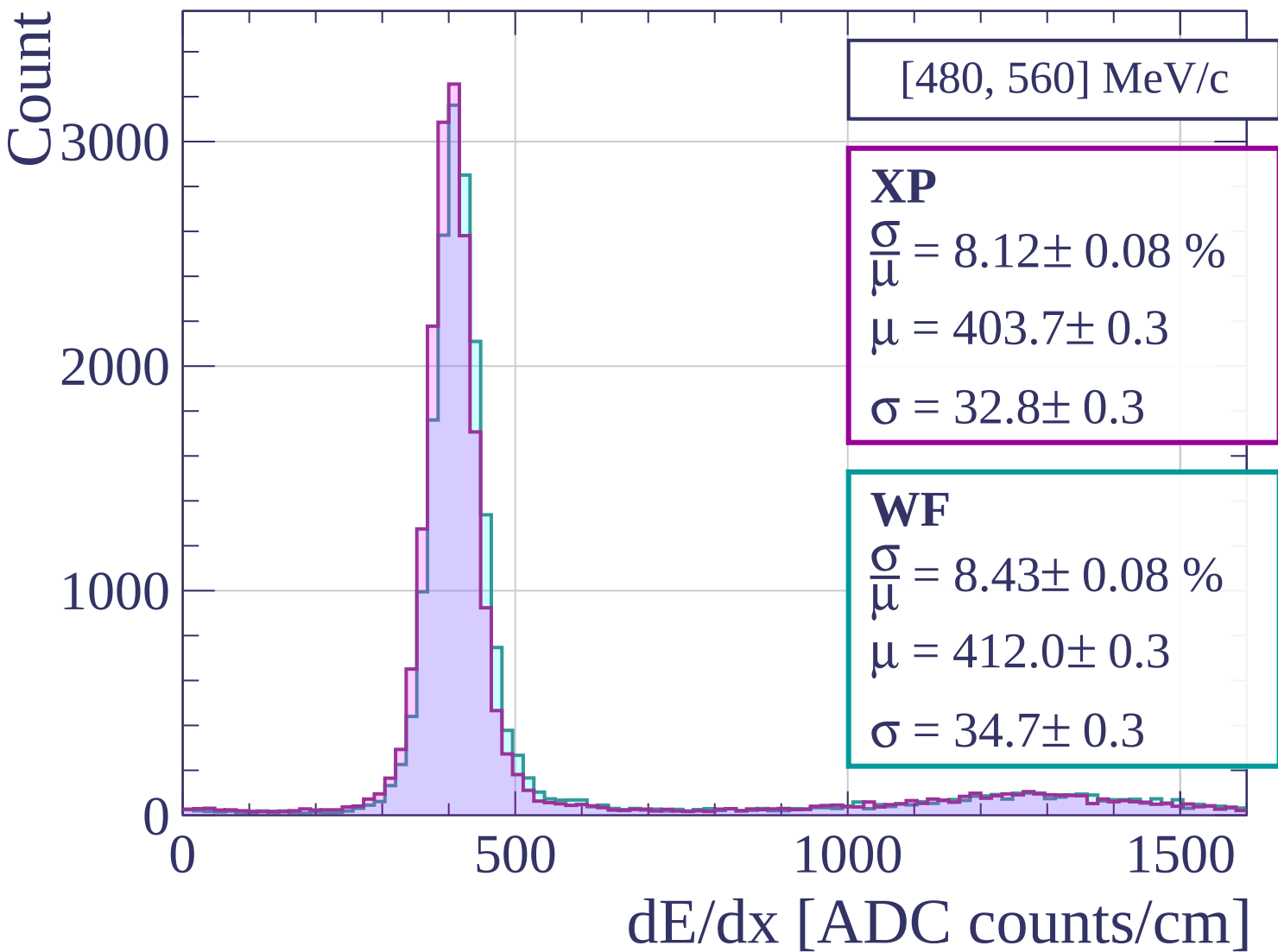


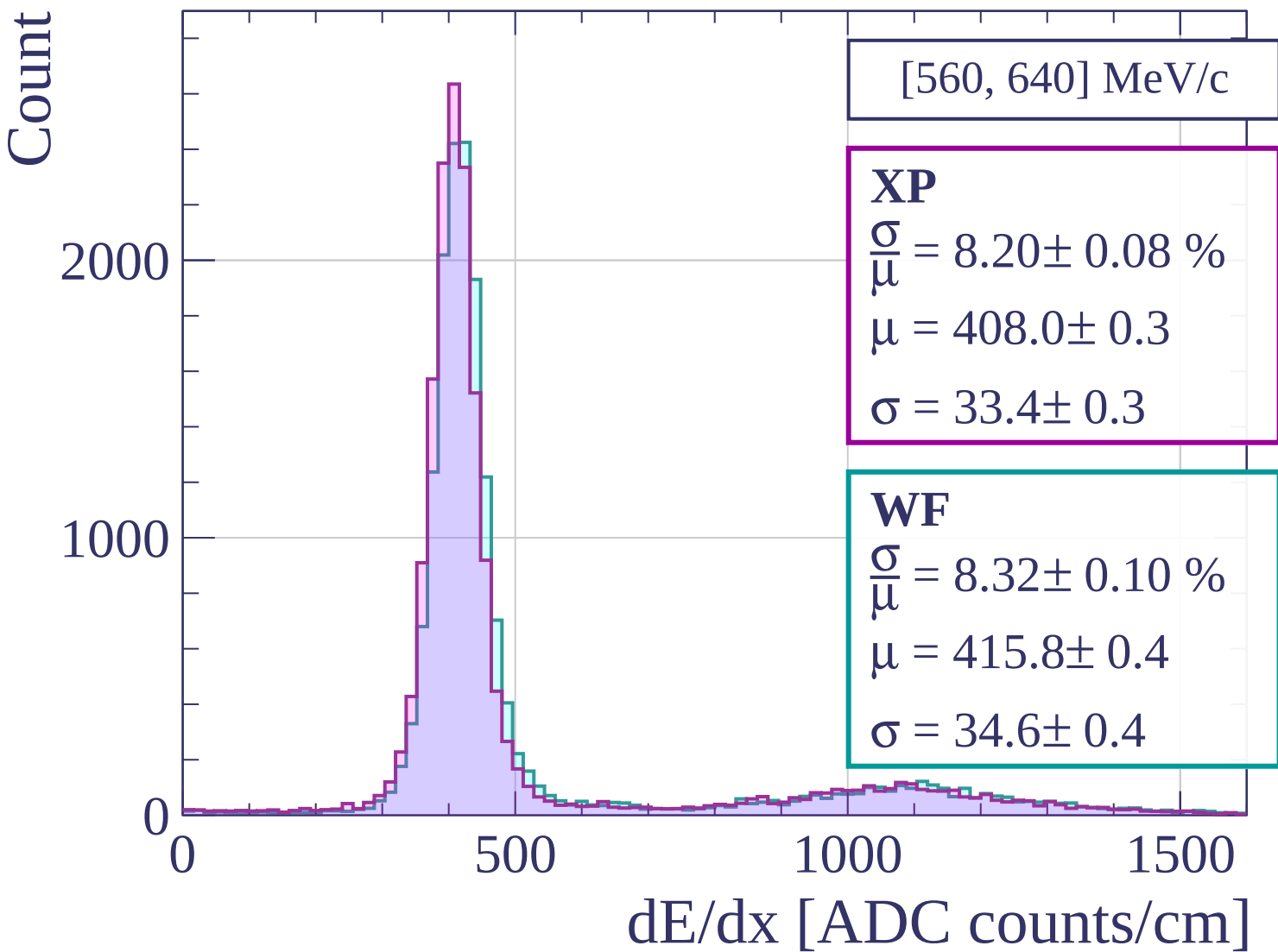


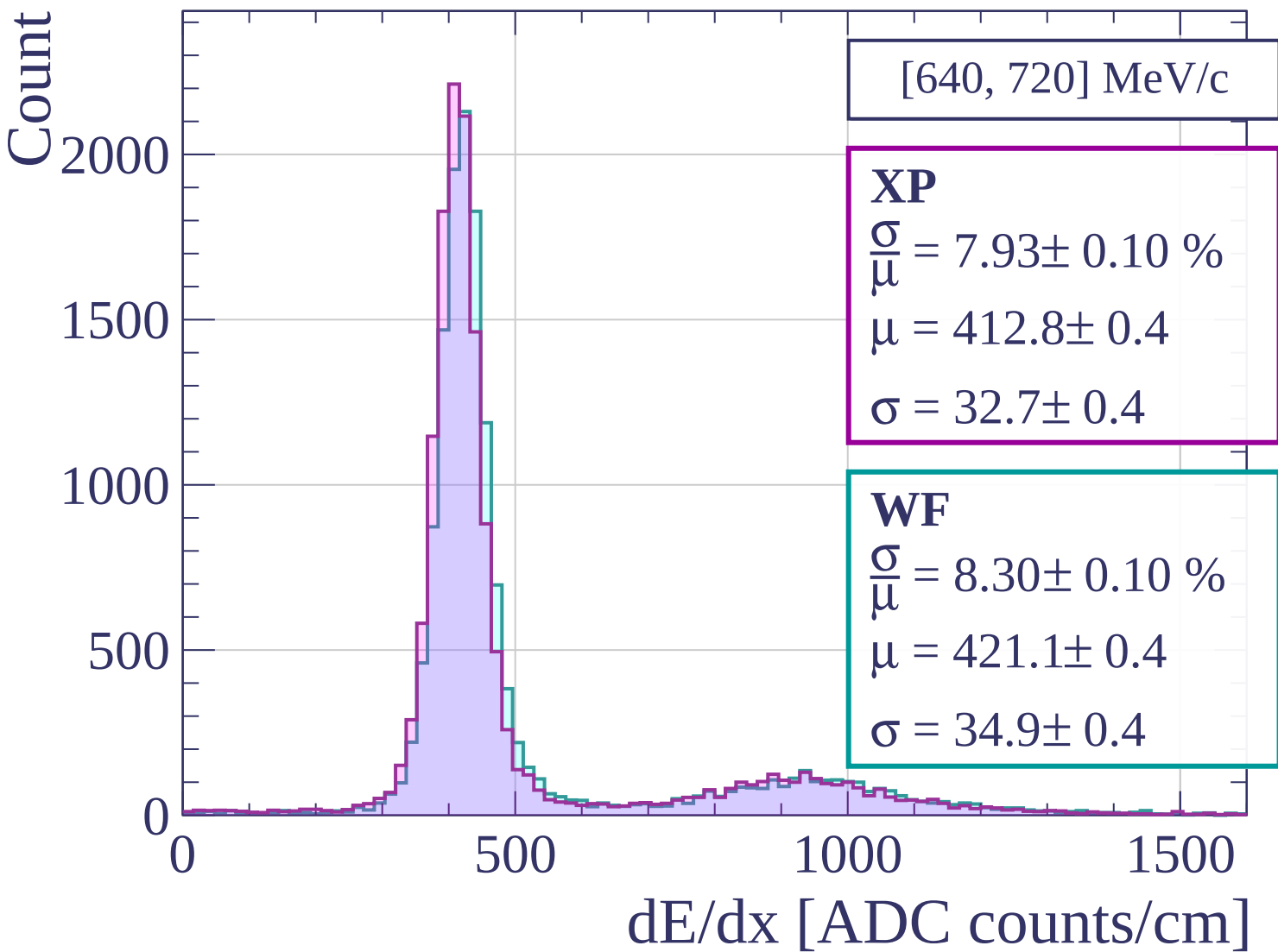


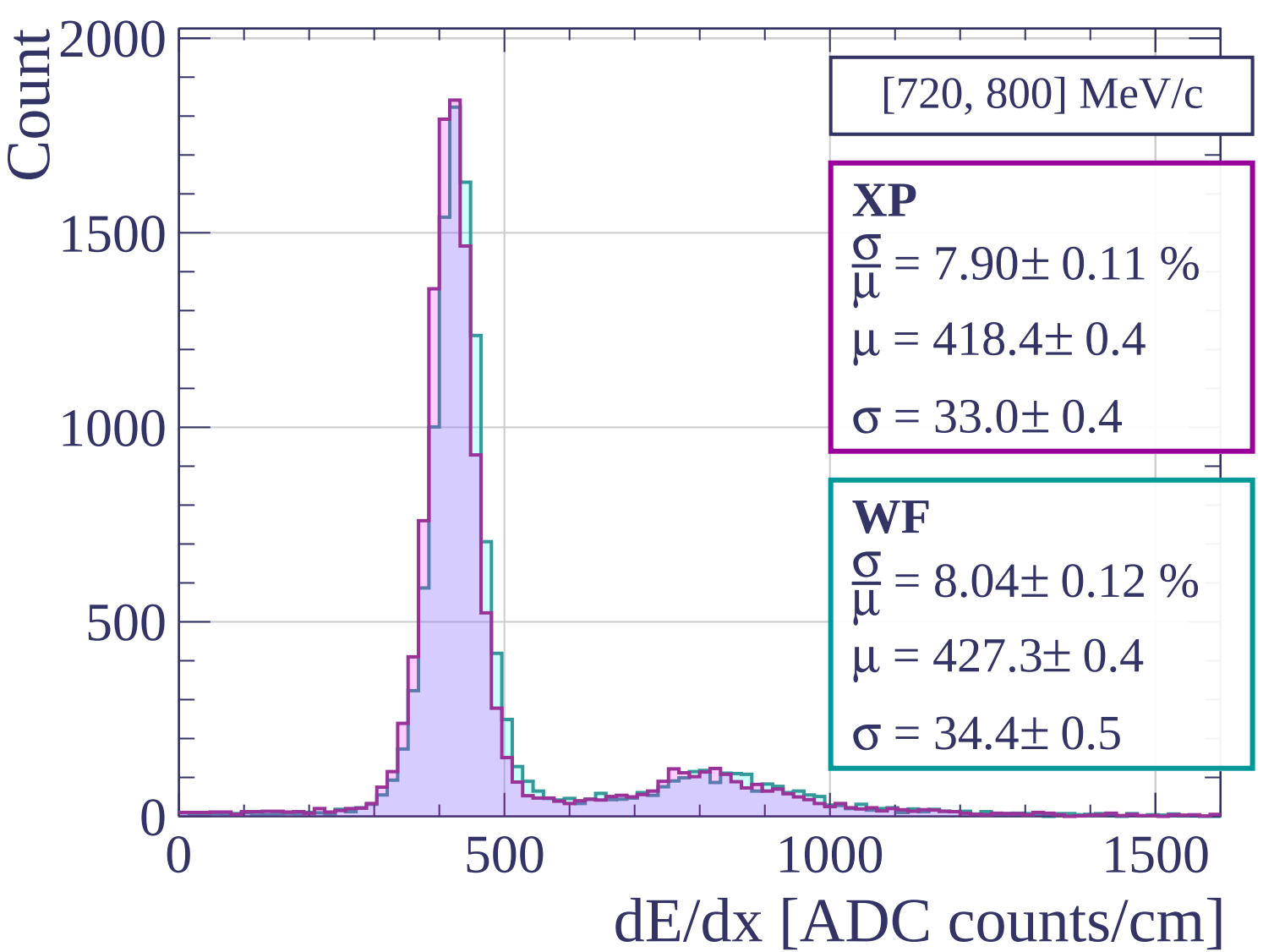


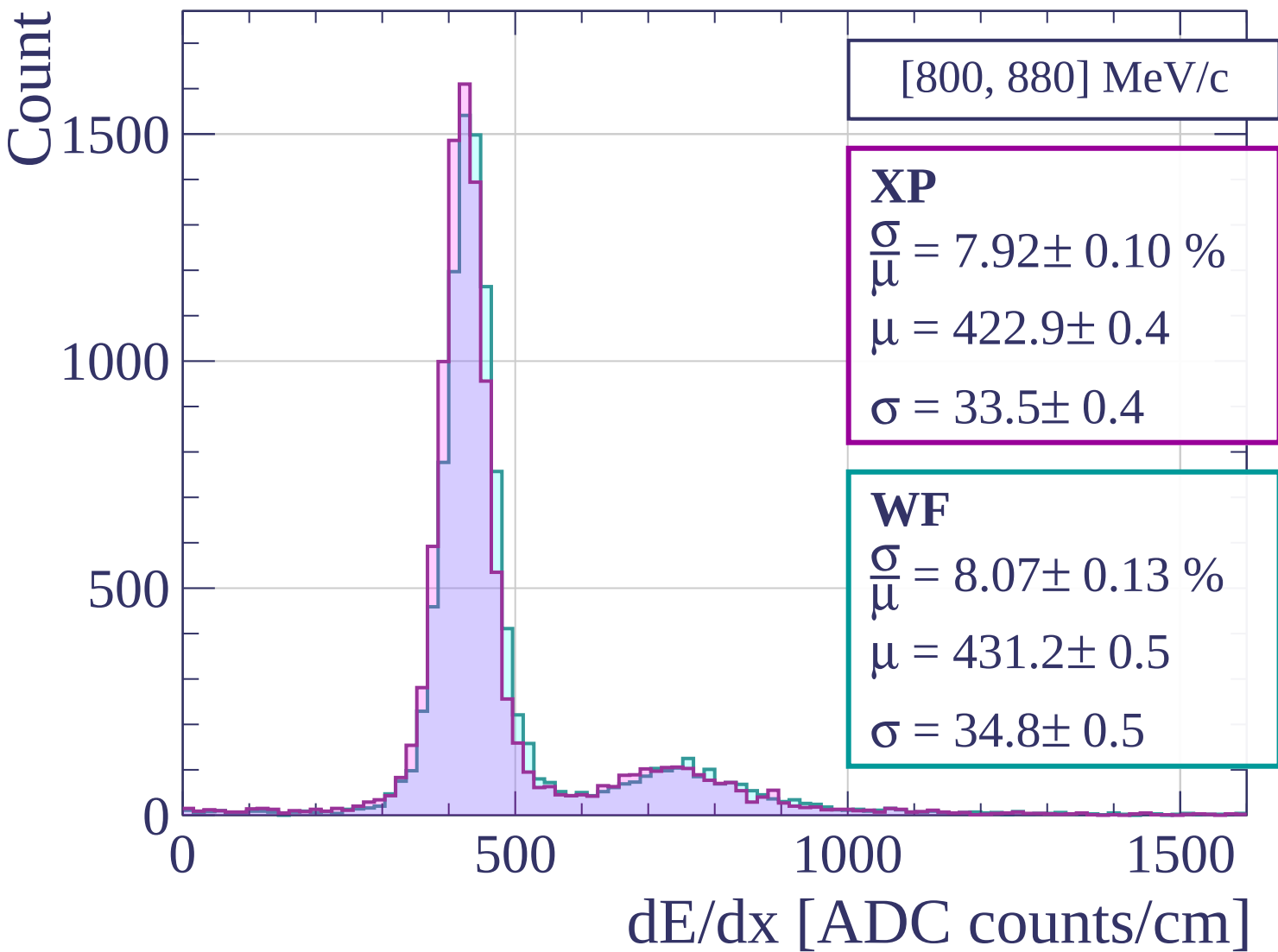


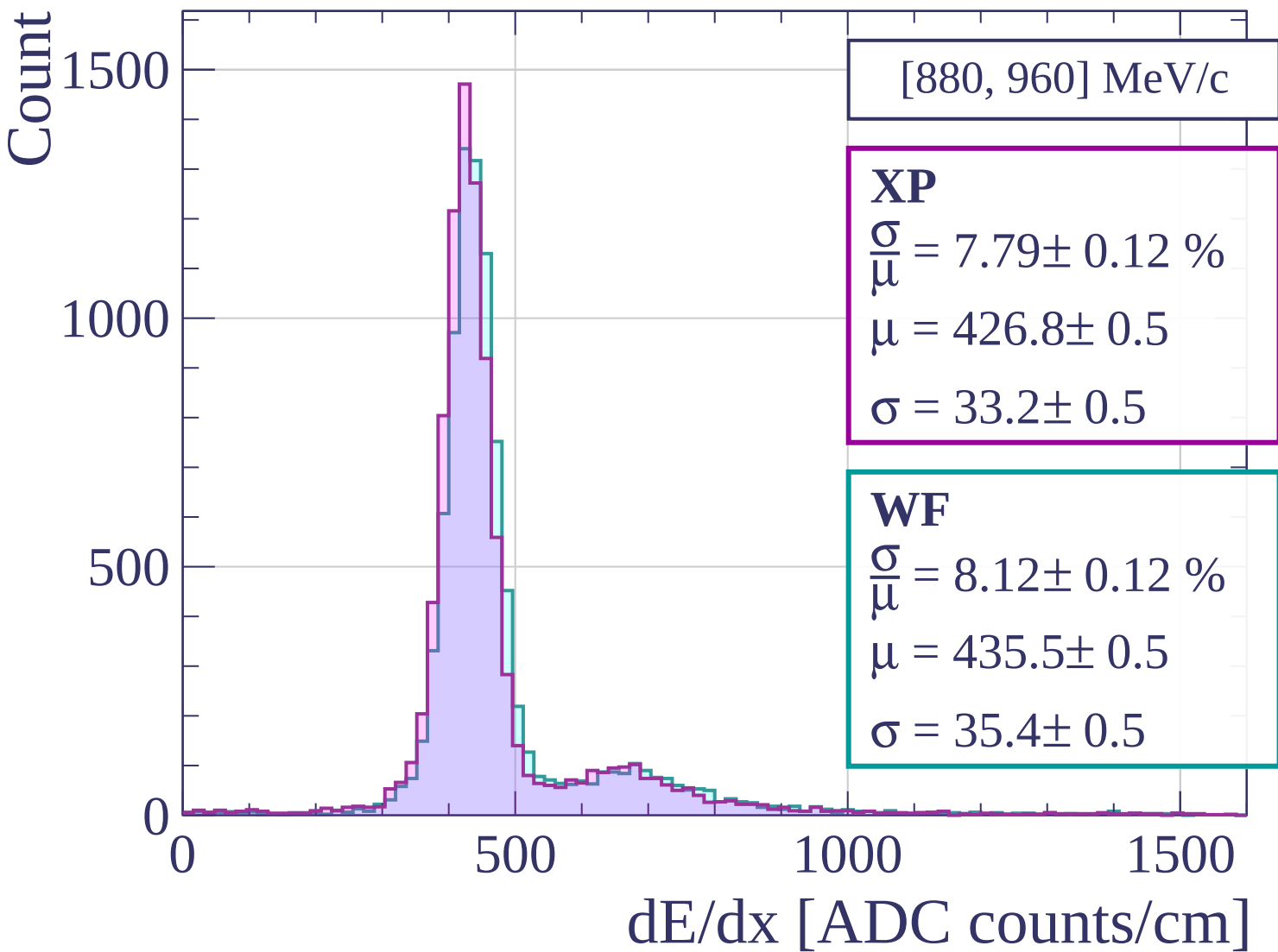


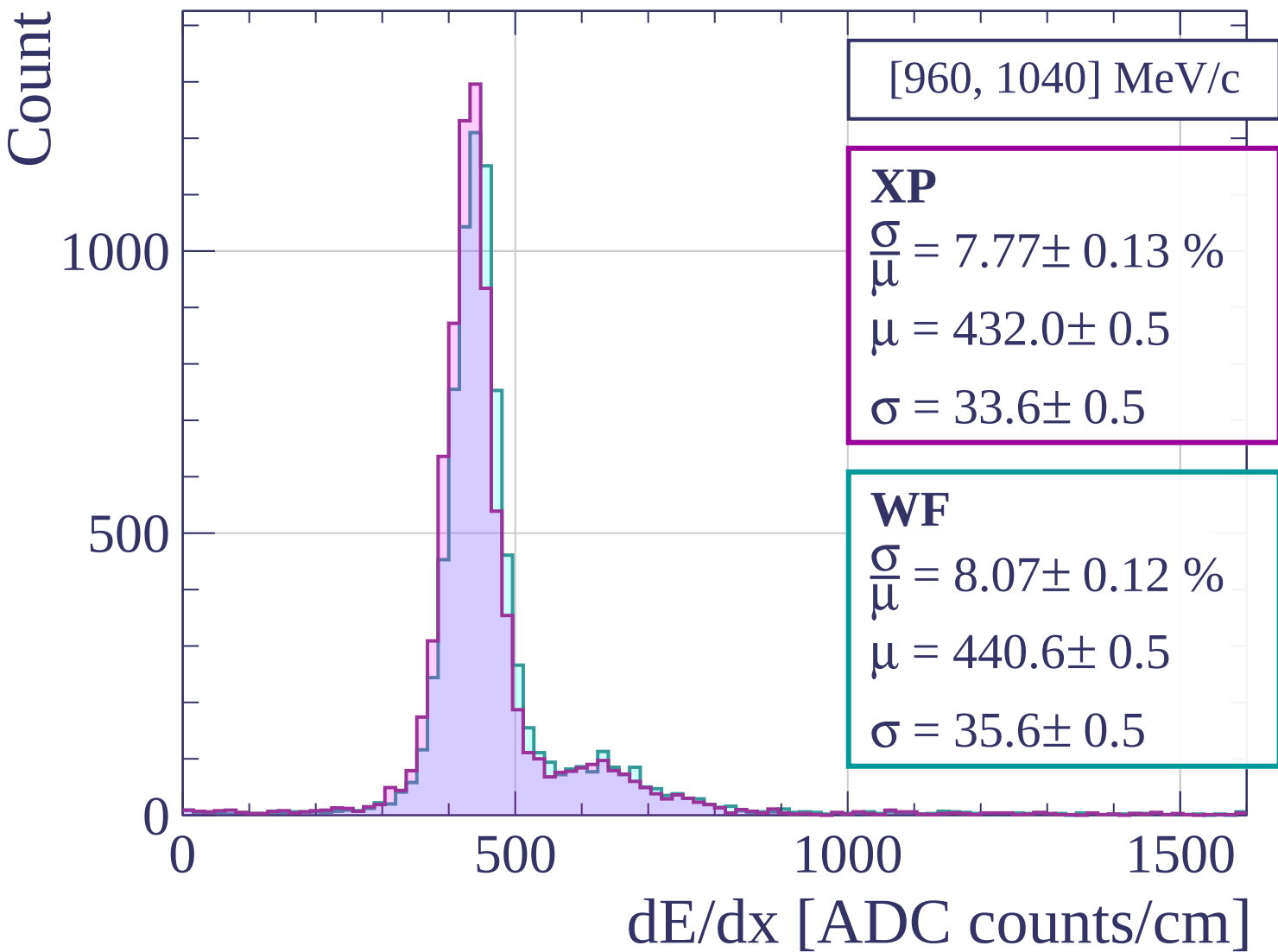


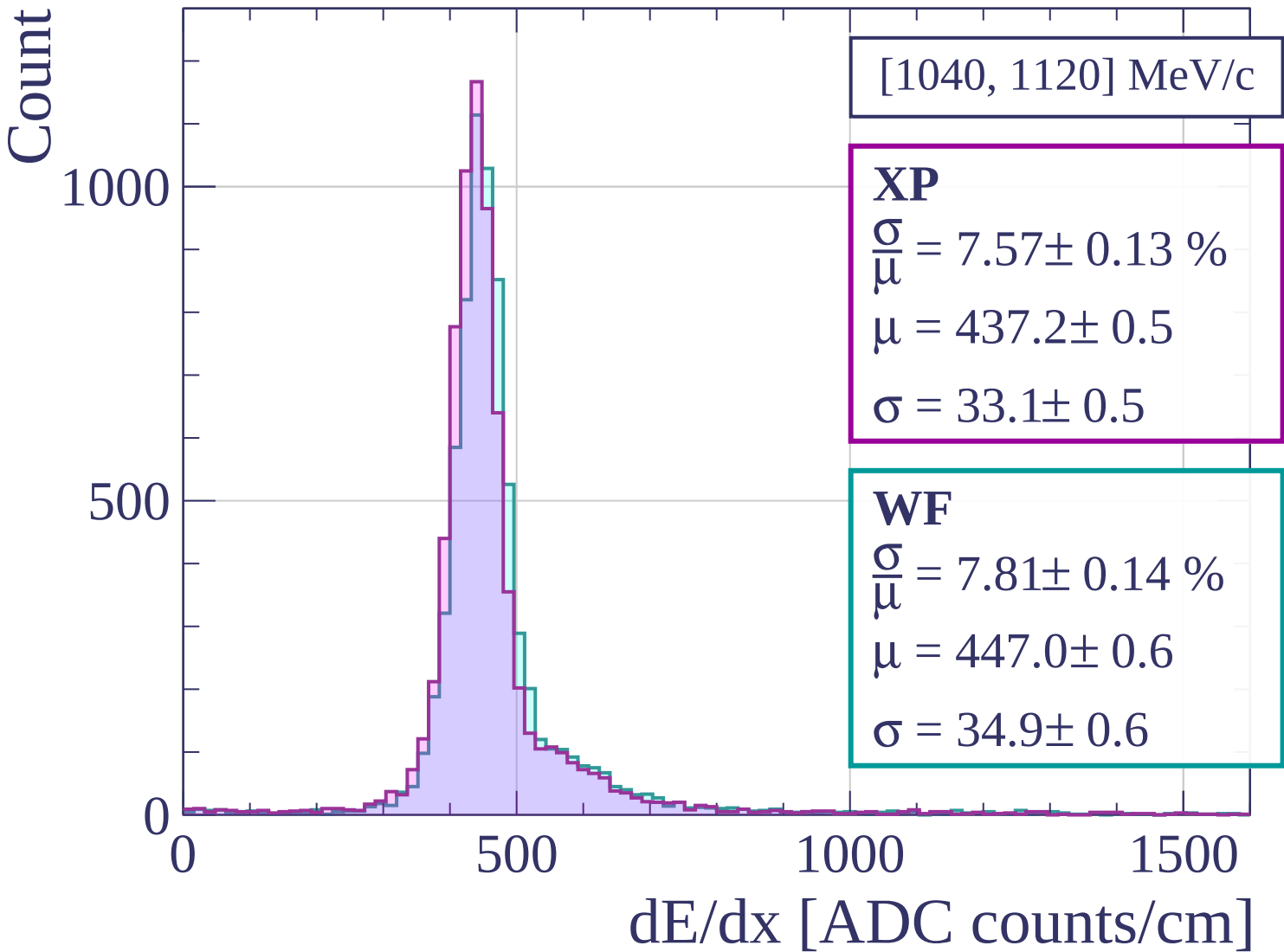


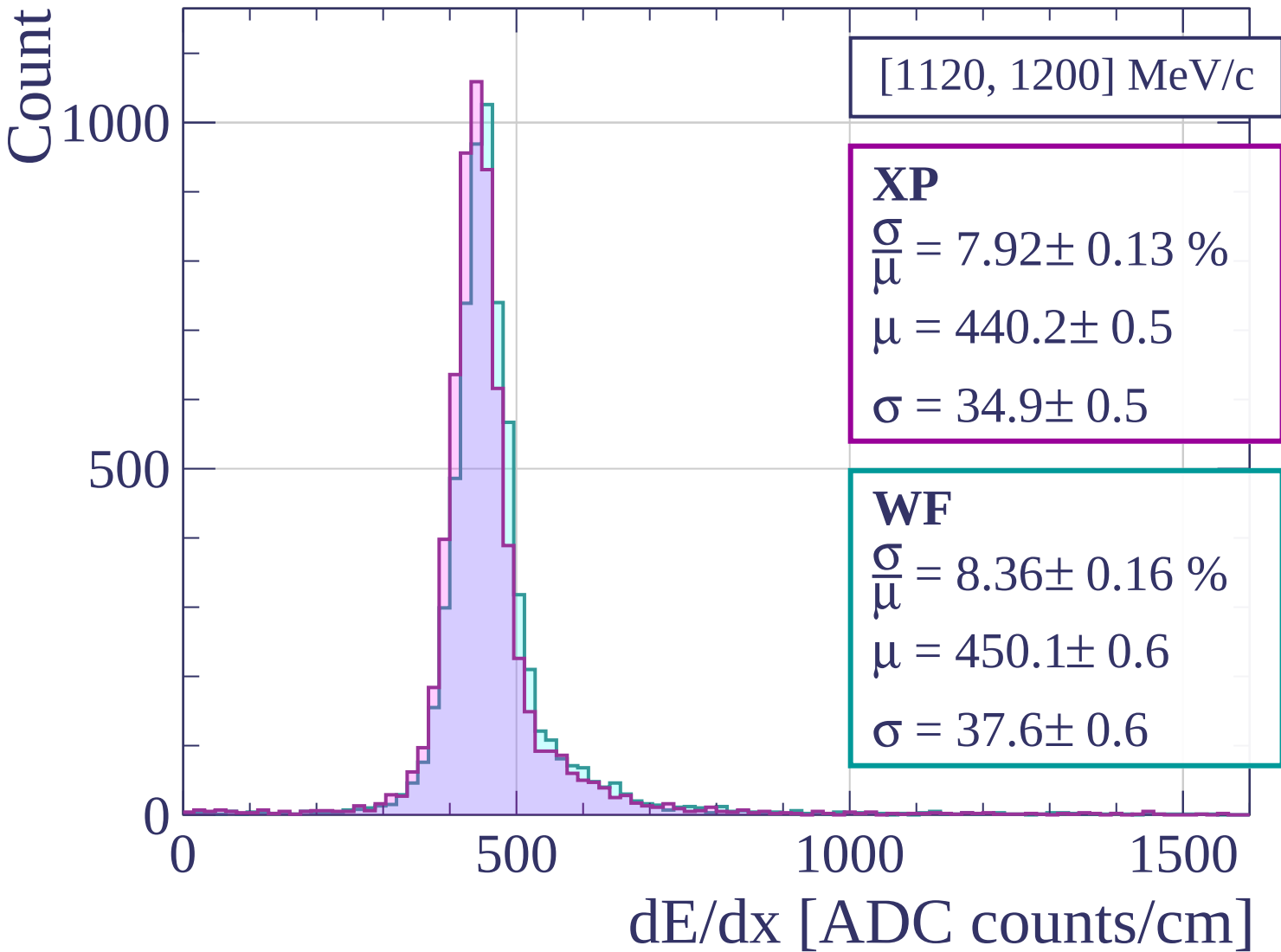


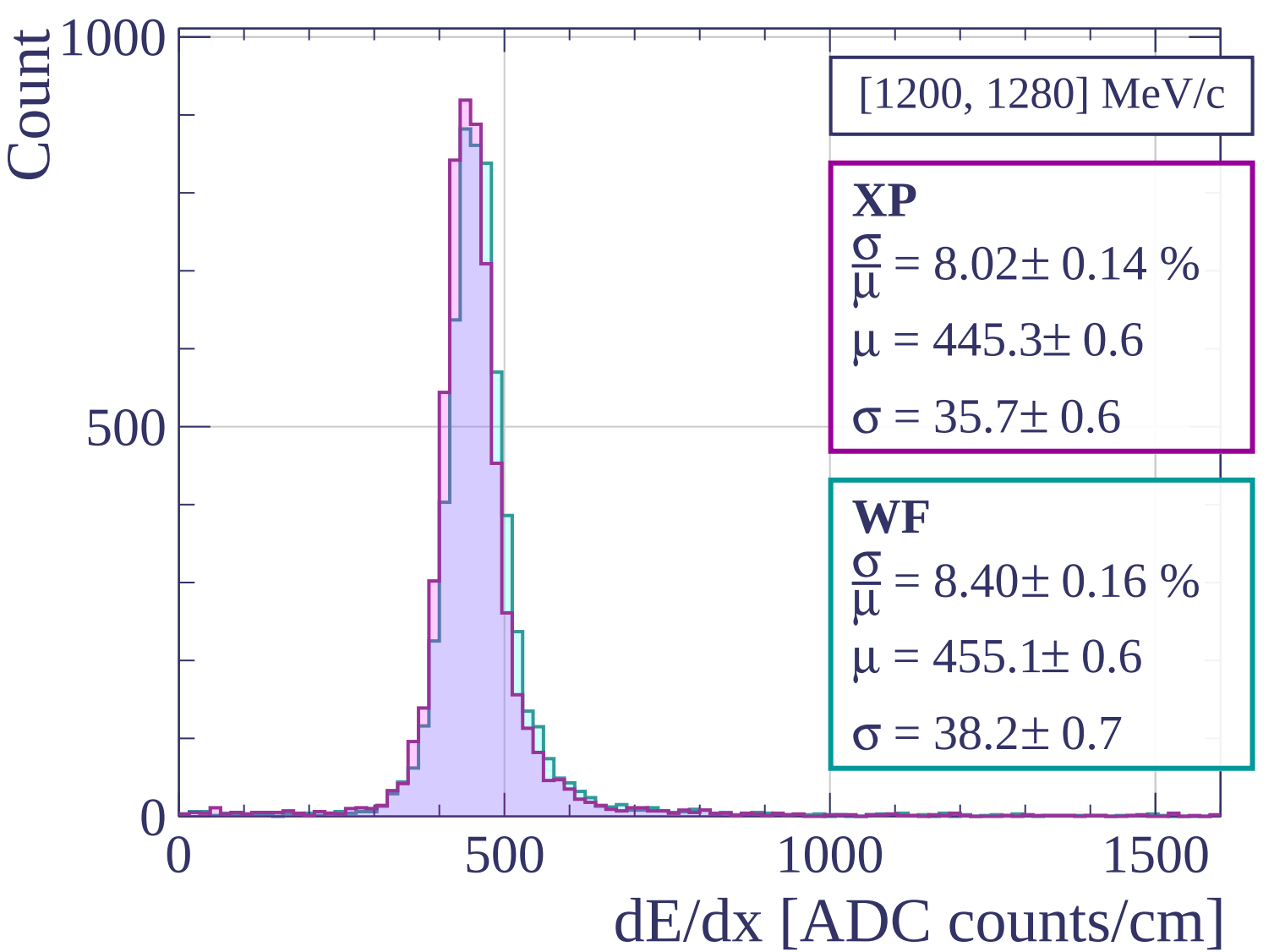


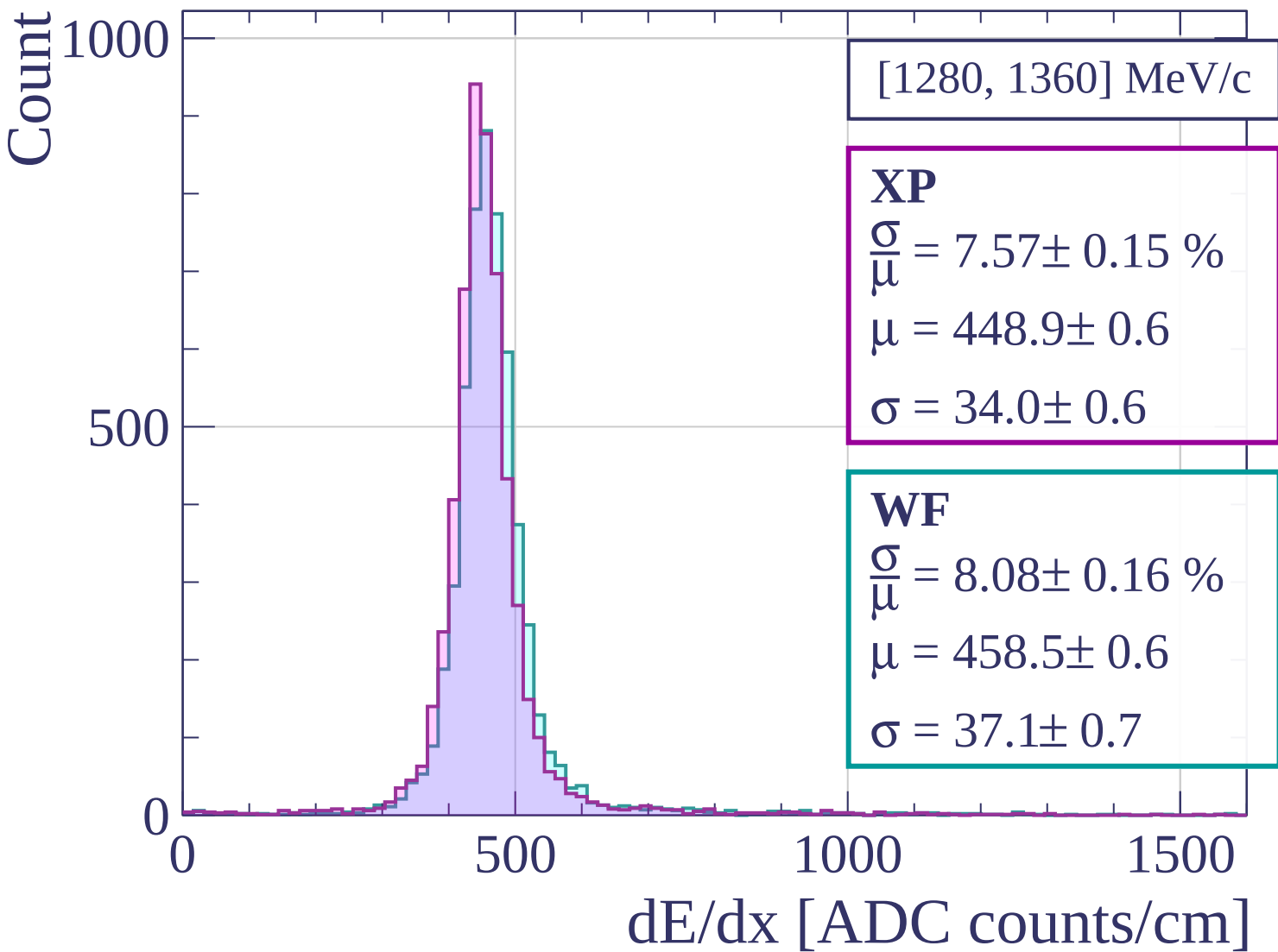




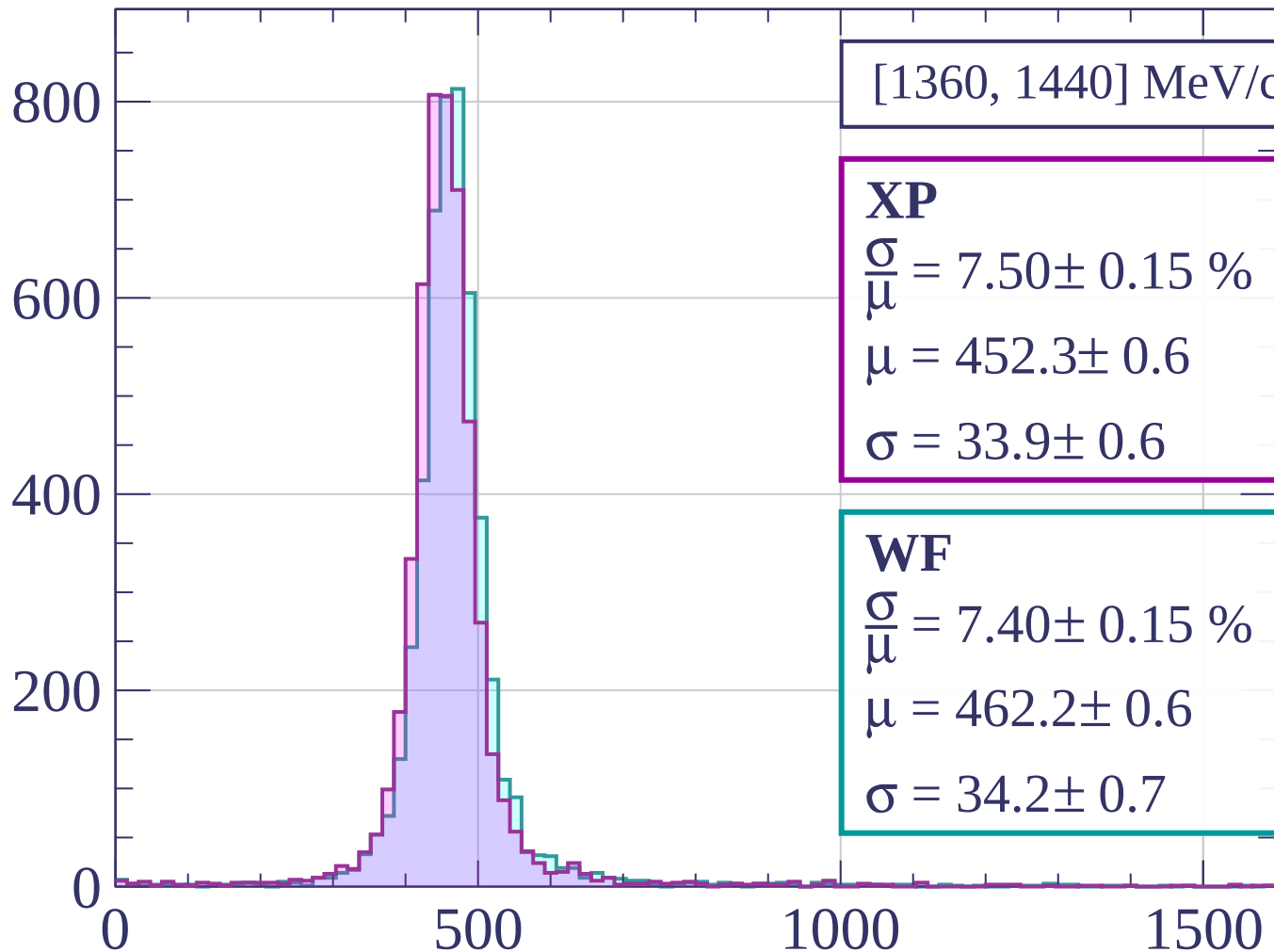






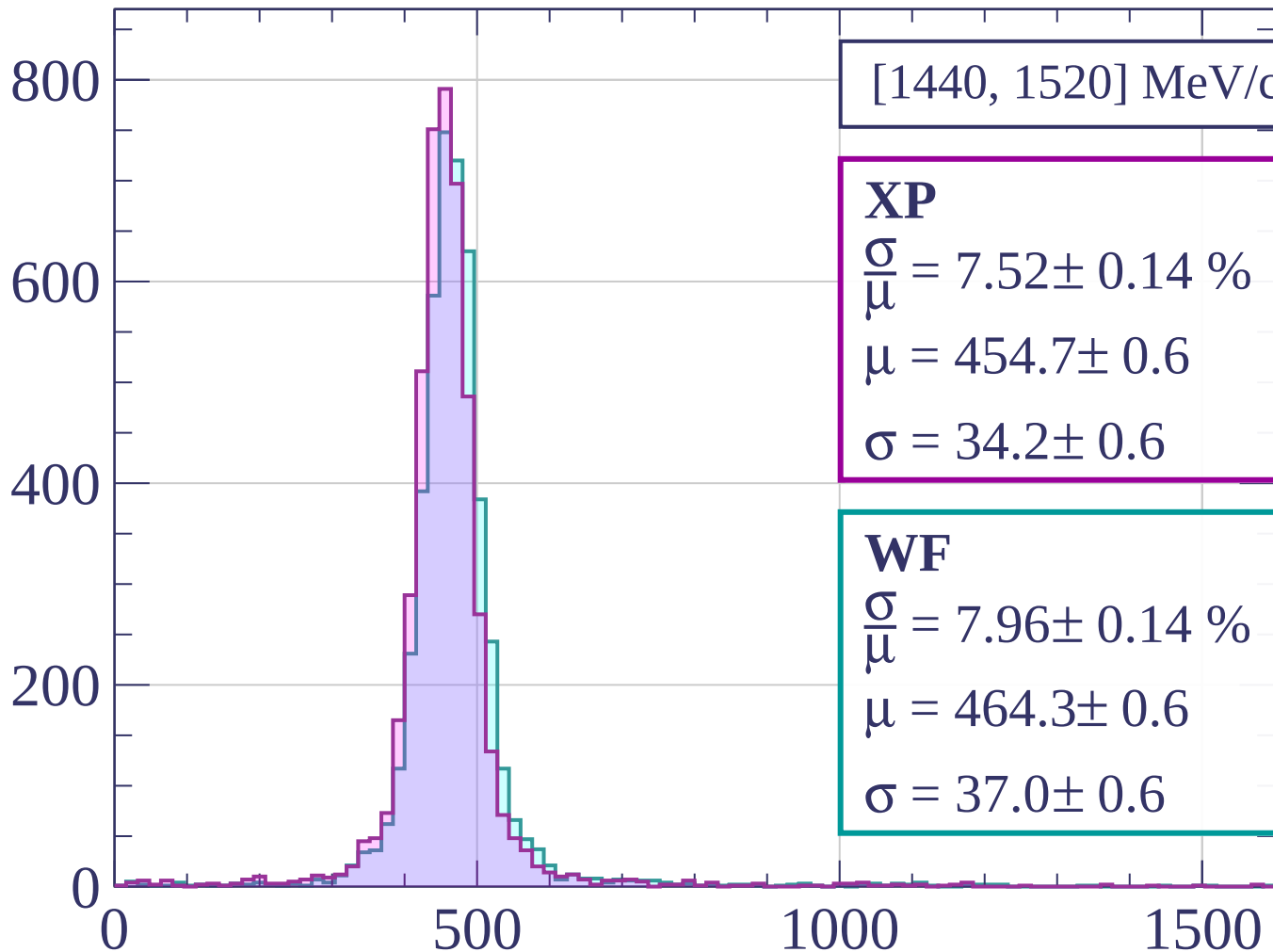


Count

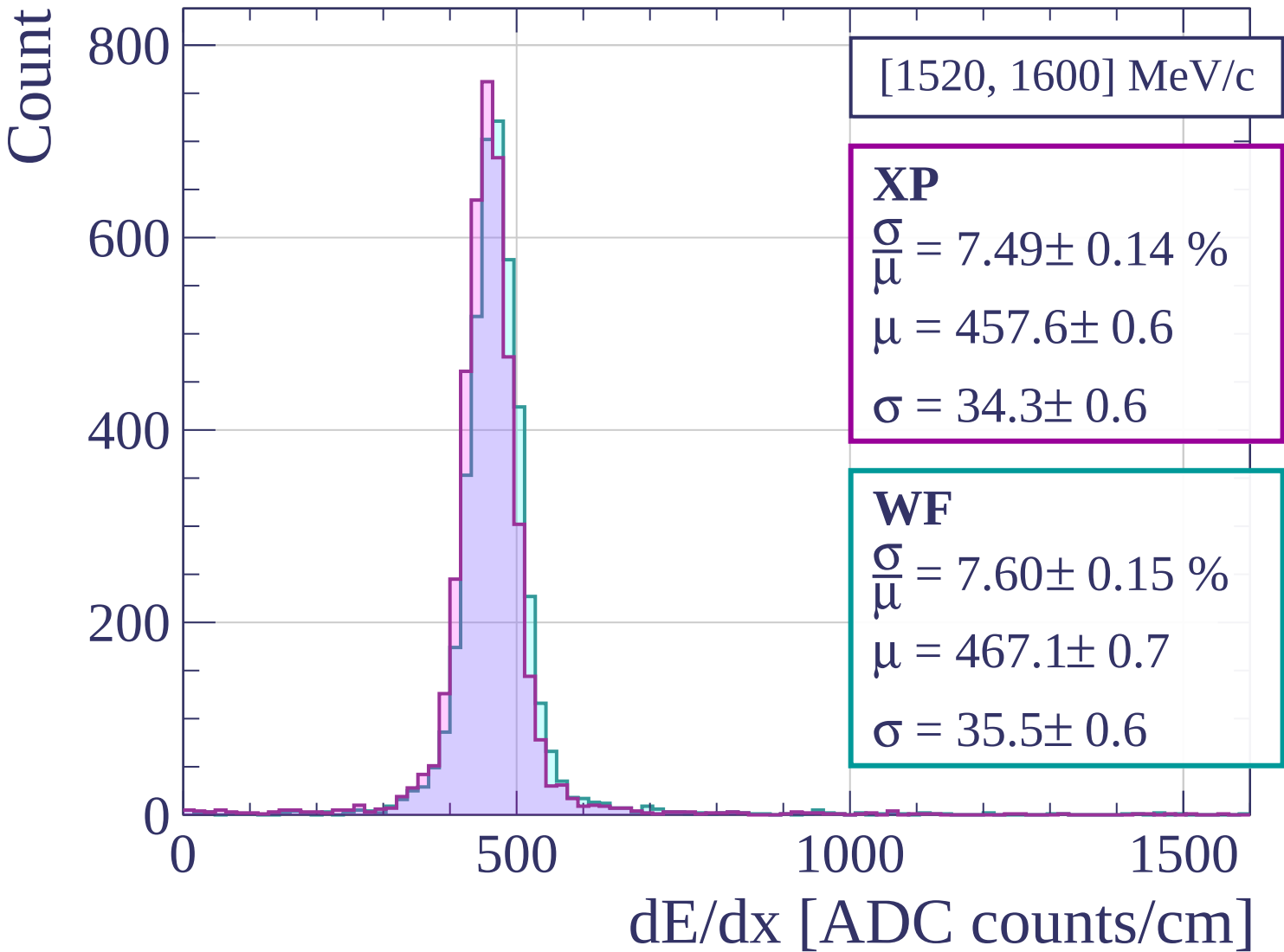


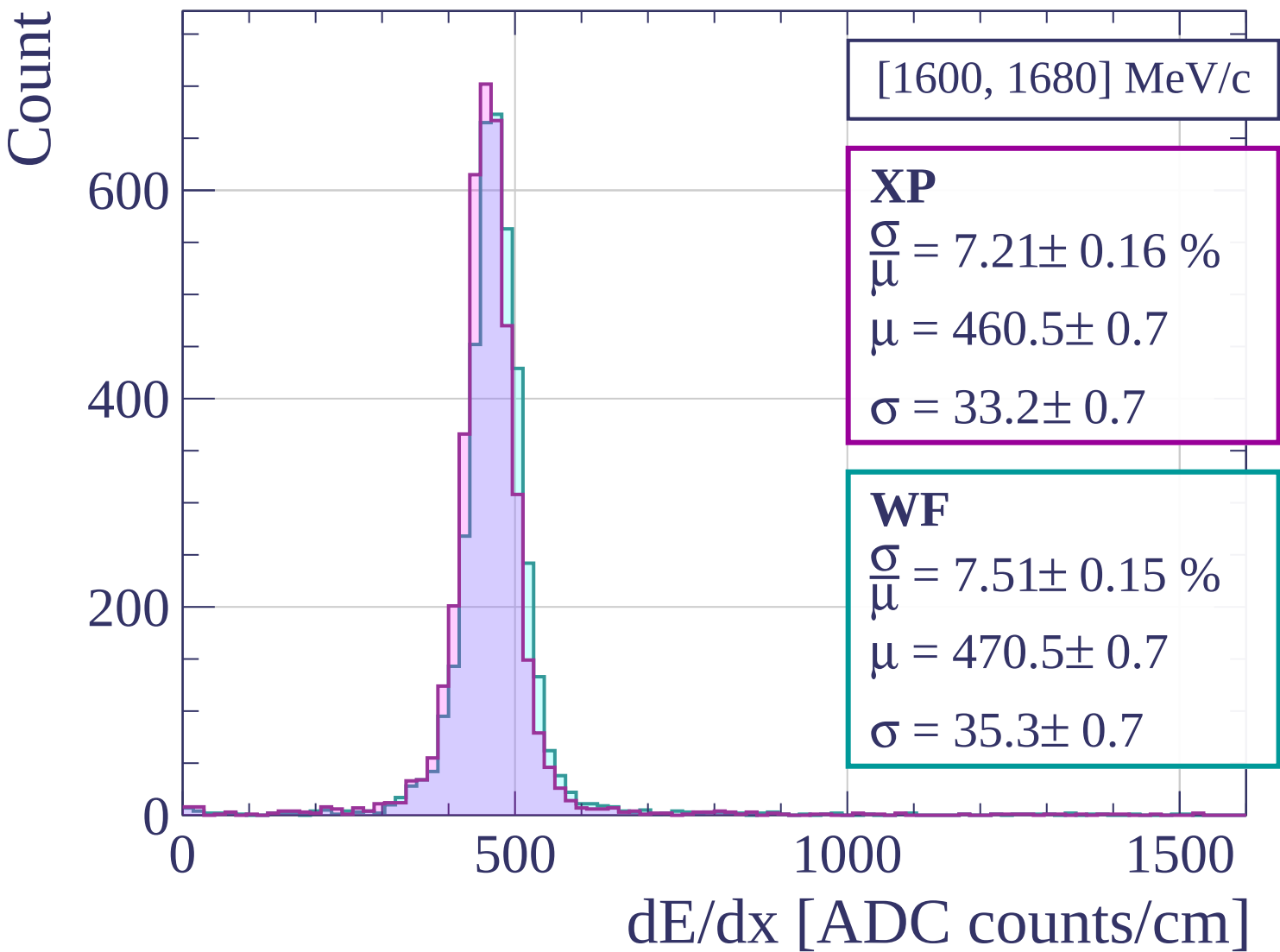
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]





Count

[1680, 1760] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.37 \pm 0.17 \%$$

$$\mu = 462.9 \pm 0.7$$

$$\sigma = 34.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.77 \pm 0.17 \%$$

$$\mu = 472.7 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

600

400

200

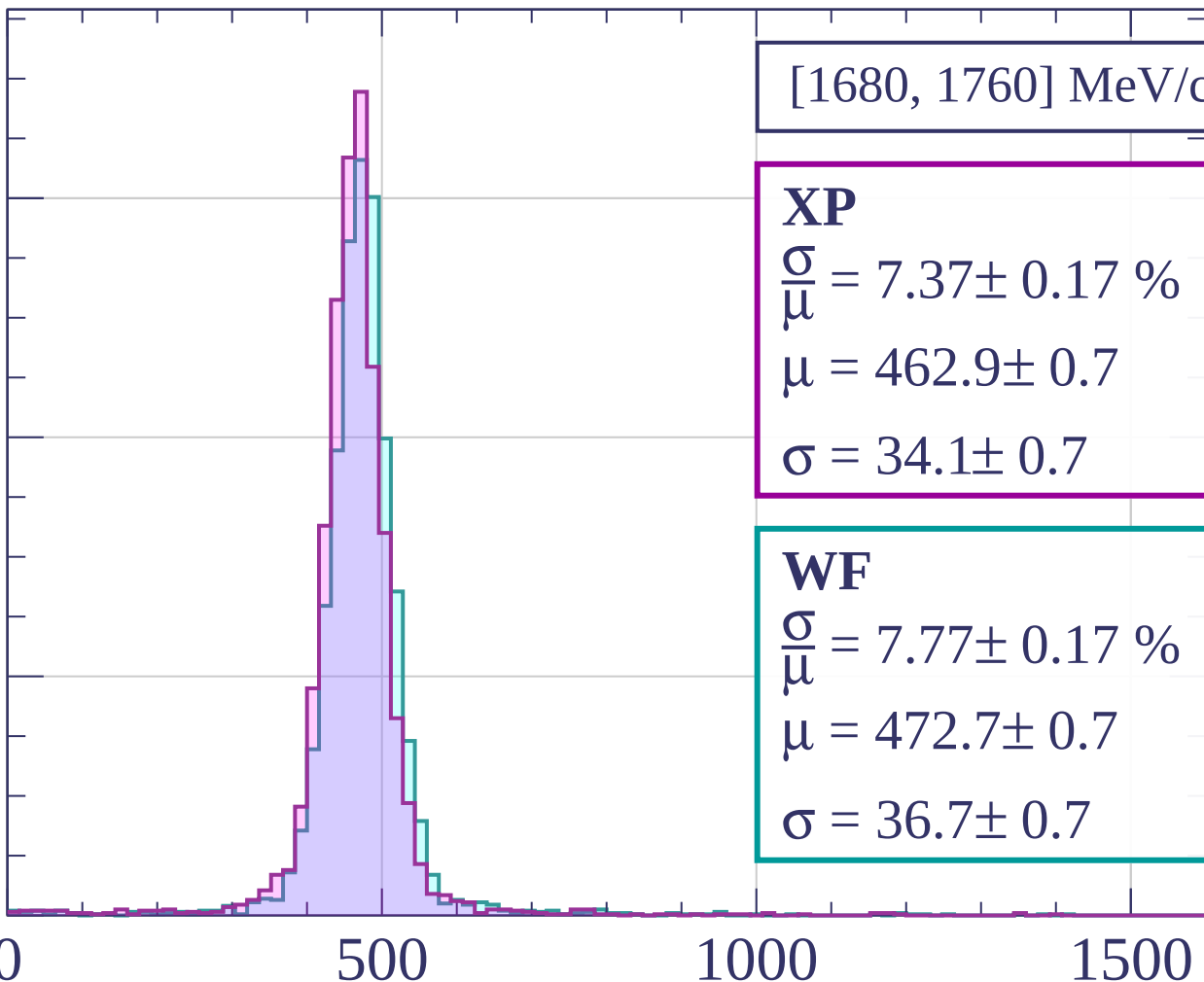
0

500

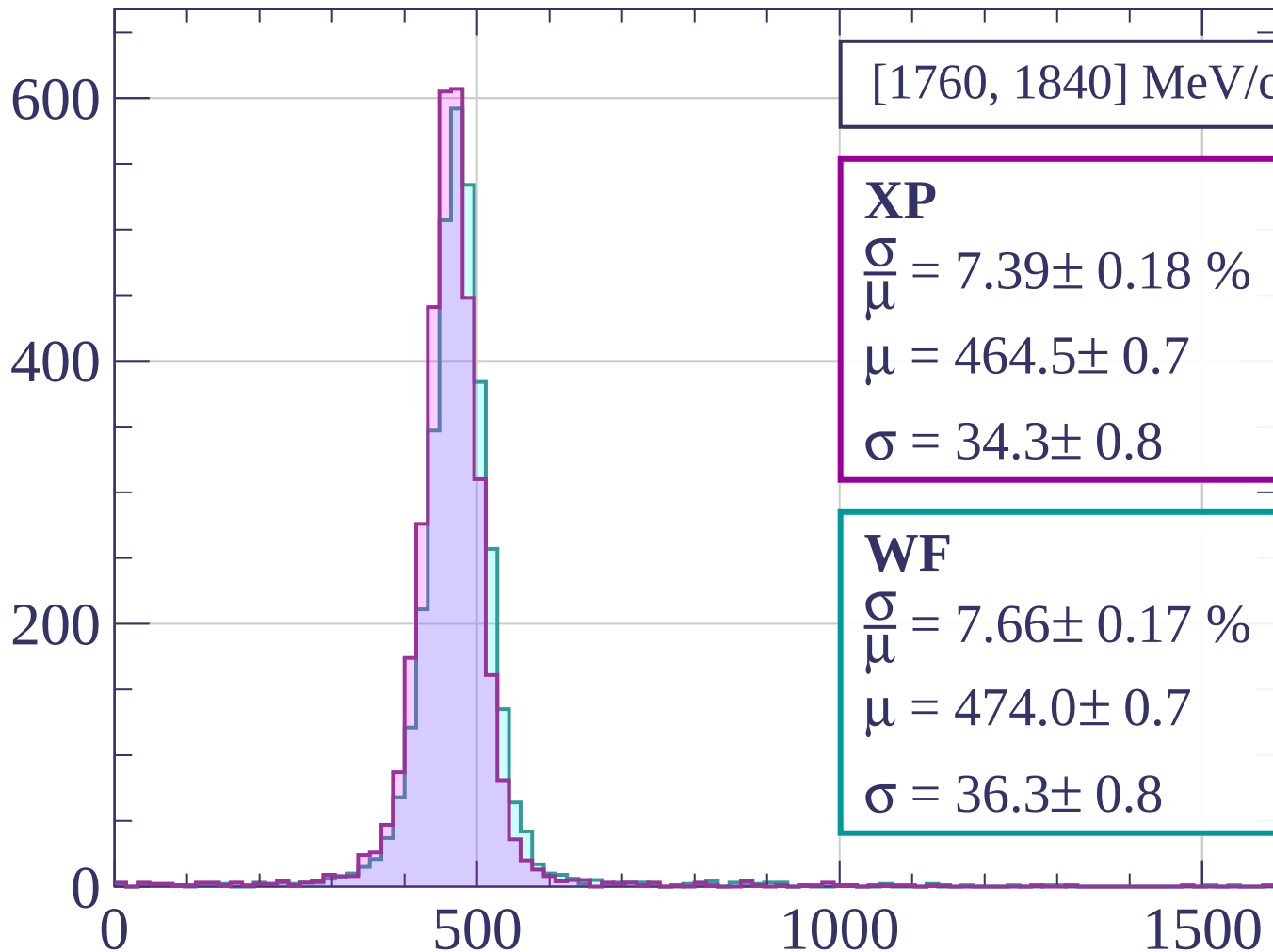
1000

1500

dE/dx [ADC counts/cm]



Count



$[1760, 1840] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 7.39 \pm 0.18 \%$$

$$\mu = 464.5 \pm 0.7$$

$$\sigma = 34.3 \pm 0.8$$

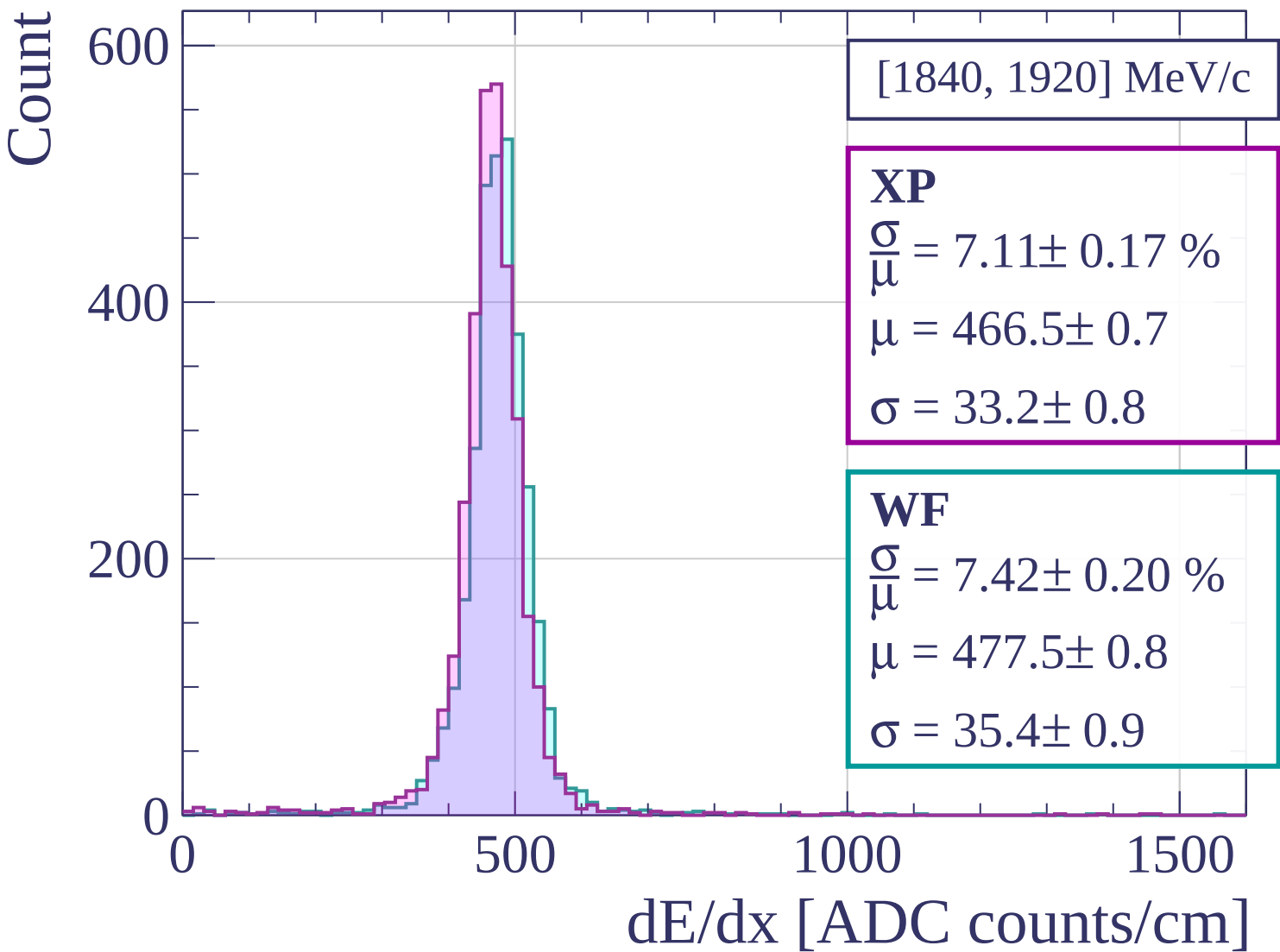
WF

$$\frac{\sigma}{\mu} = 7.66 \pm 0.17 \%$$

$$\mu = 474.0 \pm 0.7$$

$$\sigma = 36.3 \pm 0.8$$

dE/dx [ADC counts/cm]



Count

[1920, 2000] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.35 \pm 0.16 \%$$

$$\mu = 469.2 \pm 0.7$$

$$\sigma = 34.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.16 \pm 0.18 \%$$

$$\mu = 479.4 \pm 0.8$$

$$\sigma = 34.3 \pm 0.8$$

400

200

0

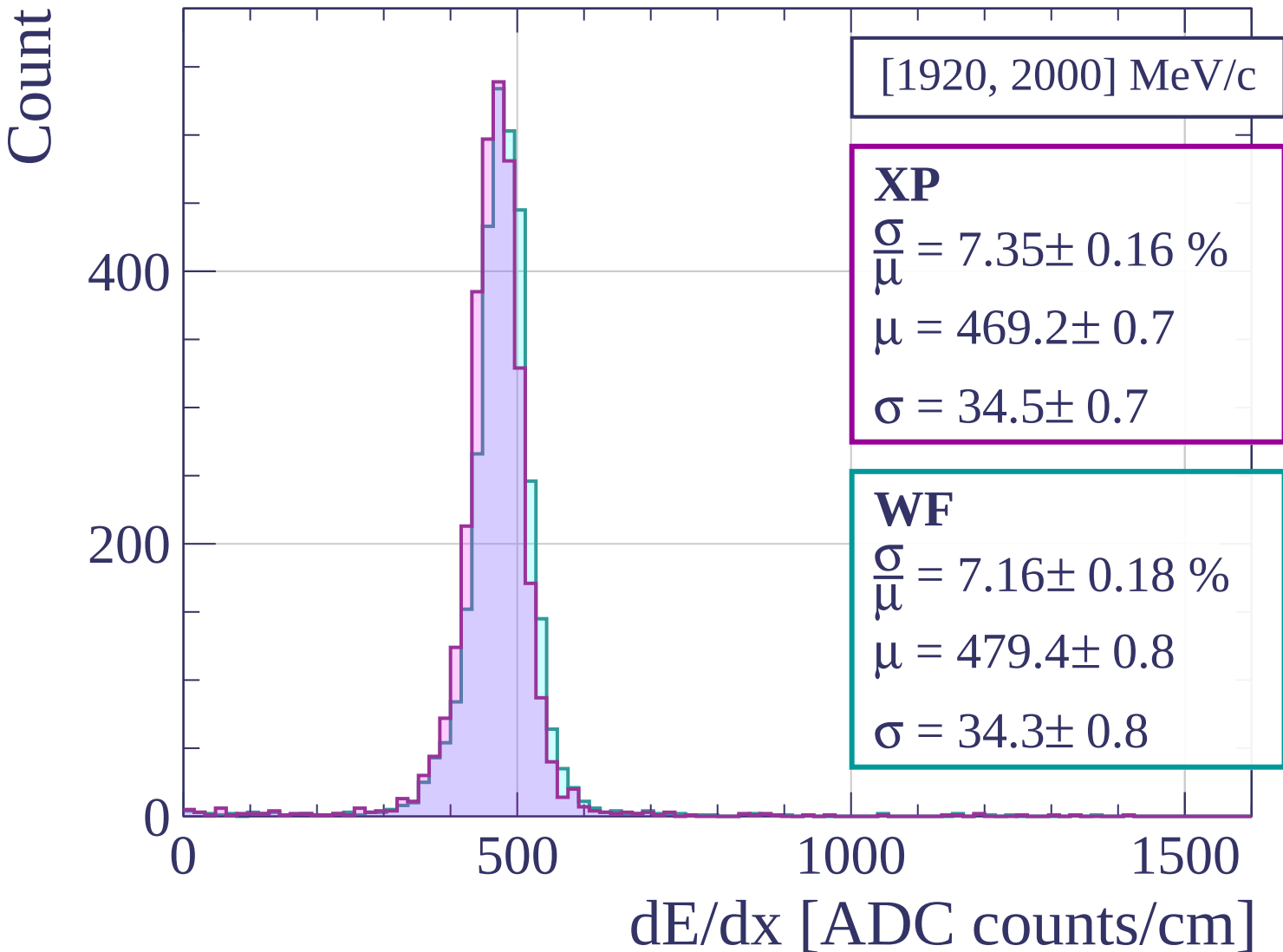
0

500

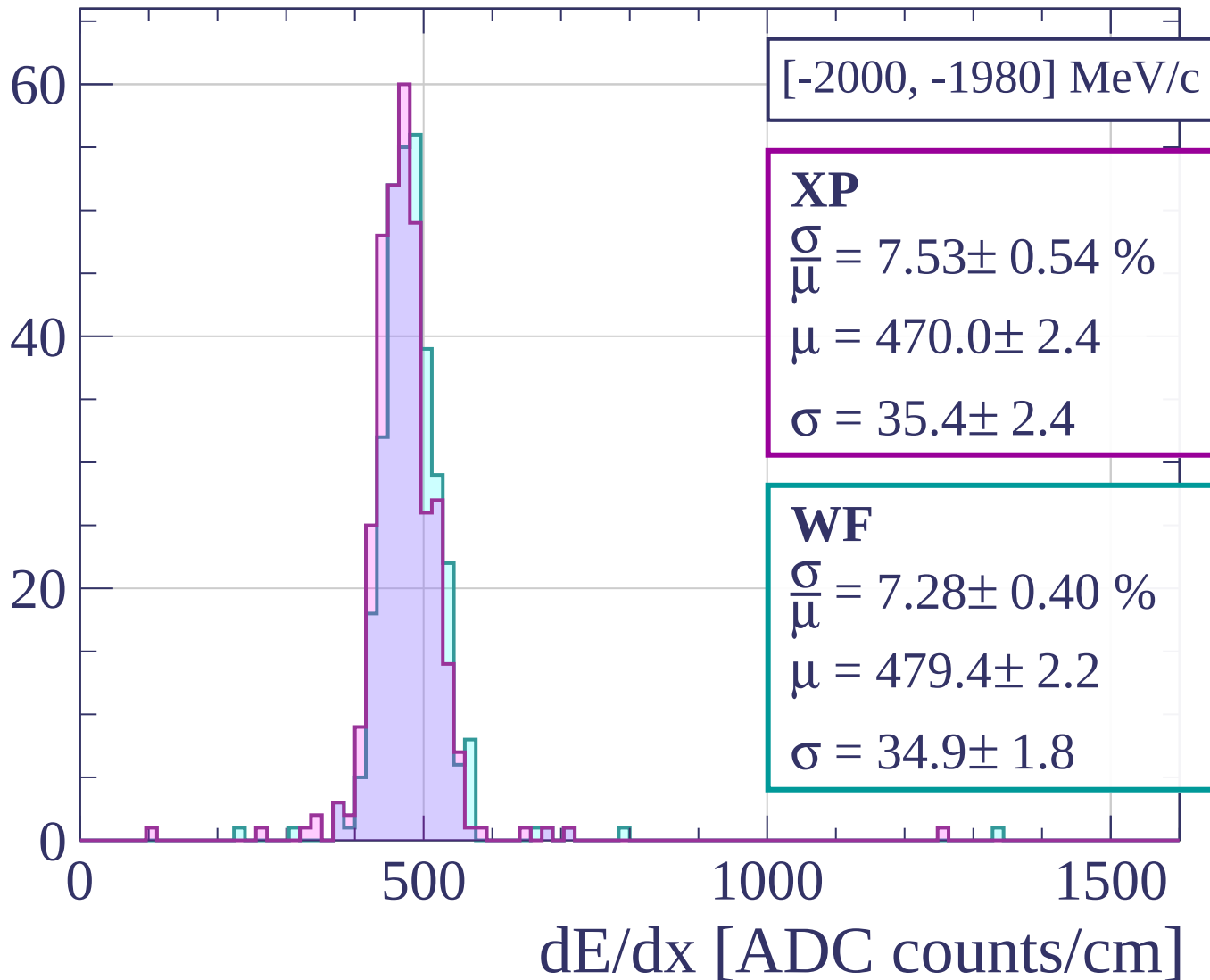
1000

1500

dE/dx [ADC counts/cm]



Count



Count

[-1980, -1960] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.03 \pm 0.36 \%$$

$$\mu = 469.0 \pm 1.5$$

$$\sigma = 33.0 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 6.82 \pm 0.34 \%$$

$$\mu = 479.5 \pm 1.5$$

$$\sigma = 32.7 \pm 1.5$$

100

50

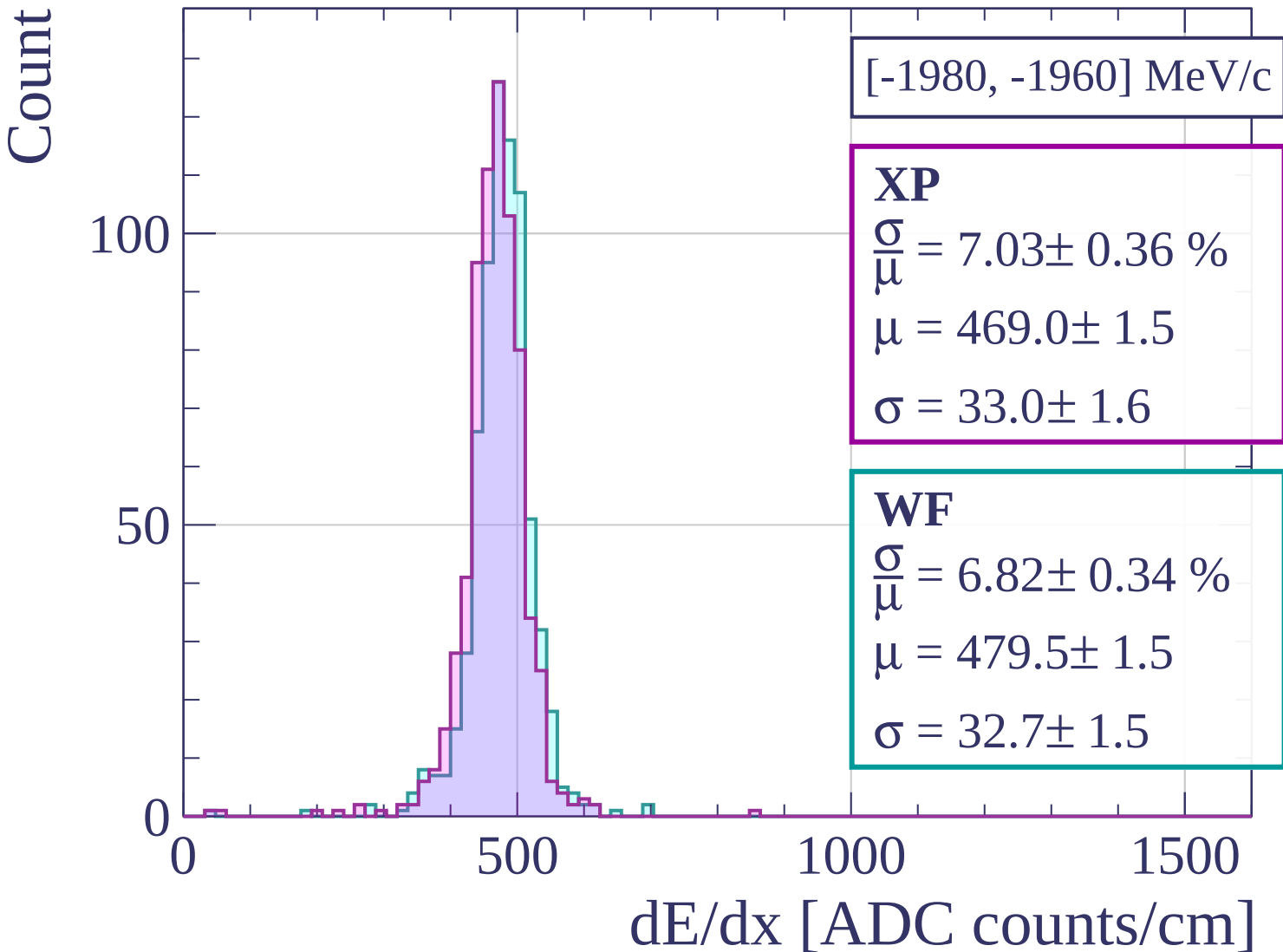
0

500

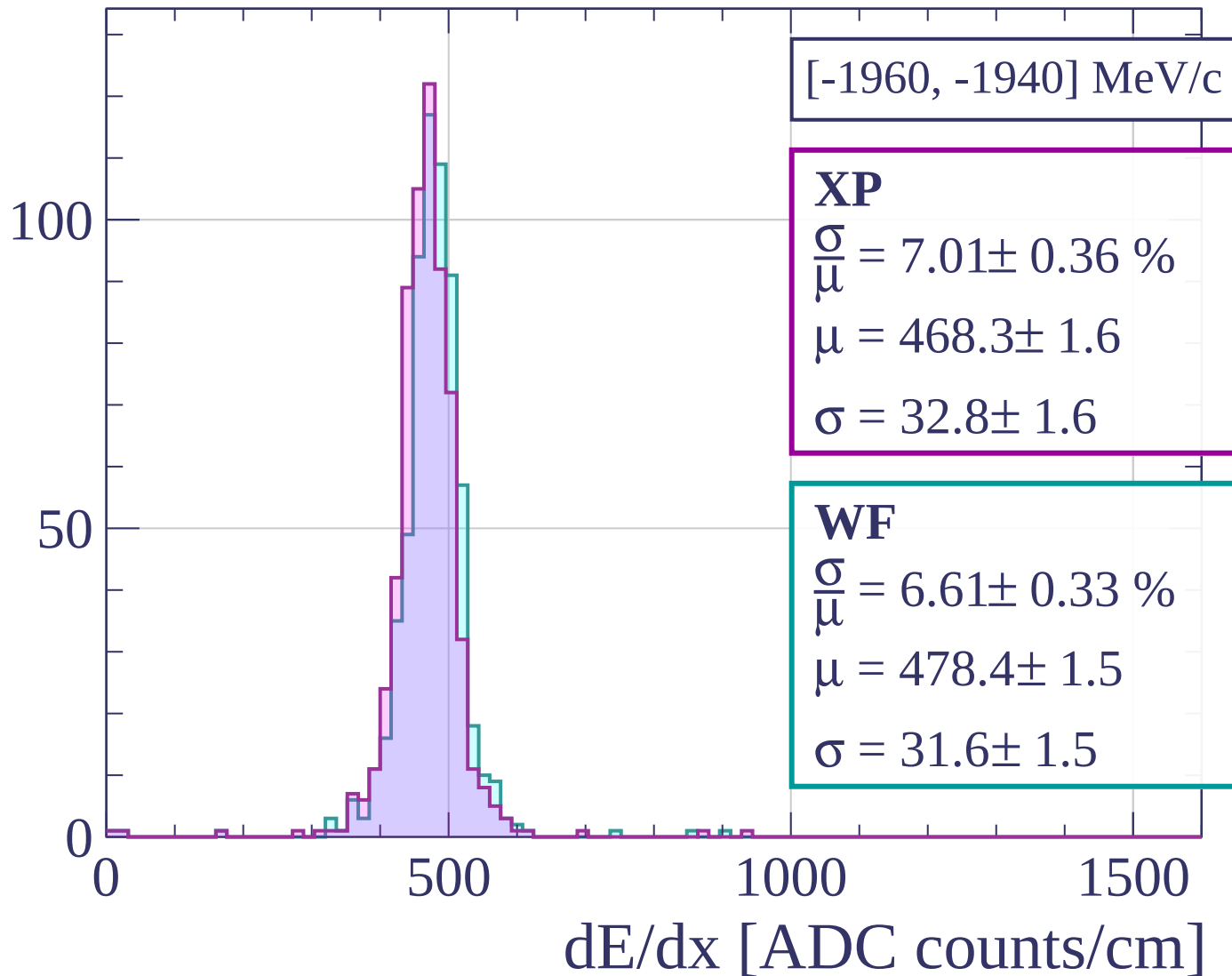
1000

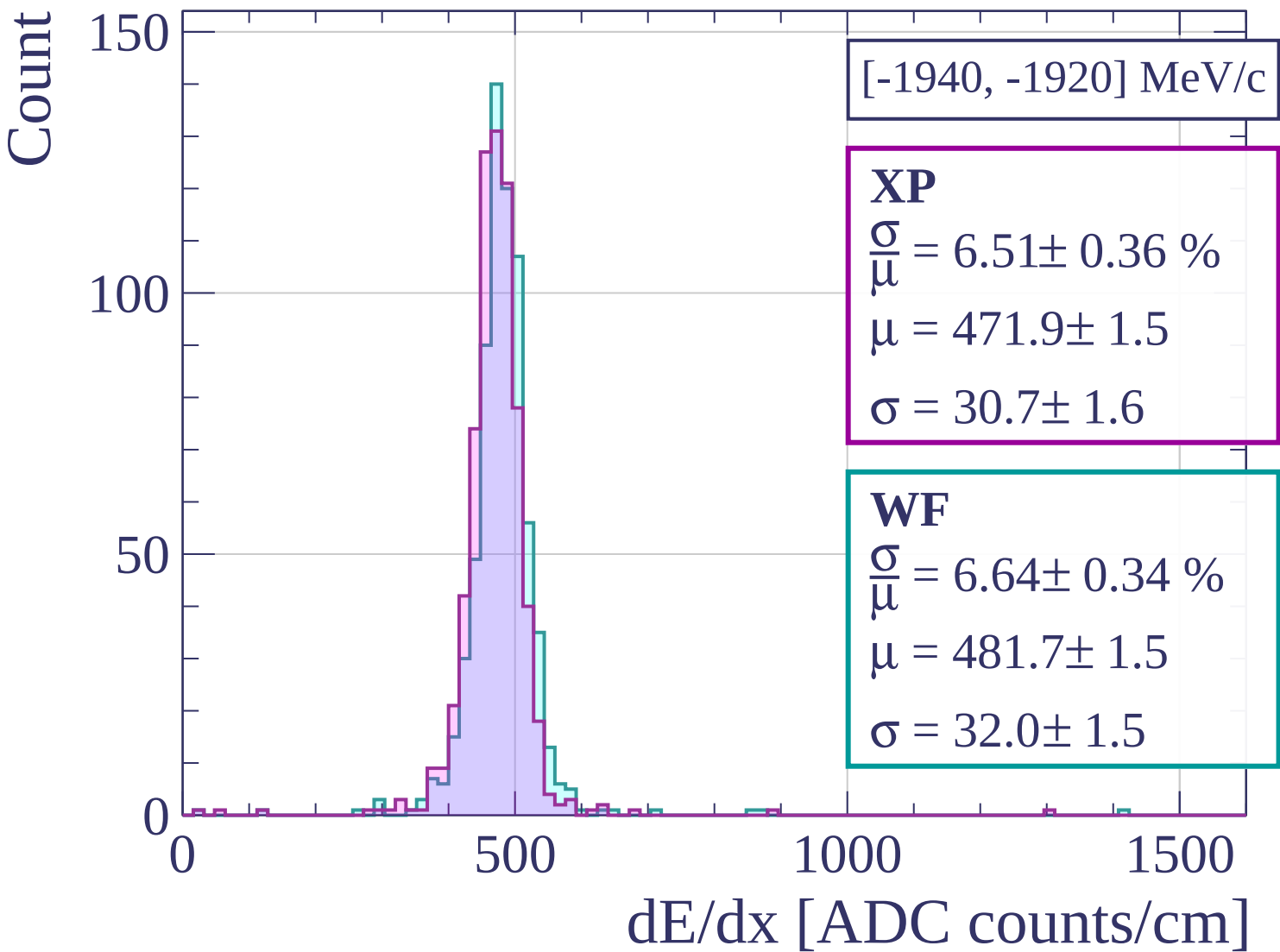
1500

dE/dx [ADC counts/cm]



Count





Count

[-1920, -1900] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.95 \pm 0.31 \%$$

$$\mu = 470.6 \pm 1.5$$

$$\sigma = 32.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.94 \pm 0.39 \%$$

$$\mu = 481.5 \pm 1.6$$

$$\sigma = 33.4 \pm 1.8$$

100

50

0

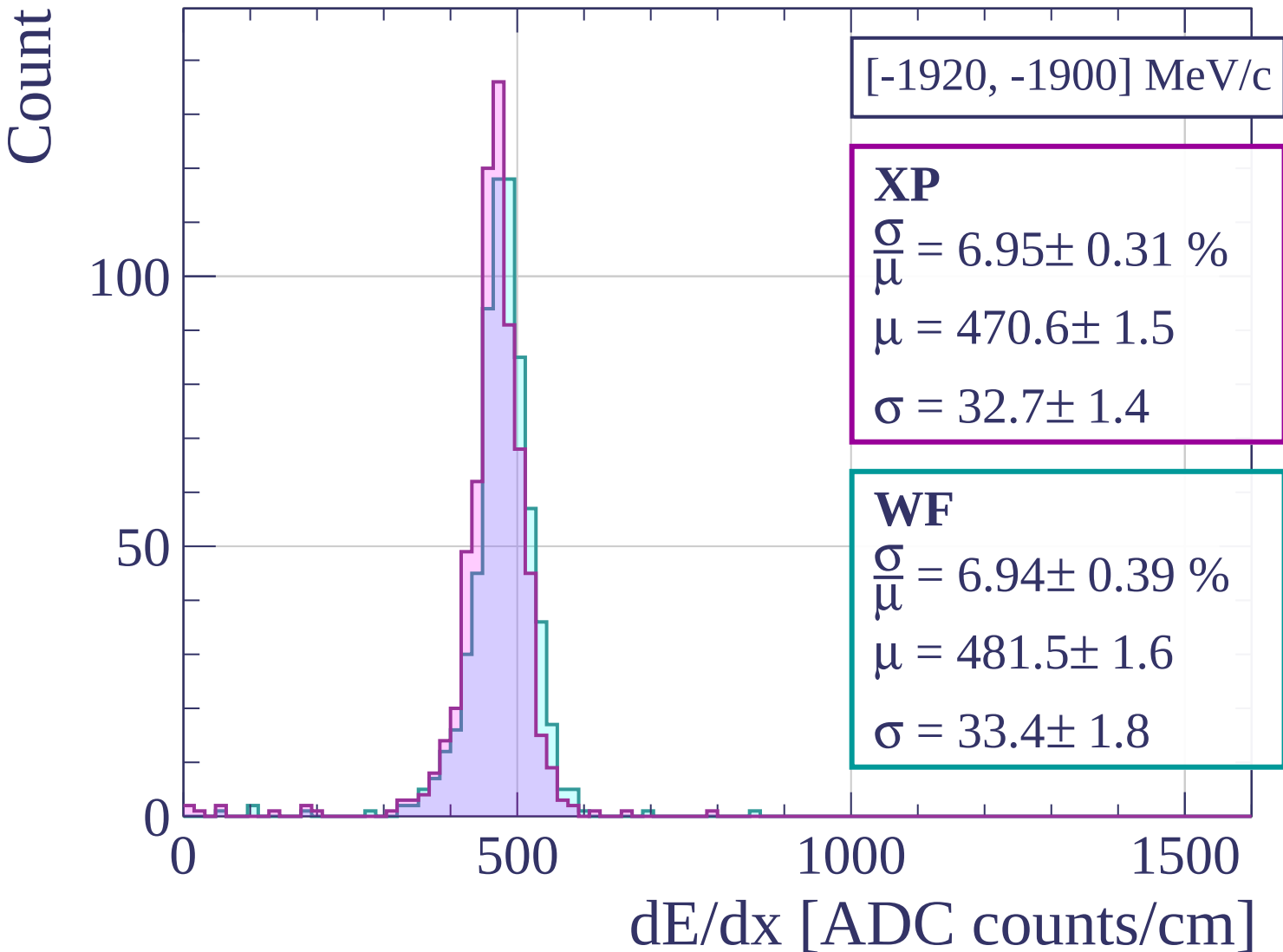
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1900, -1880] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.96 \pm 0.35 \%$$

$$\mu = 467.7 \pm 1.5$$

$$\sigma = 32.5 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.39 \pm 0.35 \%$$

$$\mu = 477.5 \pm 1.7$$

$$\sigma = 35.3 \pm 1.6$$

100

50

0

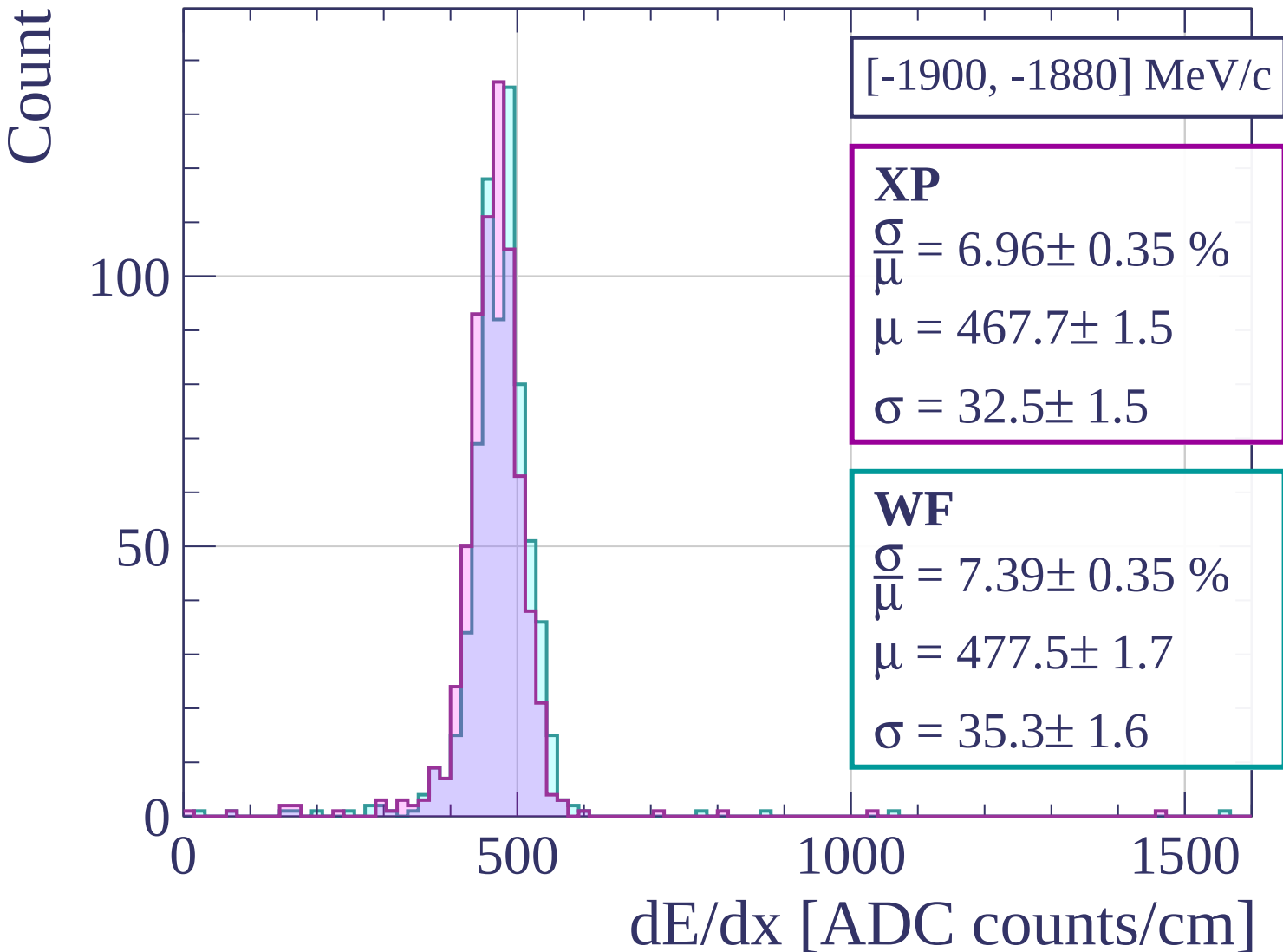
0

500

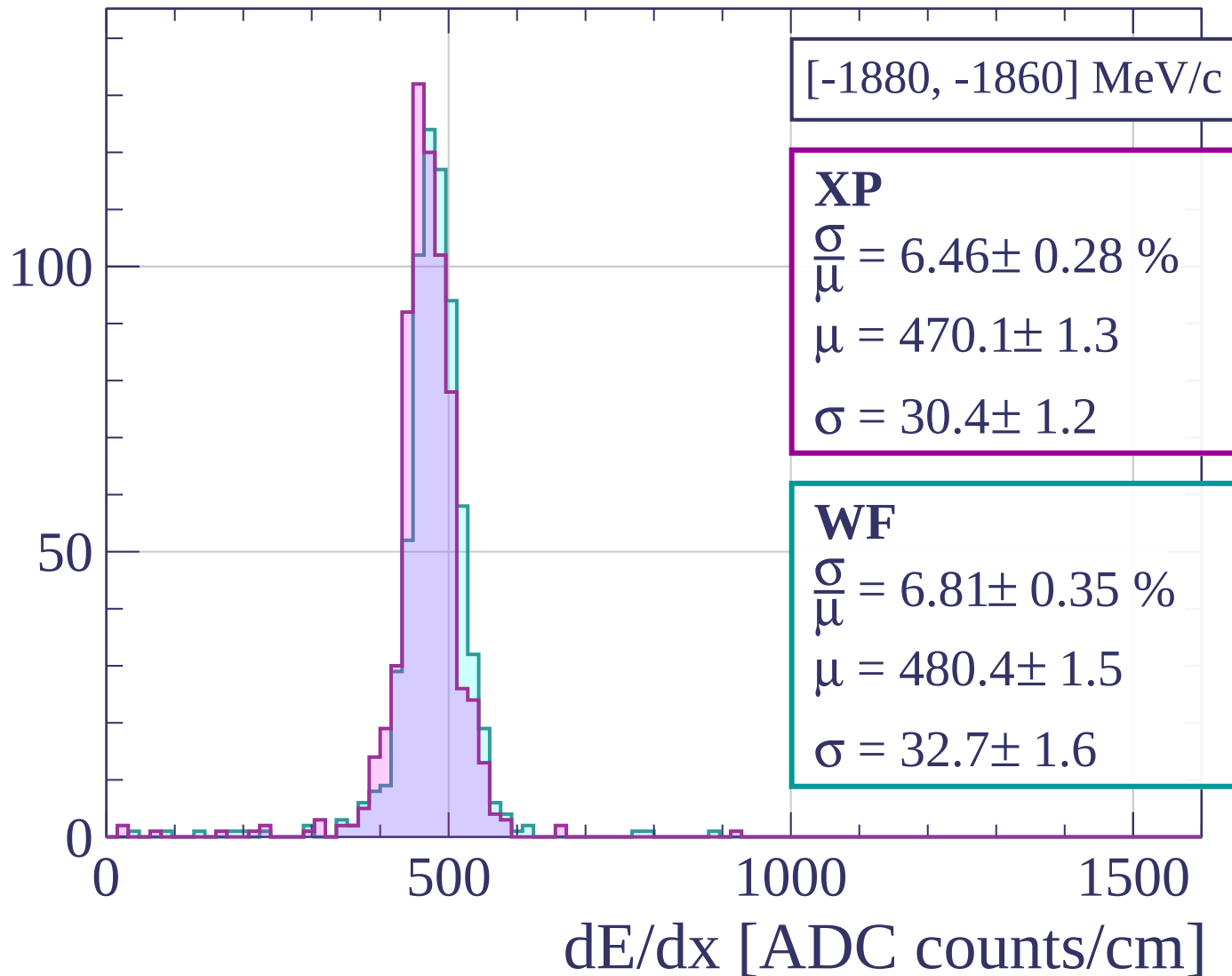
1000

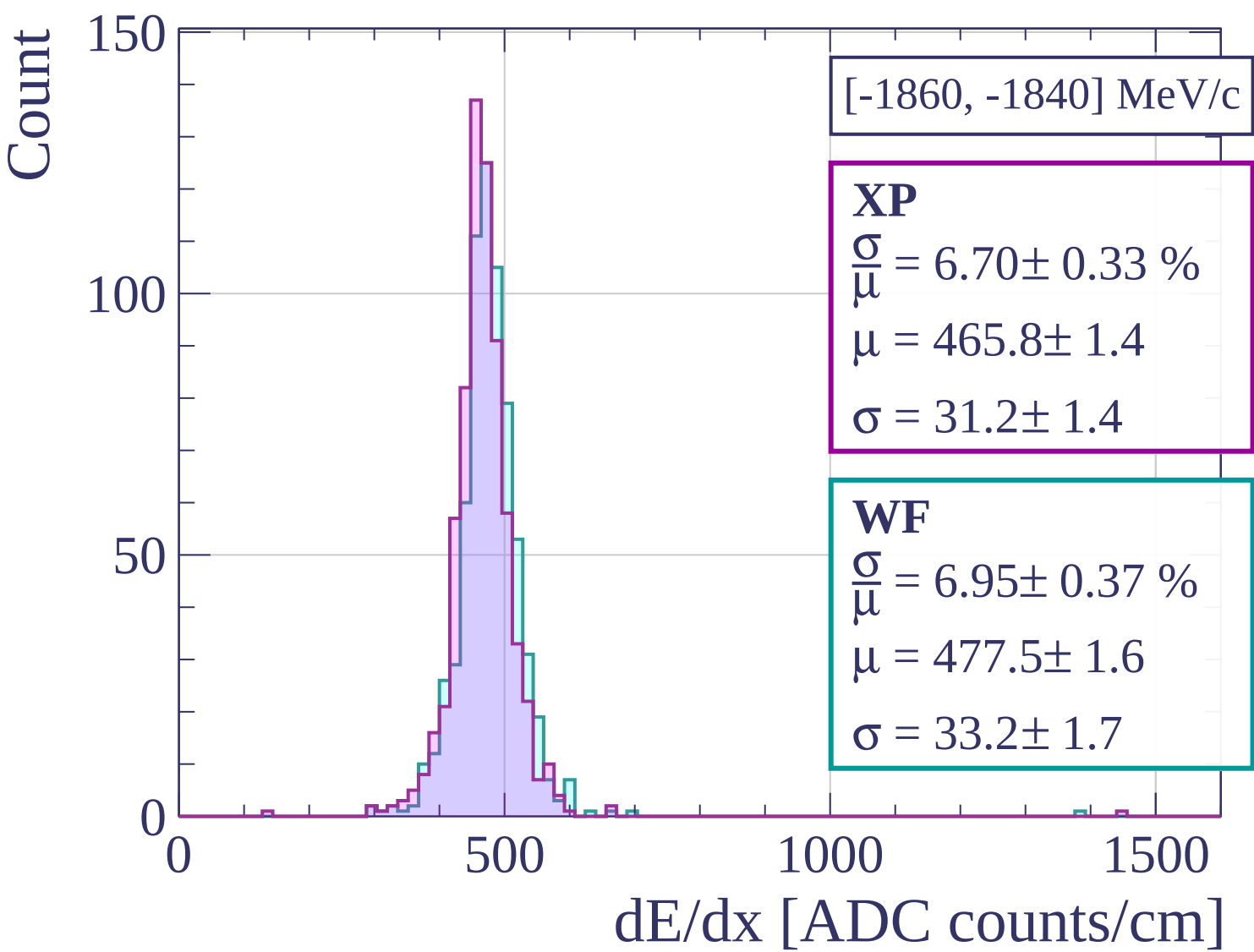
1500

dE/dx [ADC counts/cm]

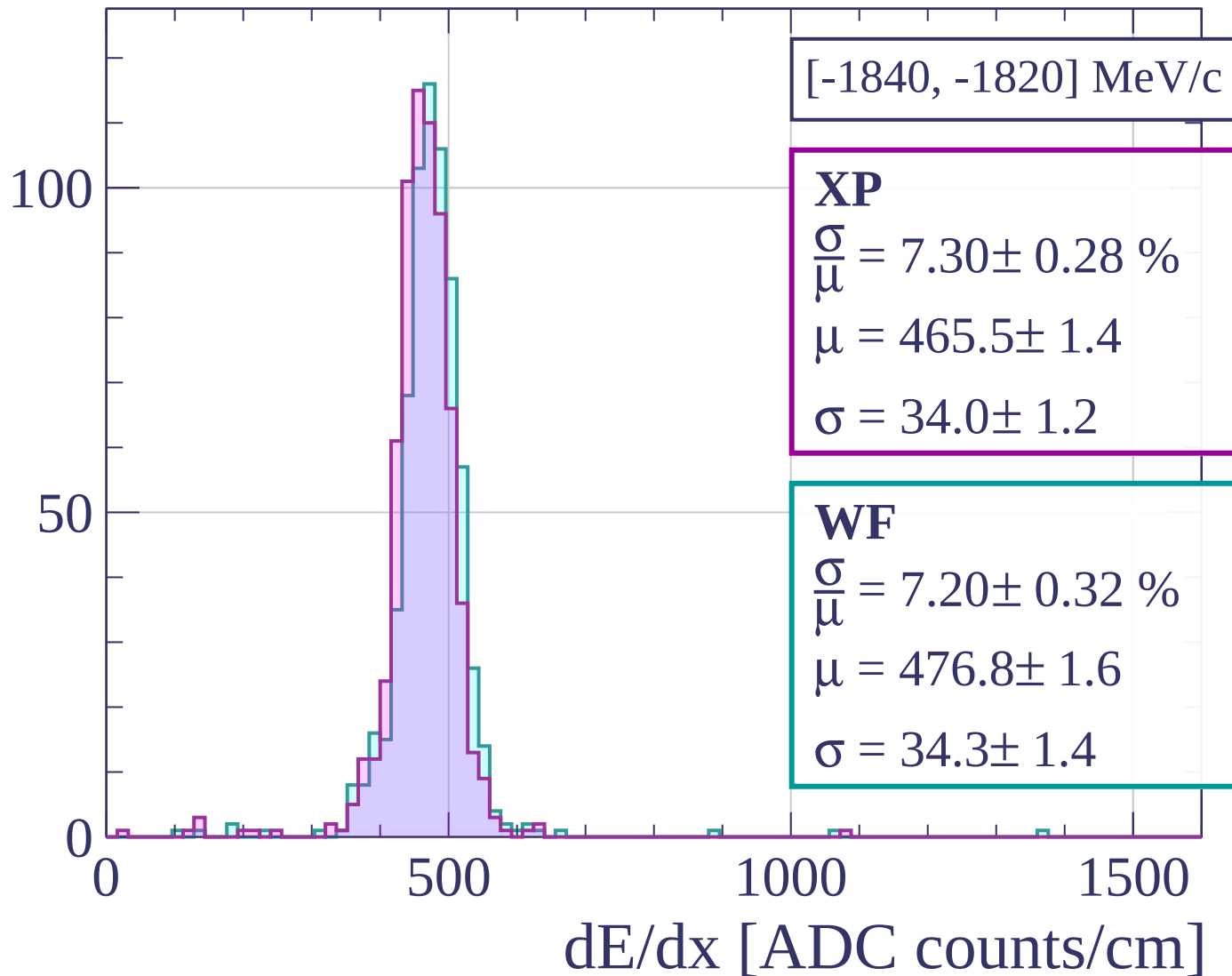


Count

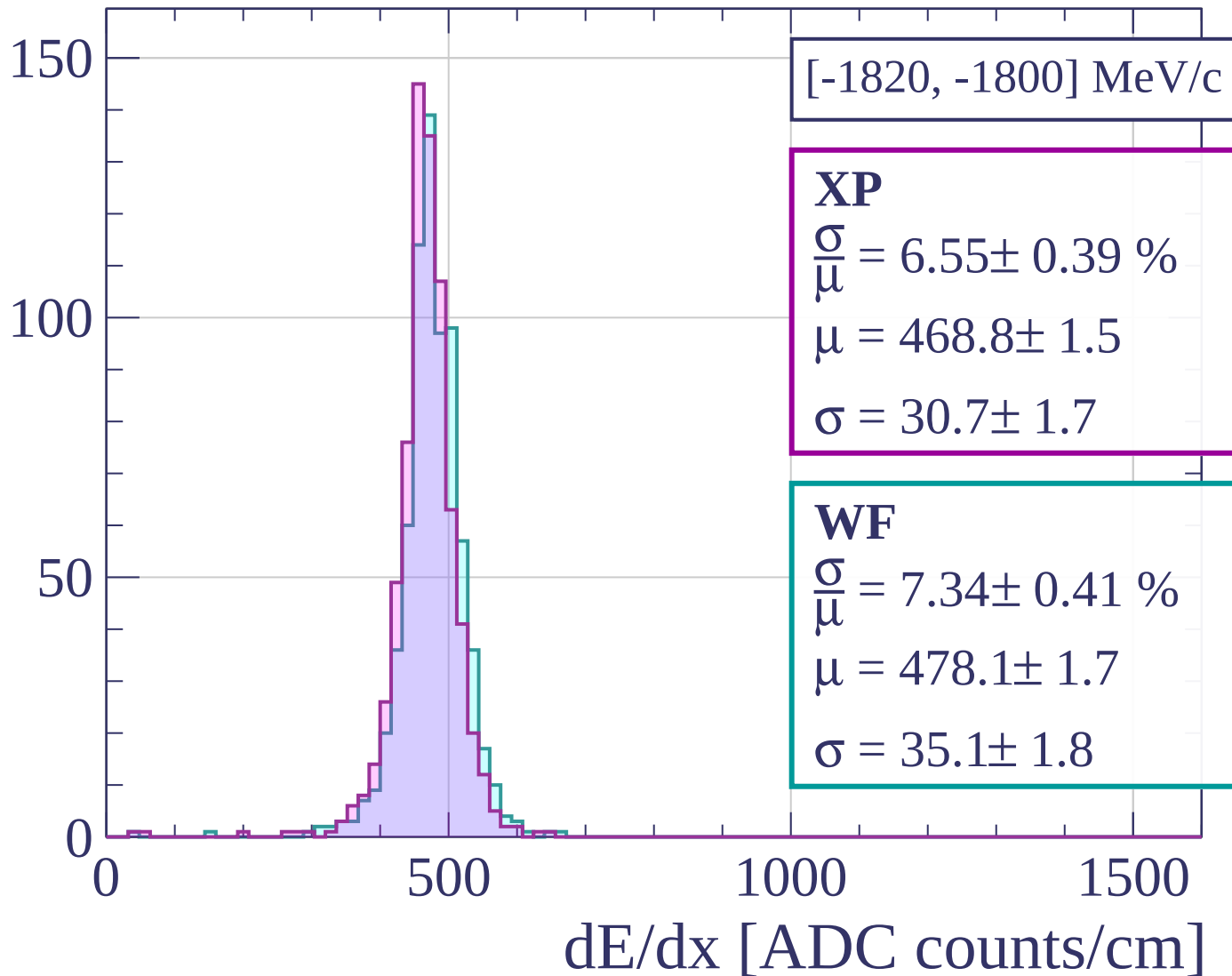


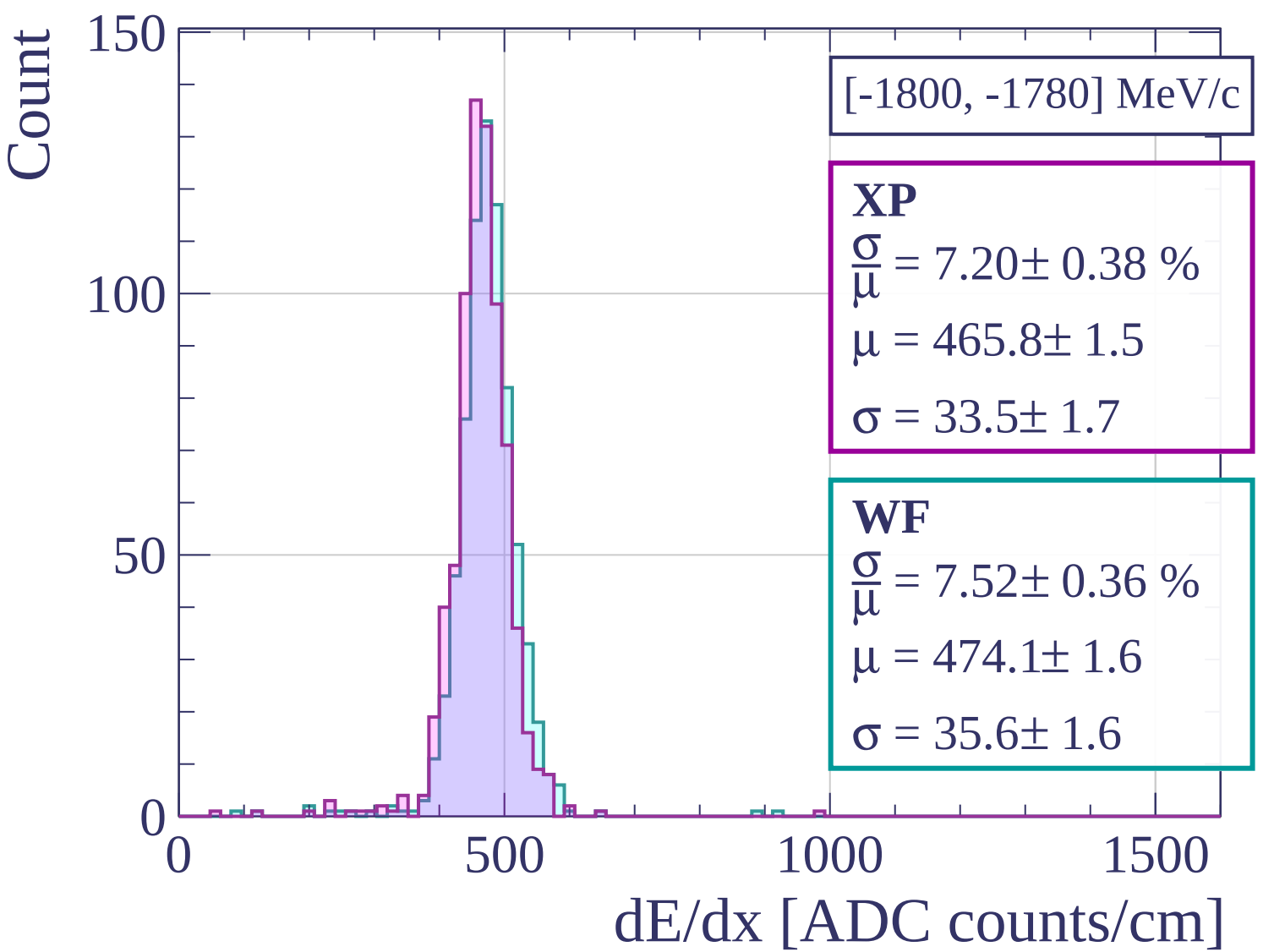


Count

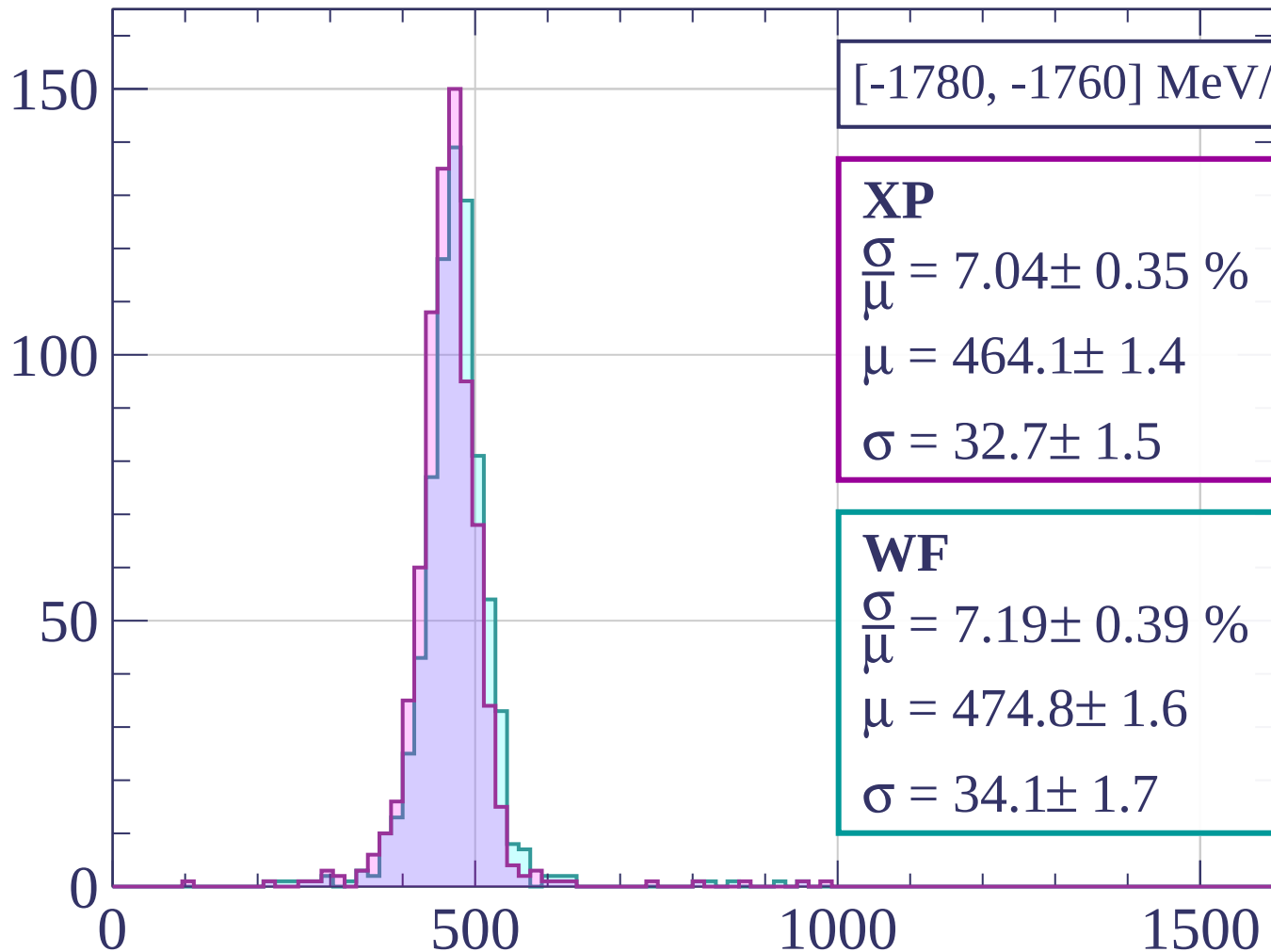


Count





Count



[-1780, -1760] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.04 \pm 0.35 \%$$

$$\mu = 464.1 \pm 1.4$$

$$\sigma = 32.7 \pm 1.5$$

WF

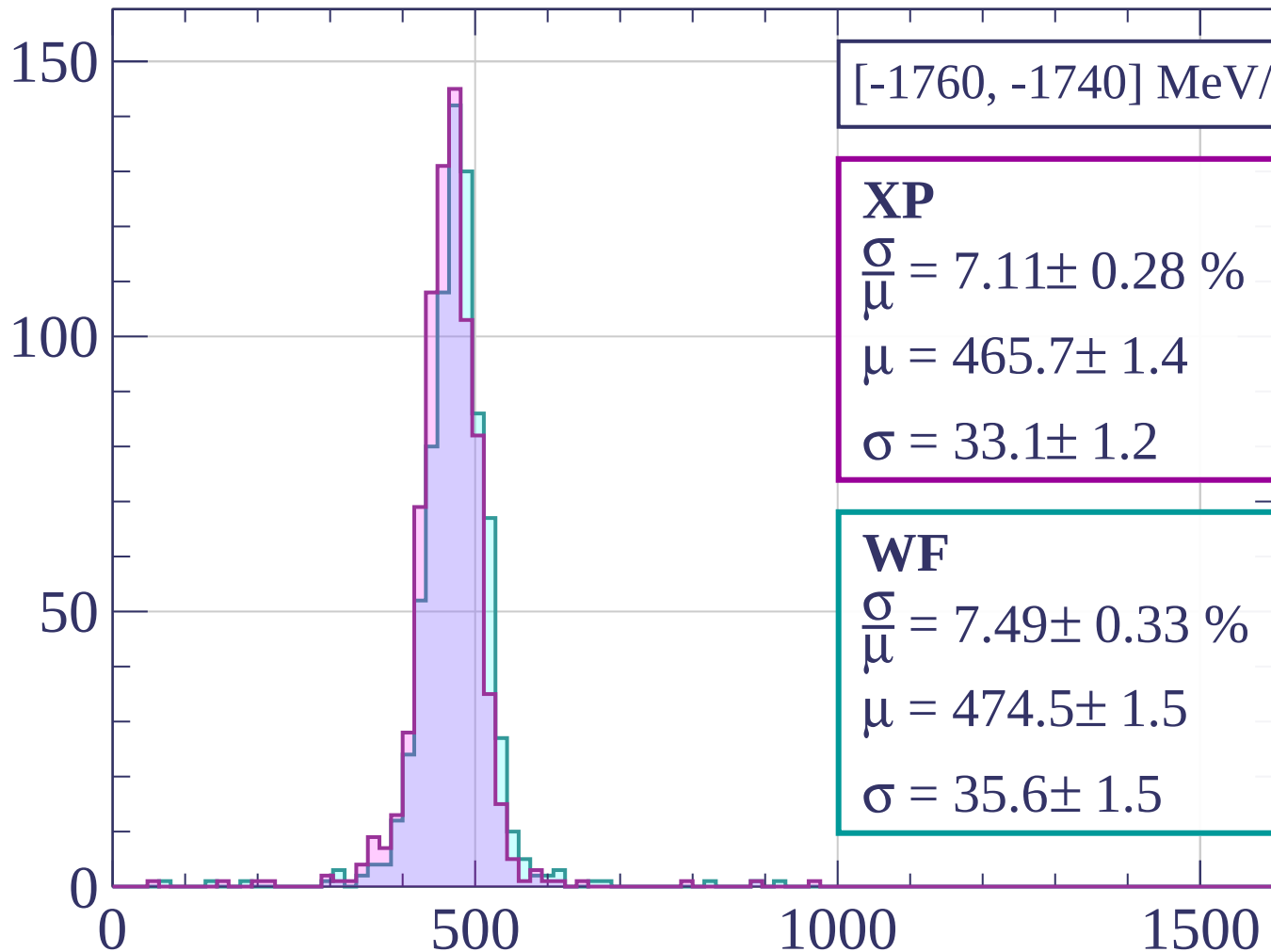
$$\frac{\sigma}{\mu} = 7.19 \pm 0.39 \%$$

$$\mu = 474.8 \pm 1.6$$

$$\sigma = 34.1 \pm 1.7$$

dE/dx [ADC counts/cm]

Count



[-1760, -1740] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.11 \pm 0.28 \%$$

$$\mu = 465.7 \pm 1.4$$

$$\sigma = 33.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.49 \pm 0.33 \%$$

$$\mu = 474.5 \pm 1.5$$

$$\sigma = 35.6 \pm 1.5$$

dE/dx [ADC counts/cm]

Count

[-1740, -1720] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.97 \pm 0.31 \%$$

$$\mu = 464.4 \pm 1.4$$

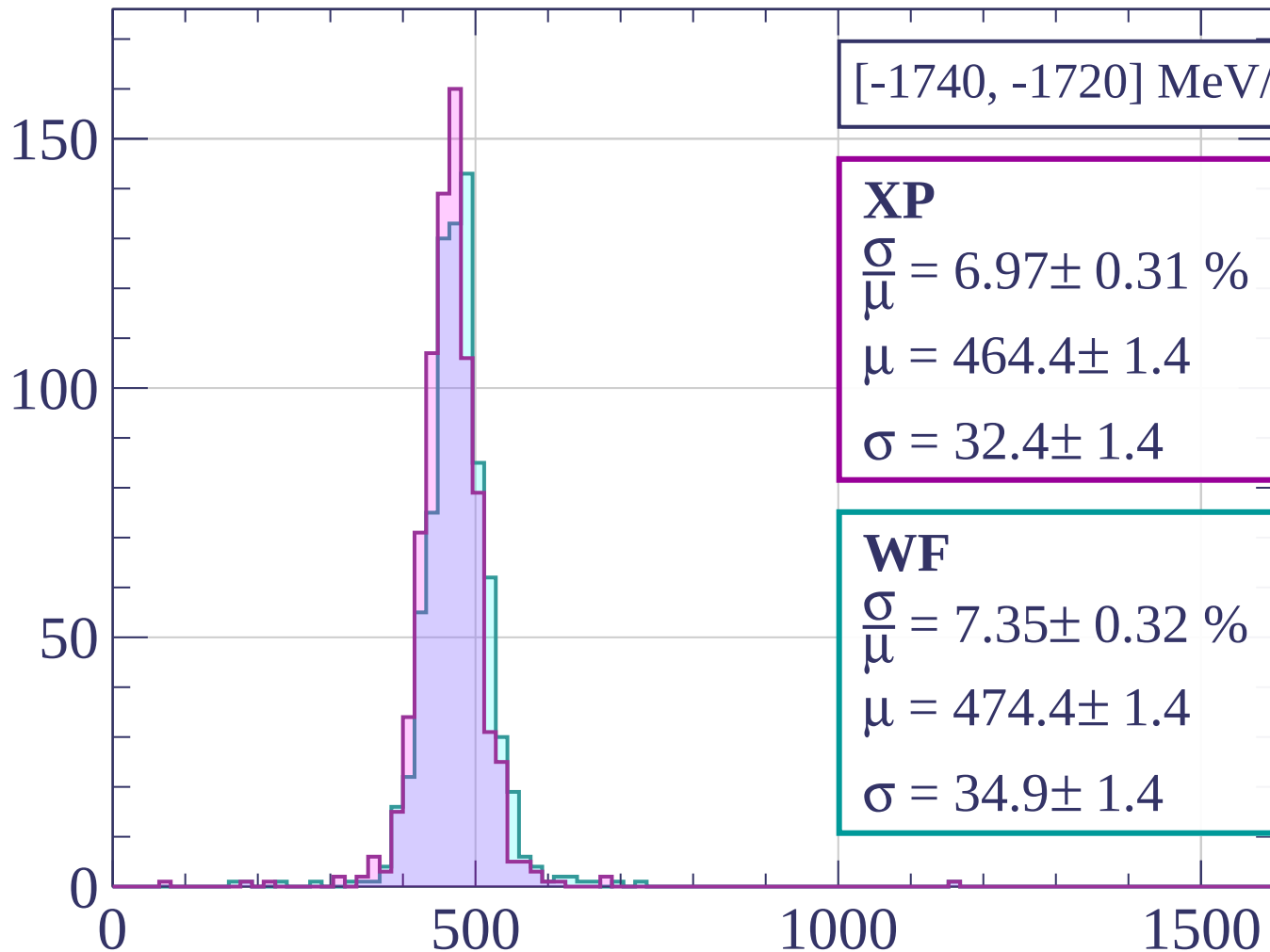
$$\sigma = 32.4 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.35 \pm 0.32 \%$$

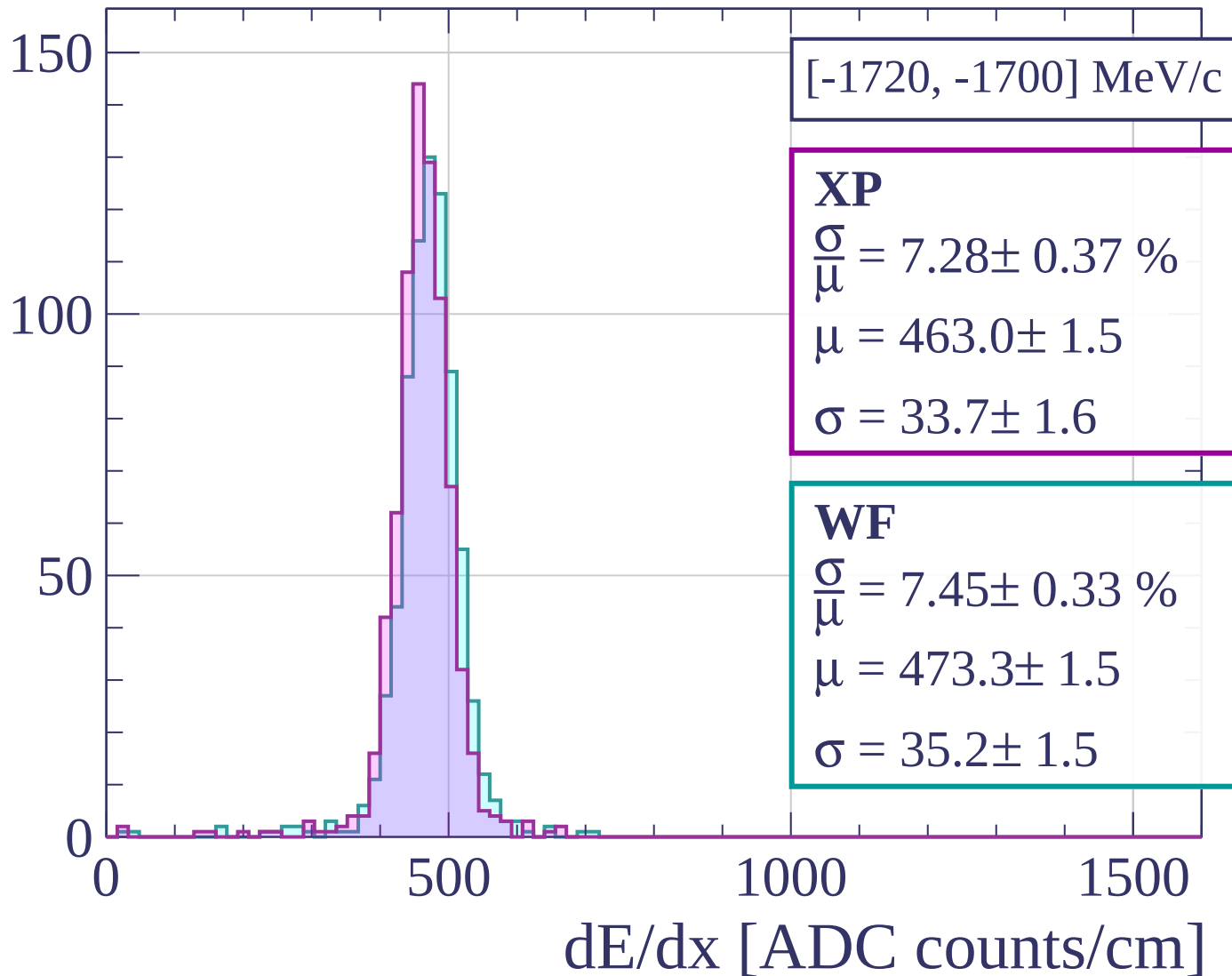
$$\mu = 474.4 \pm 1.4$$

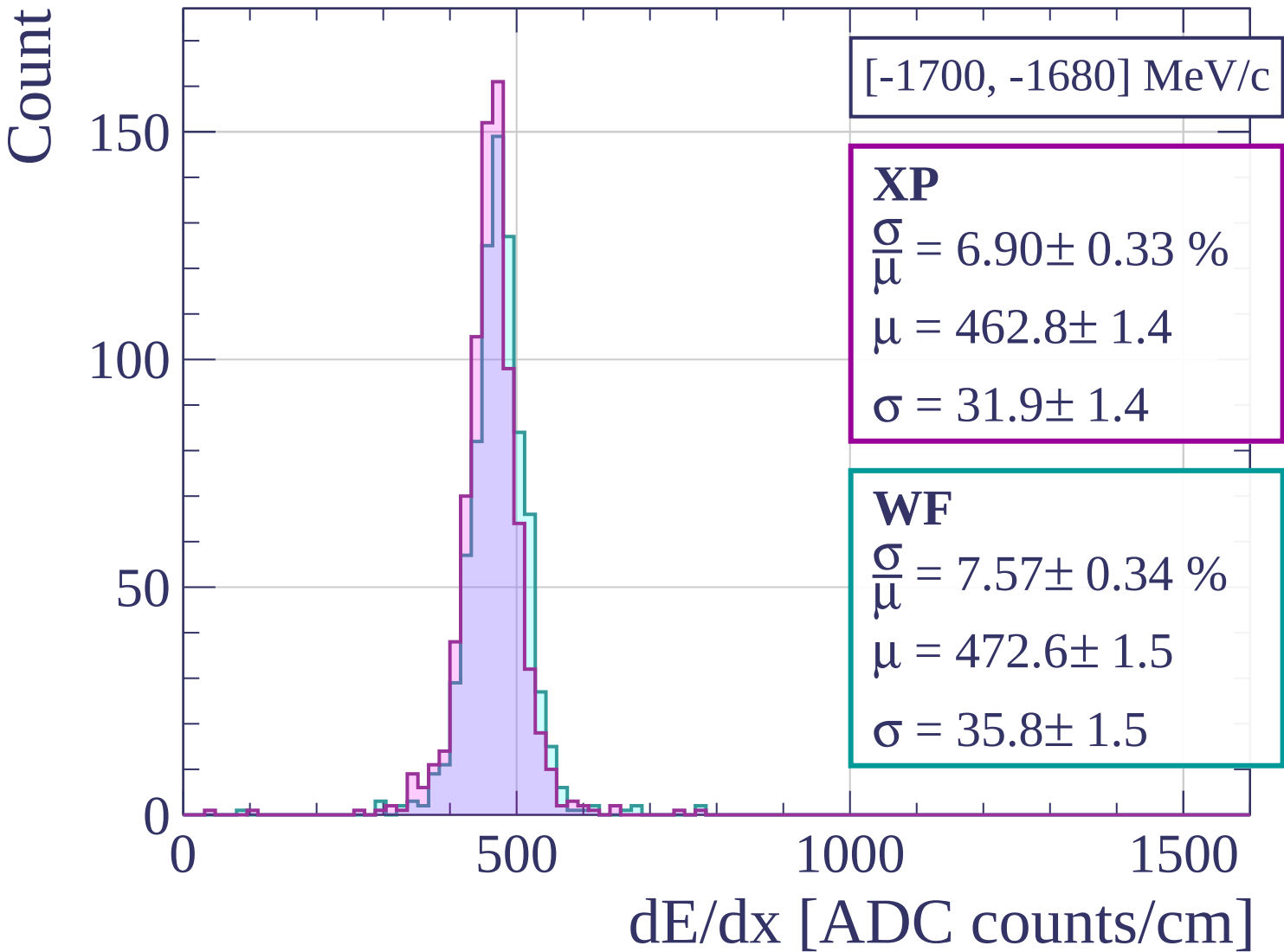
$$\sigma = 34.9 \pm 1.4$$



dE/dx [ADC counts/cm]

Count





Count

[-1680, -1660] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.97 \pm 0.32 \%$$

$$\mu = 463.4 \pm 1.4$$

$$\sigma = 32.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.59 \pm 0.34 \%$$

$$\mu = 472.3 \pm 1.5$$

$$\sigma = 31.1 \pm 1.5$$

150

100

50

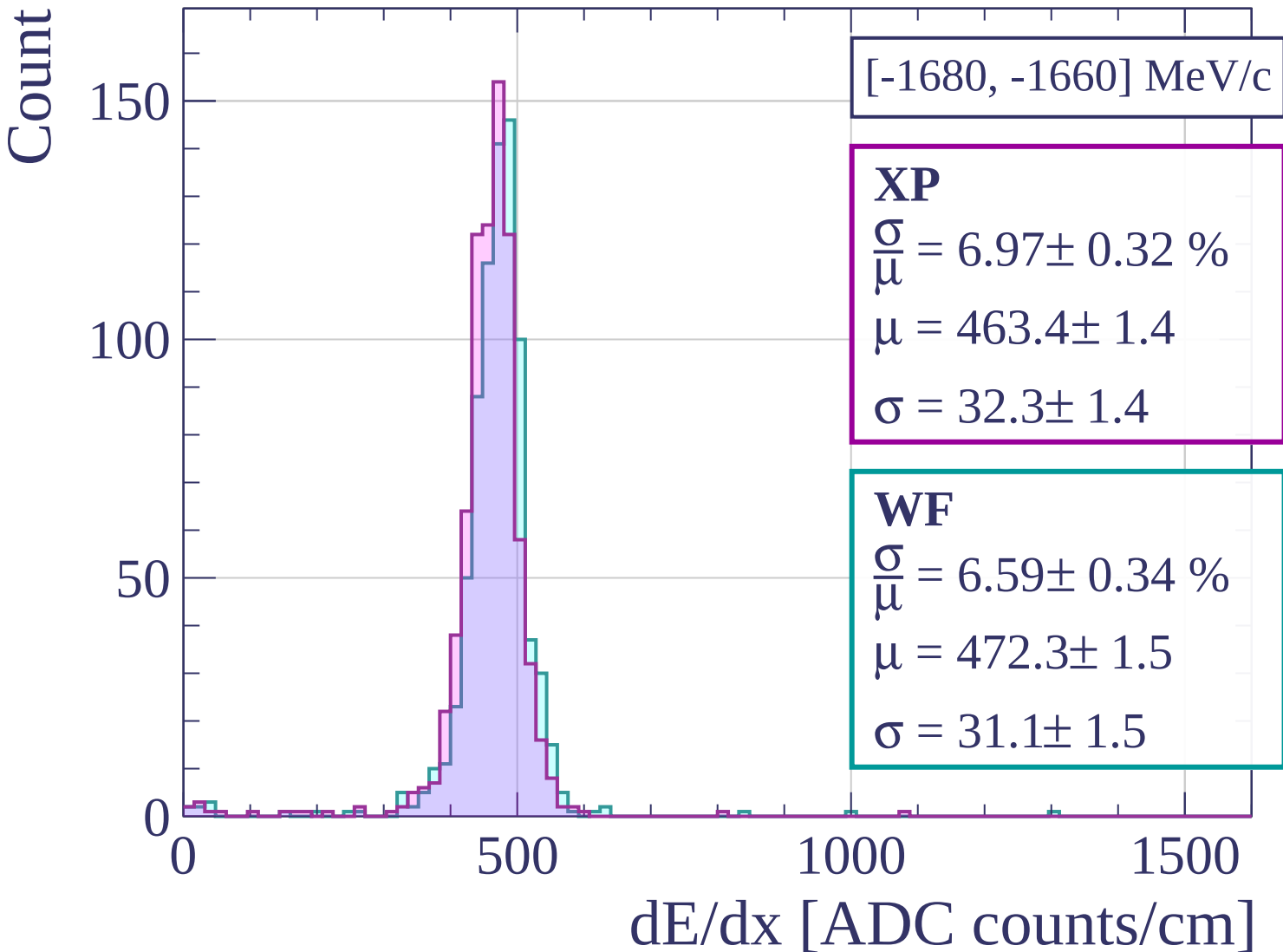
0

500

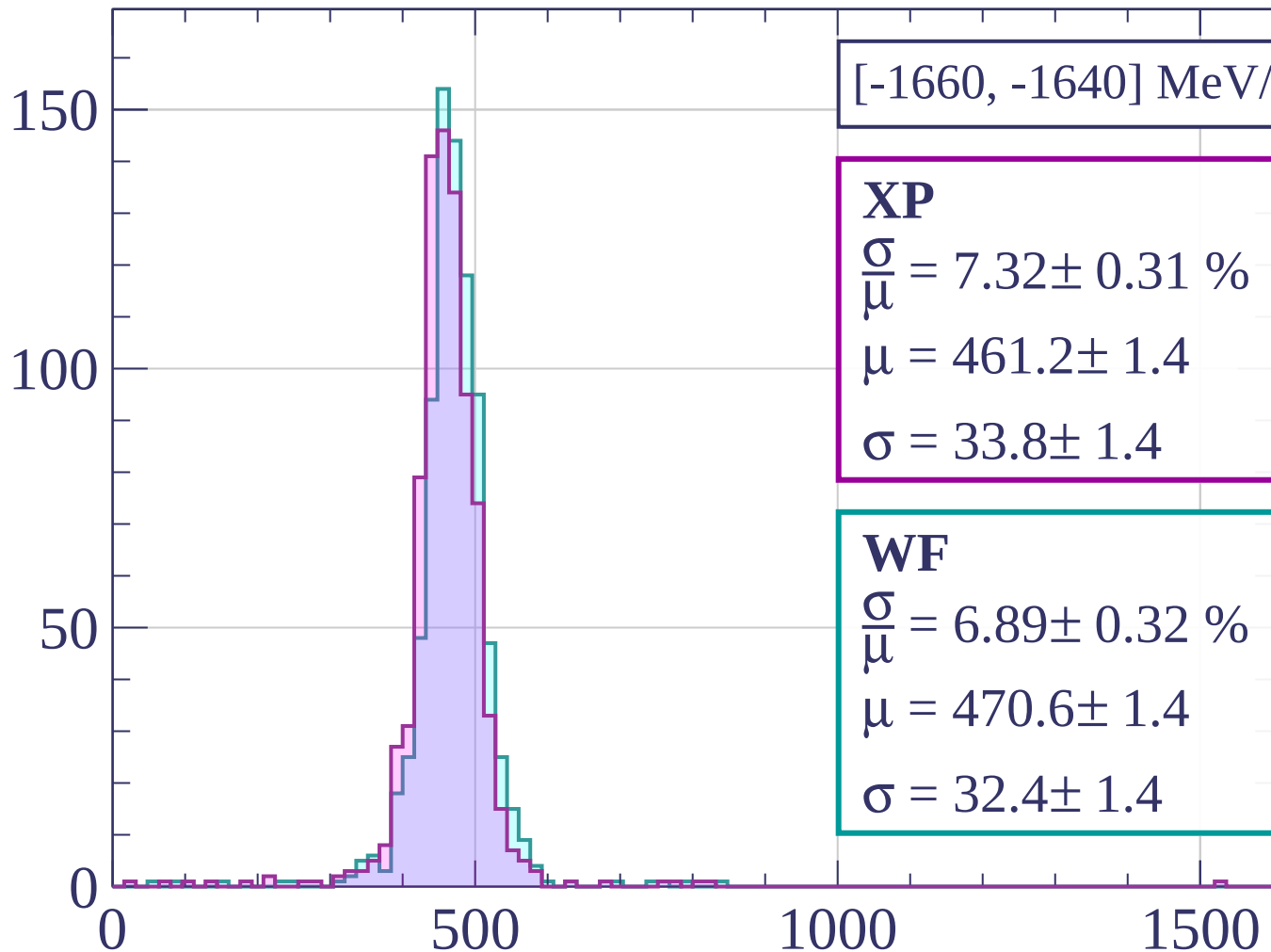
1000

1500

dE/dx [ADC counts/cm]



Count



[-1660, -1640] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.32 \pm 0.31 \%$$

$$\mu = 461.2 \pm 1.4$$

$$\sigma = 33.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.89 \pm 0.32 \%$$

$$\mu = 470.6 \pm 1.4$$

$$\sigma = 32.4 \pm 1.4$$

dE/dx [ADC counts/cm]

Count

[-1640, -1620] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.32 \pm 0.31 \%$$

$$\mu = 458.9 \pm 1.3$$

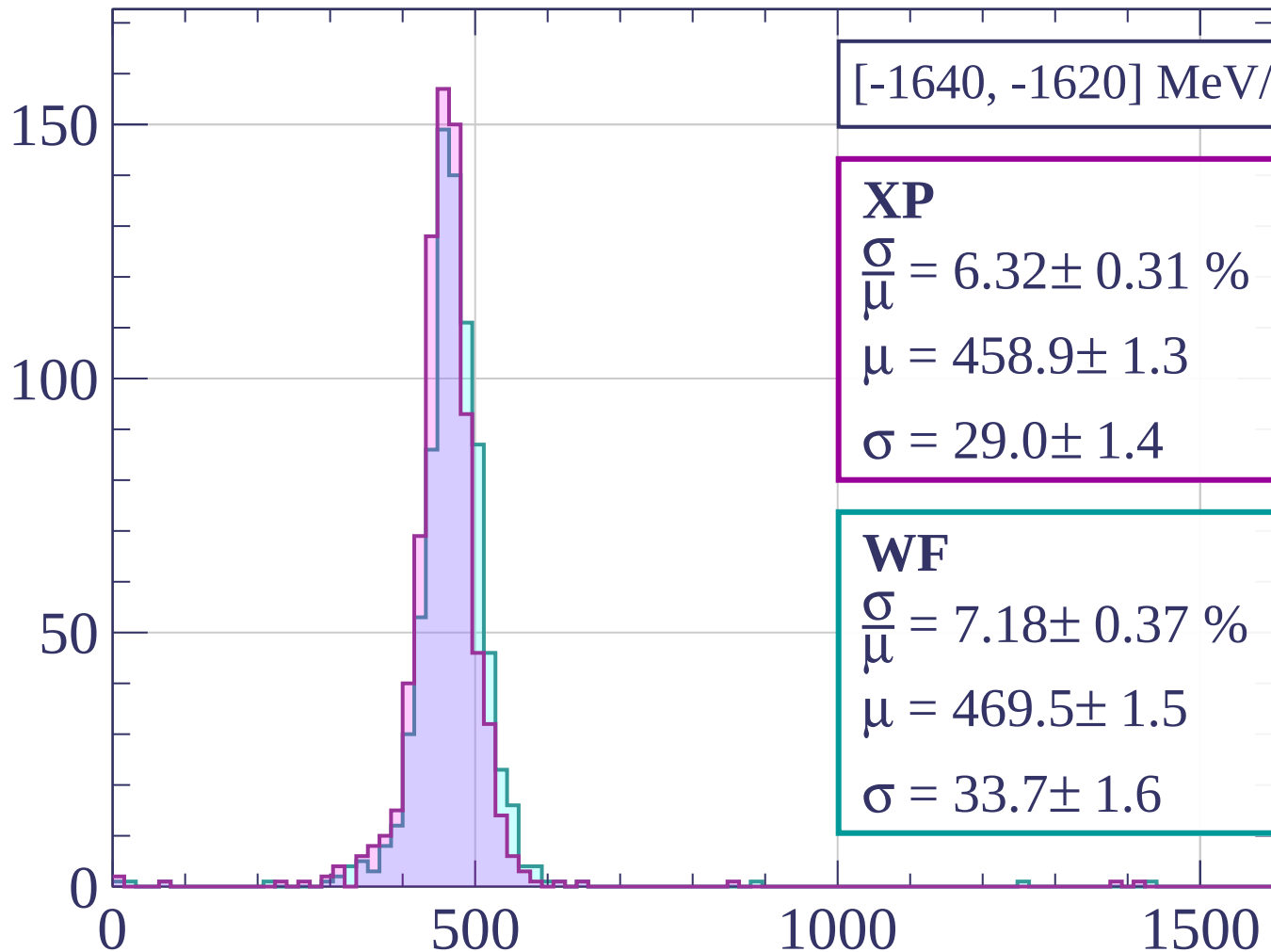
$$\sigma = 29.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.18 \pm 0.37 \%$$

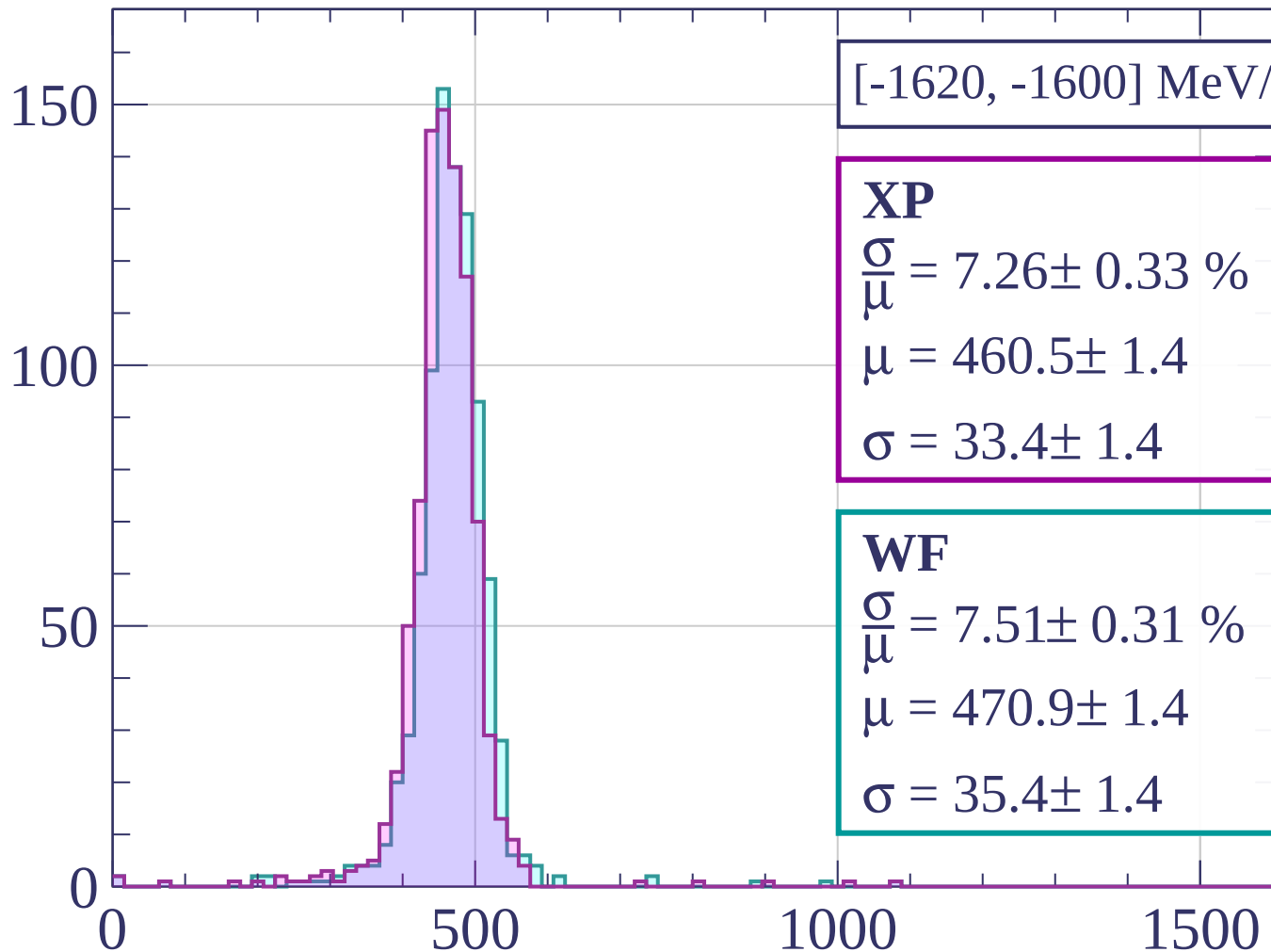
$$\mu = 469.5 \pm 1.5$$

$$\sigma = 33.7 \pm 1.6$$



dE/dx [ADC counts/cm]

Count



[-1620, -1600] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.26 \pm 0.33 \%$$

$$\mu = 460.5 \pm 1.4$$

$$\sigma = 33.4 \pm 1.4$$

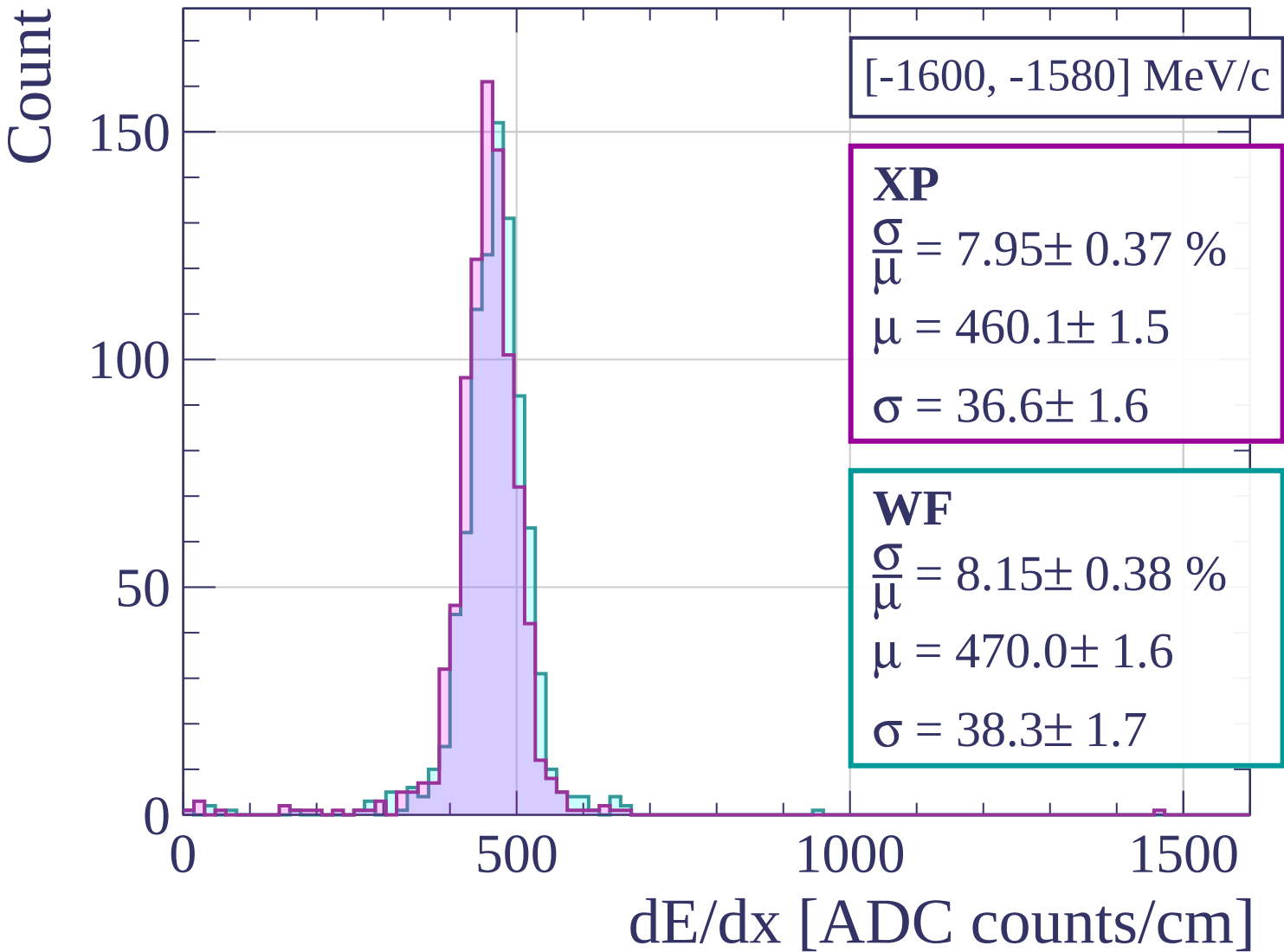
WF

$$\frac{\sigma}{\mu} = 7.51 \pm 0.31 \%$$

$$\mu = 470.9 \pm 1.4$$

$$\sigma = 35.4 \pm 1.4$$

dE/dx [ADC counts/cm]



Count

[-1580, -1560] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.09 \pm 0.33 \%$$

$$\mu = 458.5 \pm 1.4$$

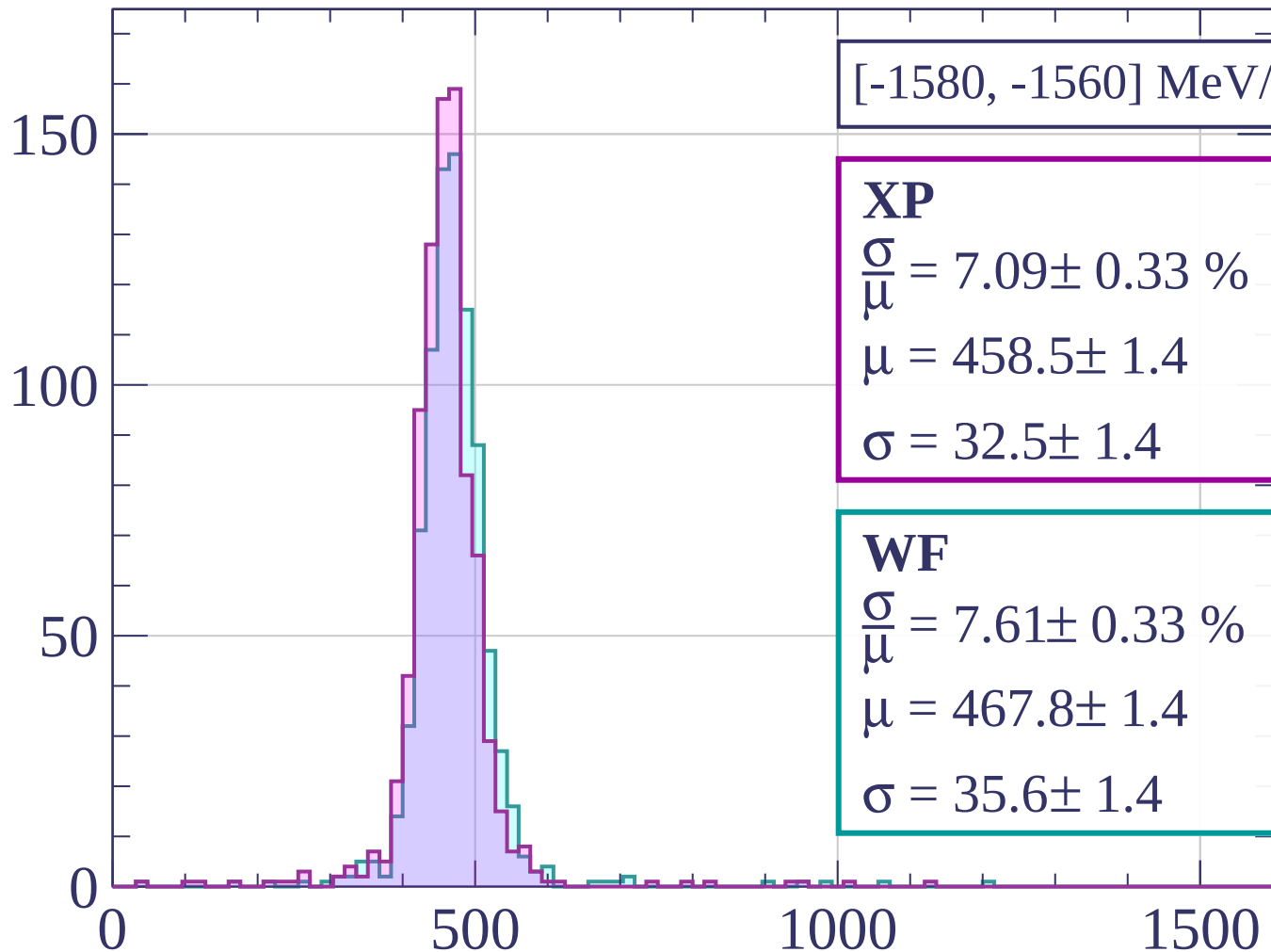
$$\sigma = 32.5 \pm 1.4$$

WF

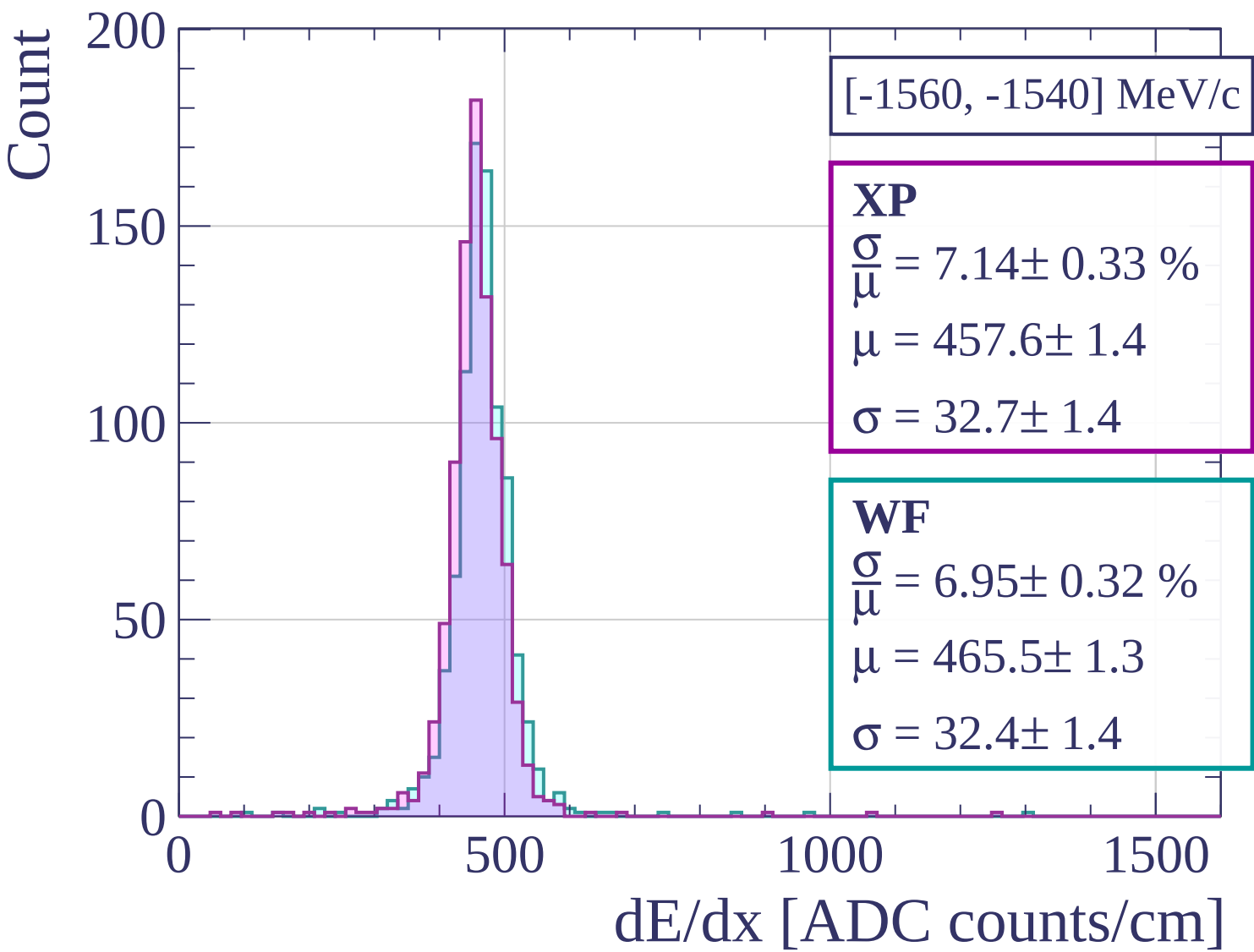
$$\frac{\sigma}{\mu} = 7.61 \pm 0.33 \%$$

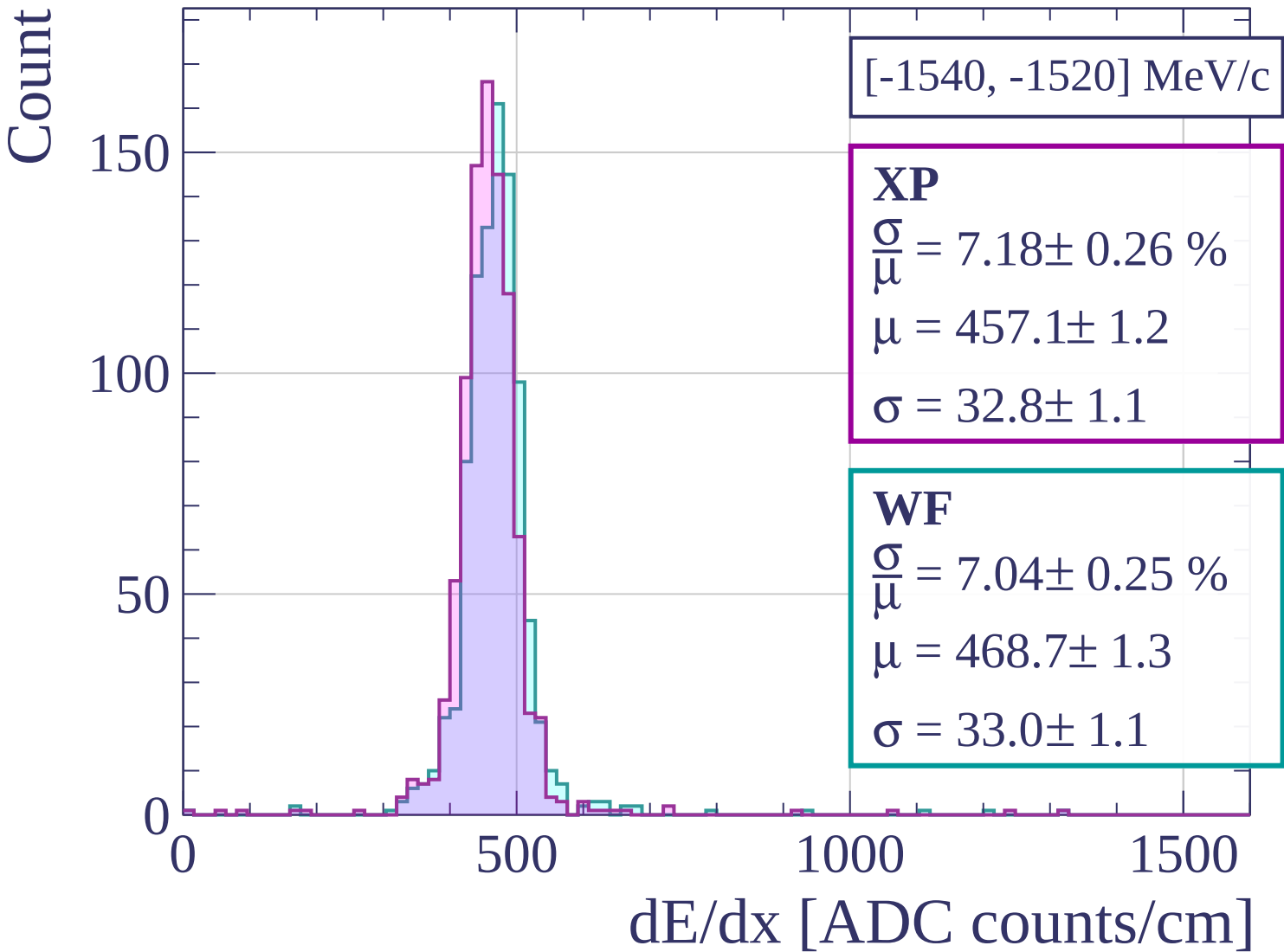
$$\mu = 467.8 \pm 1.4$$

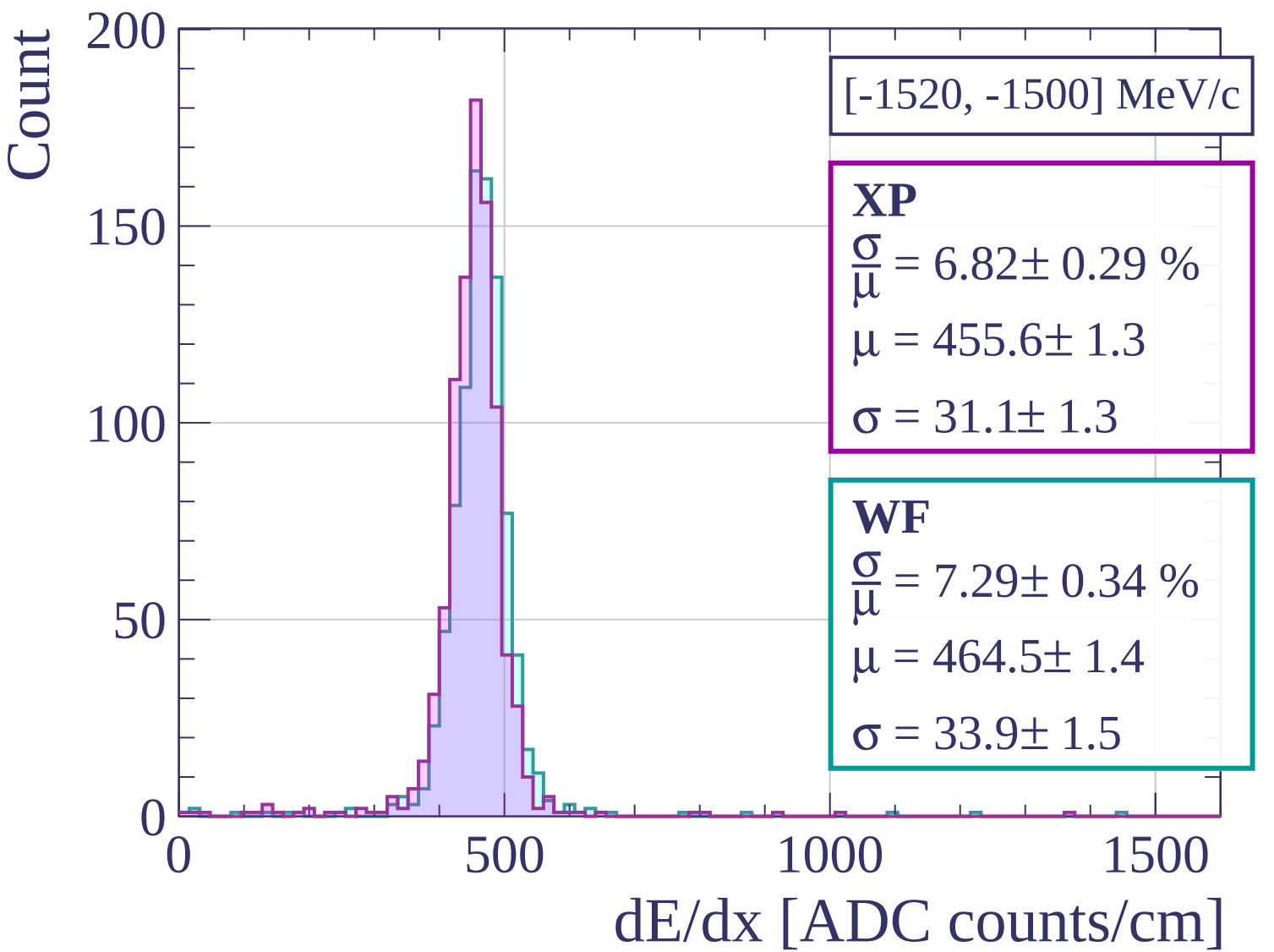
$$\sigma = 35.6 \pm 1.4$$



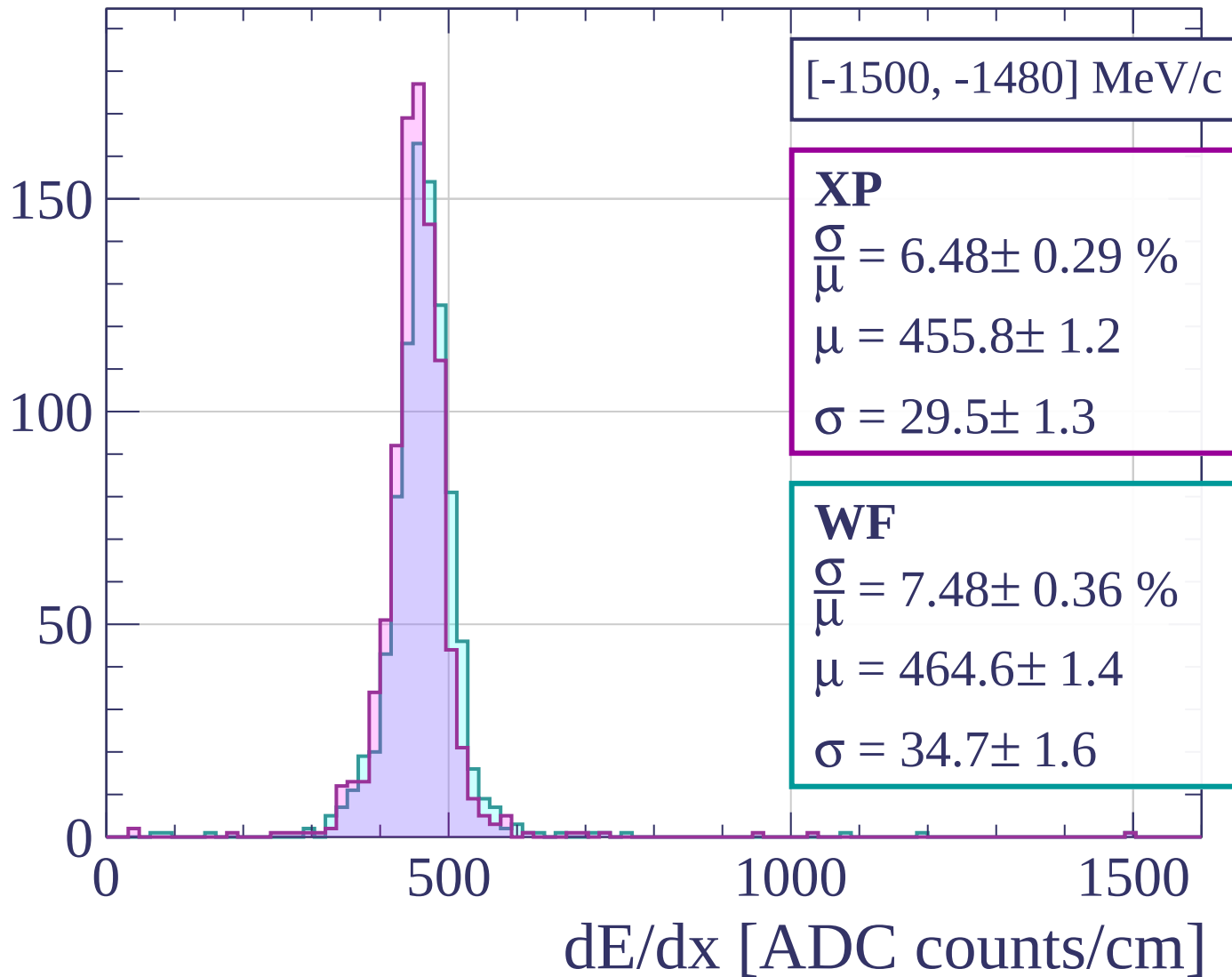
dE/dx [ADC counts/cm]







Count



Count

[-1480, -1460] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.44 \pm 0.27 \%$$

$$\mu = 455.0 \pm 1.2$$

$$\sigma = 33.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.67 \pm 0.36 \%$$

$$\mu = 465.3 \pm 1.4$$

$$\sigma = 35.7 \pm 1.6$$

150

100

50

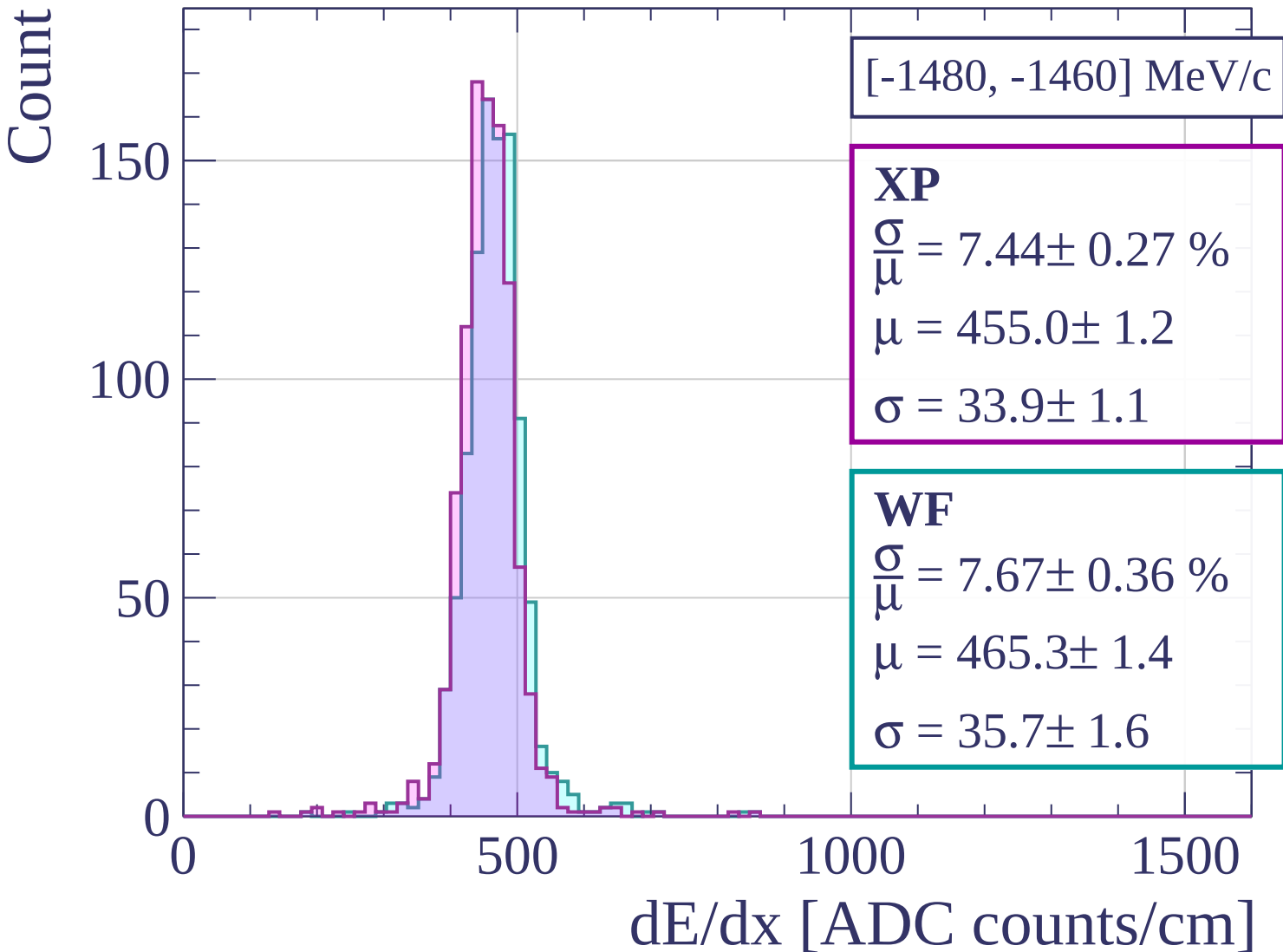
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1460, -1440] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.73 \pm 0.33 \%$$

$$\mu = 454.8 \pm 1.3$$

$$\sigma = 30.6 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.18 \pm 0.32 \%$$

$$\mu = 464.2 \pm 1.4$$

$$\sigma = 33.3 \pm 1.4$$

150

100

50

0

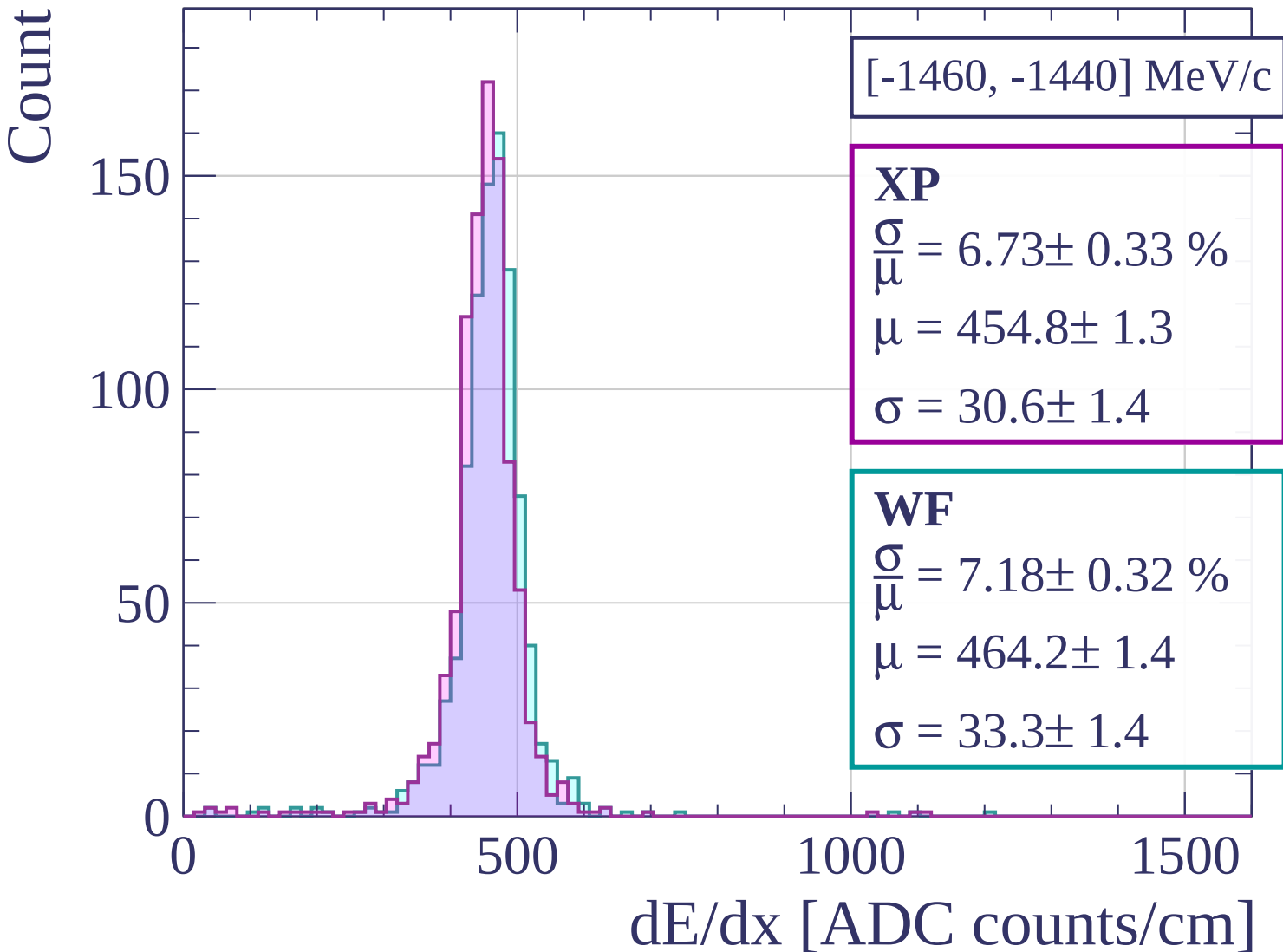
0

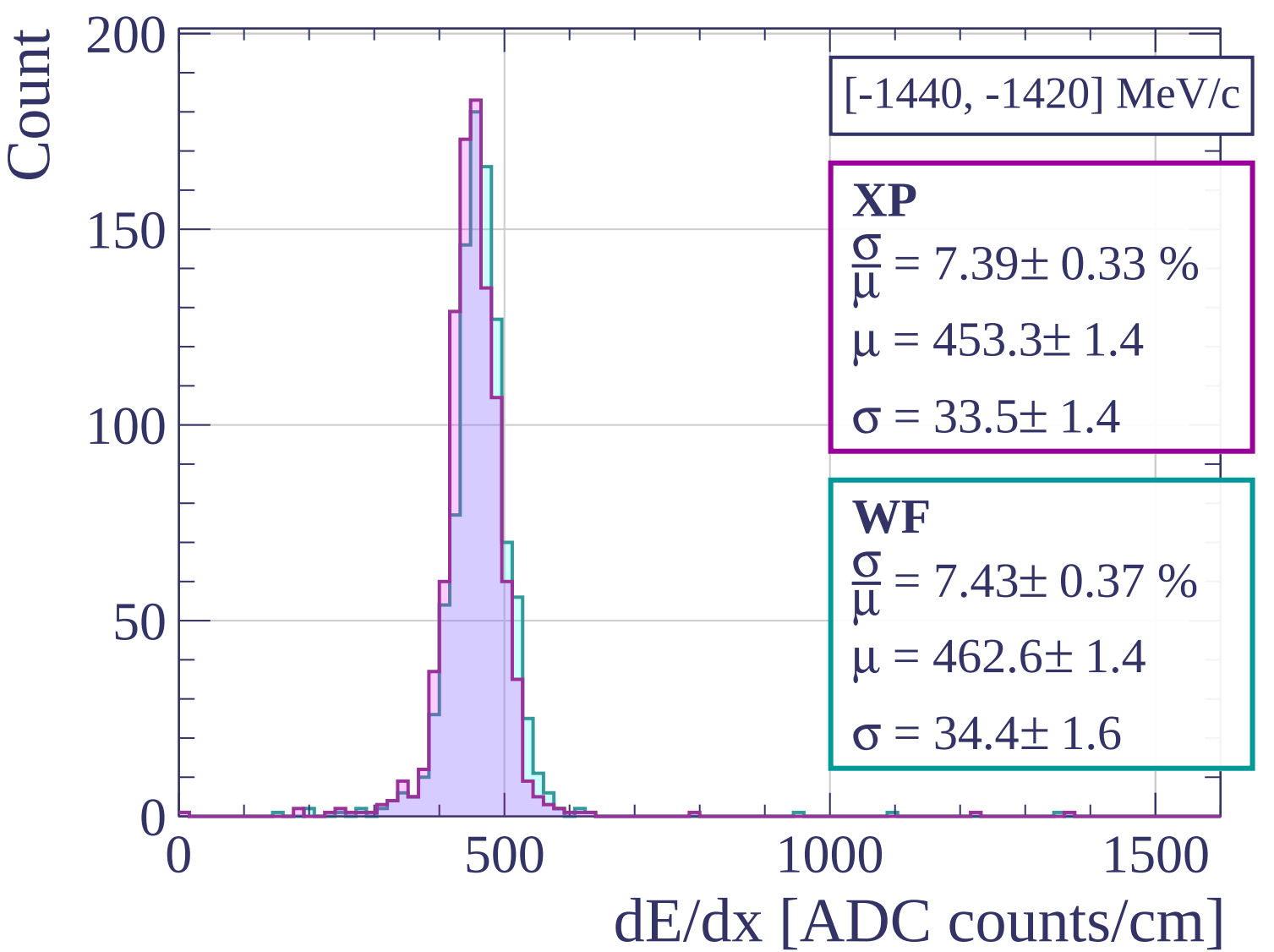
500

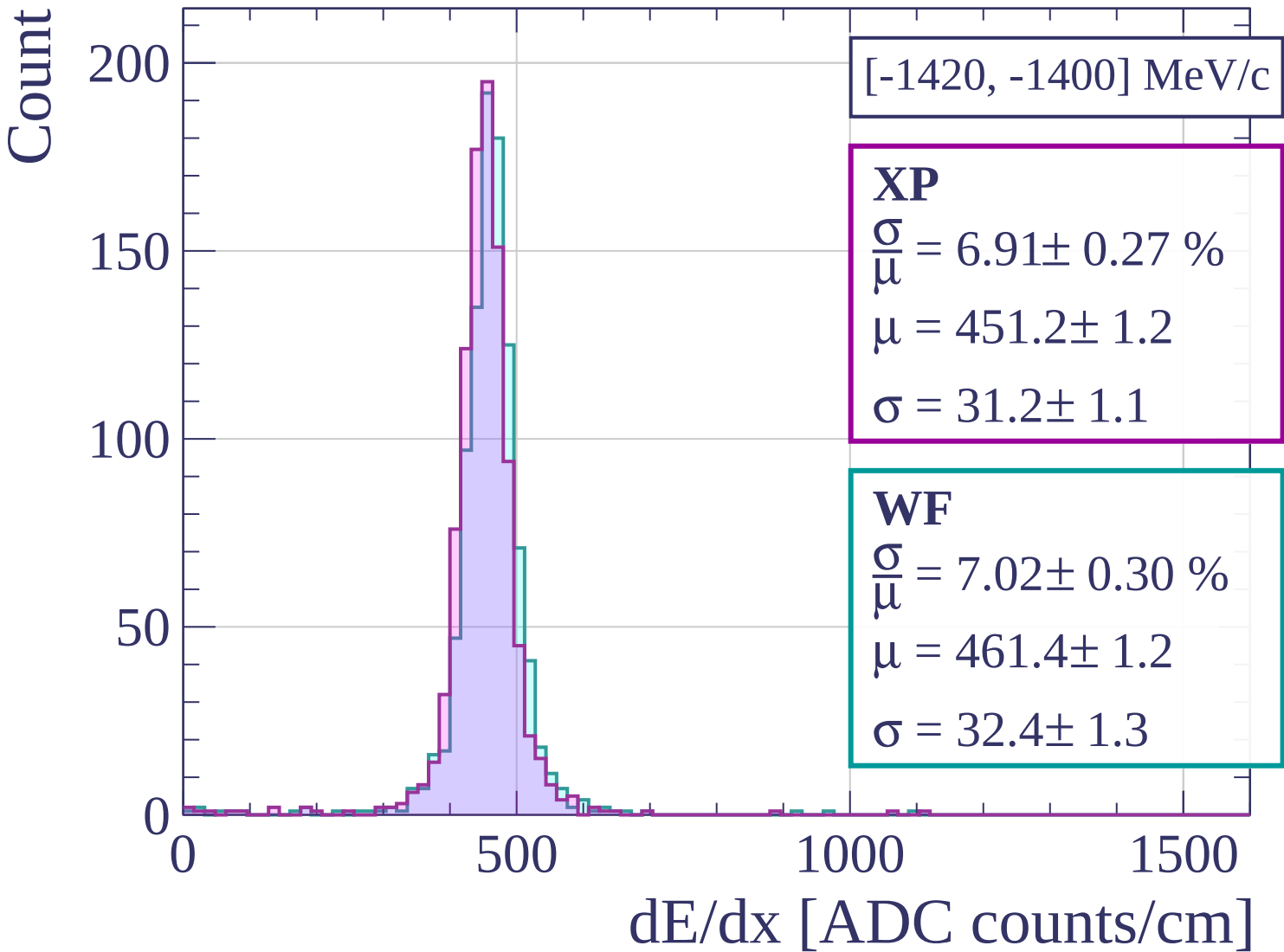
1000

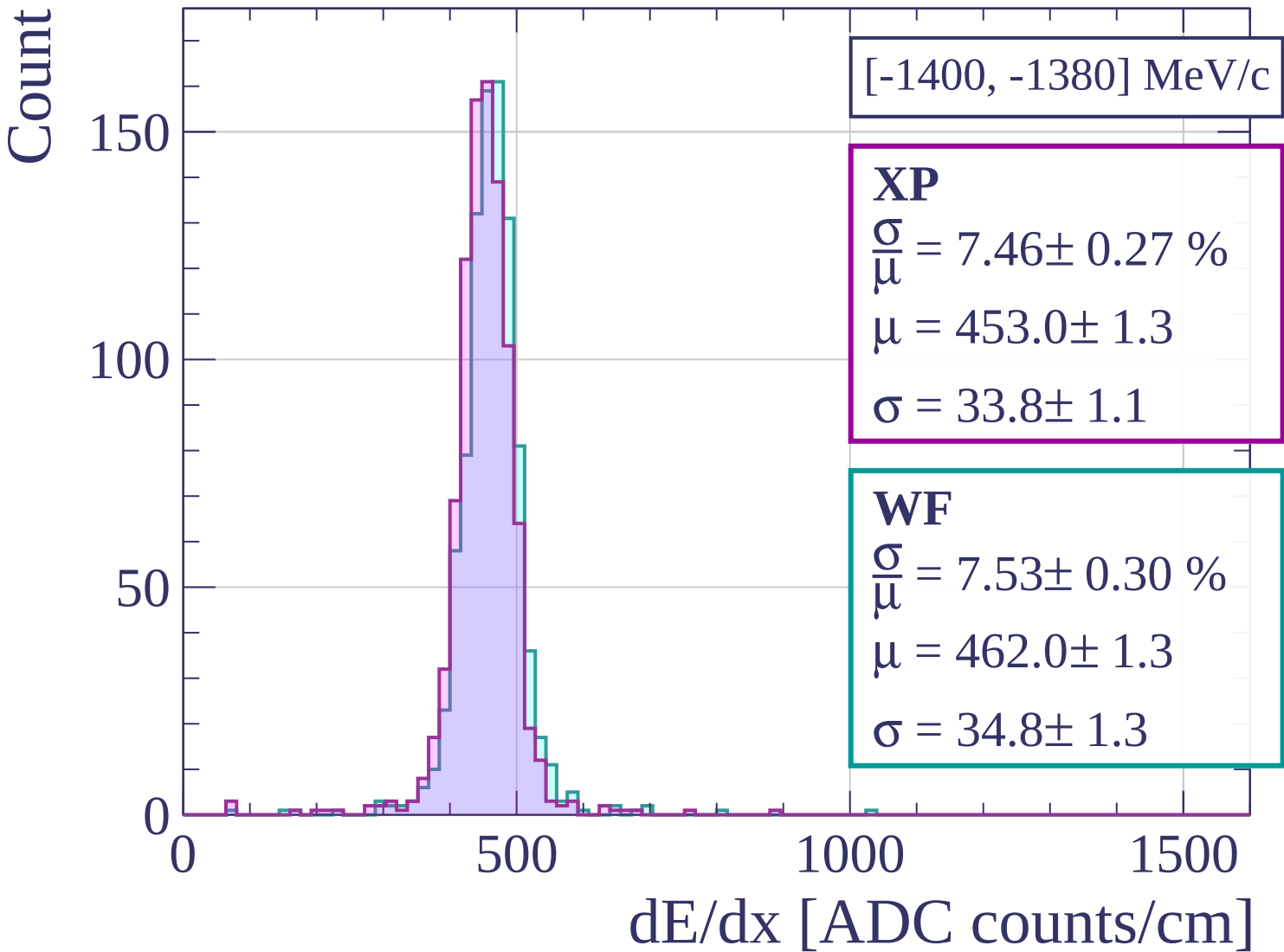
1500

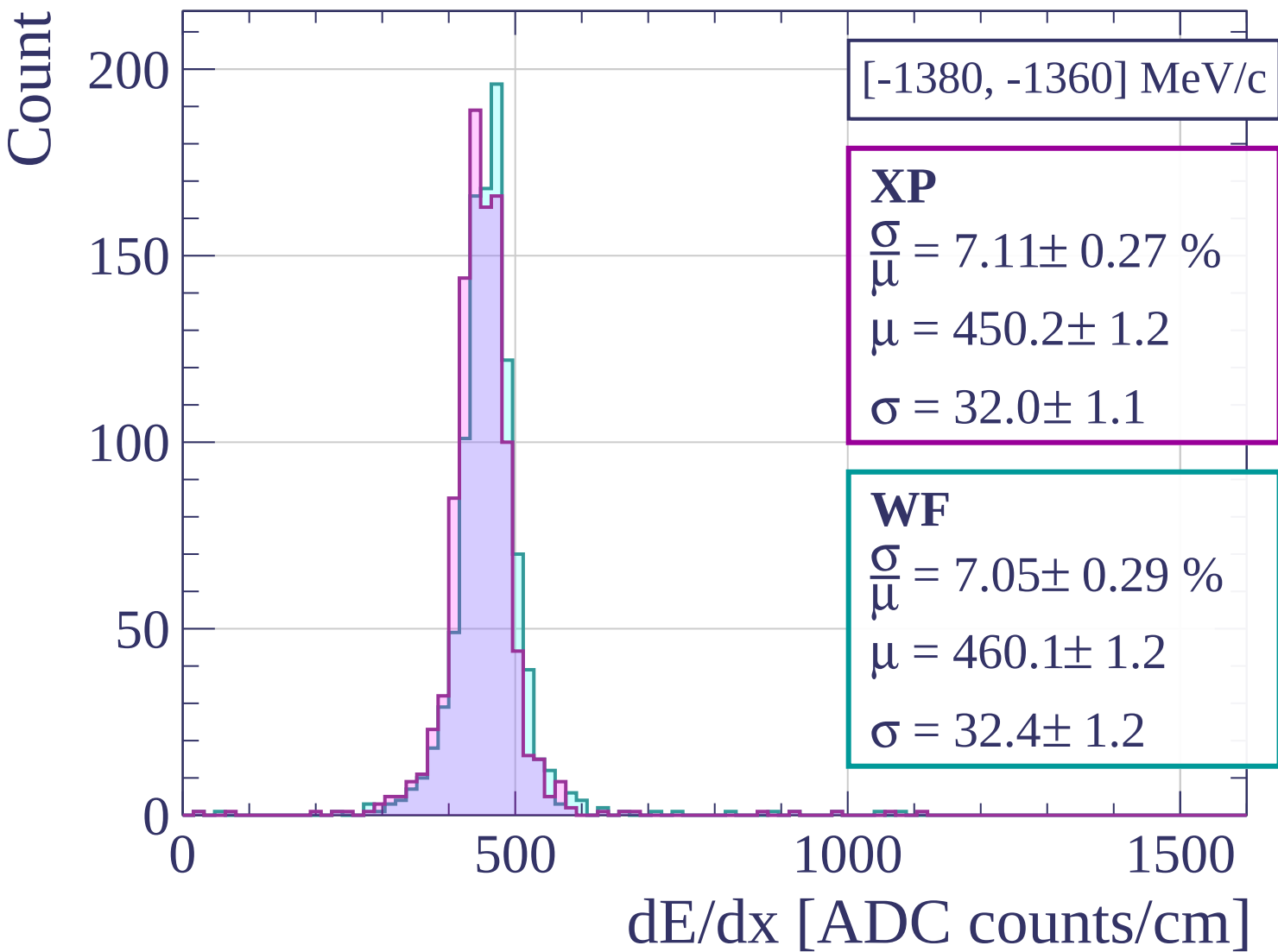
dE/dx [ADC counts/cm]











Count

[-1360, -1340] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.59 \pm 0.30 \%$$

$$\mu = 450.7 \pm 1.2$$

$$\sigma = 29.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.74 \pm 0.26 \%$$

$$\mu = 461.1 \pm 1.2$$

$$\sigma = 31.1 \pm 1.1$$

200

150

100

50

0

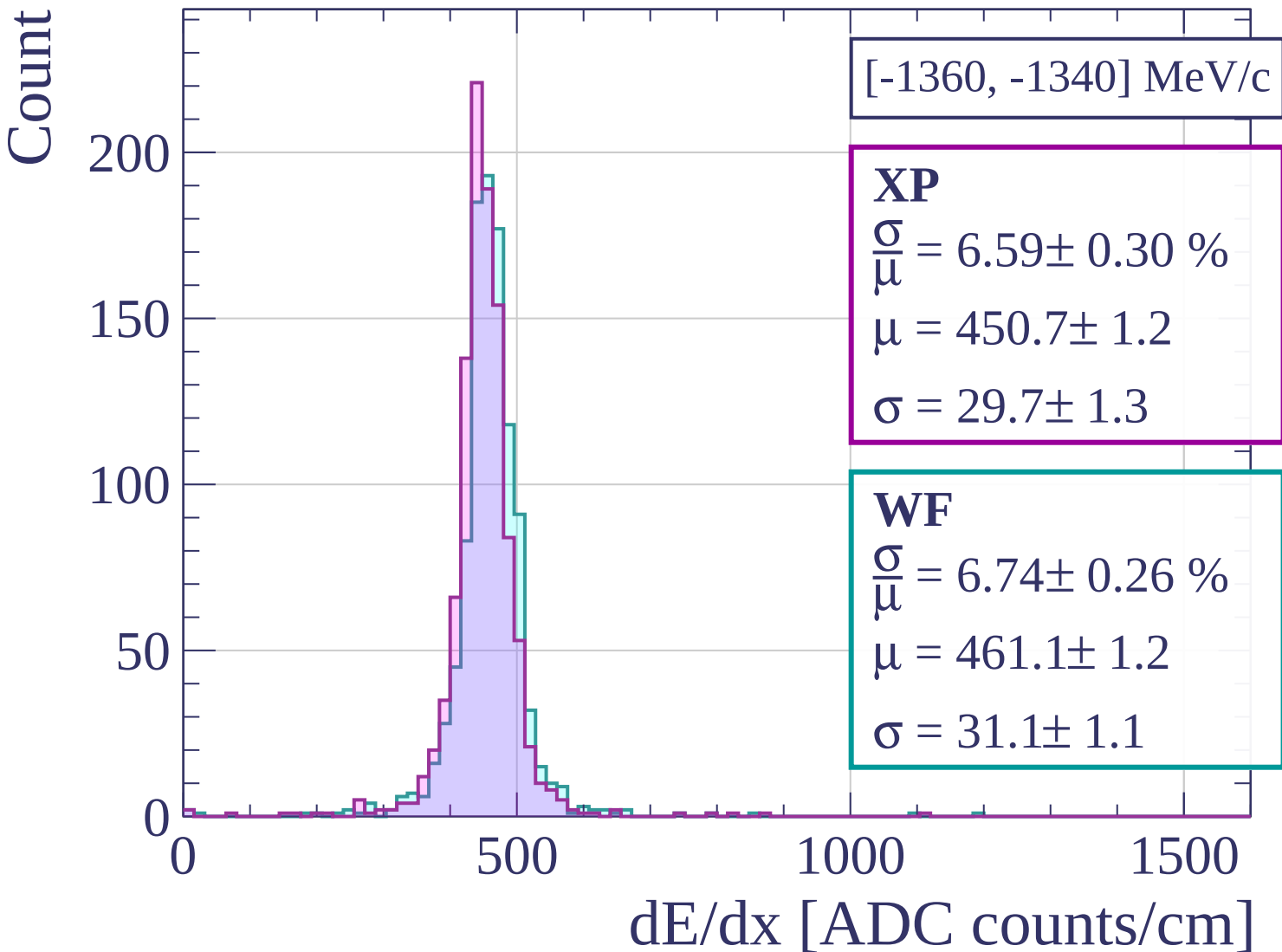
0

500

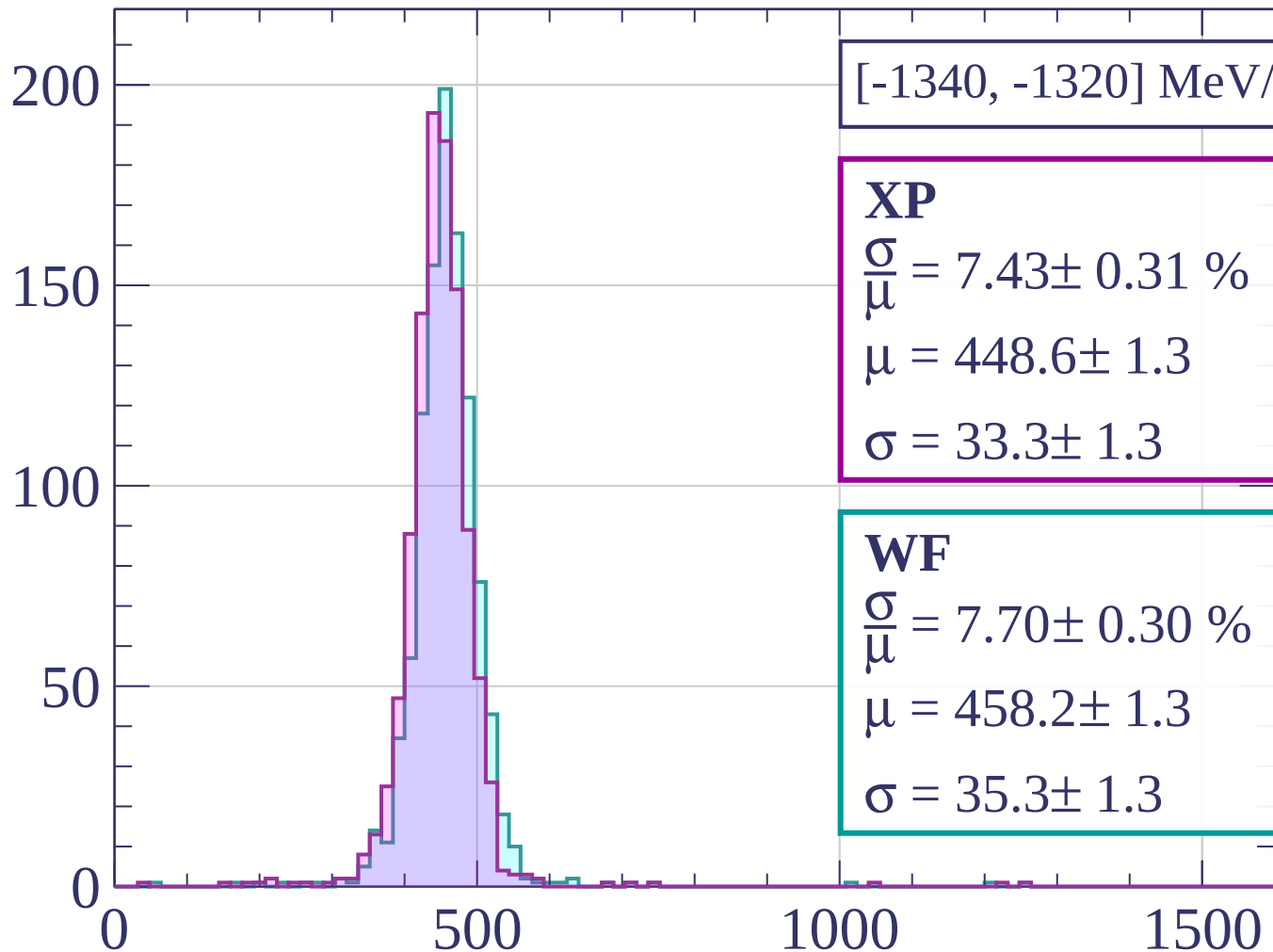
1000

1500

dE/dx [ADC counts/cm]



Count



[-1340, -1320] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.43 \pm 0.31 \%$$

$$\mu = 448.6 \pm 1.3$$

$$\sigma = 33.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.70 \pm 0.30 \%$$

$$\mu = 458.2 \pm 1.3$$

$$\sigma = 35.3 \pm 1.3$$

dE/dx [ADC counts/cm]

Count

[-1320, -1300] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.20 \pm 0.30 \%$$

$$\mu = 450.0 \pm 1.2$$

$$\sigma = 32.4 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.77 \pm 0.31 \%$$

$$\mu = 459.6 \pm 1.3$$

$$\sigma = 35.7 \pm 1.3$$

200

150

100

50

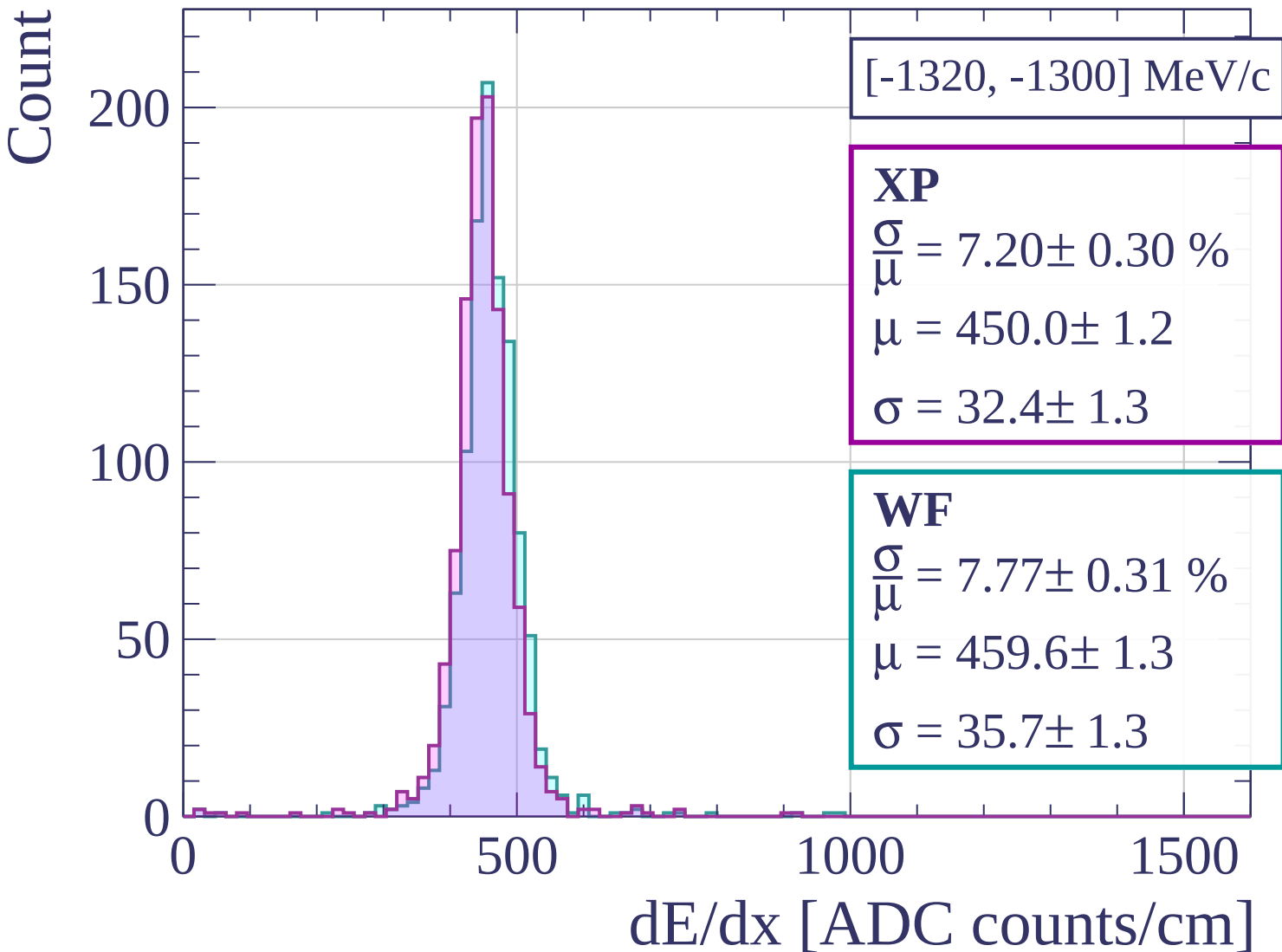
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1300, -1280] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.32 \%$$

$$\mu = 446.8 \pm 1.2$$

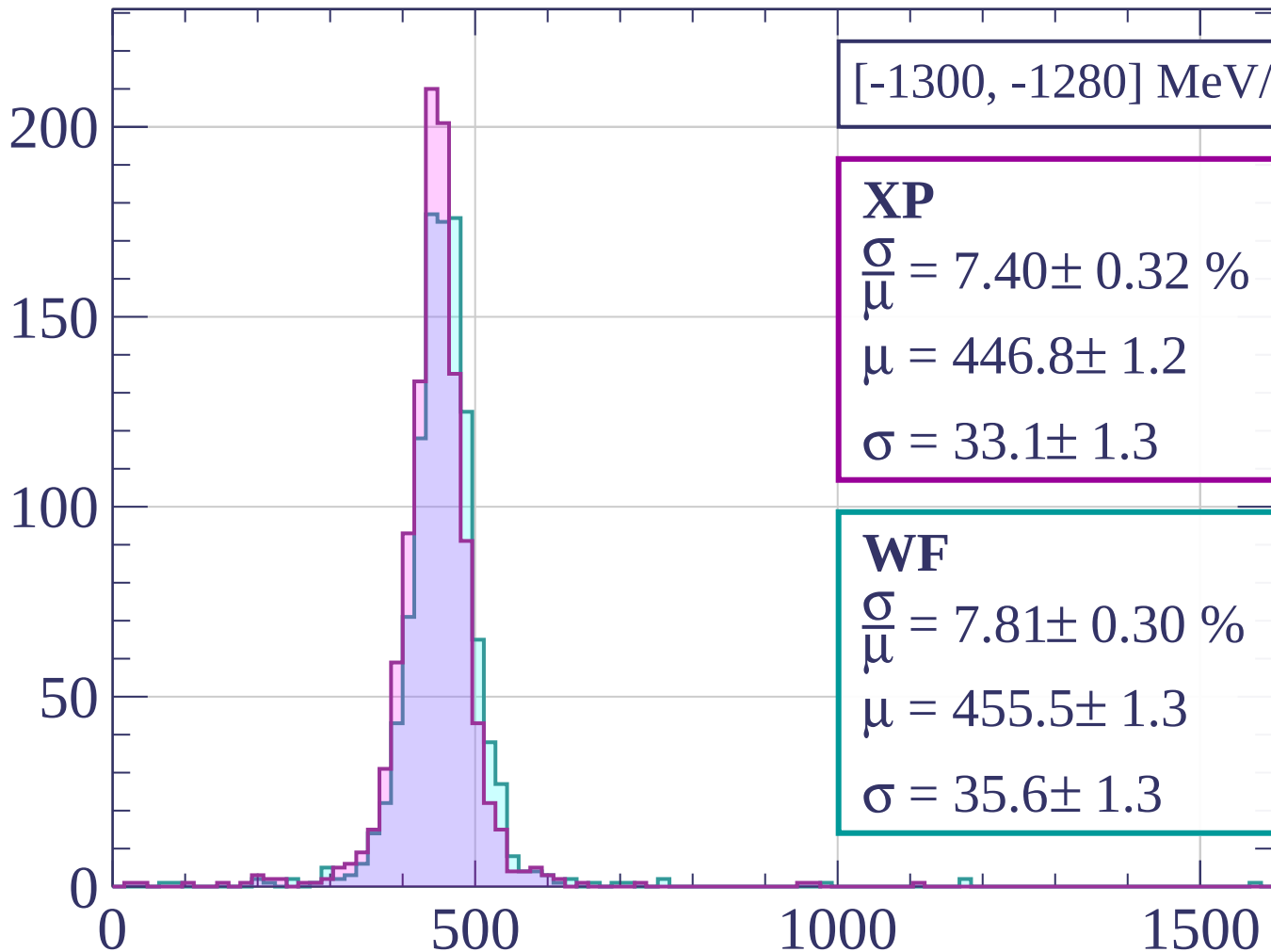
$$\sigma = 33.1 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.81 \pm 0.30 \%$$

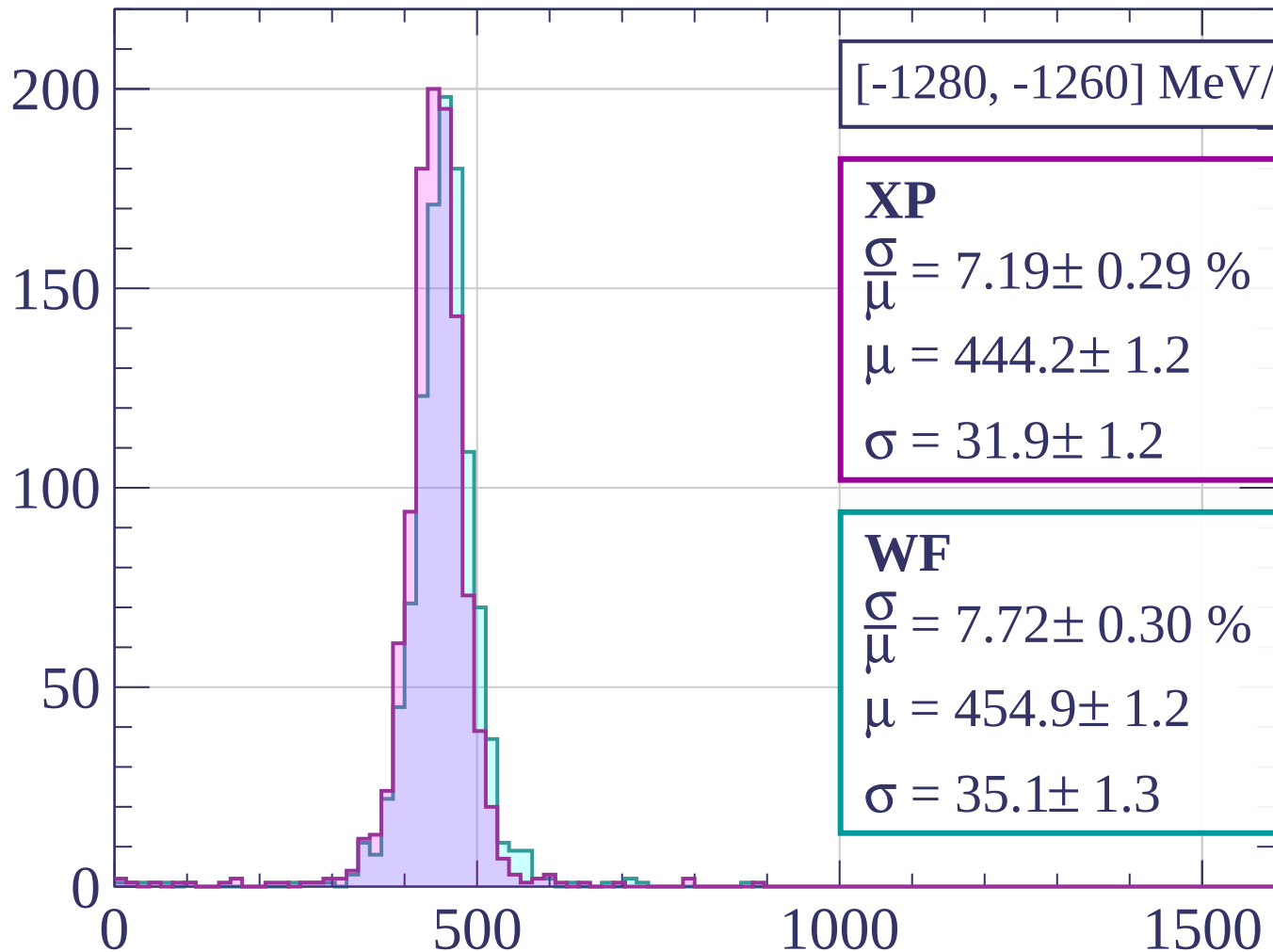
$$\mu = 455.5 \pm 1.3$$

$$\sigma = 35.6 \pm 1.3$$



dE/dx [ADC counts/cm]

Count



[-1280, -1260] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.19 \pm 0.29 \%$$

$$\mu = 444.2 \pm 1.2$$

$$\sigma = 31.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.72 \pm 0.30 \%$$

$$\mu = 454.9 \pm 1.2$$

$$\sigma = 35.1 \pm 1.3$$

dE/dx [ADC counts/cm]

Count

[-1260, -1240] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.43 \pm 0.28 \%$$

$$\mu = 444.8 \pm 1.2$$

$$\sigma = 33.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.54 \pm 0.26 \%$$

$$\mu = 454.2 \pm 1.2$$

$$\sigma = 34.2 \pm 1.1$$

200

150

100

50

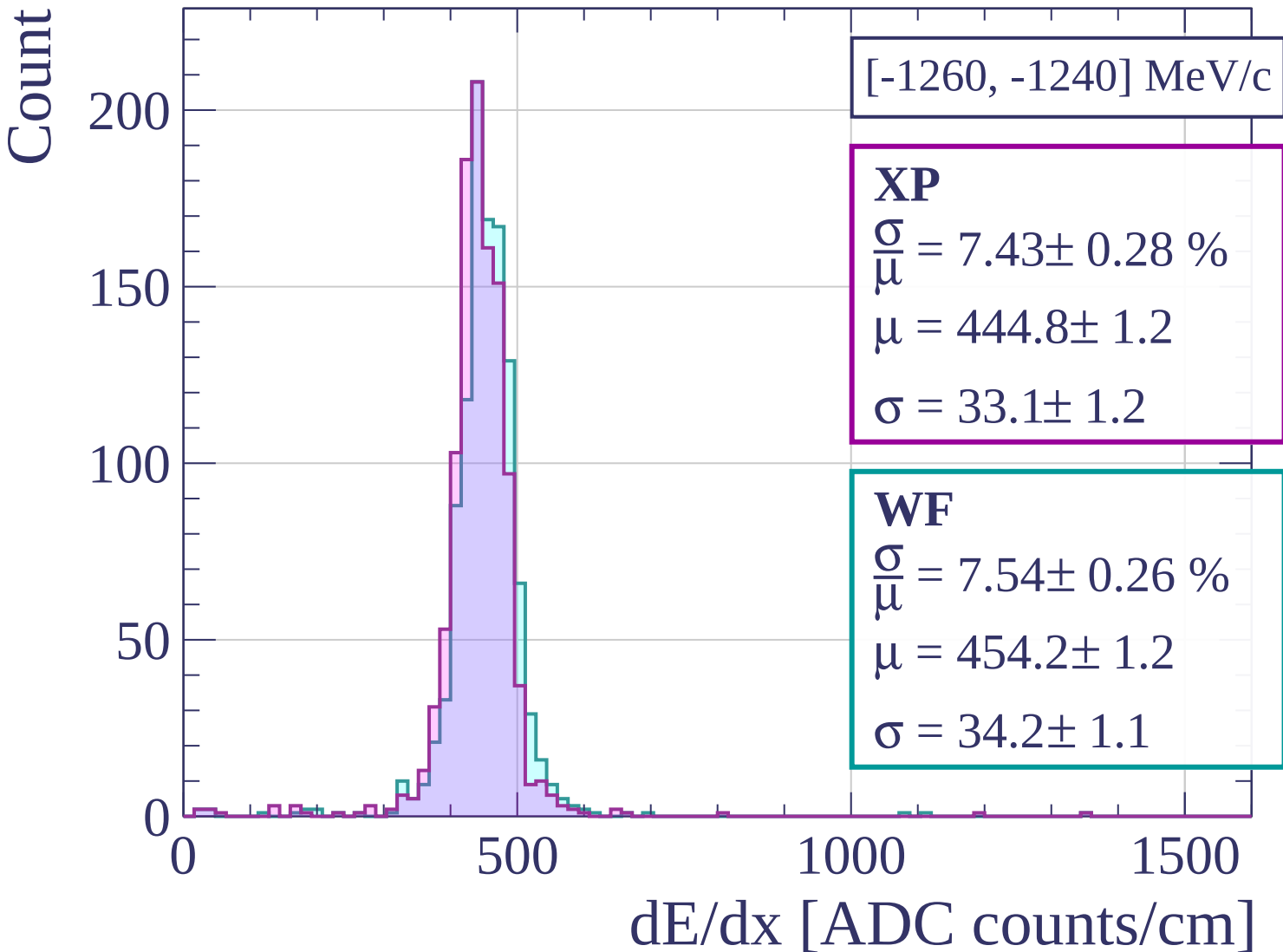
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1240, -1220] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.43 \pm 0.30 \%$$

$$\mu = 444.3 \pm 1.2$$

$$\sigma = 33.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.97 \pm 0.32 \%$$

$$\mu = 453.6 \pm 1.3$$

$$\sigma = 36.1 \pm 1.3$$

200

150

100

50

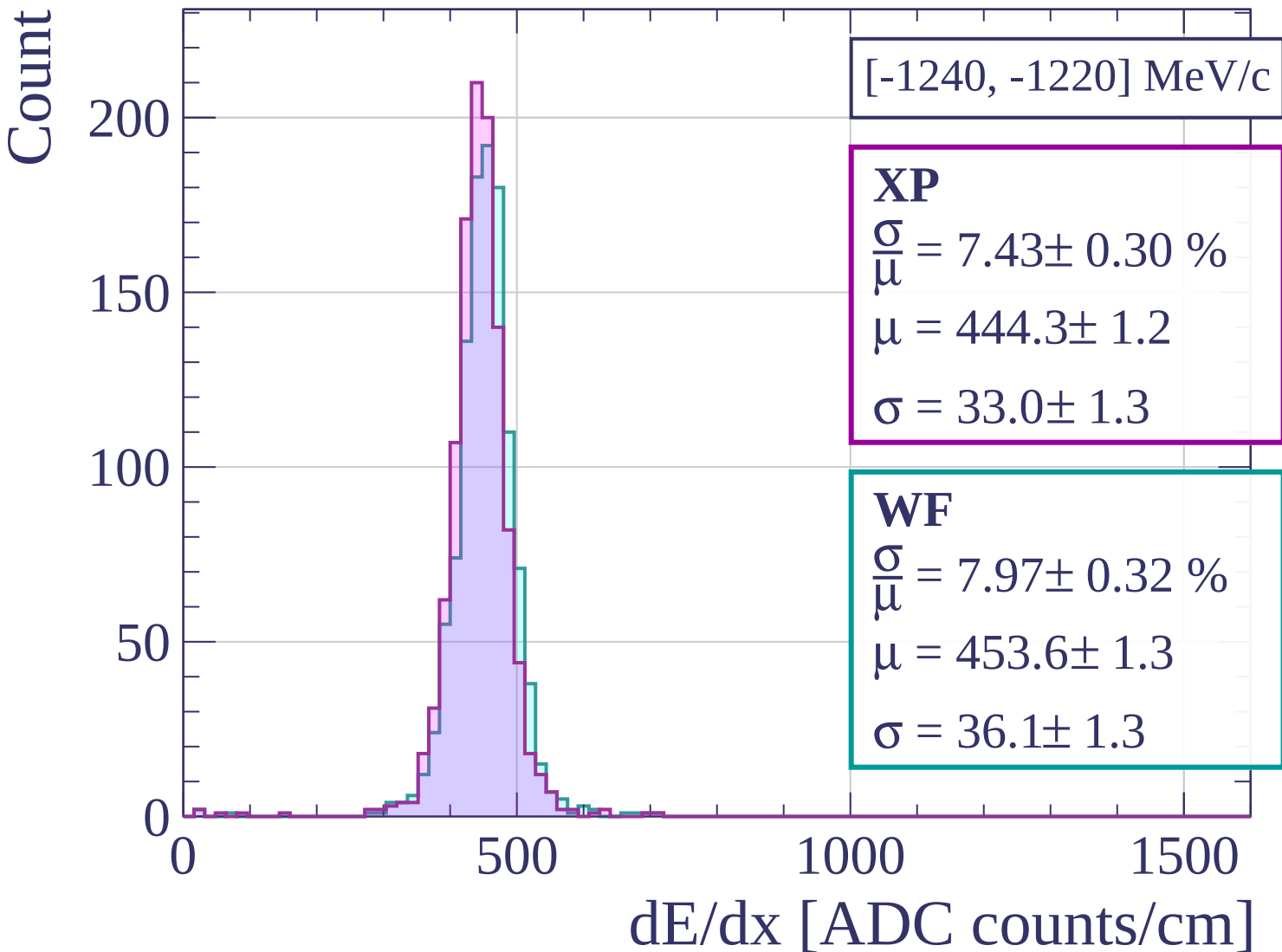
0

500

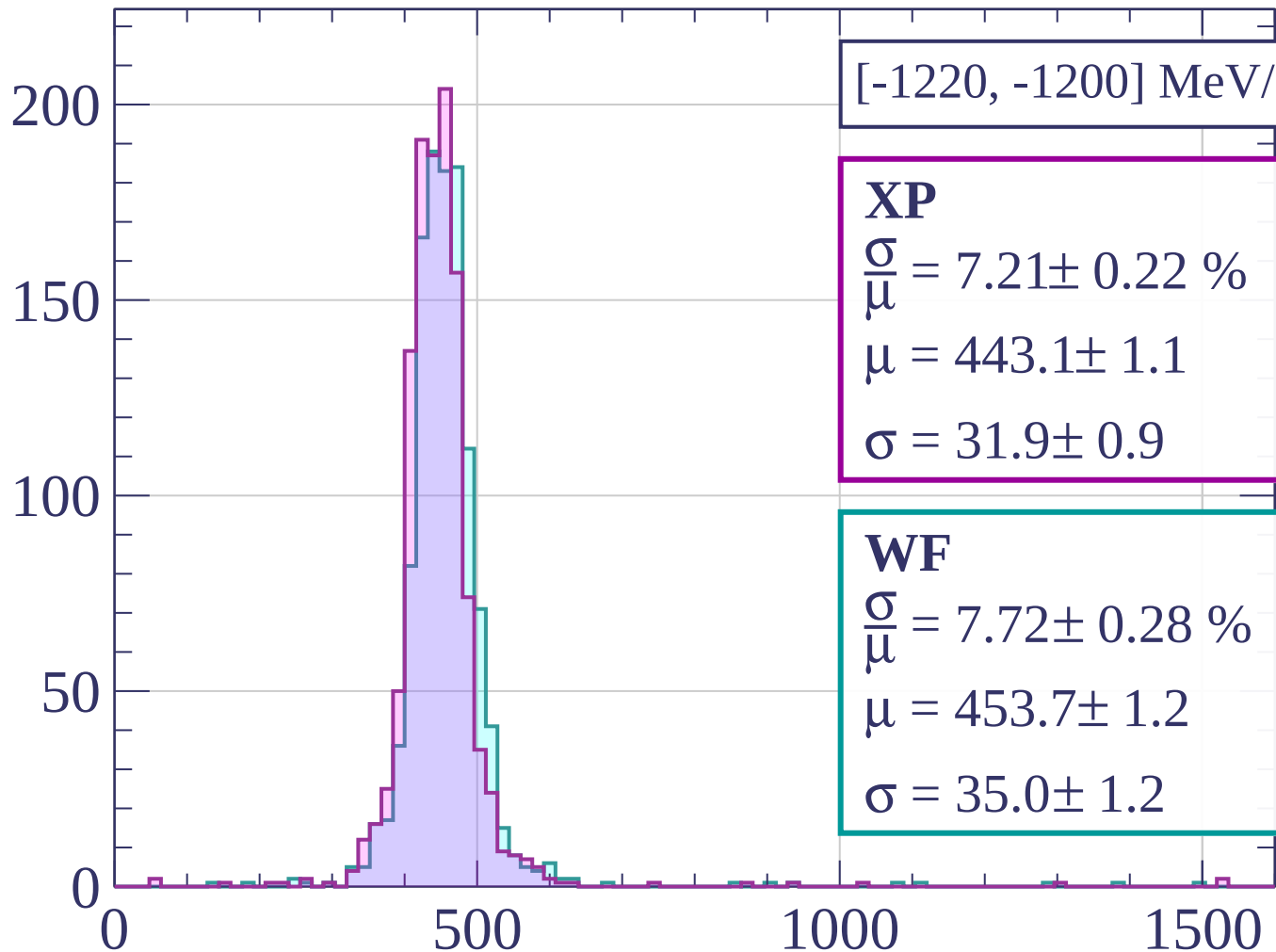
1000

1500

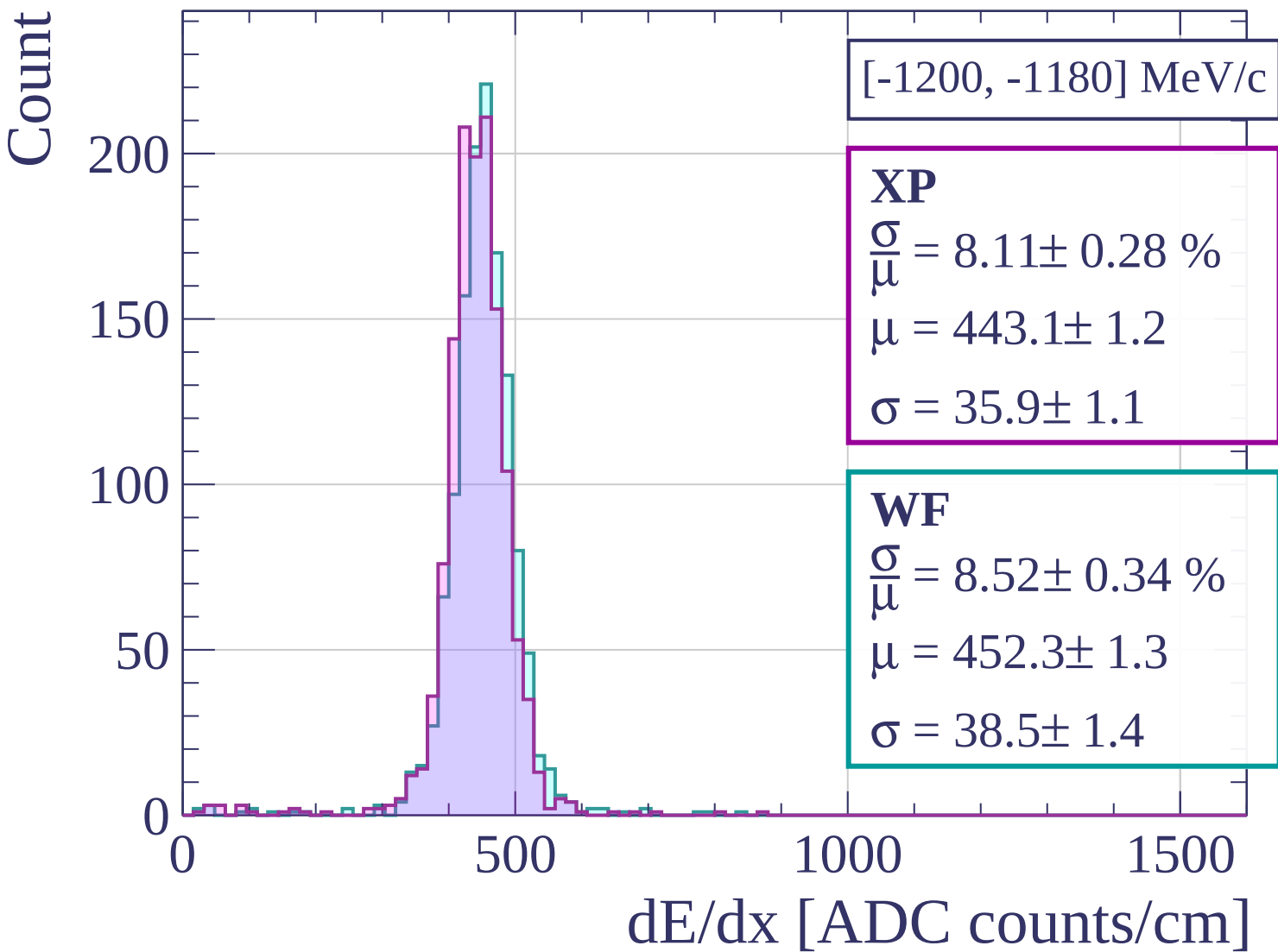
dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]



Count

[-1180, -1160] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.96 \pm 0.30 \%$$

$$\mu = 440.8 \pm 1.1$$

$$\sigma = 30.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.12 \pm 0.27 \%$$

$$\mu = 450.9 \pm 1.1$$

$$\sigma = 32.1 \pm 1.1$$

200

100

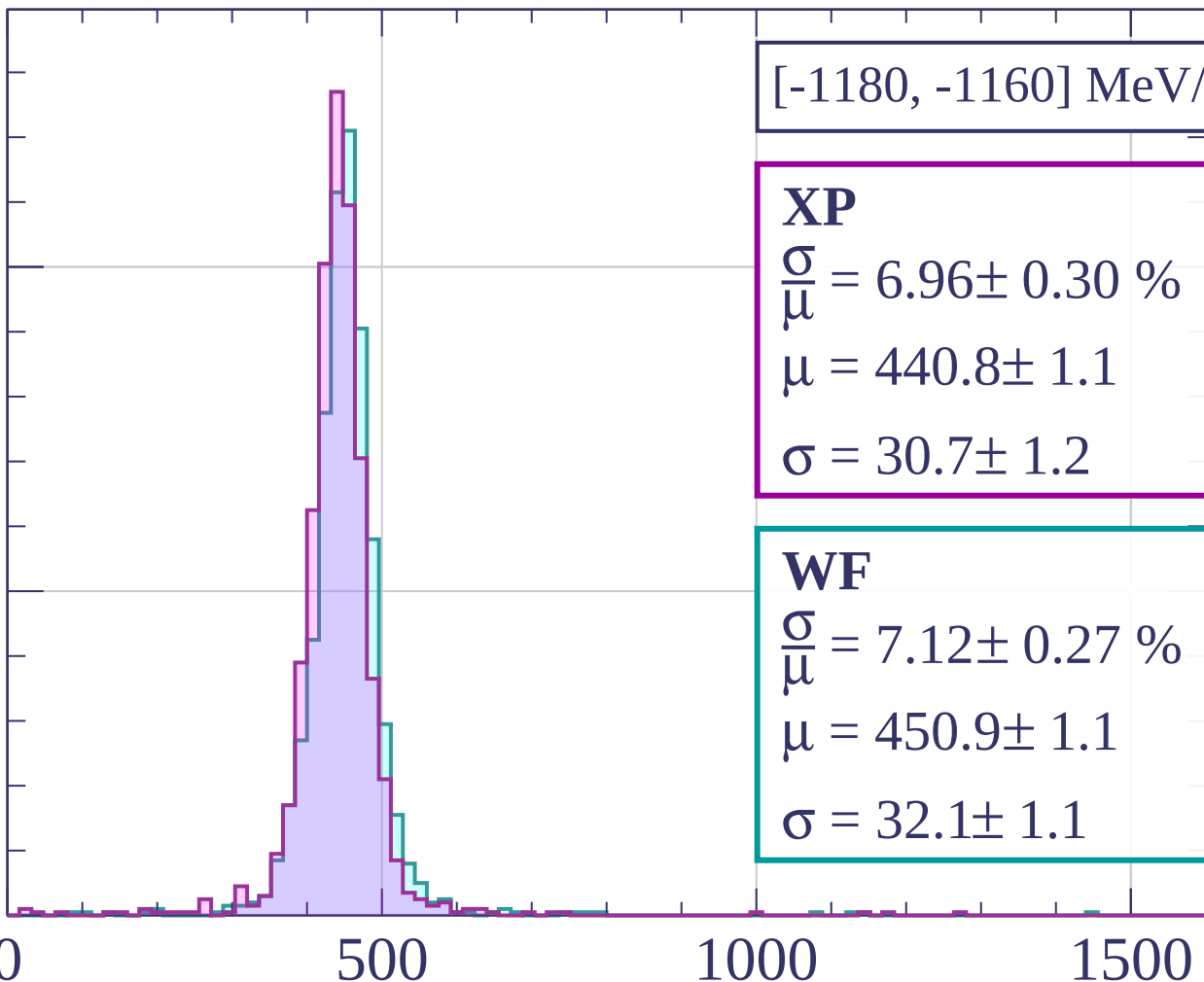
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1160, -1140] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.57 \pm 0.25 \%$$

$$\mu = 440.7 \pm 1.1$$

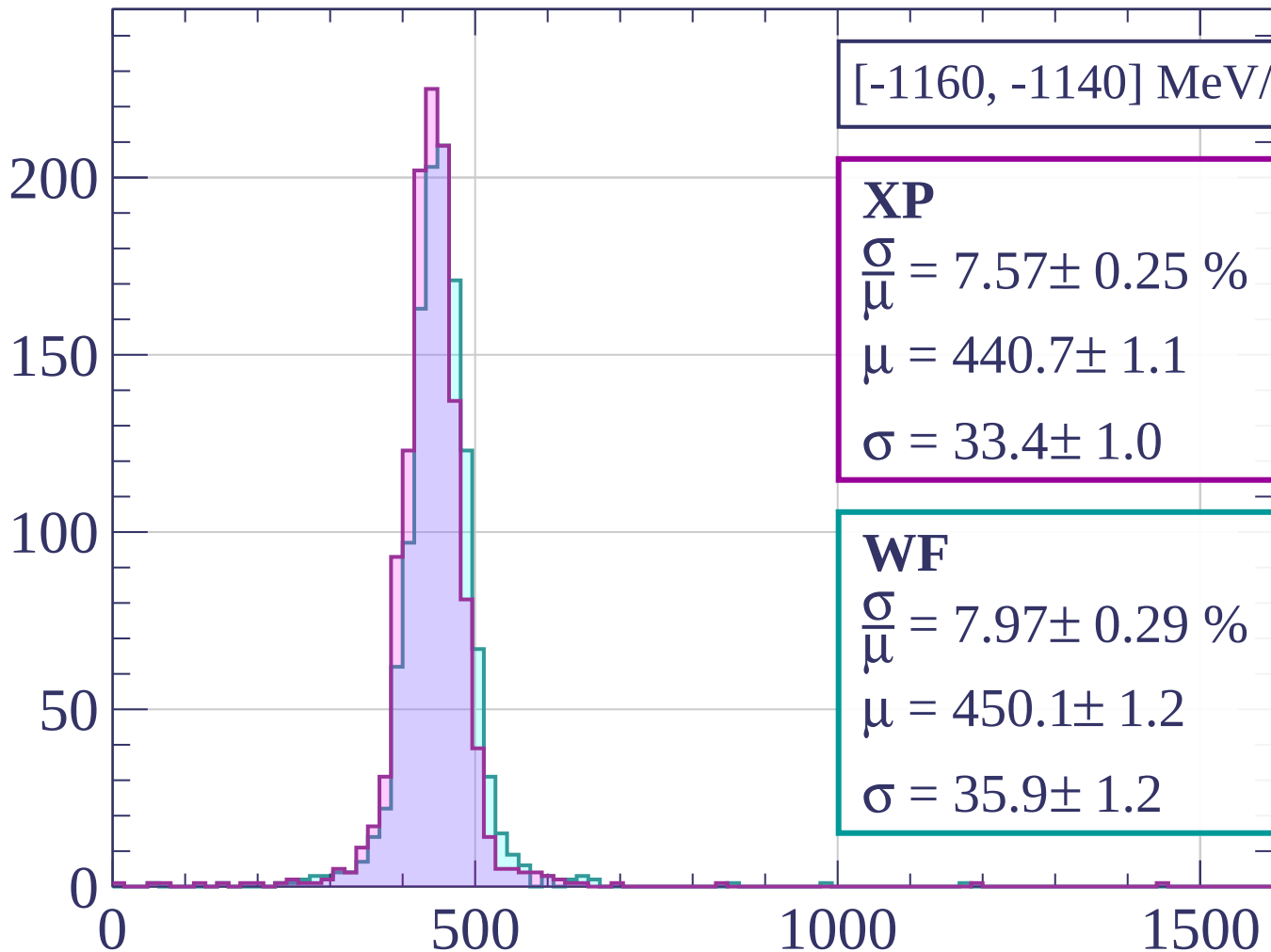
$$\sigma = 33.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.97 \pm 0.29 \%$$

$$\mu = 450.1 \pm 1.2$$

$$\sigma = 35.9 \pm 1.2$$



dE/dx [ADC counts/cm]

Count

[-1140, -1120] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.55 \pm 0.24 \%$$

$$\mu = 439.2 \pm 1.1$$

$$\sigma = 33.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.60 \pm 0.30 \%$$

$$\mu = 448.3 \pm 1.2$$

$$\sigma = 34.1 \pm 1.2$$

200

100

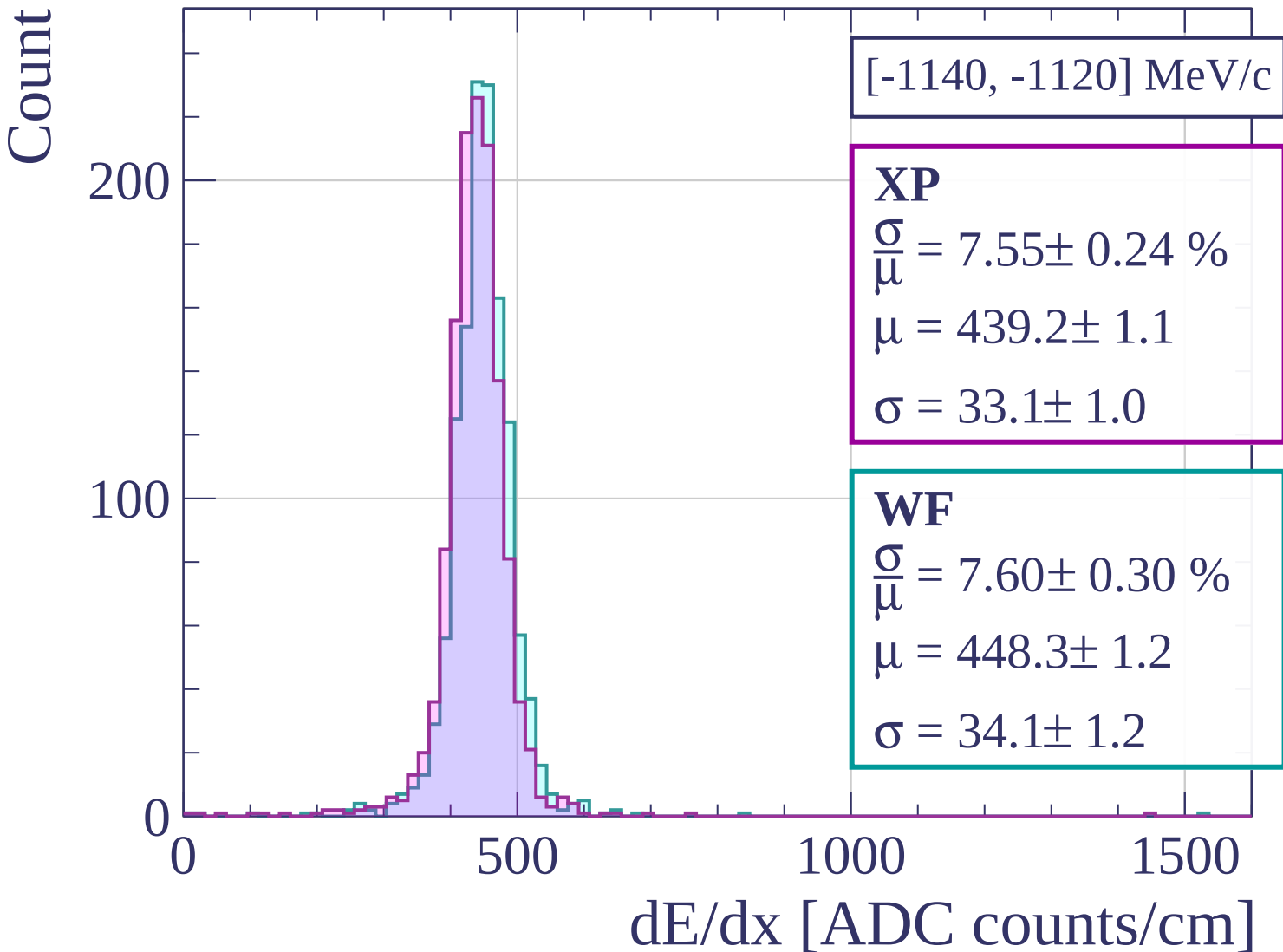
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1120, -1100] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.09 \pm 0.32 \%$$

$$\mu = 437.7 \pm 1.1$$

$$\sigma = 31.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.93 \pm 0.32 \%$$

$$\mu = 445.1 \pm 1.1$$

$$\sigma = 30.9 \pm 1.3$$

200

100

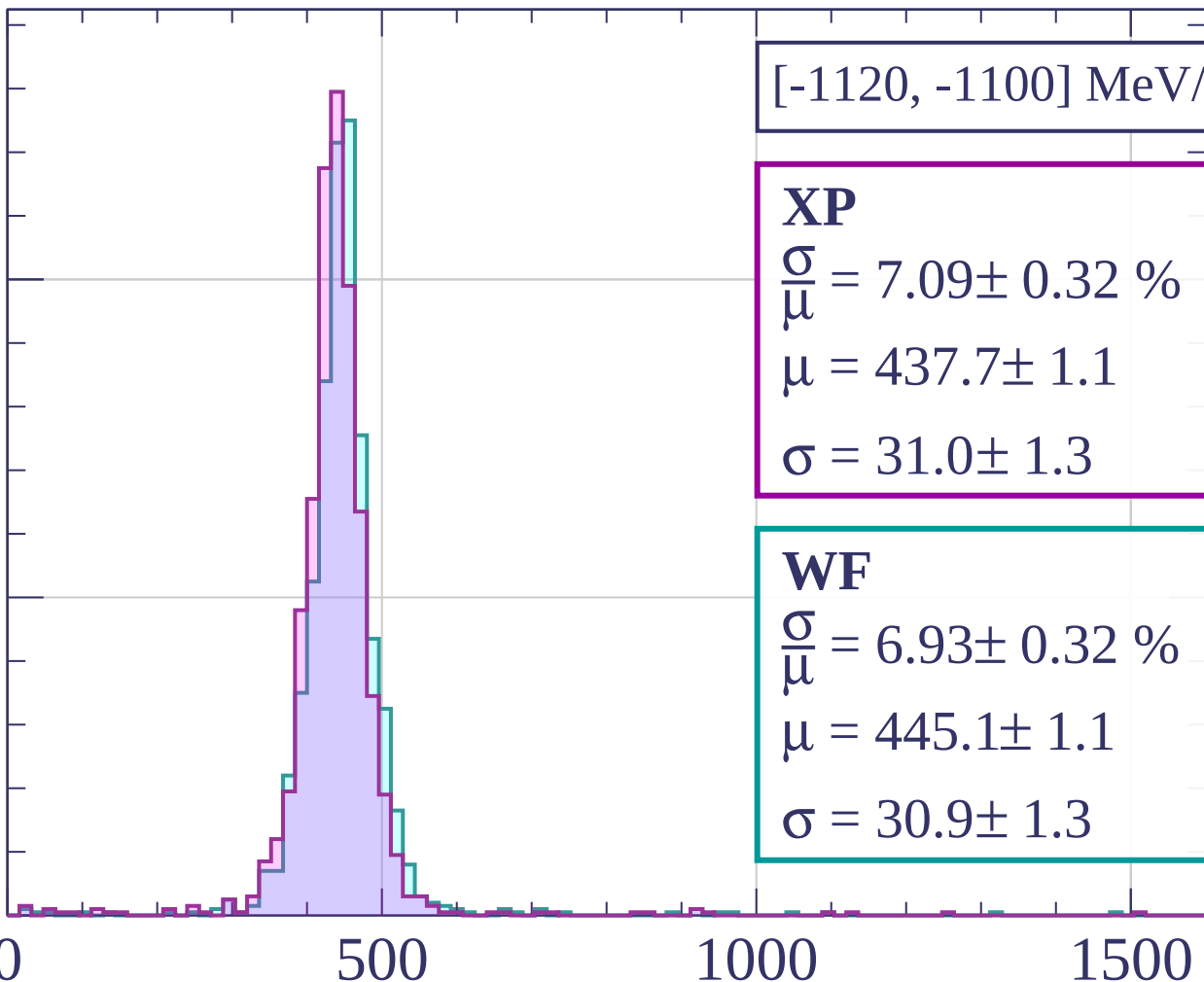
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1100, -1080] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.77 \pm 0.26 \%$$

$$\mu = 438.6 \pm 1.1$$

$$\sigma = 34.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.95 \pm 0.33 \%$$

$$\mu = 447.5 \pm 1.3$$

$$\sigma = 35.6 \pm 1.4$$

200

100

0

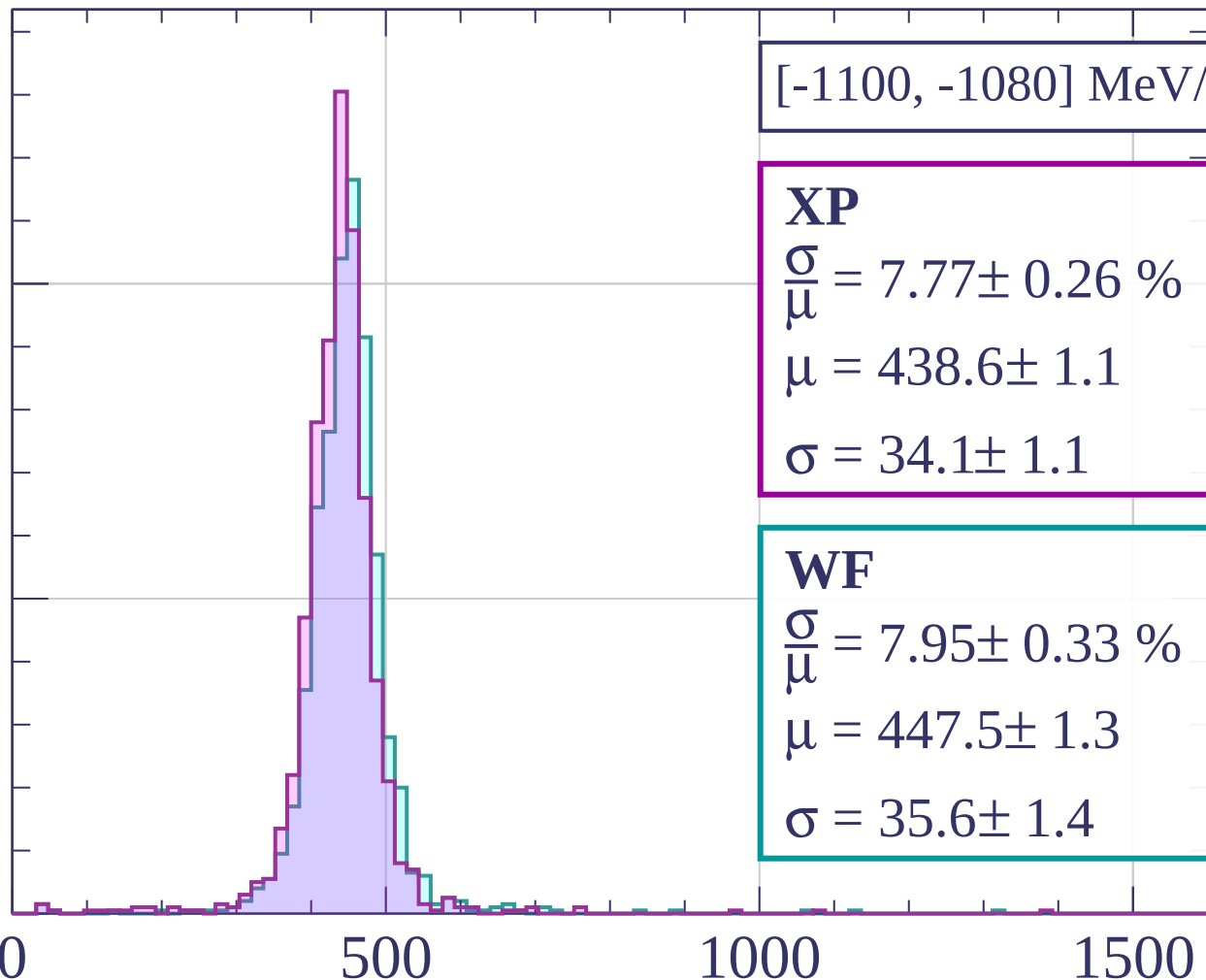
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1080, -1060] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.38 \pm 0.25 \%$$

$$\mu = 437.3 \pm 1.1$$

$$\sigma = 32.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.75 \pm 0.24 \%$$

$$\mu = 446.5 \pm 1.1$$

$$\sigma = 34.6 \pm 1.0$$

200

100

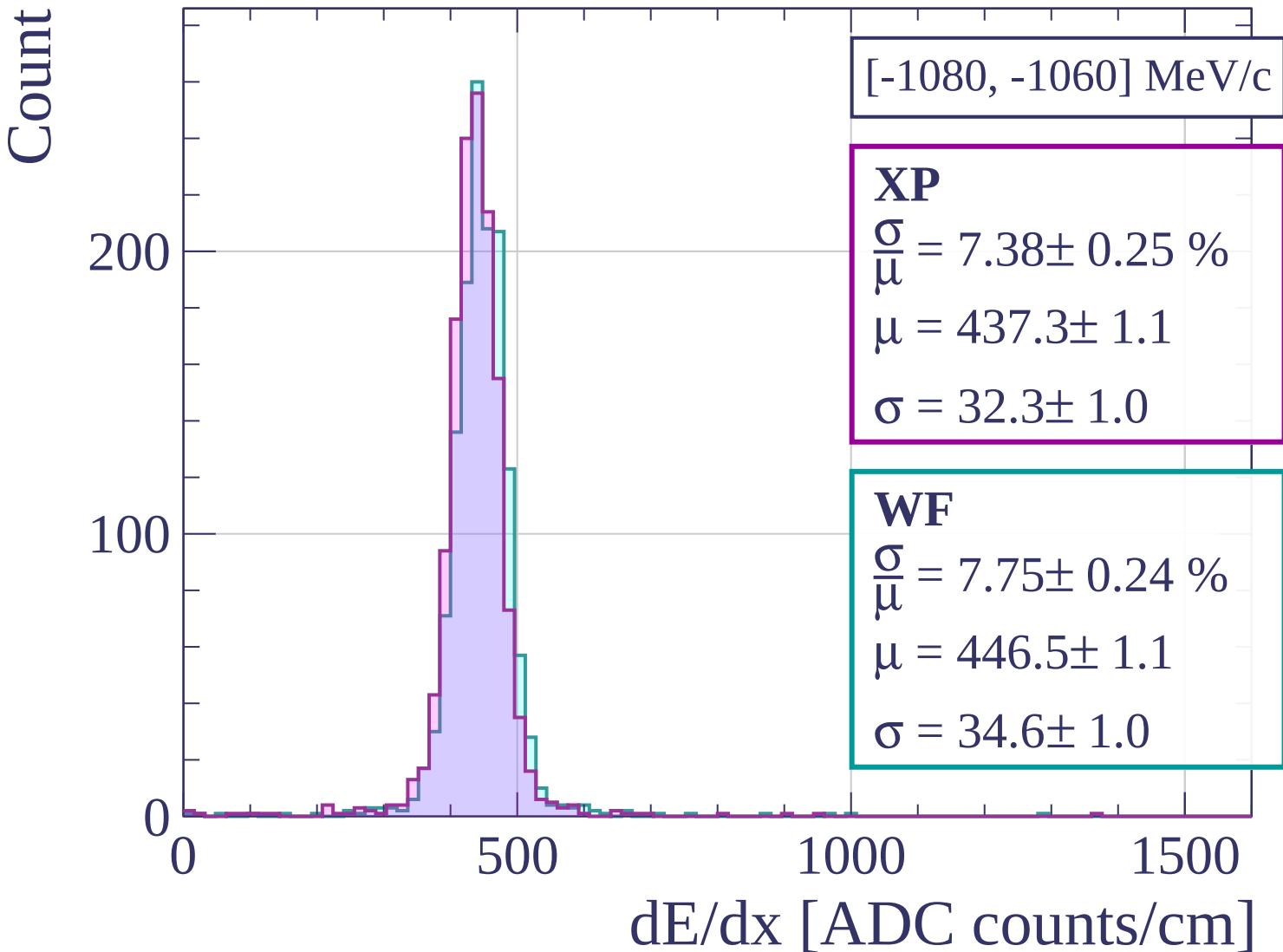
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1060, -1040] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.11 \pm 0.29 \%$$

$$\mu = 437.1 \pm 1.1$$

$$\sigma = 31.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.37 \pm 0.27 \%$$

$$\mu = 446.6 \pm 1.1$$

$$\sigma = 32.9 \pm 1.1$$

200

100

0

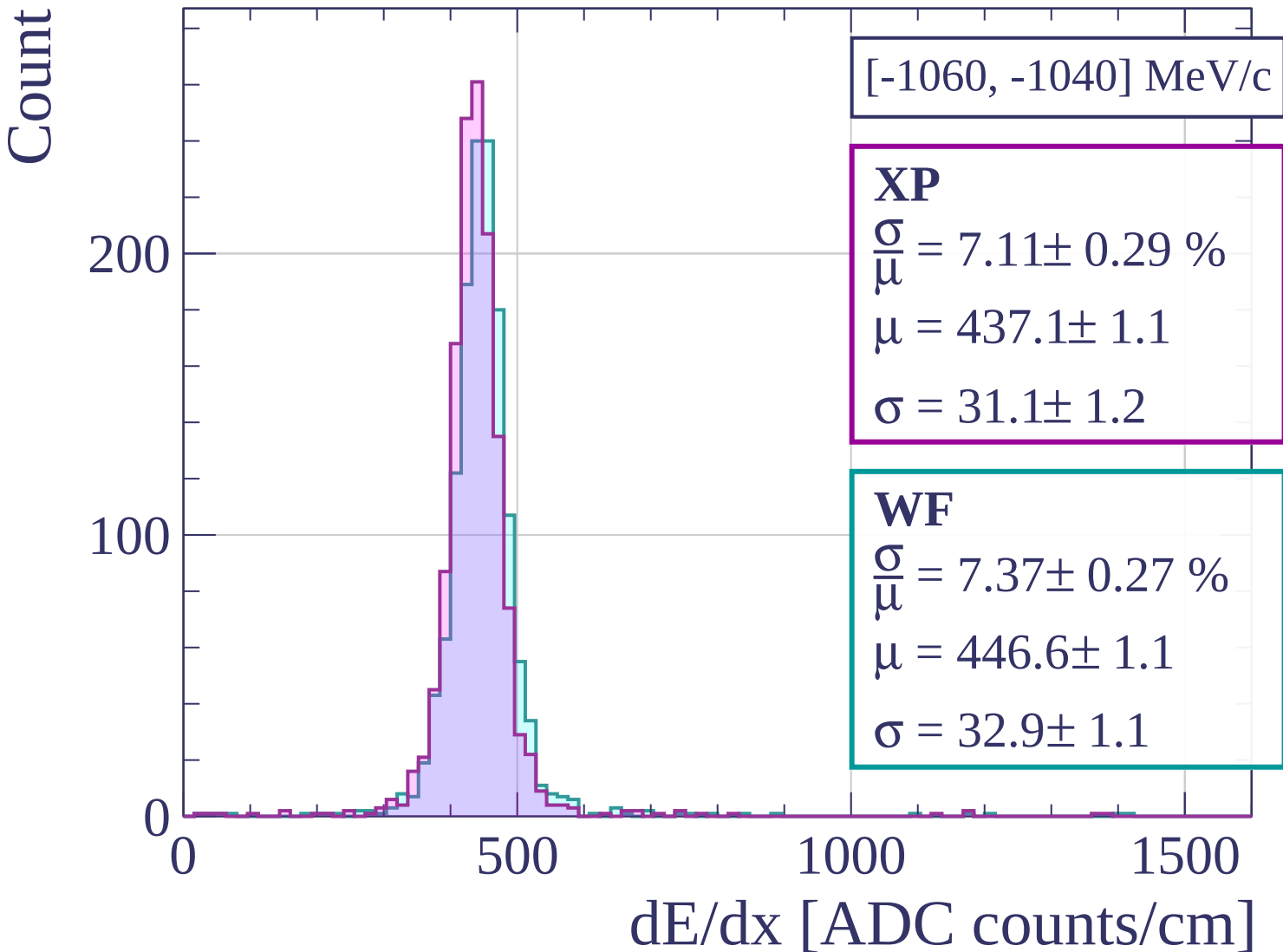
0

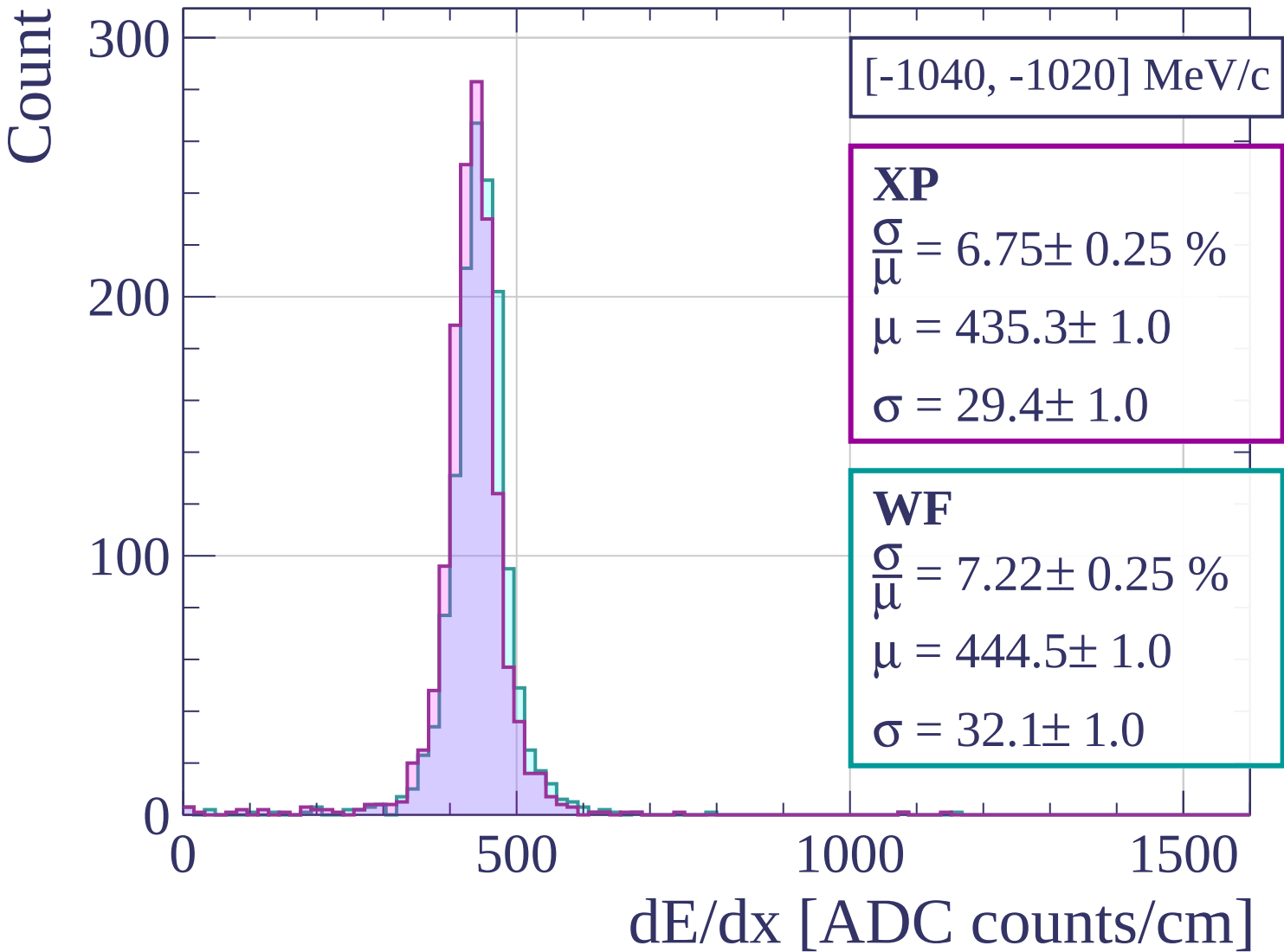
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-1020, -1000] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.26 \pm 0.24 \%$$

$$\mu = 432.9 \pm 1.0$$

$$\sigma = 31.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.44 \pm 0.26 \%$$

$$\mu = 441.0 \pm 1.1$$

$$\sigma = 32.8 \pm 1.1$$

300

200

100

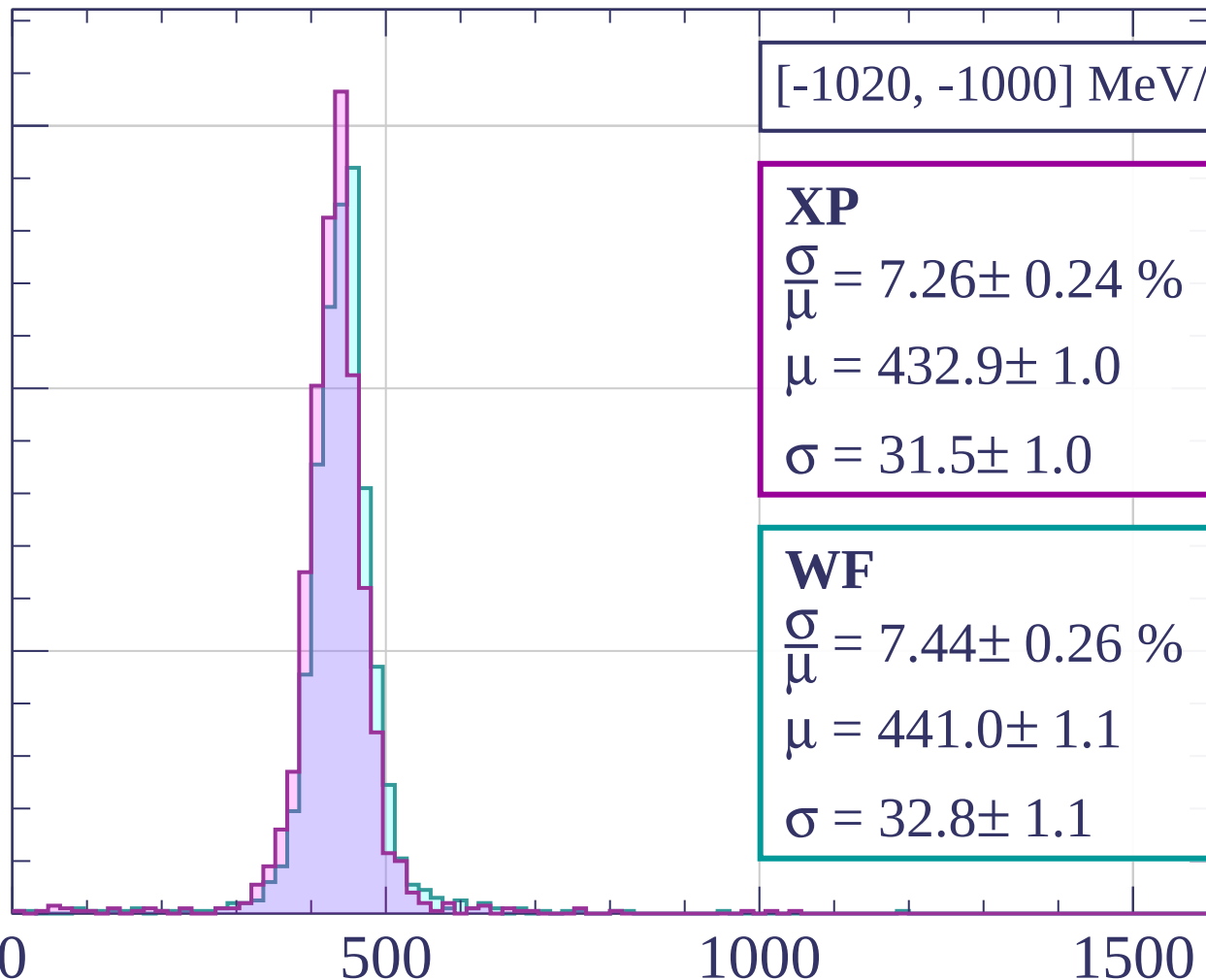
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1000, -980] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.51 \pm 0.26 \%$$

$$\mu = 432.7 \pm 1.0$$

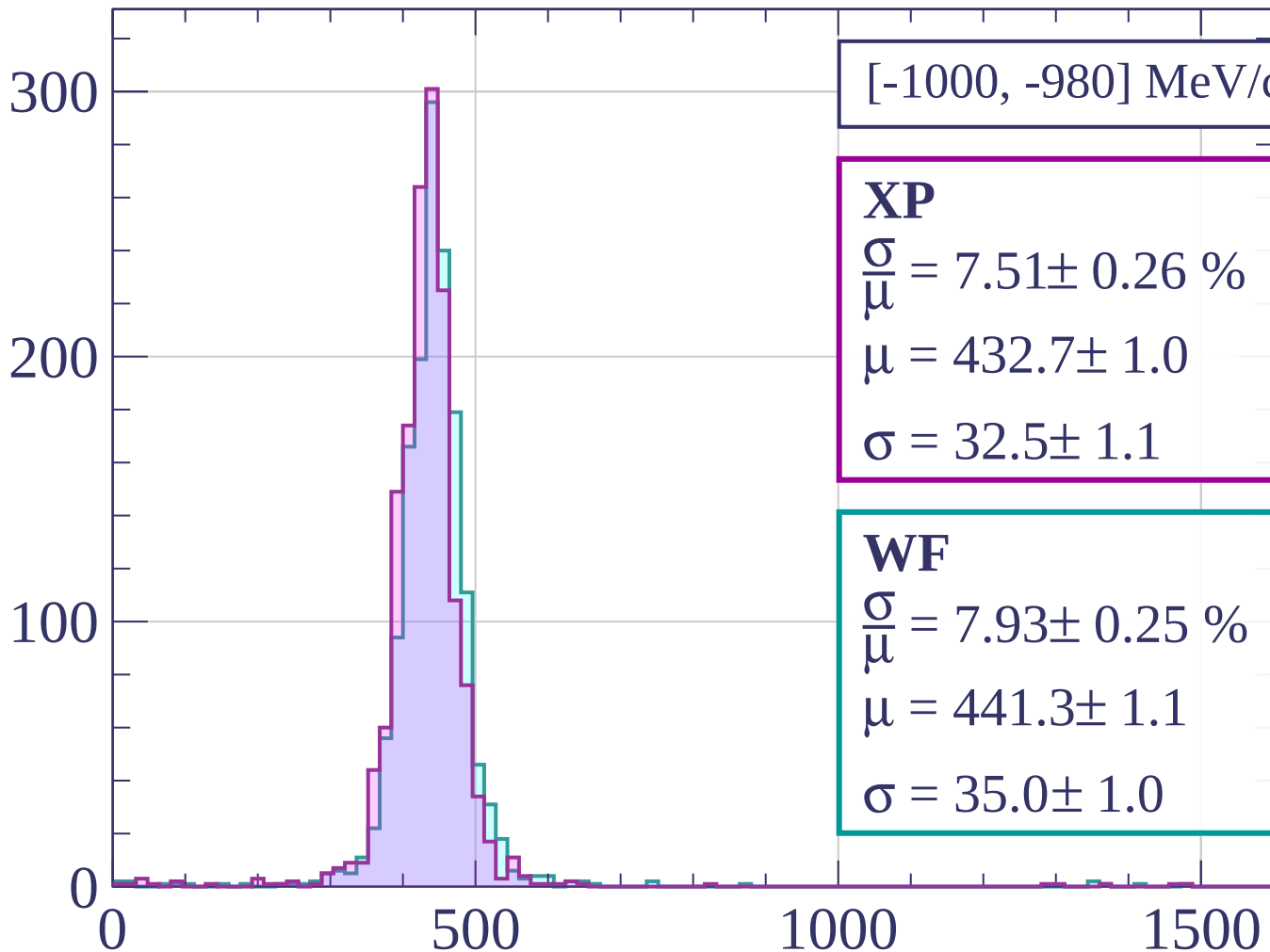
$$\sigma = 32.5 \pm 1.1$$

WF

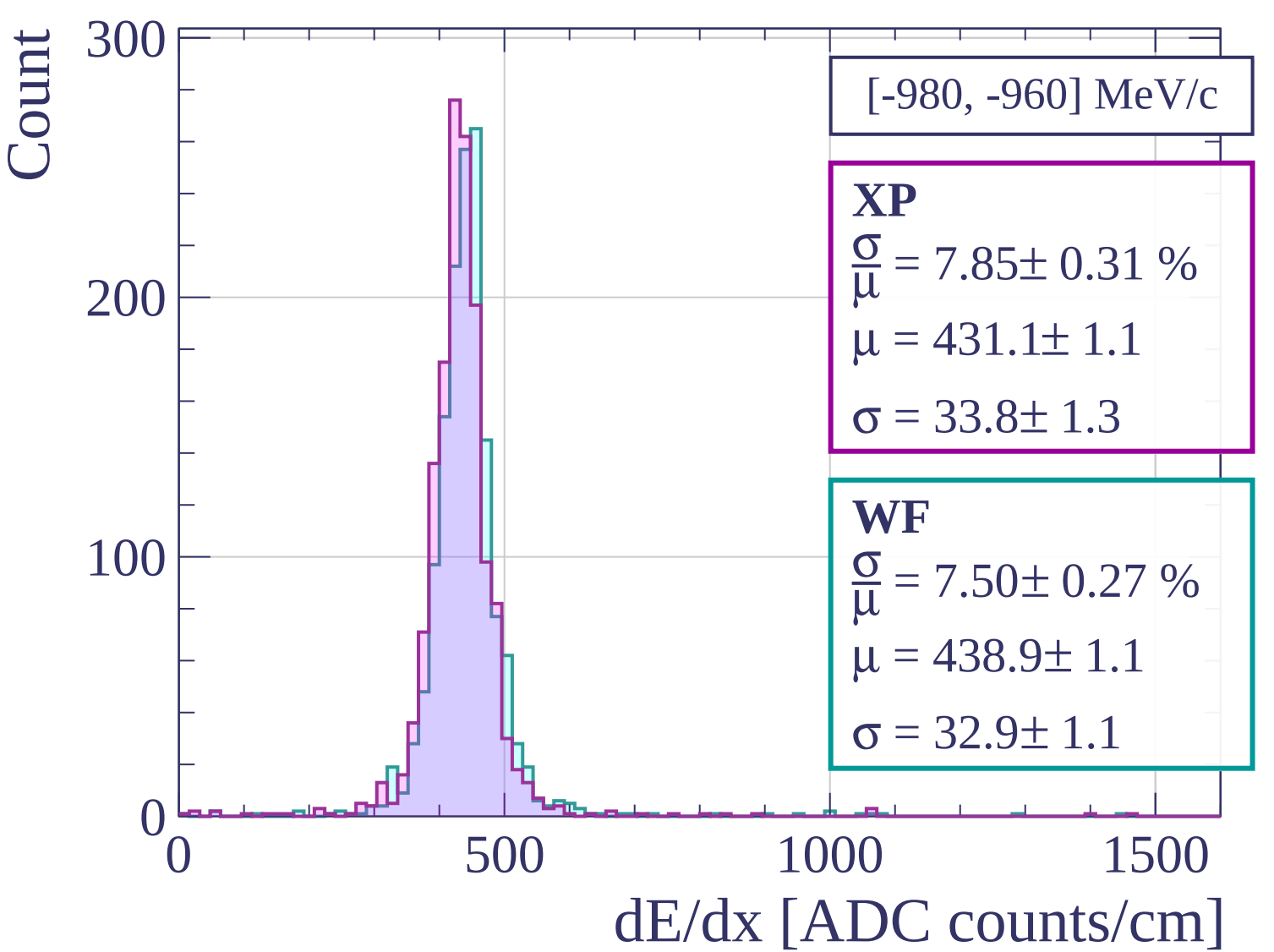
$$\frac{\sigma}{\mu} = 7.93 \pm 0.25 \%$$

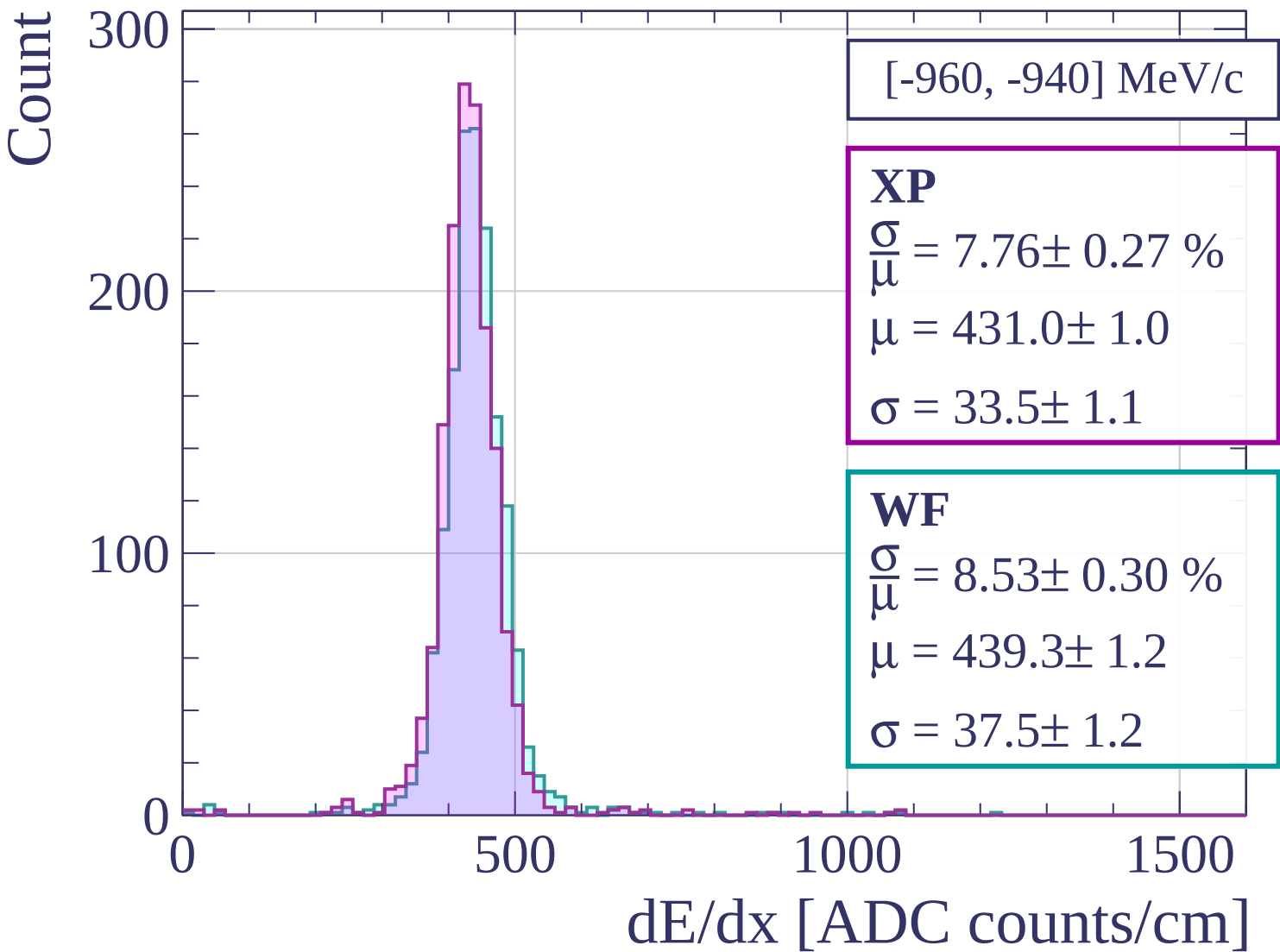
$$\mu = 441.3 \pm 1.1$$

$$\sigma = 35.0 \pm 1.0$$



dE/dx [ADC counts/cm]





Count

[-940, -920] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.54 \pm 0.24 \%$$

$$\mu = 428.9 \pm 1.0$$

$$\sigma = 32.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.88 \pm 0.23 \%$$

$$\mu = 437.2 \pm 1.0$$

$$\sigma = 34.4 \pm 0.9$$

300

200

100

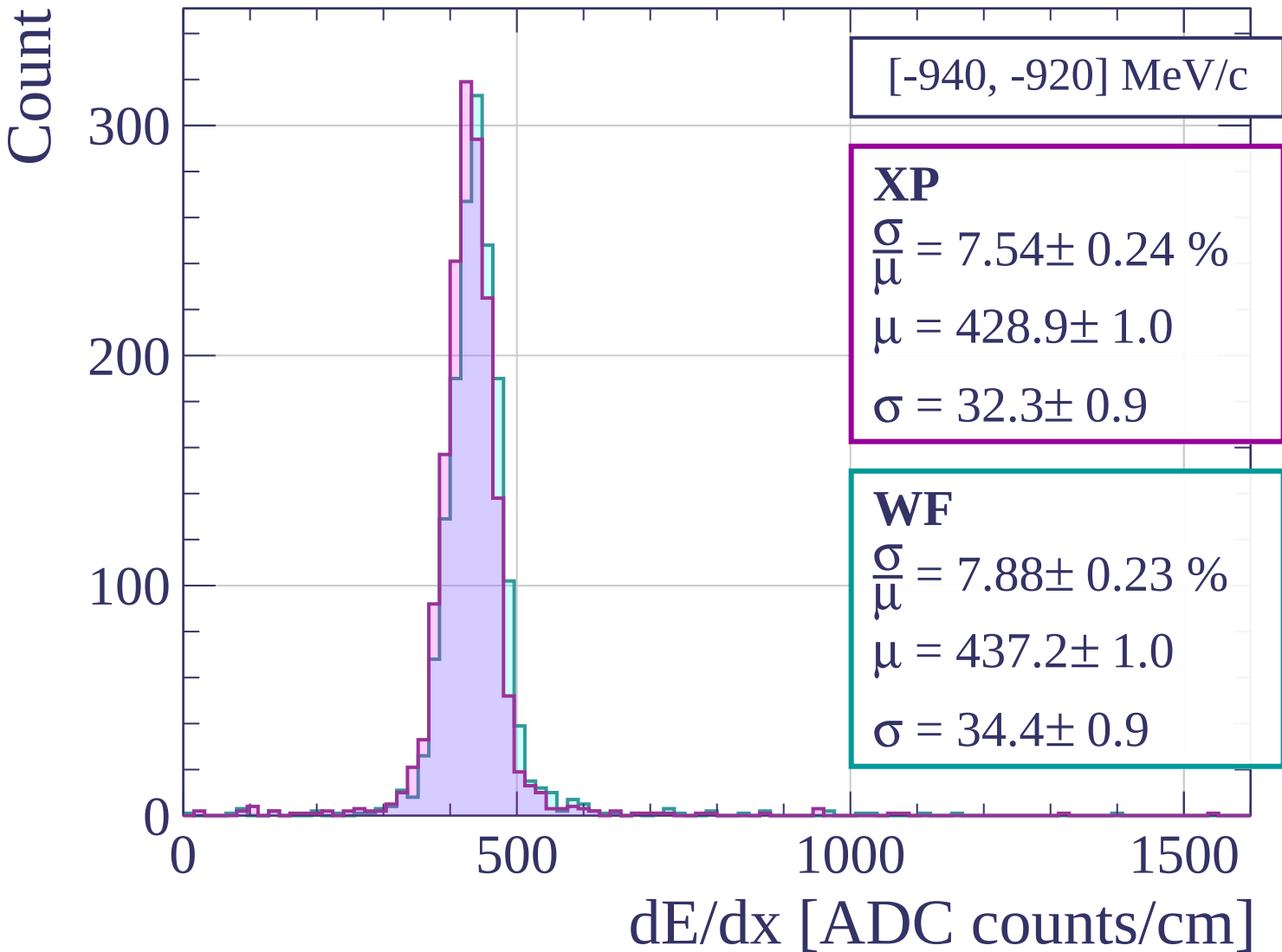
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-920, -900] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.78 \pm 0.26 \%$$

$$\mu = 426.3 \pm 1.0$$

$$\sigma = 33.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.81 \pm 0.27 \%$$

$$\mu = 435.0 \pm 1.0$$

$$\sigma = 34.0 \pm 1.1$$

300

200

100

0

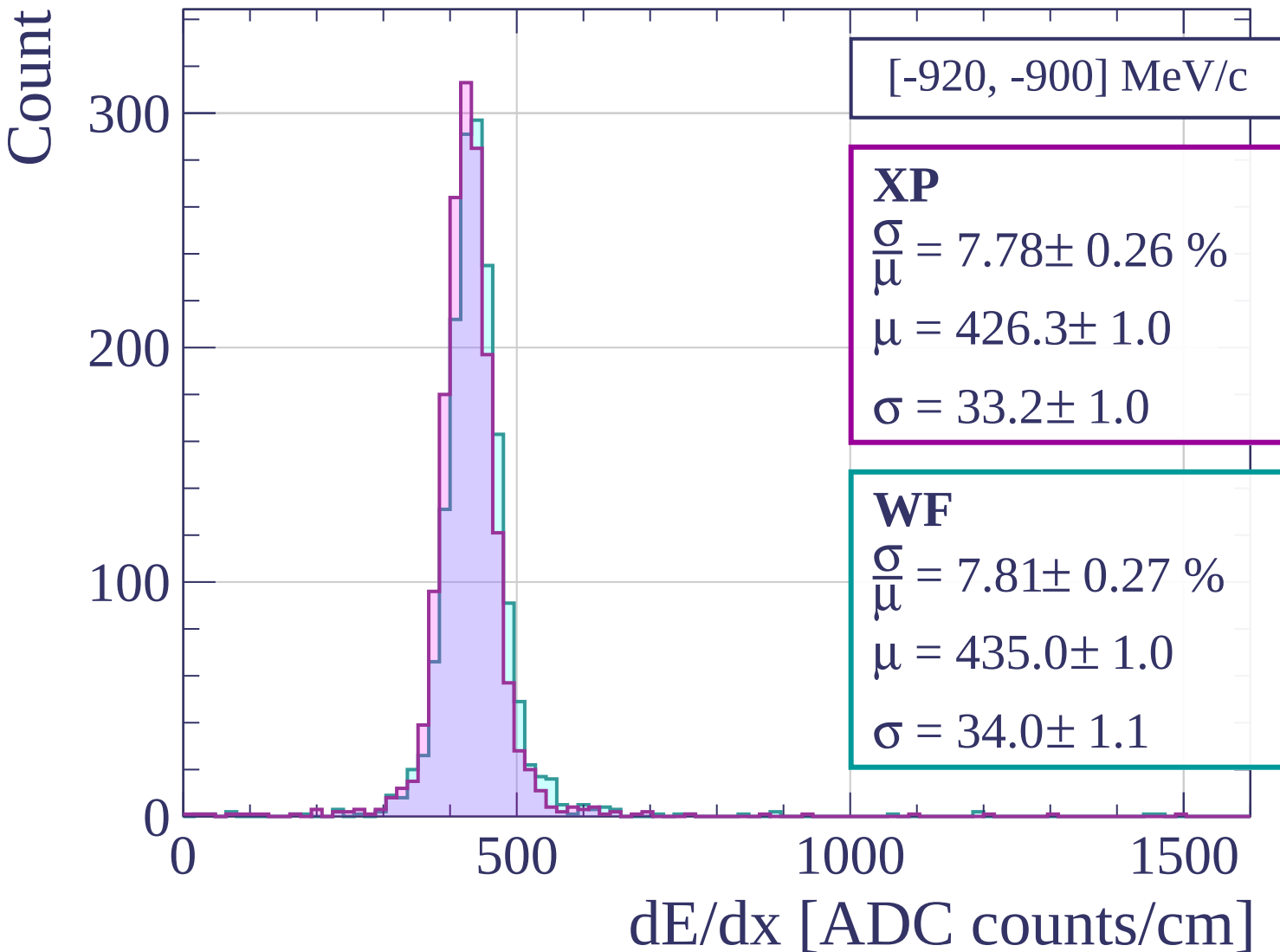
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-900, -880] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.88 \pm 0.22 \%$$

$$\mu = 424.8 \pm 0.9$$

$$\sigma = 29.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.47 \pm 0.23 \%$$

$$\mu = 434.6 \pm 1.0$$

$$\sigma = 32.5 \pm 0.9$$

300

200

100

0

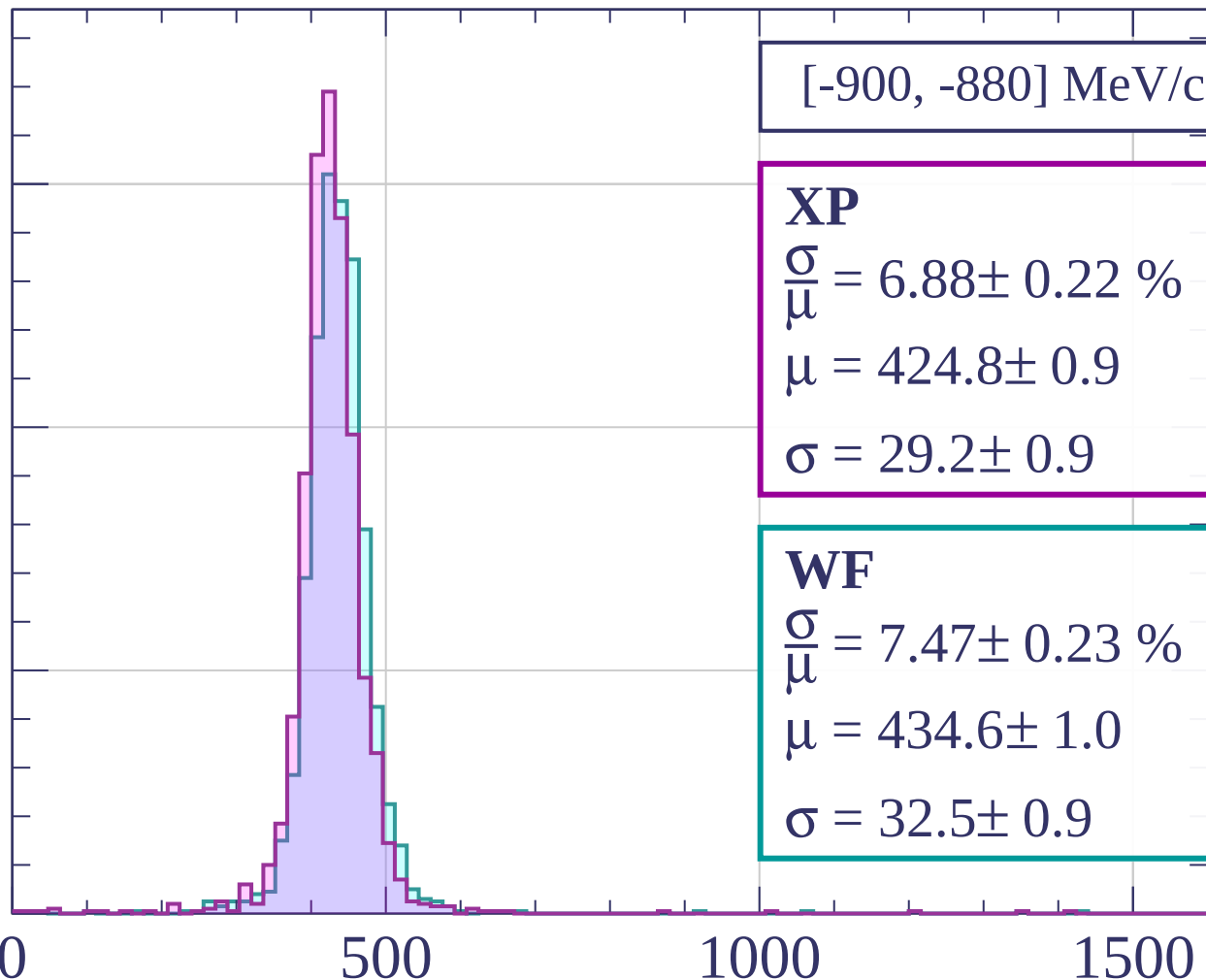
0

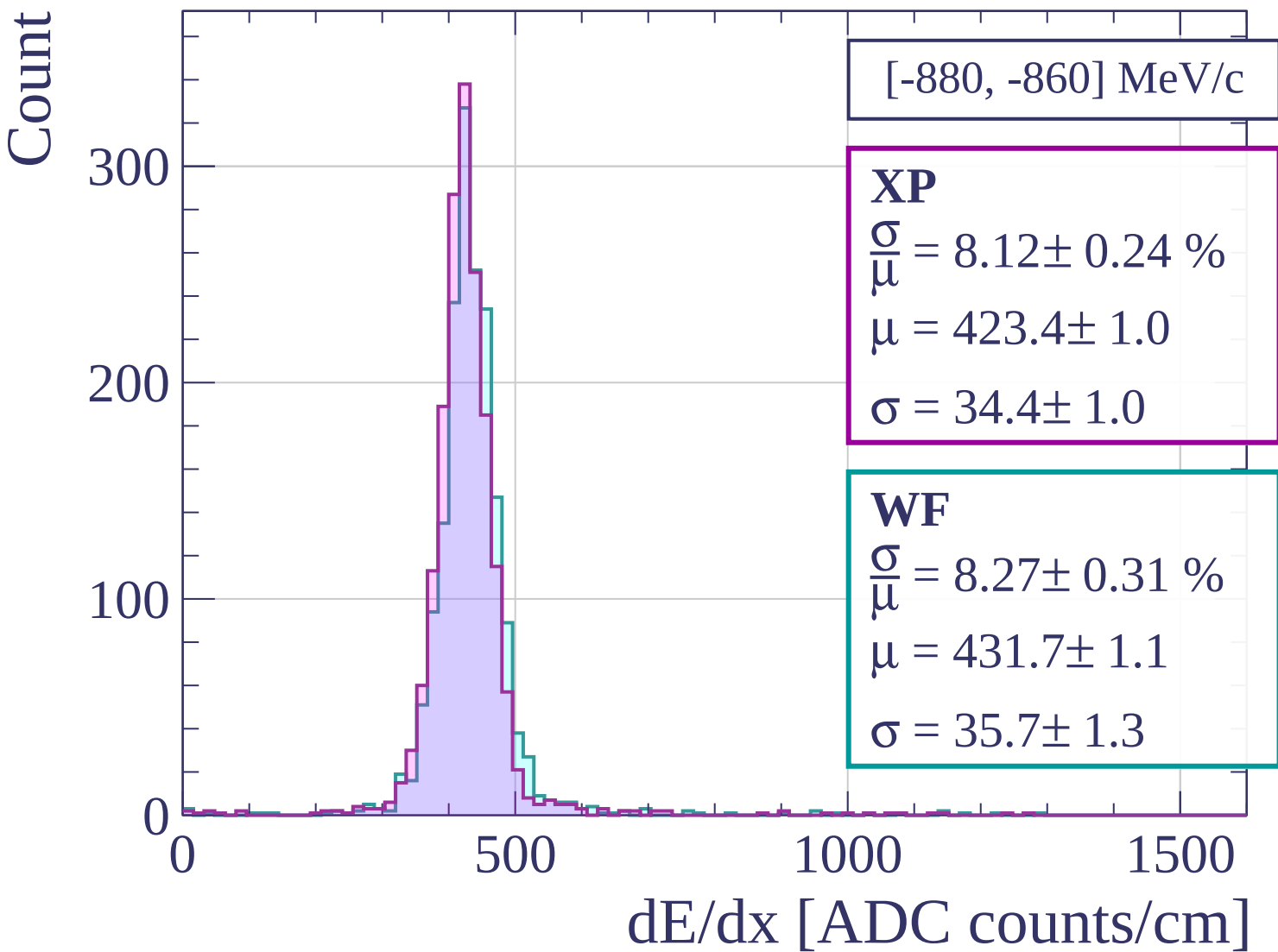
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-860, -840] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.51 \pm 0.23 \%$$

$$\mu = 424.7 \pm 0.9$$

$$\sigma = 31.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.78 \pm 0.21 \%$$

$$\mu = 432.6 \pm 0.9$$

$$\sigma = 33.7 \pm 0.8$$

300

200

100

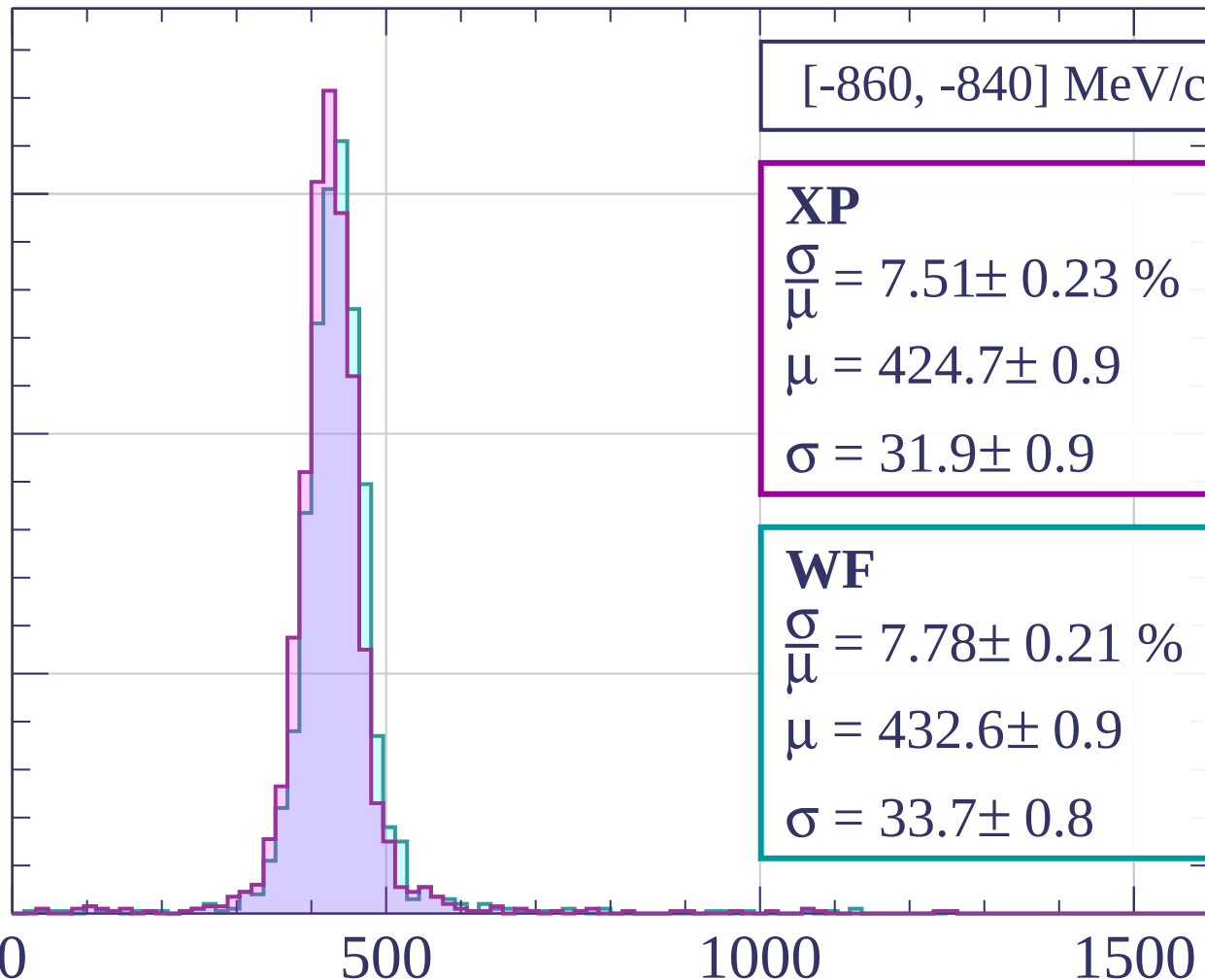
0

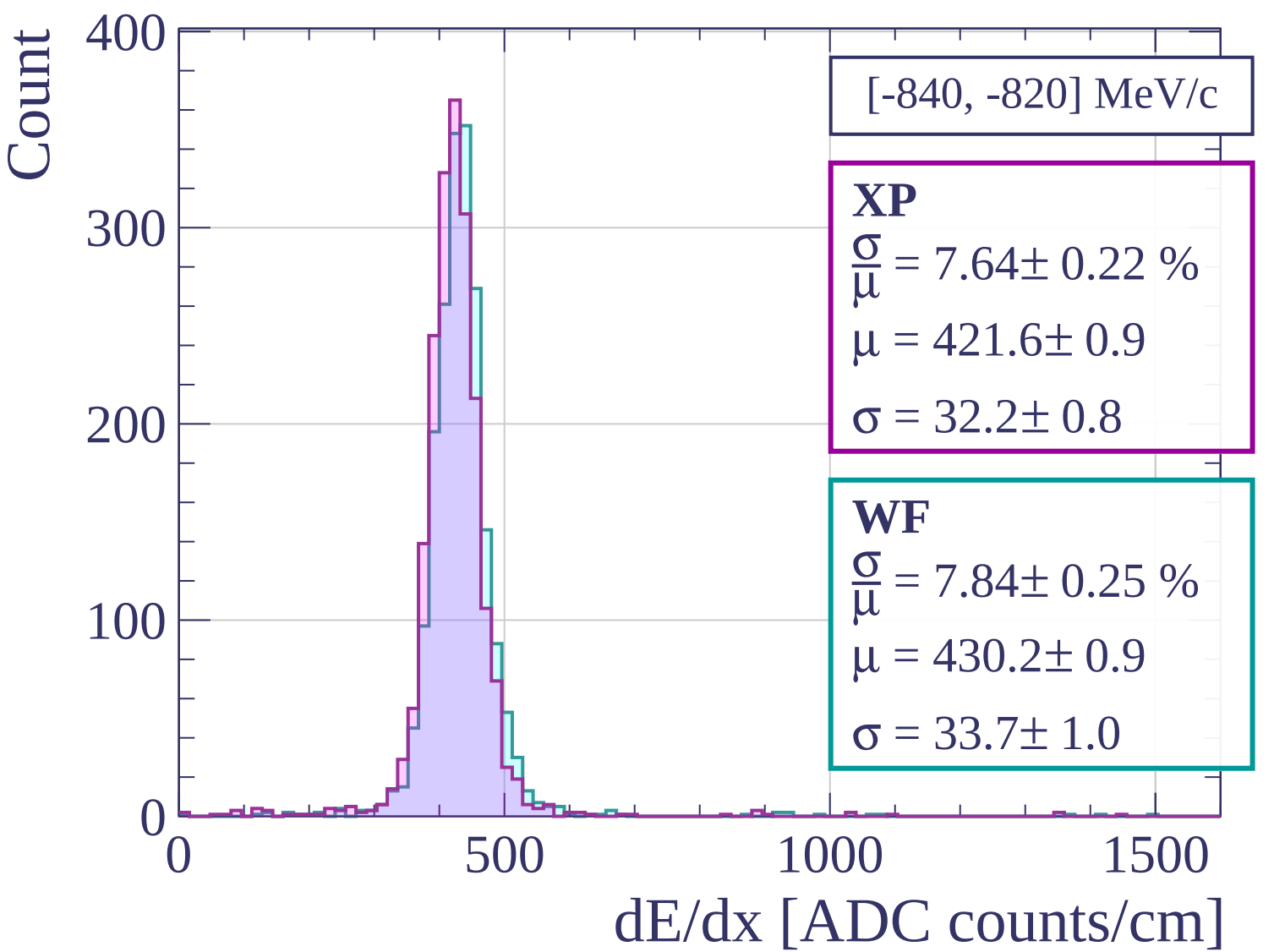
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-820, -800] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.70 \pm 0.23 \%$$

$$\mu = 422.3 \pm 0.9$$

$$\sigma = 32.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.83 \pm 0.25 \%$$

$$\mu = 430.5 \pm 0.9$$

$$\sigma = 33.7 \pm 1.0$$

300

200

100

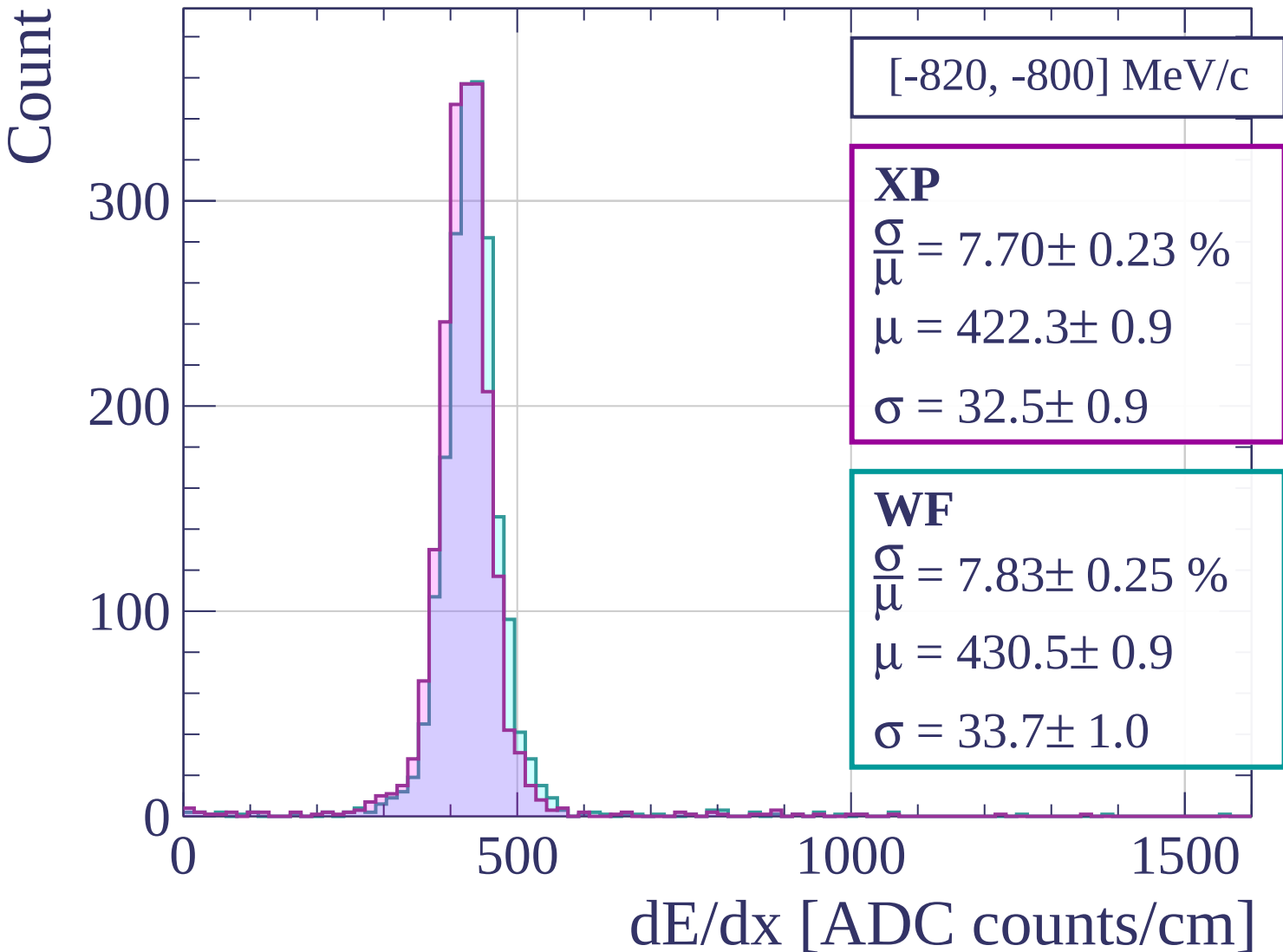
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-800, -780] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.44 \pm 0.26 \%$$

$$\mu = 421.1 \pm 0.9$$

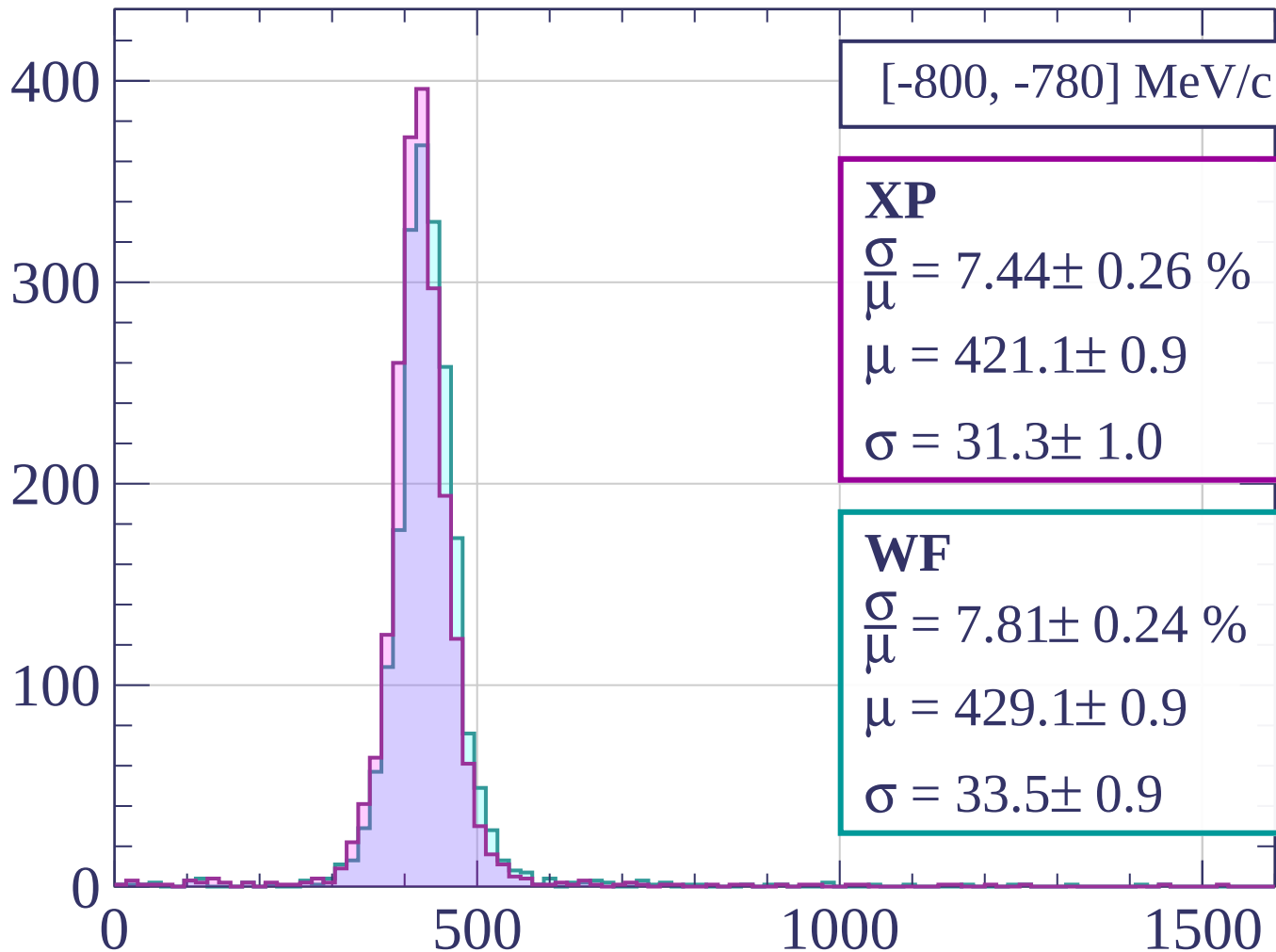
$$\sigma = 31.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.81 \pm 0.24 \%$$

$$\mu = 429.1 \pm 0.9$$

$$\sigma = 33.5 \pm 0.9$$



dE/dx [ADC counts/cm]

Count

[-780, -760] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.78 \pm 0.23 \%$$

$$\mu = 419.8 \pm 0.9$$

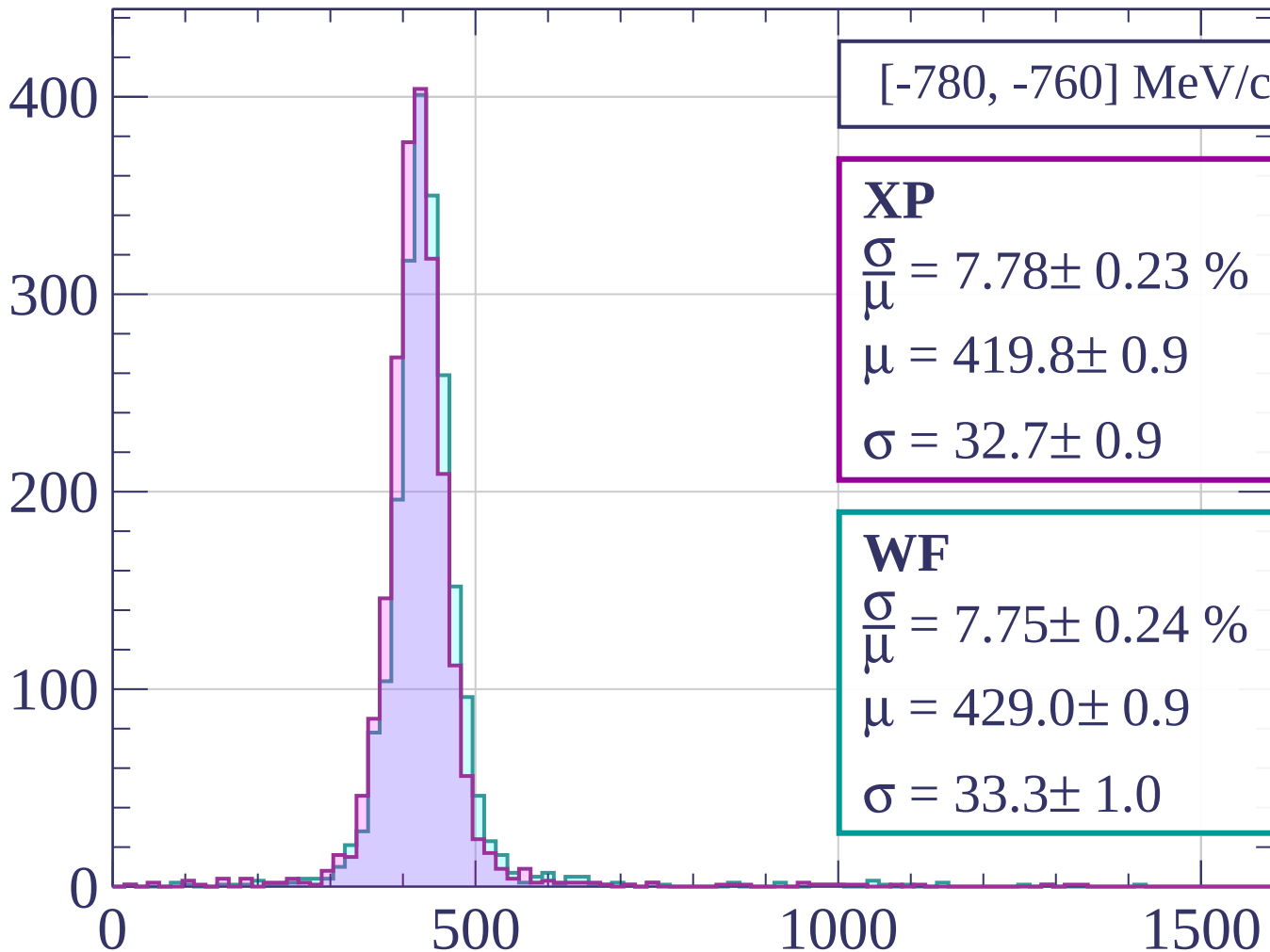
$$\sigma = 32.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.75 \pm 0.24 \%$$

$$\mu = 429.0 \pm 0.9$$

$$\sigma = 33.3 \pm 1.0$$



dE/dx [ADC counts/cm]

Count

[-760, -740] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.56 \pm 0.20 \%$$

$$\mu = 417.9 \pm 0.8$$

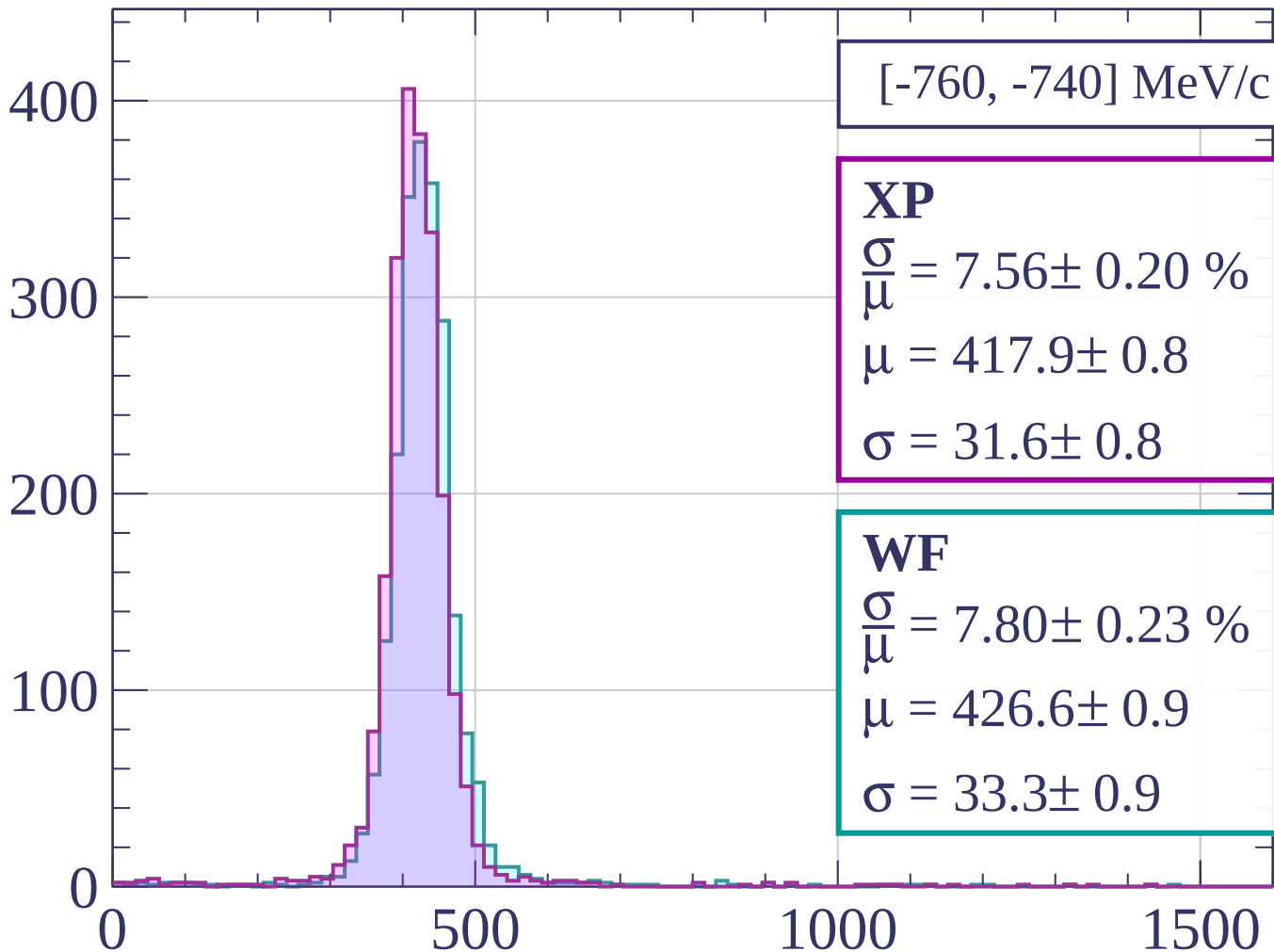
$$\sigma = 31.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.80 \pm 0.23 \%$$

$$\mu = 426.6 \pm 0.9$$

$$\sigma = 33.3 \pm 0.9$$



dE/dx [ADC counts/cm]

Count

[-740, -720] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.46 \pm 0.20 \%$$

$$\mu = 418.6 \pm 0.8$$

$$\sigma = 31.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.66 \pm 0.21 \%$$

$$\mu = 427.2 \pm 0.8$$

$$\sigma = 32.7 \pm 0.8$$

400

300

200

100

0

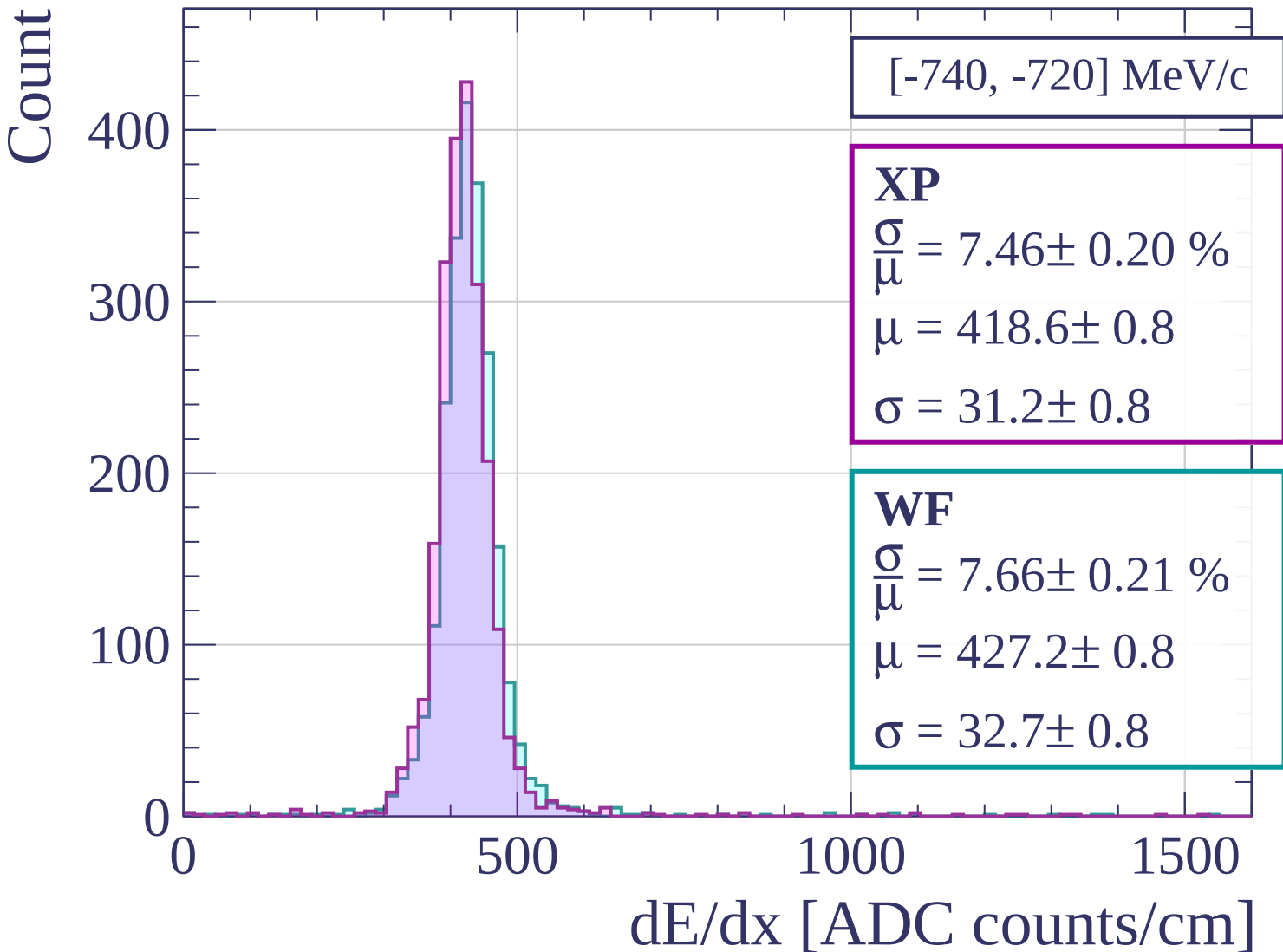
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-720, -700] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.61 \pm 0.19 \%$$

$$\mu = 416.1 \pm 0.8$$

$$\sigma = 31.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.08 \pm 0.20 \%$$

$$\mu = 424.8 \pm 0.8$$

$$\sigma = 34.3 \pm 0.8$$

400

200

0

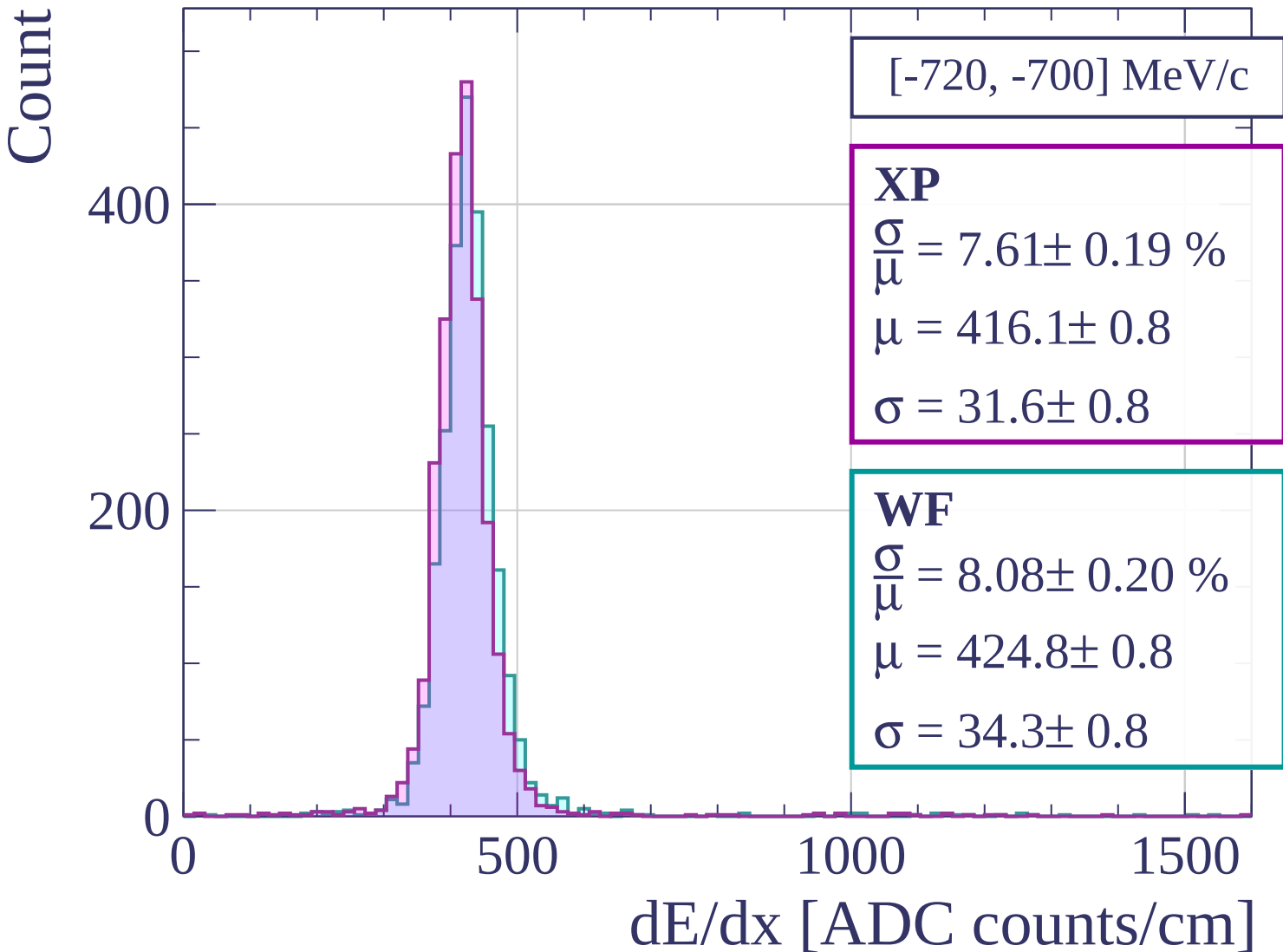
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-700, -680] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.93 \pm 0.22 \%$$

$$\mu = 413.0 \pm 0.8$$

$$\sigma = 32.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.08 \pm 0.19 \%$$

$$\mu = 421.5 \pm 0.8$$

$$\sigma = 34.1 \pm 0.7$$

400

200

0

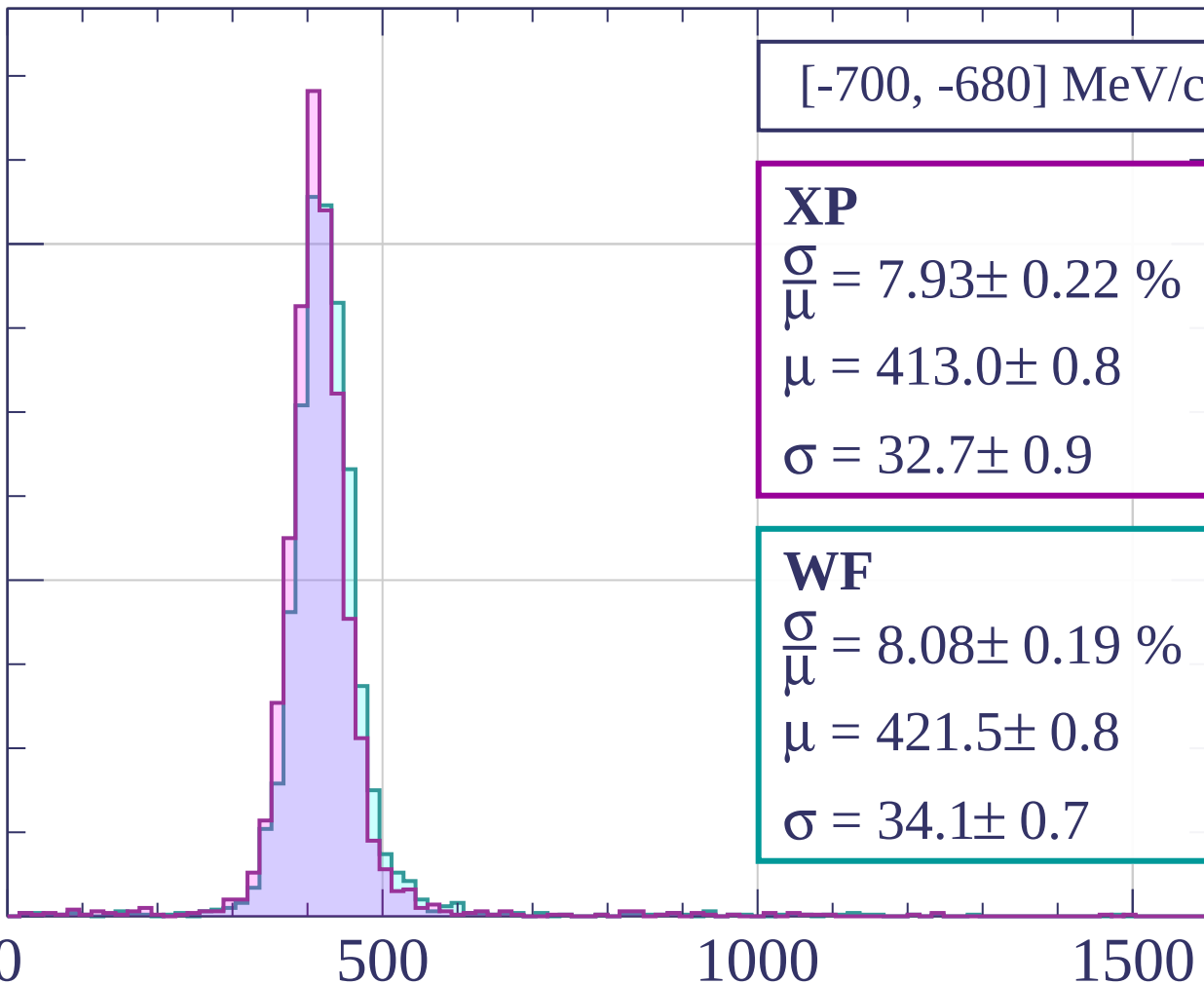
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-680, -660] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.78 \pm 0.20 \%$$

$$\mu = 412.7 \pm 0.7$$

$$\sigma = 32.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.90 \pm 0.21 \%$$

$$\mu = 420.3 \pm 0.8$$

$$\sigma = 33.2 \pm 0.8$$

400

200

0

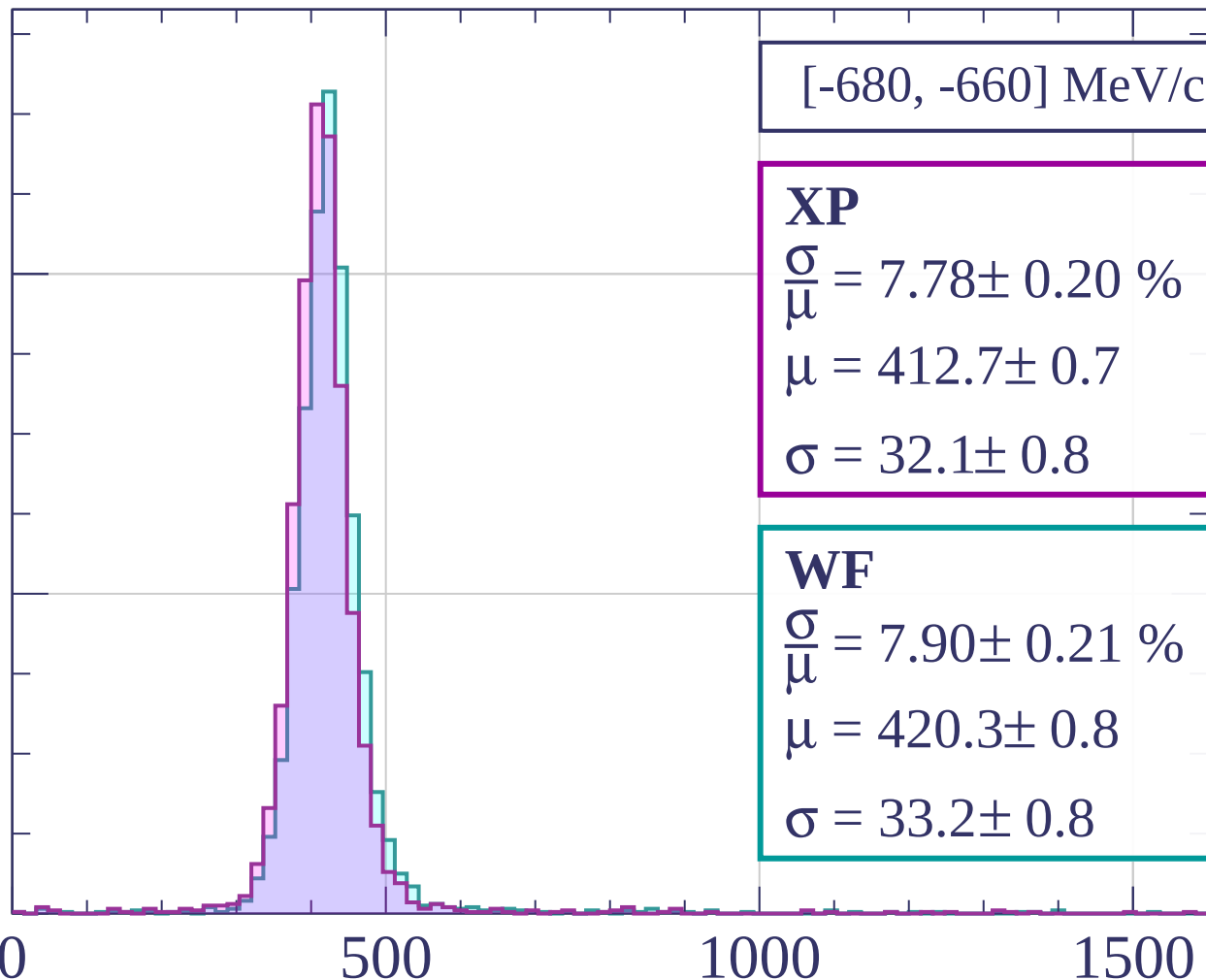
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-660, -640] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.74 \pm 0.20 \%$$

$$\mu = 412.9 \pm 0.7$$

$$\sigma = 32.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.02 \pm 0.22 \%$$

$$\mu = 420.5 \pm 0.8$$

$$\sigma = 33.7 \pm 0.9$$

400

200

0

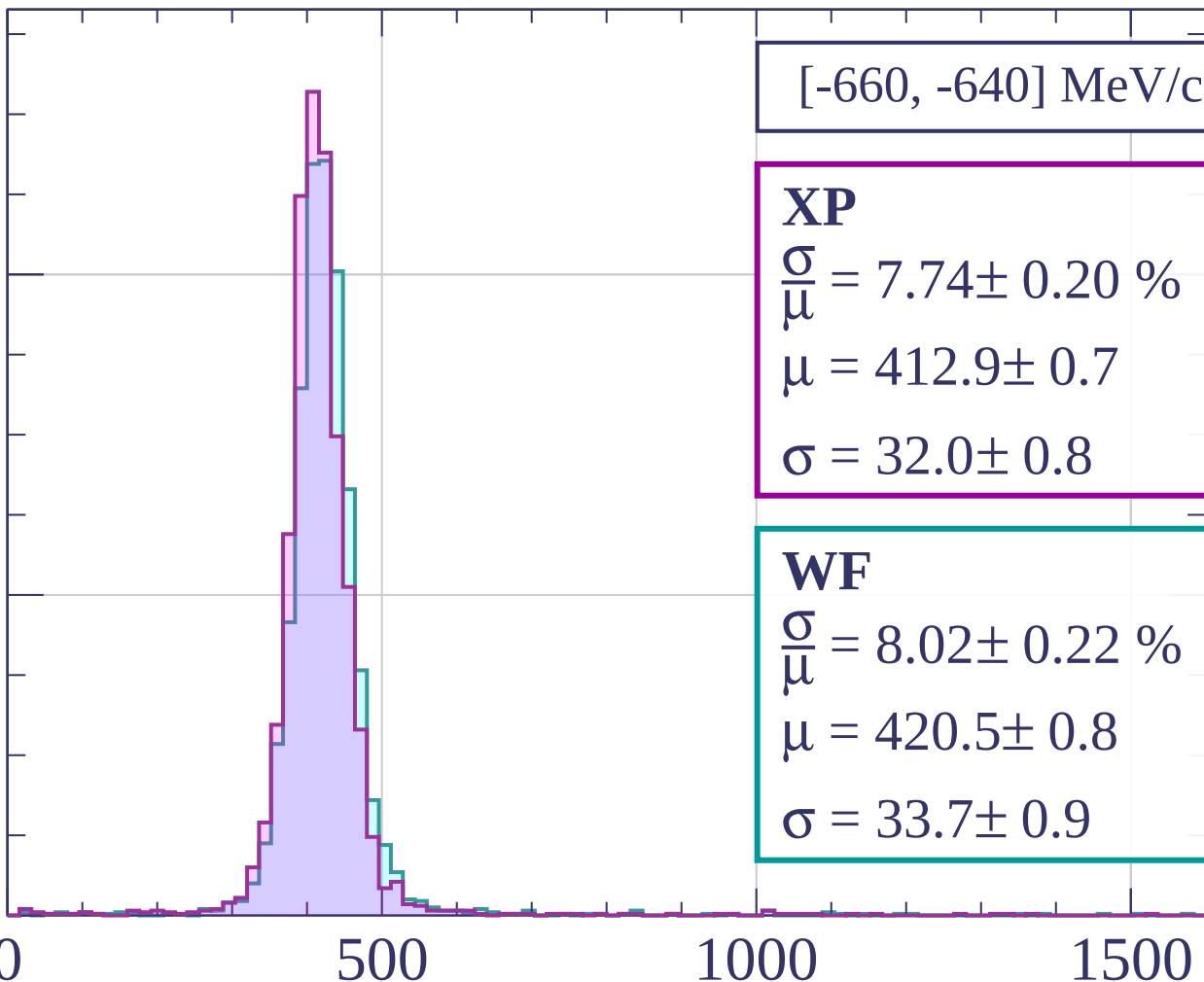
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-640, -620] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.89 \pm 0.20 \%$$

$$\mu = 411.9 \pm 0.7$$

$$\sigma = 32.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.95 \pm 0.21 \%$$

$$\mu = 419.6 \pm 0.8$$

$$\sigma = 33.3 \pm 0.8$$

400

200

0

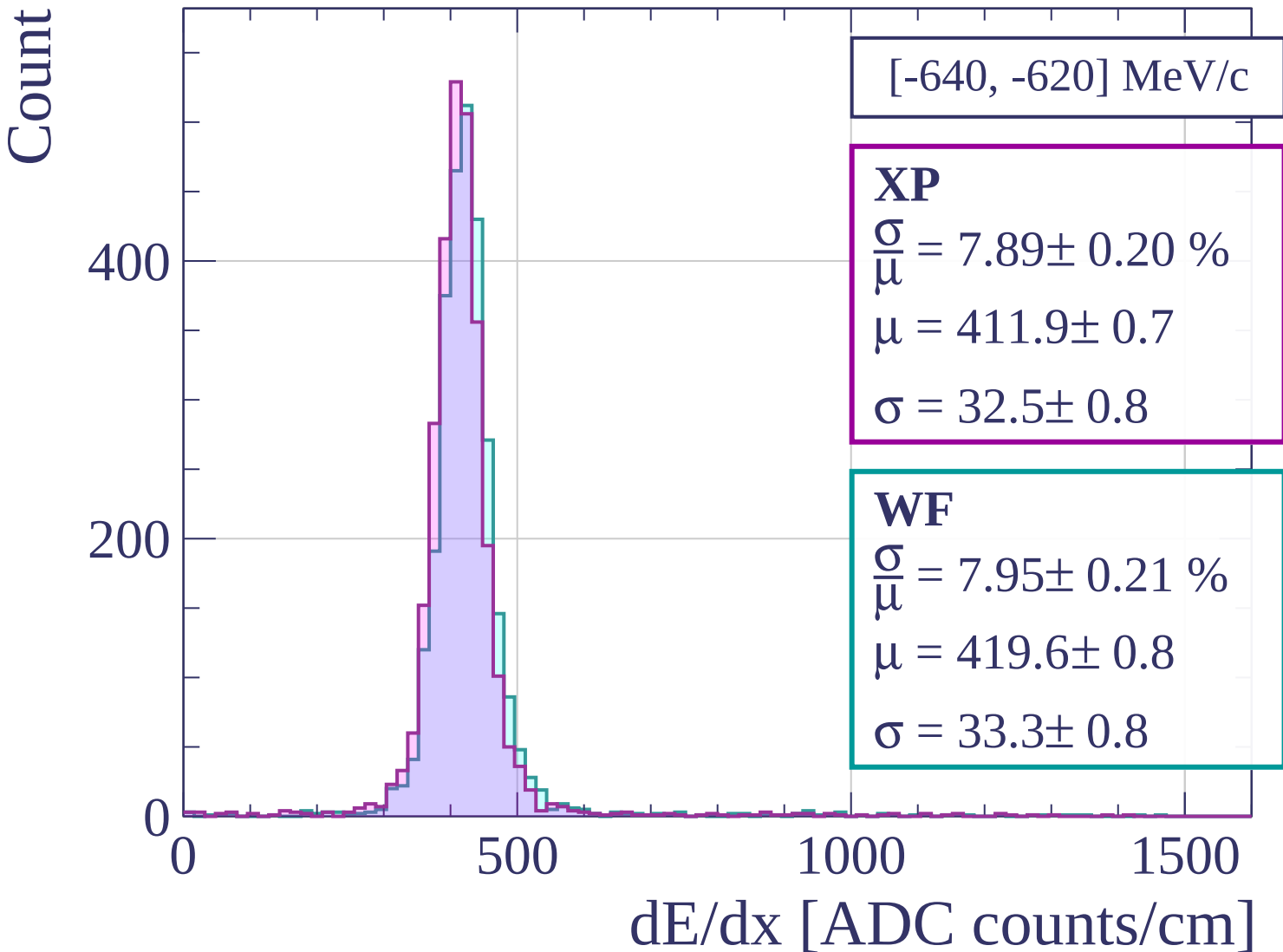
0

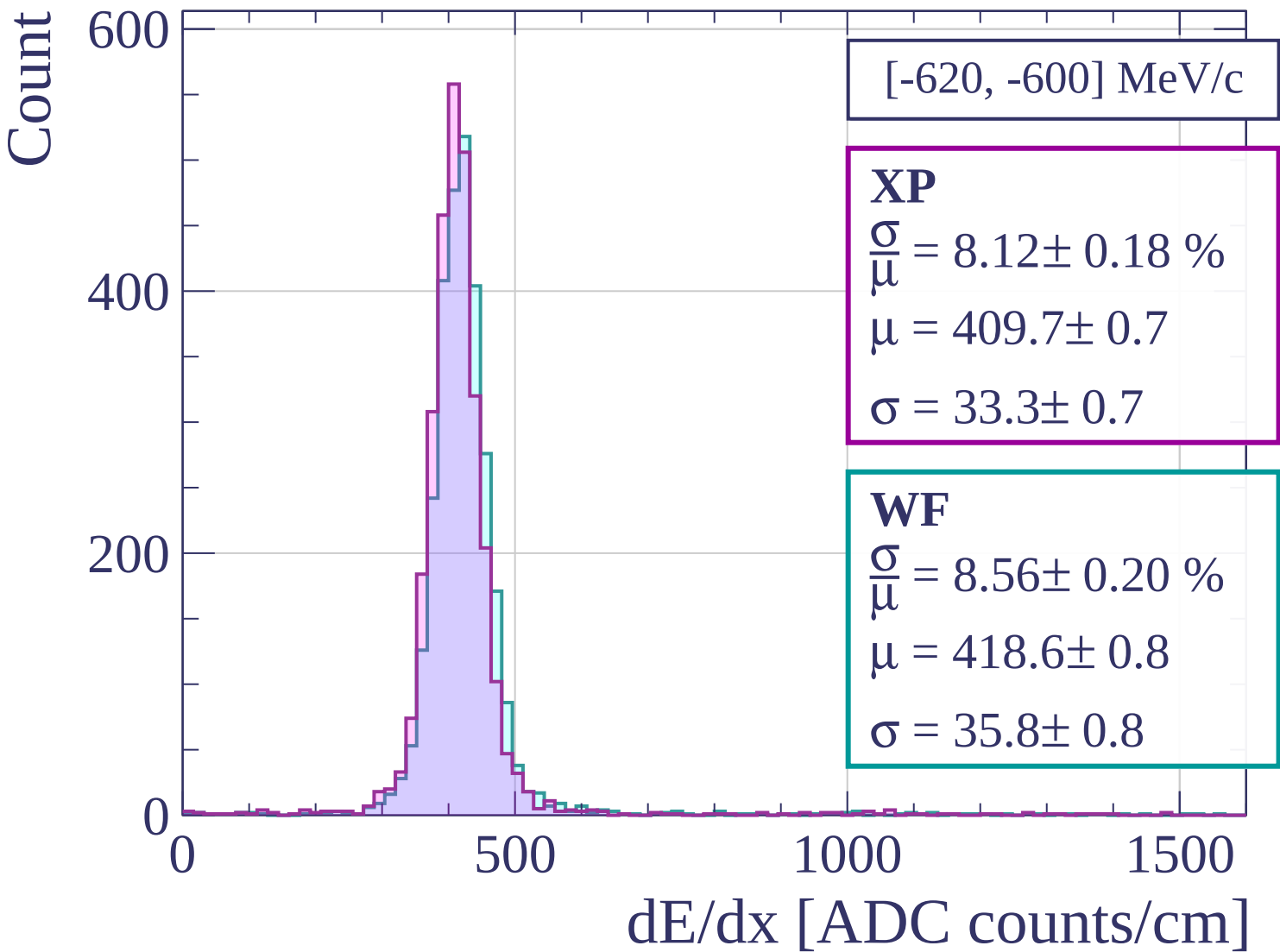
500

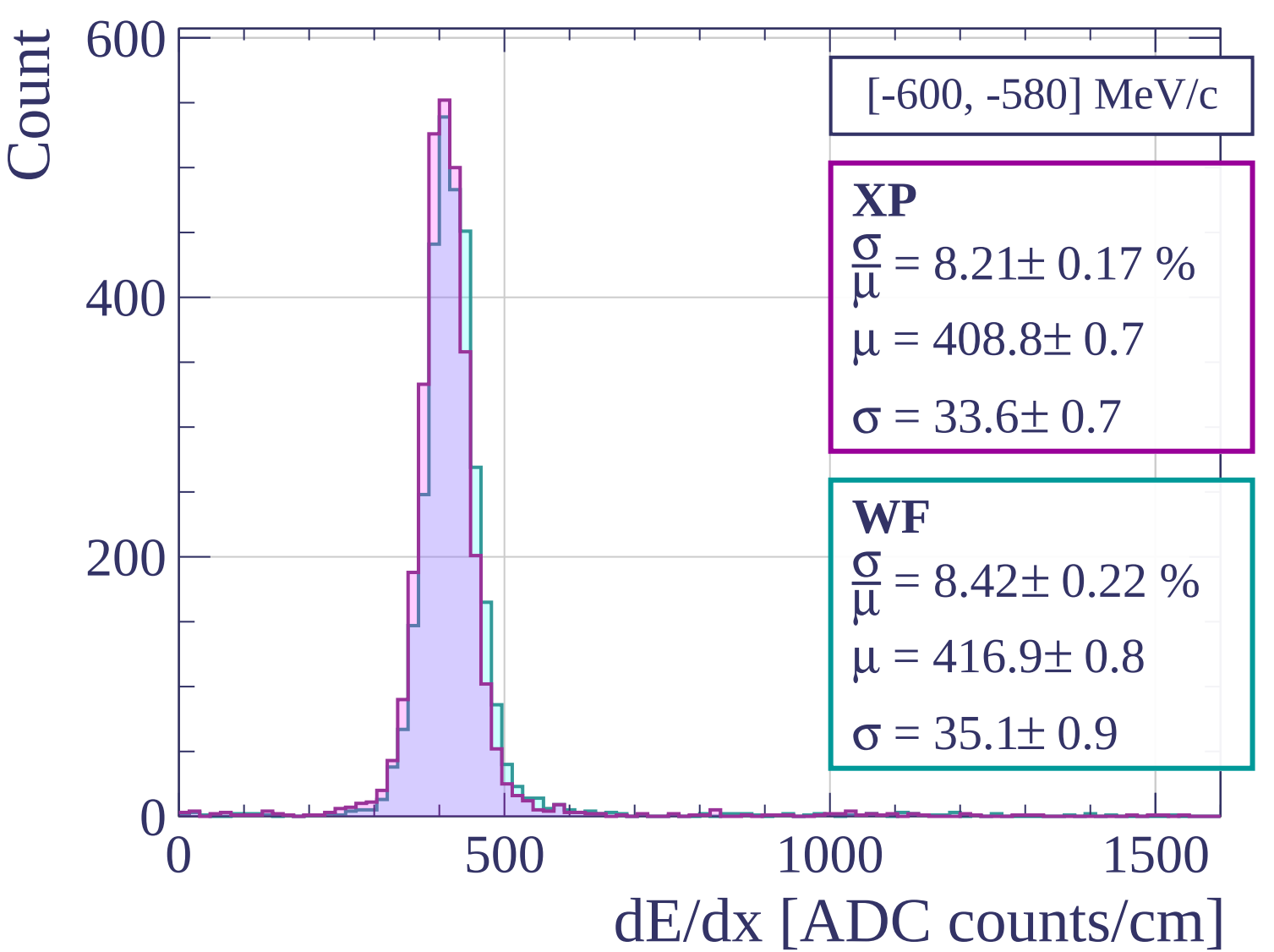
1000

1500

dE/dx [ADC counts/cm]







Count

[-580, -560] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.08 \pm 0.16 \%$$

$$\mu = 407.5 \pm 0.7$$

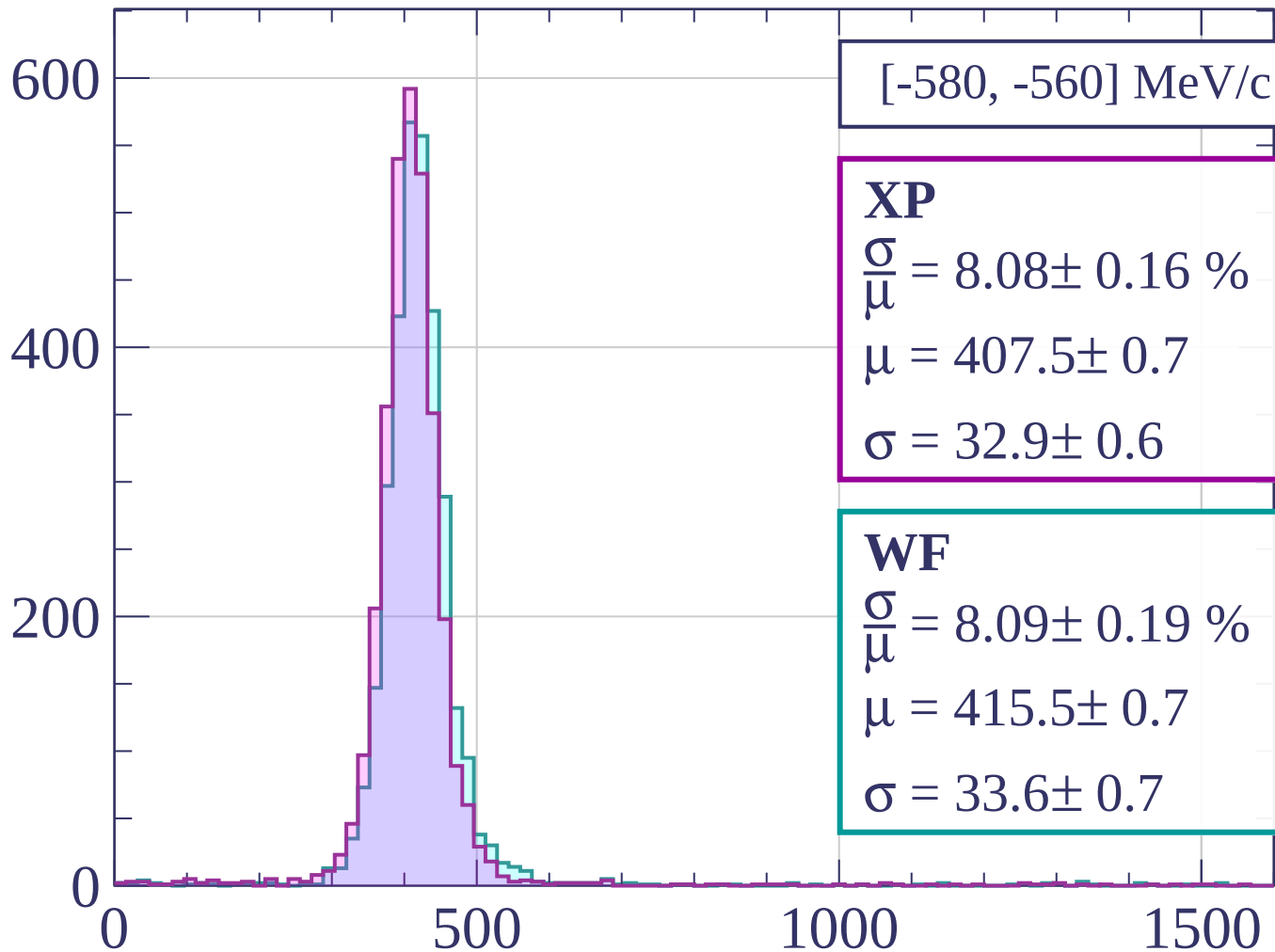
$$\sigma = 32.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.09 \pm 0.19 \%$$

$$\mu = 415.5 \pm 0.7$$

$$\sigma = 33.6 \pm 0.7$$



dE/dx [ADC counts/cm]

Count

[-560, -540] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.62 \pm 0.20 \%$$

$$\mu = 404.9 \pm 0.7$$

$$\sigma = 30.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.85 \pm 0.18 \%$$

$$\mu = 413.4 \pm 0.7$$

$$\sigma = 32.5 \pm 0.7$$

600

400

200

0

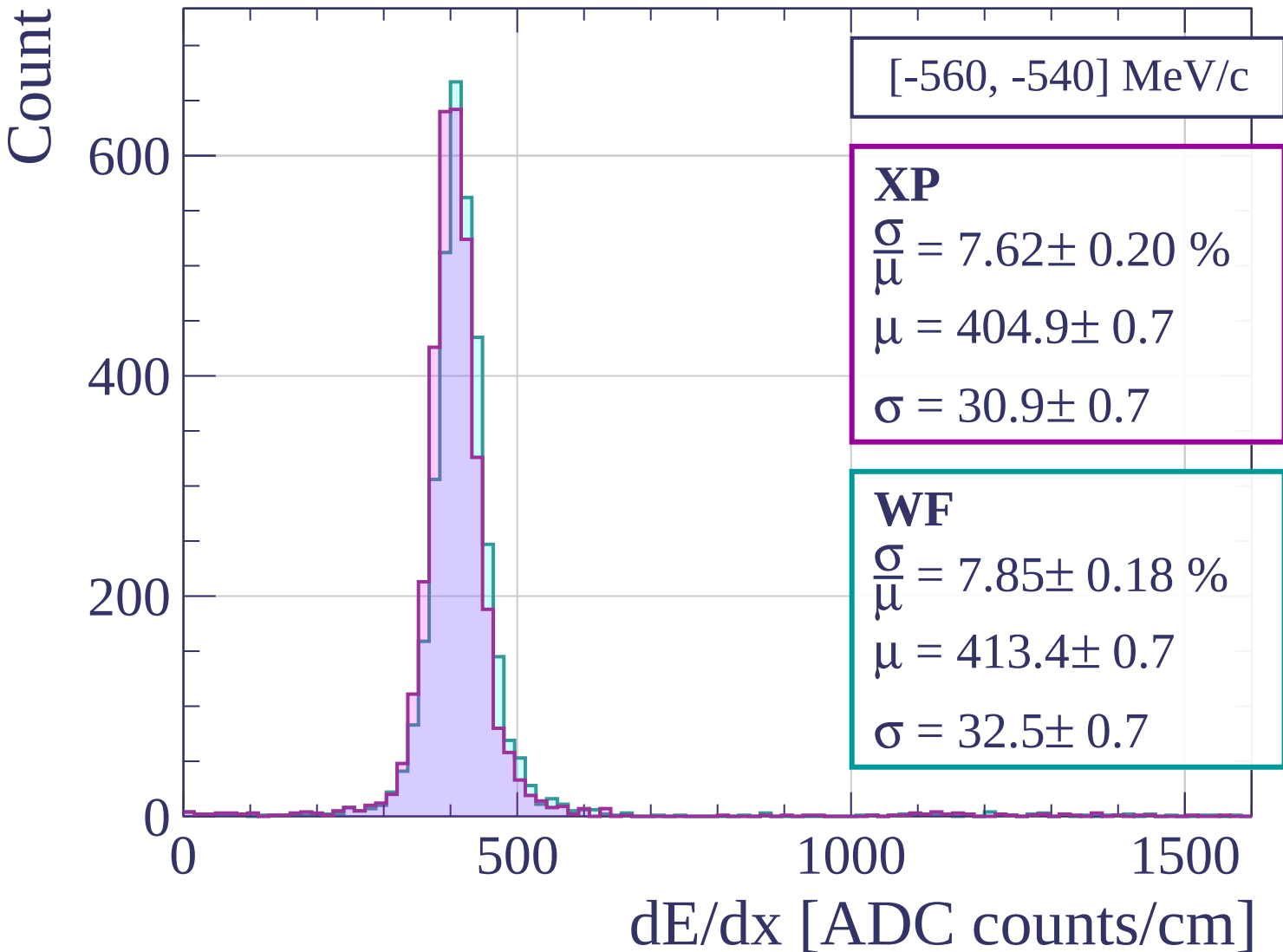
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-540, -520] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.07 \pm 0.18 \%$$

$$\mu = 405.1 \pm 0.7$$

$$\sigma = 32.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.25 \pm 0.20 \%$$

$$\mu = 413.1 \pm 0.7$$

$$\sigma = 34.1 \pm 0.8$$

600

400

200

0

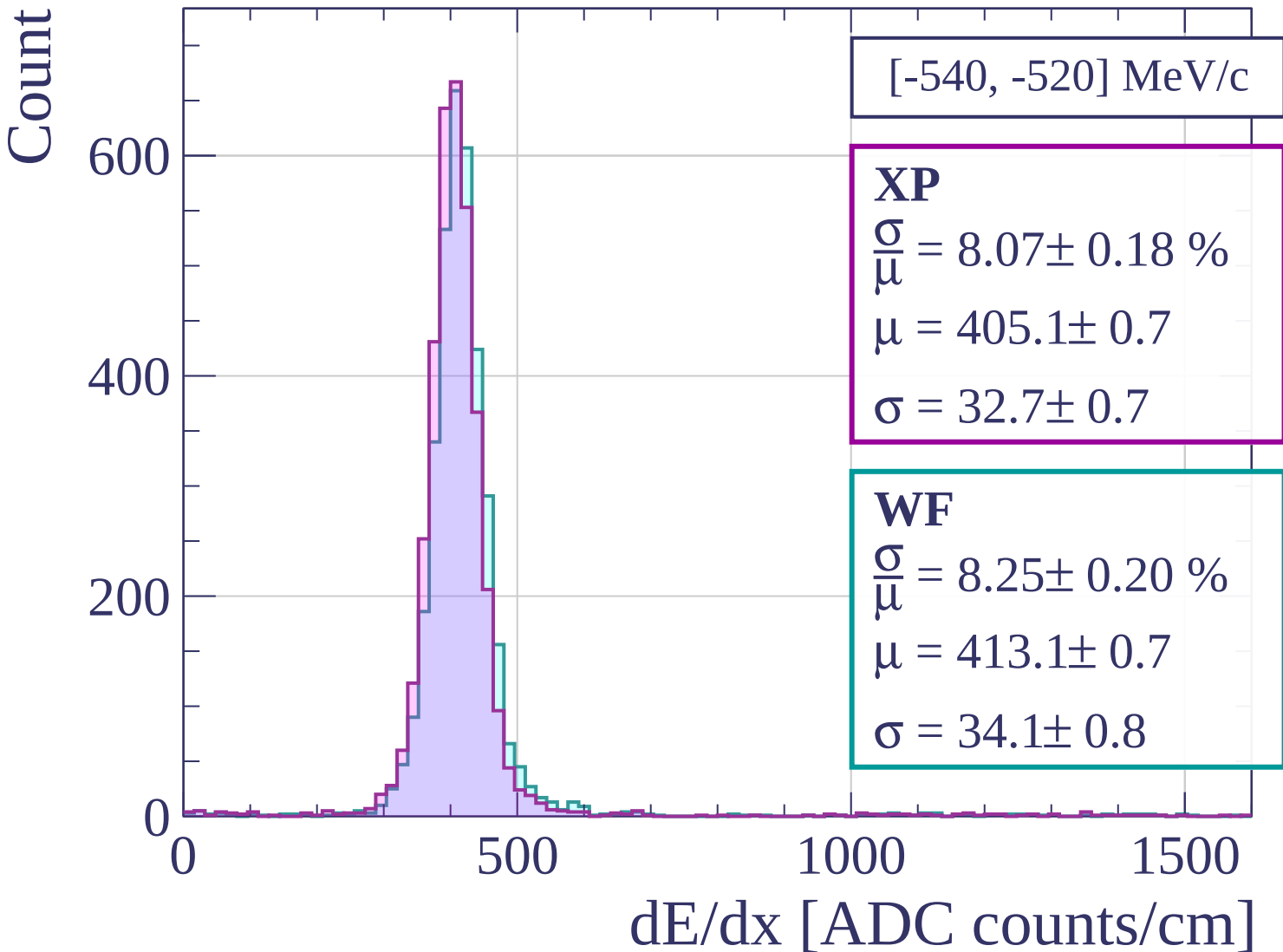
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-520, -500] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.88 \pm 0.17 \%$$

$$\mu = 404.9 \pm 0.6$$

$$\sigma = 31.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.37 \pm 0.20 \%$$

$$\mu = 412.4 \pm 0.7$$

$$\sigma = 34.5 \pm 0.8$$

600

400

200

0

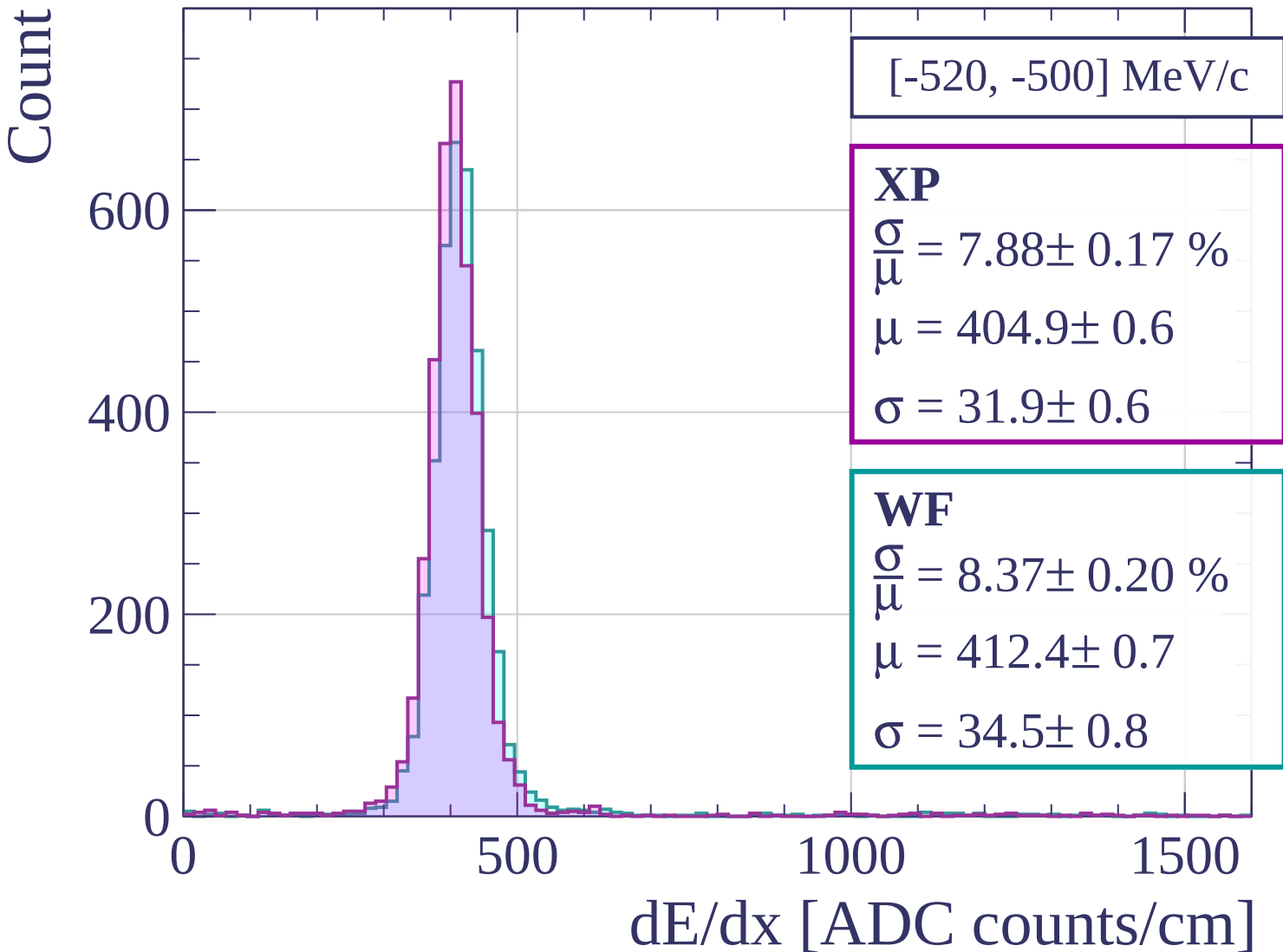
0

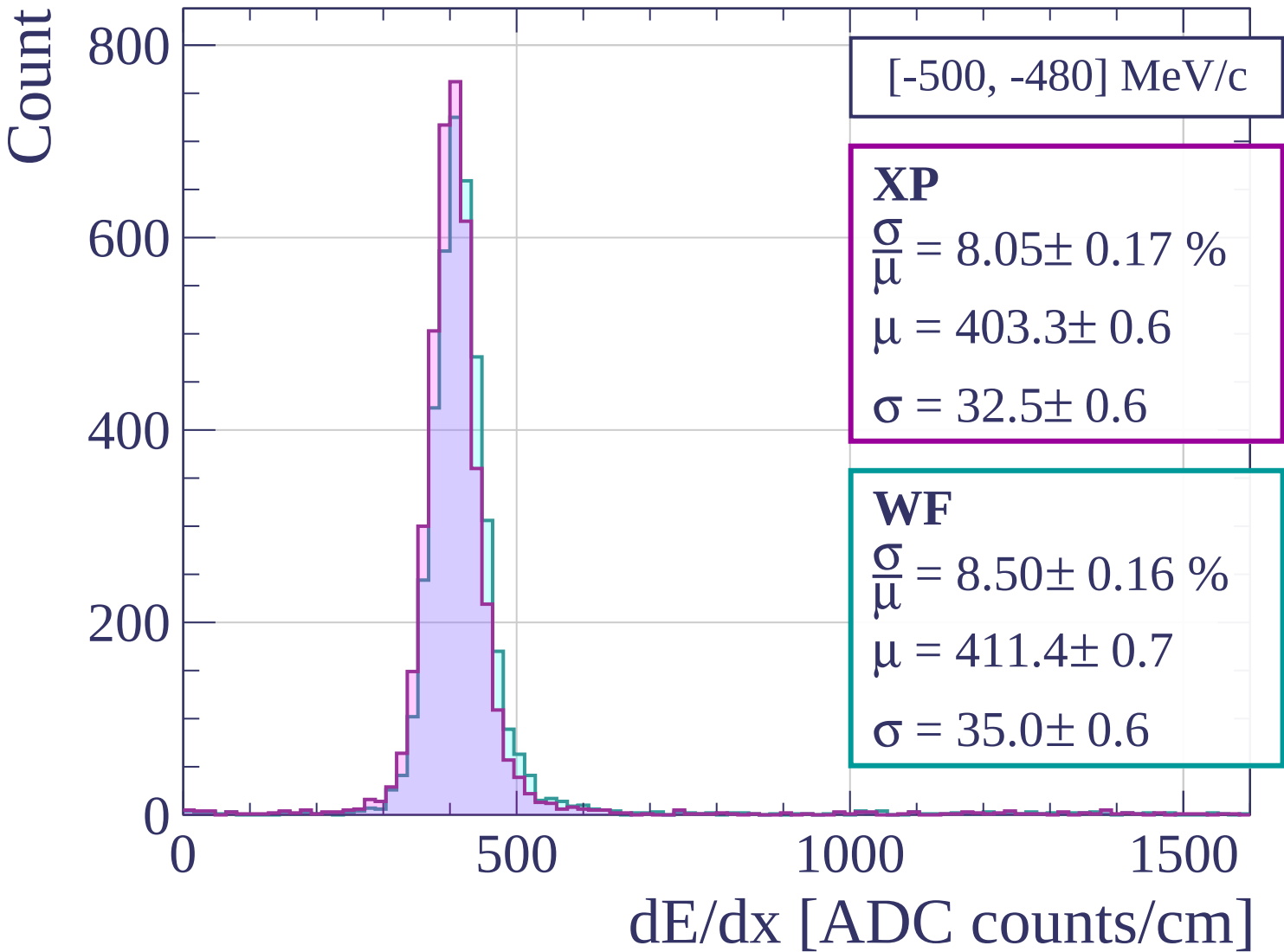
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-480, -460] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.86 \pm 0.15 \%$$

$$\mu = 401.6 \pm 0.6$$

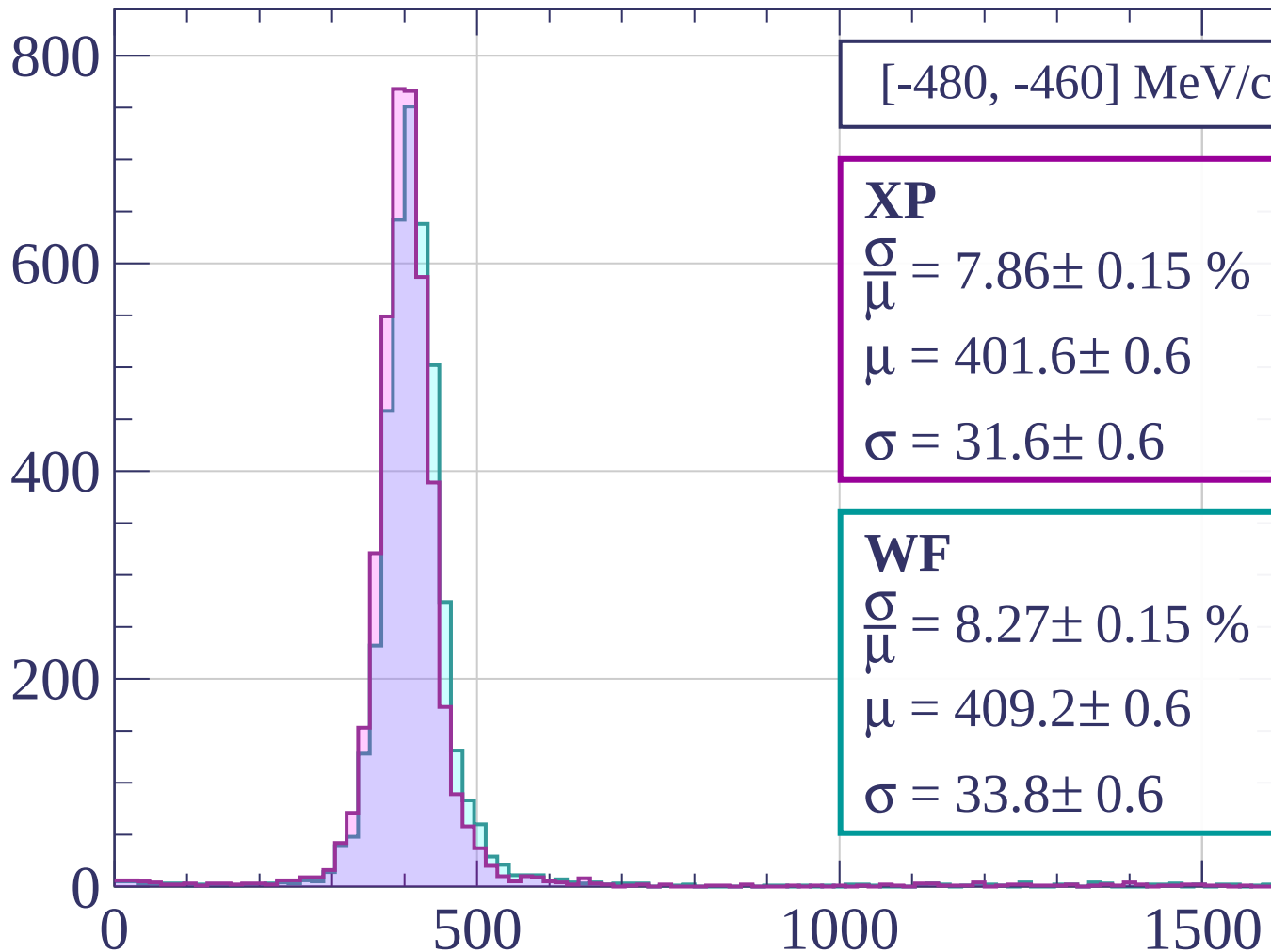
$$\sigma = 31.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.27 \pm 0.15 \%$$

$$\mu = 409.2 \pm 0.6$$

$$\sigma = 33.8 \pm 0.6$$



dE/dx [ADC counts/cm]

Count

[-460, -440] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.00 \pm 0.16 \%$$

$$\mu = 401.1 \pm 0.6$$

$$\sigma = 32.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.46 \pm 0.16 \%$$

$$\mu = 408.9 \pm 0.6$$

$$\sigma = 34.6 \pm 0.6$$

800

600

400

200

0

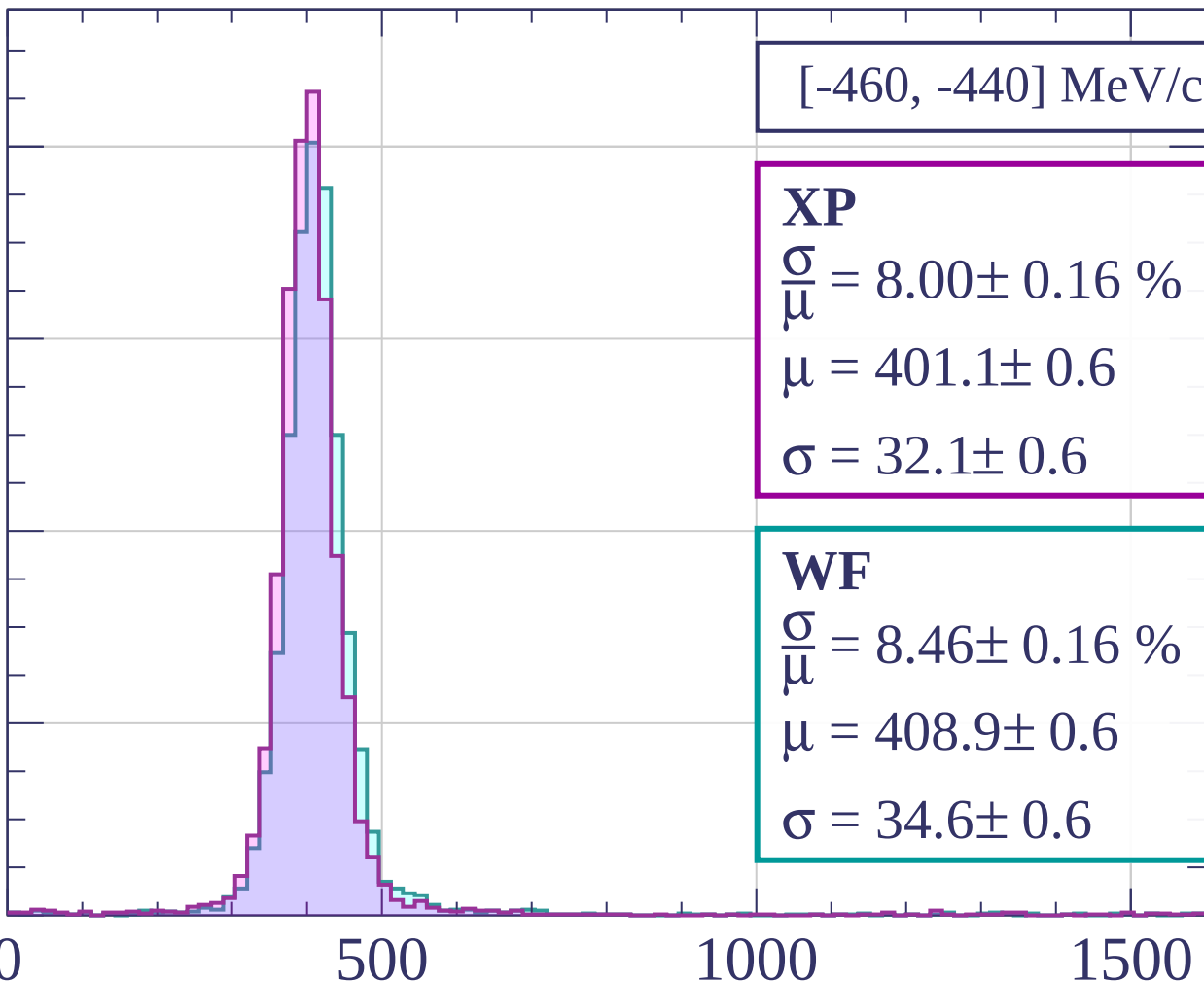
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-440, -420] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.02 \pm 0.15 \%$$

$$\mu = 400.3 \pm 0.6$$

$$\sigma = 32.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.50 \pm 0.16 \%$$

$$\mu = 407.8 \pm 0.6$$

$$\sigma = 34.6 \pm 0.6$$

800

600

400

200

0

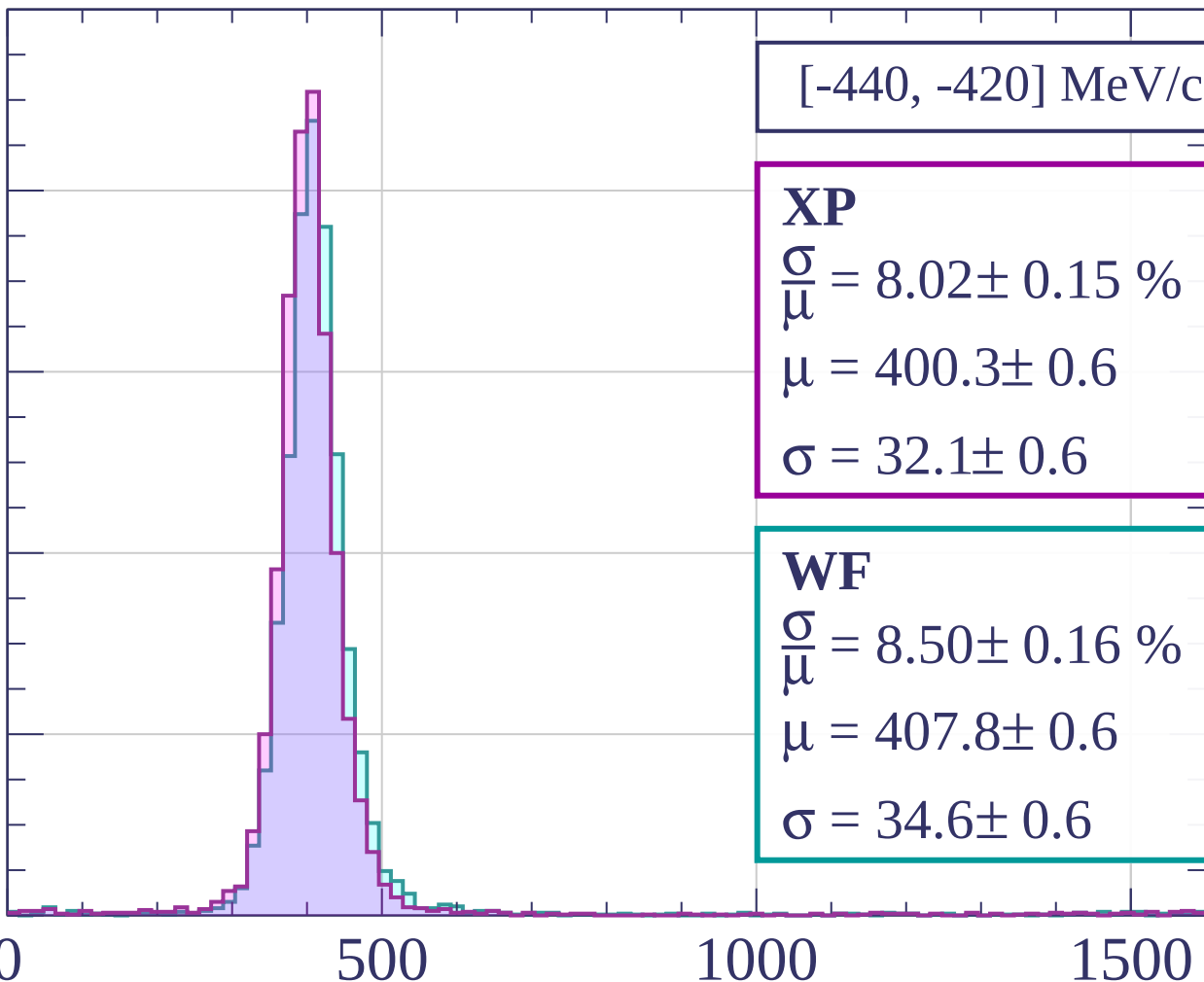
0

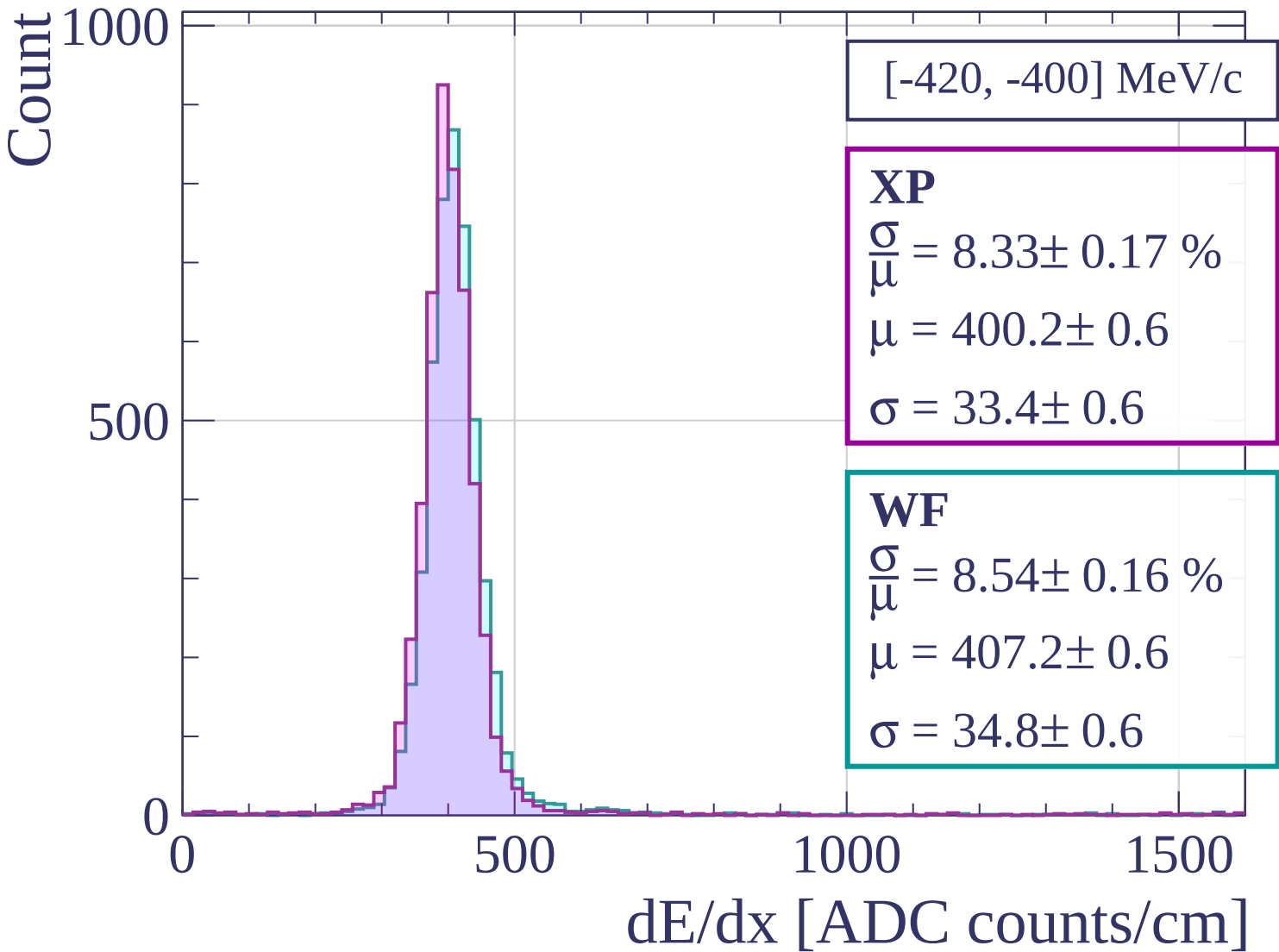
500

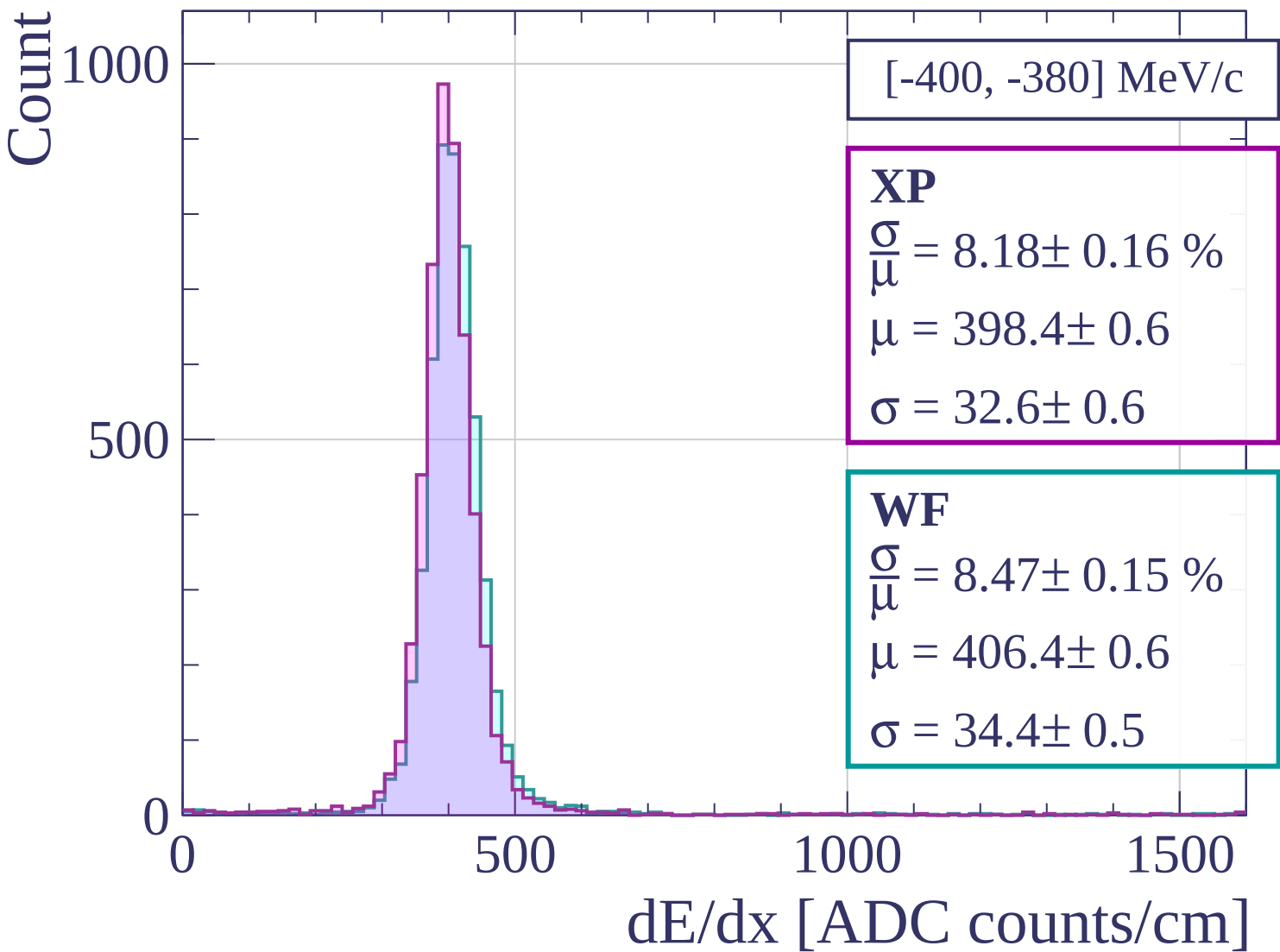
1000

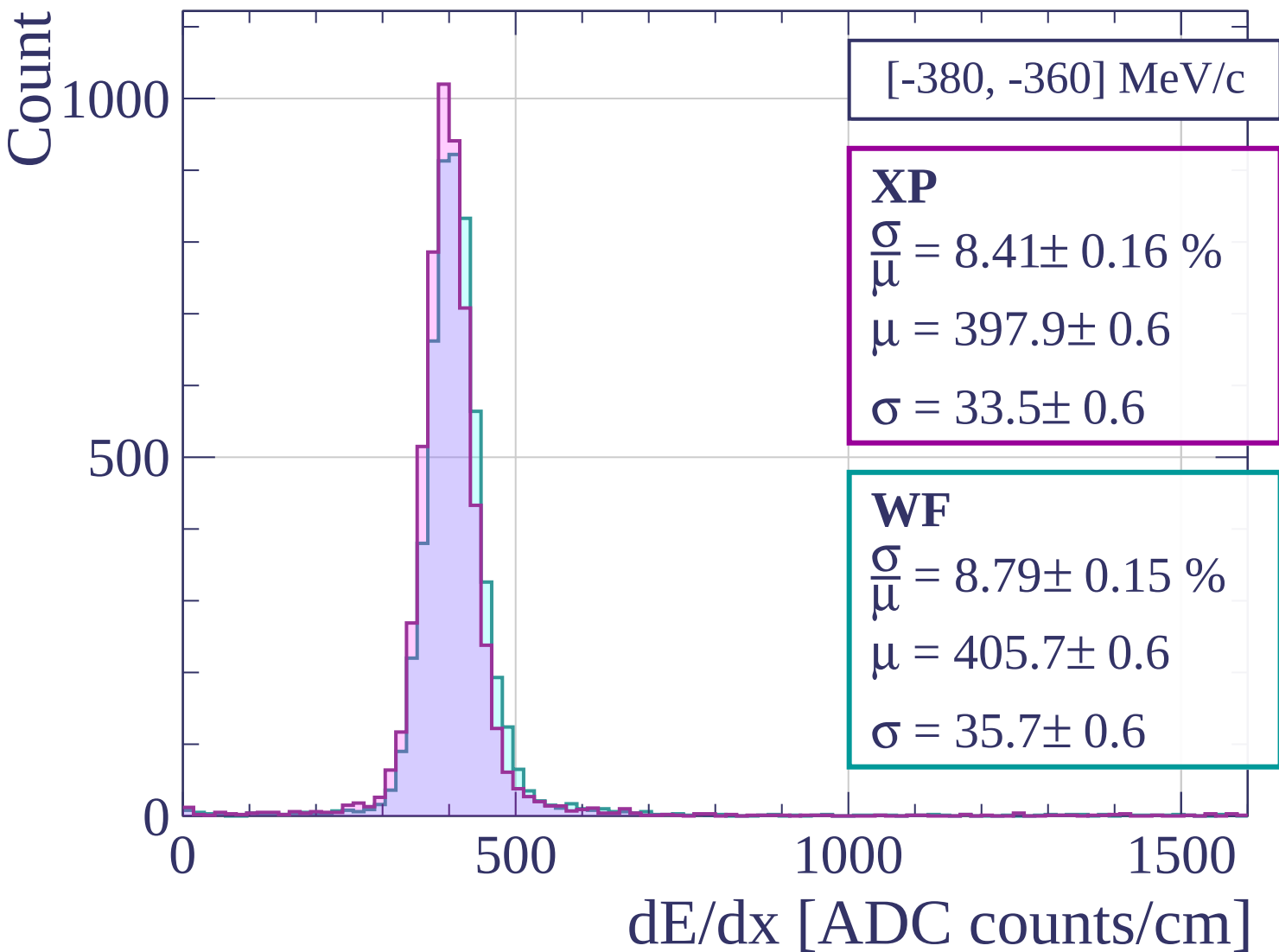
1500

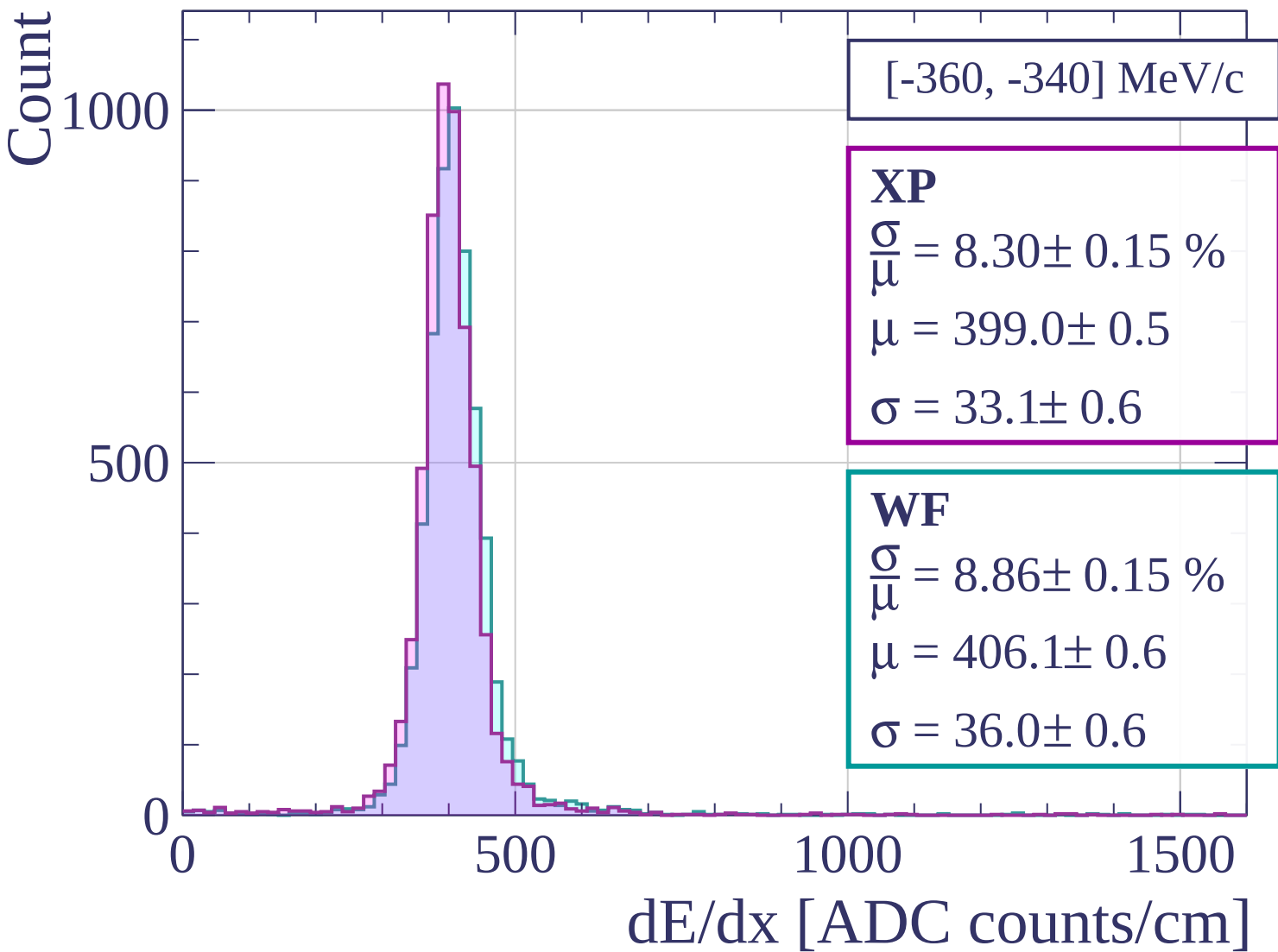
dE/dx [ADC counts/cm]

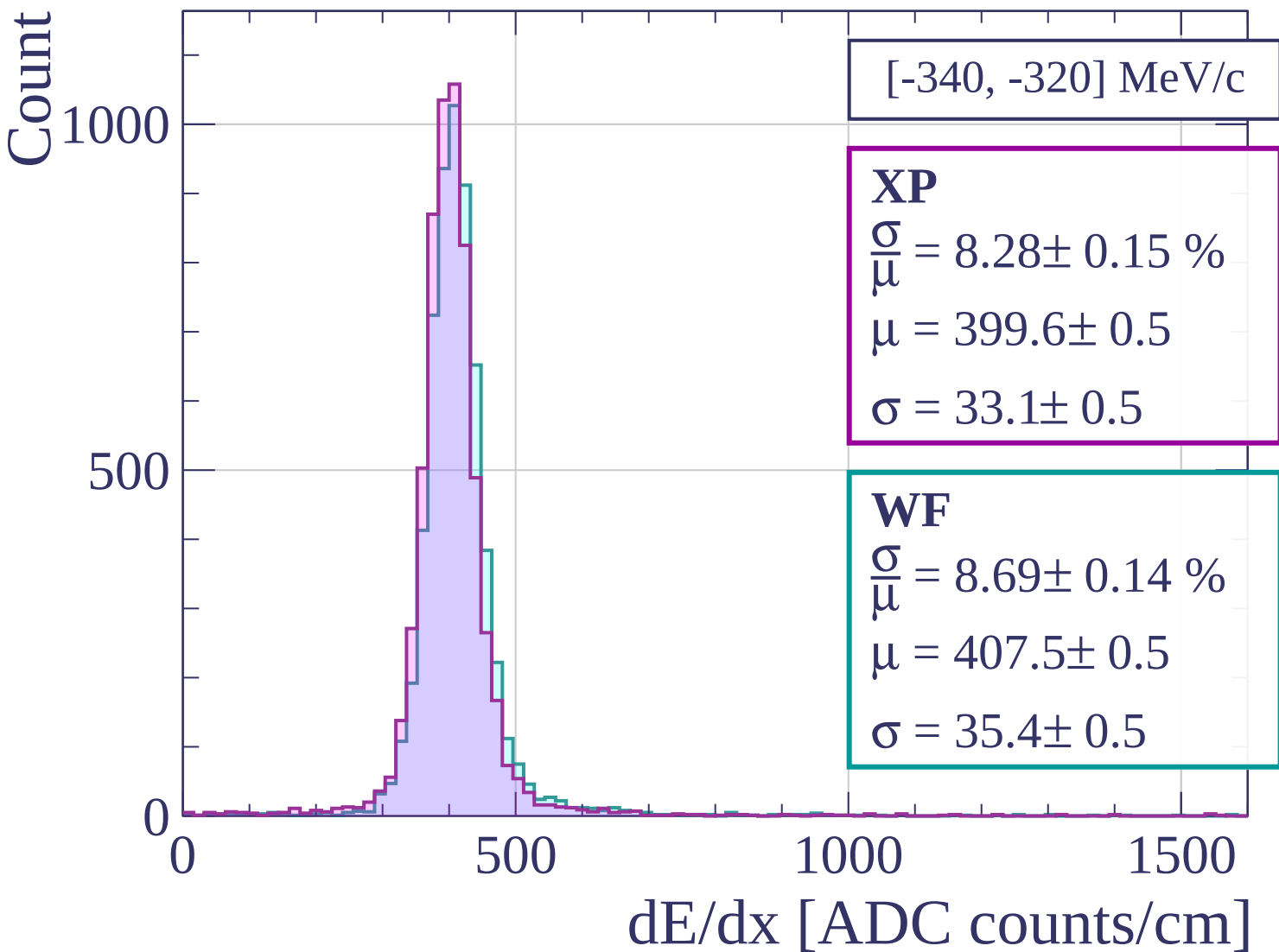


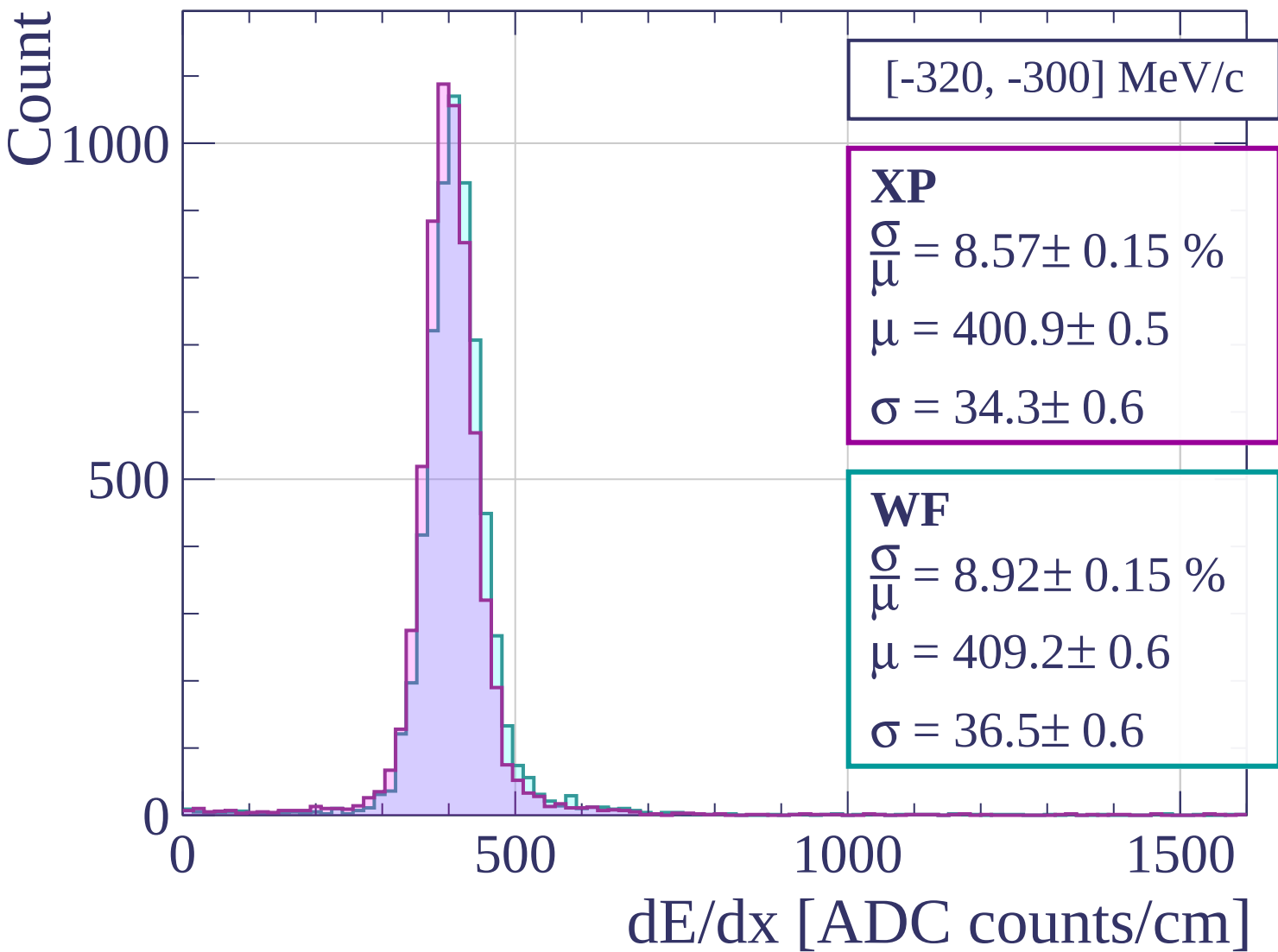


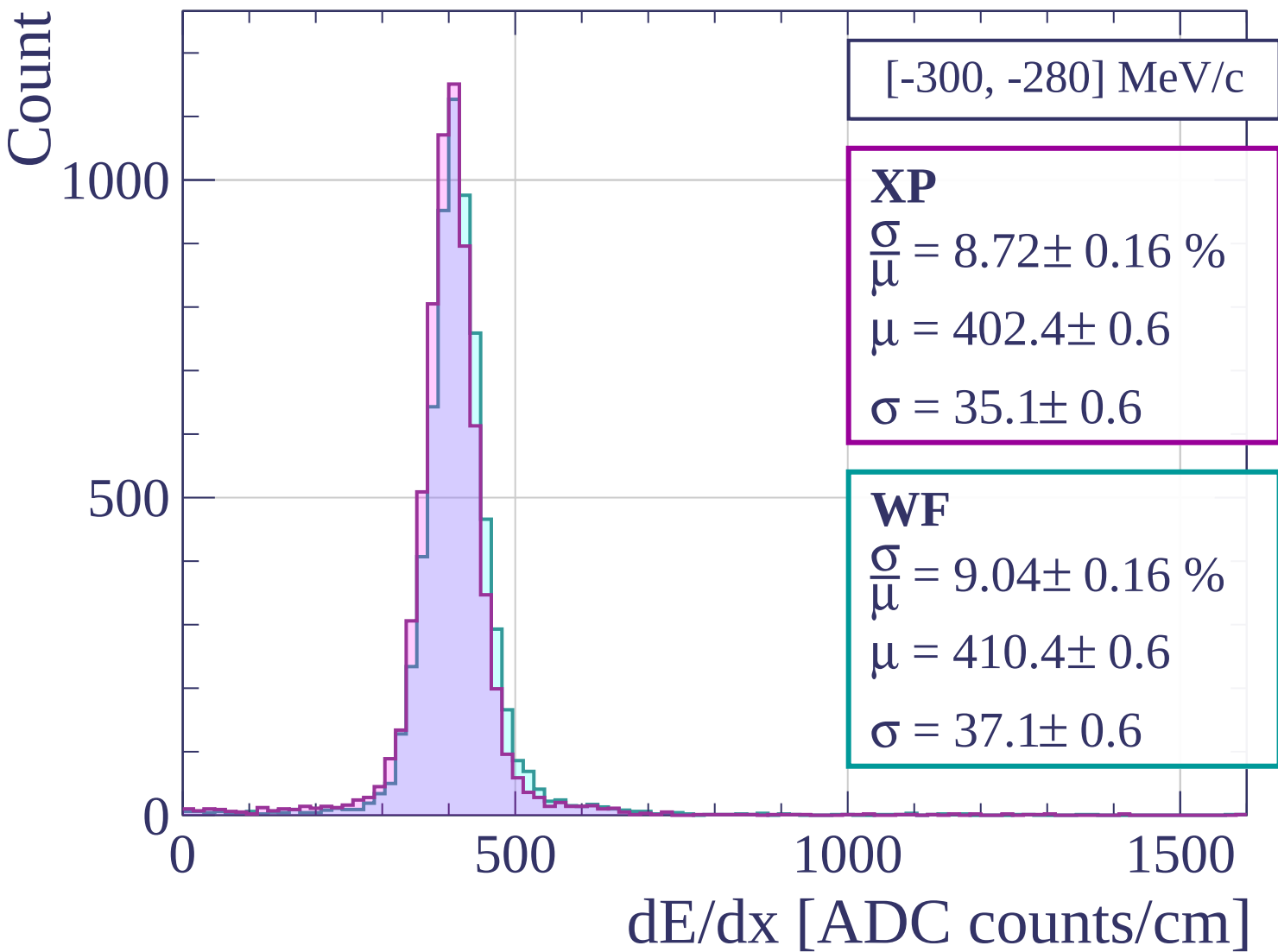


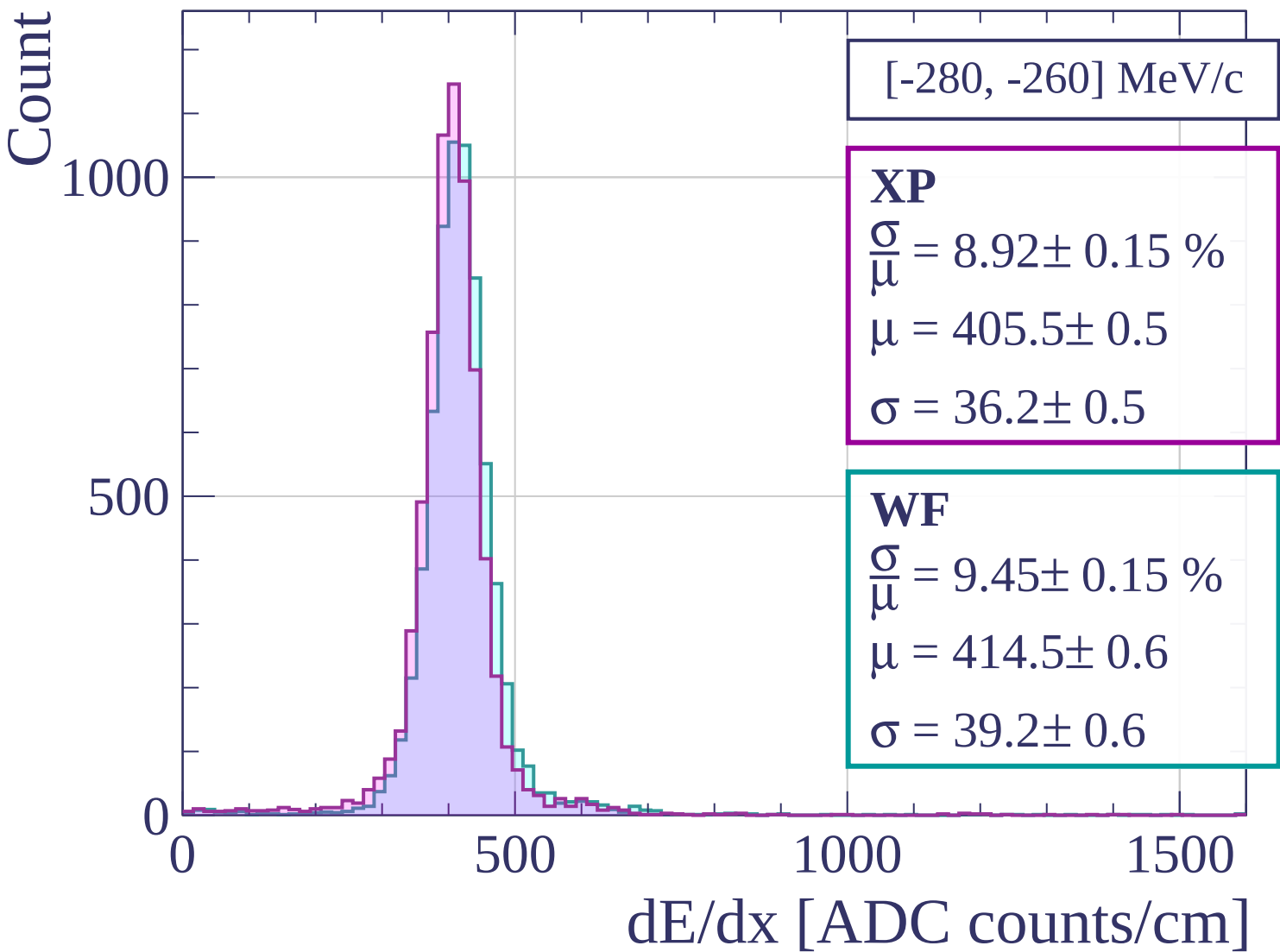


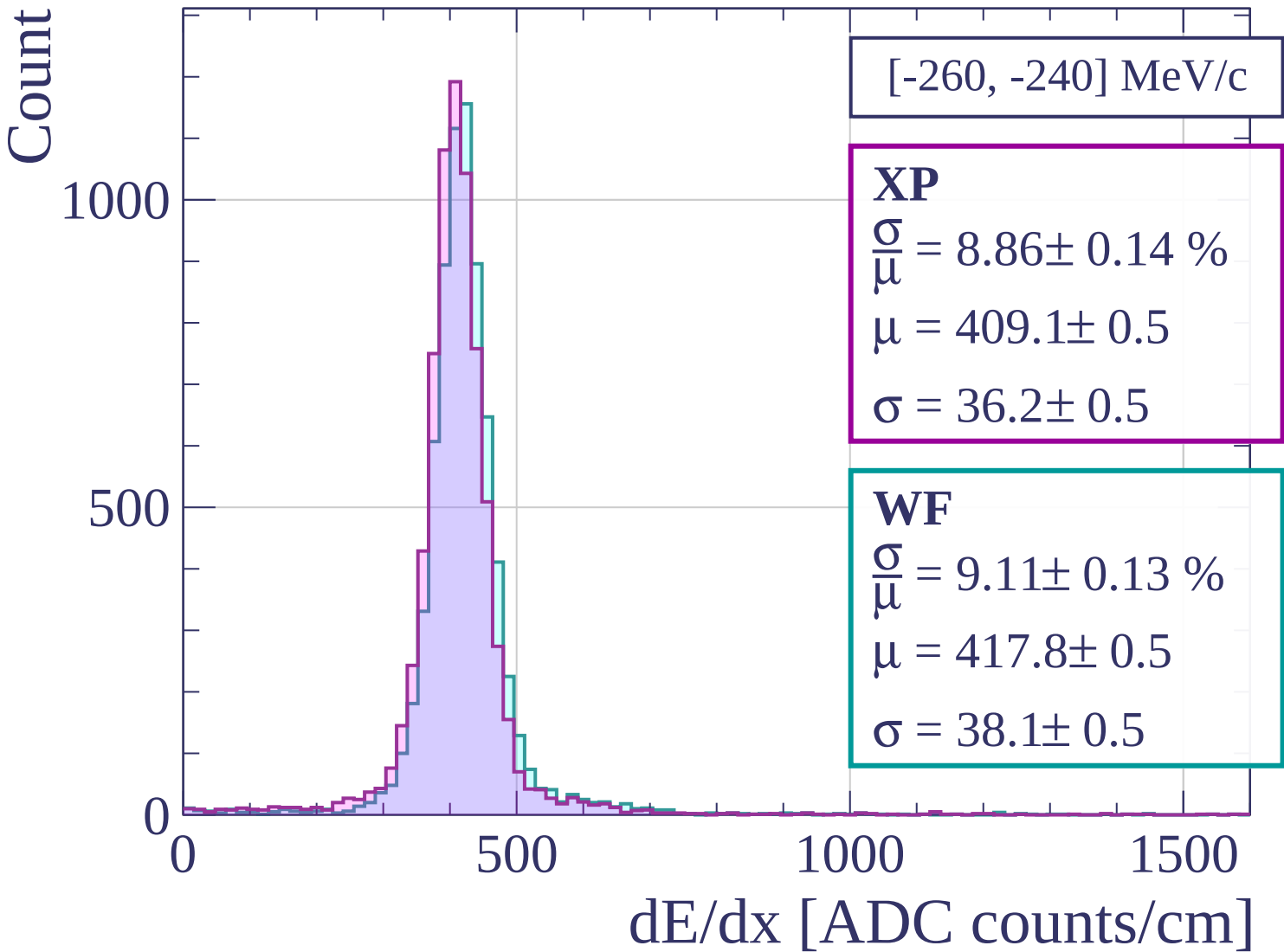


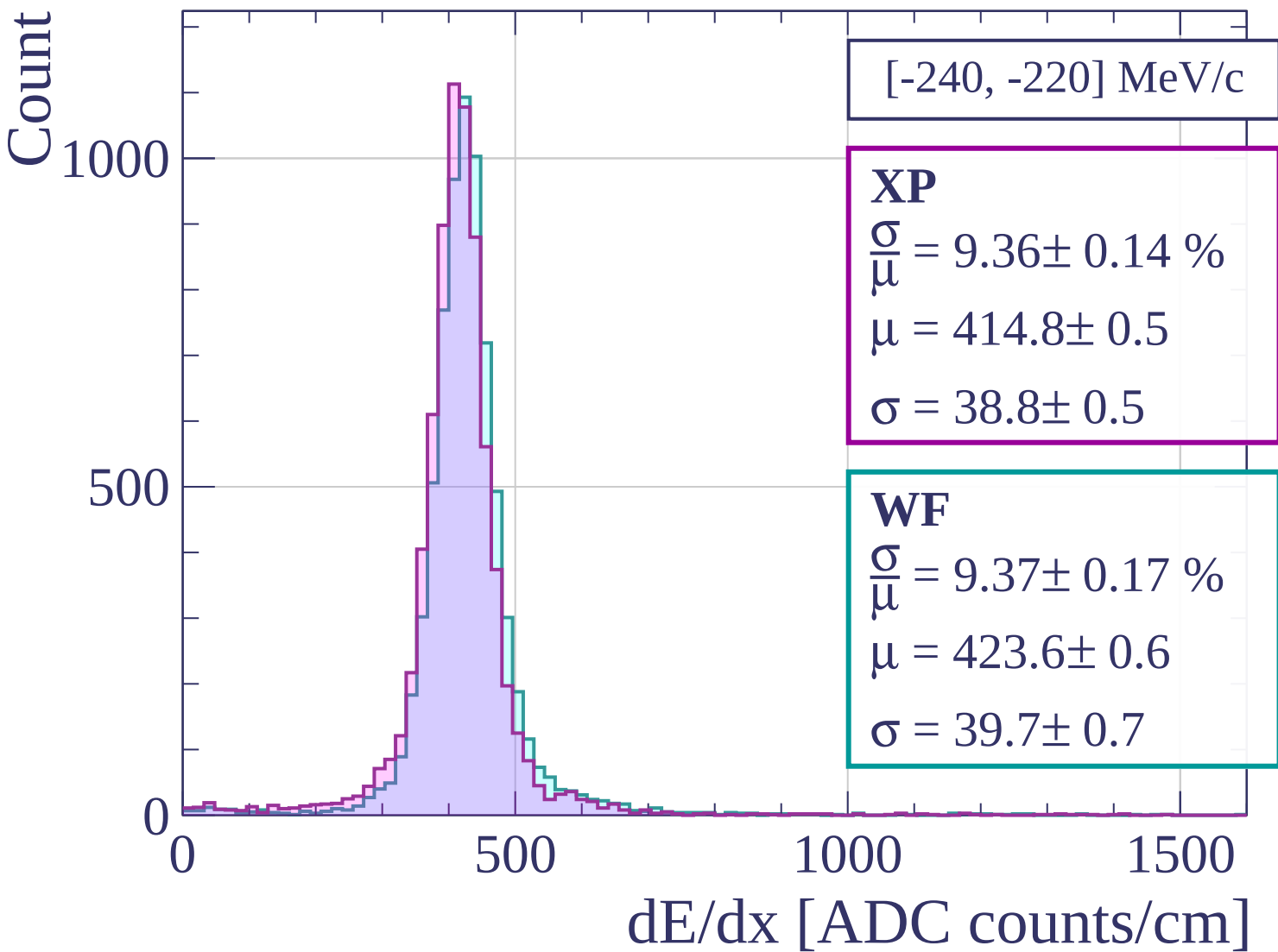


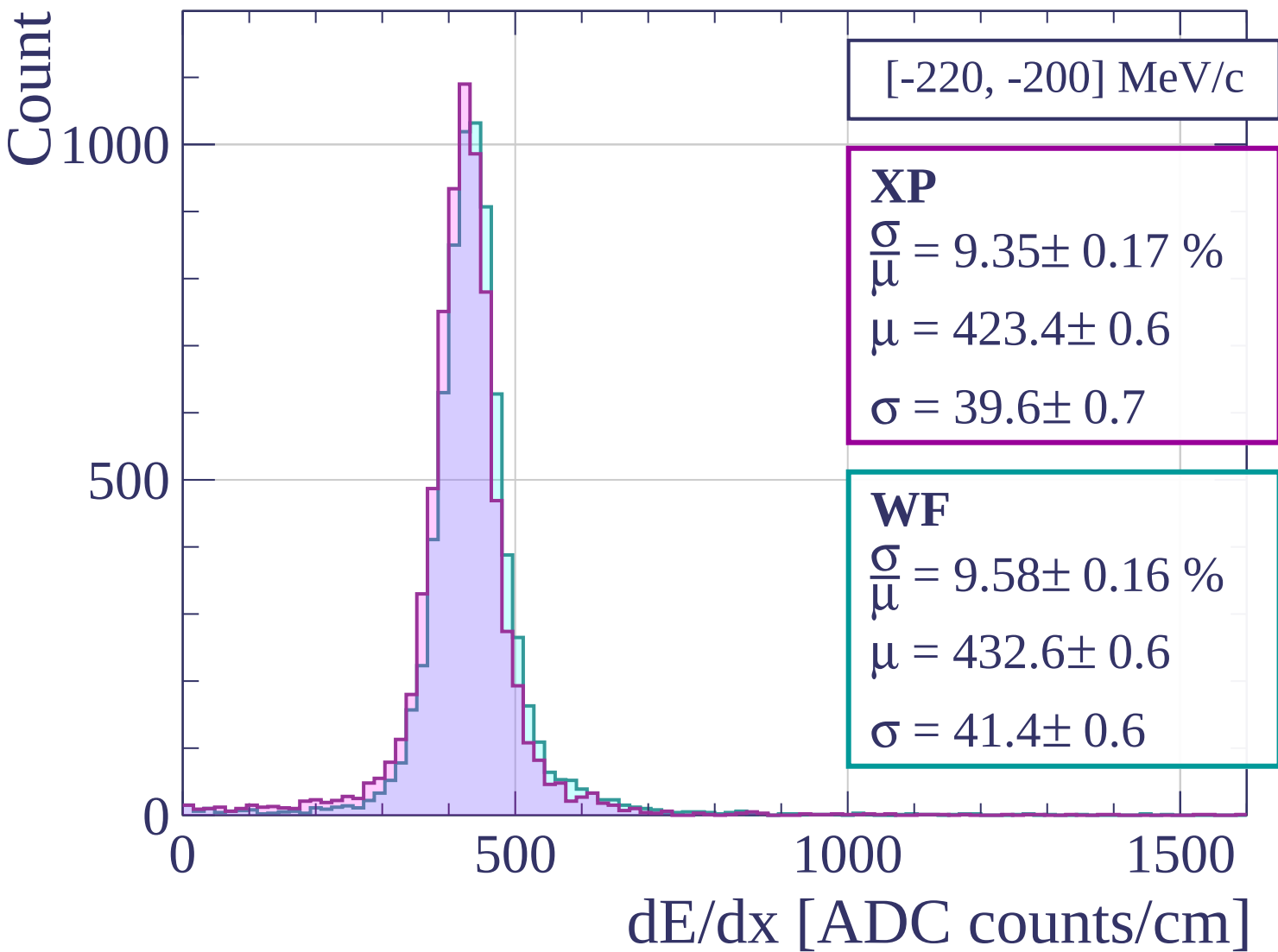


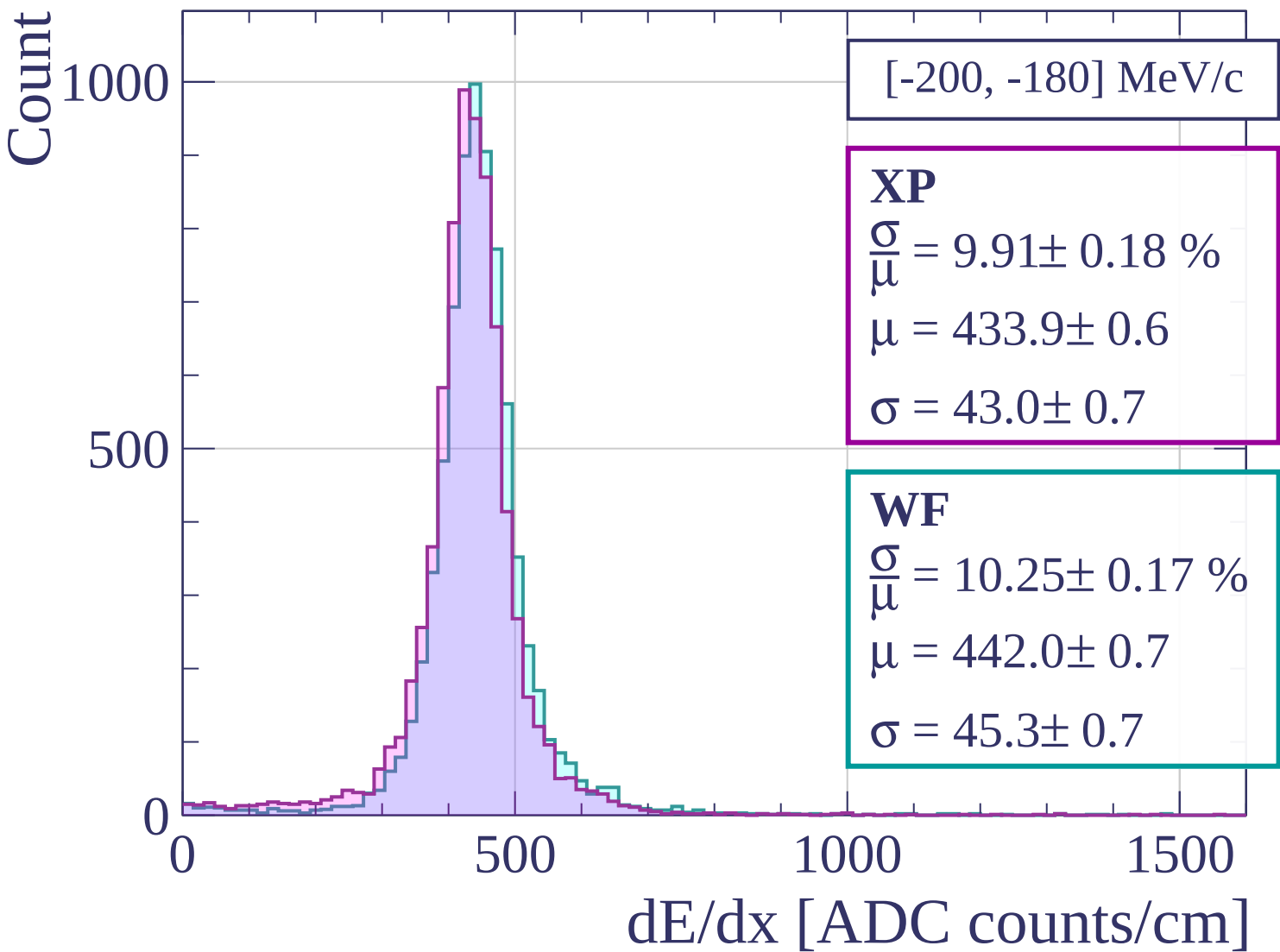












Count

[-180, -160] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.21 \pm 0.18 \%$$

$$\mu = 451.2 \pm 0.7$$

$$\sigma = 46.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.76 \pm 0.18 \%$$

$$\mu = 460.6 \pm 0.7$$

$$\sigma = 49.6 \pm 0.8$$

800

600

400

200

0

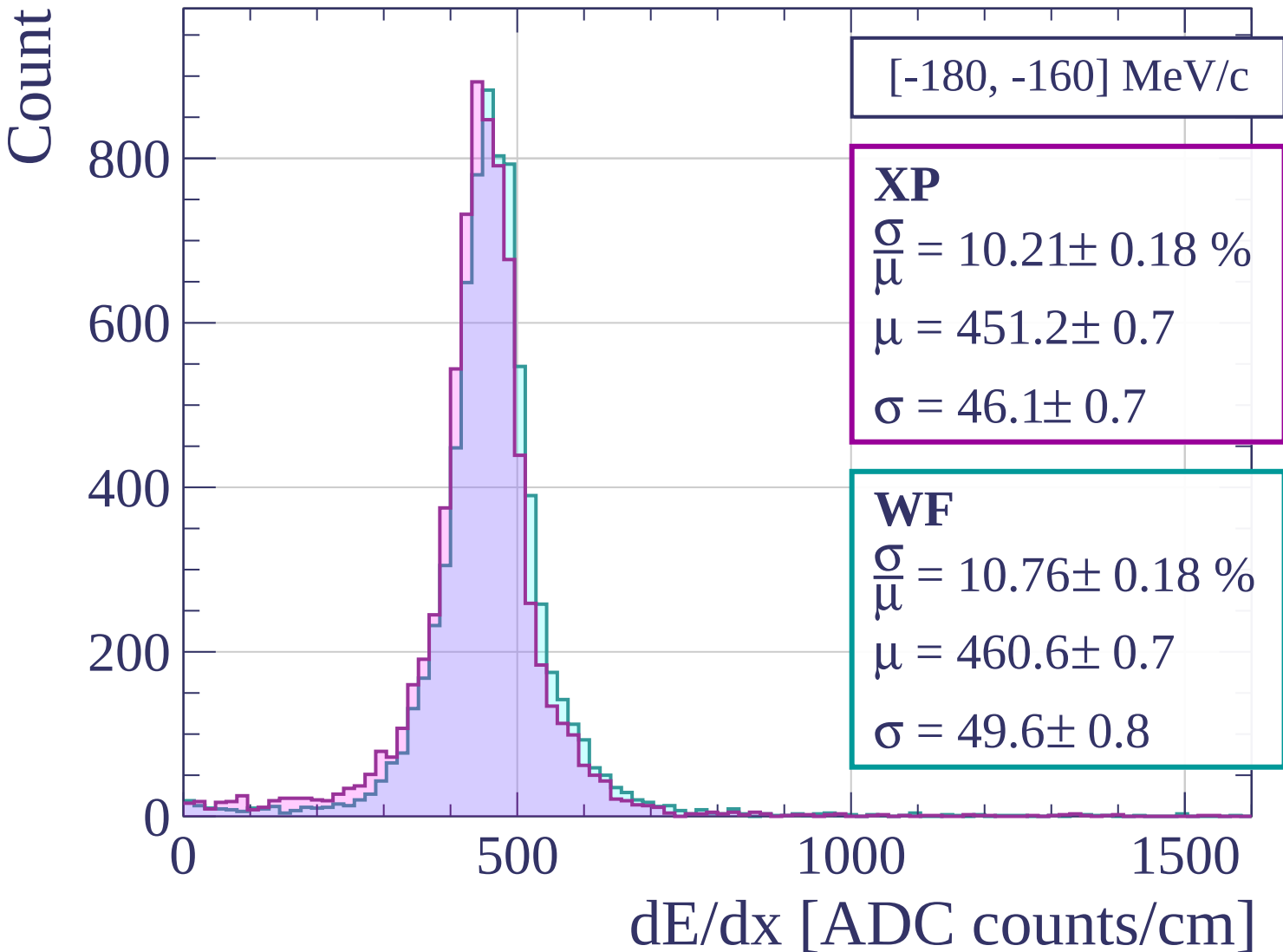
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-160, -140] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.45 \pm 0.21 \%$$

$$\mu = 474.5 \pm 0.8$$

$$\sigma = 54.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 12.19 \pm 0.22 \%$$

$$\mu = 483.1 \pm 0.9$$

$$\sigma = 58.9 \pm 1.0$$

800

600

400

200

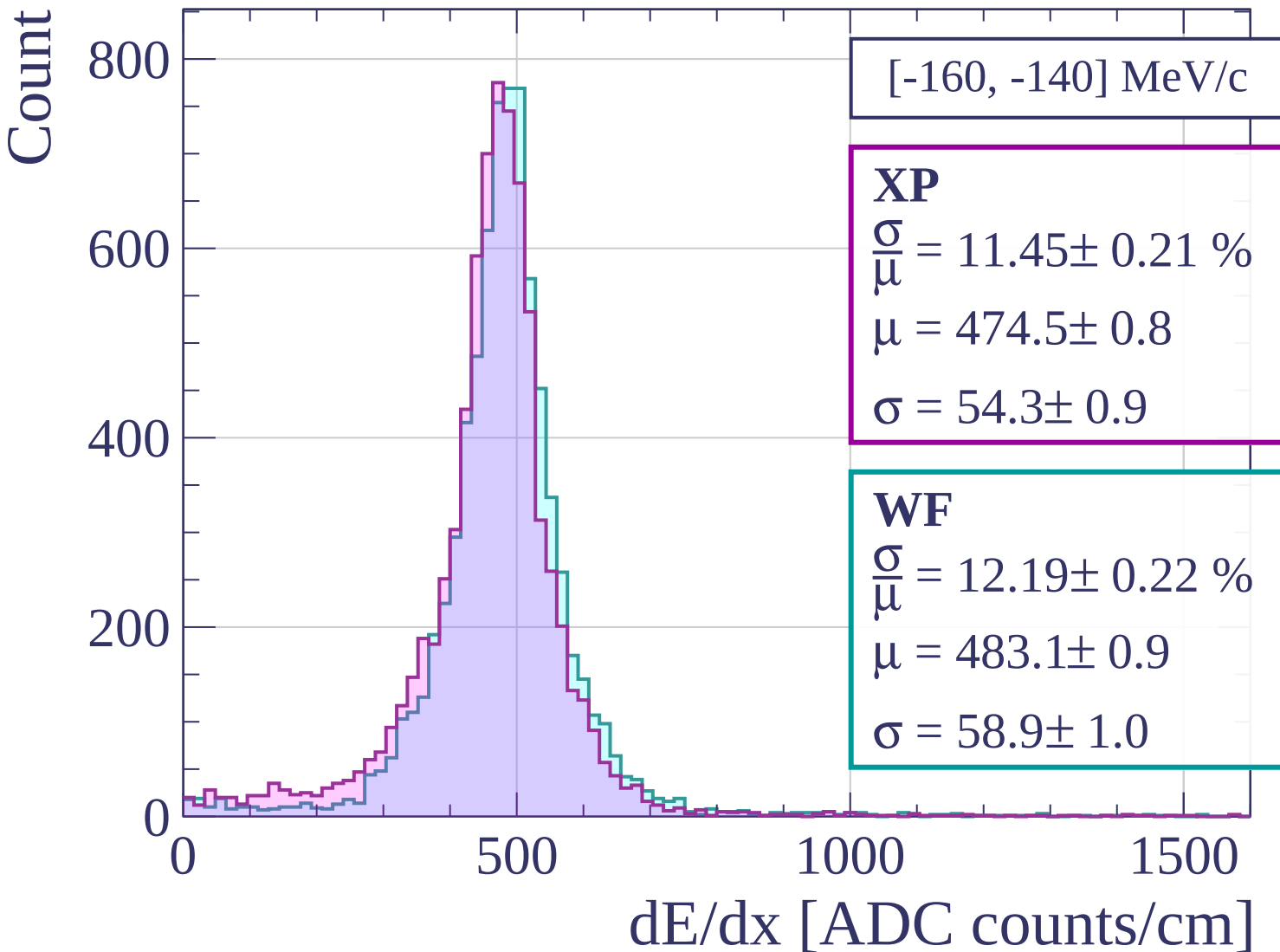
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-140, -120] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.40 \pm 0.27 \%$$

$$\mu = 512.9 \pm 1.1$$

$$\sigma = 68.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 13.89 \pm 0.27 \%$$

$$\mu = 522.0 \pm 1.1$$

$$\sigma = 72.5 \pm 1.3$$

600

400

200

0

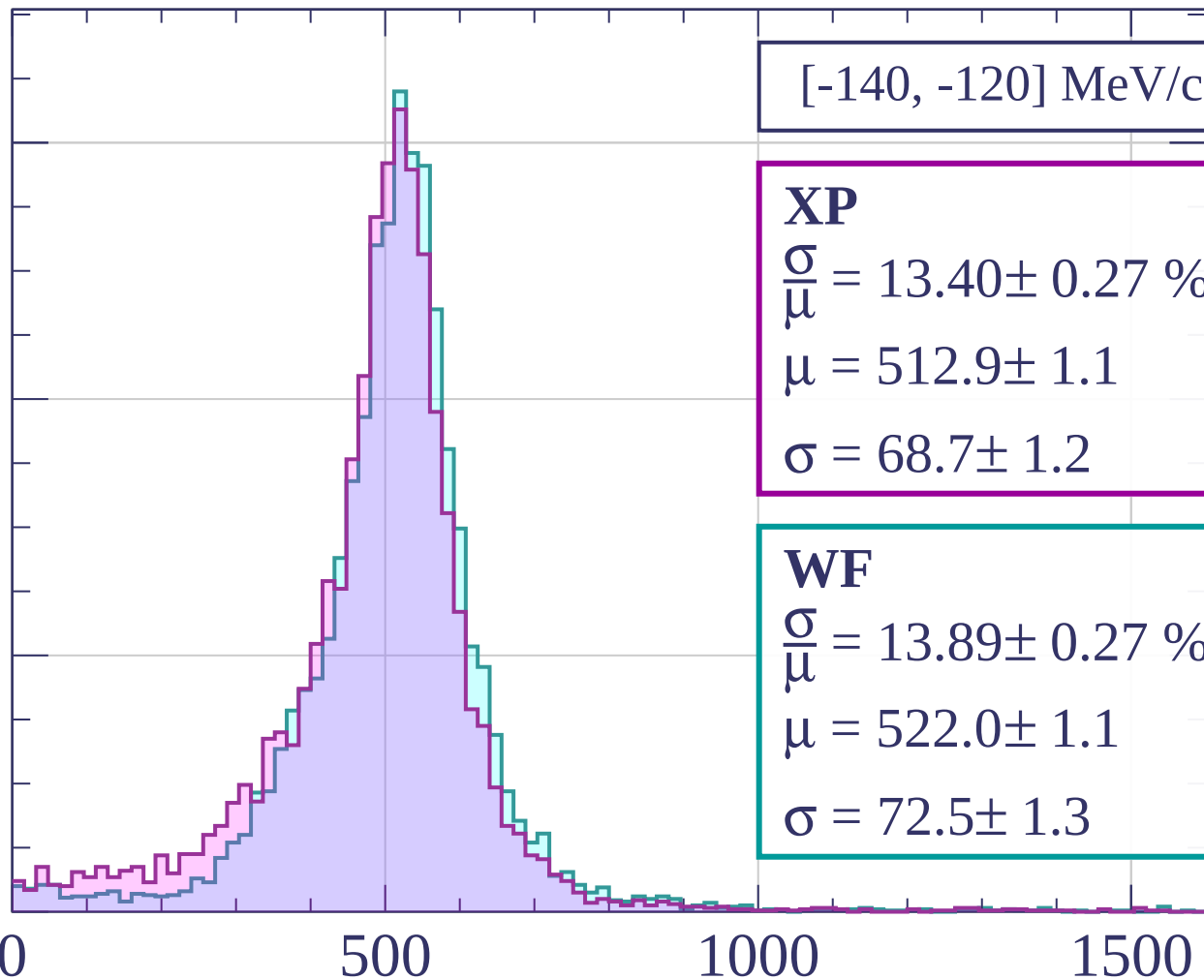
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-120, -100] MeV/c

XP

$$\frac{\sigma}{\mu} = 14.48 \pm 0.29 \%$$

$$\mu = 567.0 \pm 1.3$$

$$\sigma = 82.1 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 15.79 \pm 0.34 \%$$

$$\mu = 573.7 \pm 1.5$$

$$\sigma = 90.6 \pm 1.7$$

400

200

0

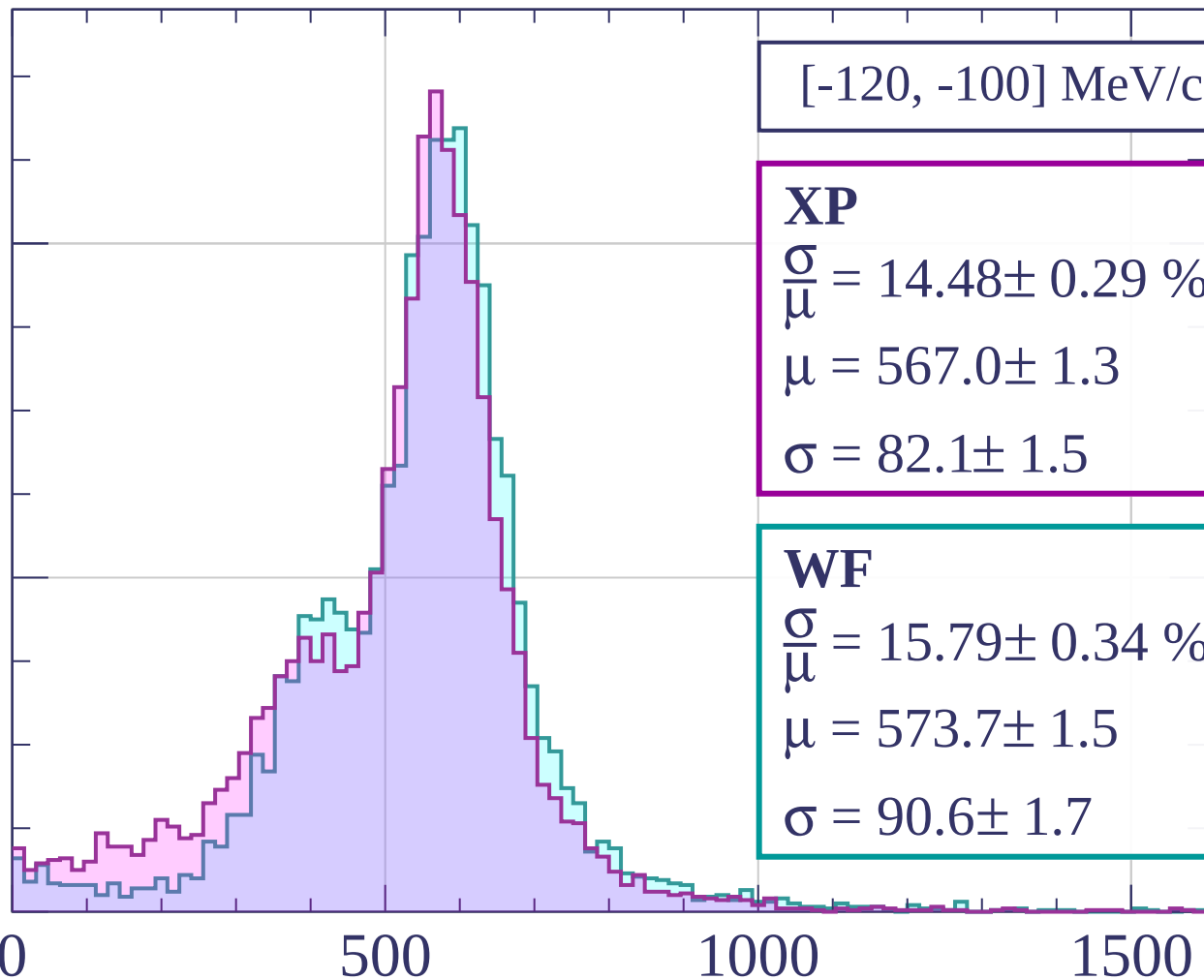
0

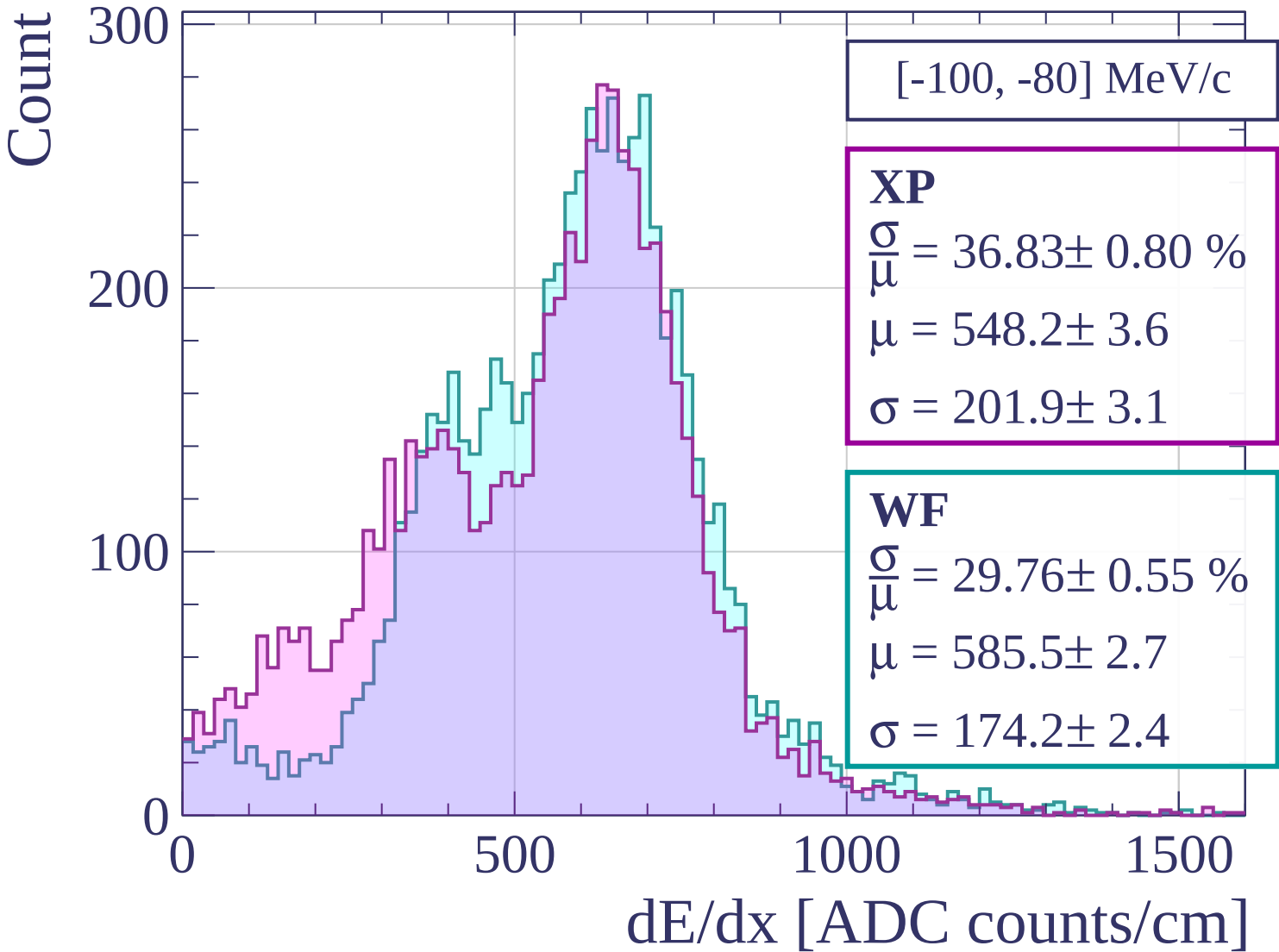
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-80, -60] MeV/c

XP

$$\frac{\sigma}{\mu} = 70.32 \pm 1.90 \%$$

$$\mu = 486.3 \pm 5.7$$

$$\sigma = 341.9 \pm 5.2$$

WF

$$\frac{\sigma}{\mu} = 46.06 \pm 1.23 \%$$

$$\mu = 593.2 \pm 4.9$$

$$\sigma = 273.2 \pm 5.0$$

0

100

200

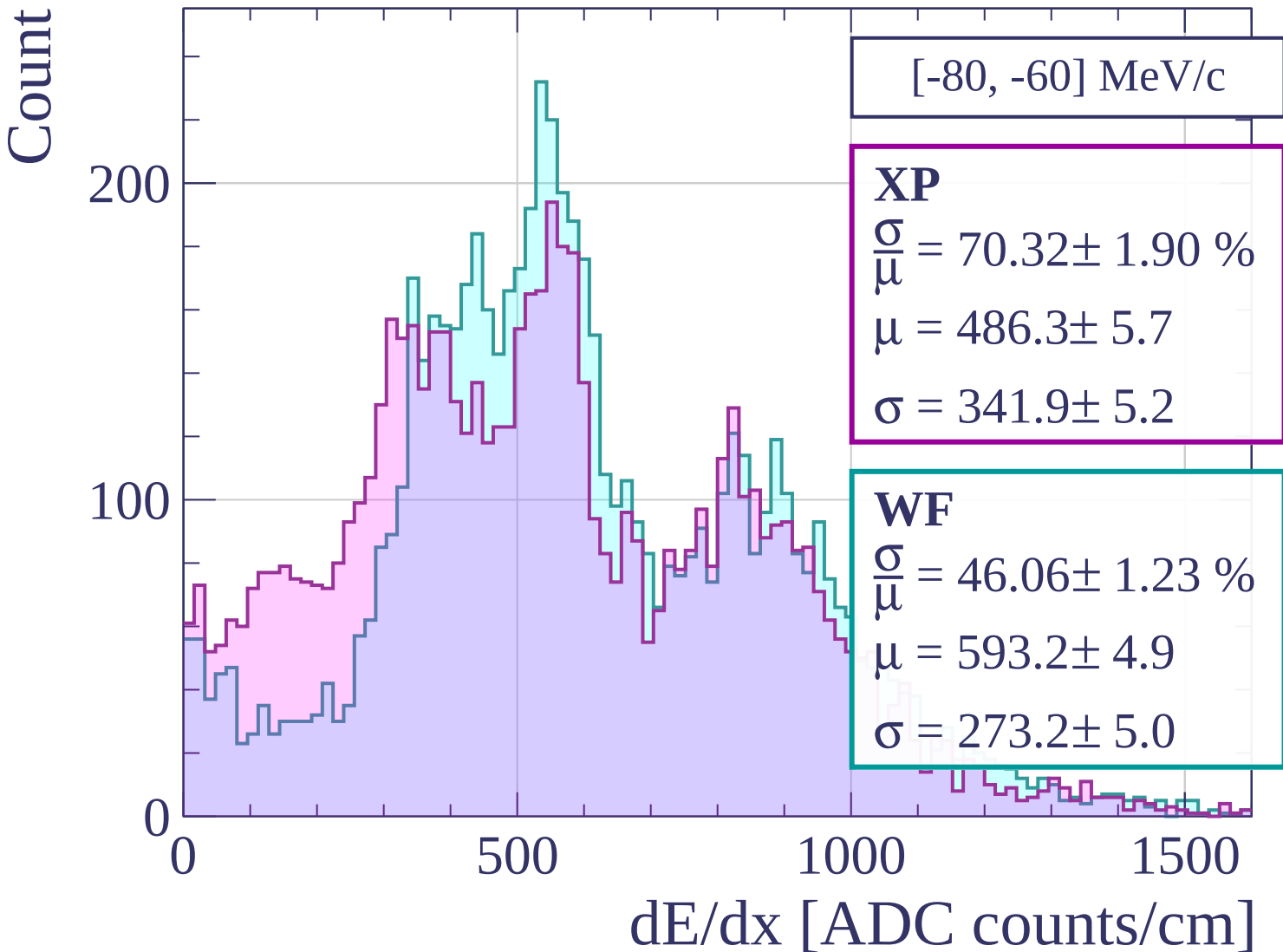
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-60, -40] MeV/c

XP

$$\frac{\sigma}{\mu} = 58.62 \pm 1.59 \%$$

$$\mu = 350.4 \pm 4.1$$

$$\sigma = 205.4 \pm 3.1$$

WF

$$\frac{\sigma}{\mu} = 26.76 \pm 0.51 \%$$

$$\mu = 484.3 \pm 2.1$$

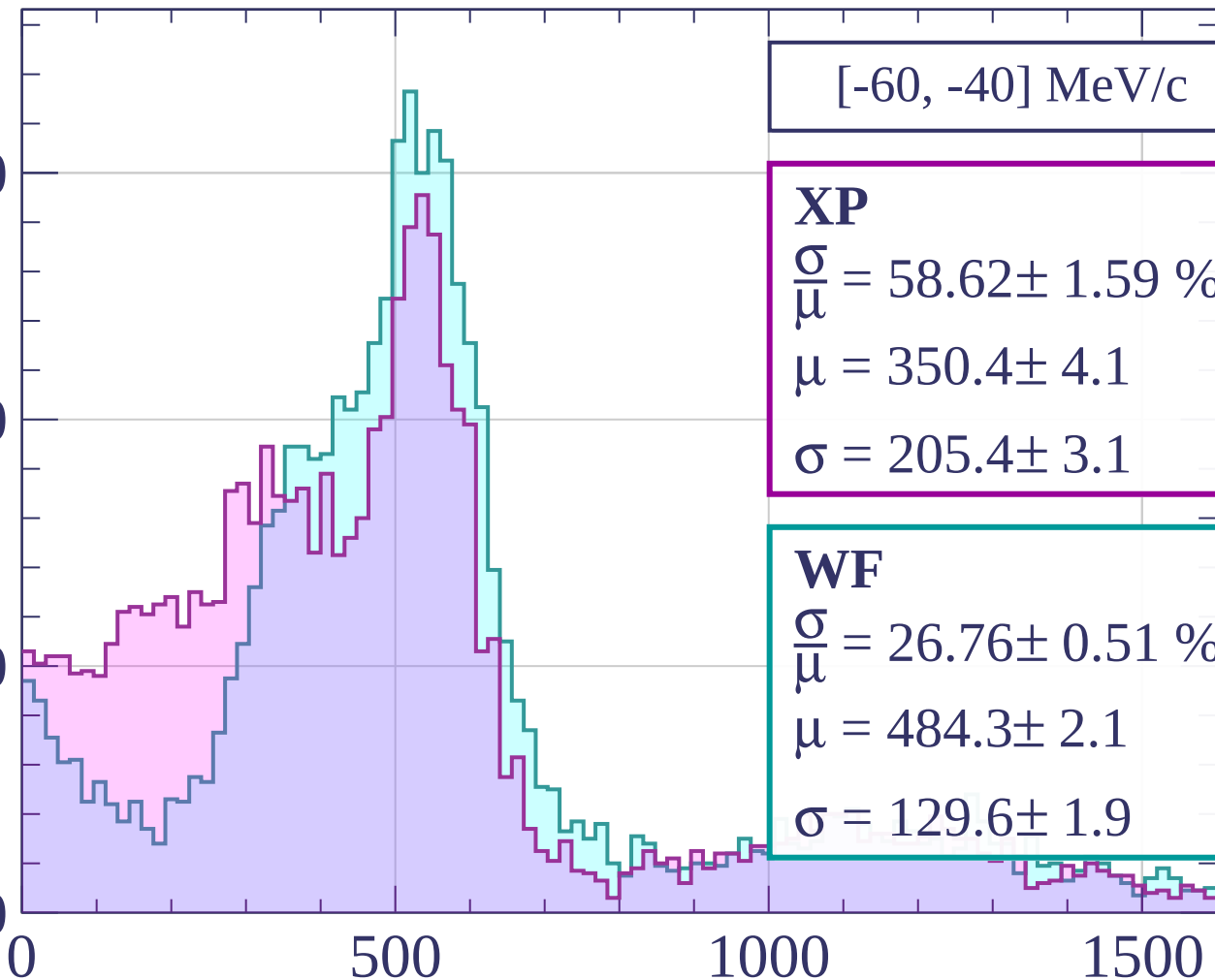
$$\sigma = 129.6 \pm 1.9$$

300

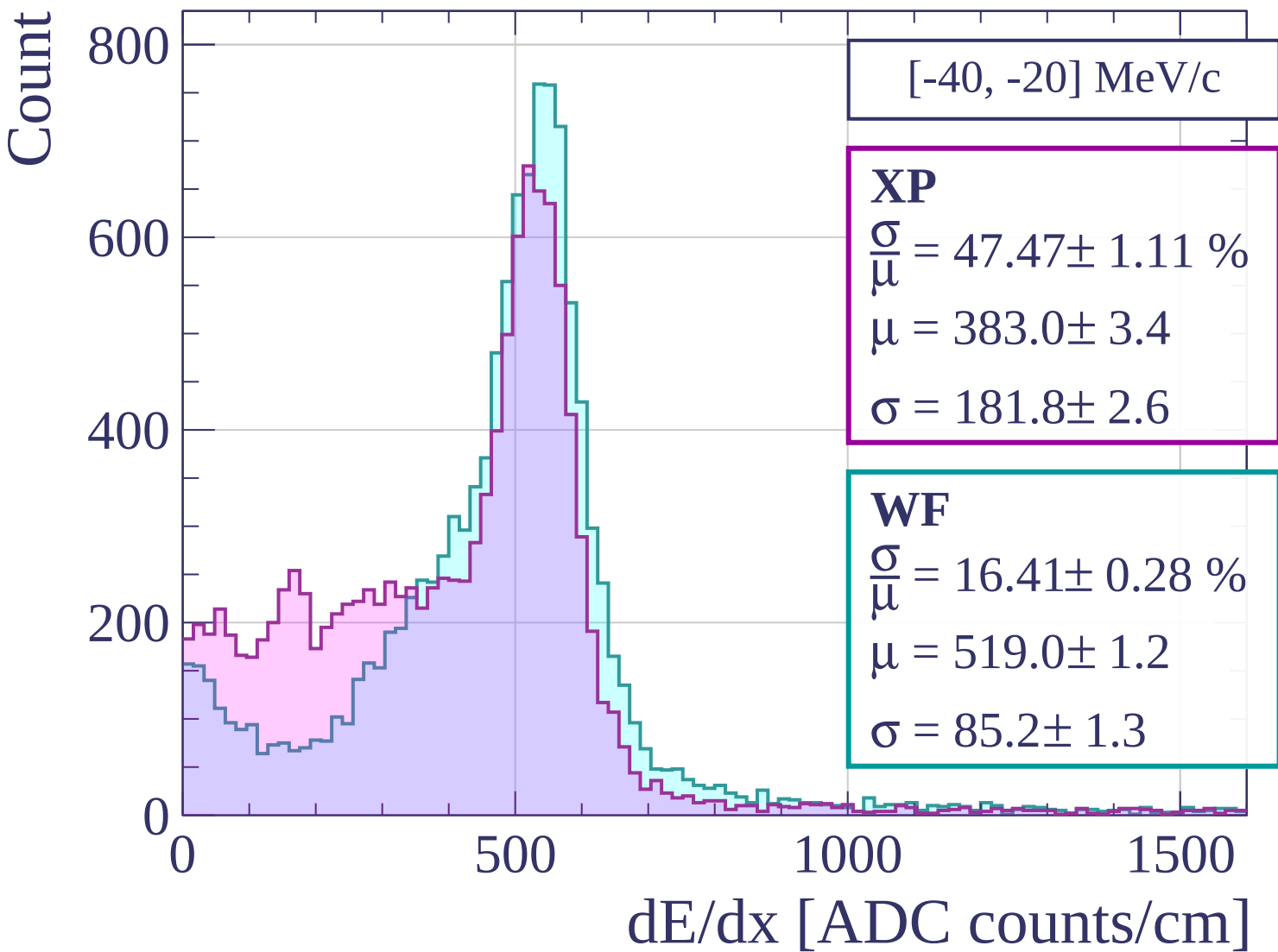
200

100

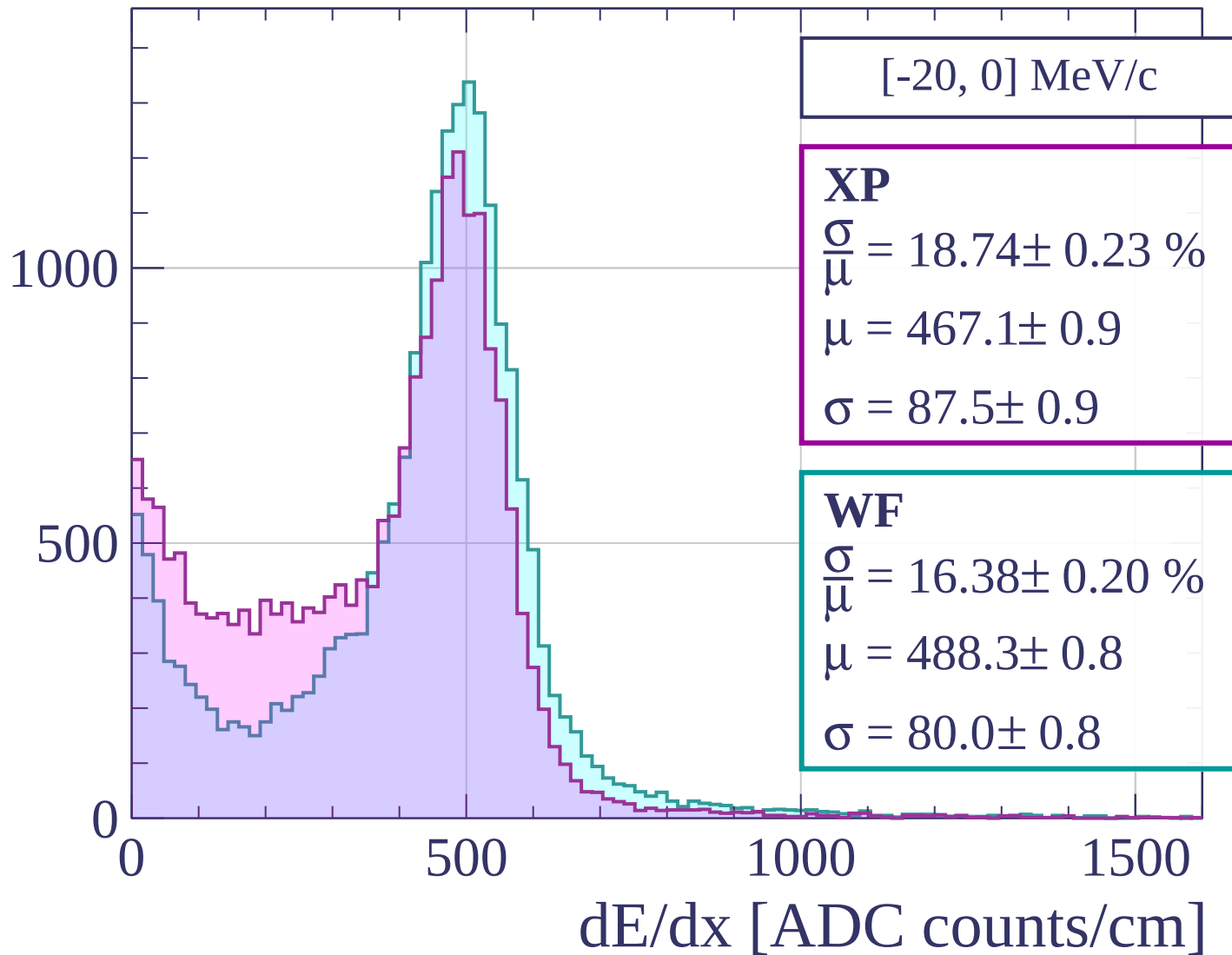
0

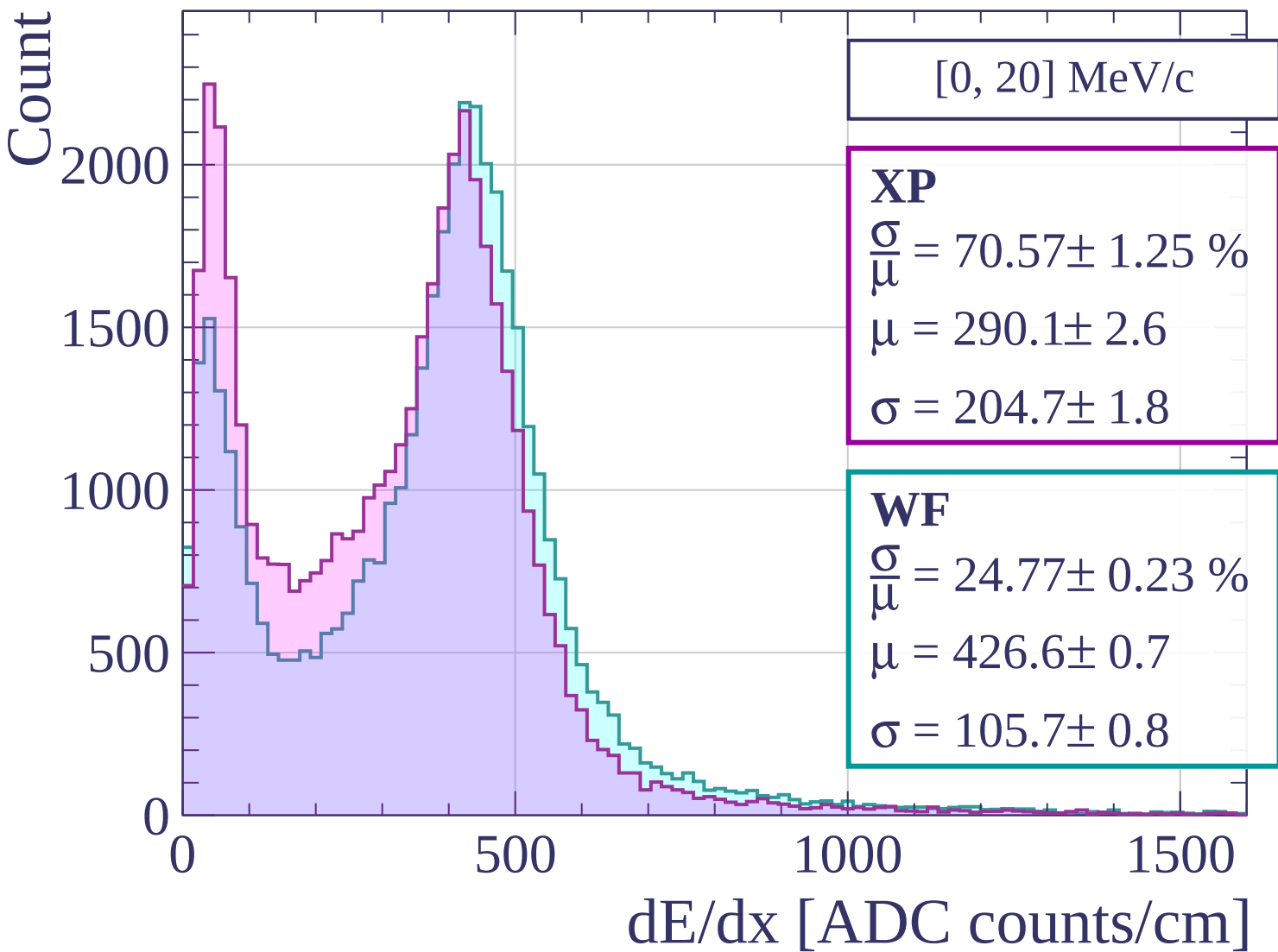


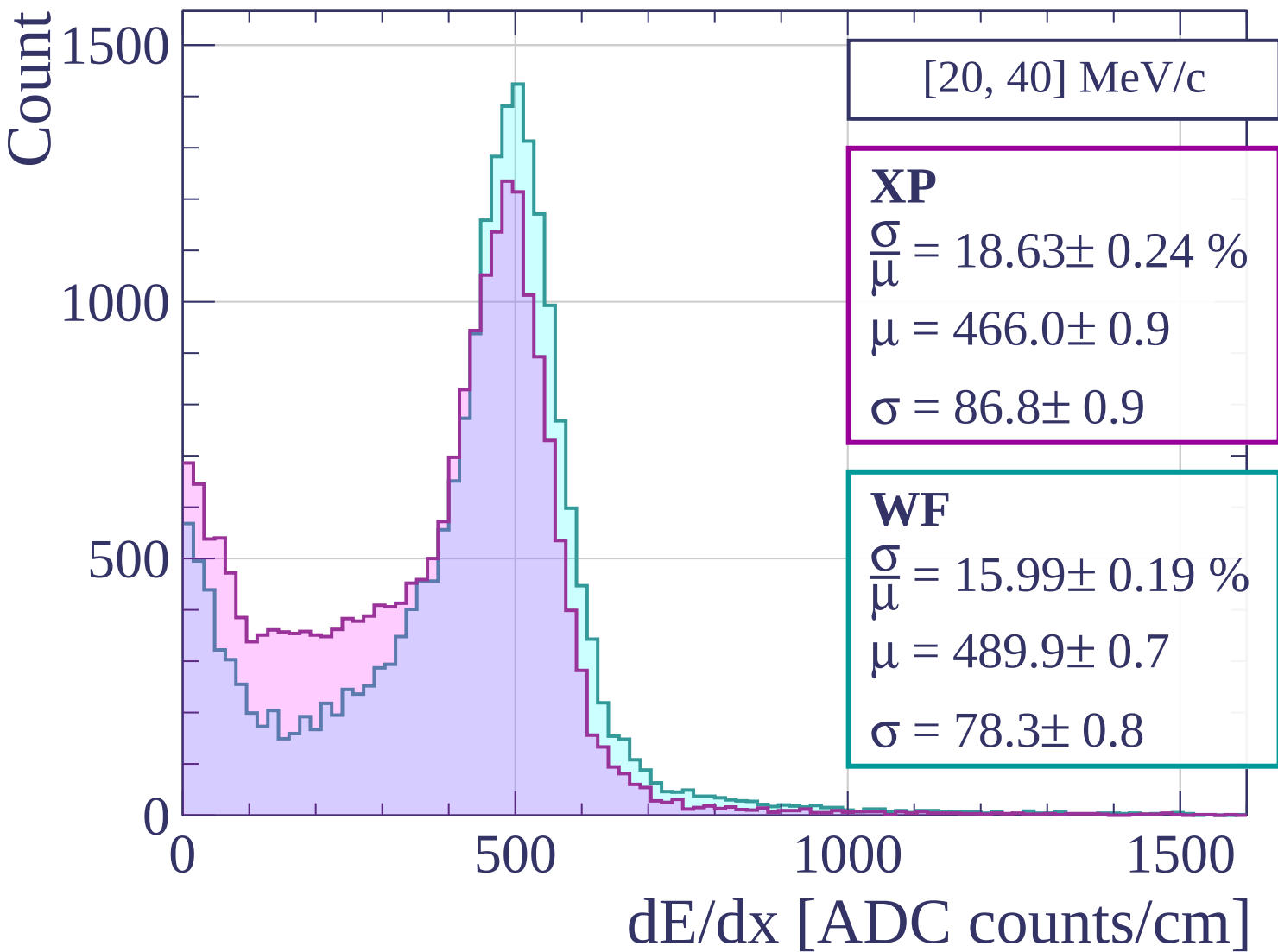
dE/dx [ADC counts/cm]



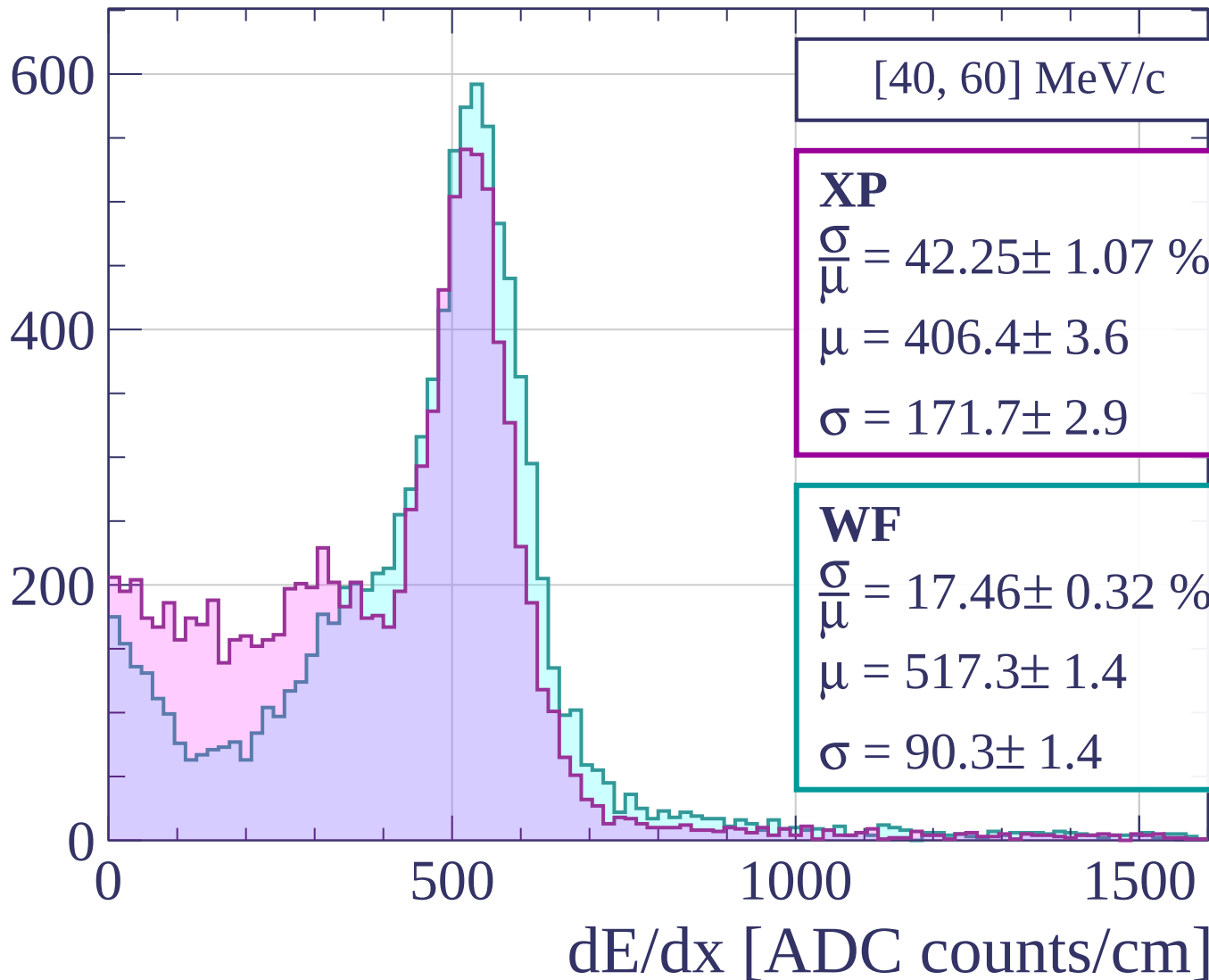
Count



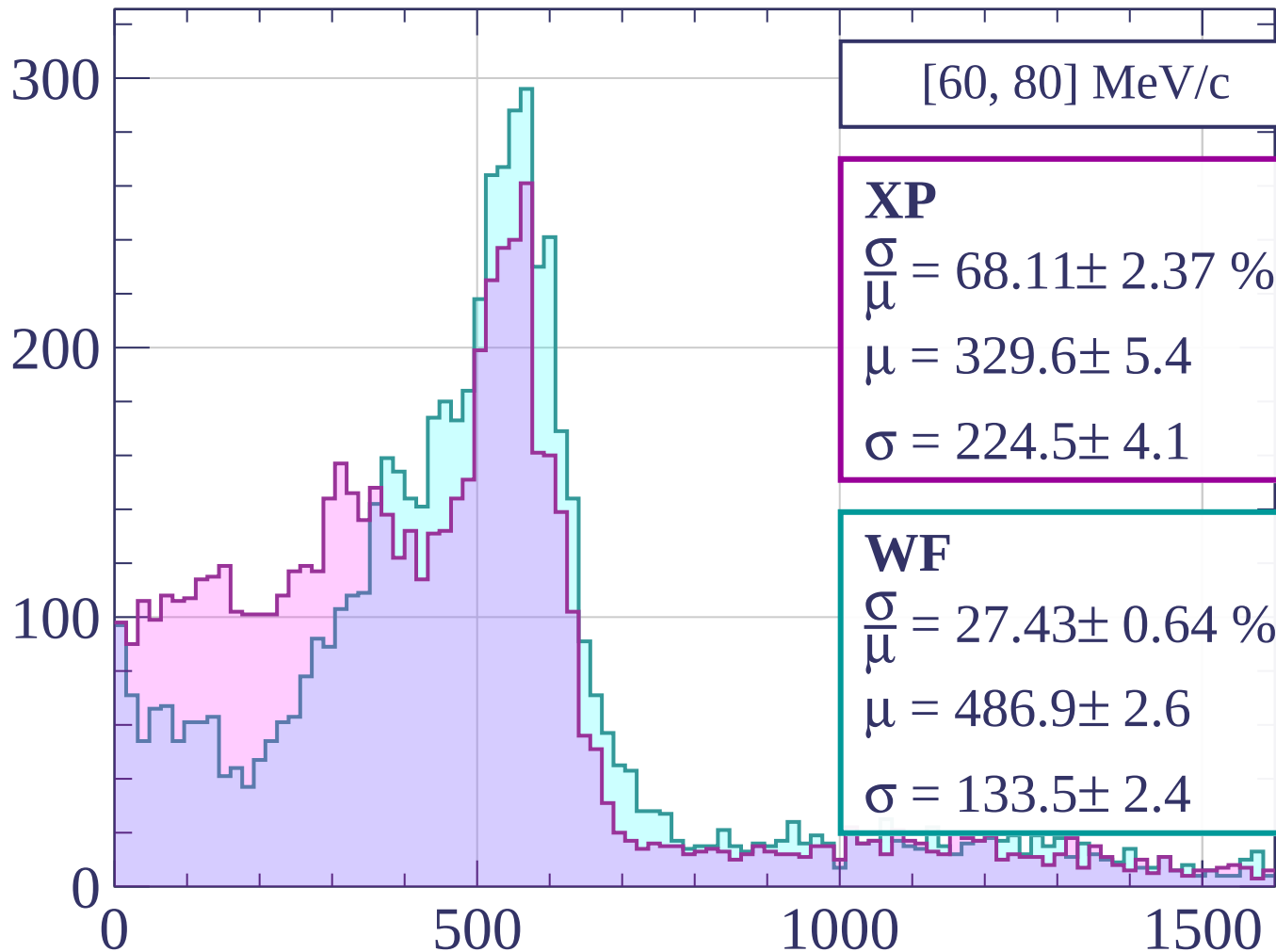




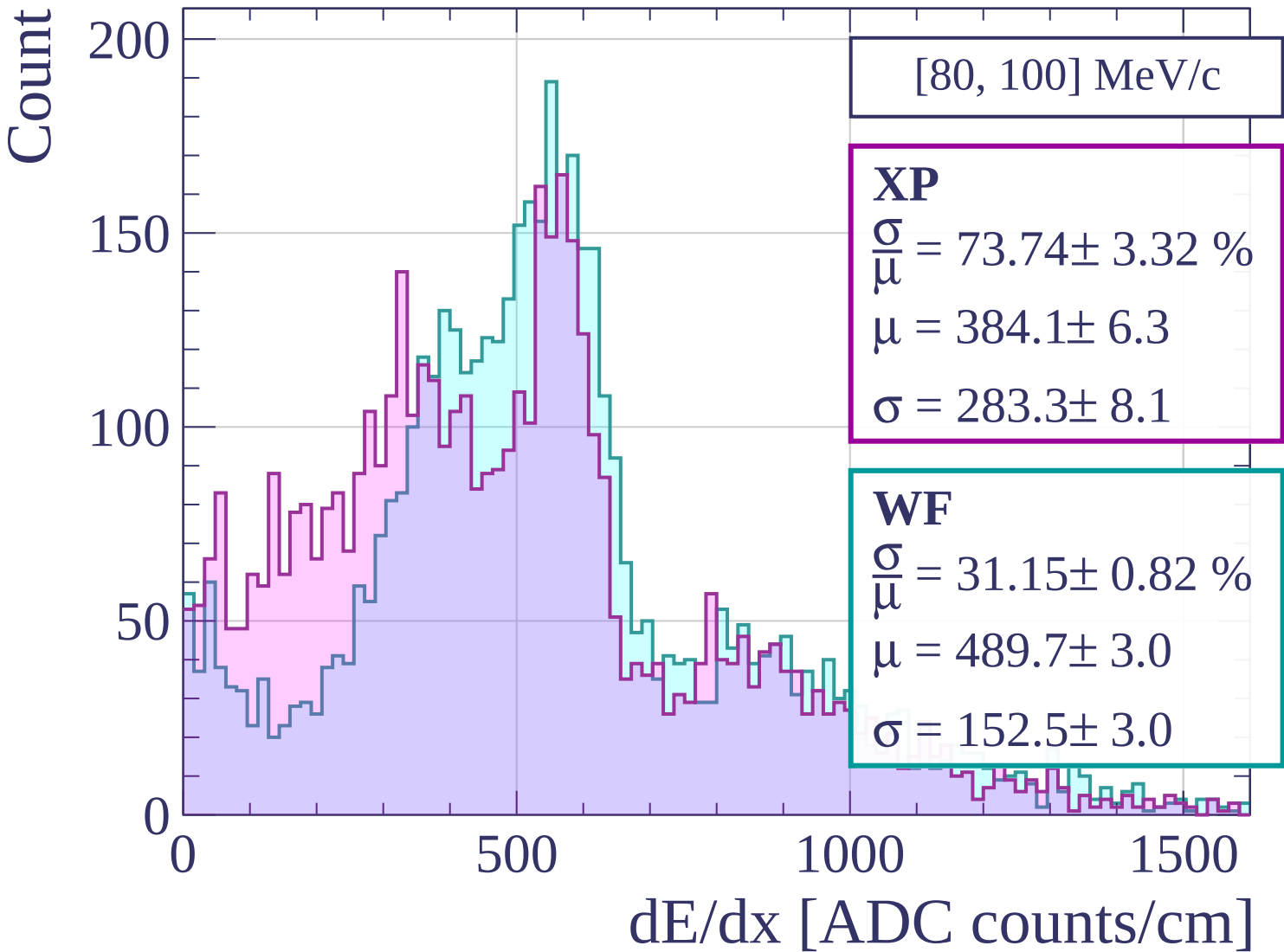
Count



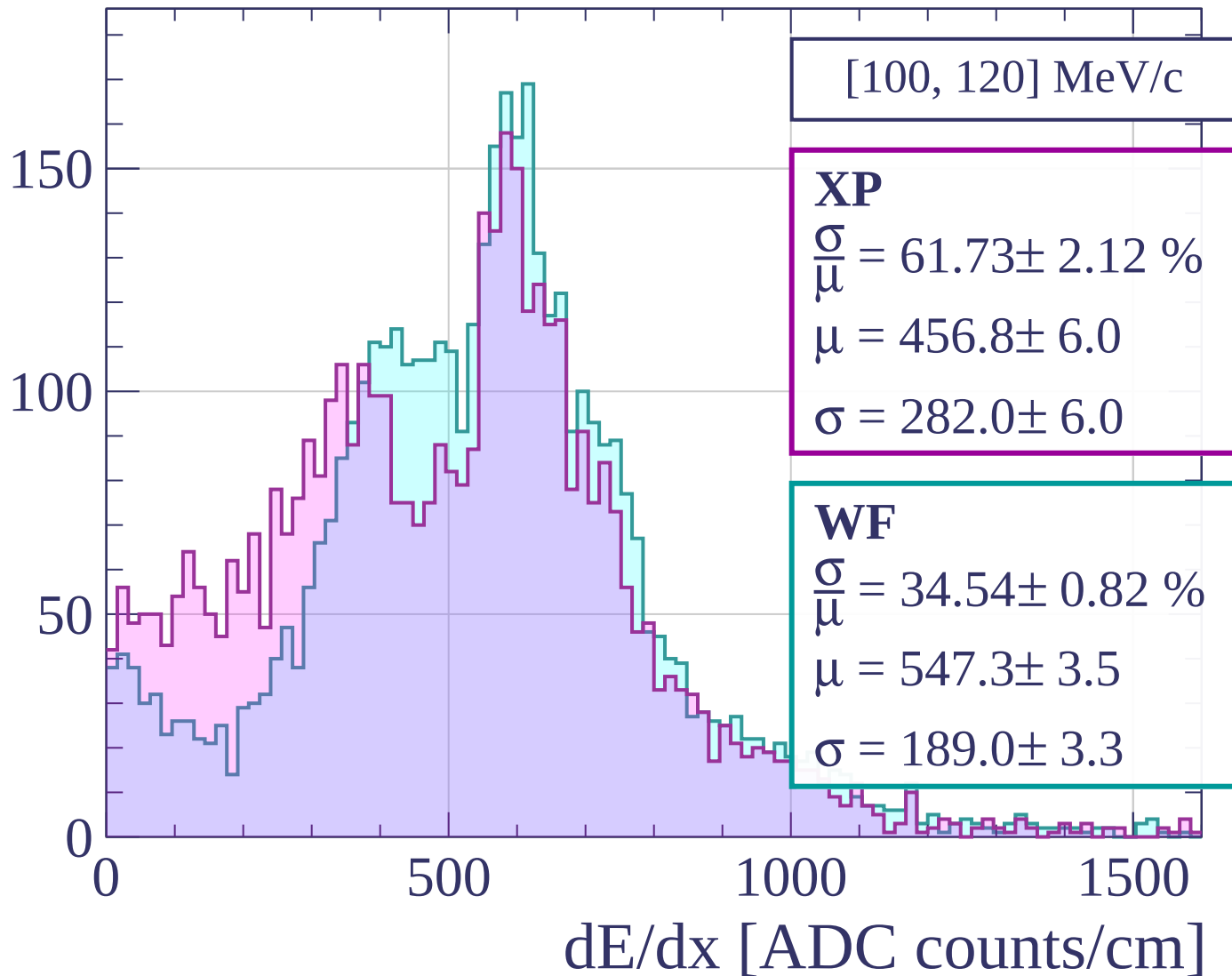
Count



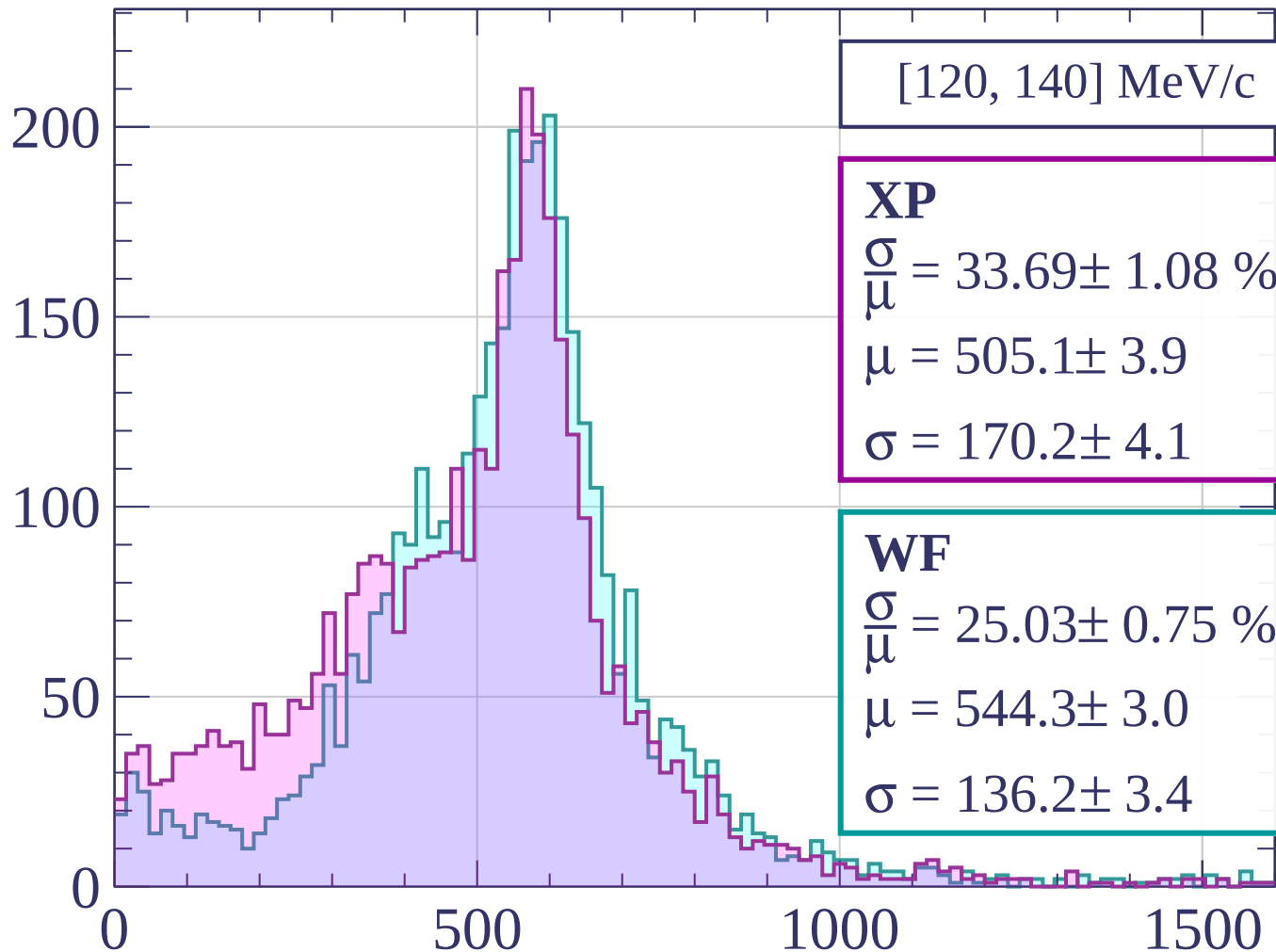
dE/dx [ADC counts/cm]



Count



Count



[120, 140] MeV/c

XP

$$\frac{\sigma}{\mu} = 33.69 \pm 1.08 \%$$

$$\mu = 505.1 \pm 3.9$$

$$\sigma = 170.2 \pm 4.1$$

WF

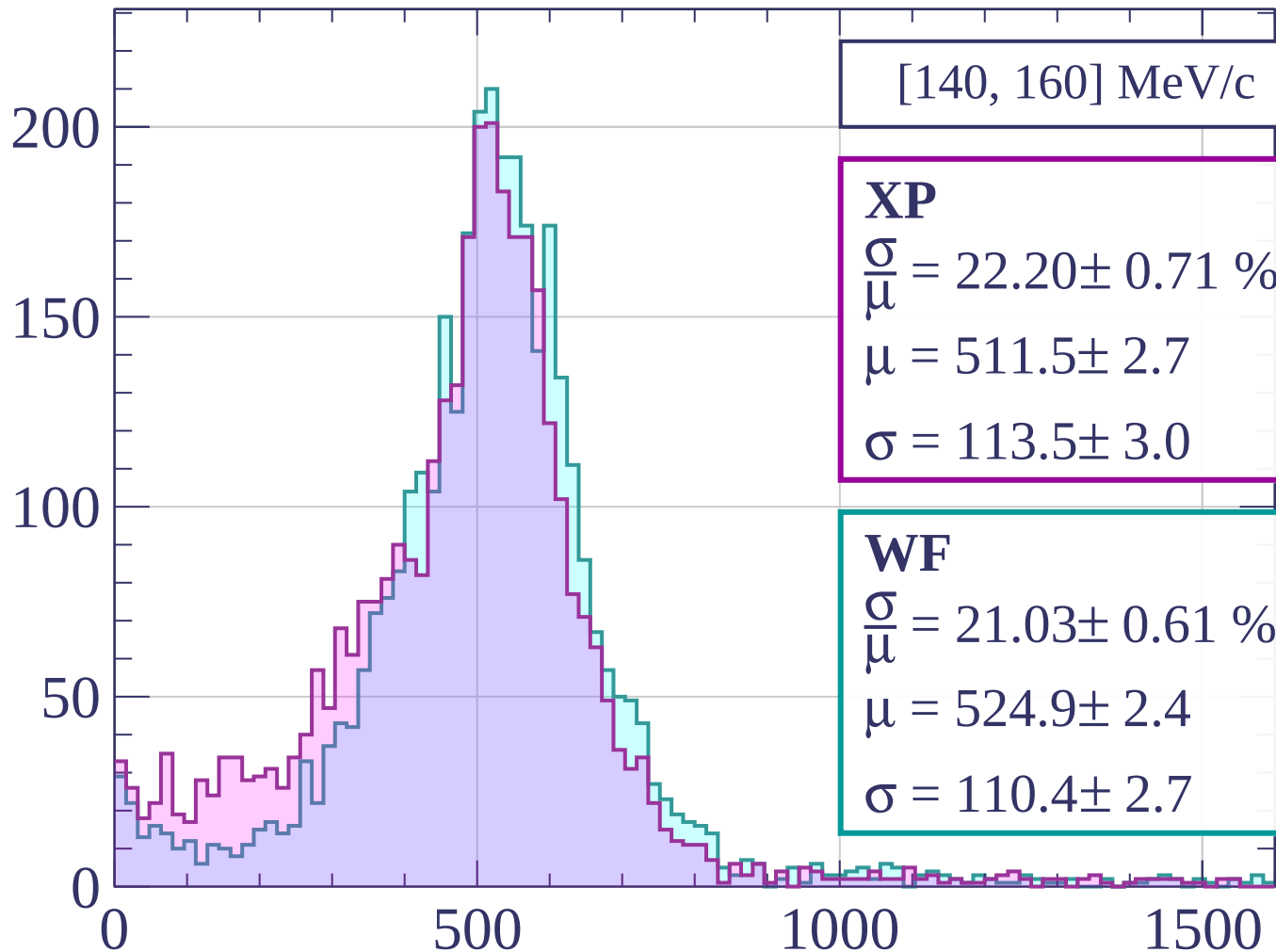
$$\frac{\sigma}{\mu} = 25.03 \pm 0.75 \%$$

$$\mu = 544.3 \pm 3.0$$

$$\sigma = 136.2 \pm 3.4$$

dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

[160, 180] MeV/c

XP

$$\frac{\sigma}{\mu} = 20.71 \pm 0.68 \%$$

$$\mu = 484.6 \pm 2.3$$

$$\sigma = 100.4 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 20.10 \pm 0.57 \%$$

$$\mu = 500.6 \pm 2.2$$

$$\sigma = 100.6 \pm 2.4$$

200

100

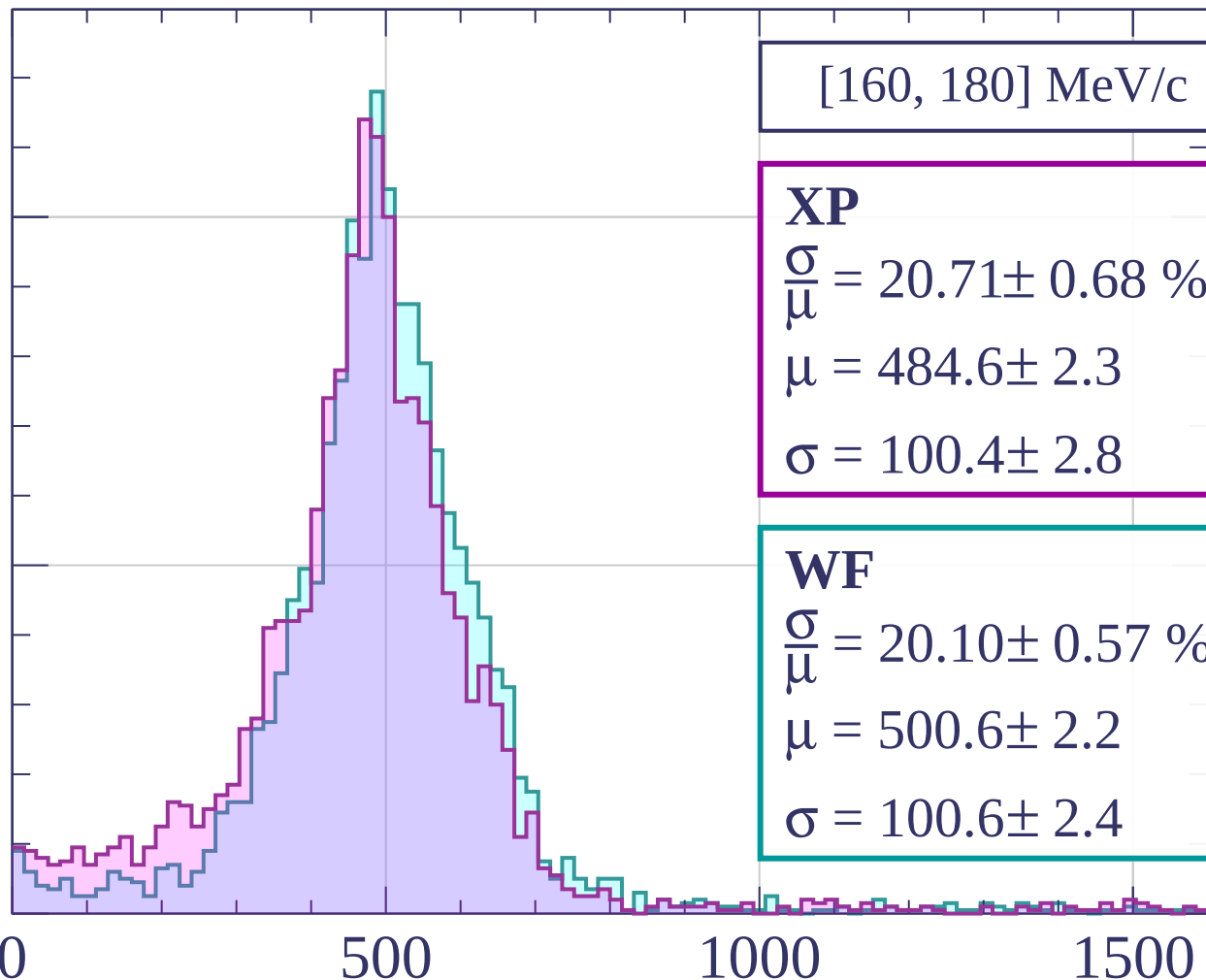
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[180, 200] MeV/c

XP

$$\frac{\sigma}{\mu} = 15.64 \pm 0.51 \%$$

$$\mu = 461.6 \pm 1.8$$

$$\sigma = 72.2 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 15.71 \pm 0.51 \%$$

$$\mu = 472.7 \pm 1.8$$

$$\sigma = 74.3 \pm 2.1$$

200

100

0

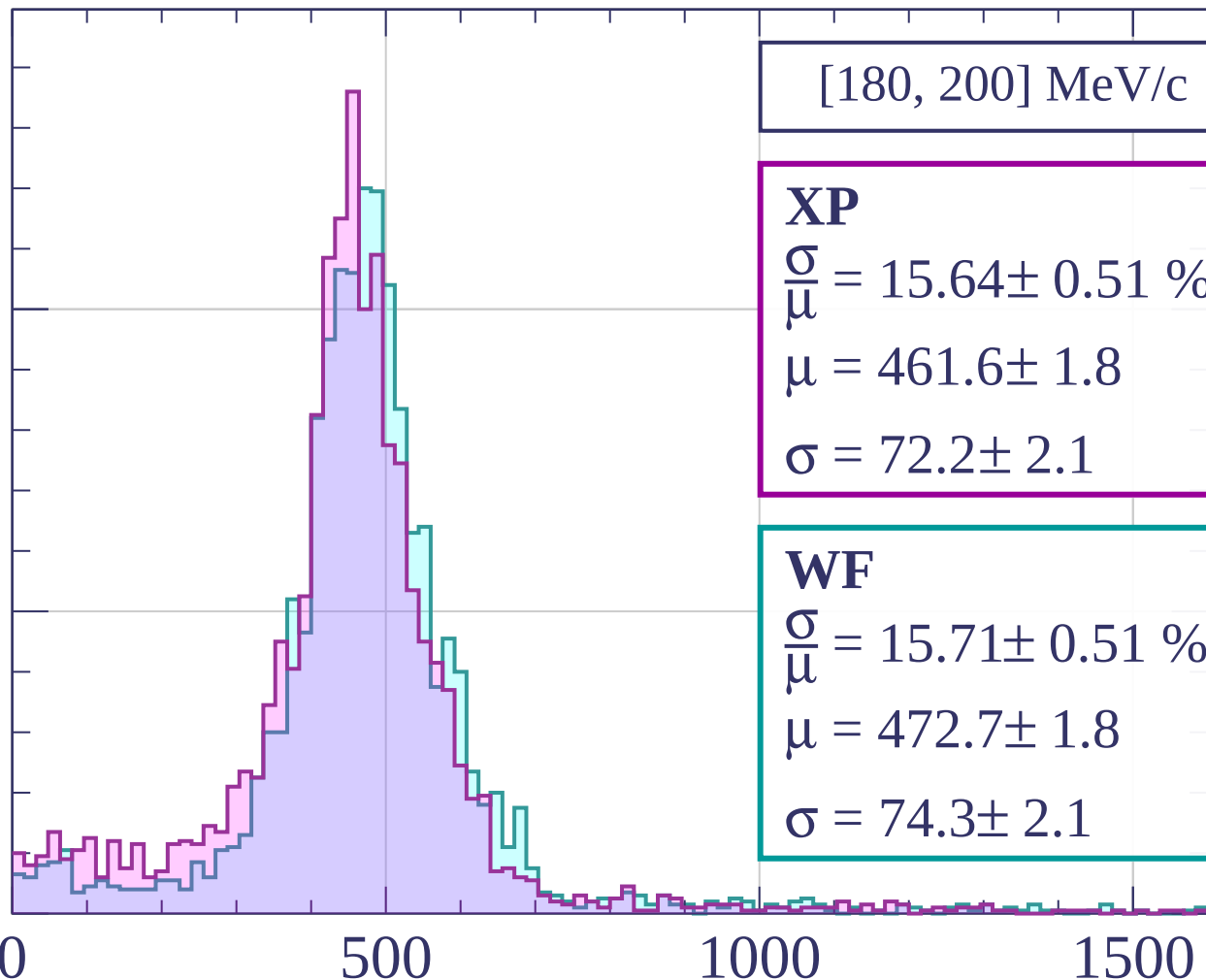
0

500

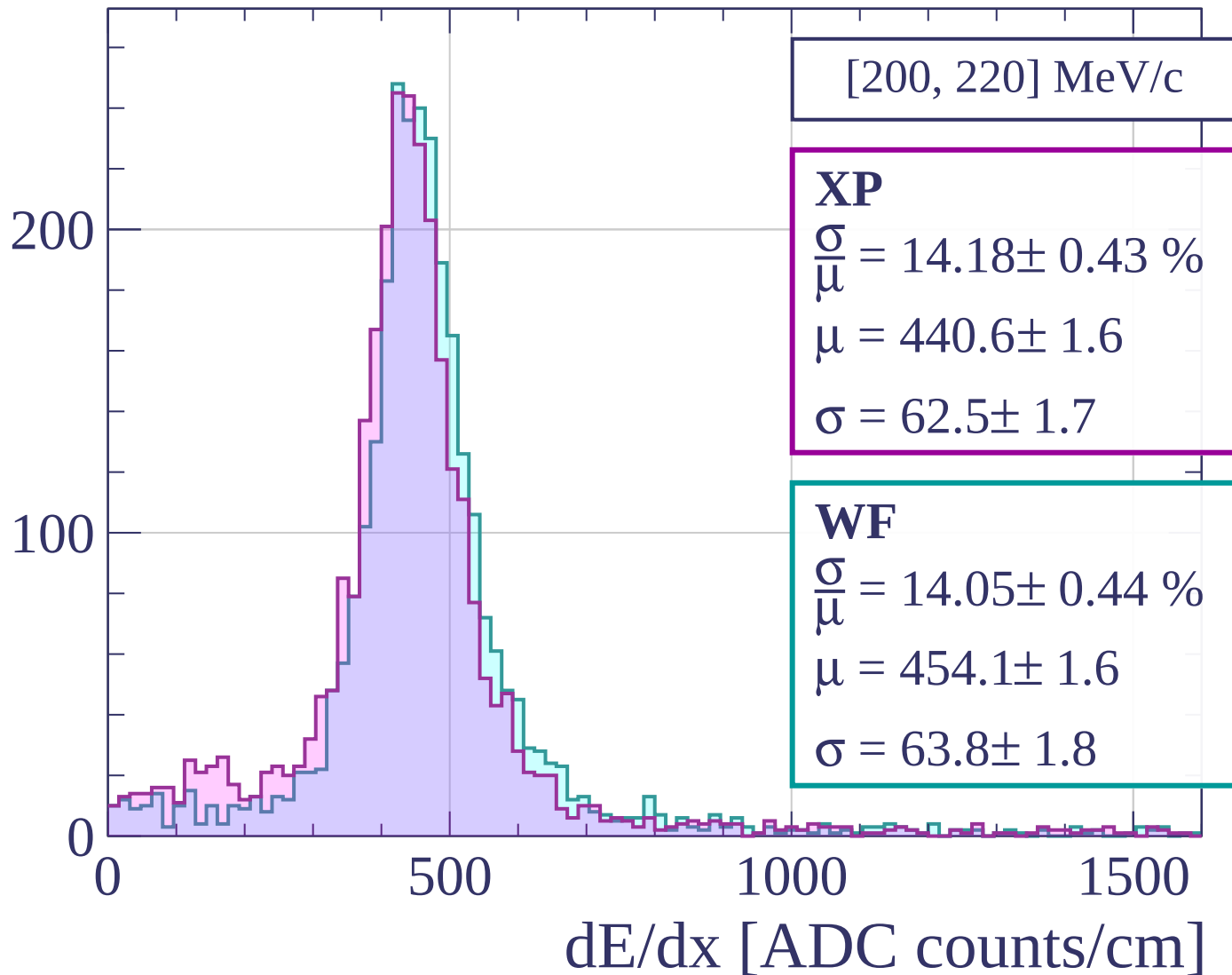
1000

1500

dE/dx [ADC counts/cm]



Count



Count

[220, 240] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.36 \pm 0.39 \%$$

$$\mu = 429.8 \pm 1.4$$

$$\sigma = 53.1 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 12.66 \pm 0.41 \%$$

$$\mu = 441.3 \pm 1.5$$

$$\sigma = 55.9 \pm 1.6$$

200

100

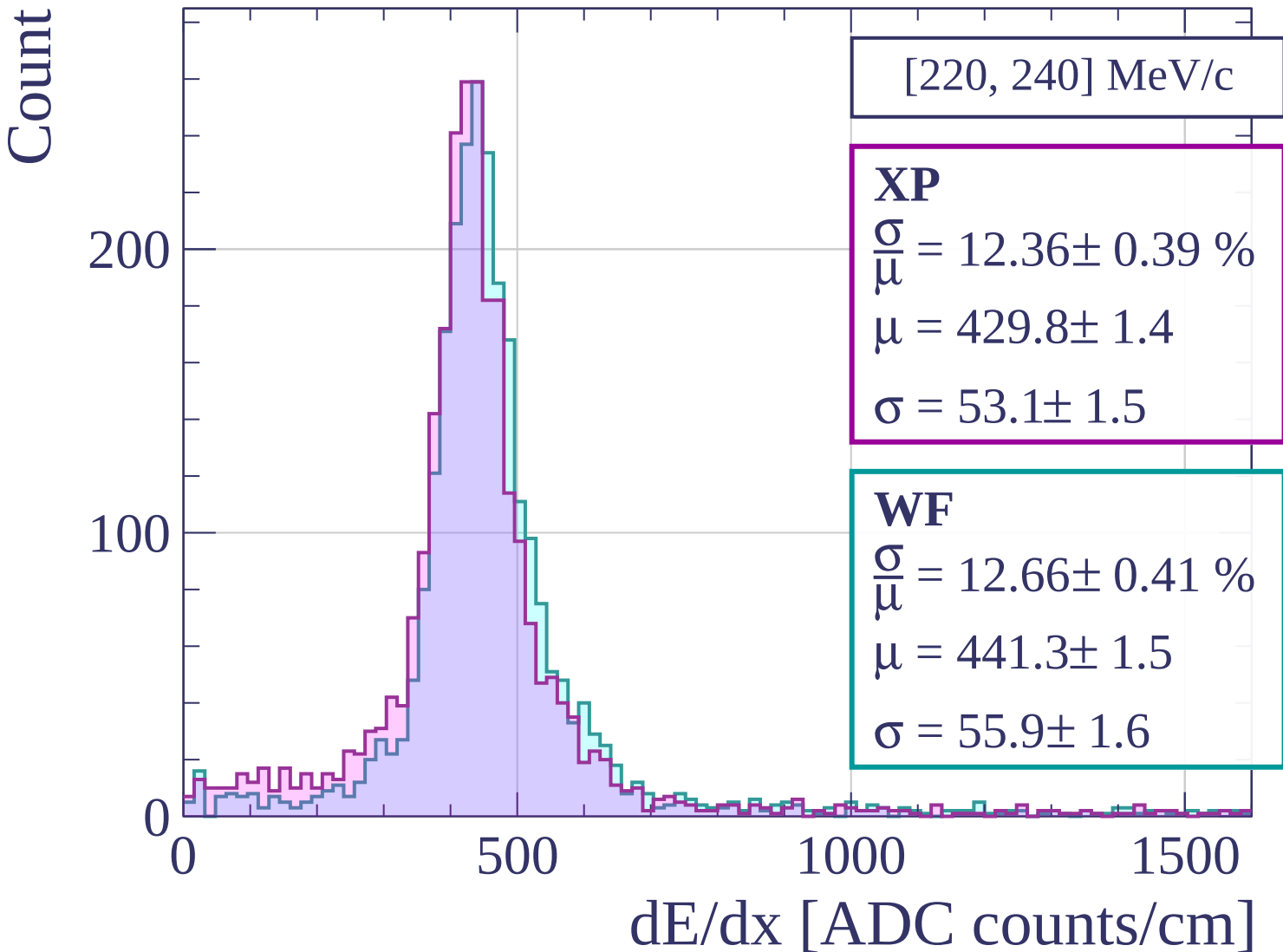
0

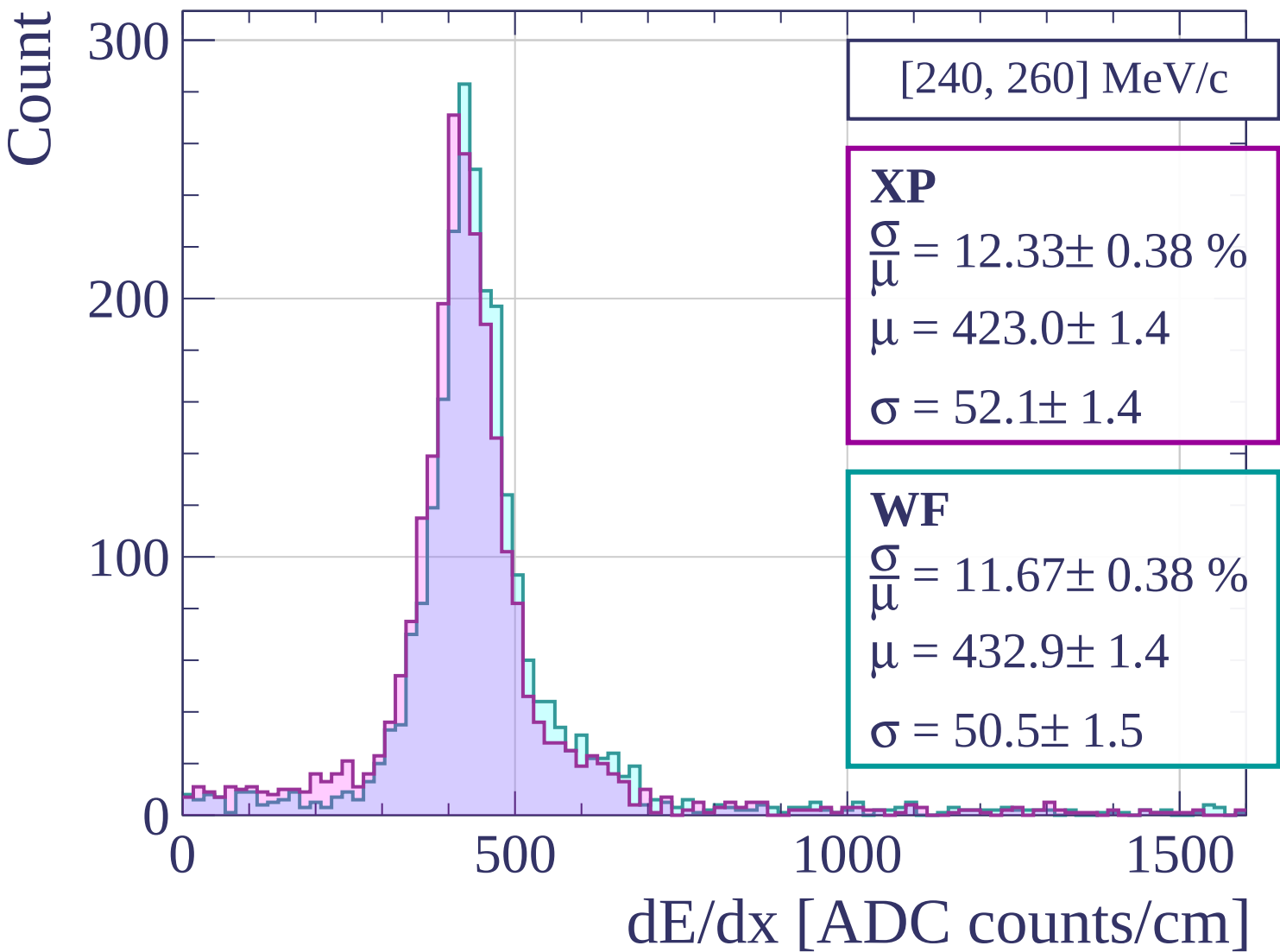
500

1000

1500

dE/dx [ADC counts/cm]





Count

[260, 280] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.99 \pm 0.38 \%$$

$$\mu = 413.4 \pm 1.3$$

$$\sigma = 45.4 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.41 \pm 0.43 \%$$

$$\mu = 423.9 \pm 1.5$$

$$\sigma = 48.4 \pm 1.7$$

200

100

0

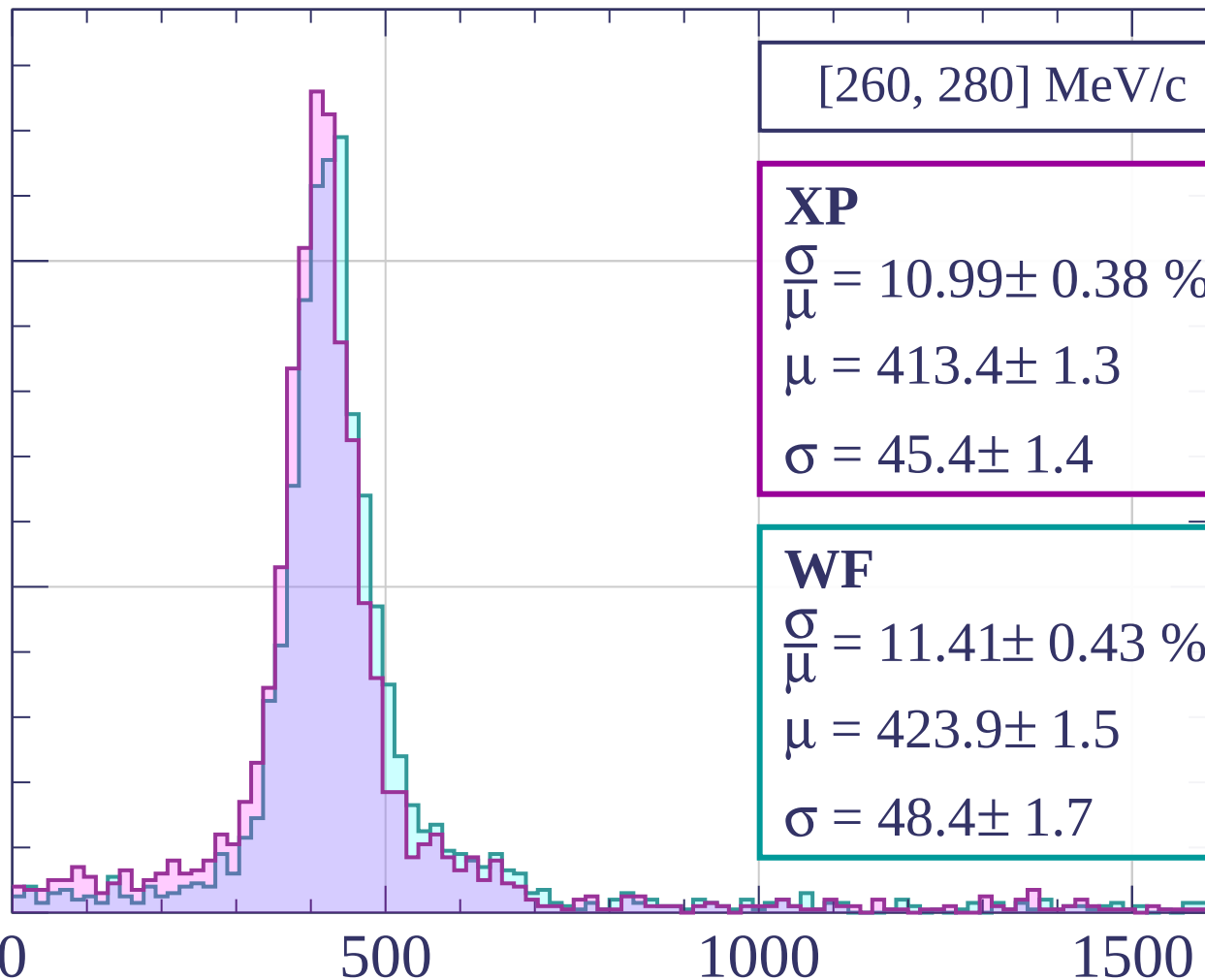
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[280, 300] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.66 \pm 0.36 \%$$

$$\mu = 409.7 \pm 1.3$$

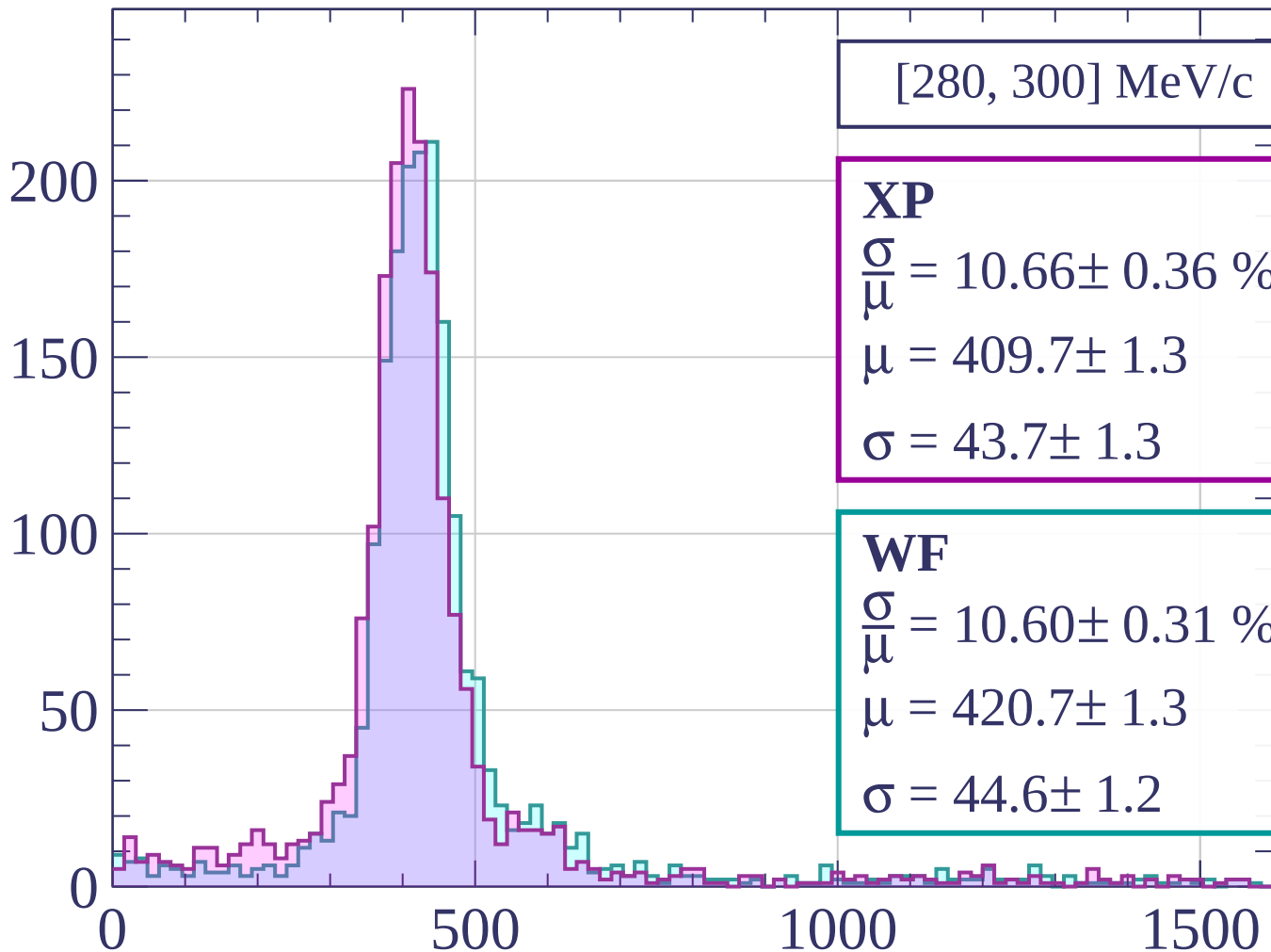
$$\sigma = 43.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.60 \pm 0.31 \%$$

$$\mu = 420.7 \pm 1.3$$

$$\sigma = 44.6 \pm 1.2$$



dE/dx [ADC counts/cm]

Count

[300, 320] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.53 \pm 0.39 \%$$

$$\mu = 406.9 \pm 1.3$$

$$\sigma = 38.8 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.85 \pm 0.37 \%$$

$$\mu = 415.5 \pm 1.3$$

$$\sigma = 40.9 \pm 1.4$$

200

100

0

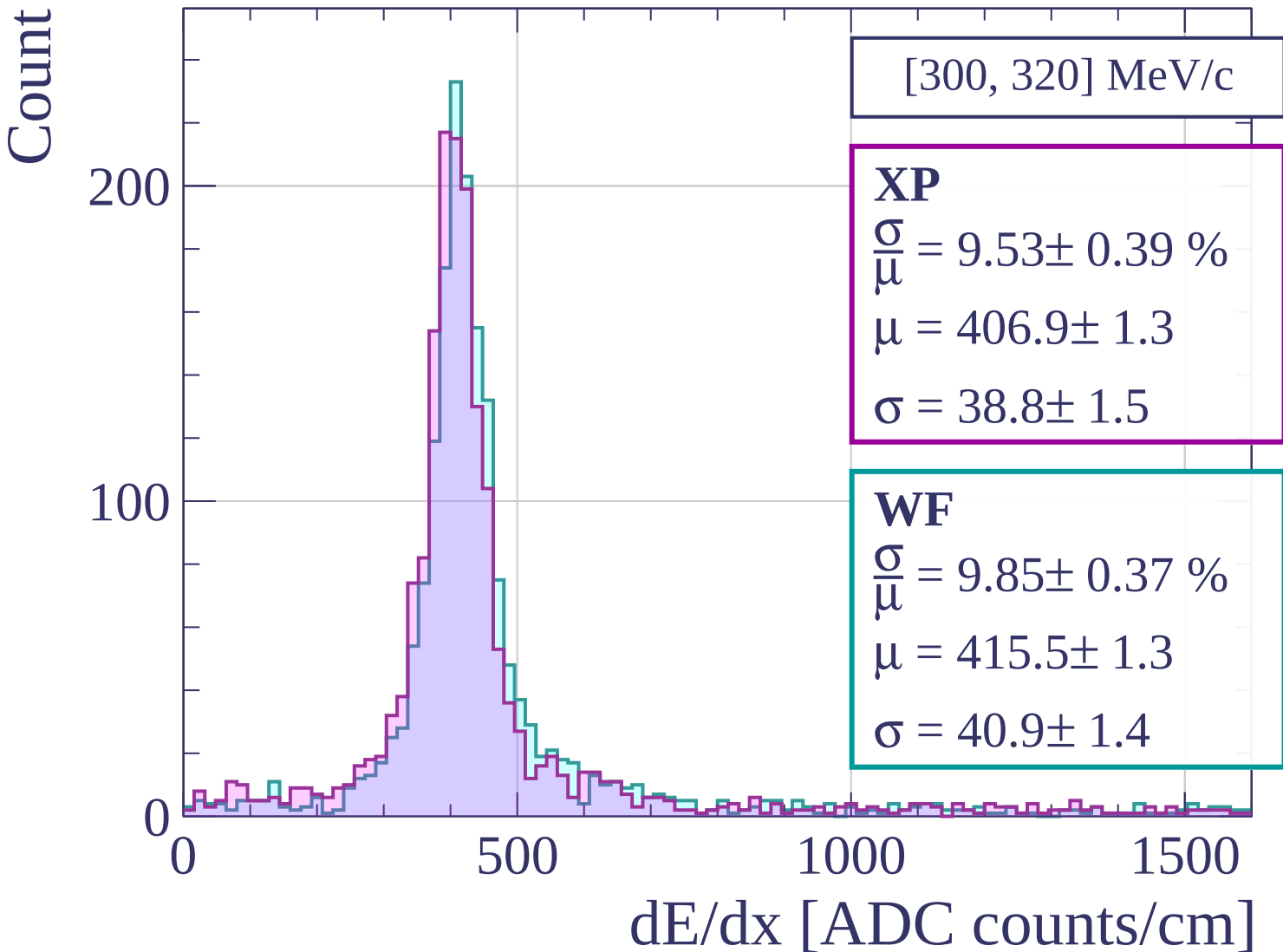
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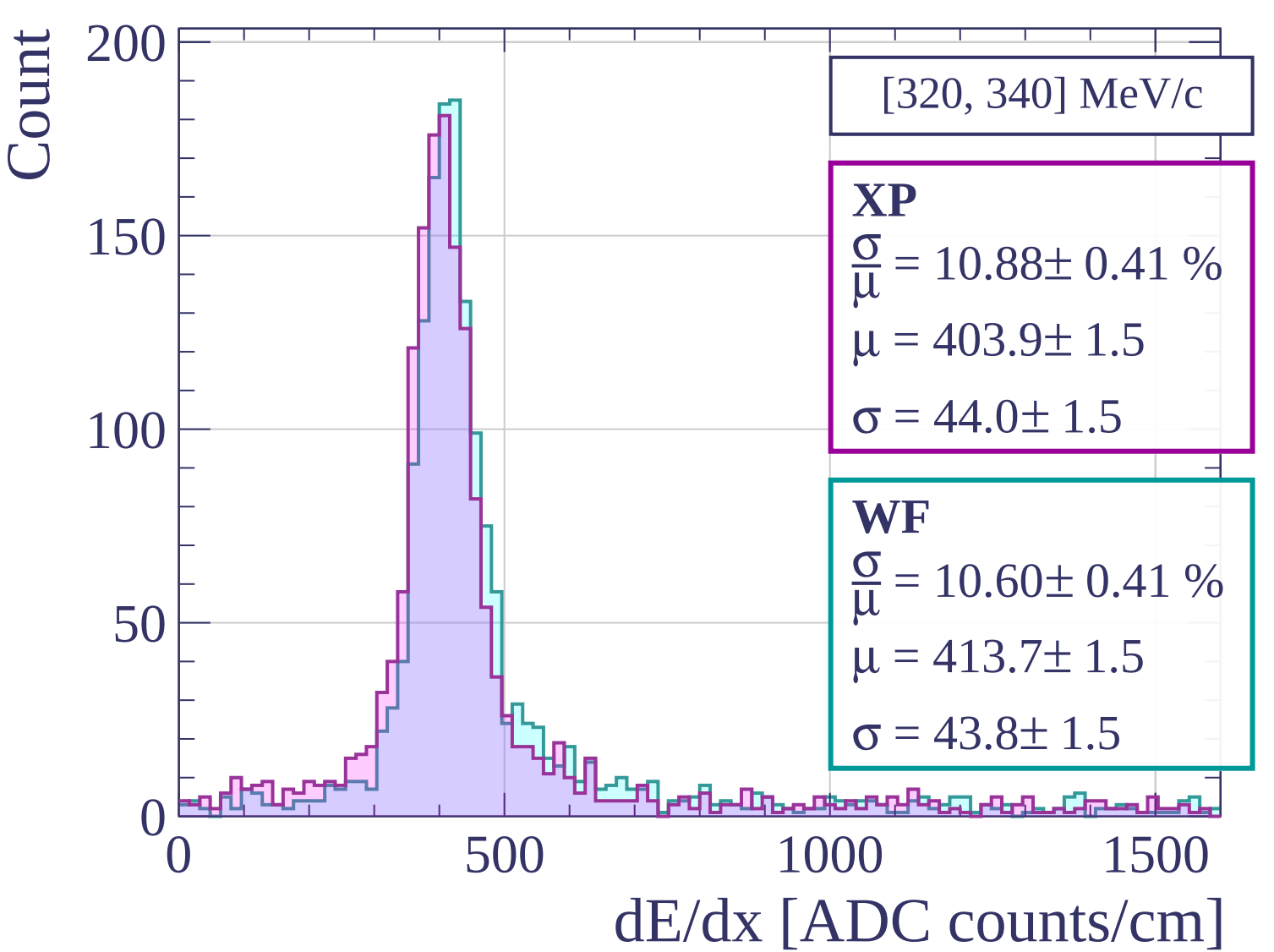
500

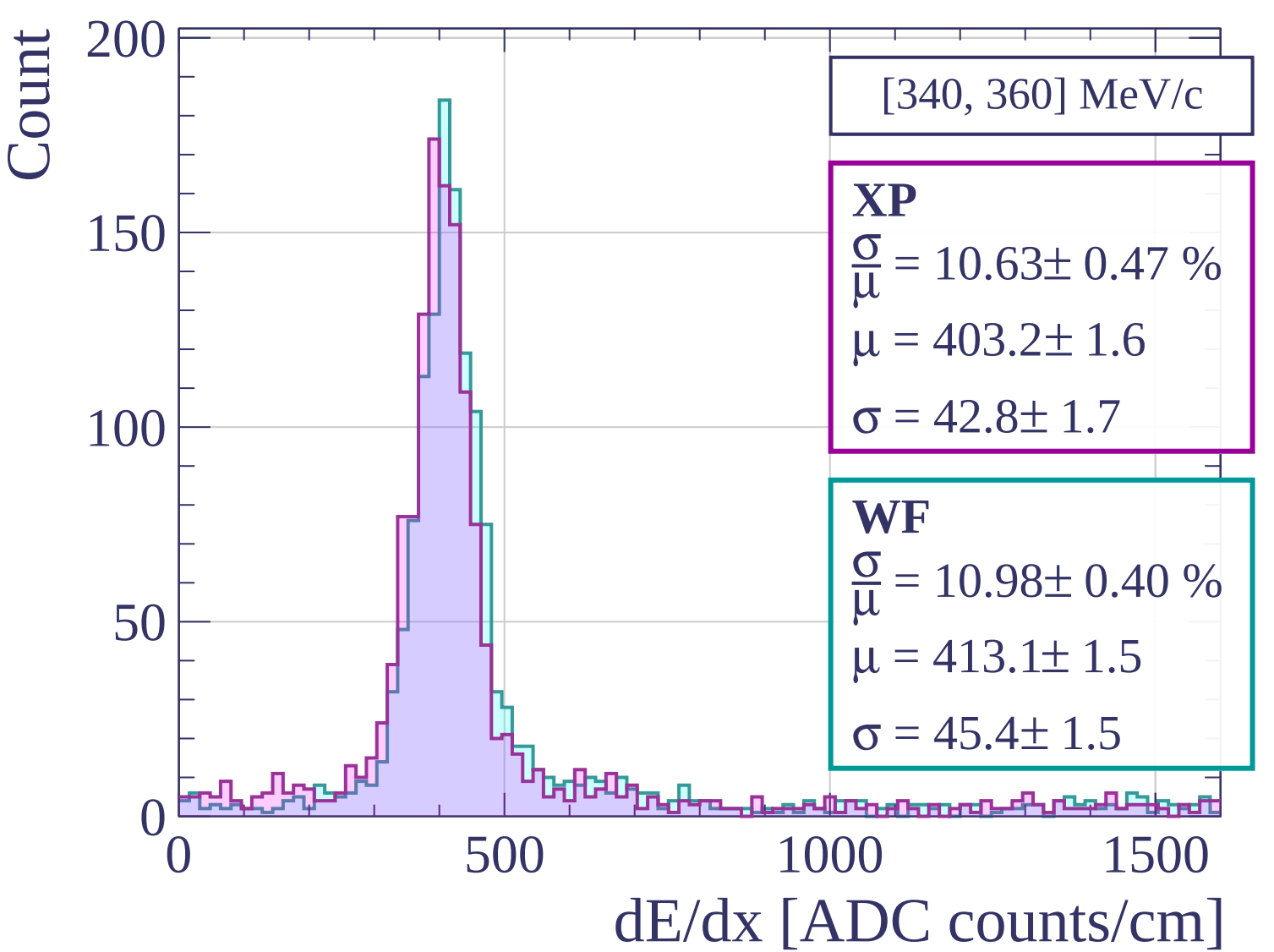
1000

1500

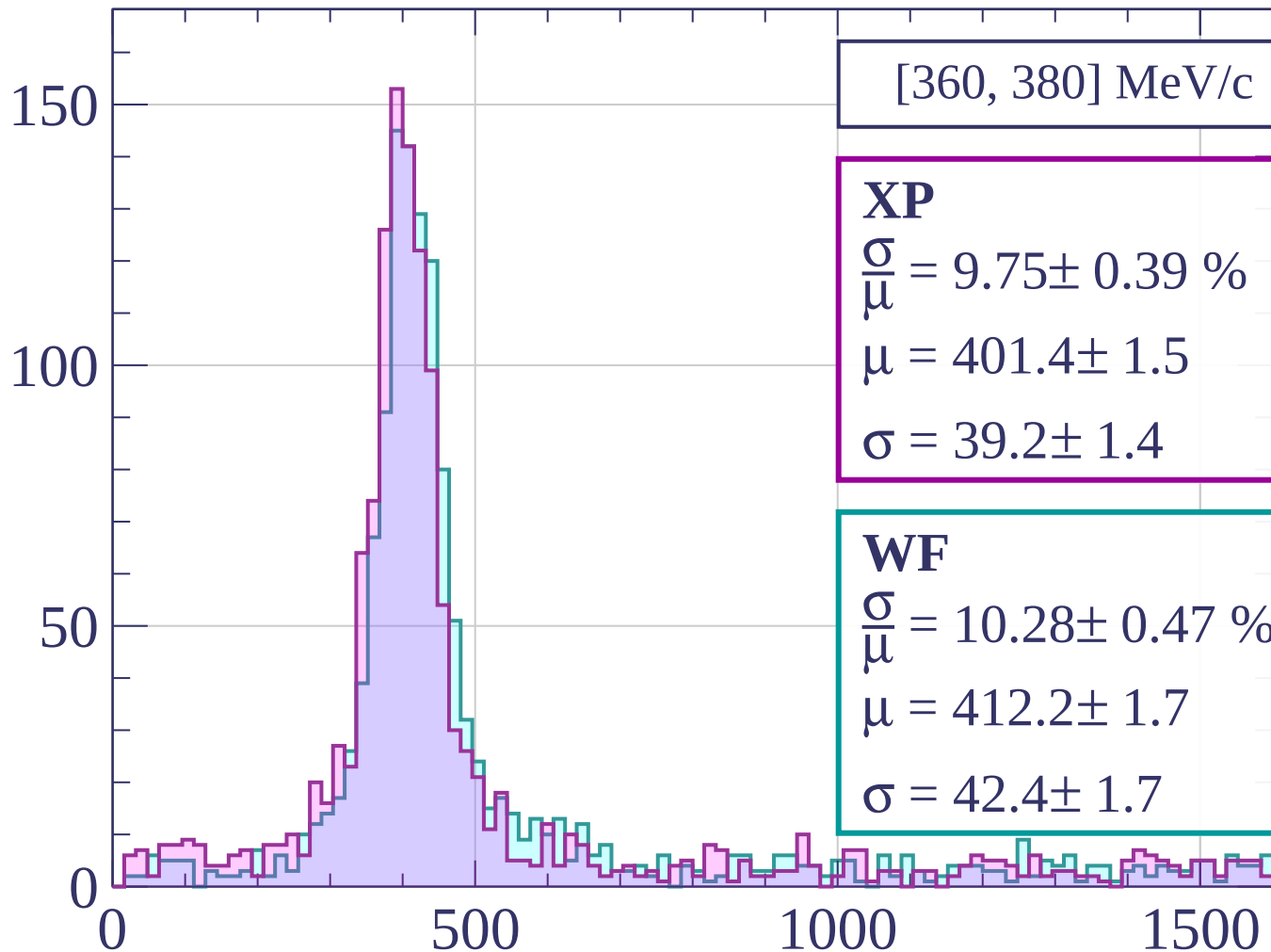
dE/dx [ADC counts/cm]



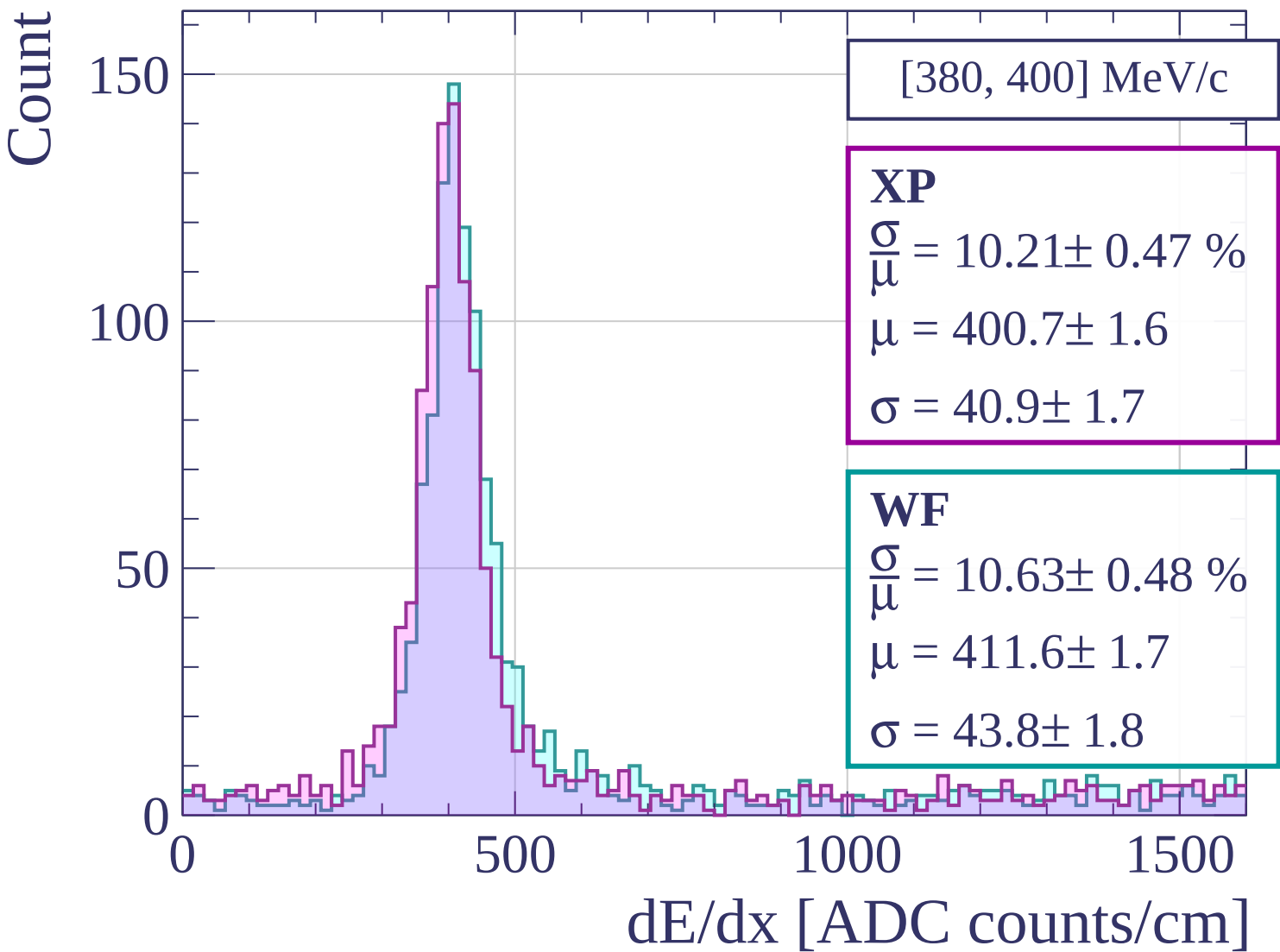




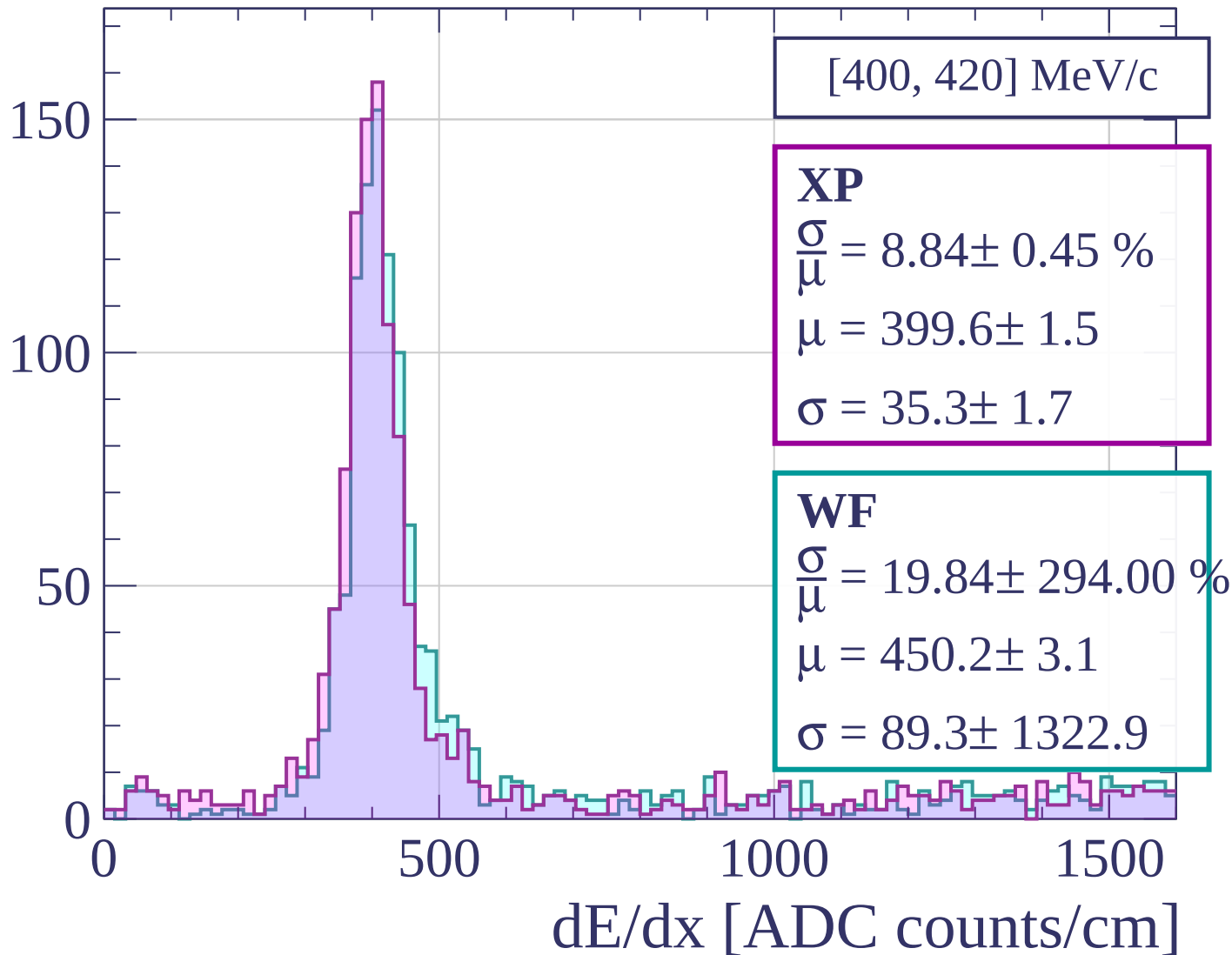
Count



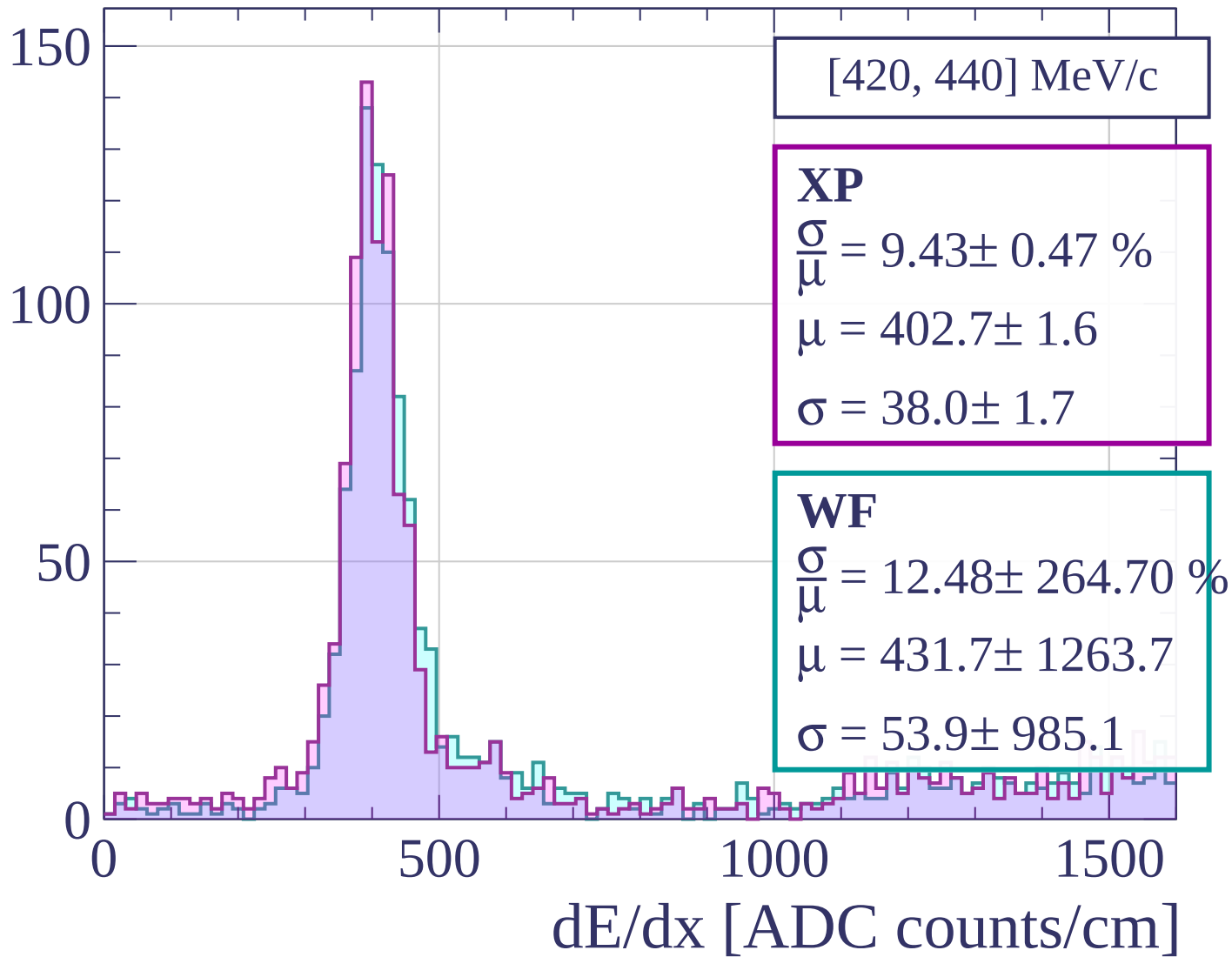
dE/dx [ADC counts/cm]



Count



Count



Count

[440, 460] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.67 \pm 0.45 \%$$

$$\mu = 400.0 \pm 1.7$$

$$\sigma = 38.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 10.61 \pm 0.50 \%$$

$$\mu = 410.8 \pm 1.9$$

$$\sigma = 43.6 \pm 1.9$$

0

500

1000

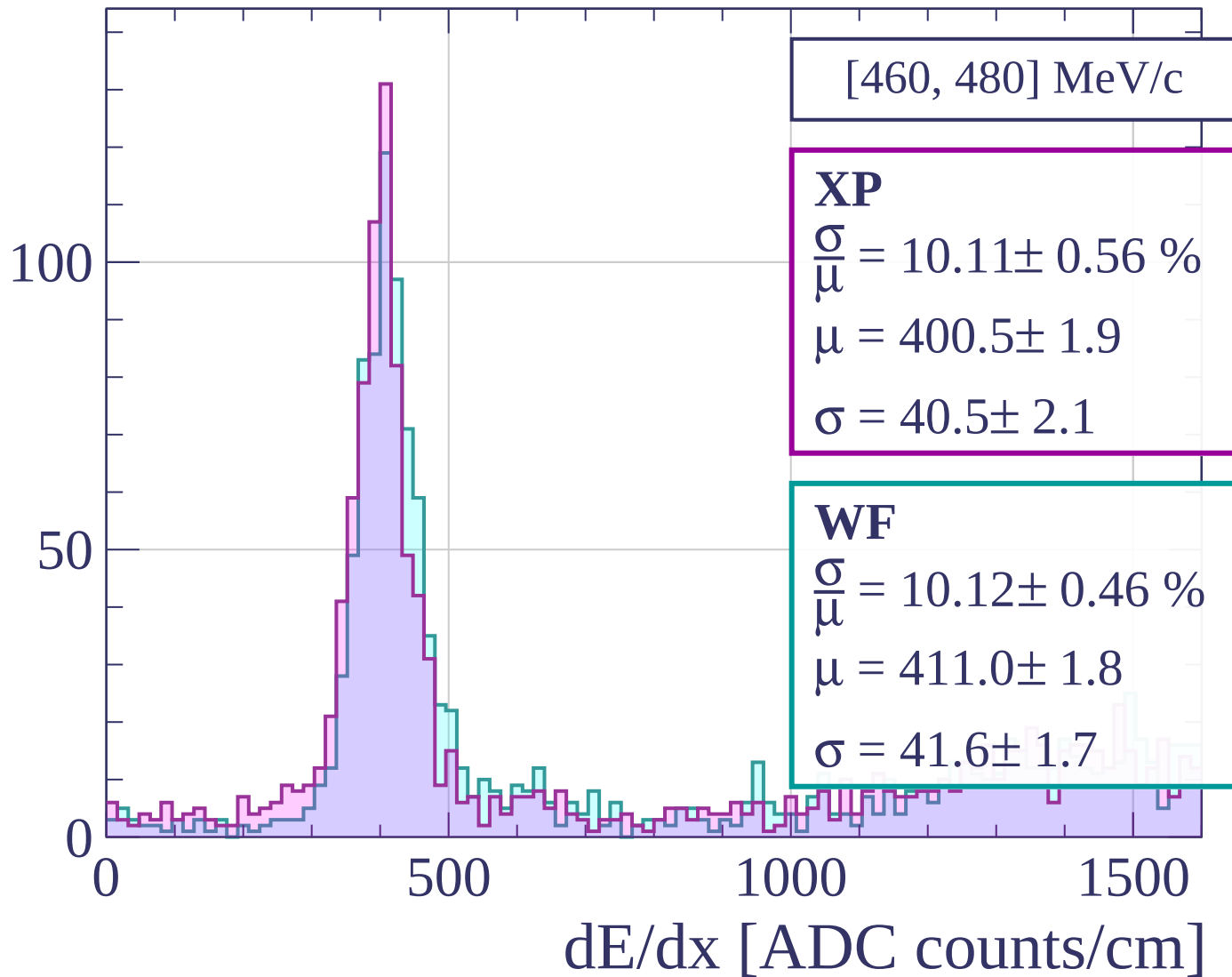
1500

dE/dx [ADC counts/cm]

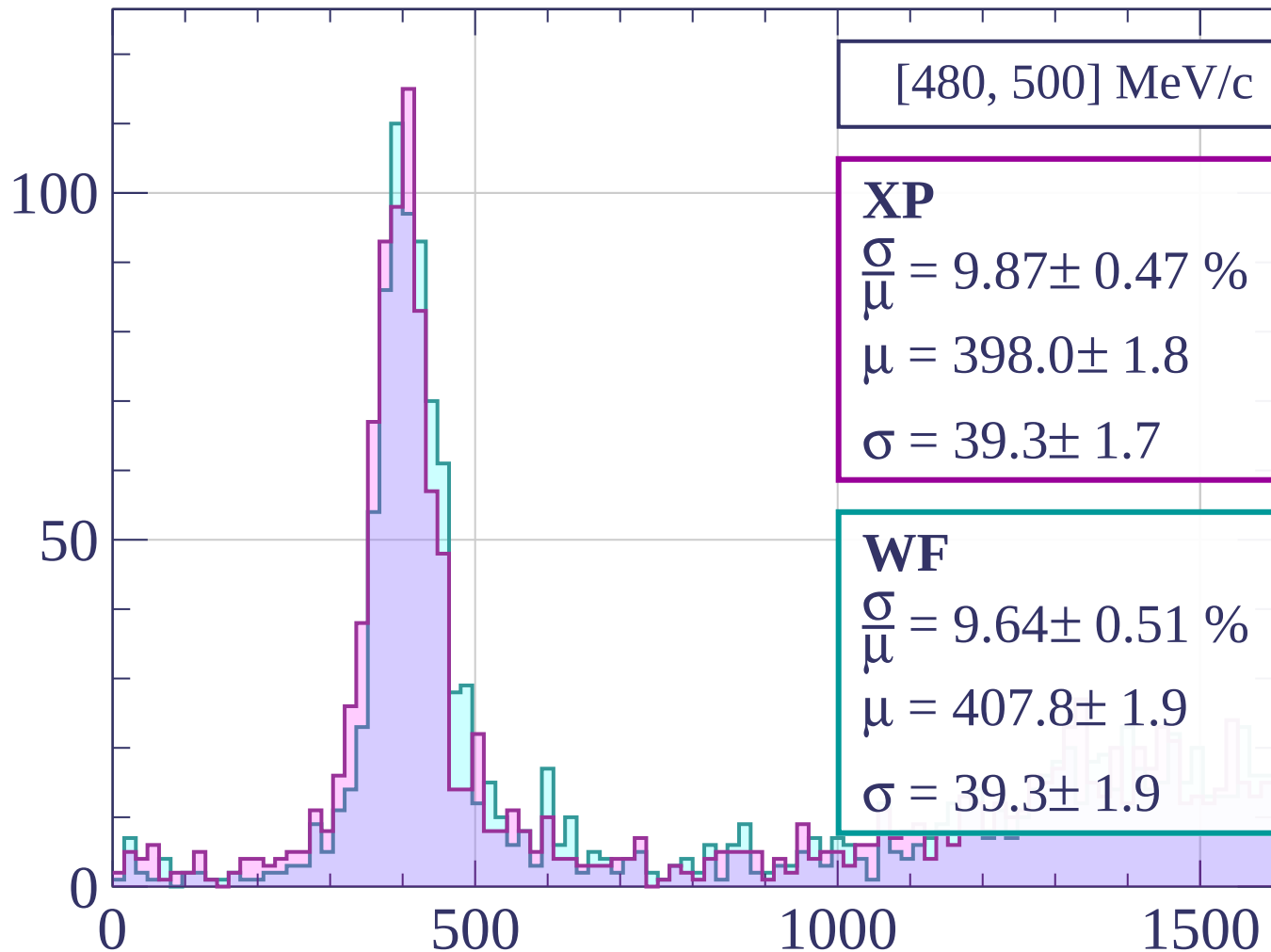
100

50

Count



Count



dE/dx [ADC counts/cm]

Count

[500, 520] MeV/c

XP

$$\frac{\sigma}{\mu} = 14.09 \pm 8.12 \%$$

$$\mu = 1350.5 \pm 119.2$$

$$\sigma = 190.2 \pm 92.8$$

WF

$$\frac{\sigma}{\mu} = 28.56 \pm 19.10 \%$$

$$\mu = 8563.0 \pm 3737.5$$

$$\sigma = 2445.8 \pm 568.1$$

0

500

1000

1500

dE/dx [ADC counts/cm]

100

50

Count

[520, 540] MeV/c

XP

$$\frac{\sigma}{\mu} = 35.07 \pm 36.16 \%$$

$$\mu = 8419.2 \pm 1040.1$$

$$\sigma = 2952.9 \pm 2679.8$$

WF

$$\frac{\sigma}{\mu} = 22.88 \pm 534.79 \%$$

$$\mu = 435.2 \pm 3.6$$

$$\sigma = 99.6 \pm 2326.5$$

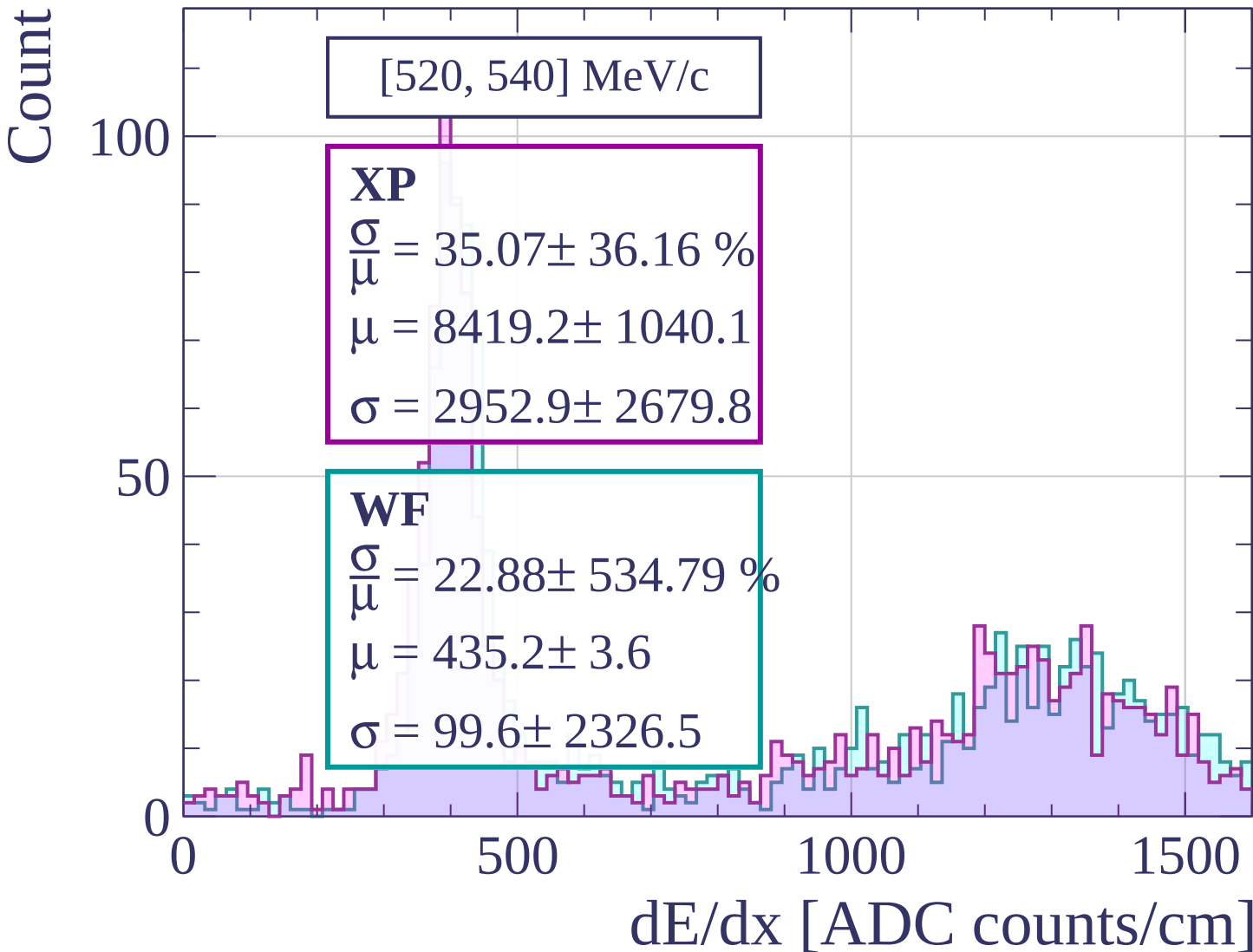
0

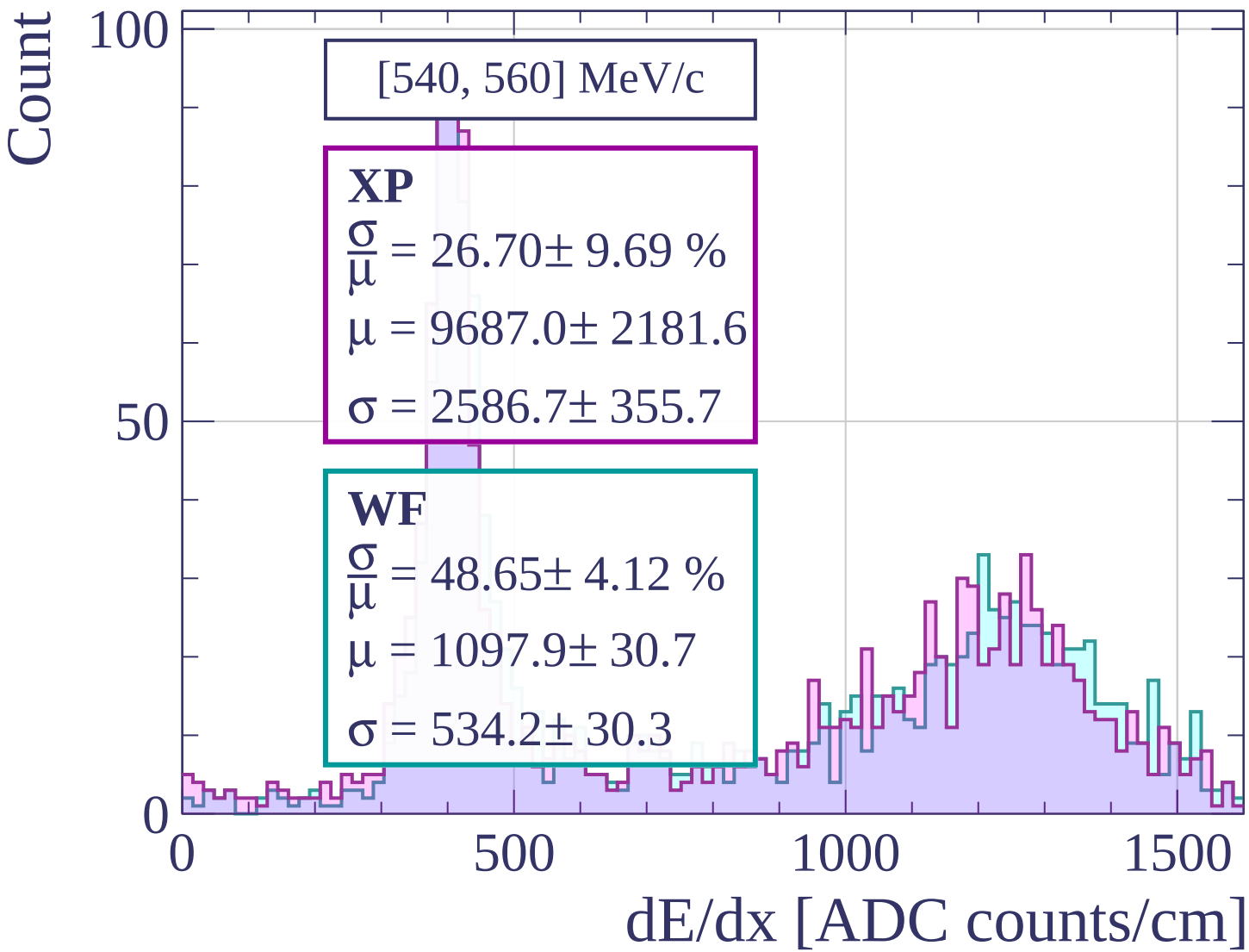
500

1000

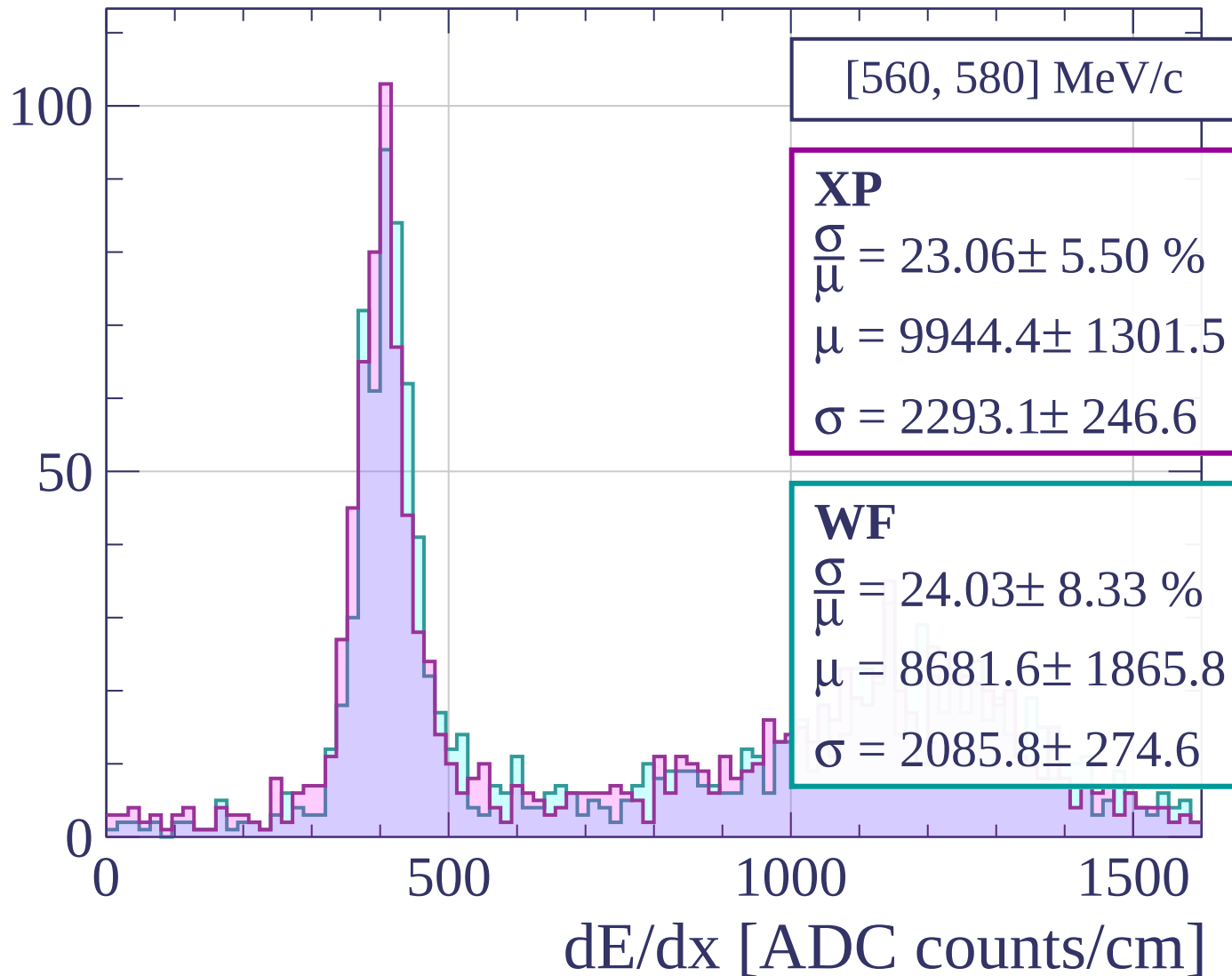
1500

dE/dx [ADC counts/cm]

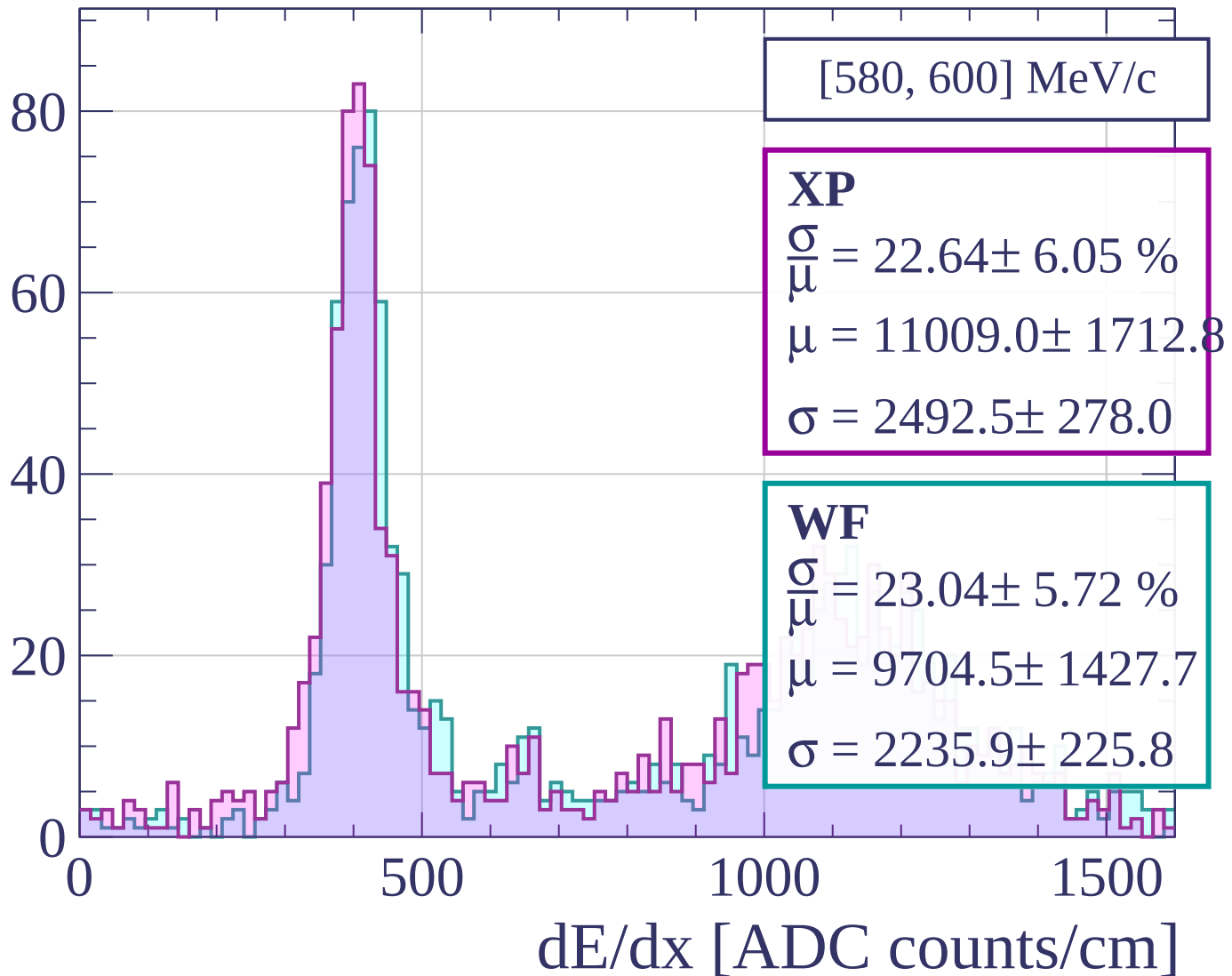




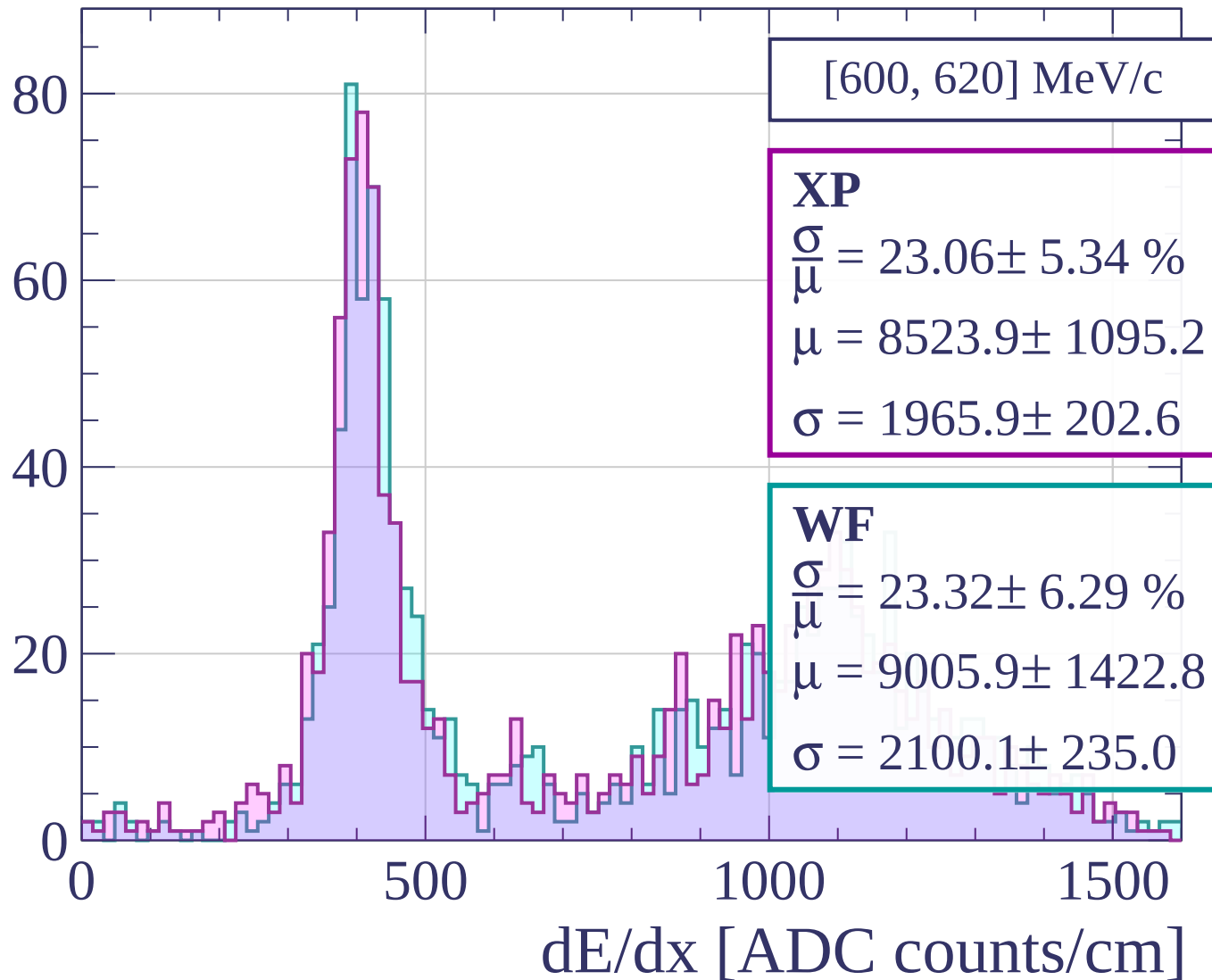
Count



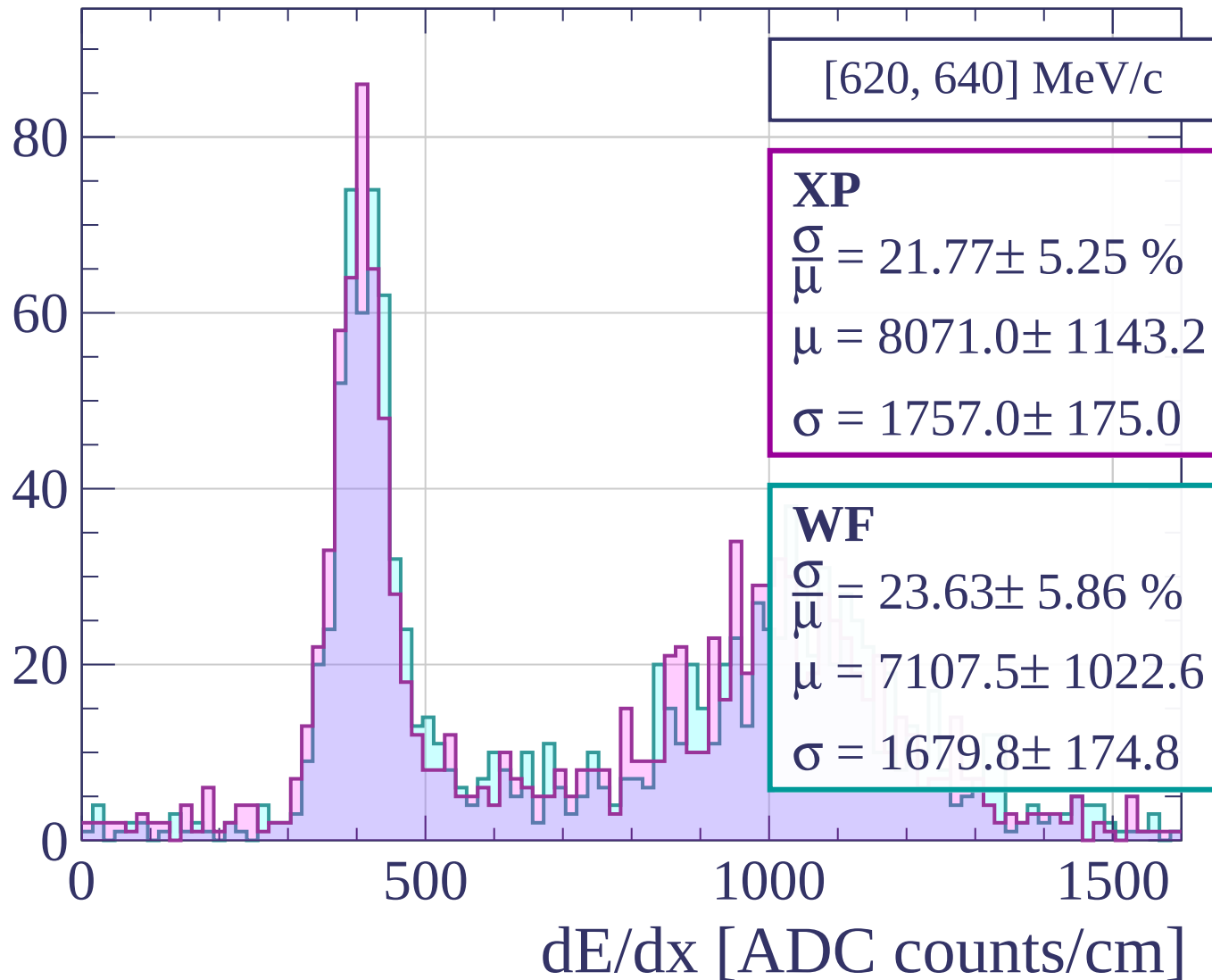
Count



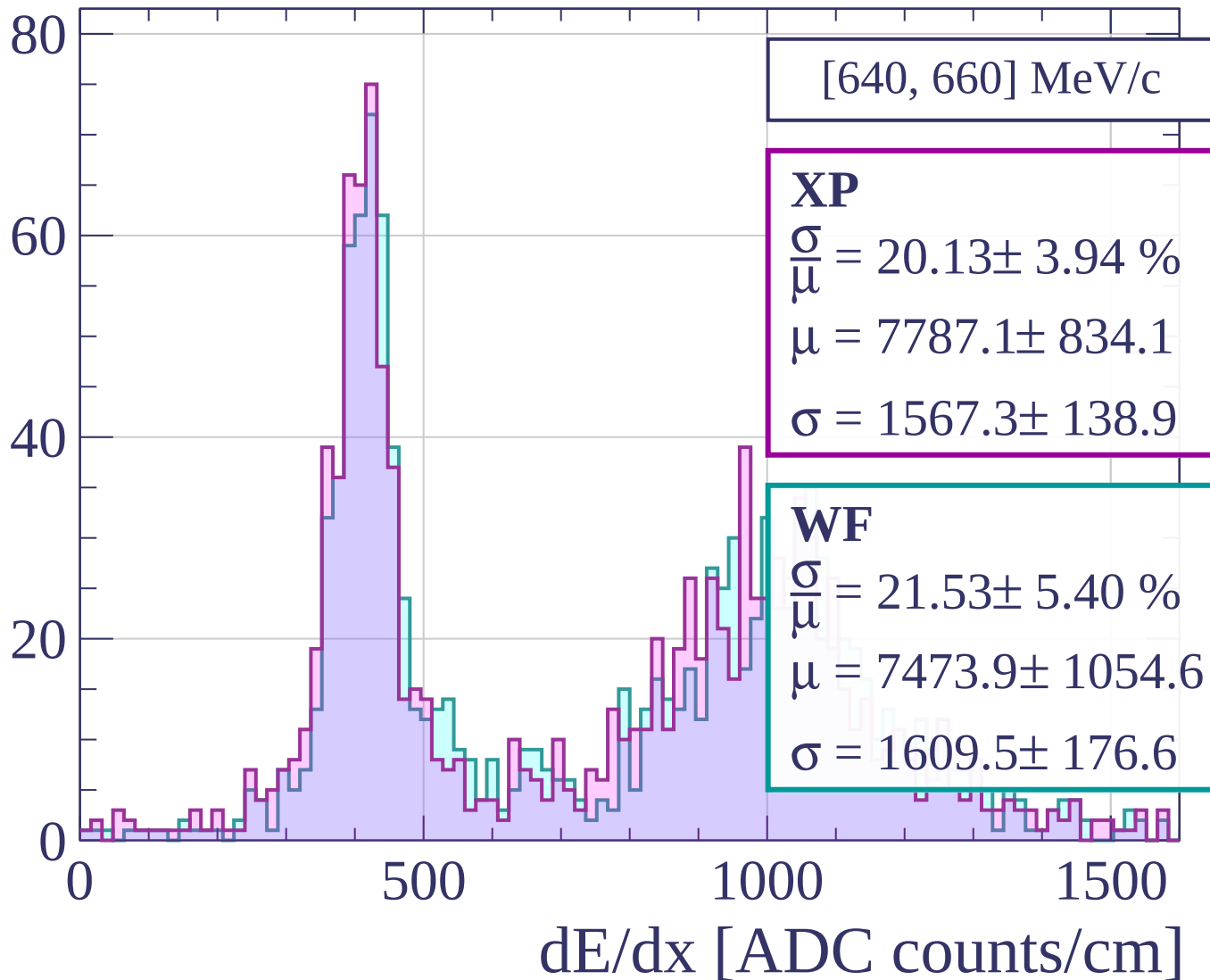
Count



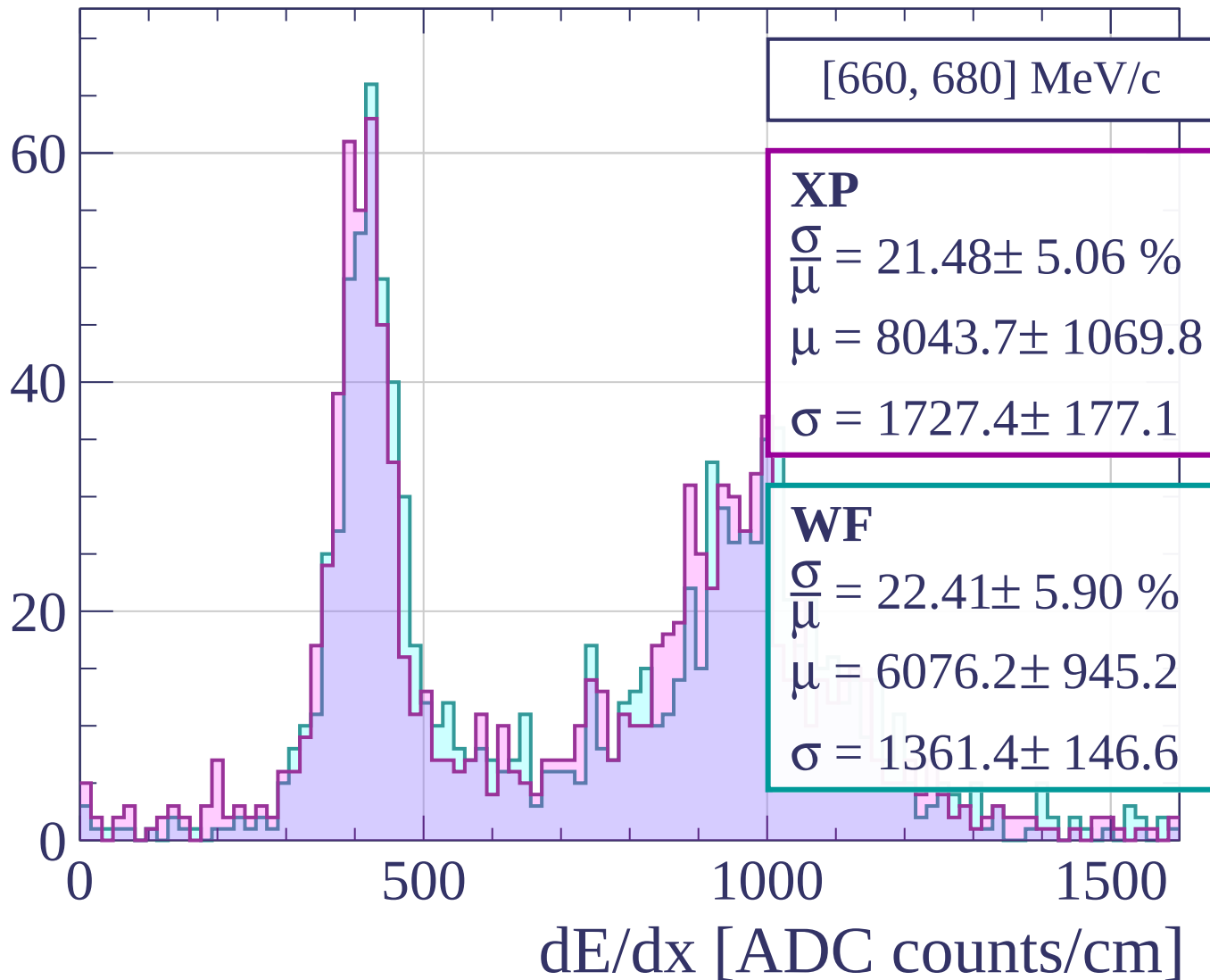
Count



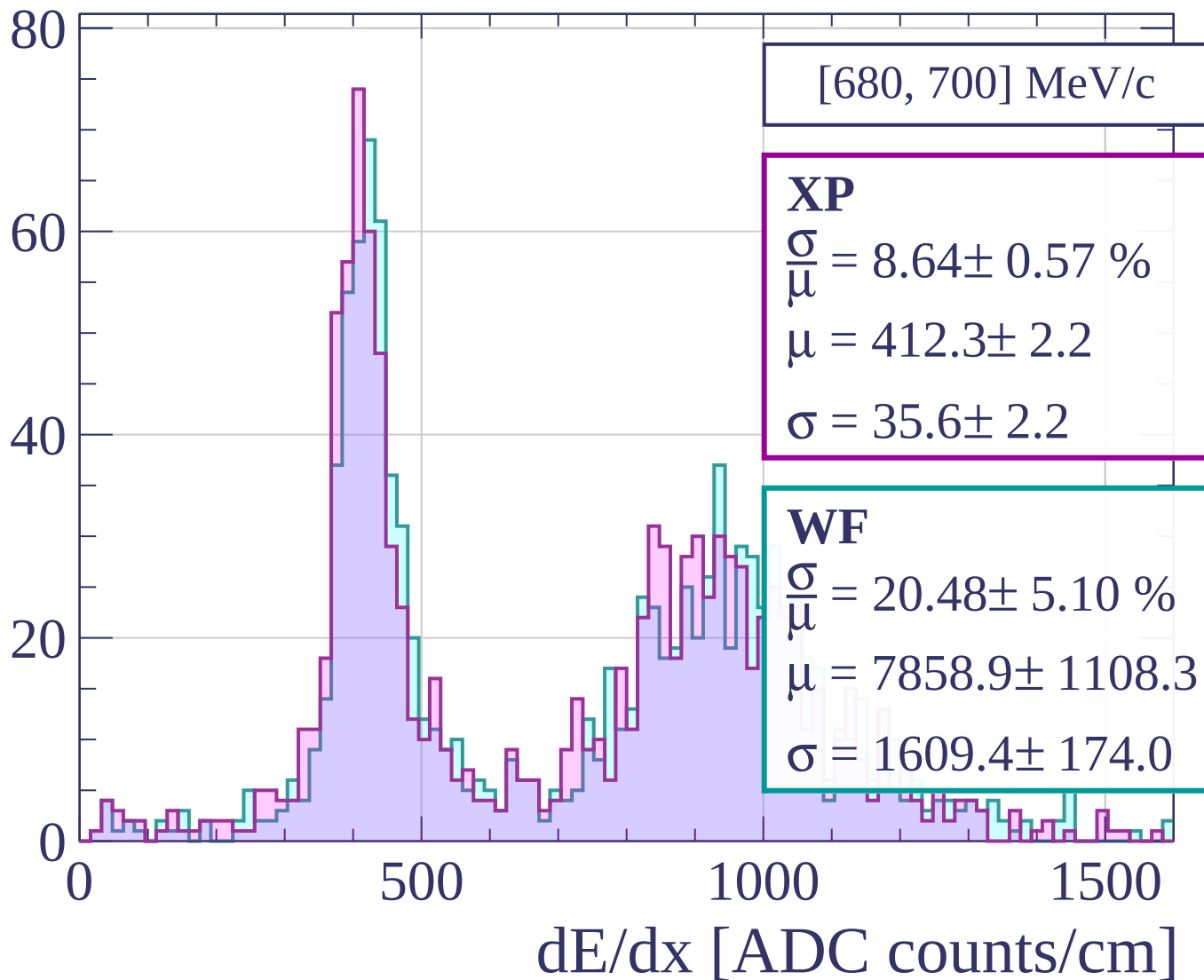
Count



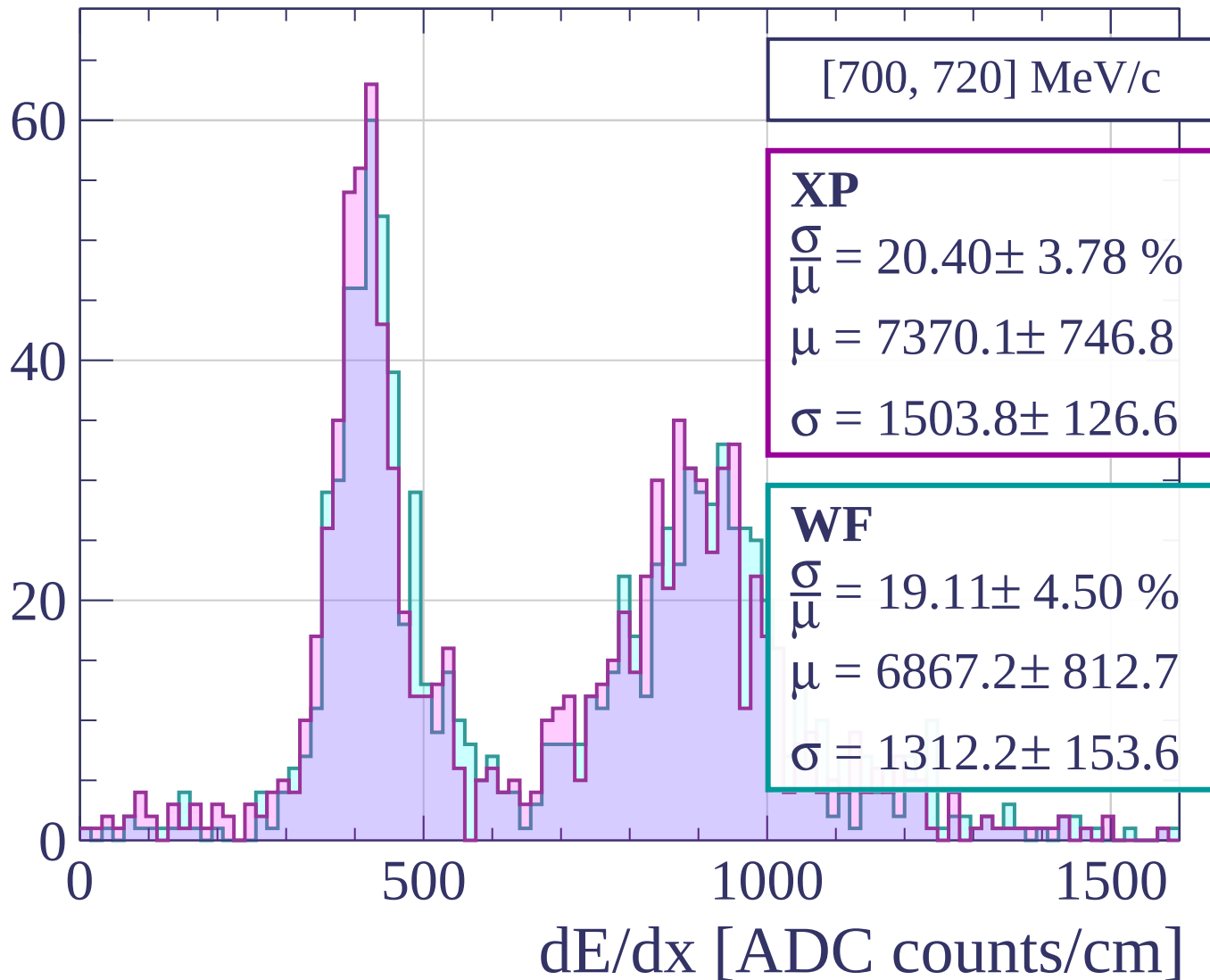
Count



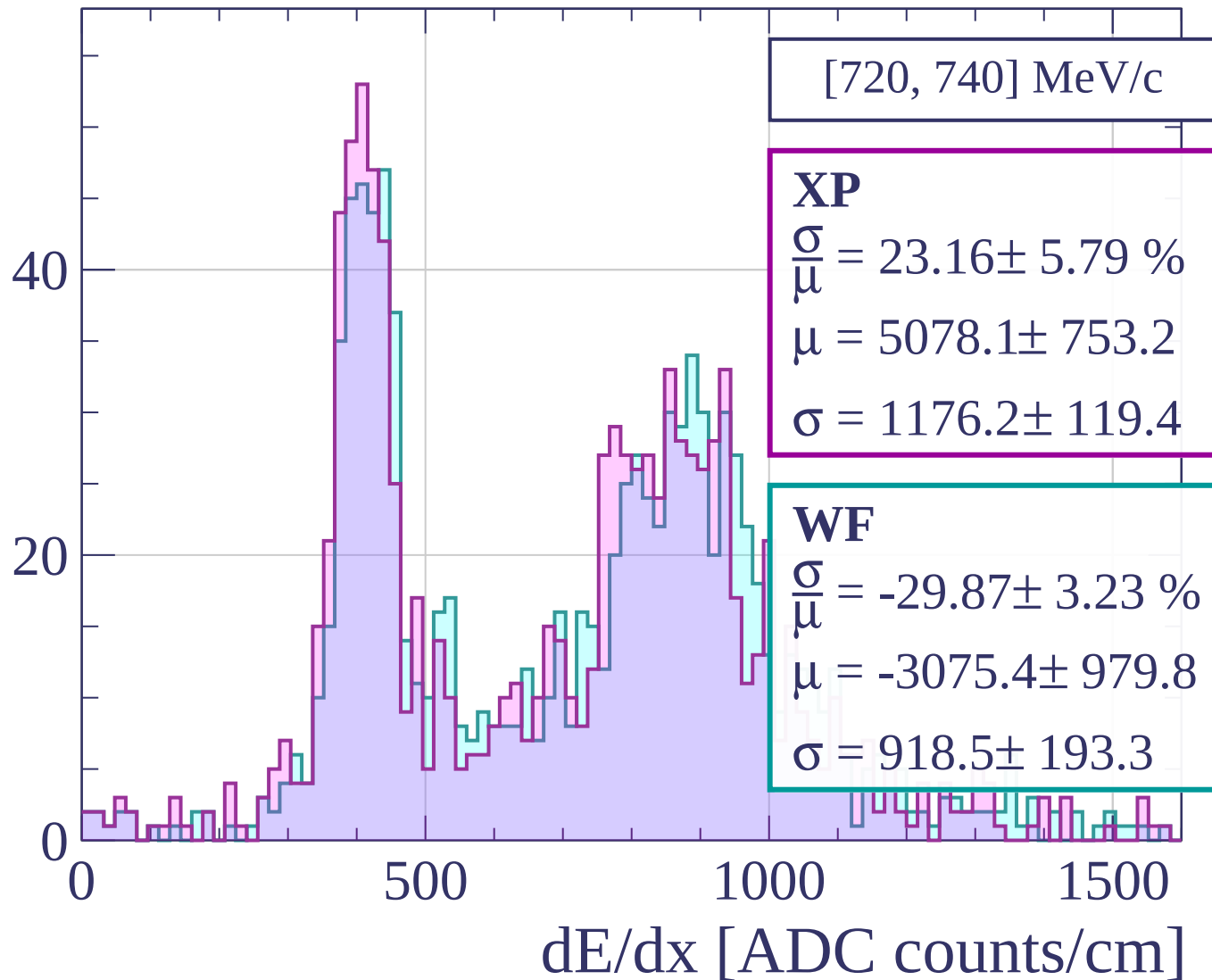
Count



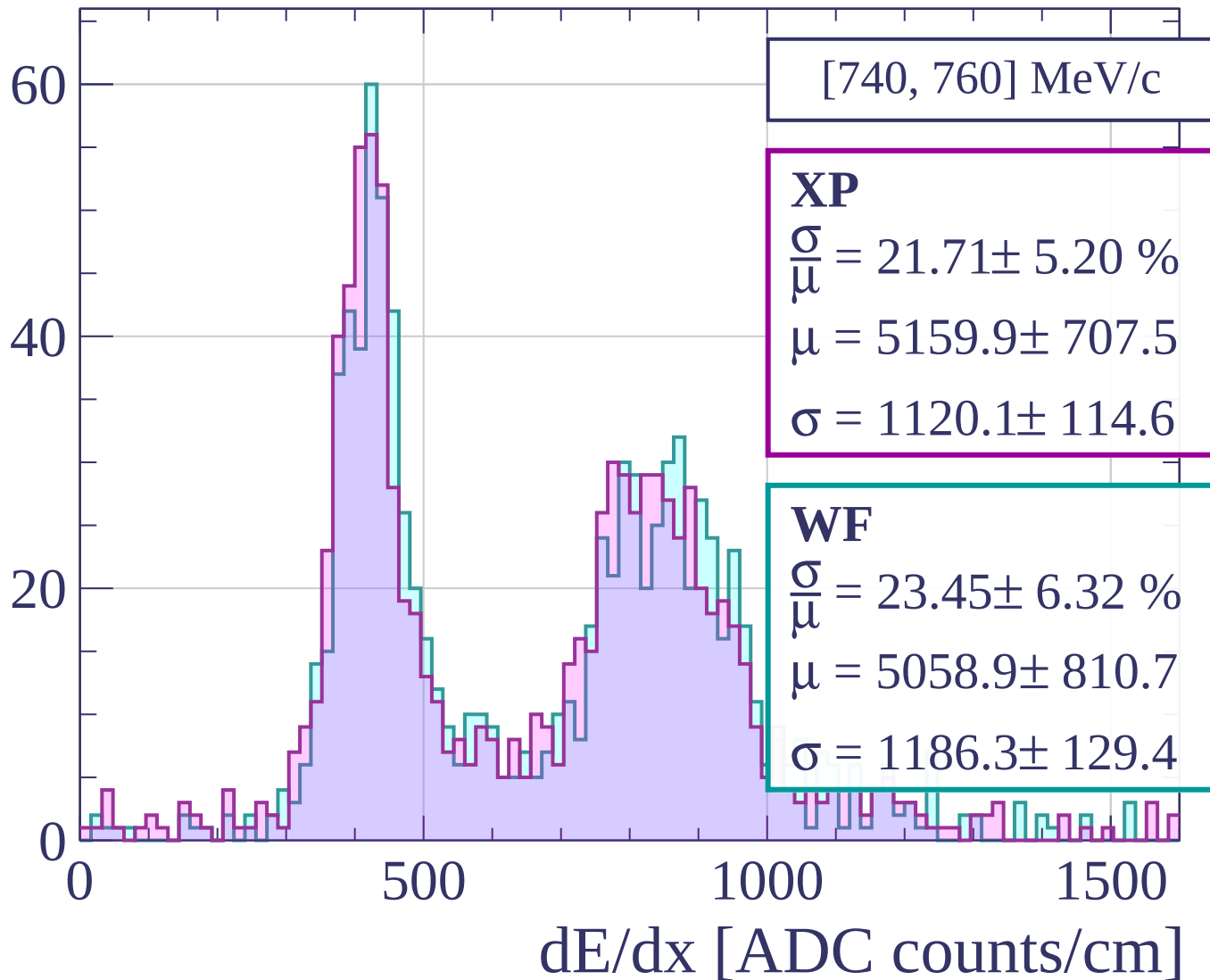
Count



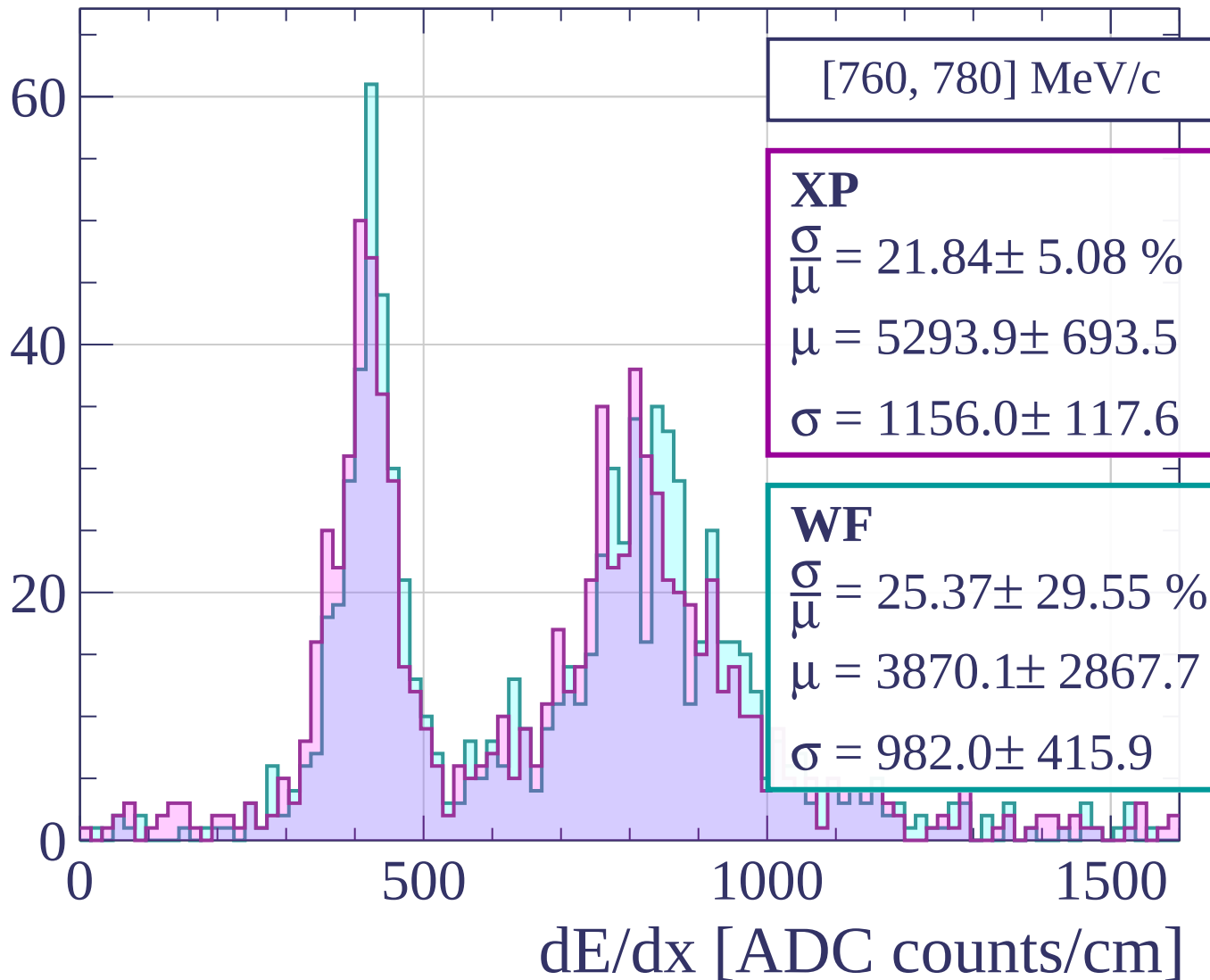
Count



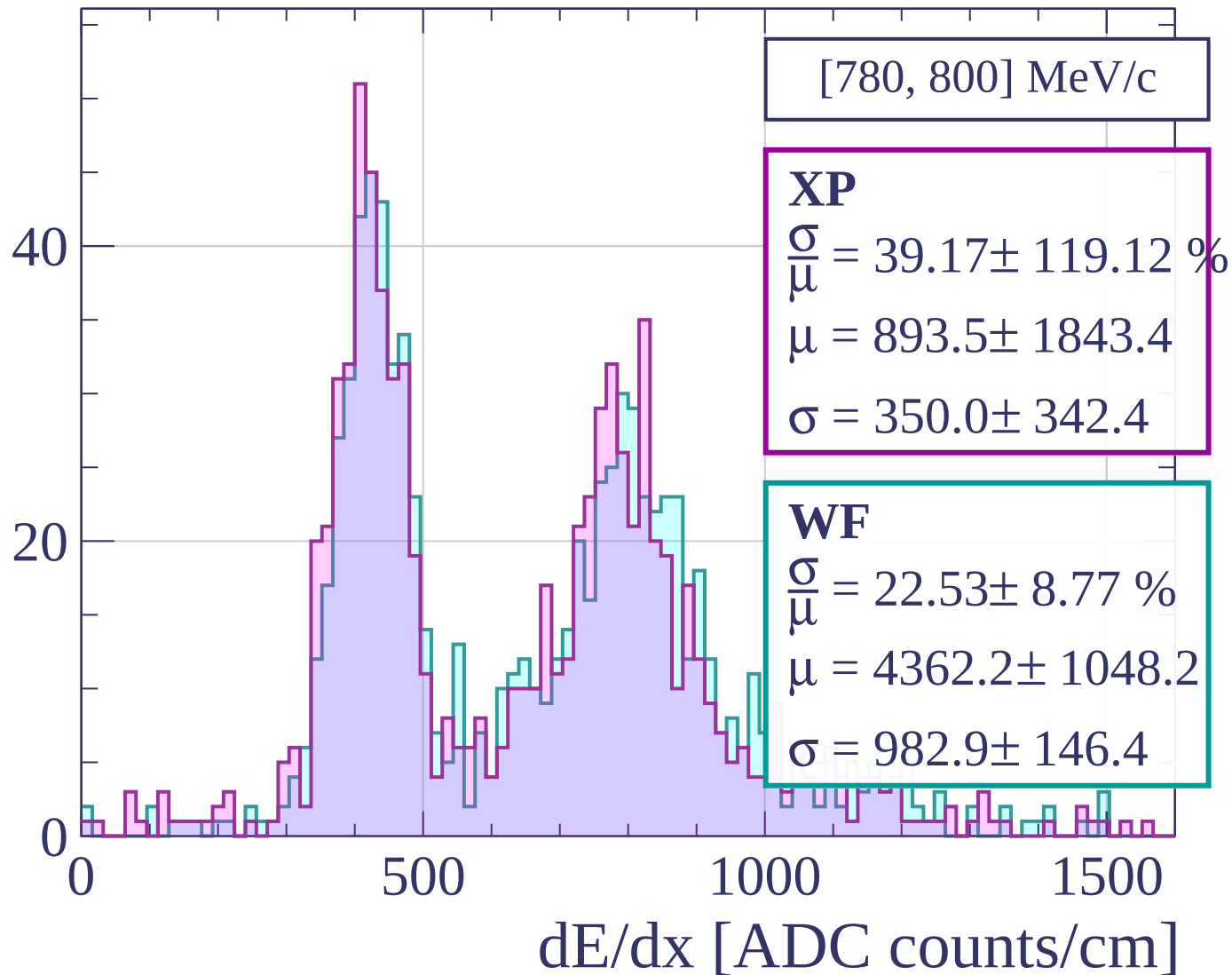
Count



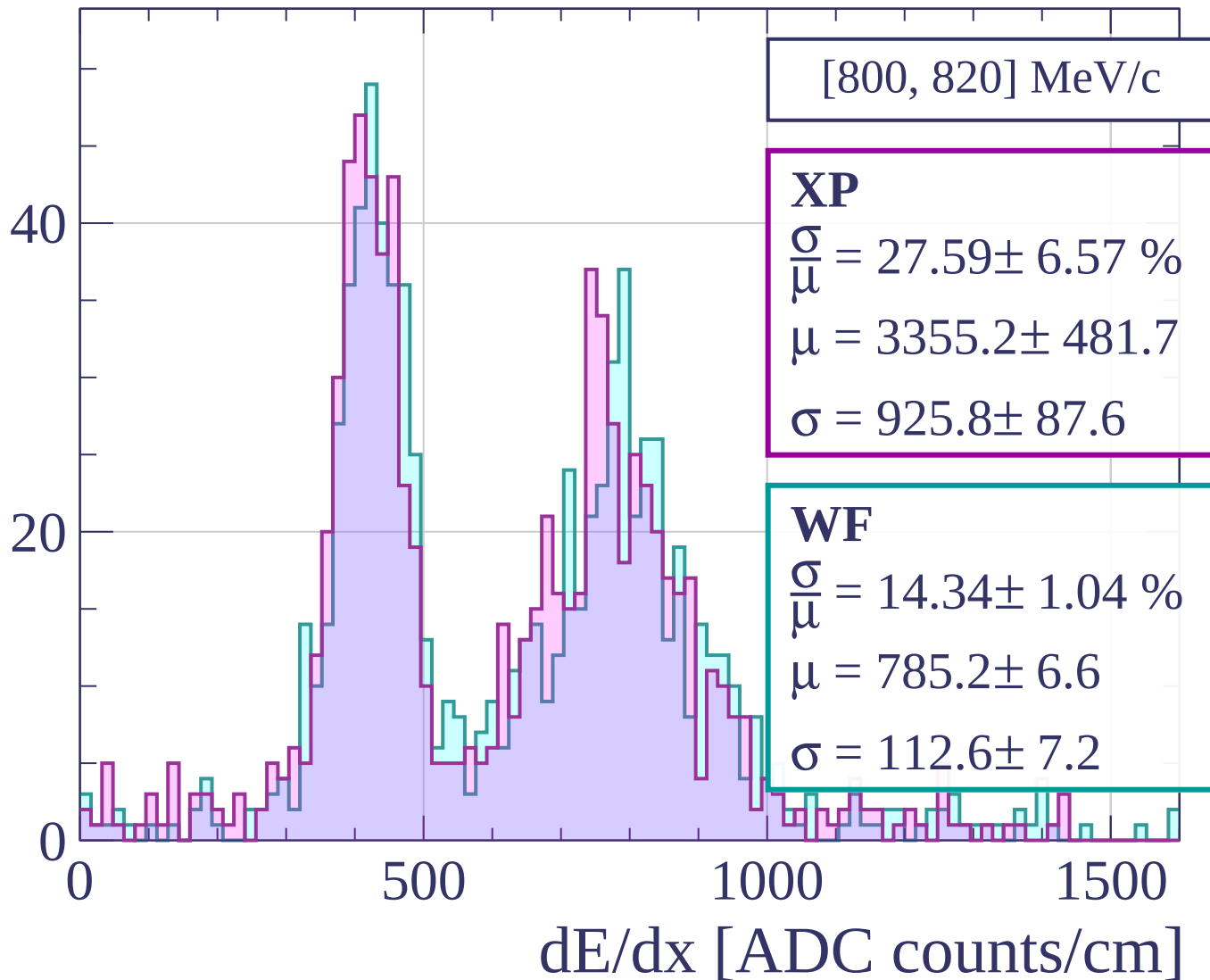
Count



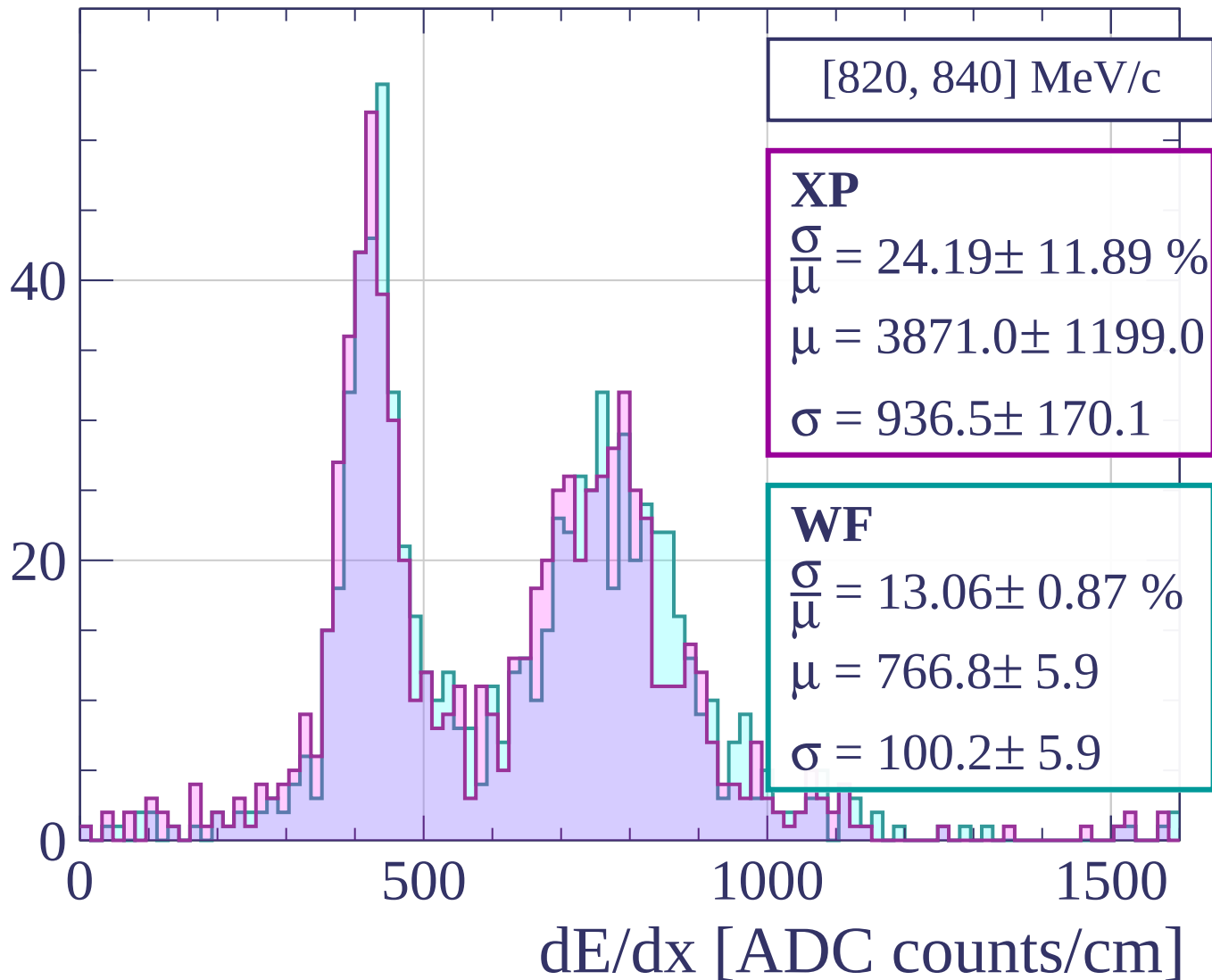
Count



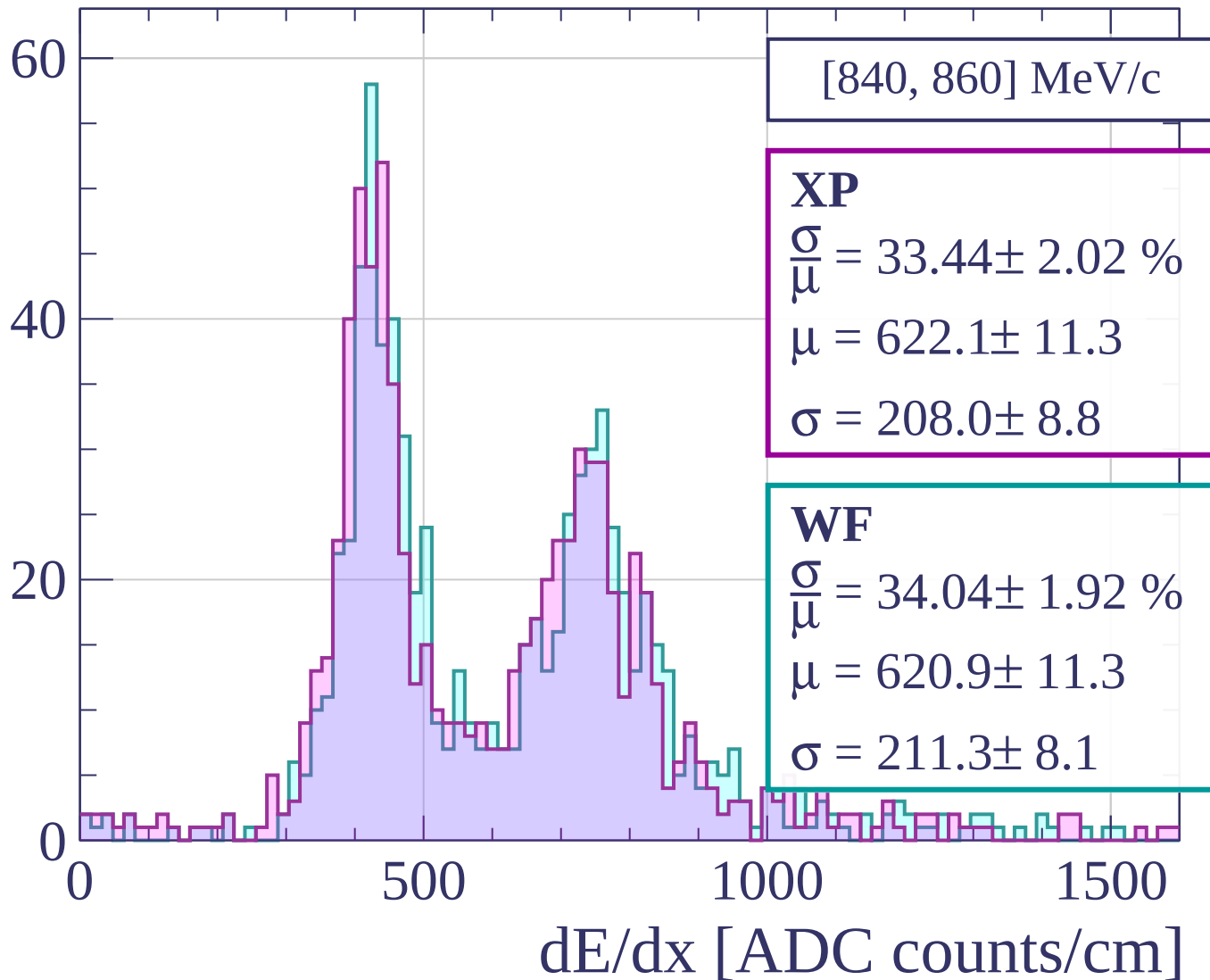
Count



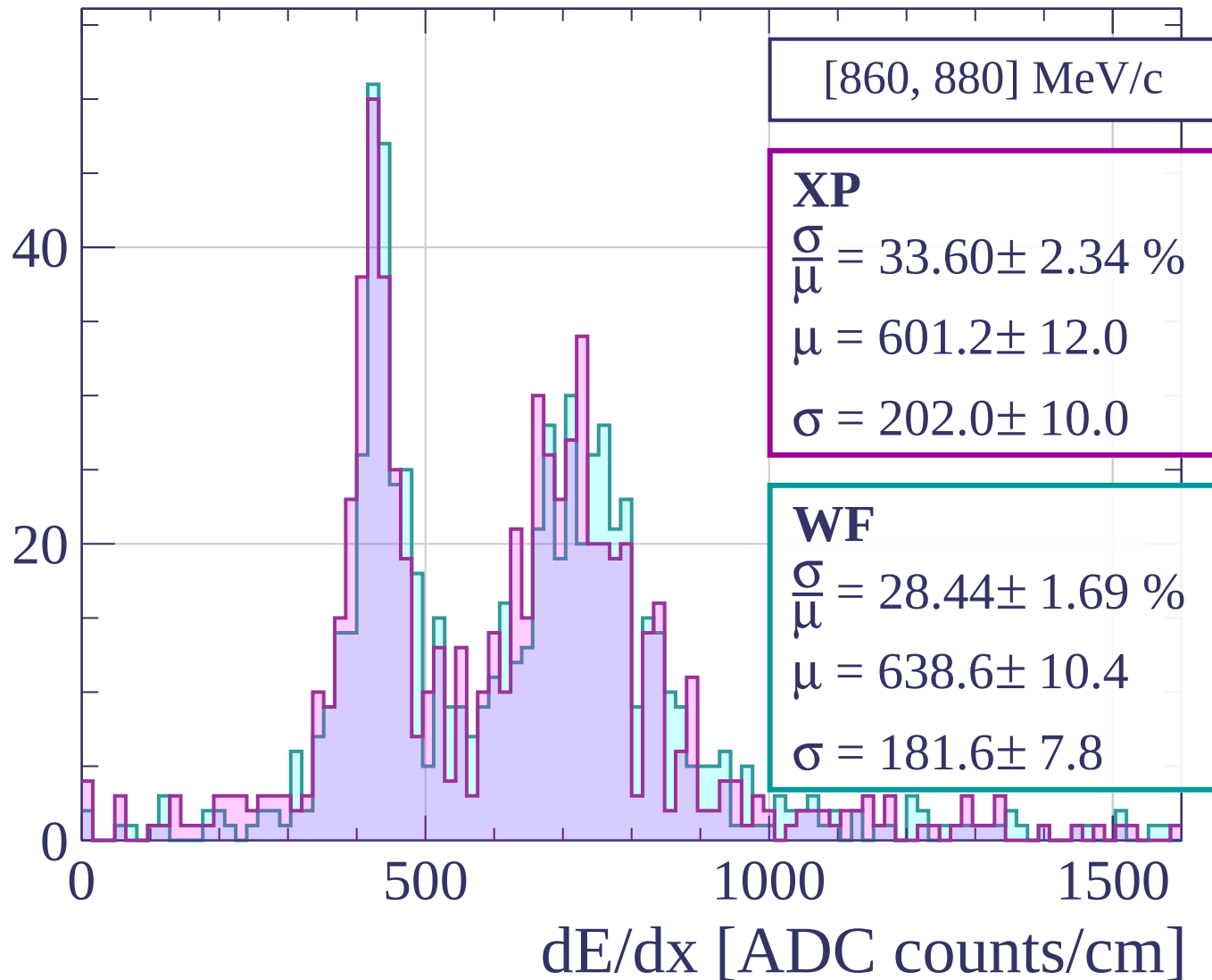
Count



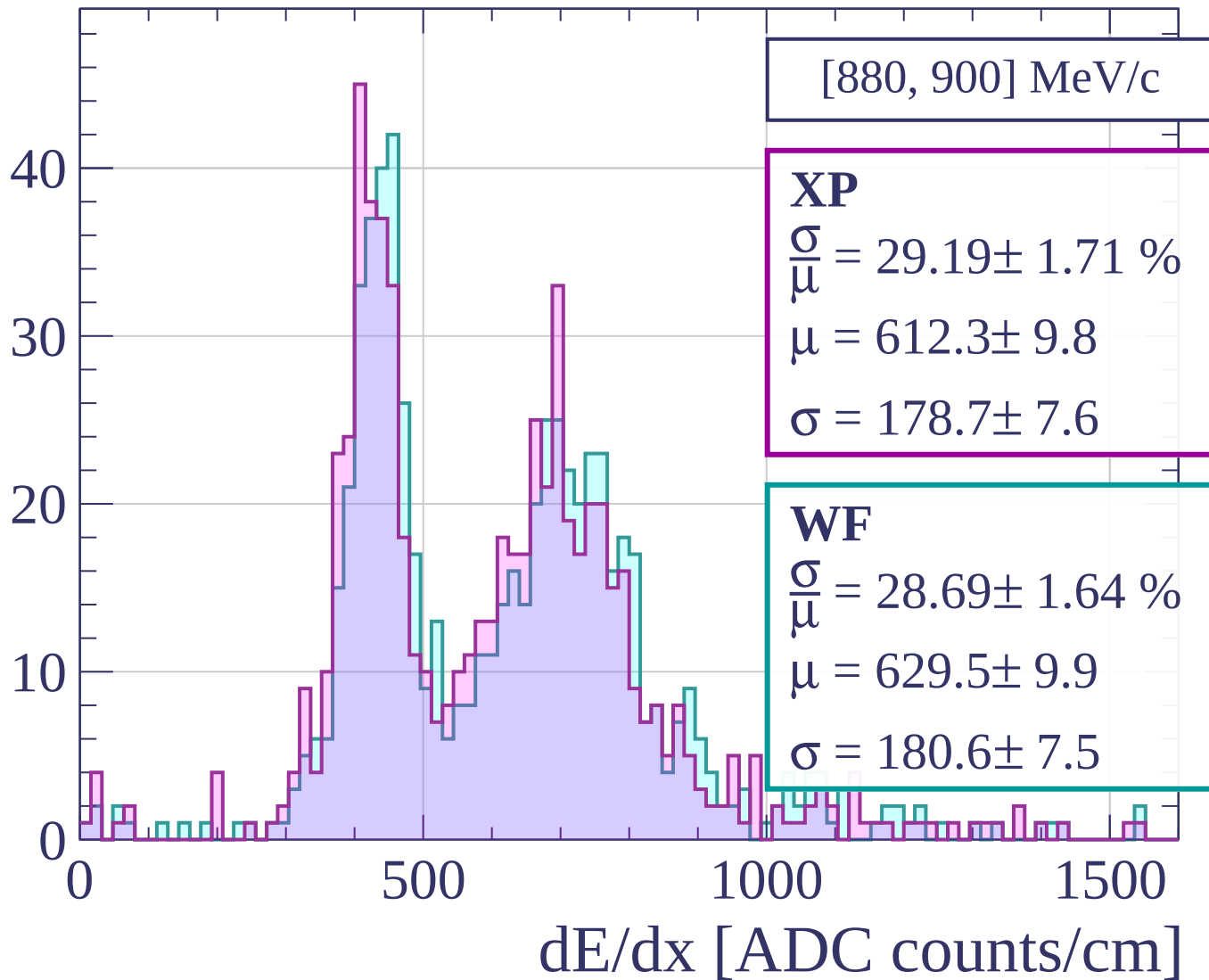
Count



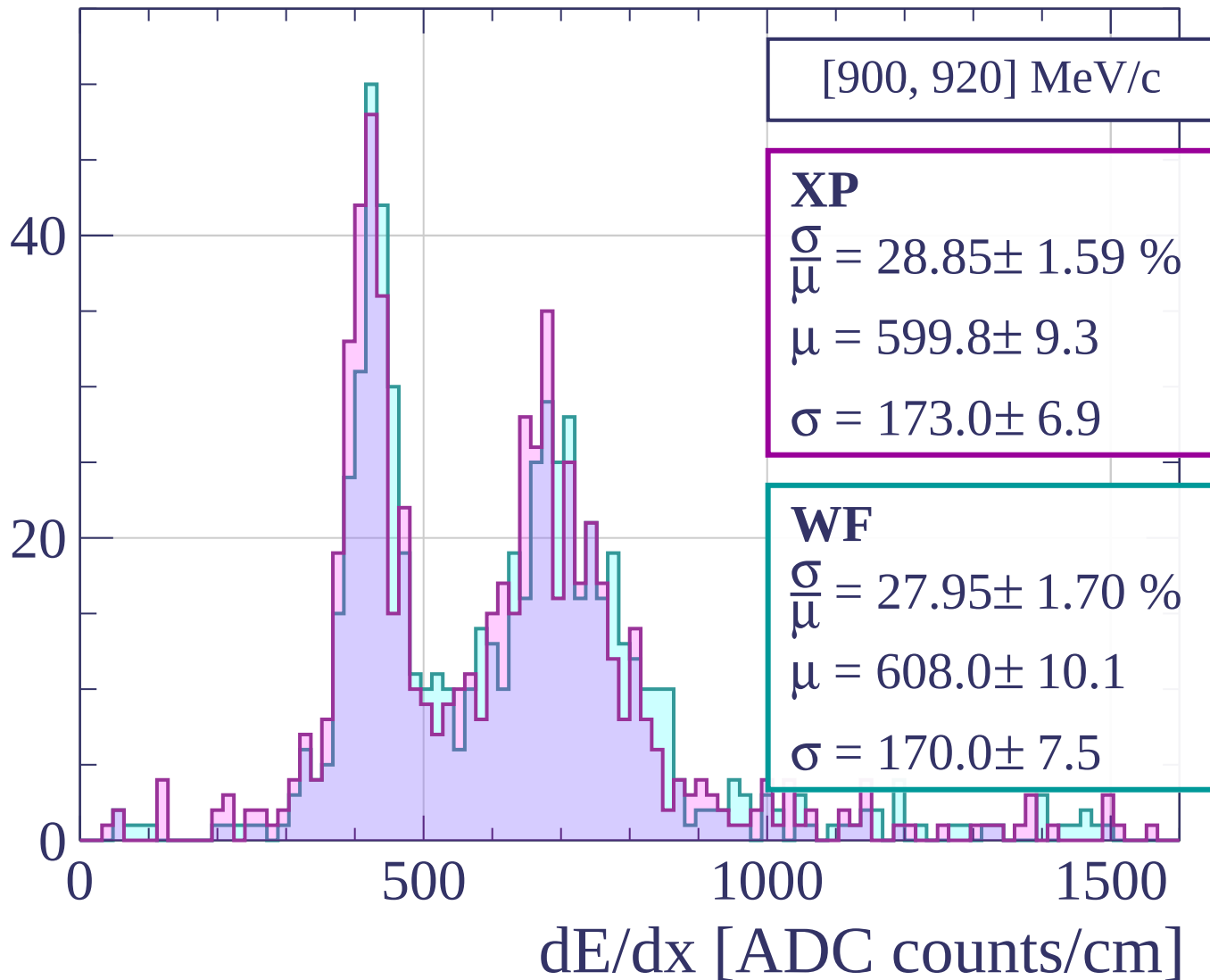
Count



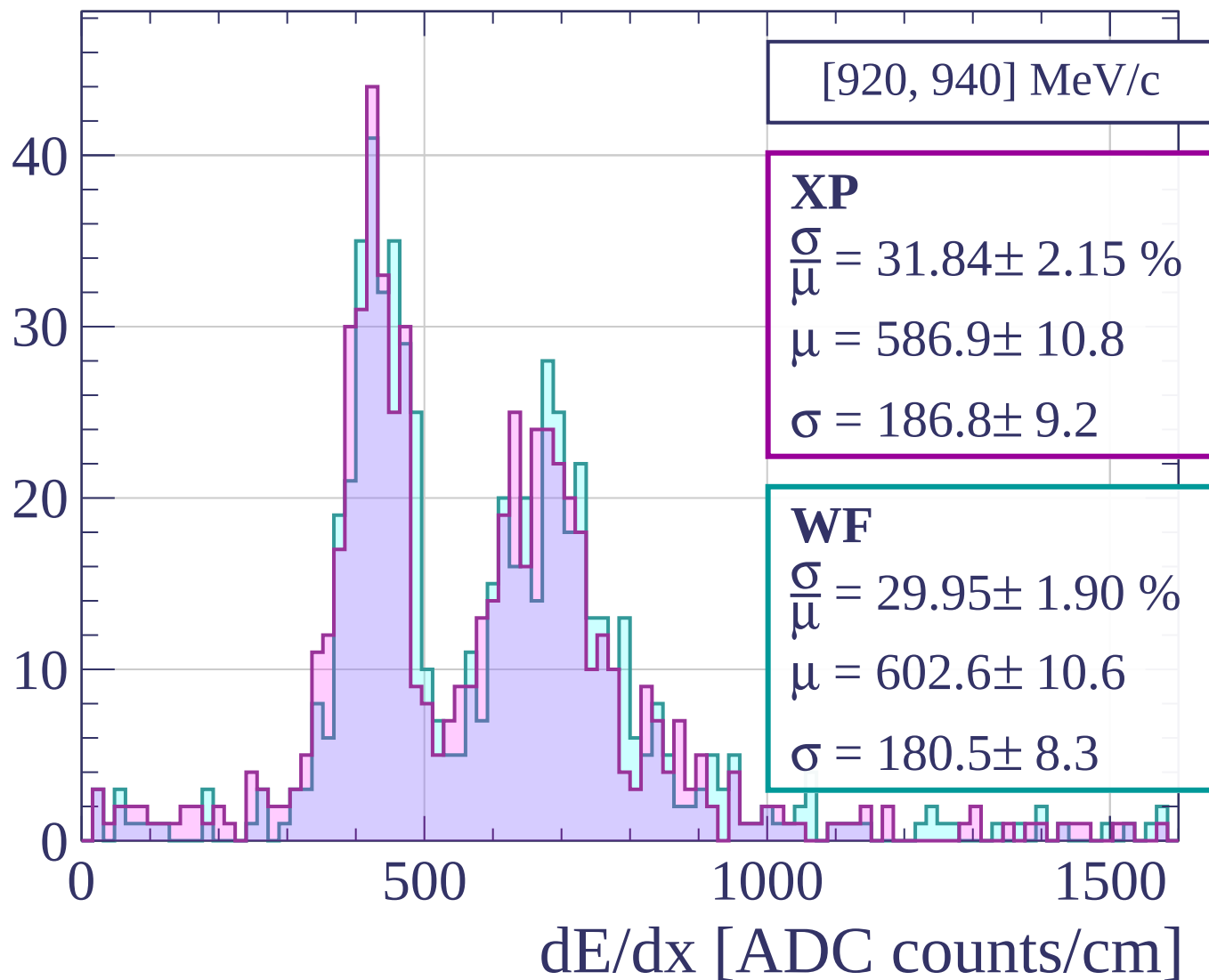
Count



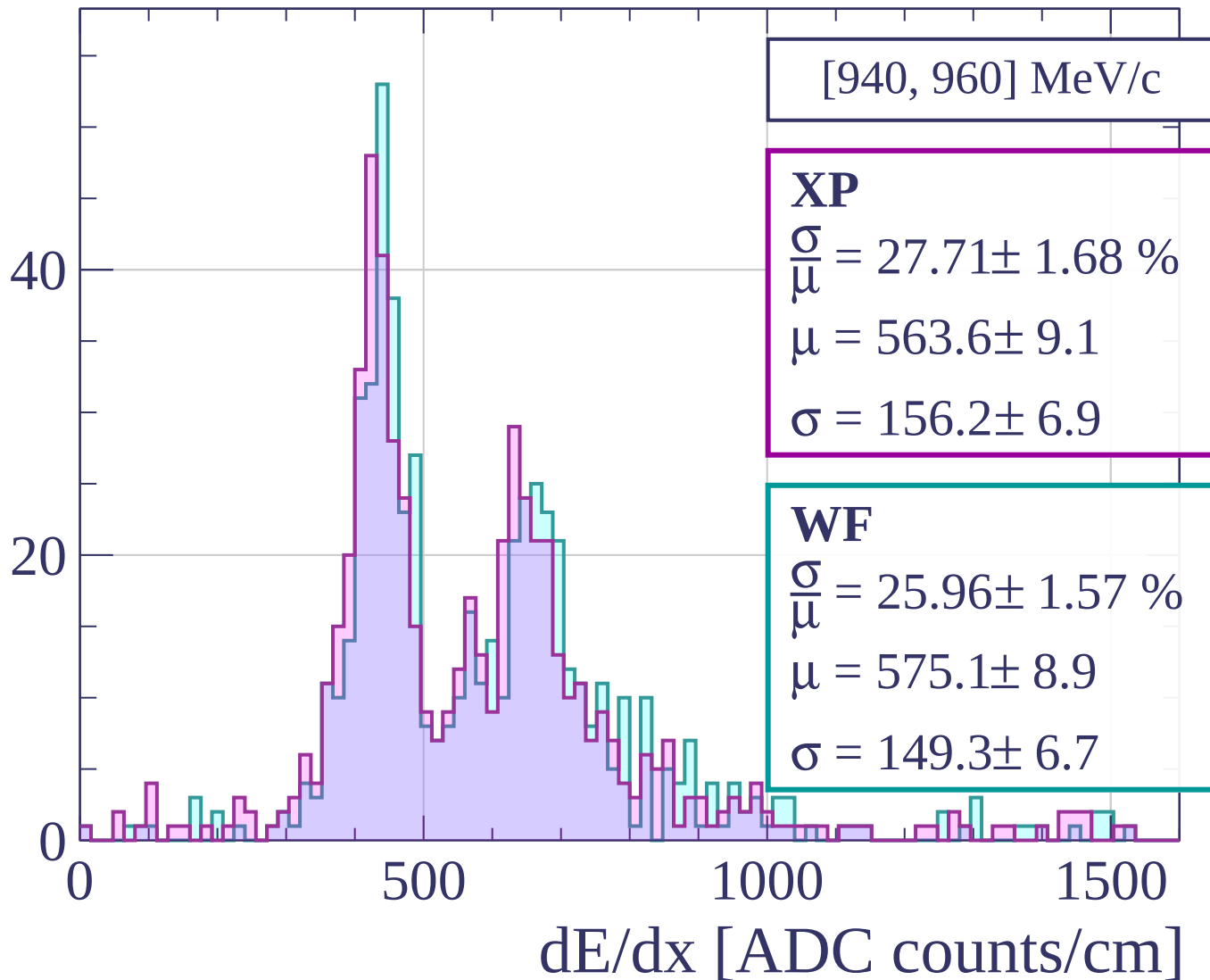
Count



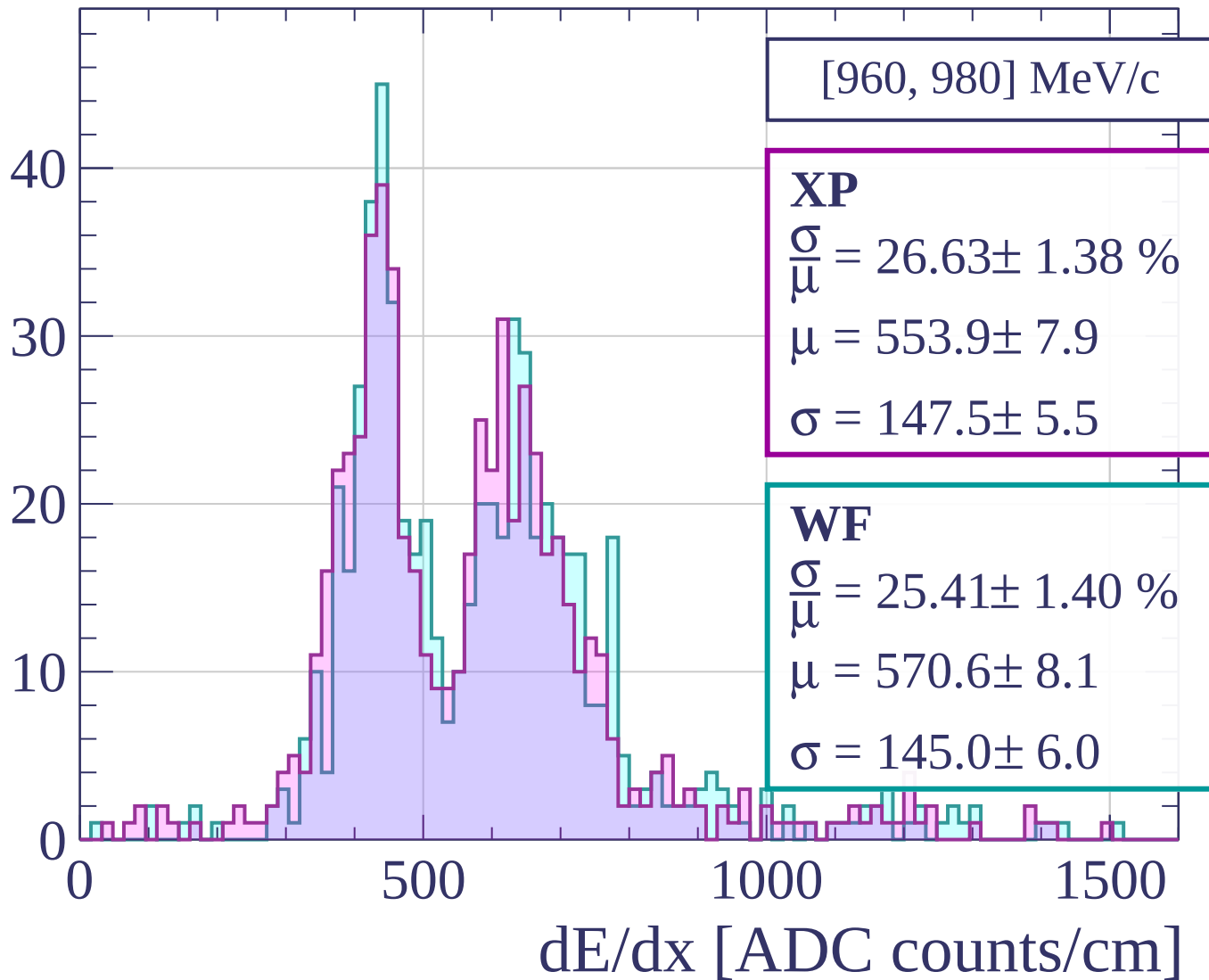
Count



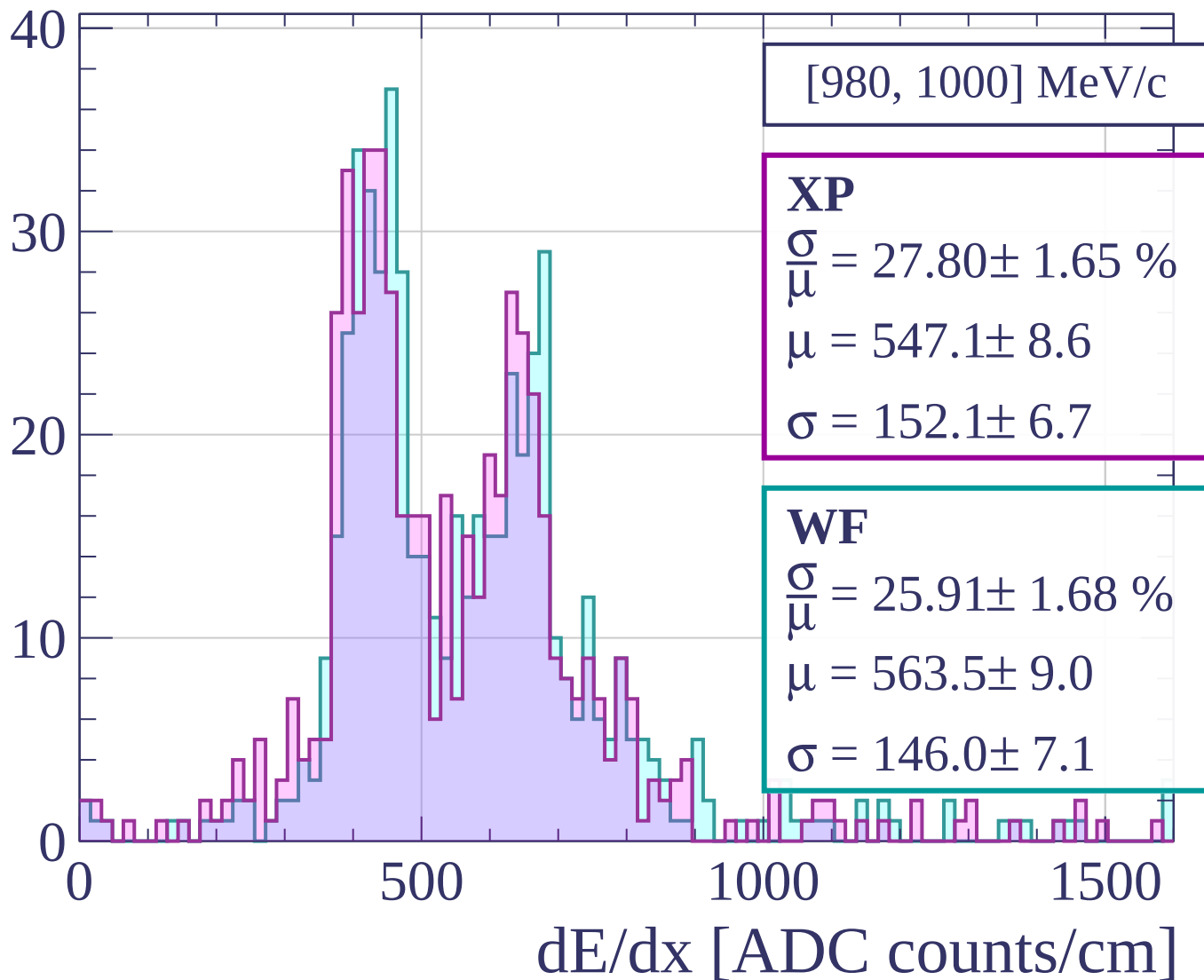
Count



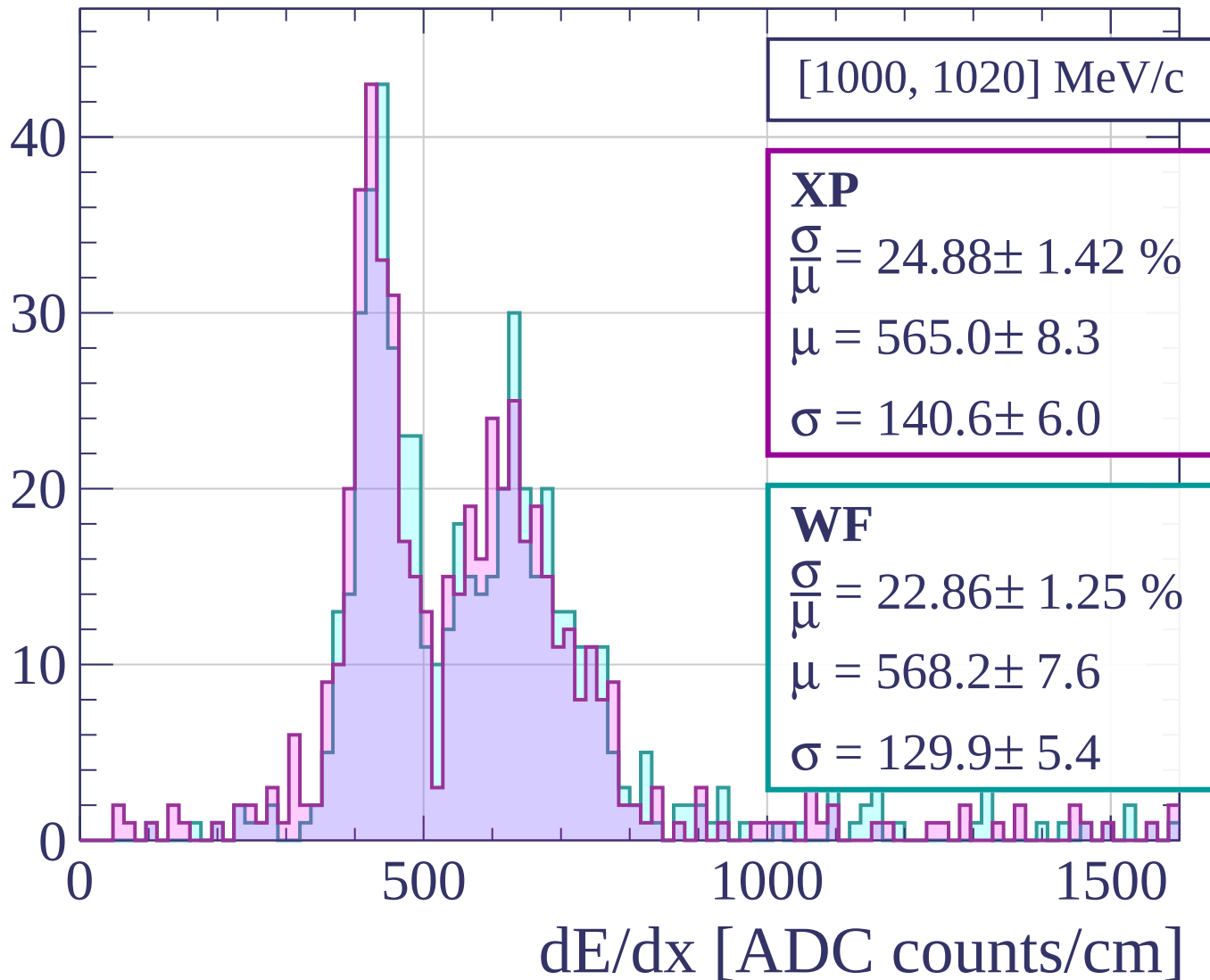
Count



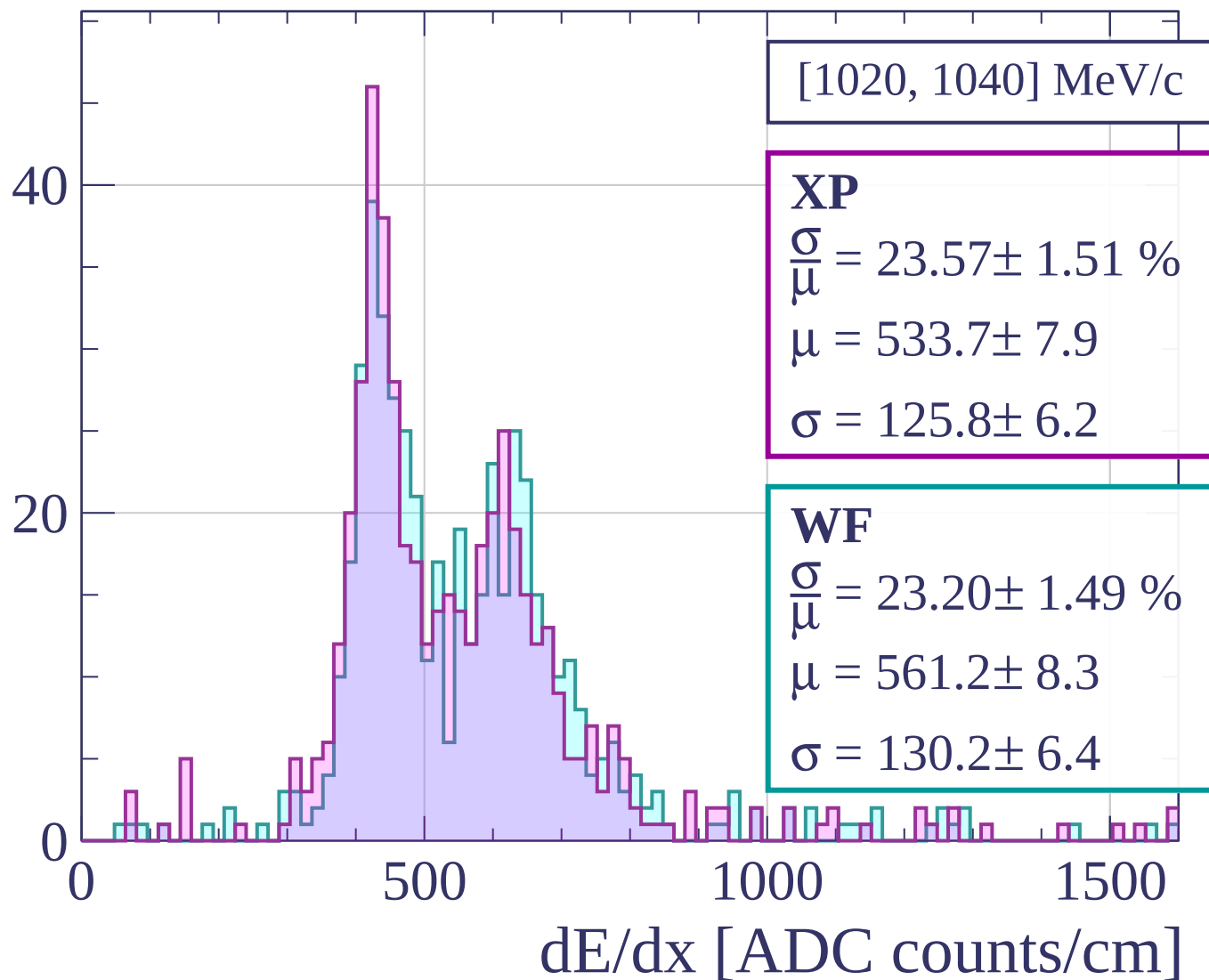
Count



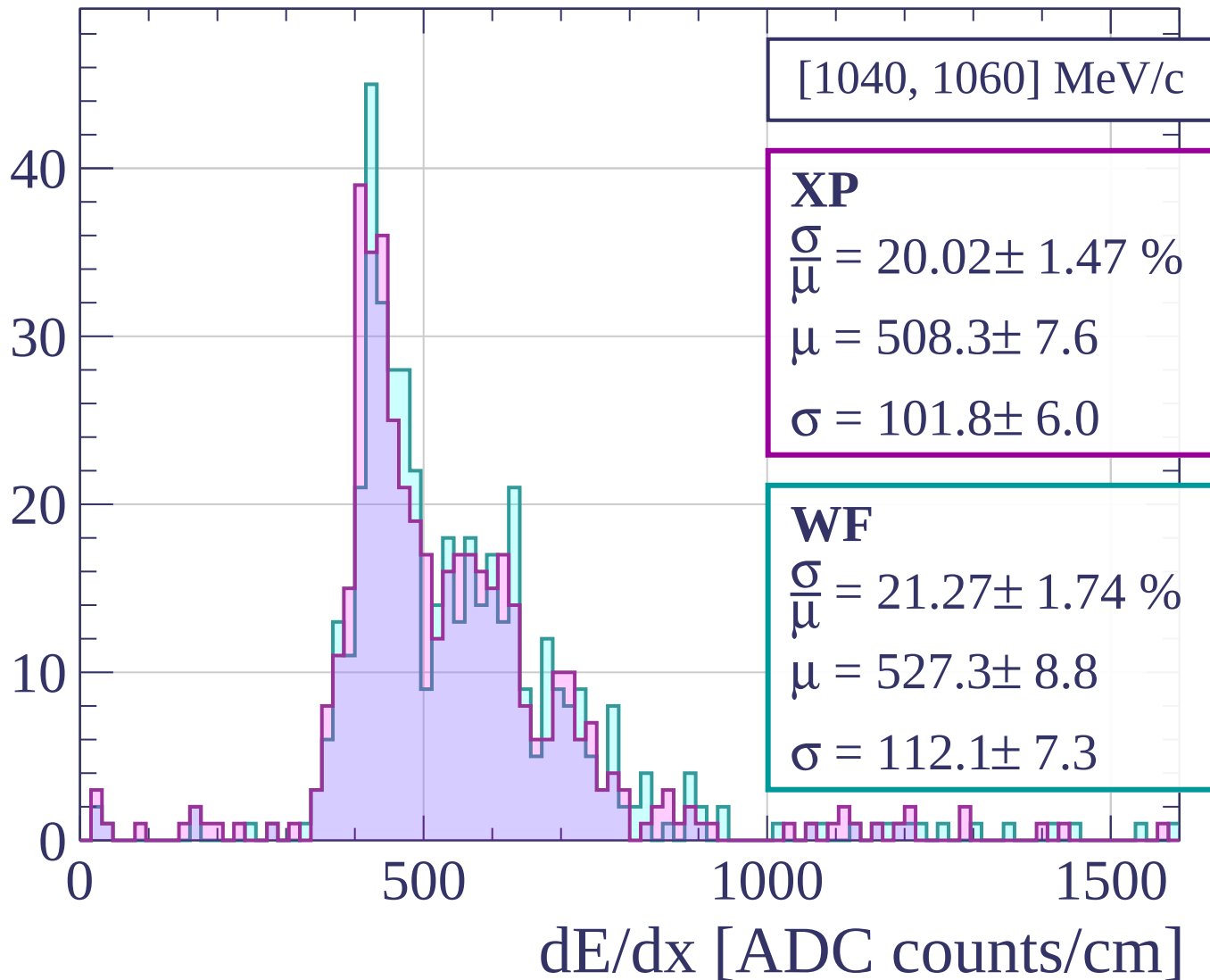
Count



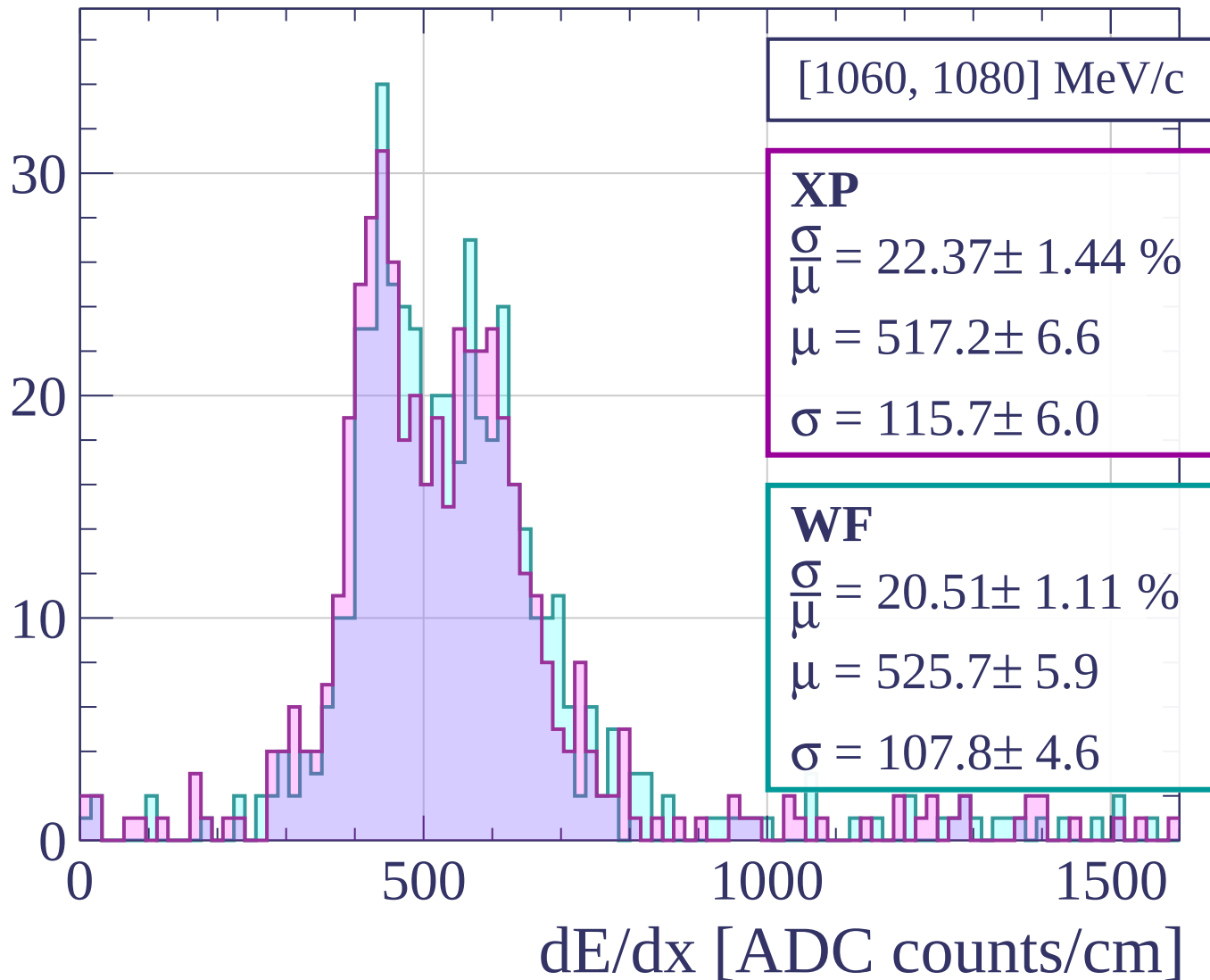
Count



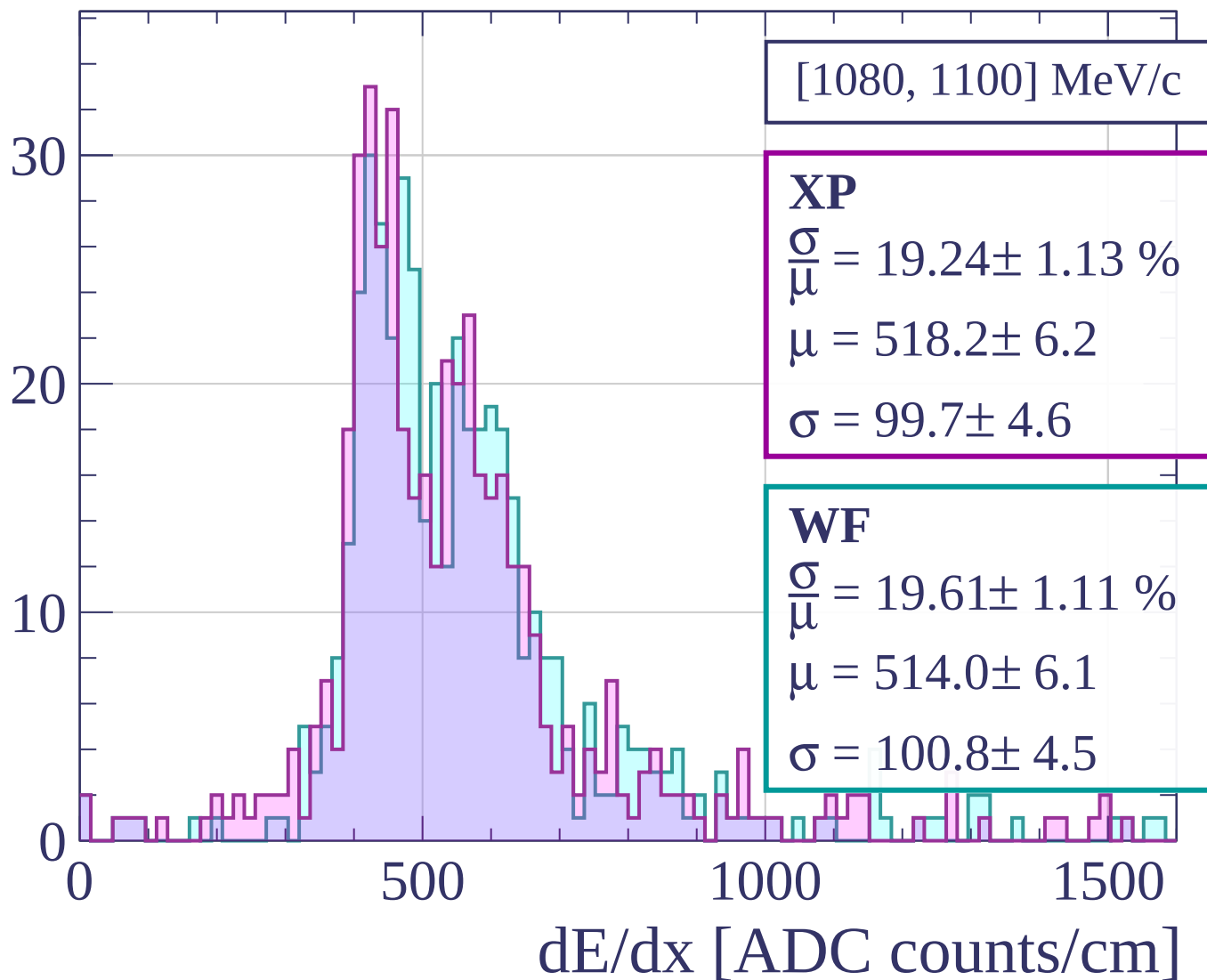
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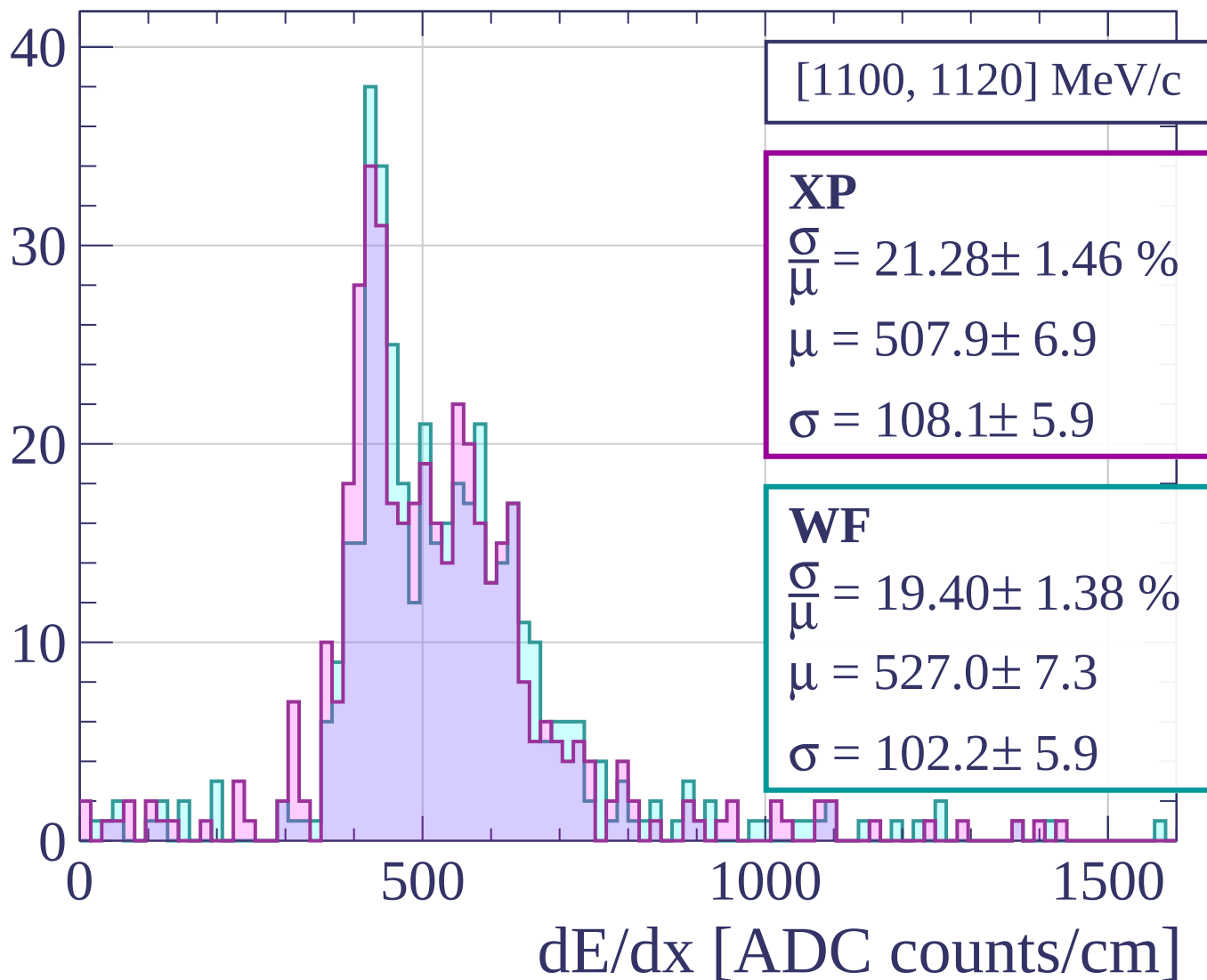
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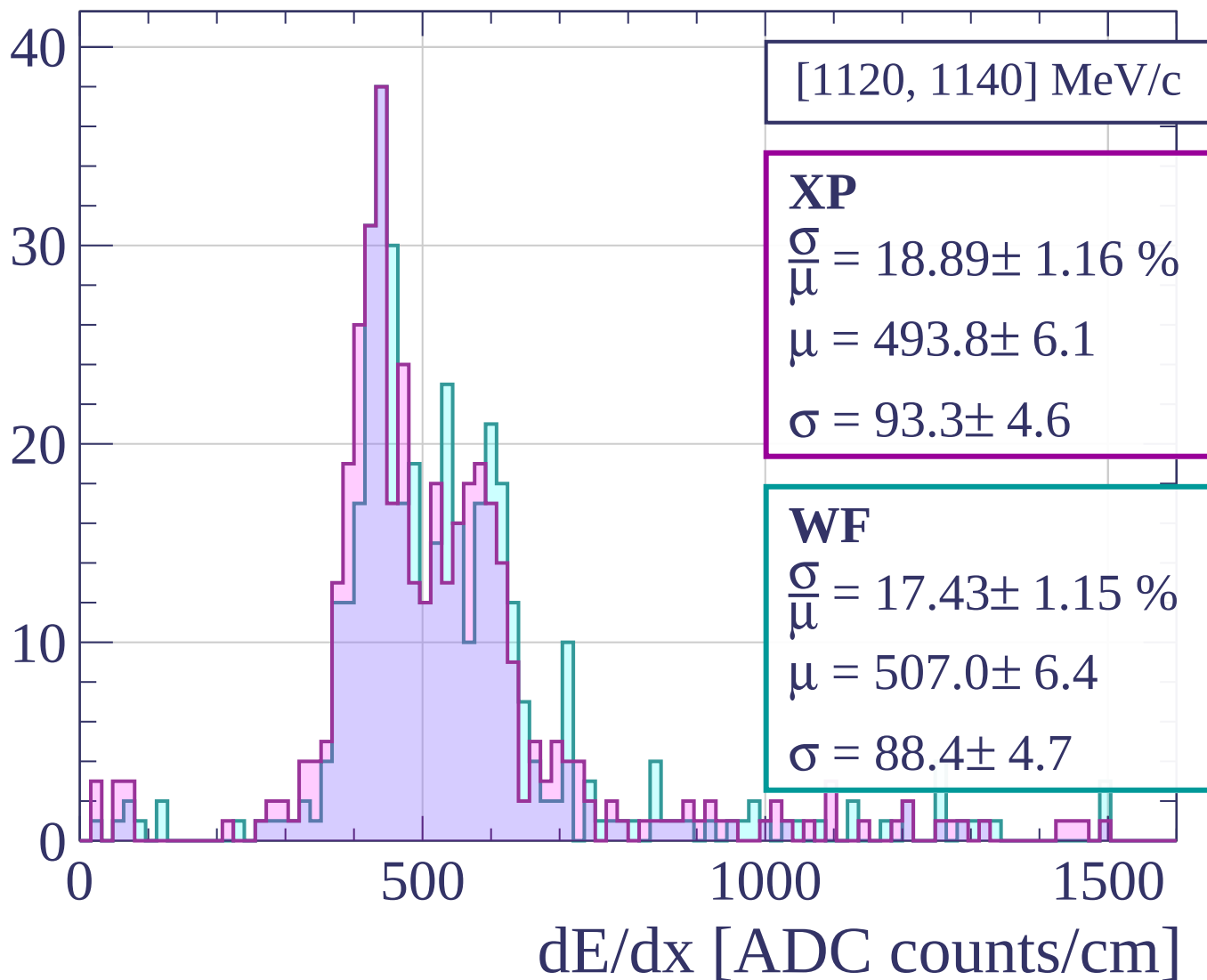
Count



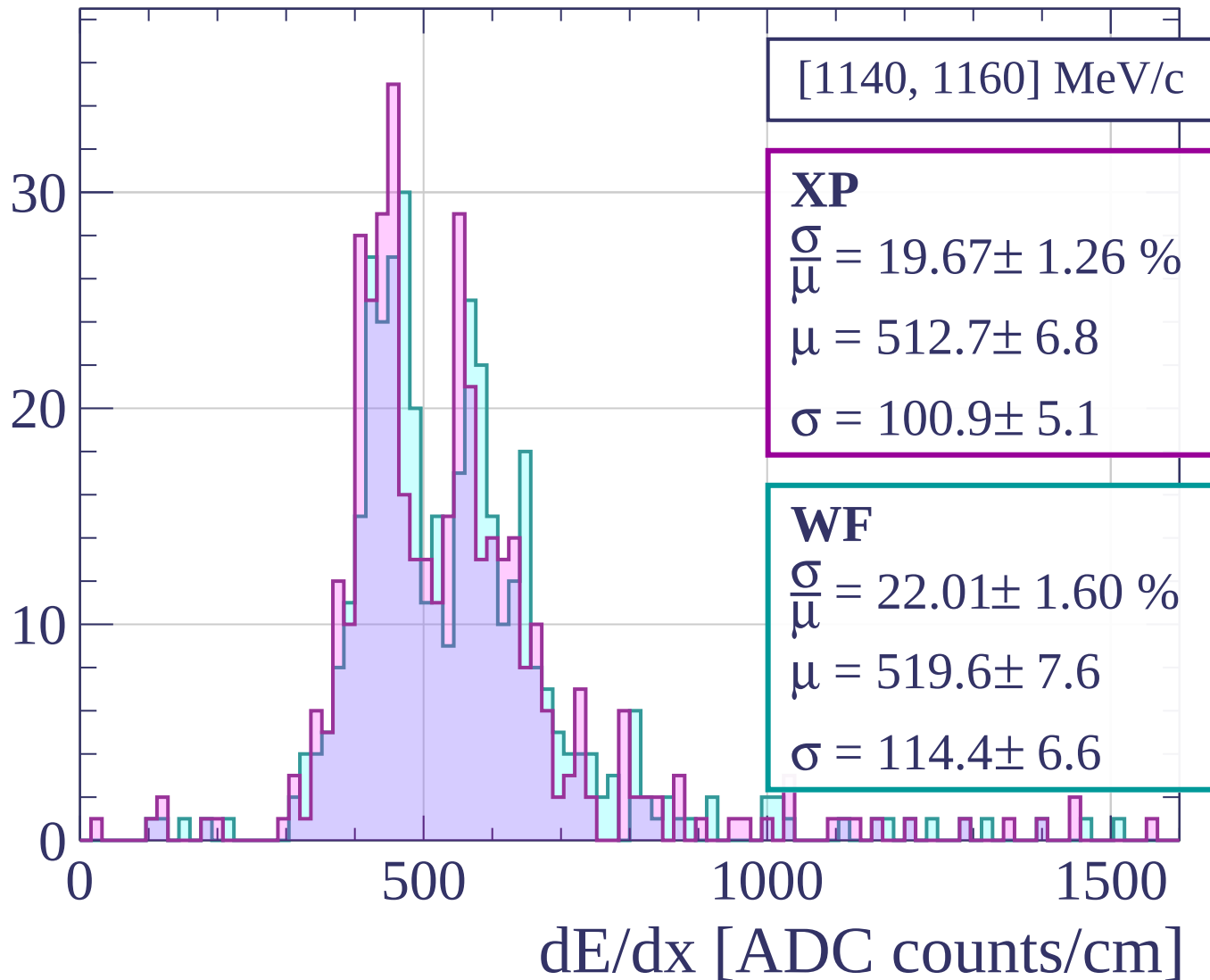
Count



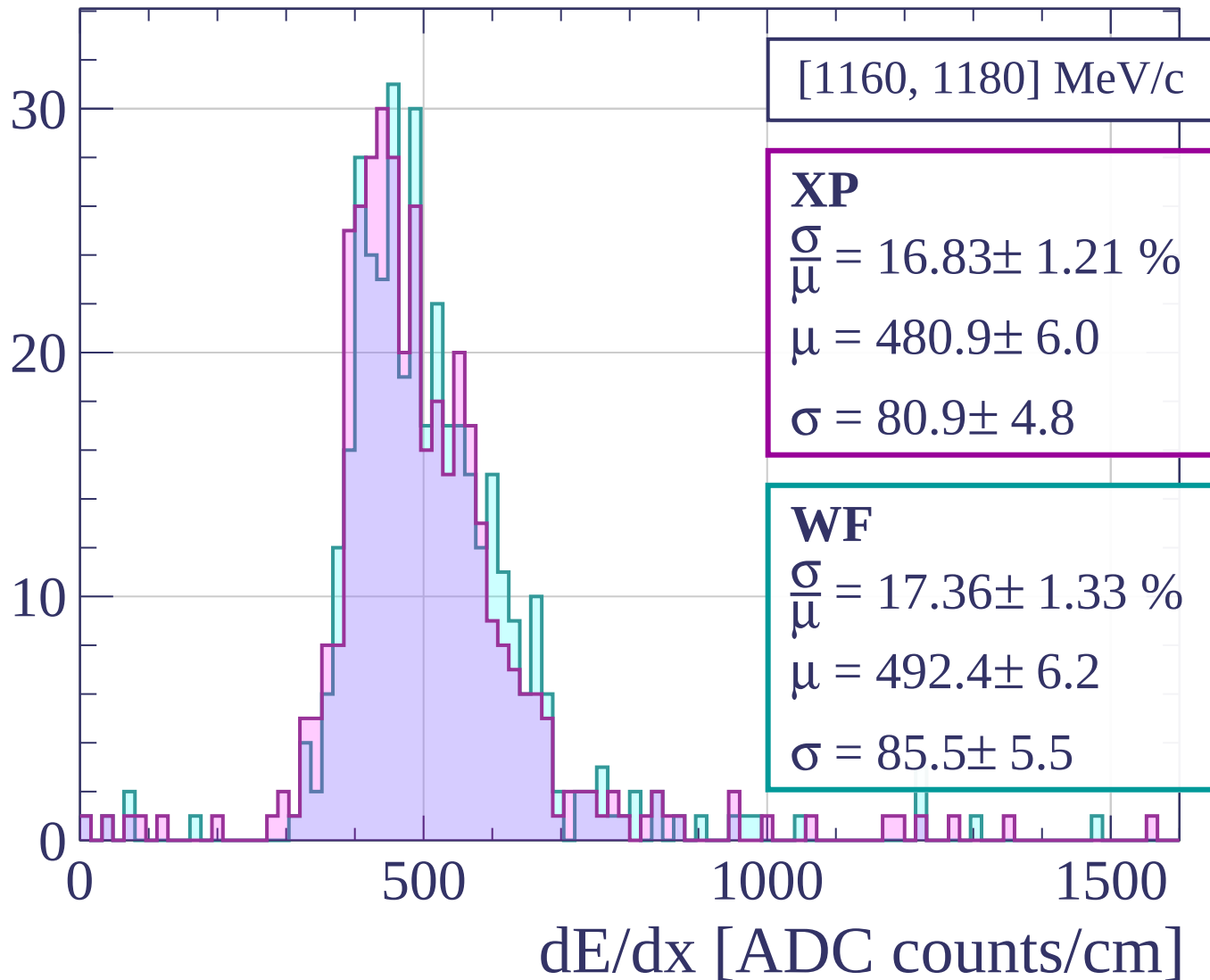
Count



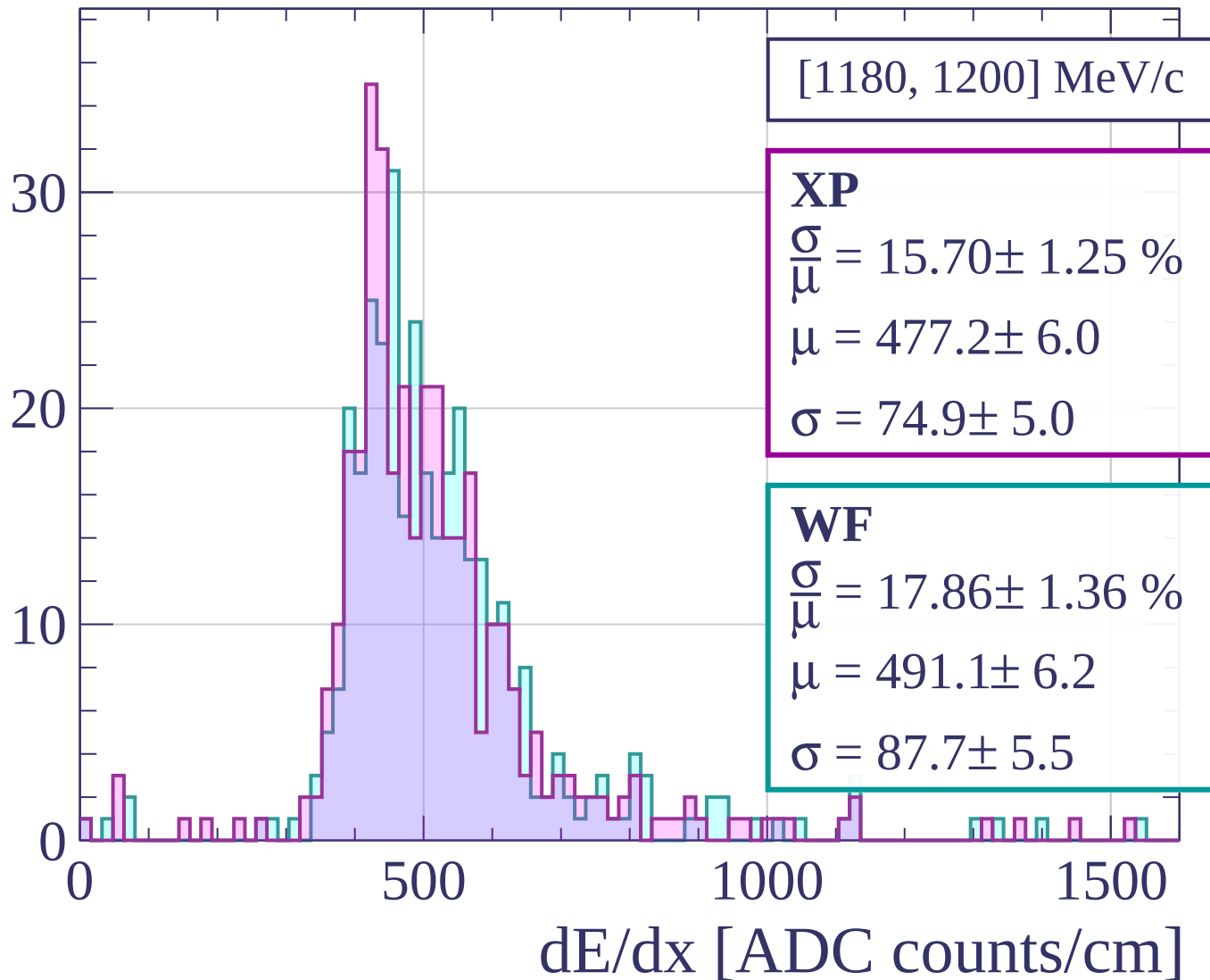
Count



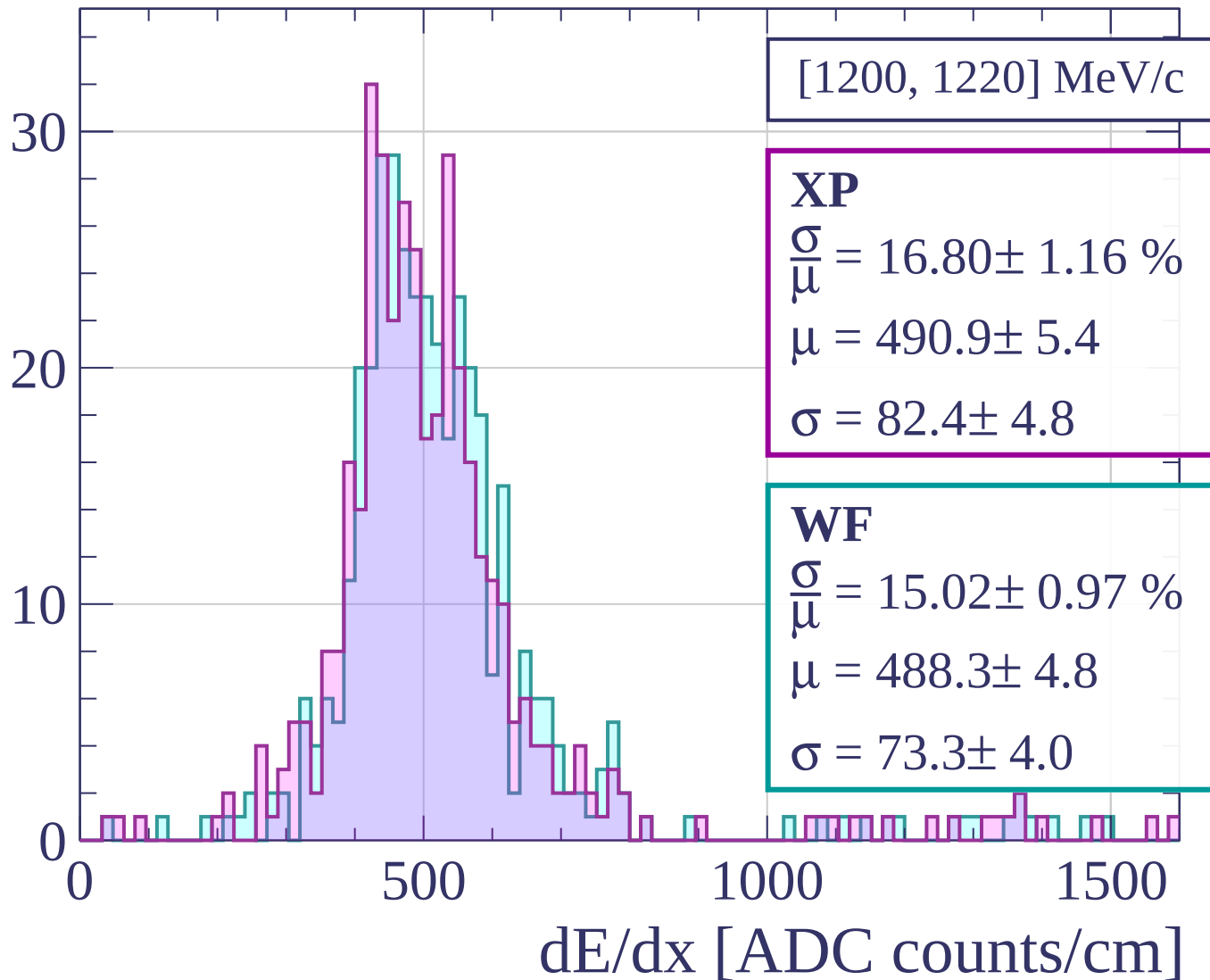
Count



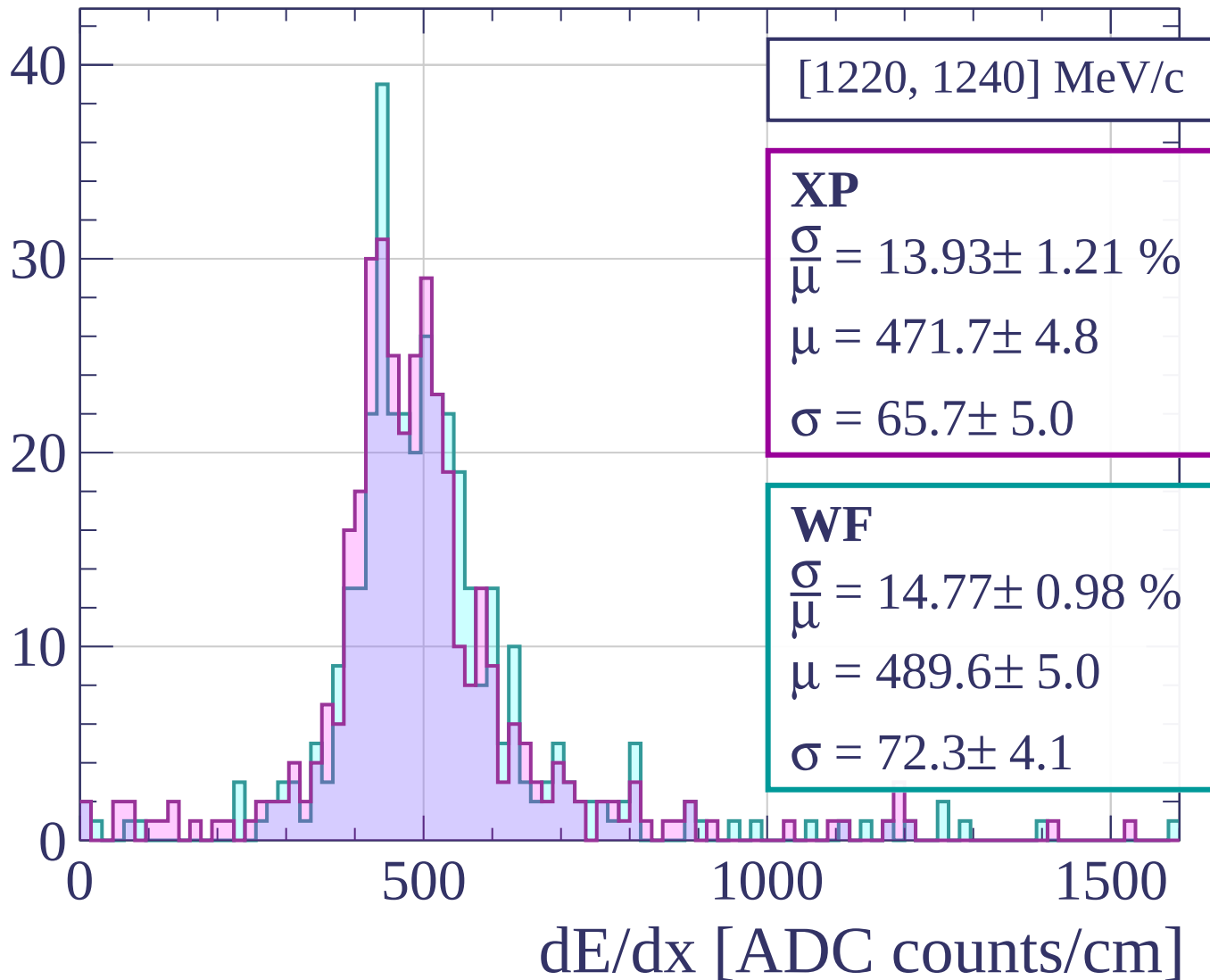
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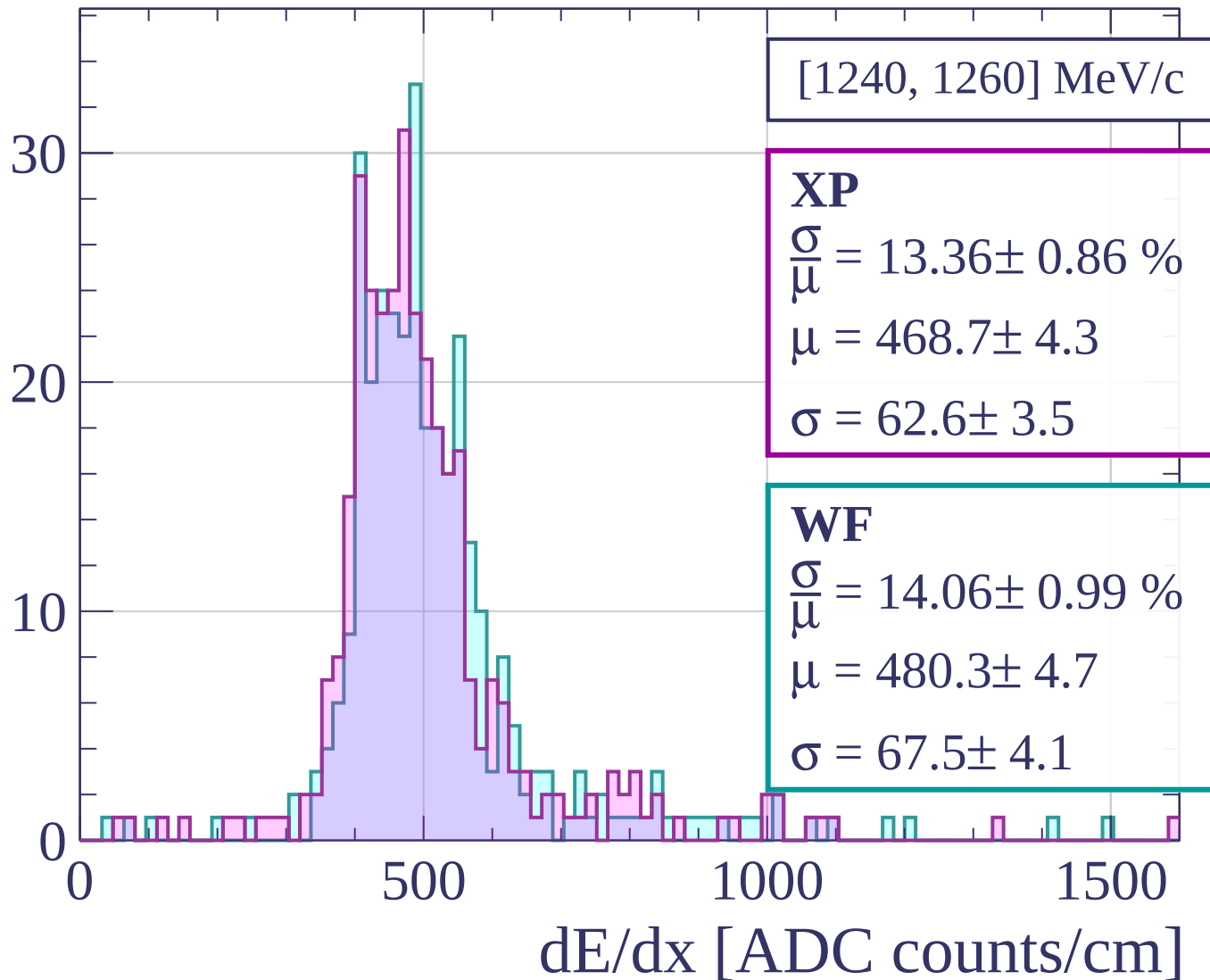
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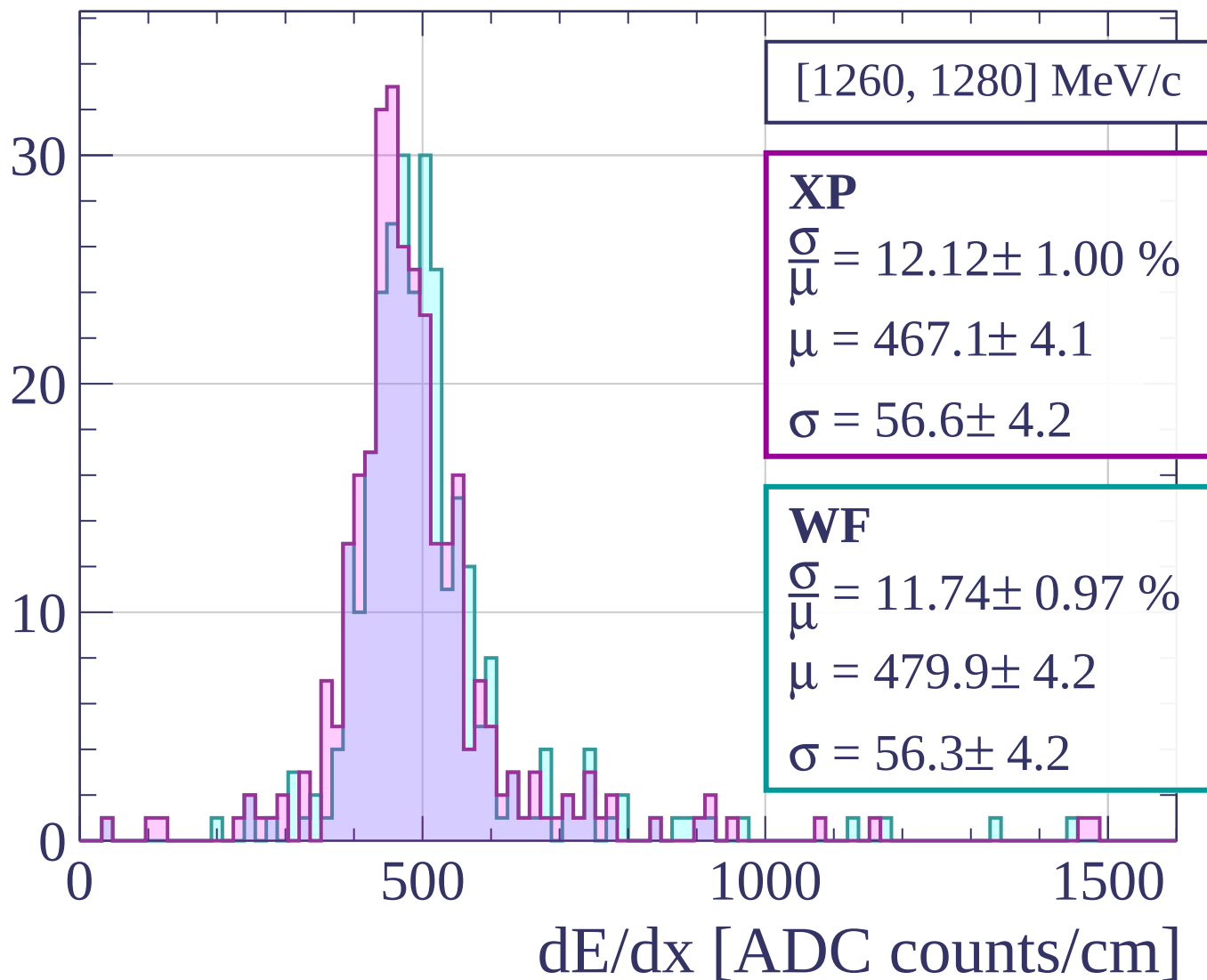
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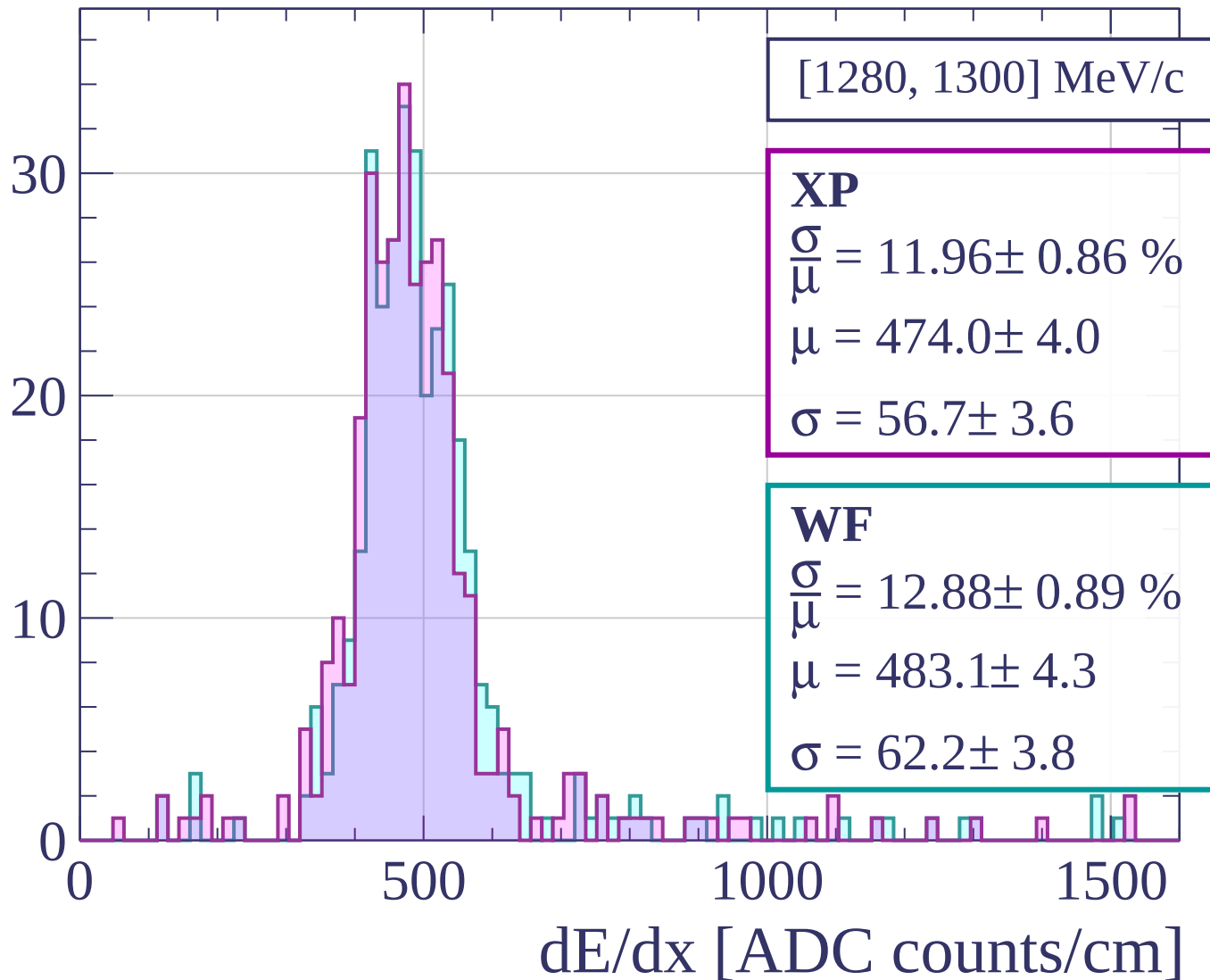
Count



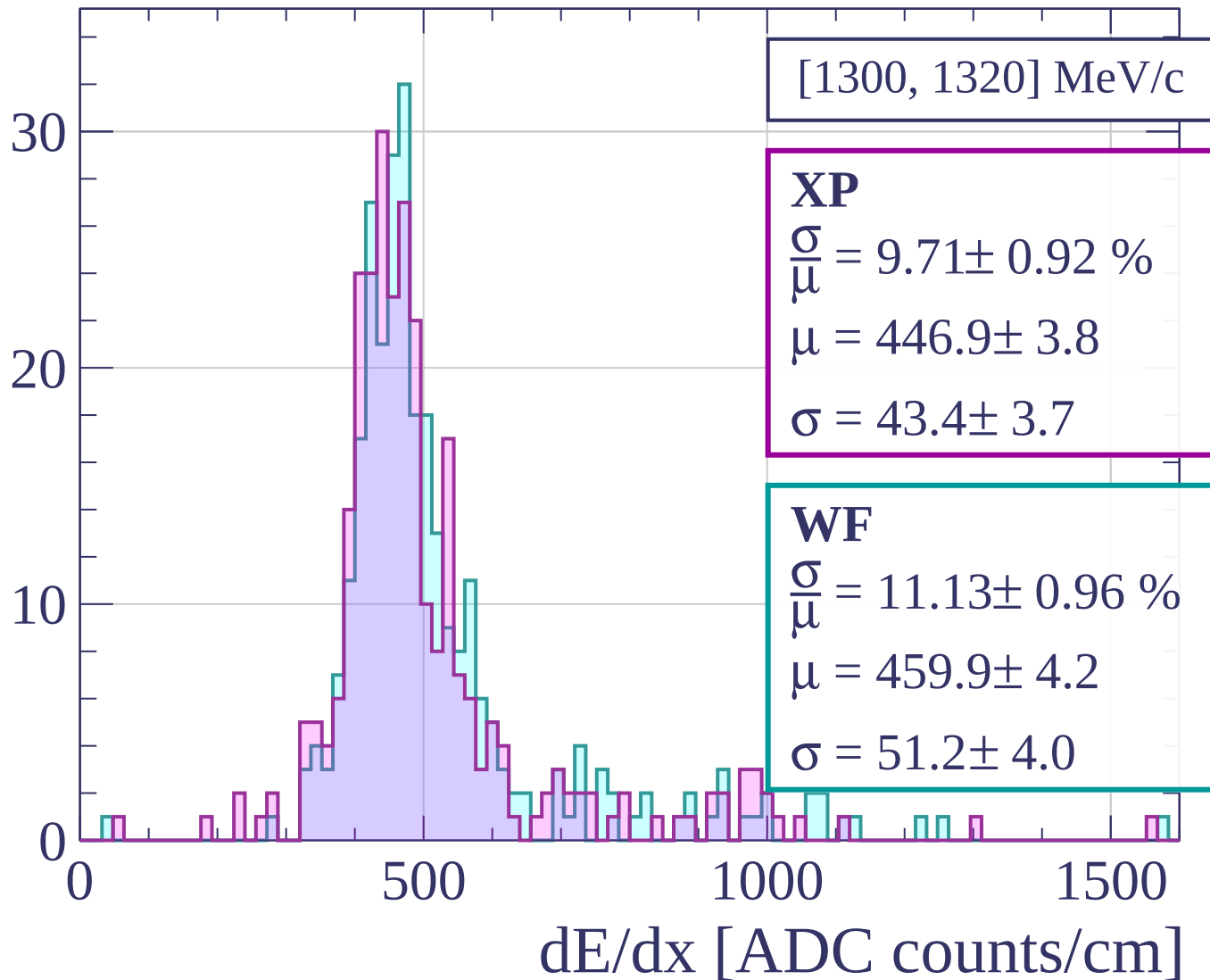
Count



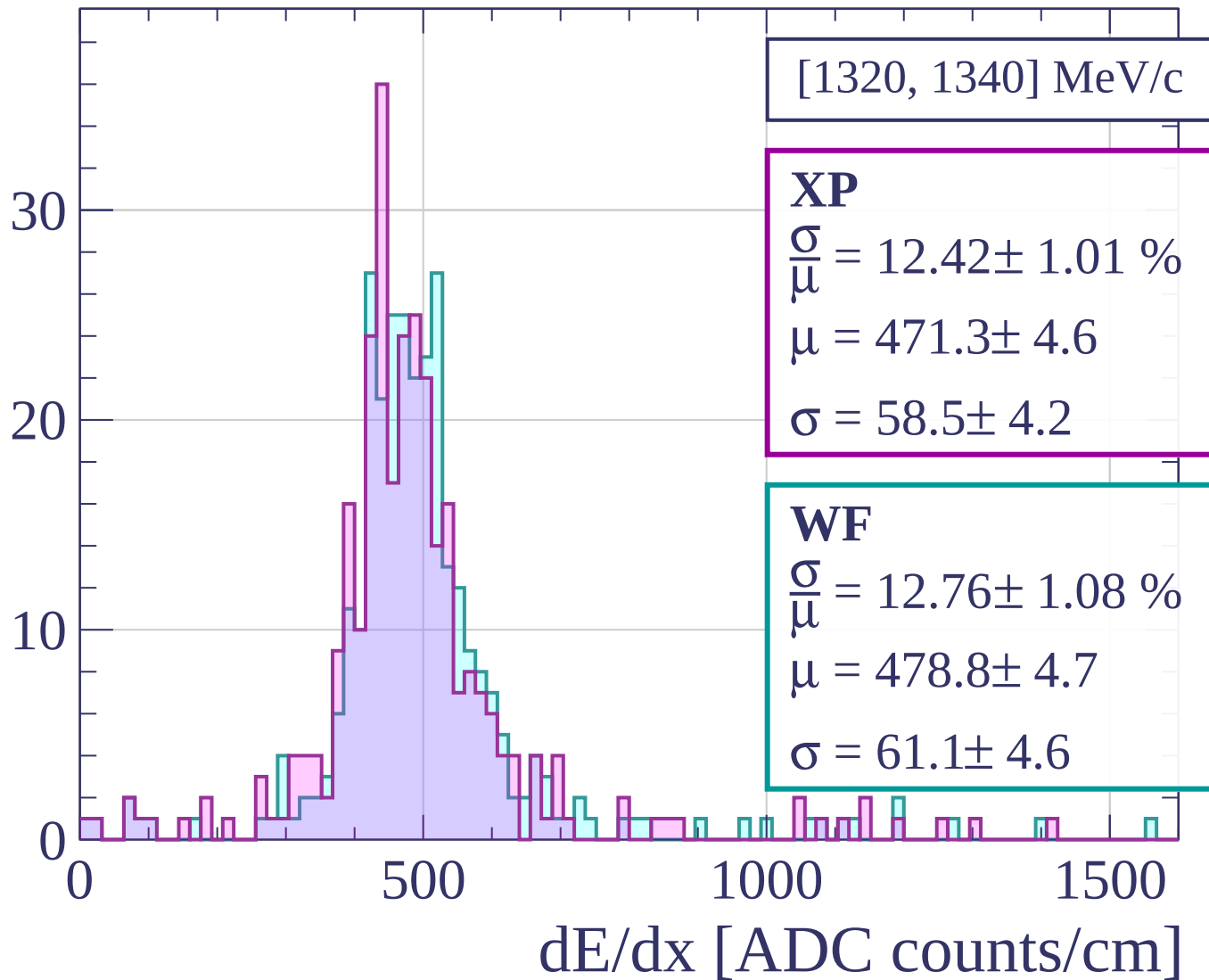
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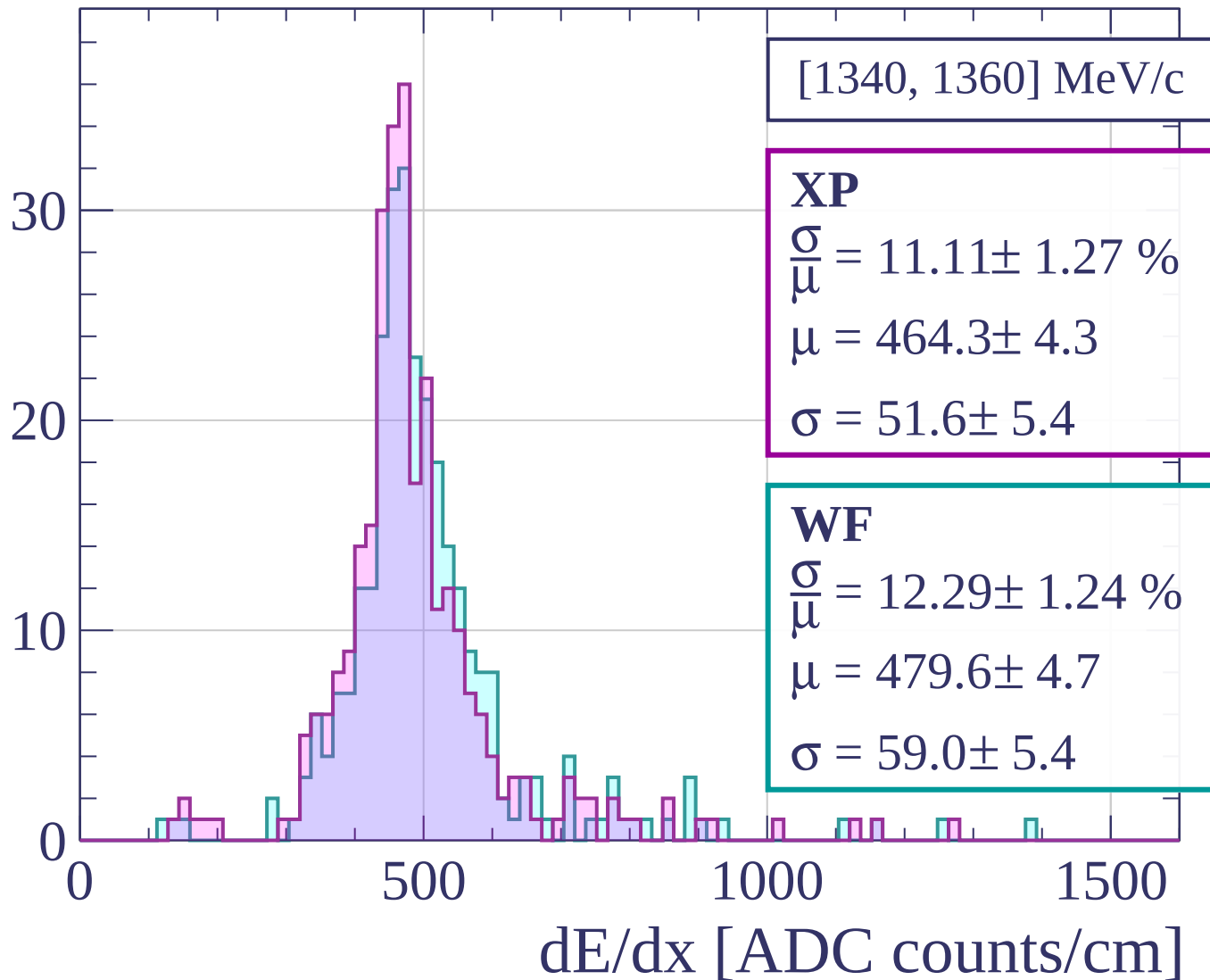
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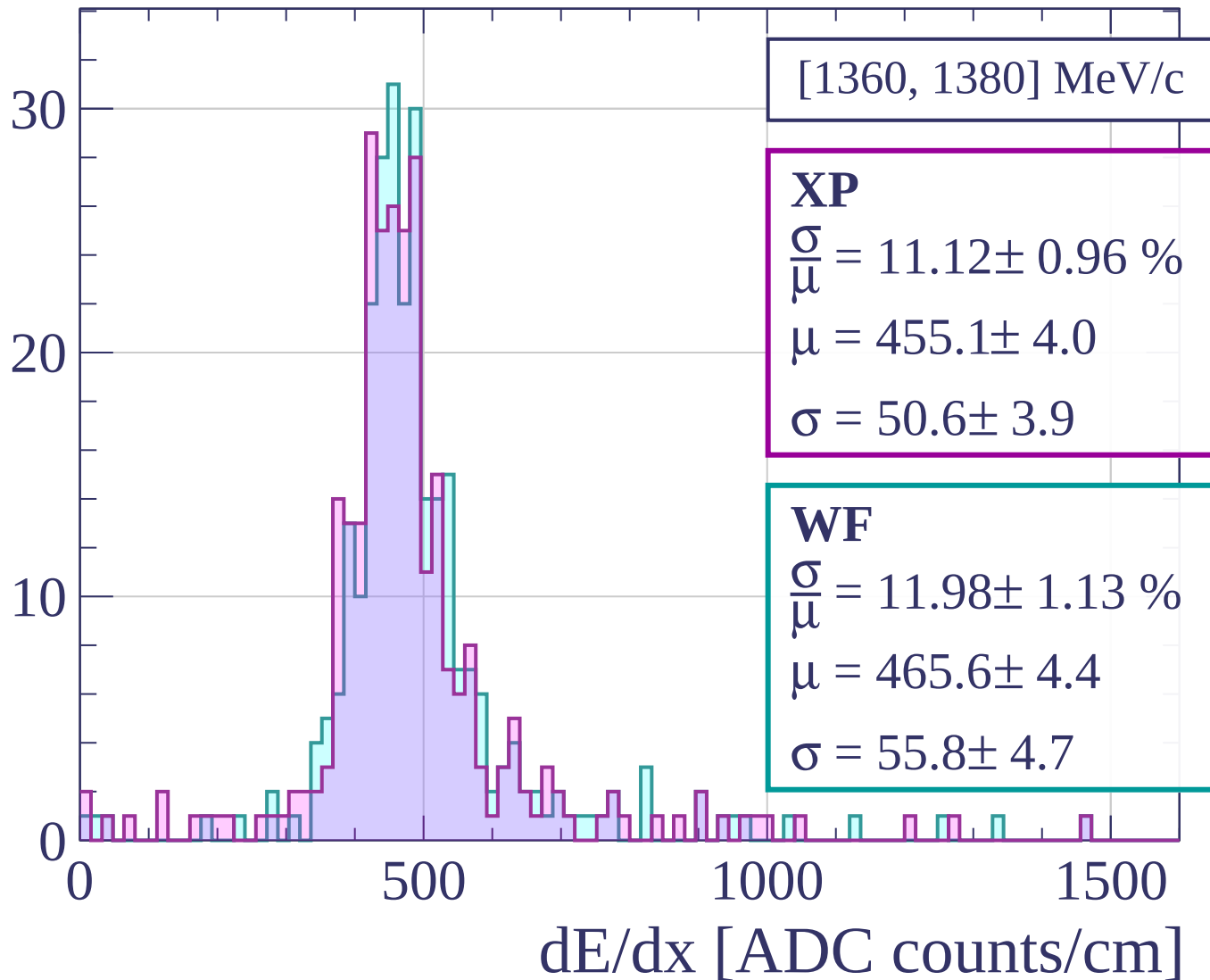
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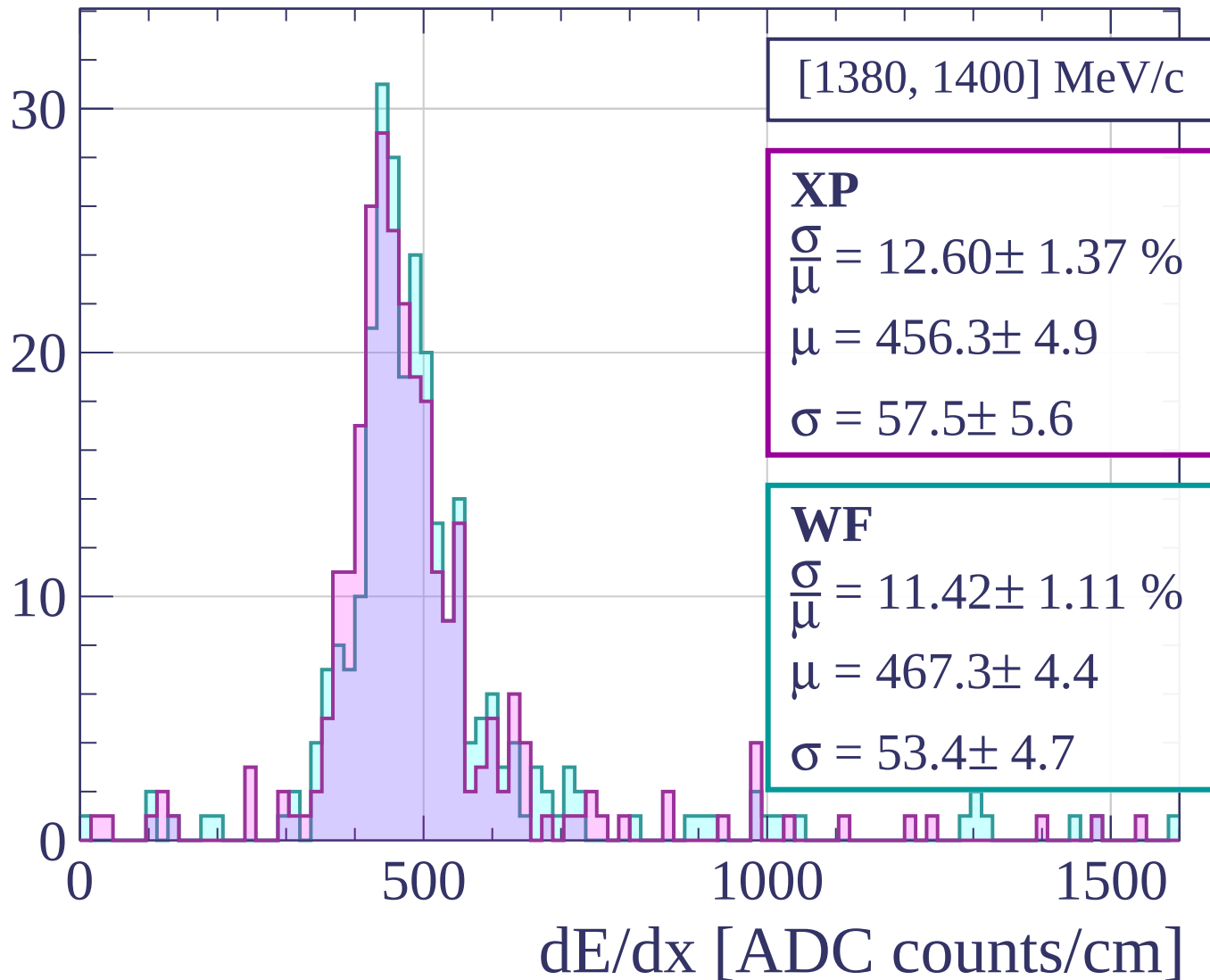
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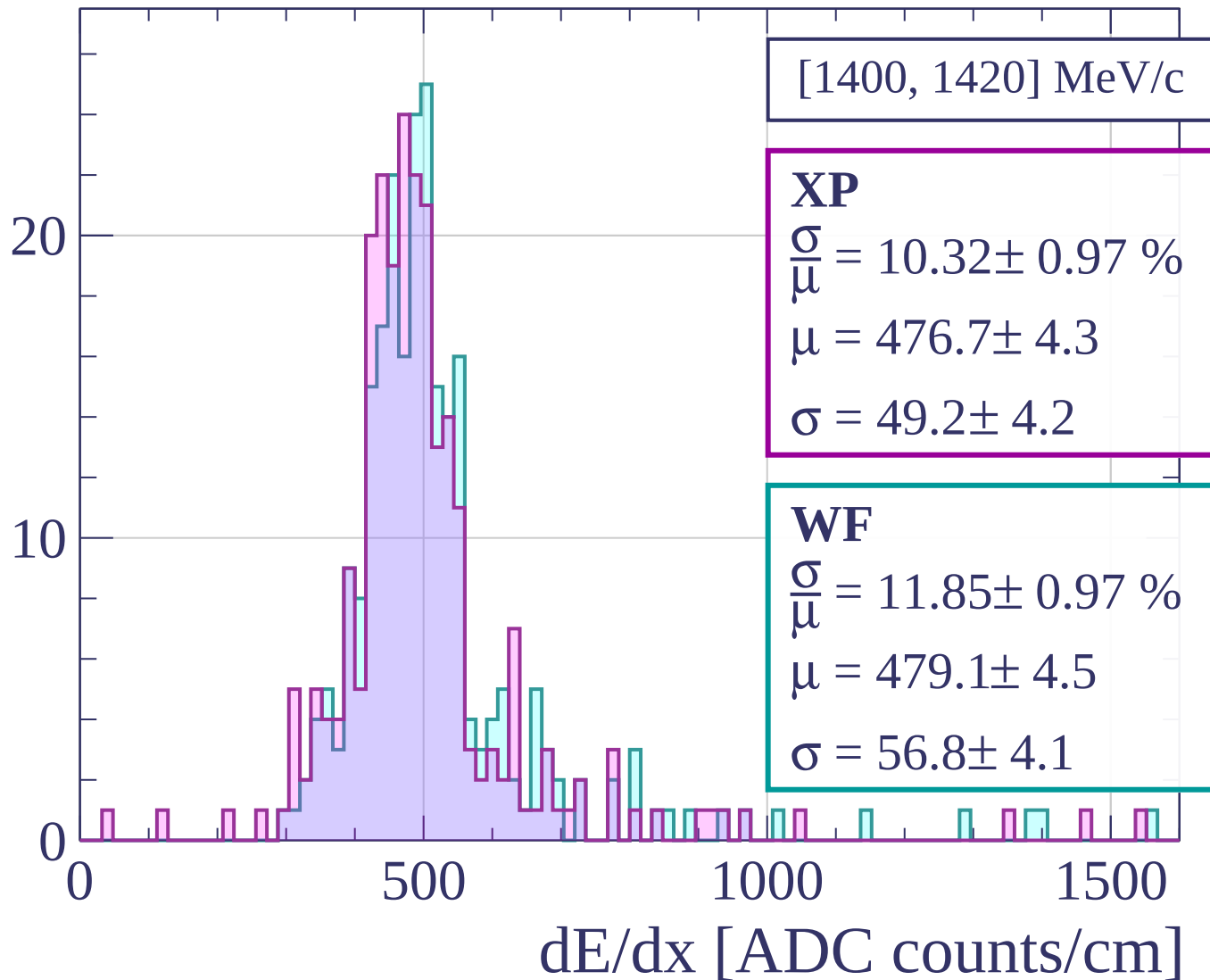
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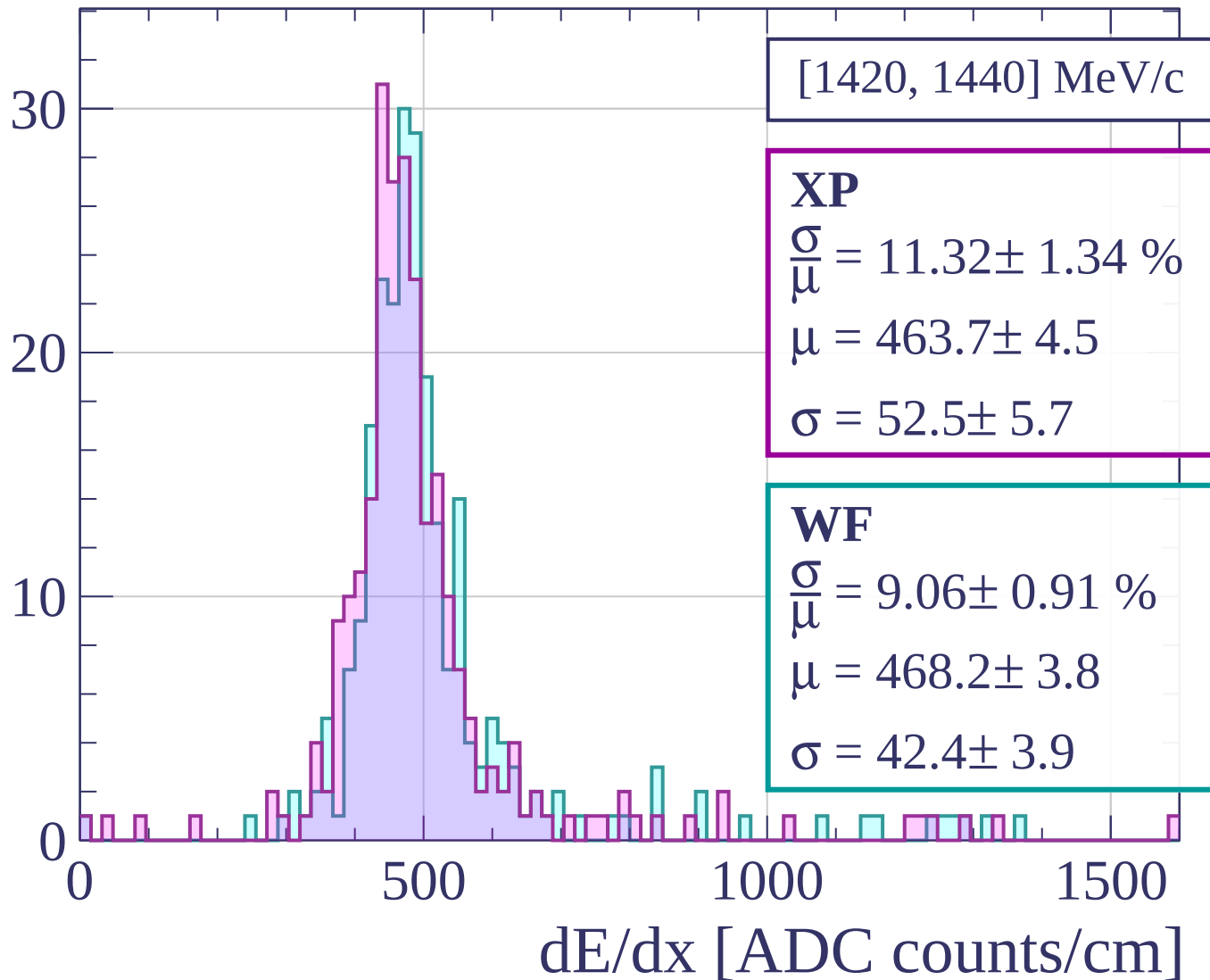
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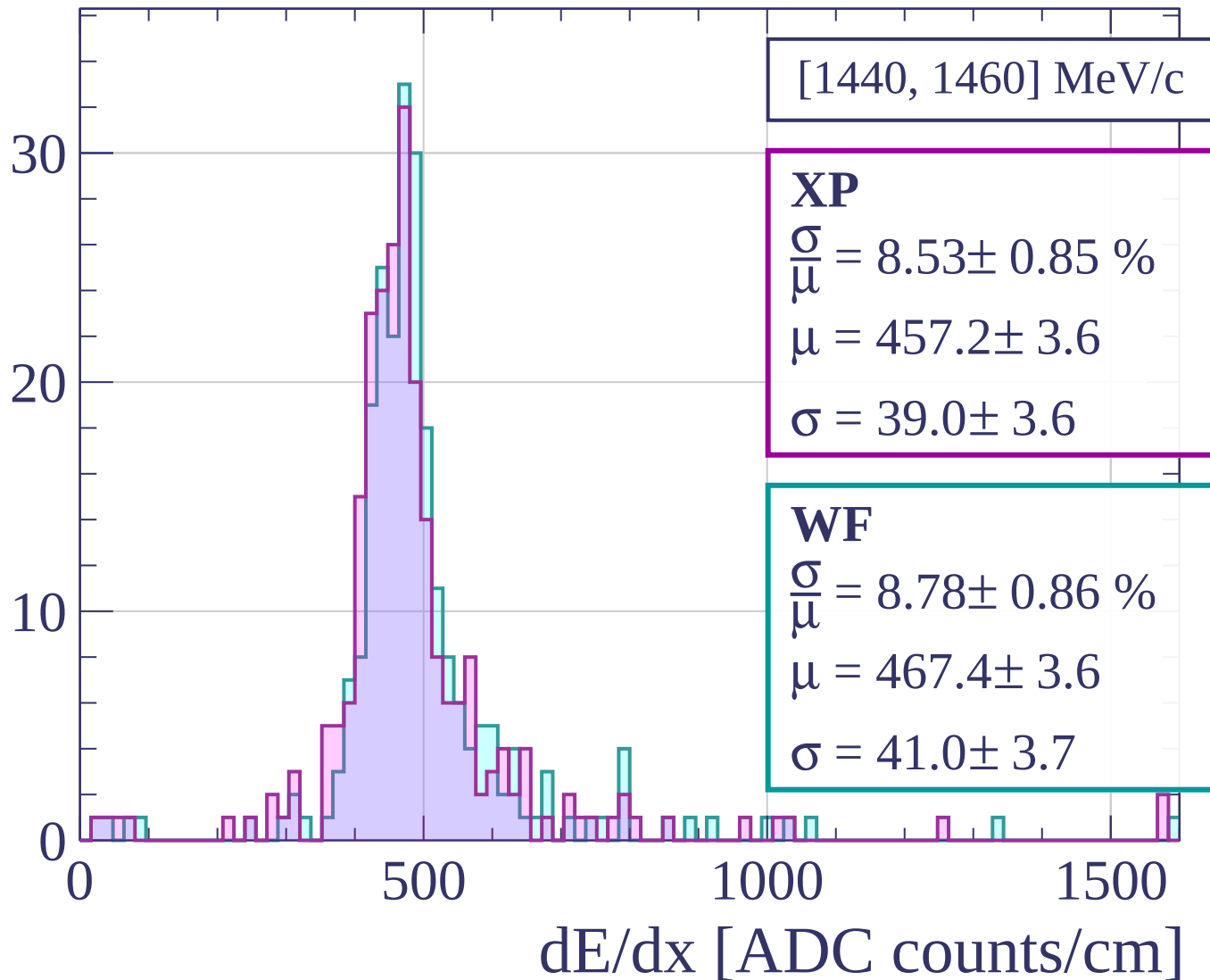
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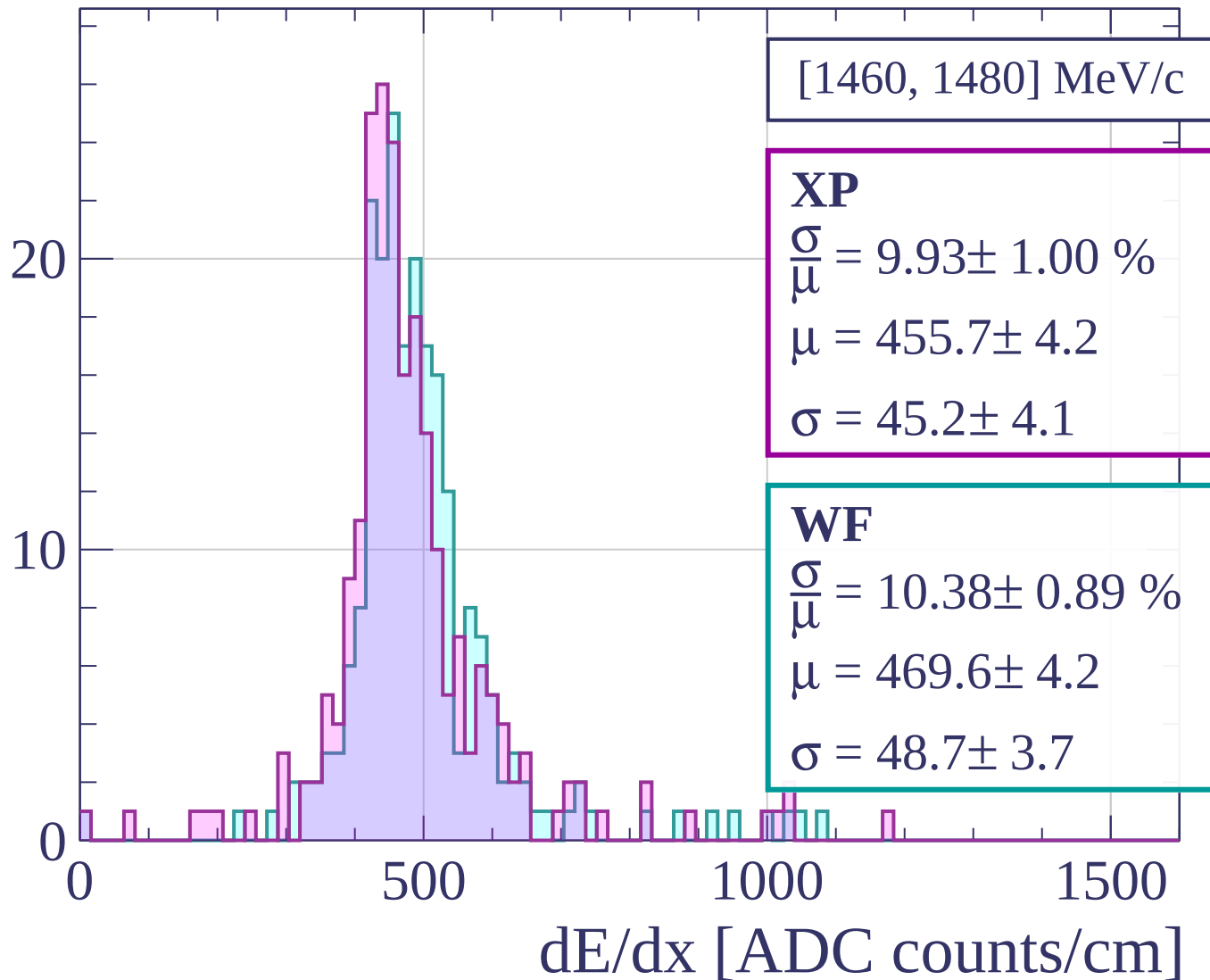
Count



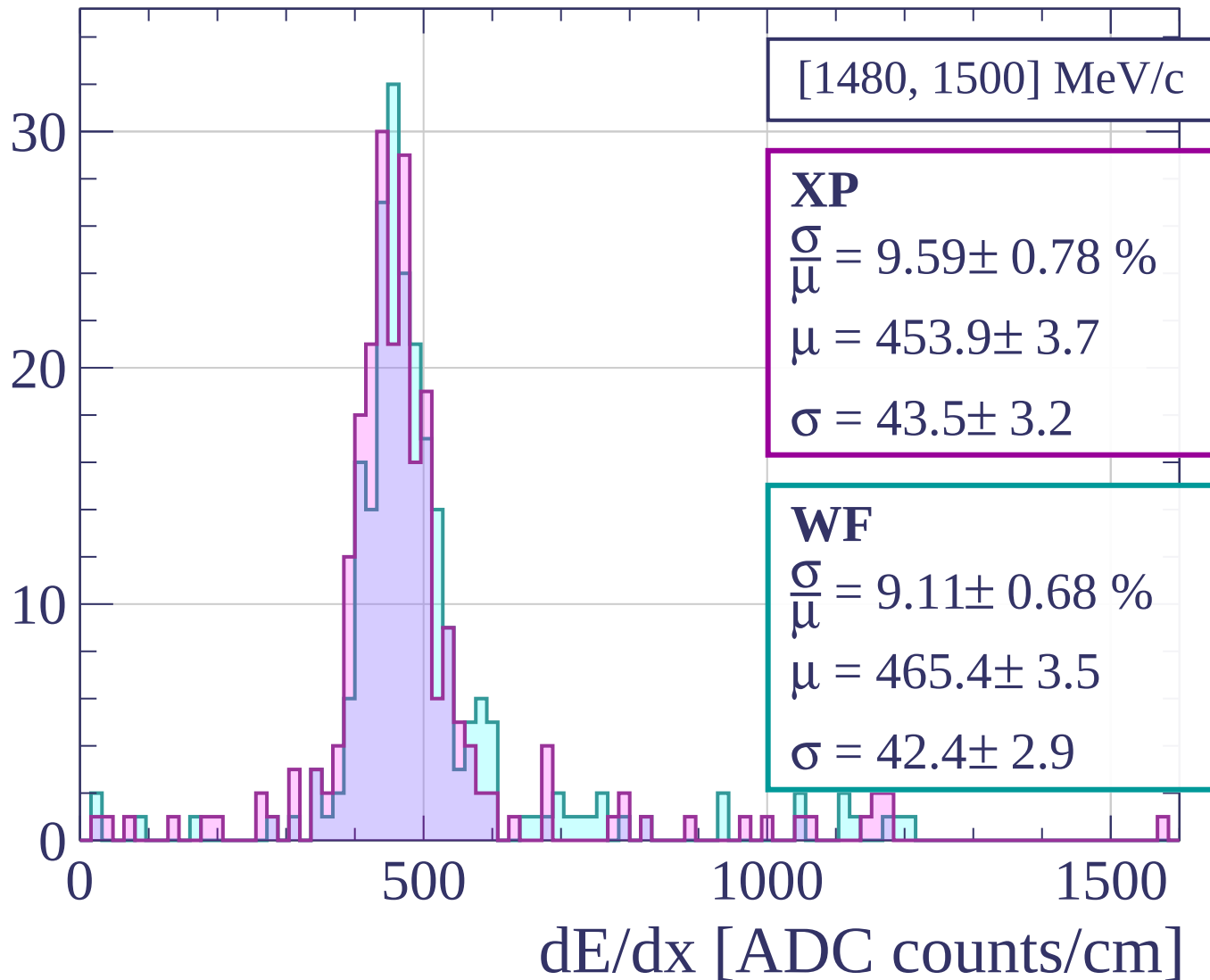
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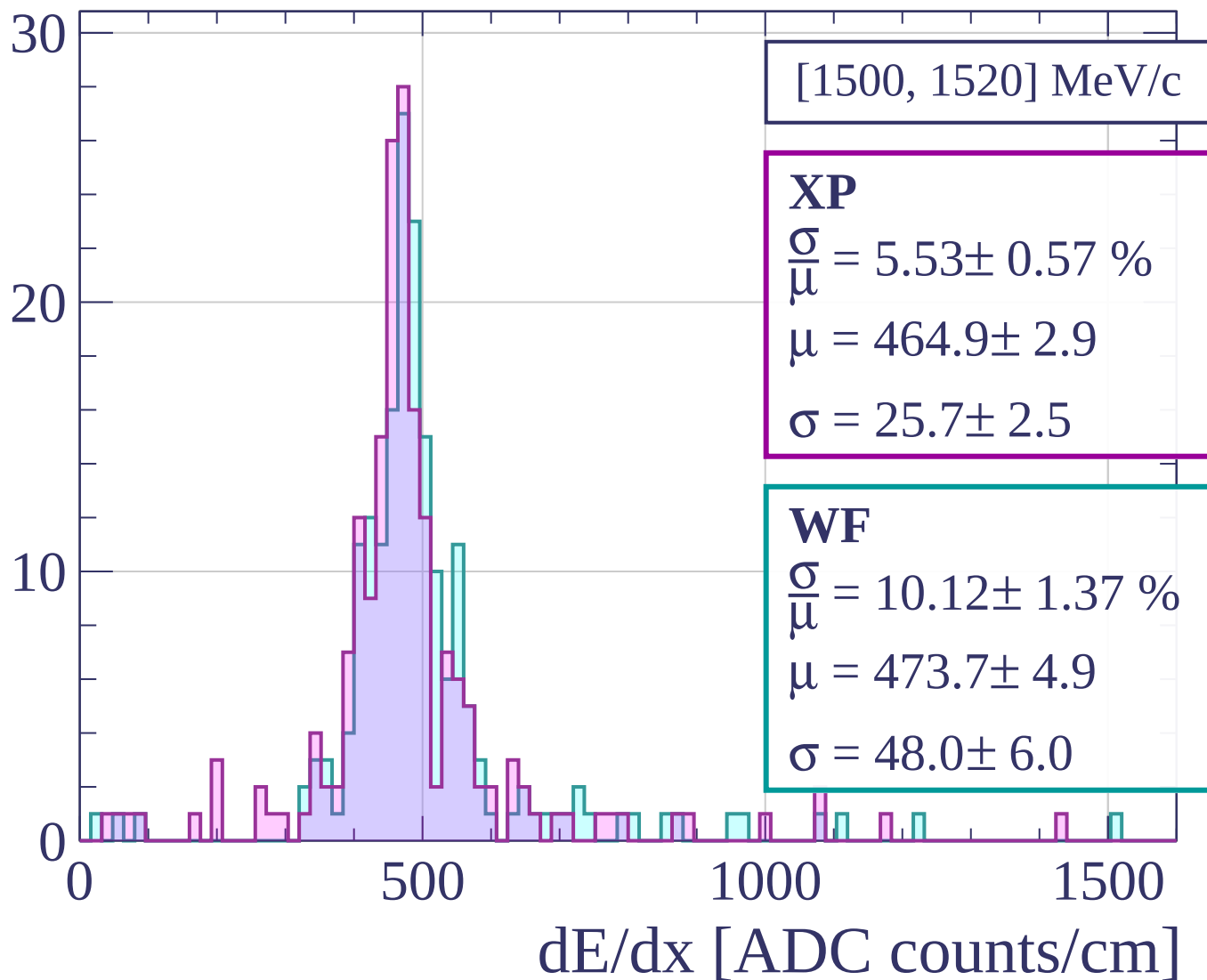
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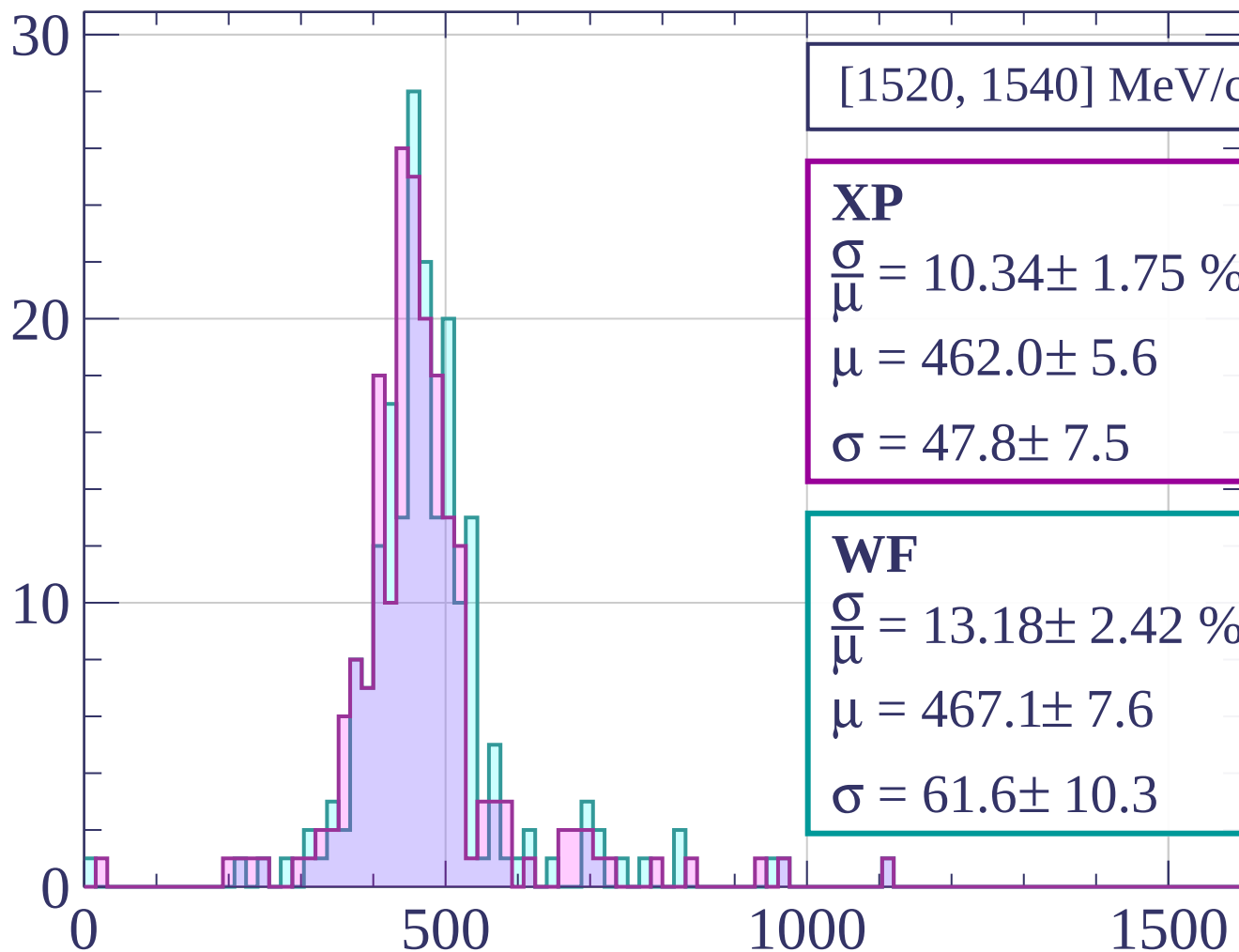
Count



Count



Count



[1520, 1540] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.34 \pm 1.75 \%$$

$$\mu = 462.0 \pm 5.6$$

$$\sigma = 47.8 \pm 7.5$$

WF

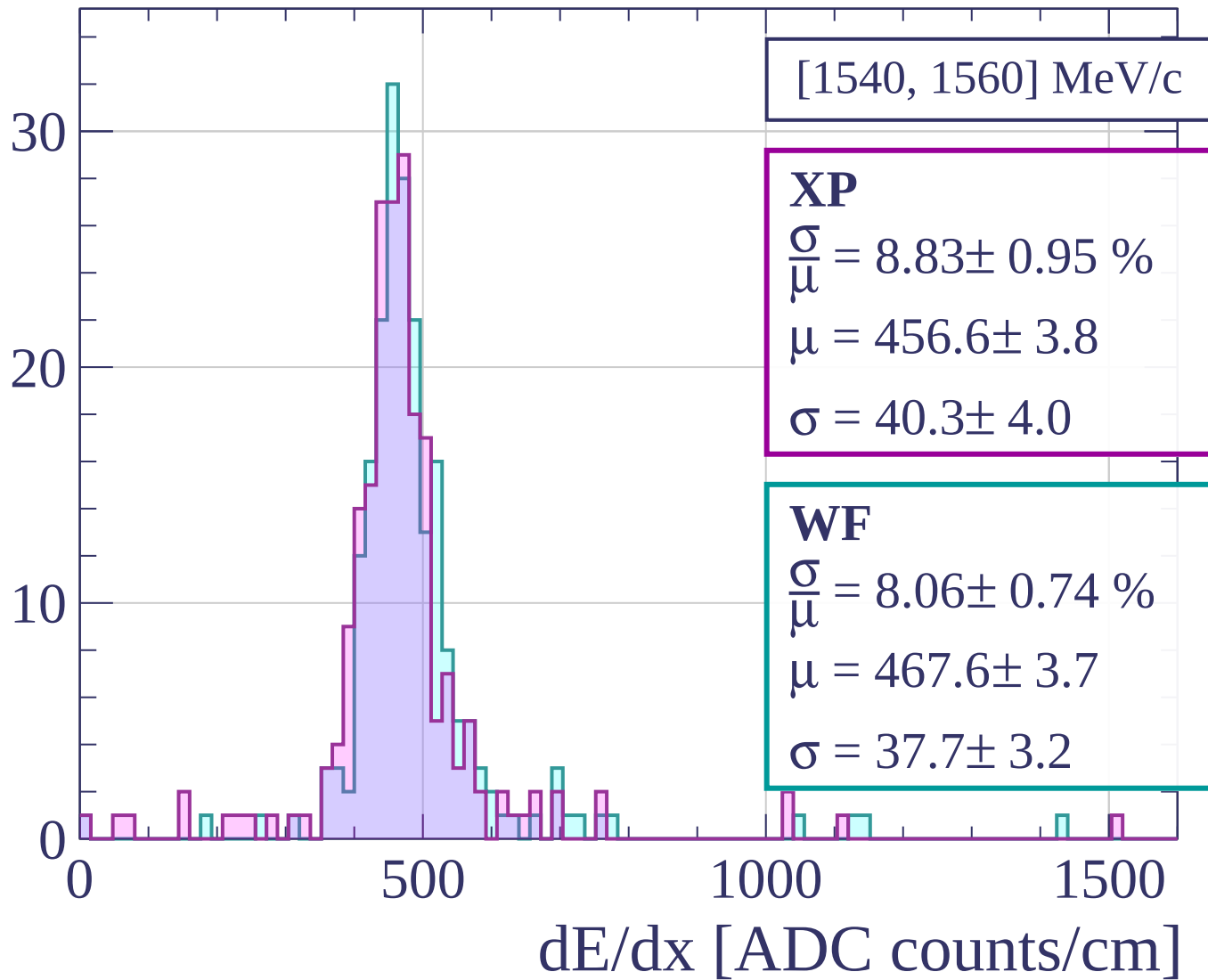
$$\frac{\sigma}{\mu} = 13.18 \pm 2.42 \%$$

$$\mu = 467.1 \pm 7.6$$

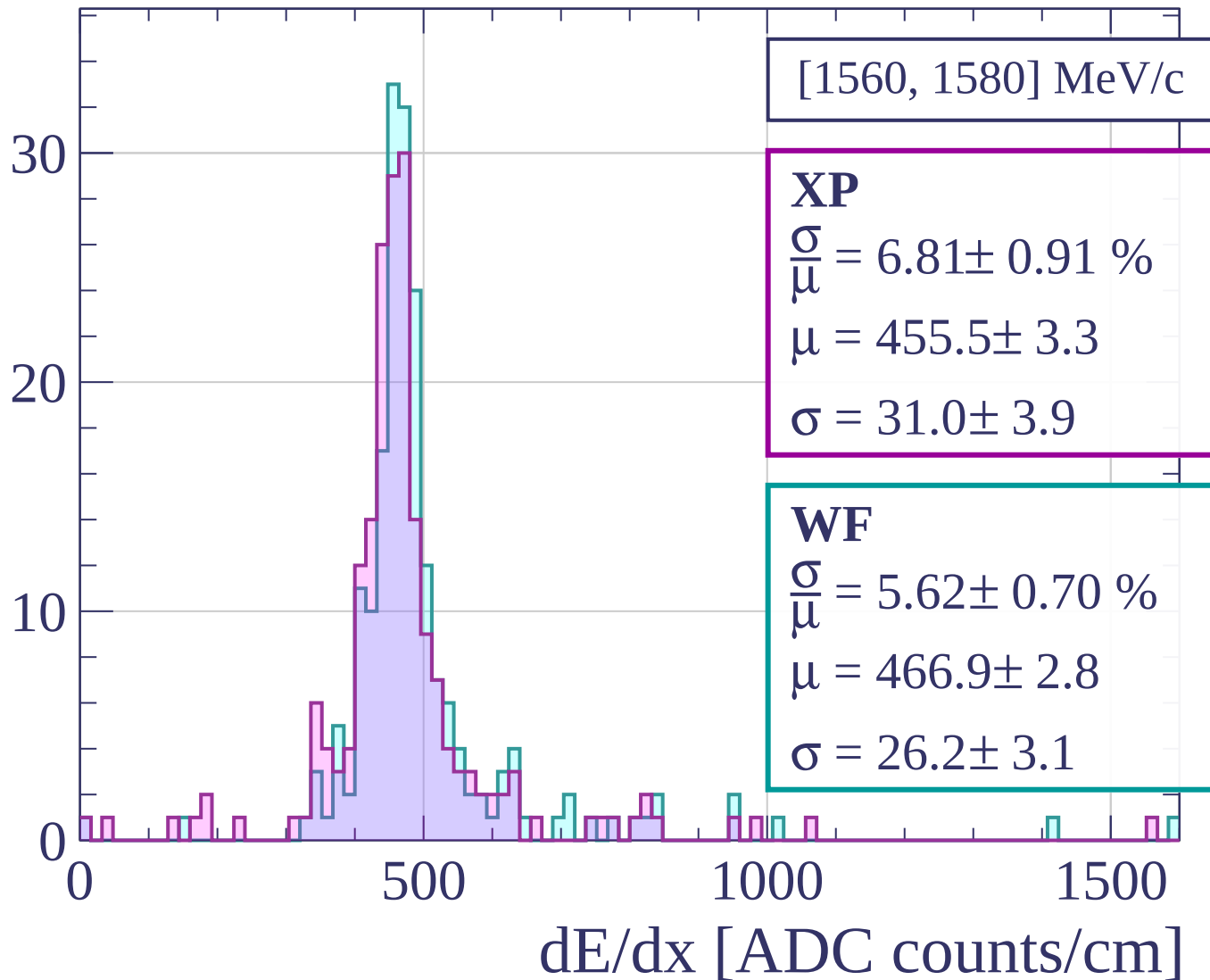
$$\sigma = 61.6 \pm 10.3$$

dE/dx [ADC counts/cm]

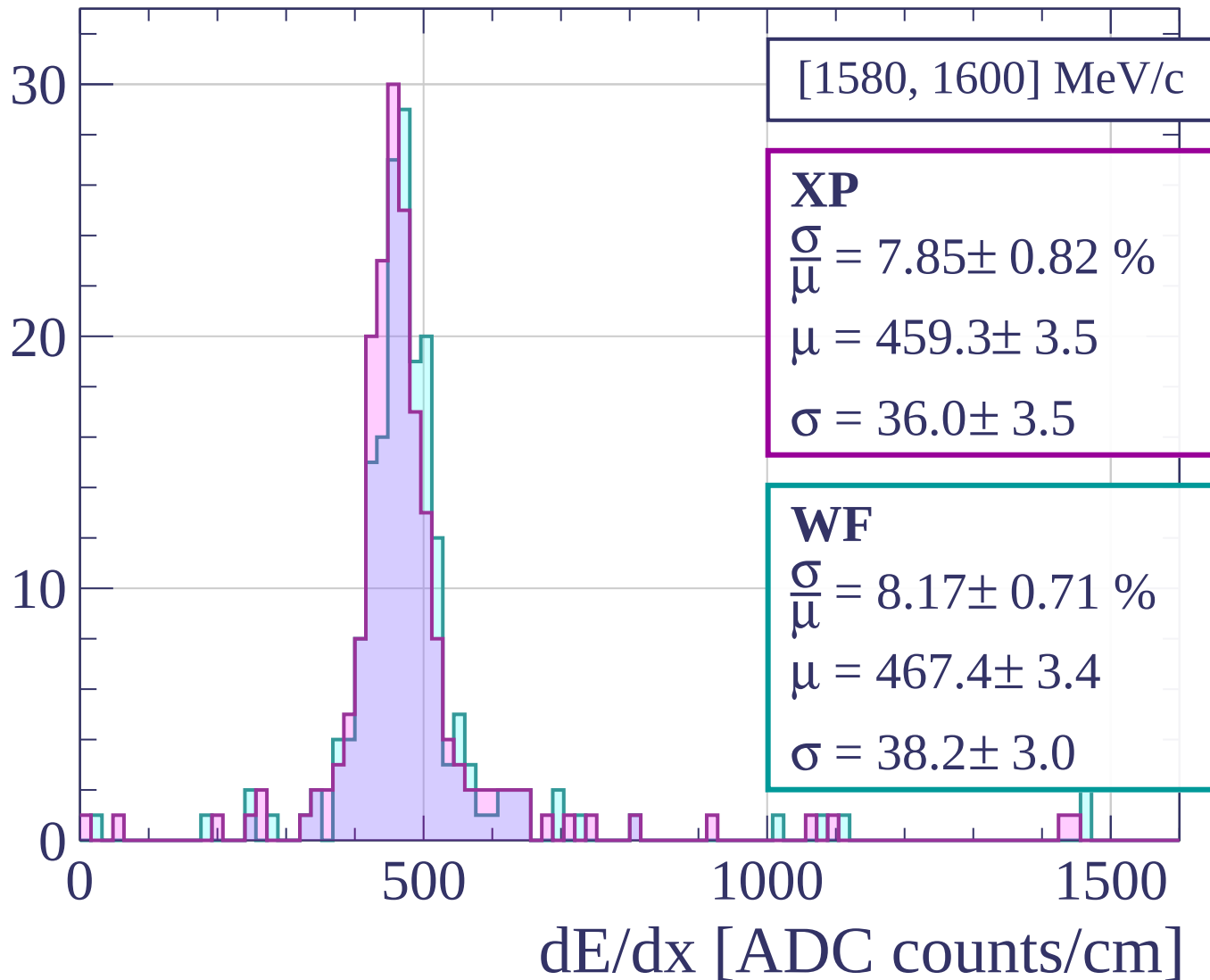
Count



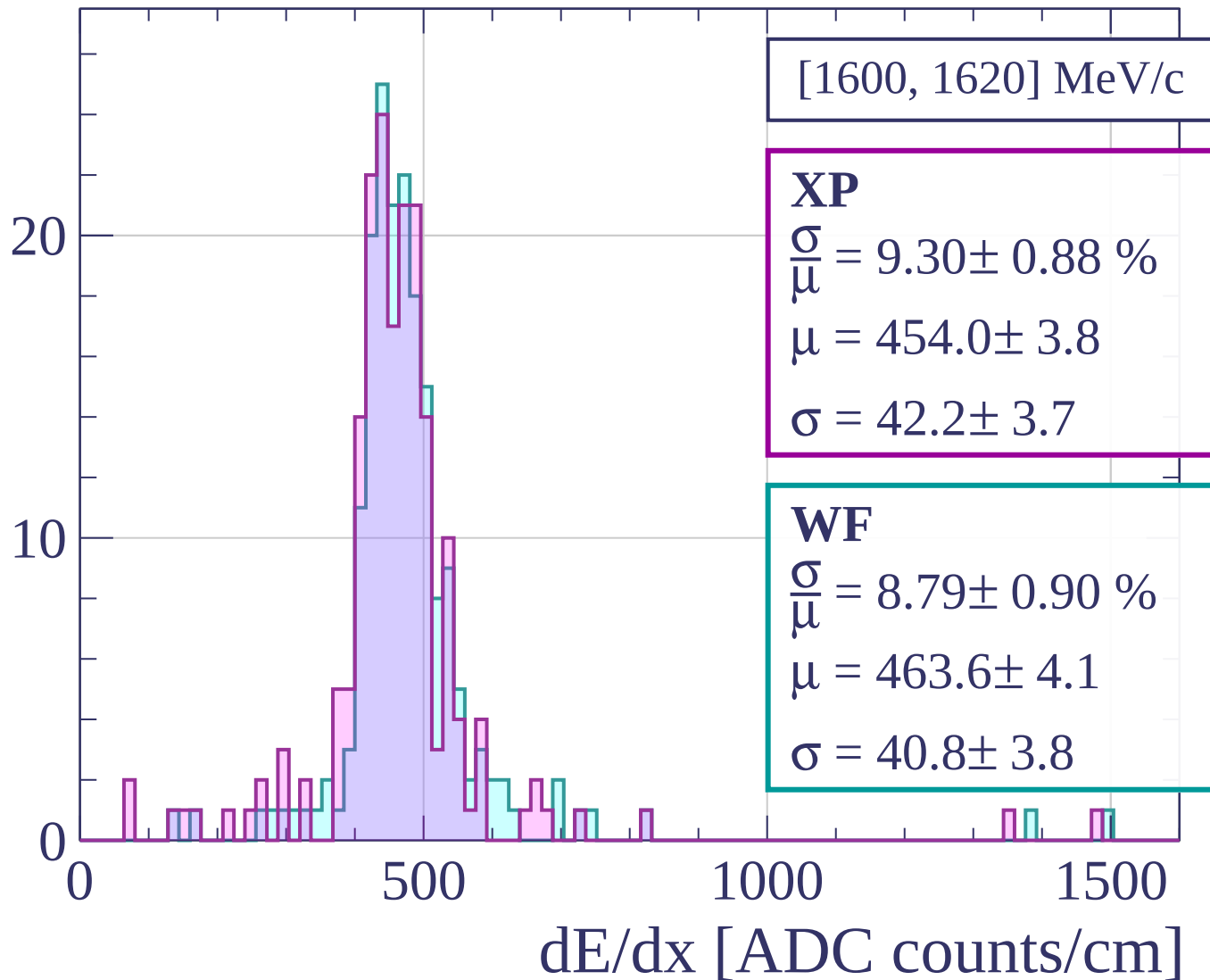
Count



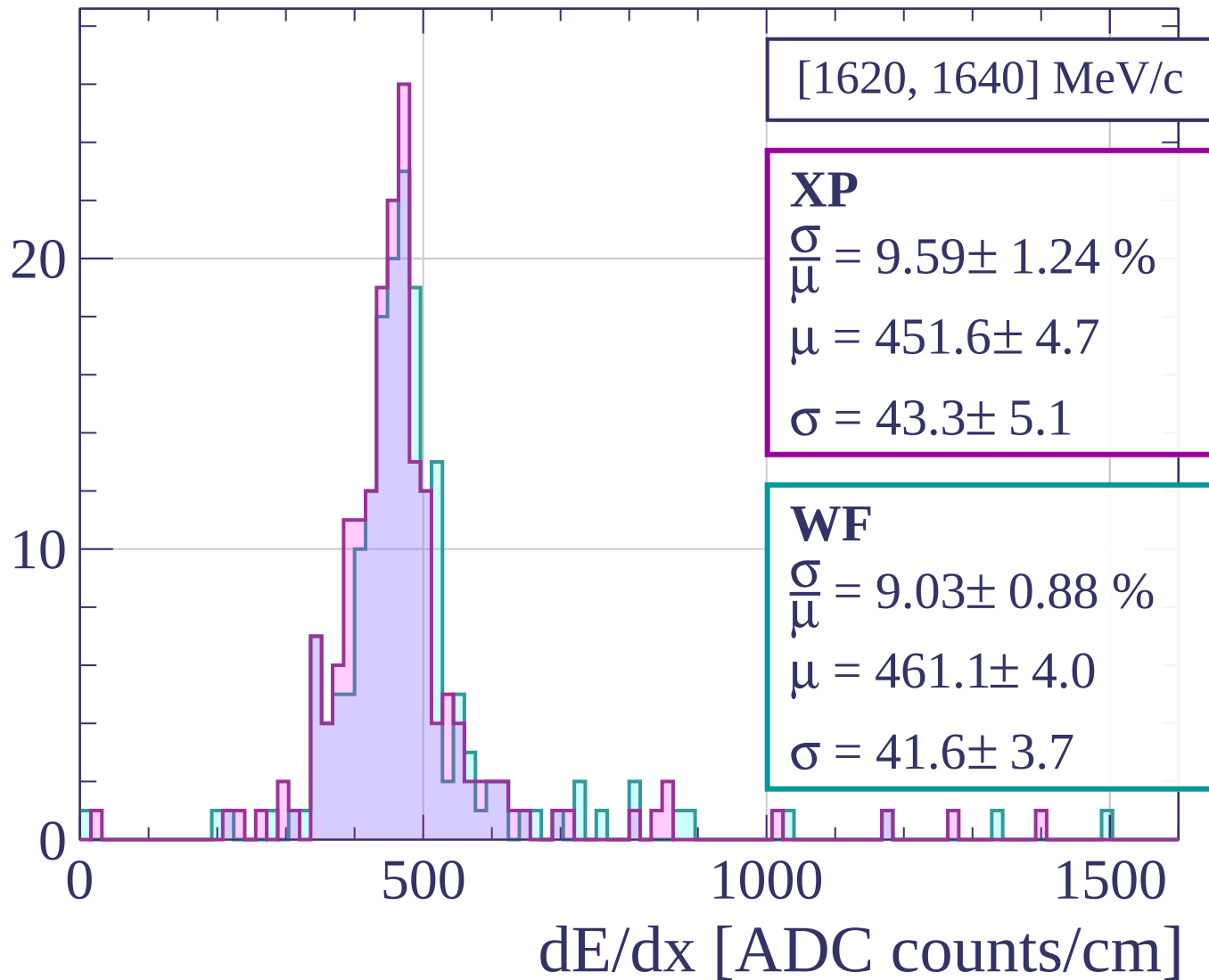
Count



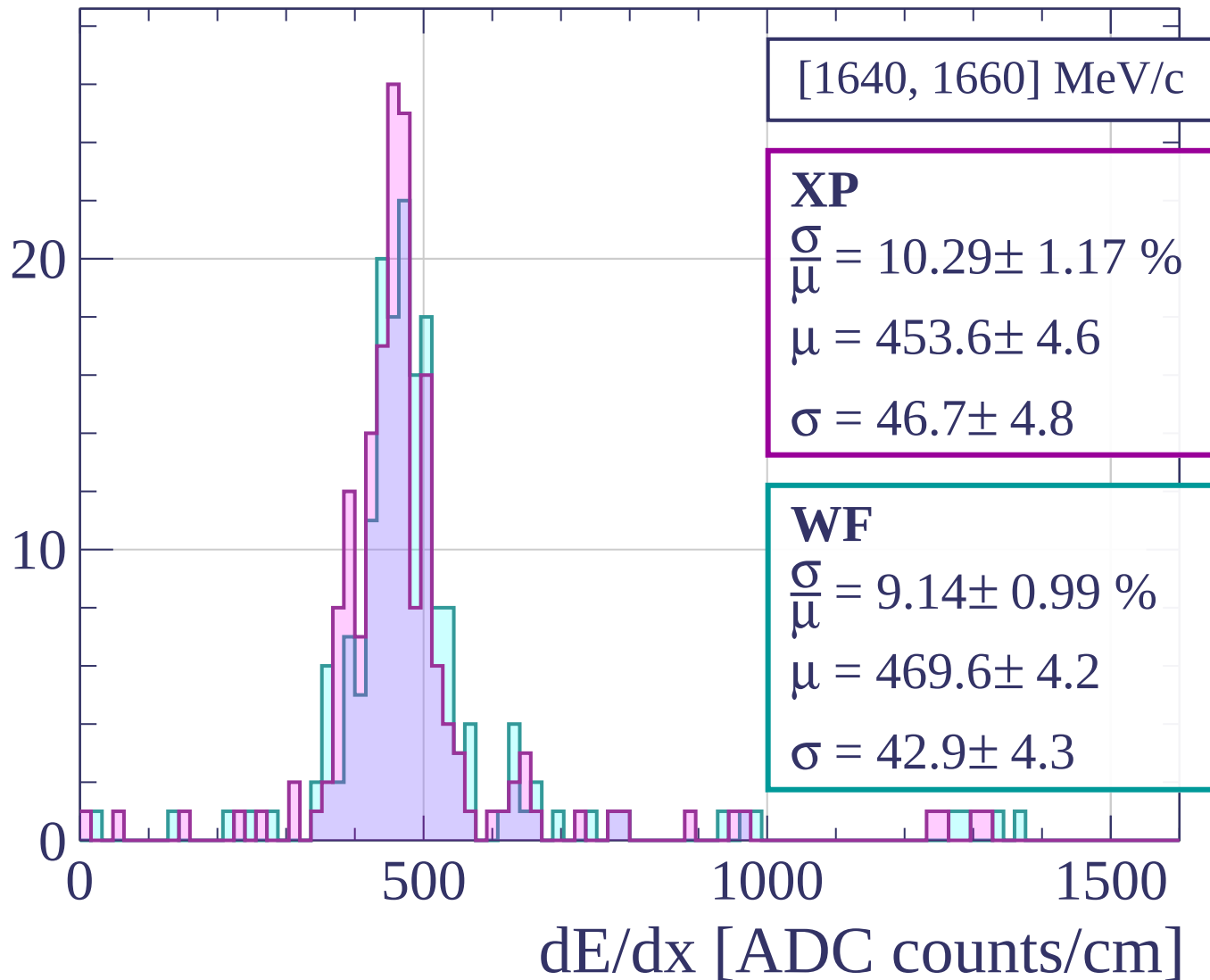
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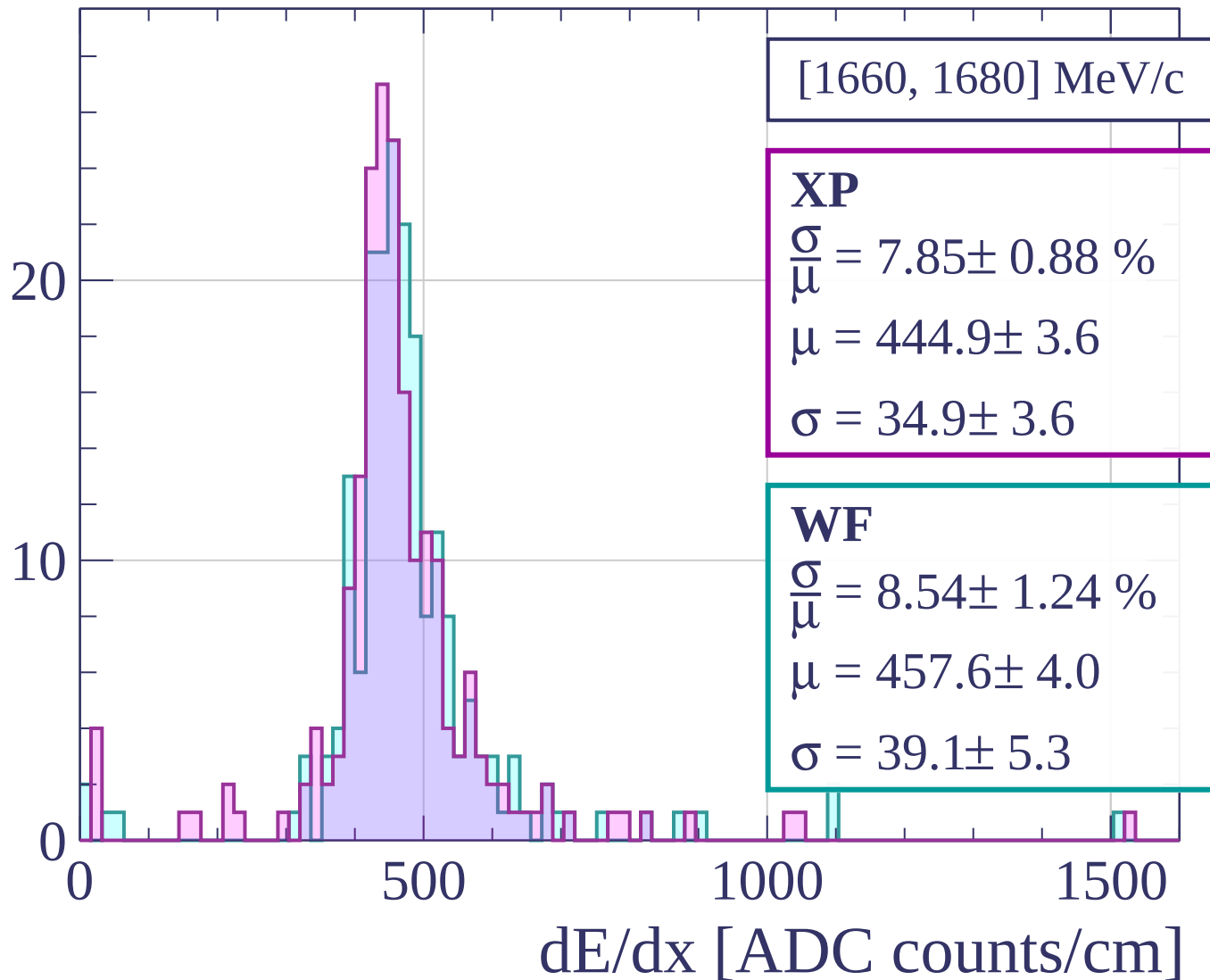
Count



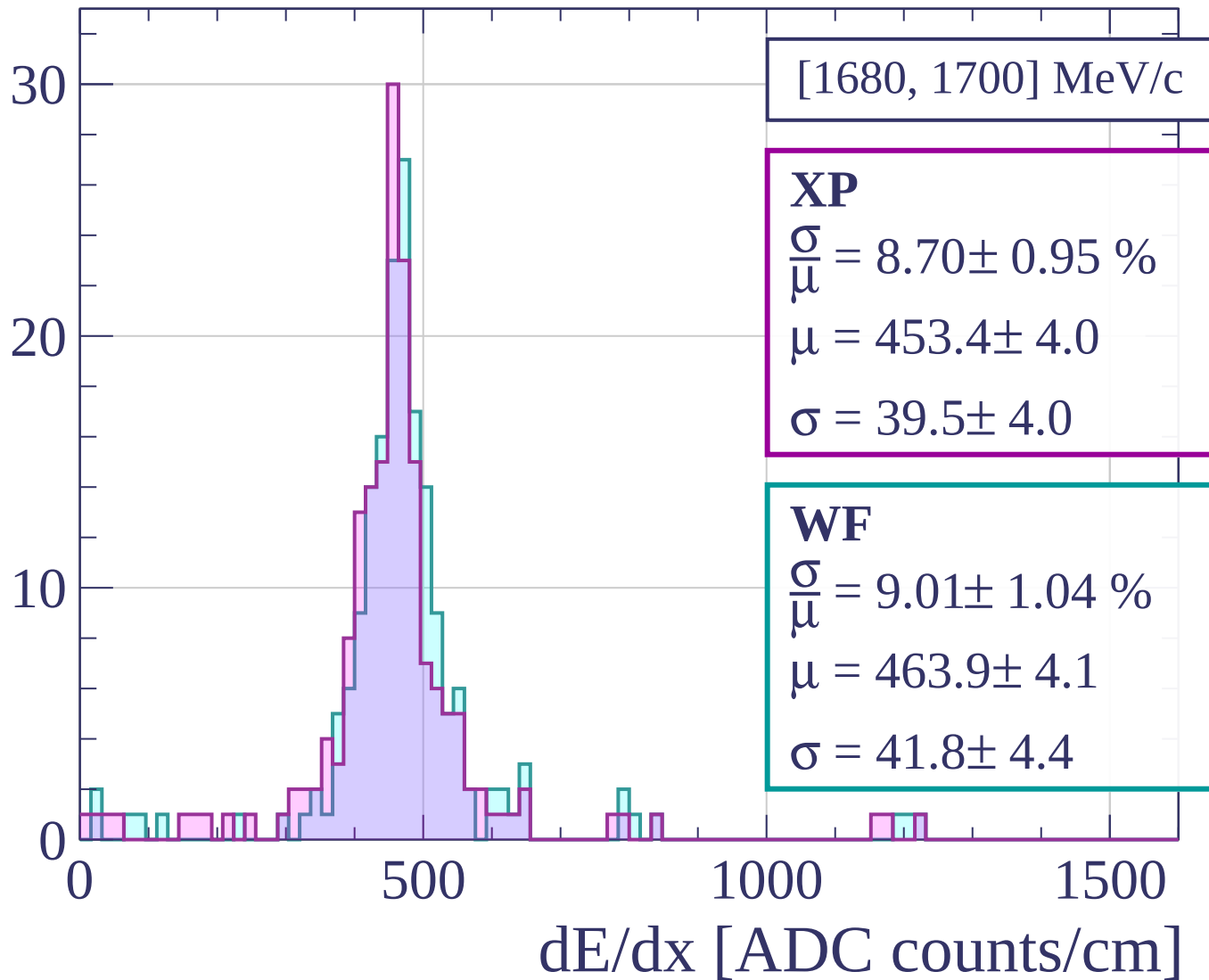
Count



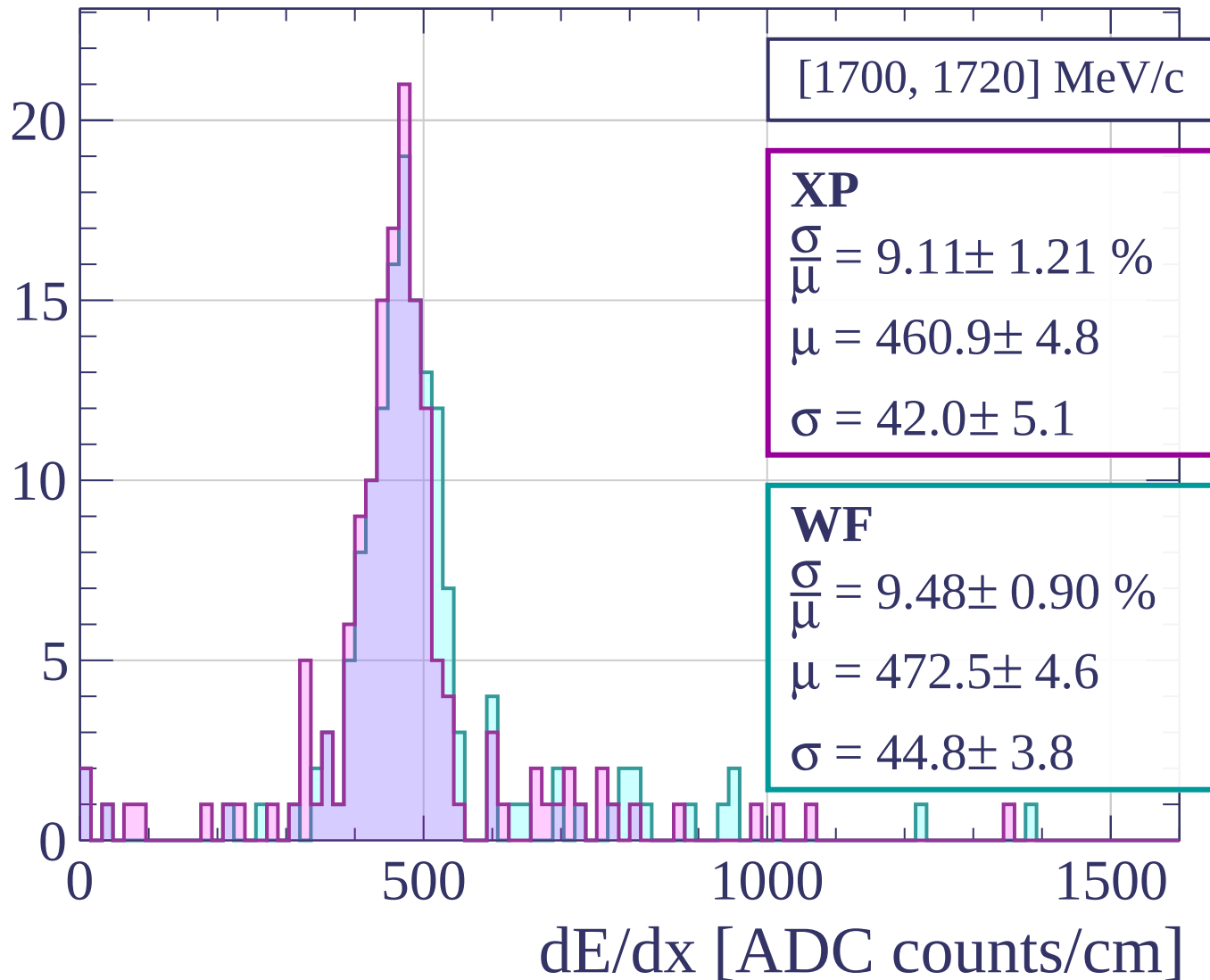
Count



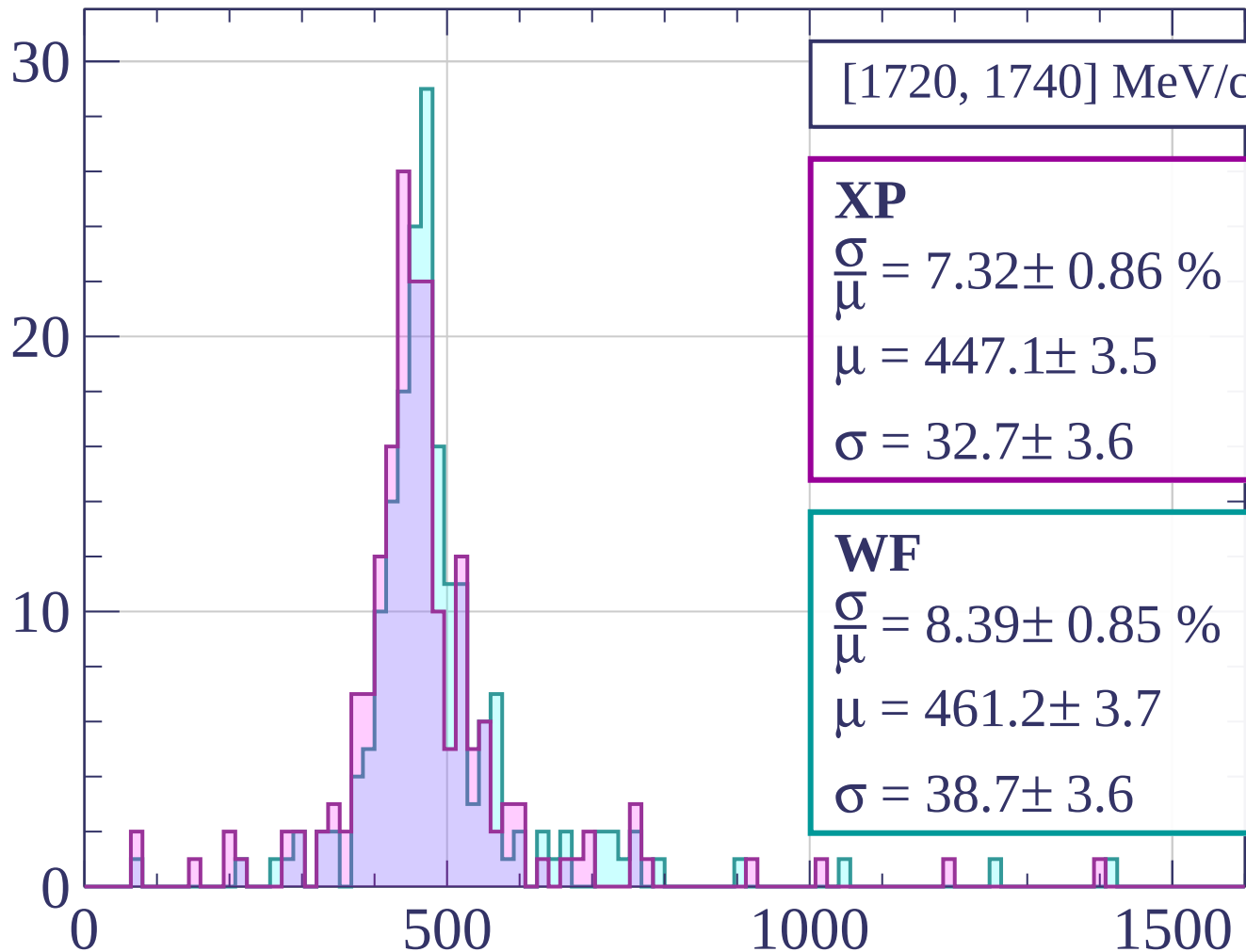
Count



Count

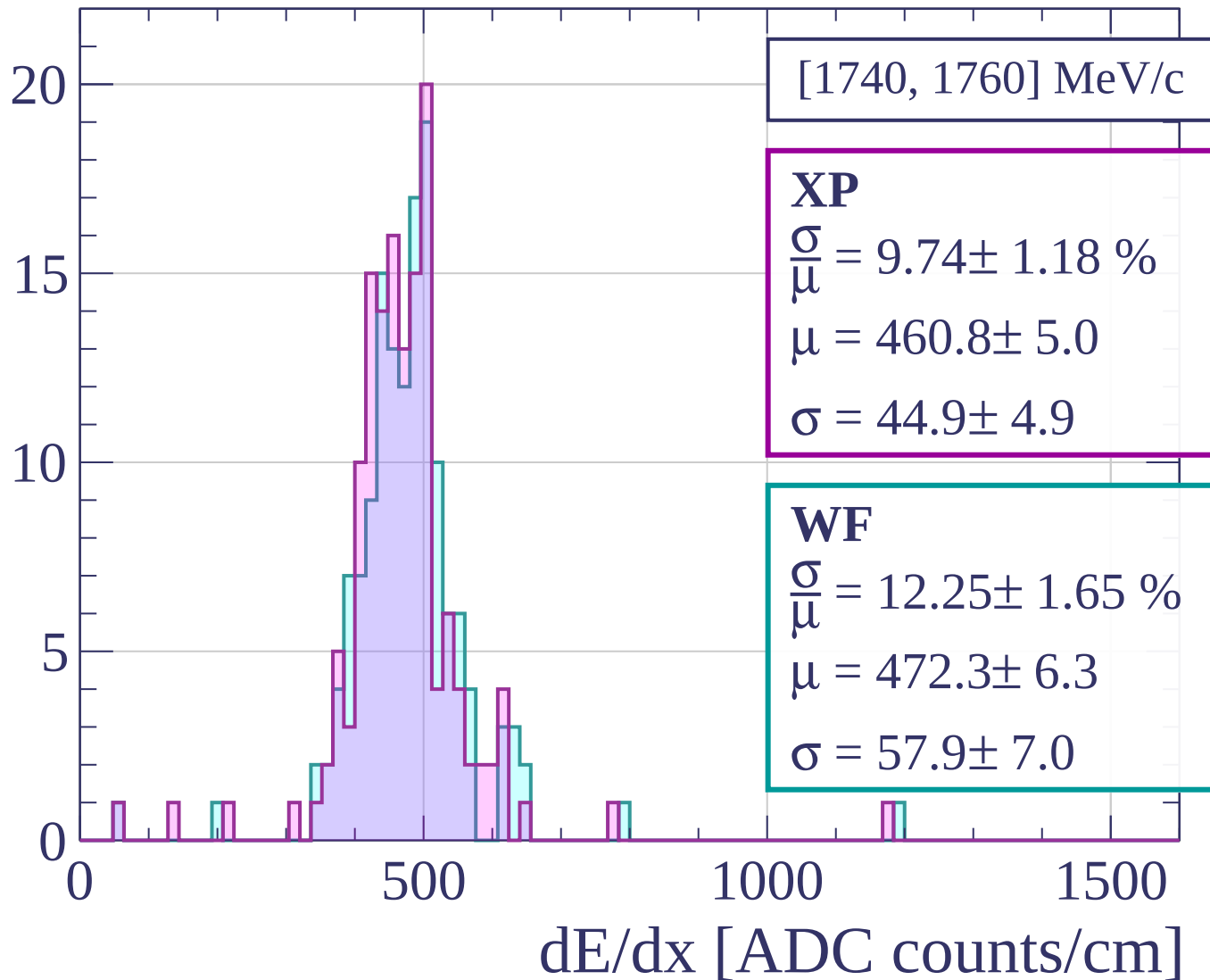


Count

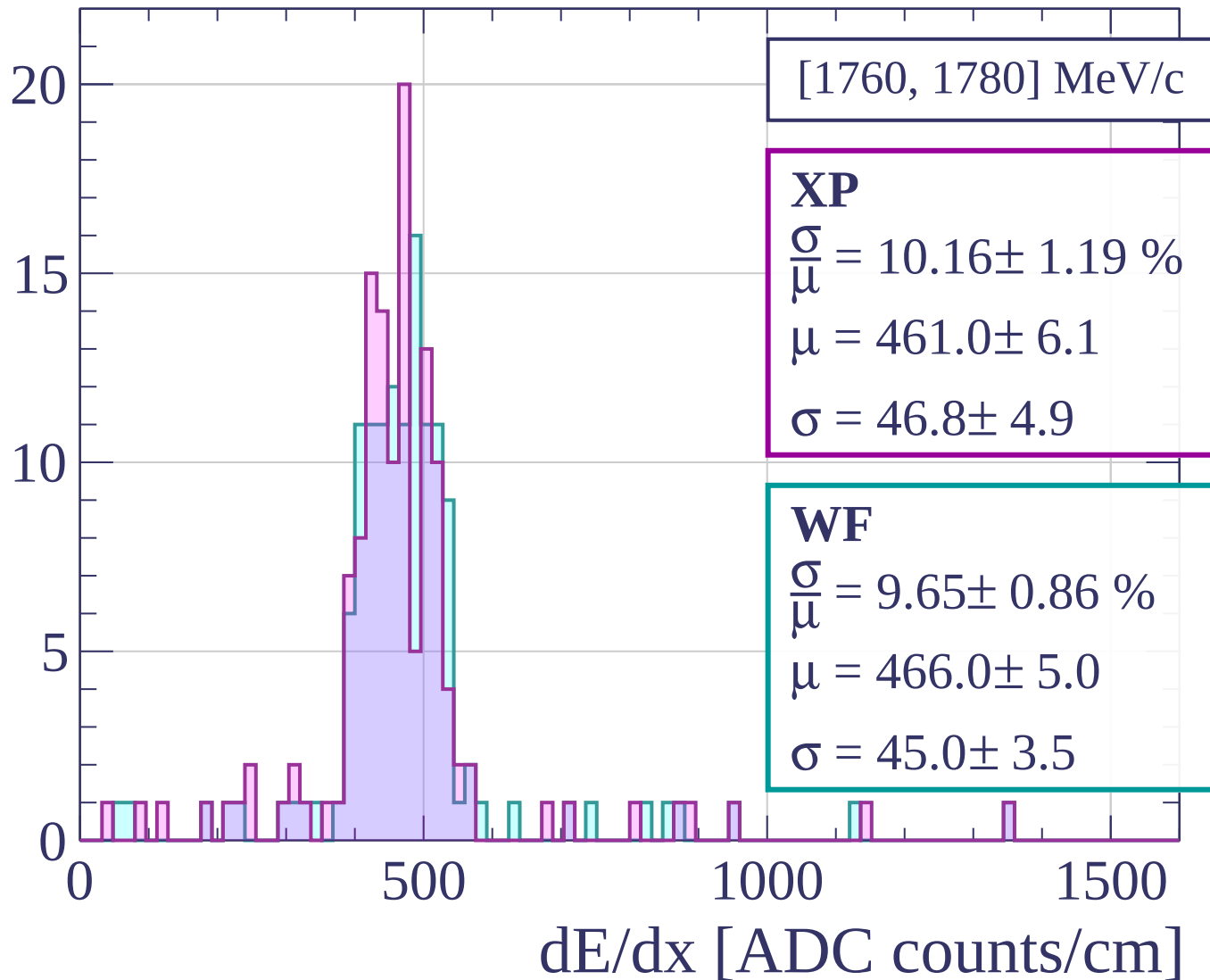


dE/dx [ADC counts/cm]

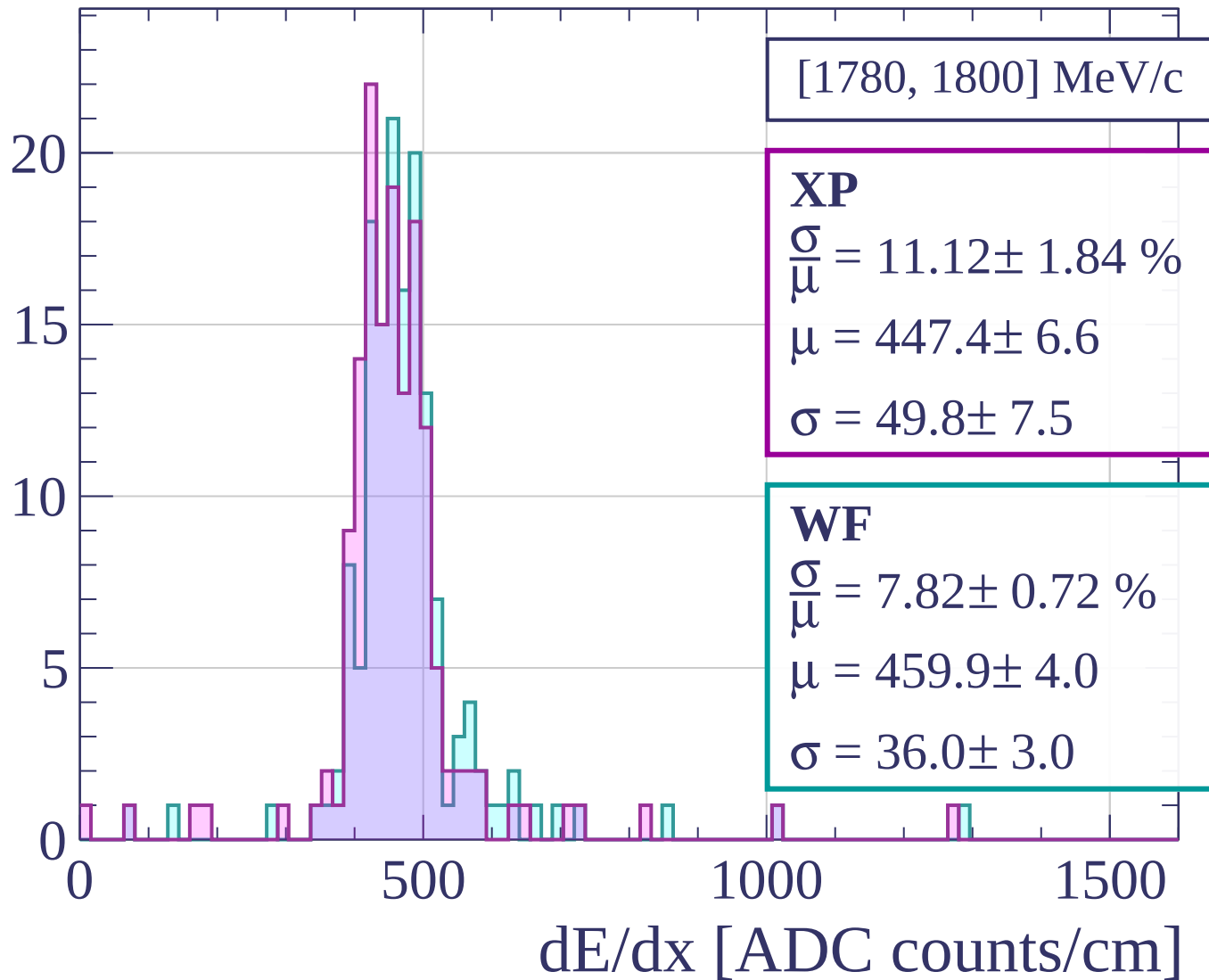
Count



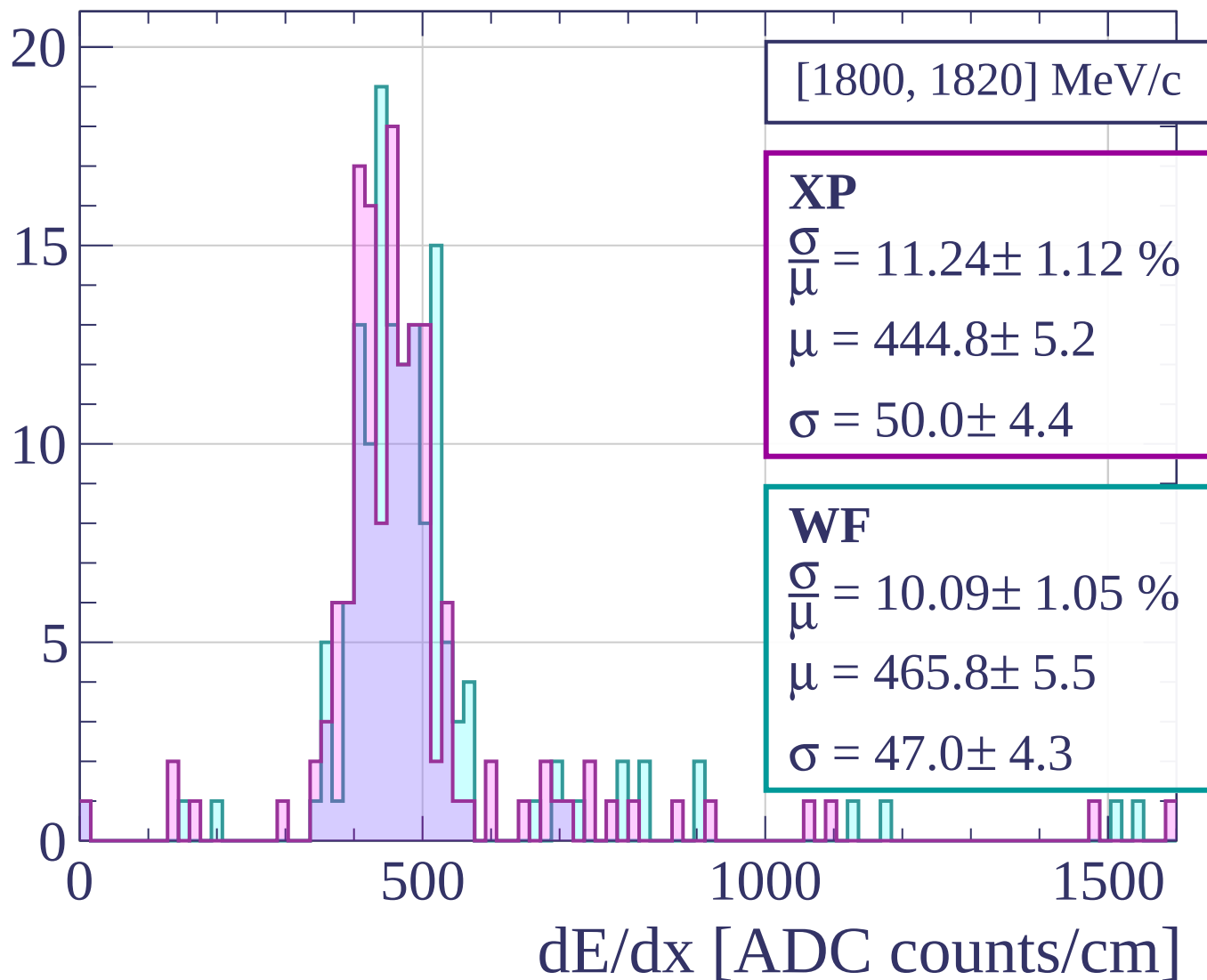
Count



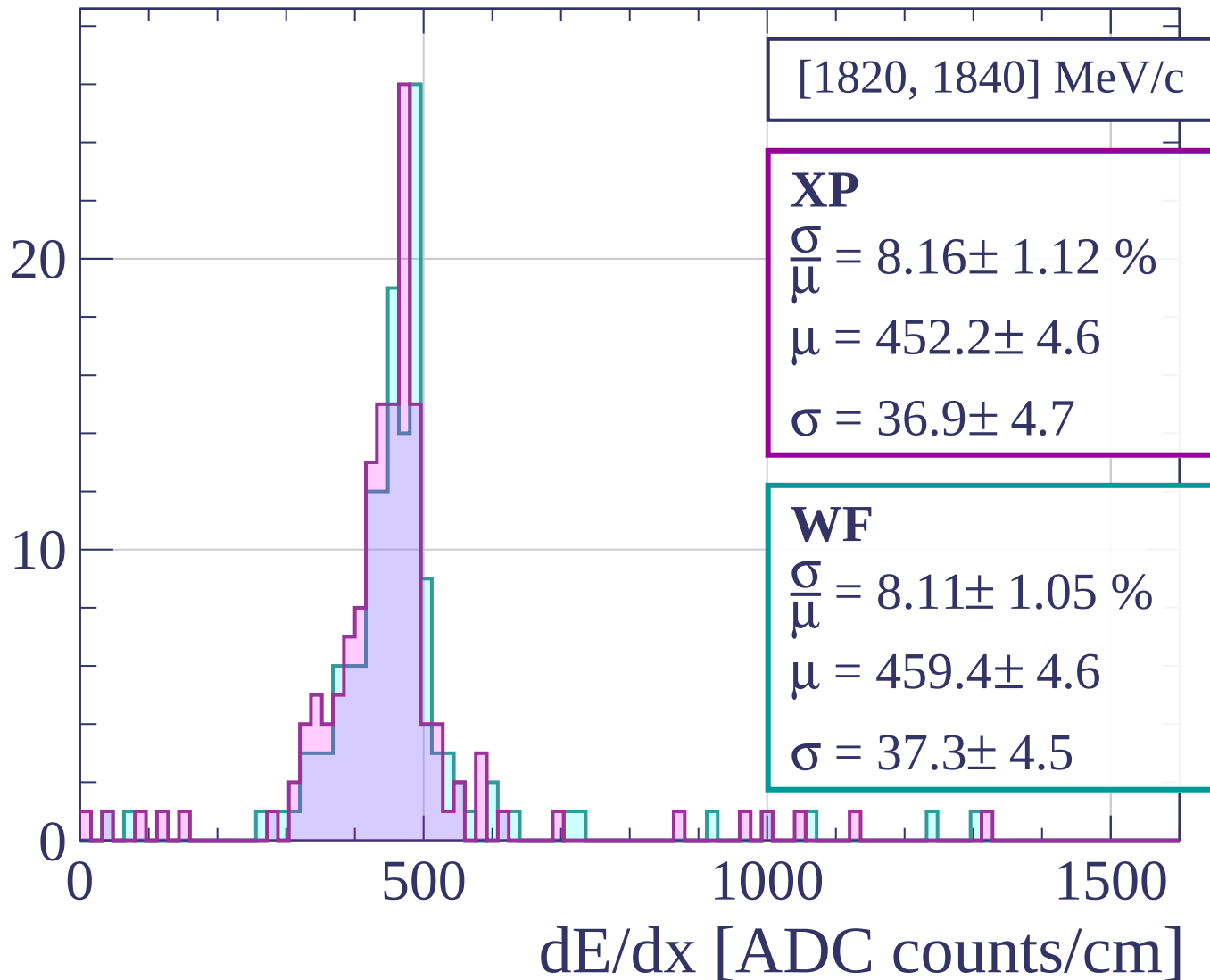
Count



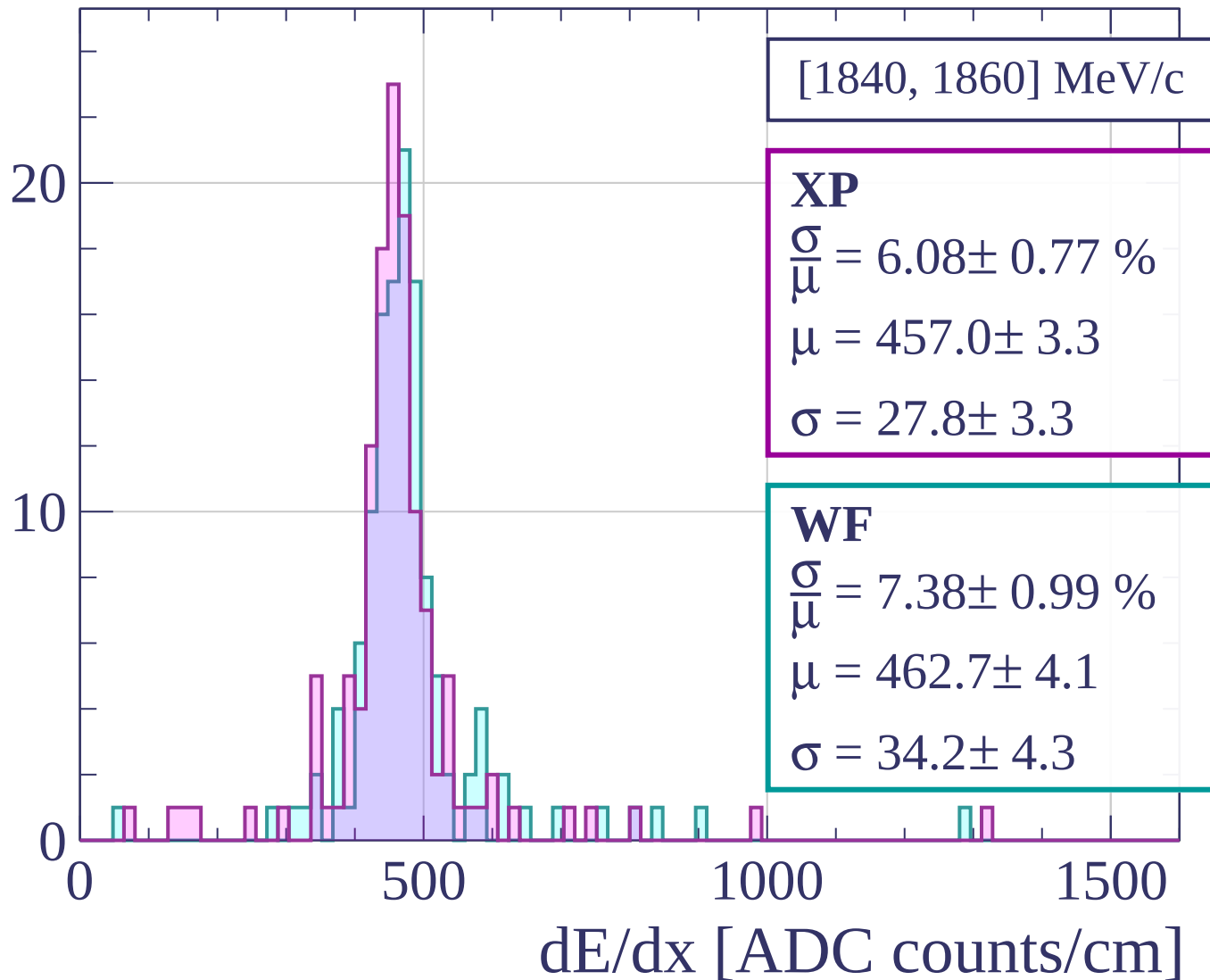
Count



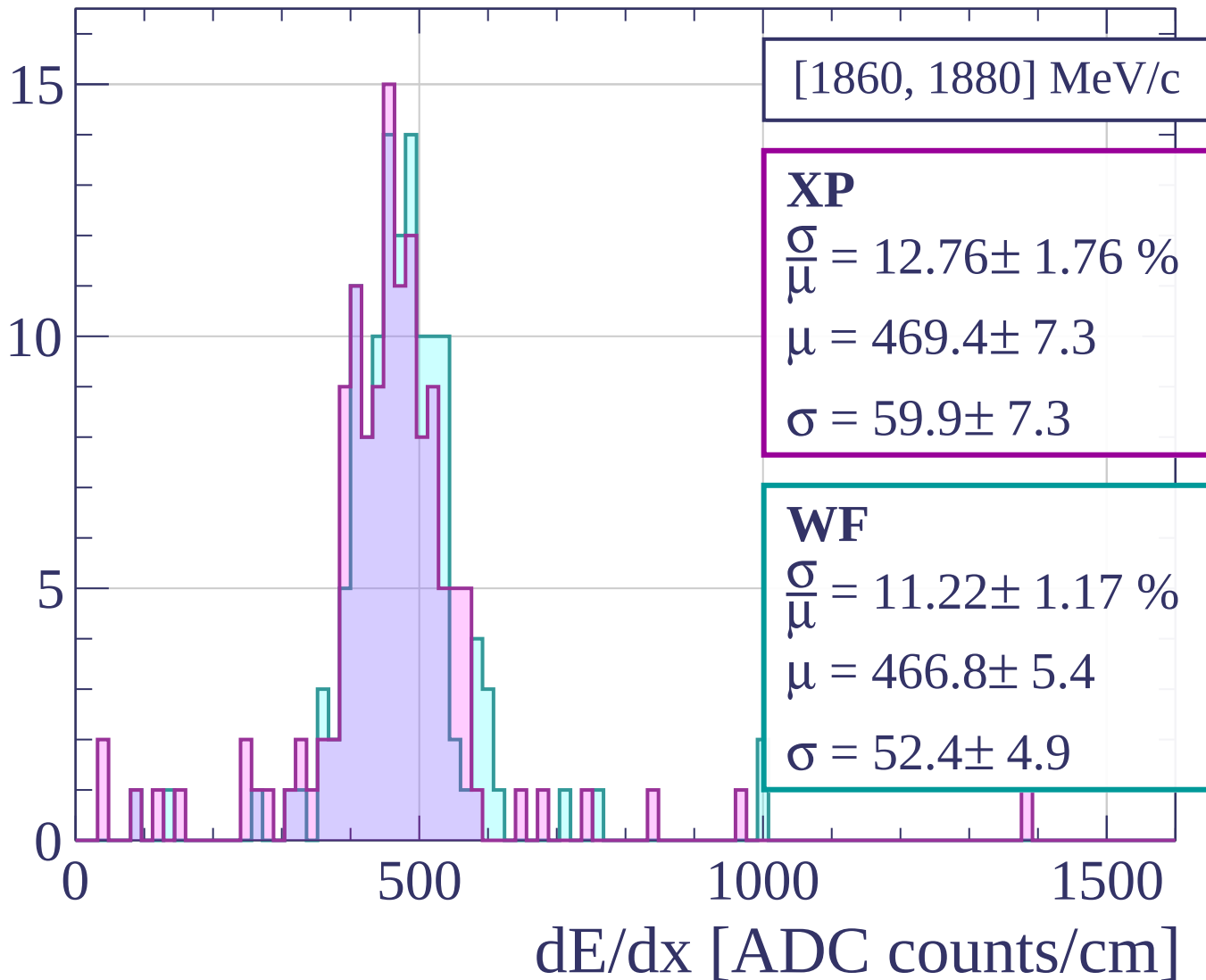
Count



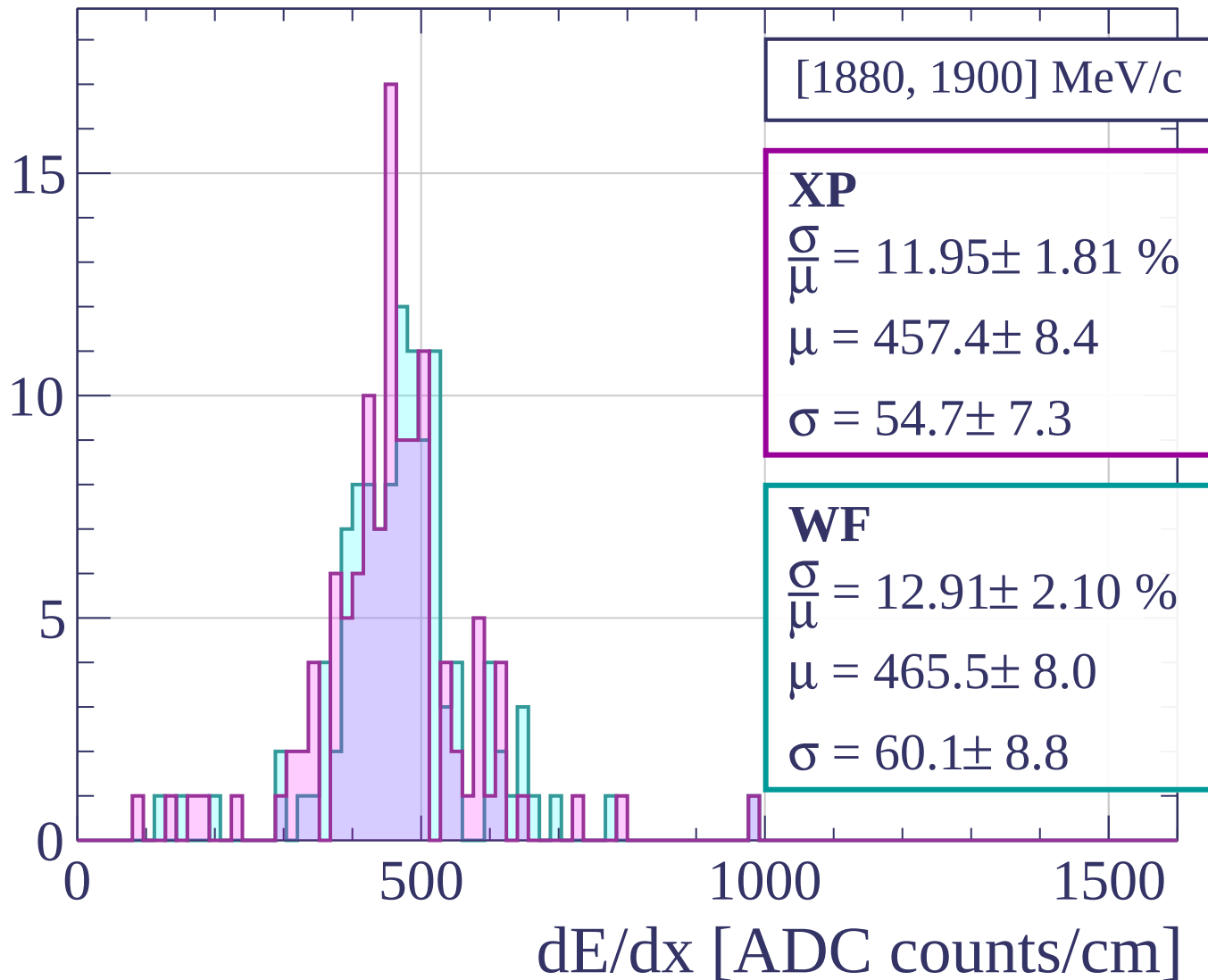
Count



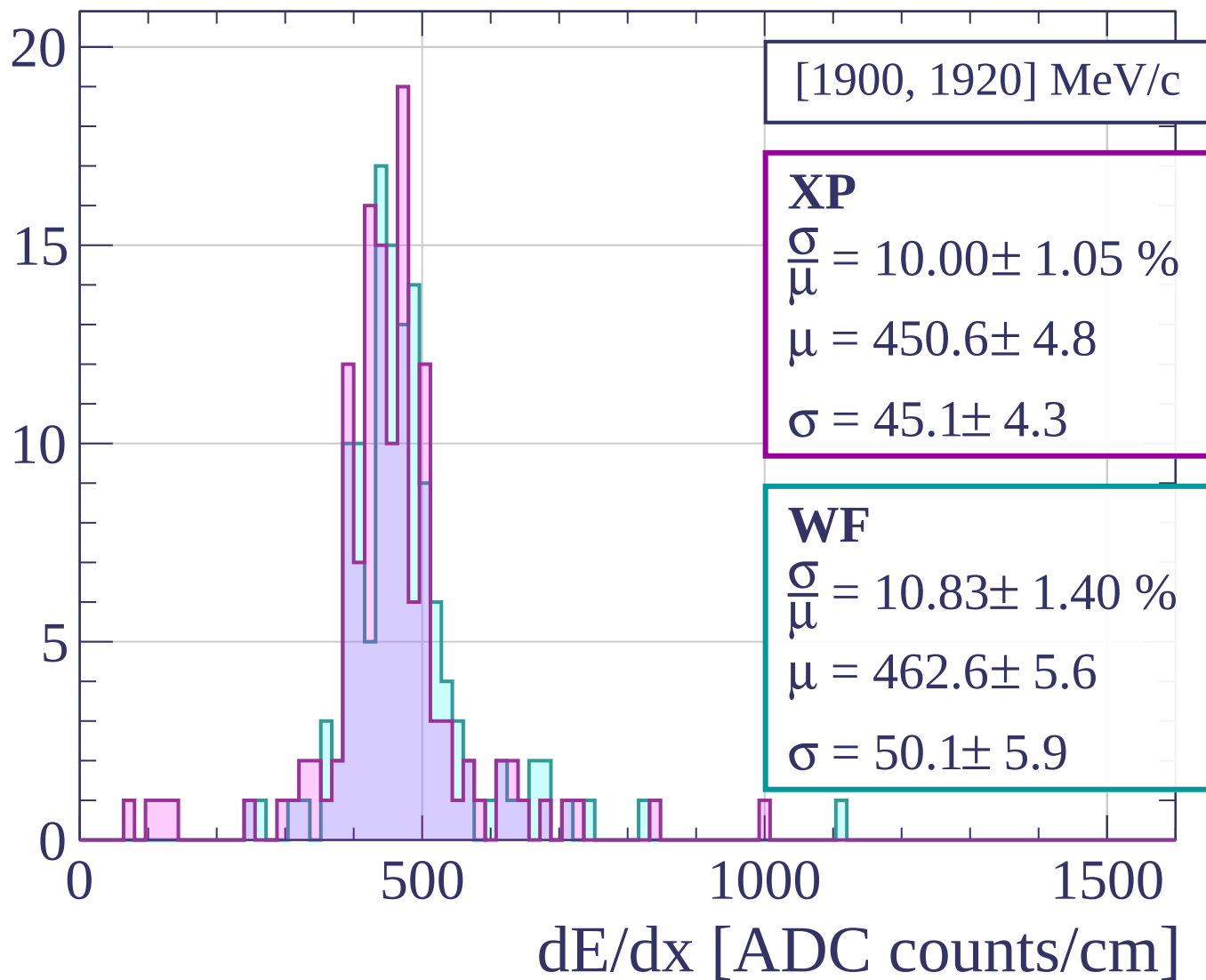
Count



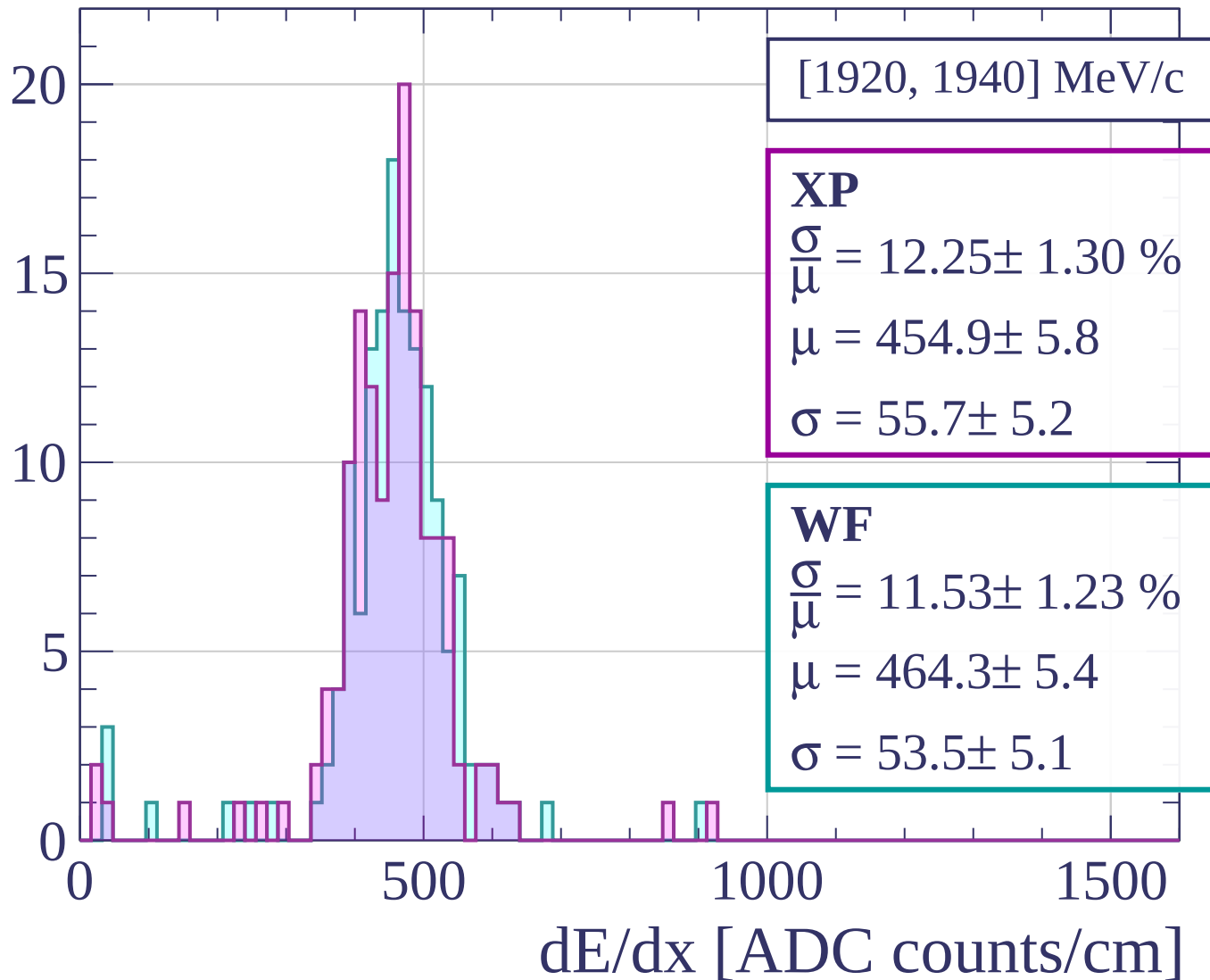
Count



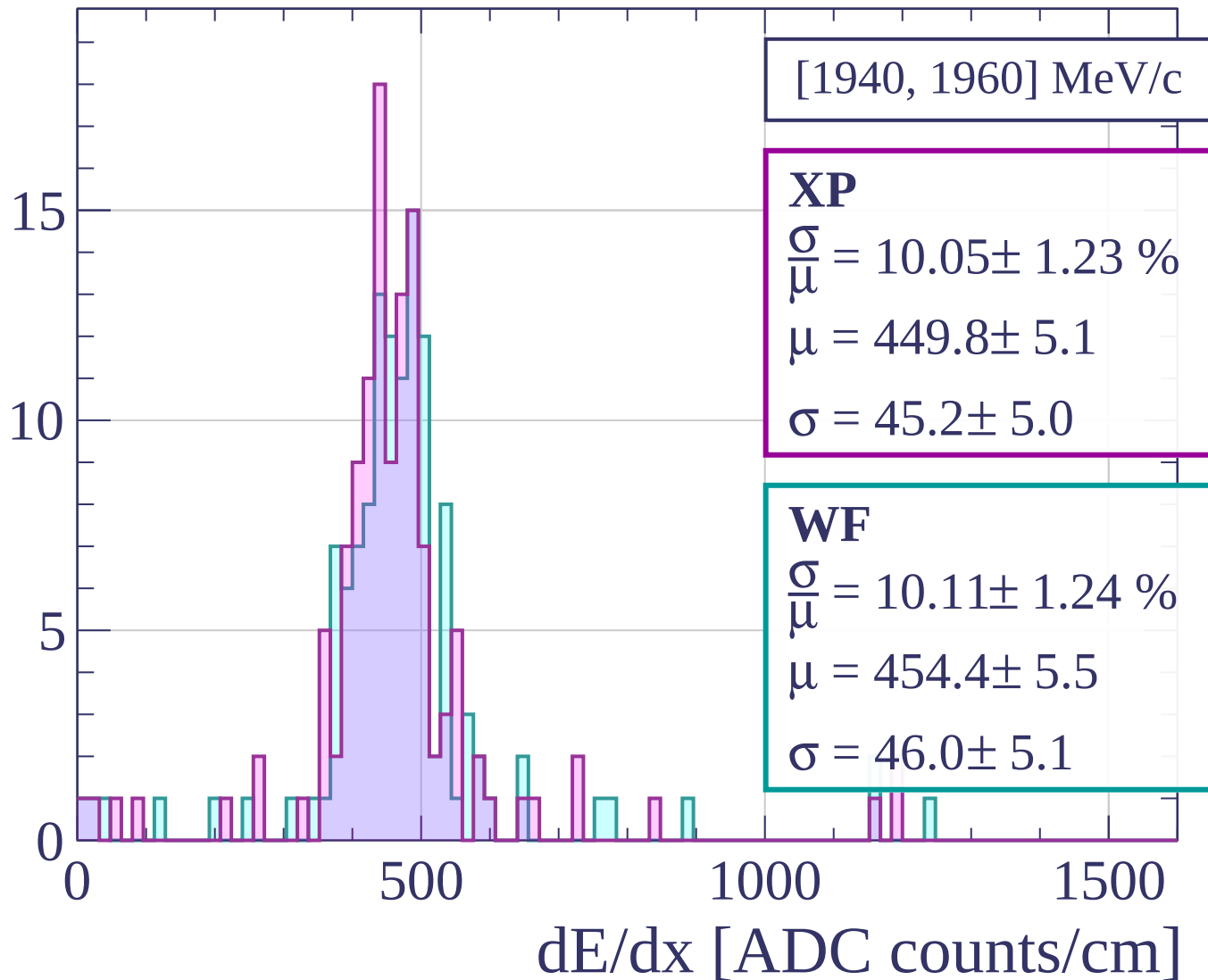
Count



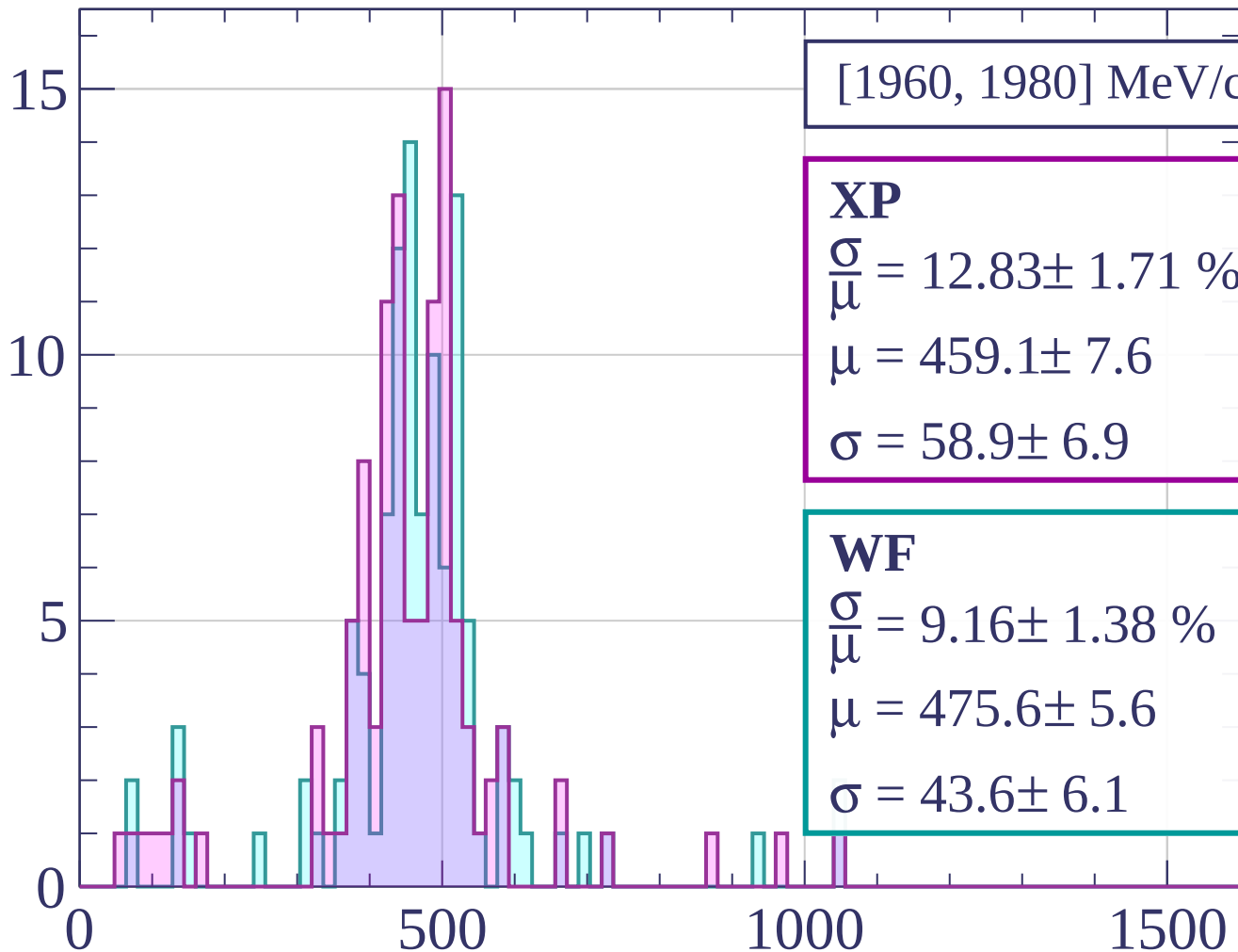
Count



Count

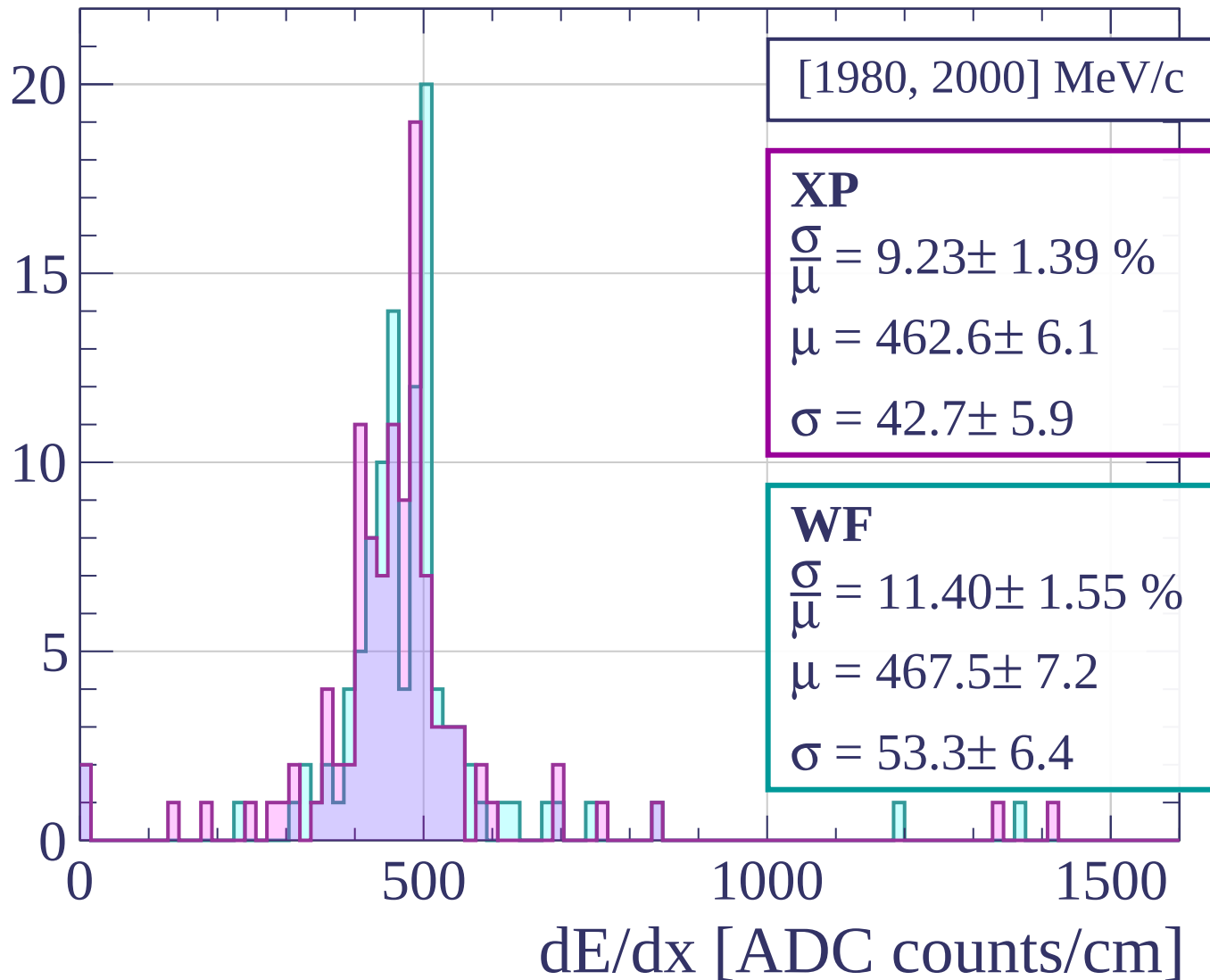


Count

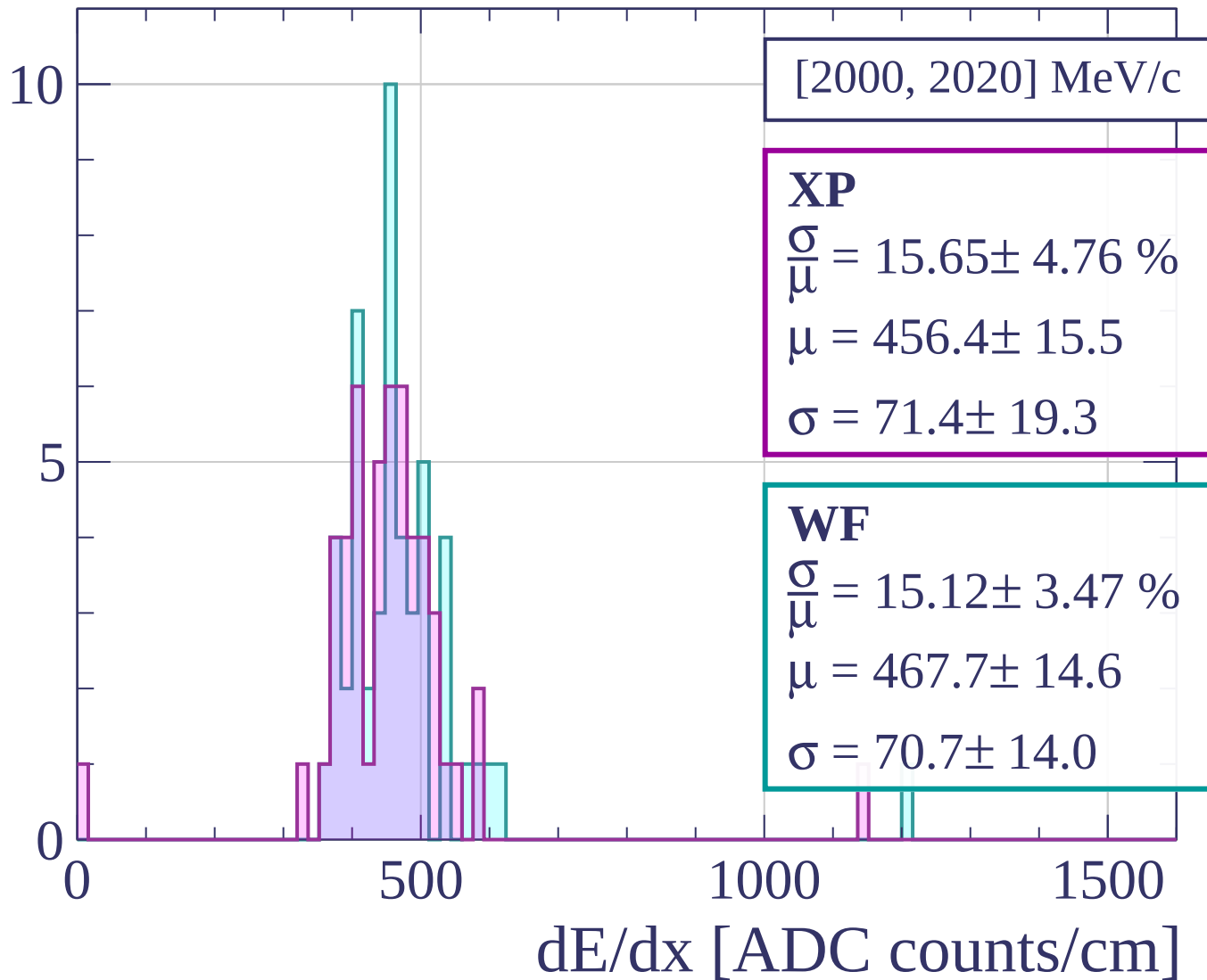


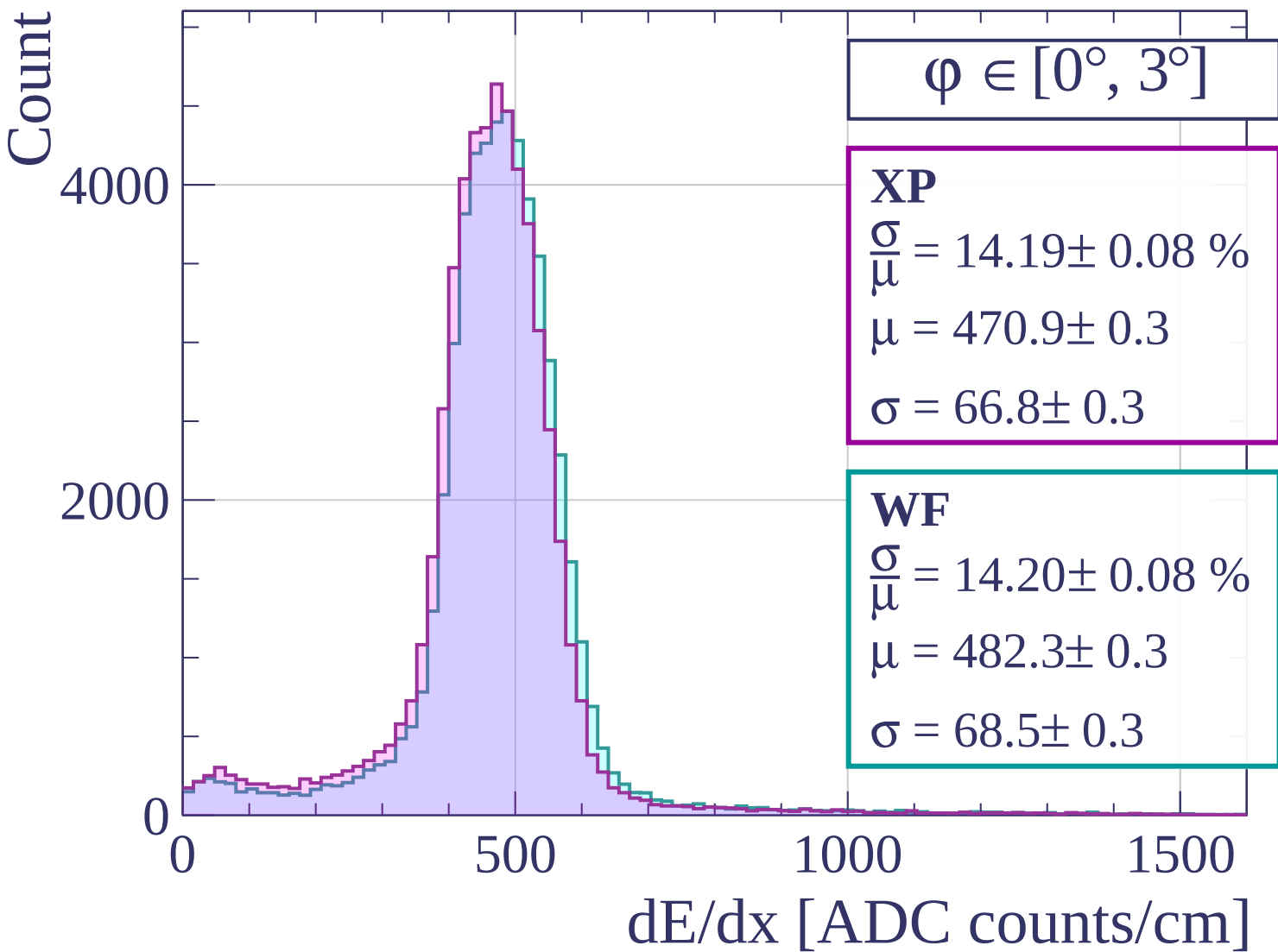
dE/dx [ADC counts/cm]

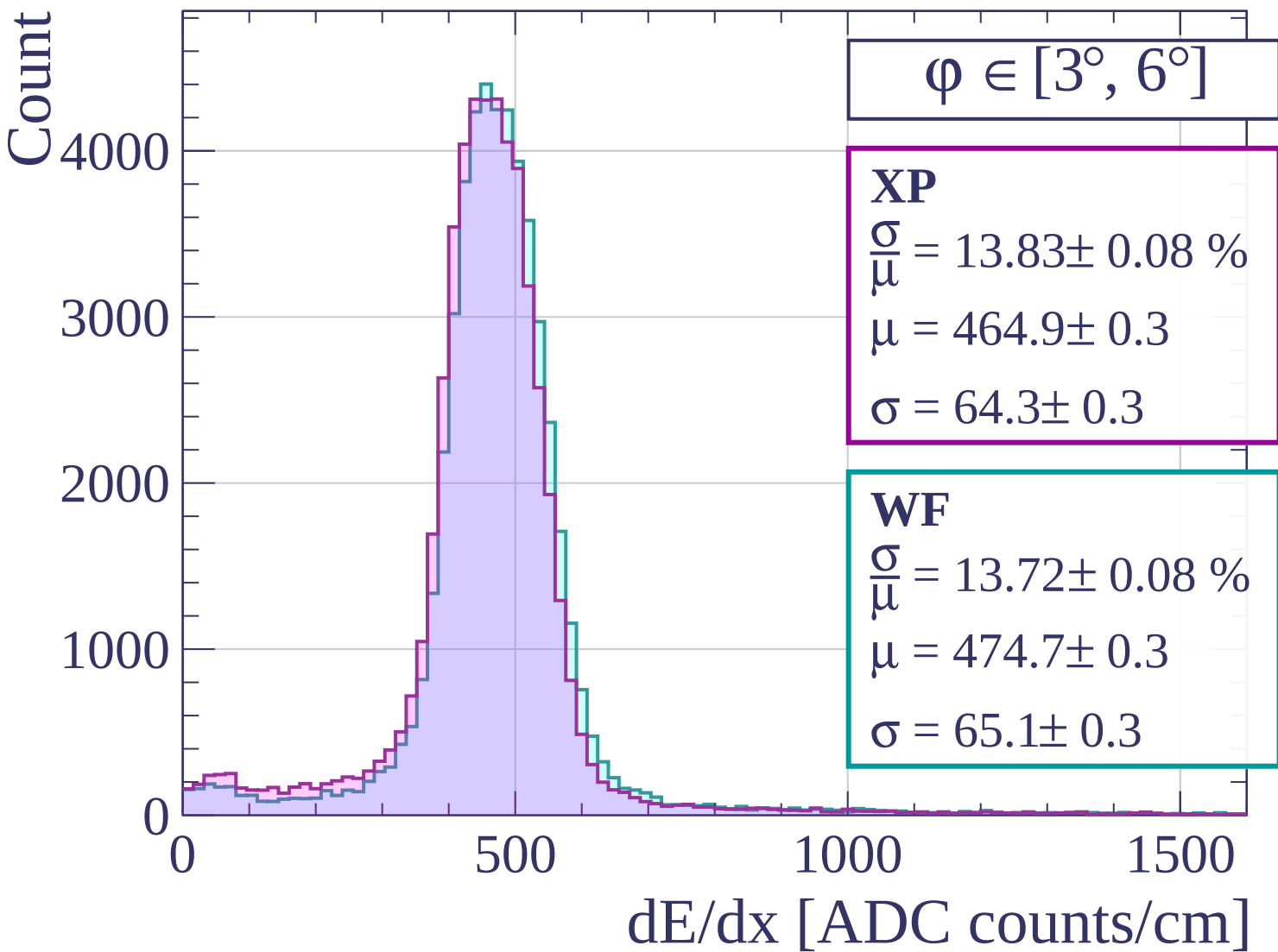
Count

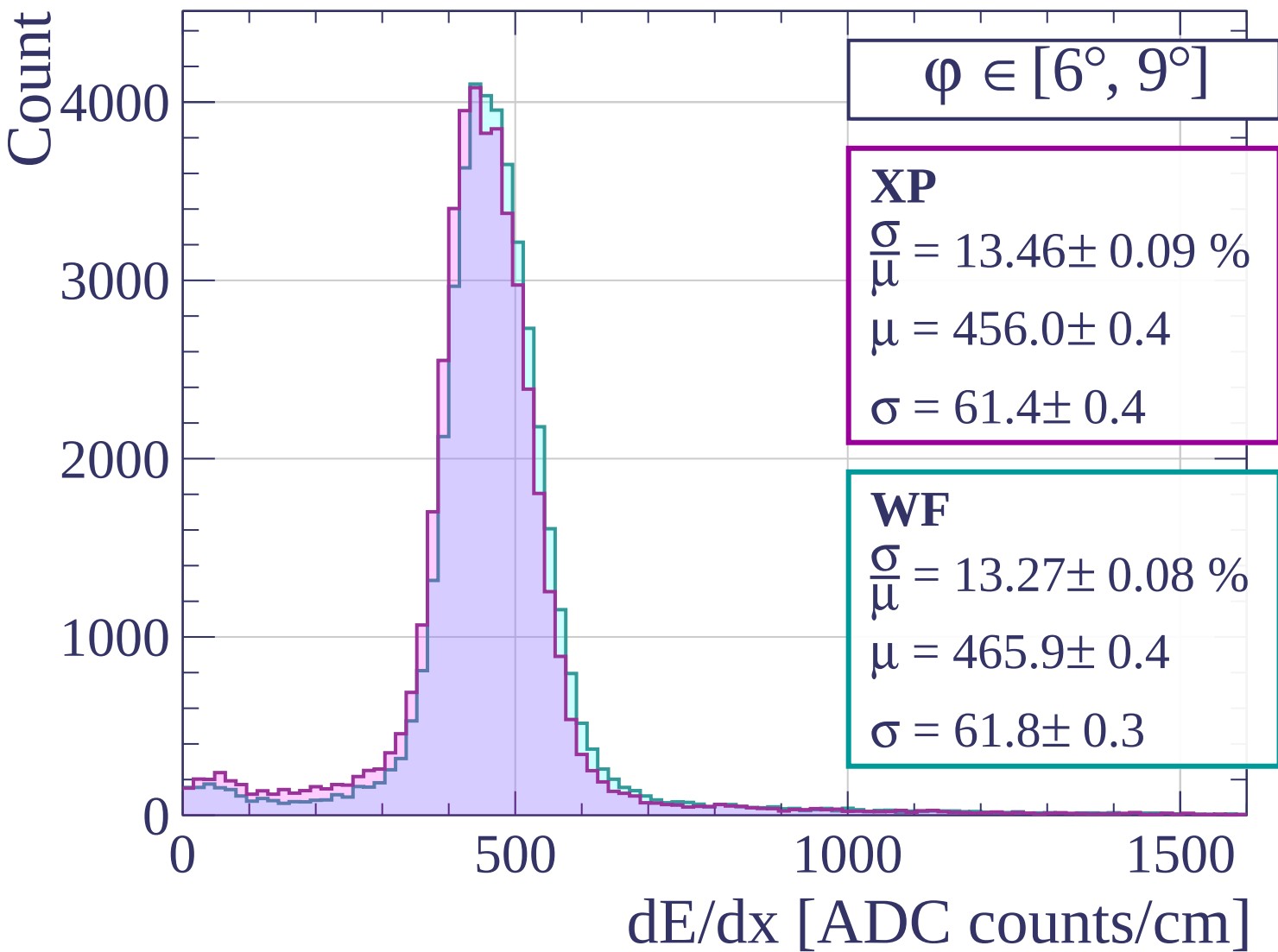


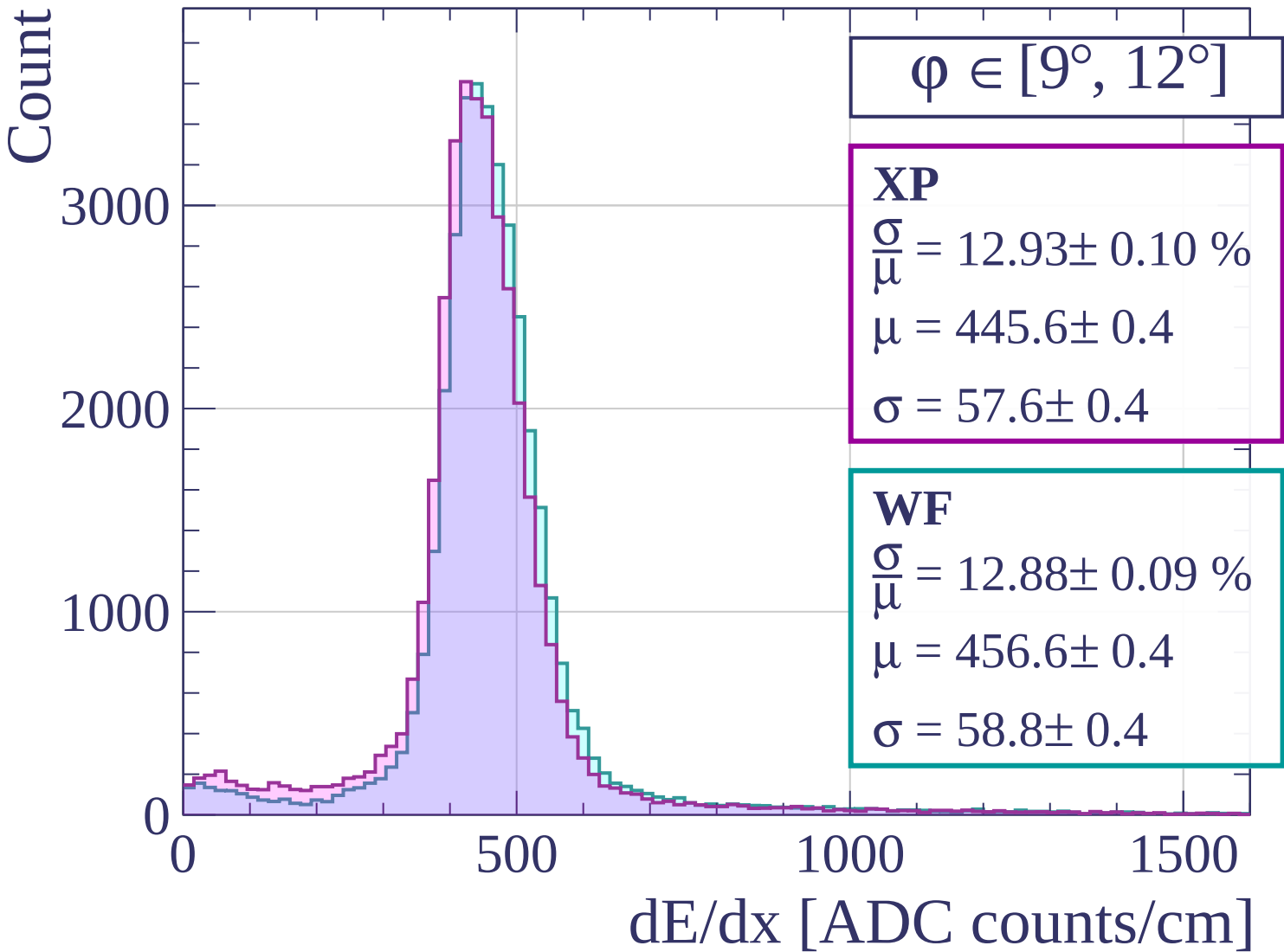
Count

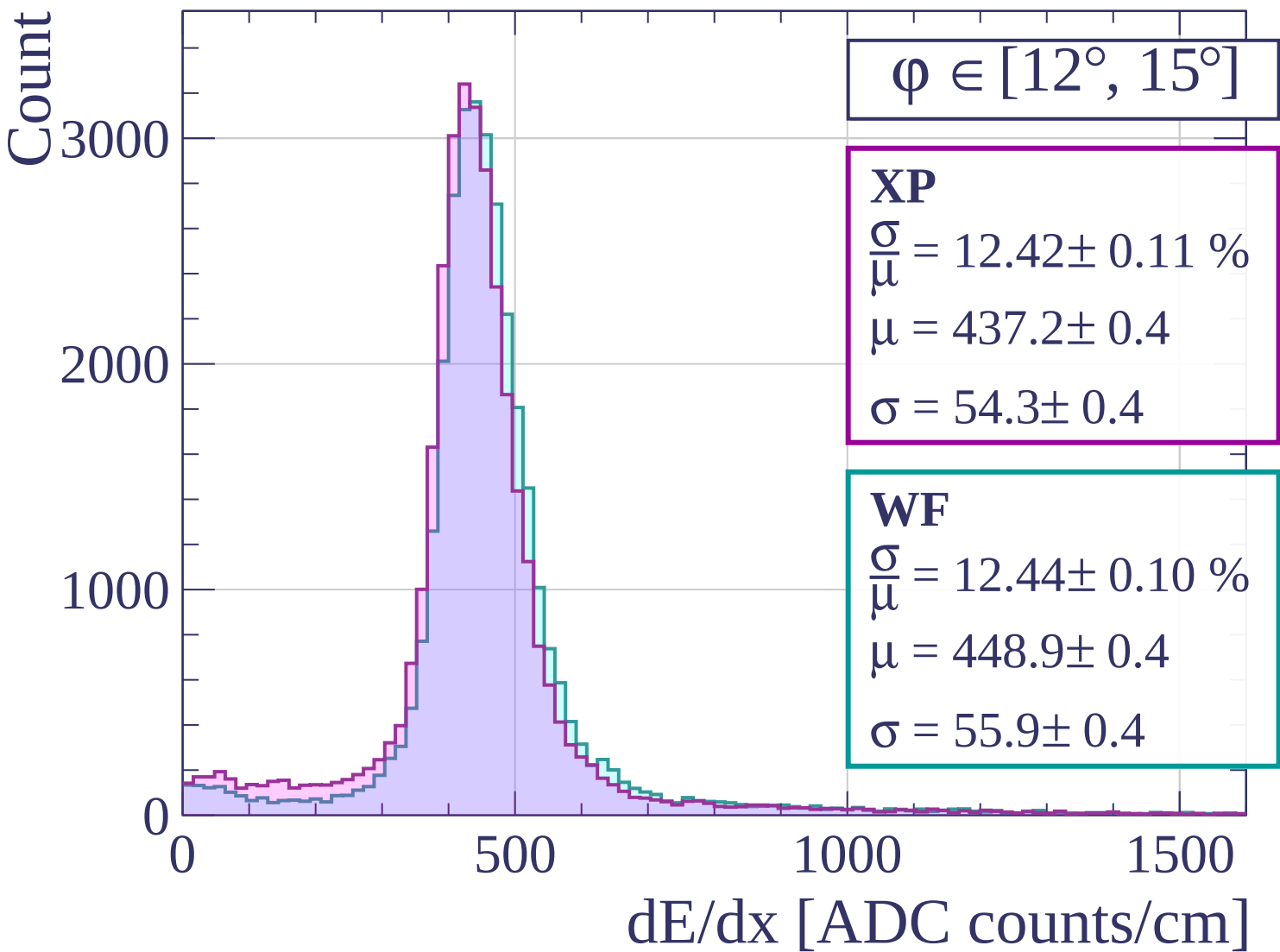


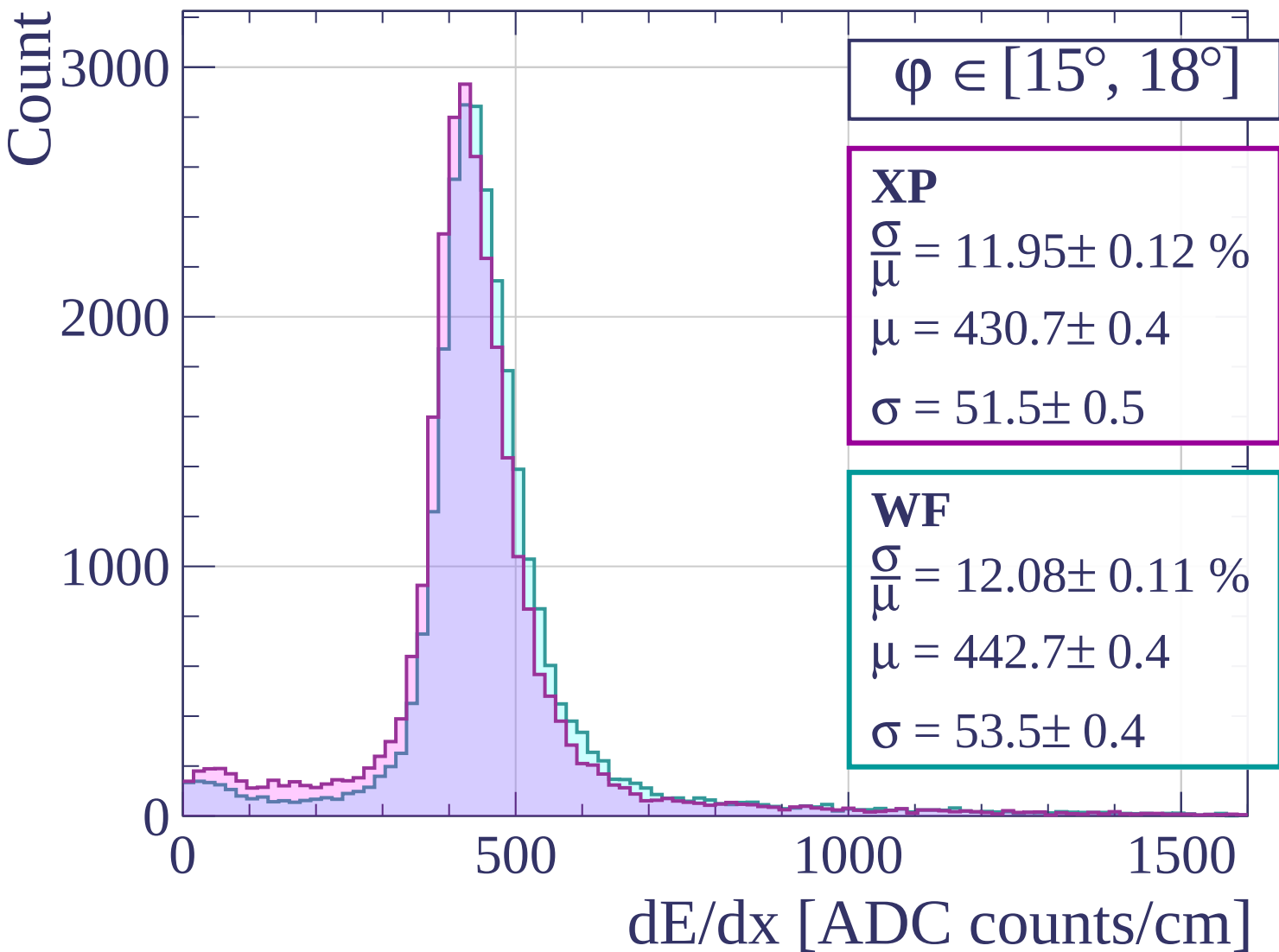


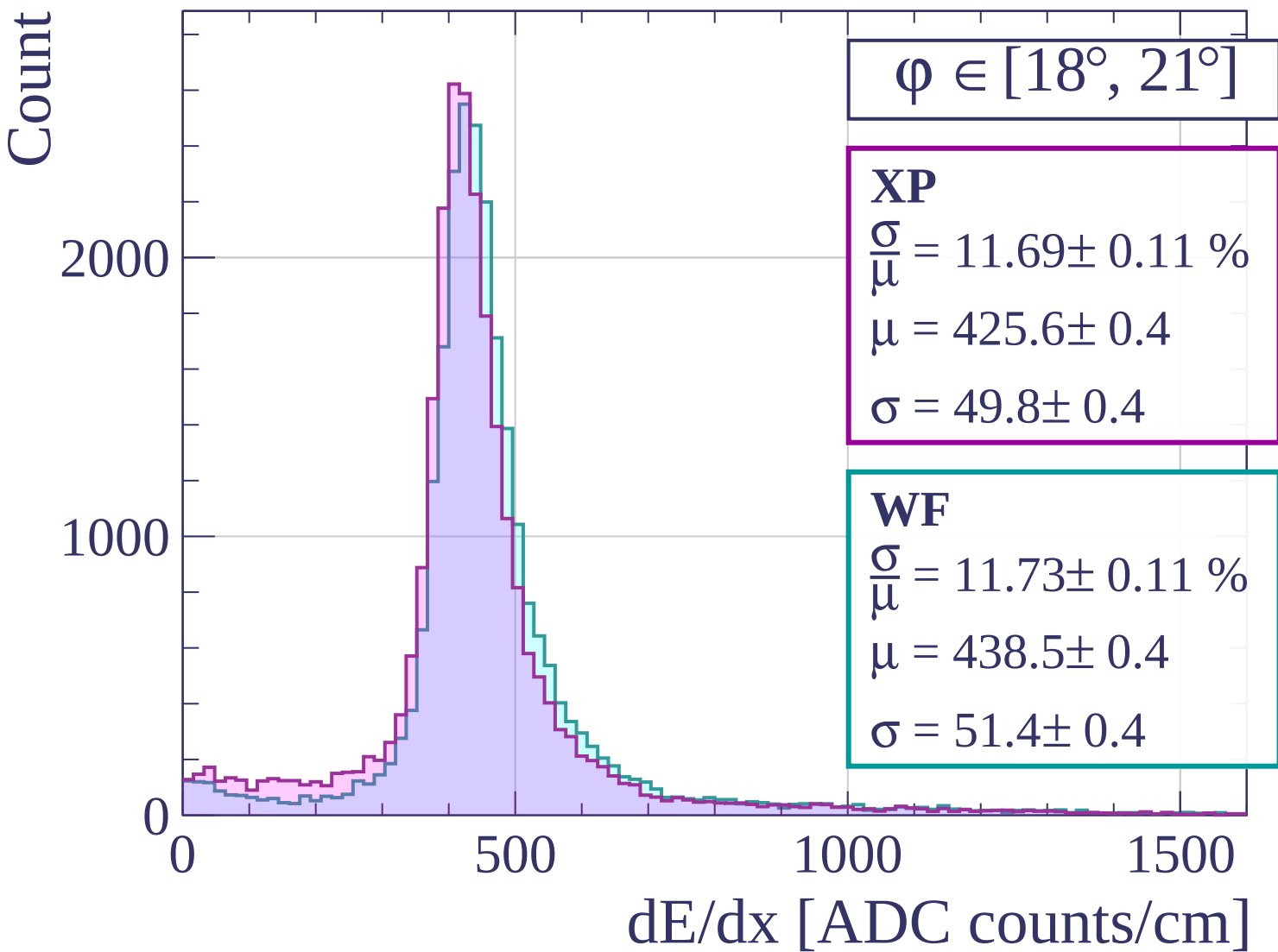


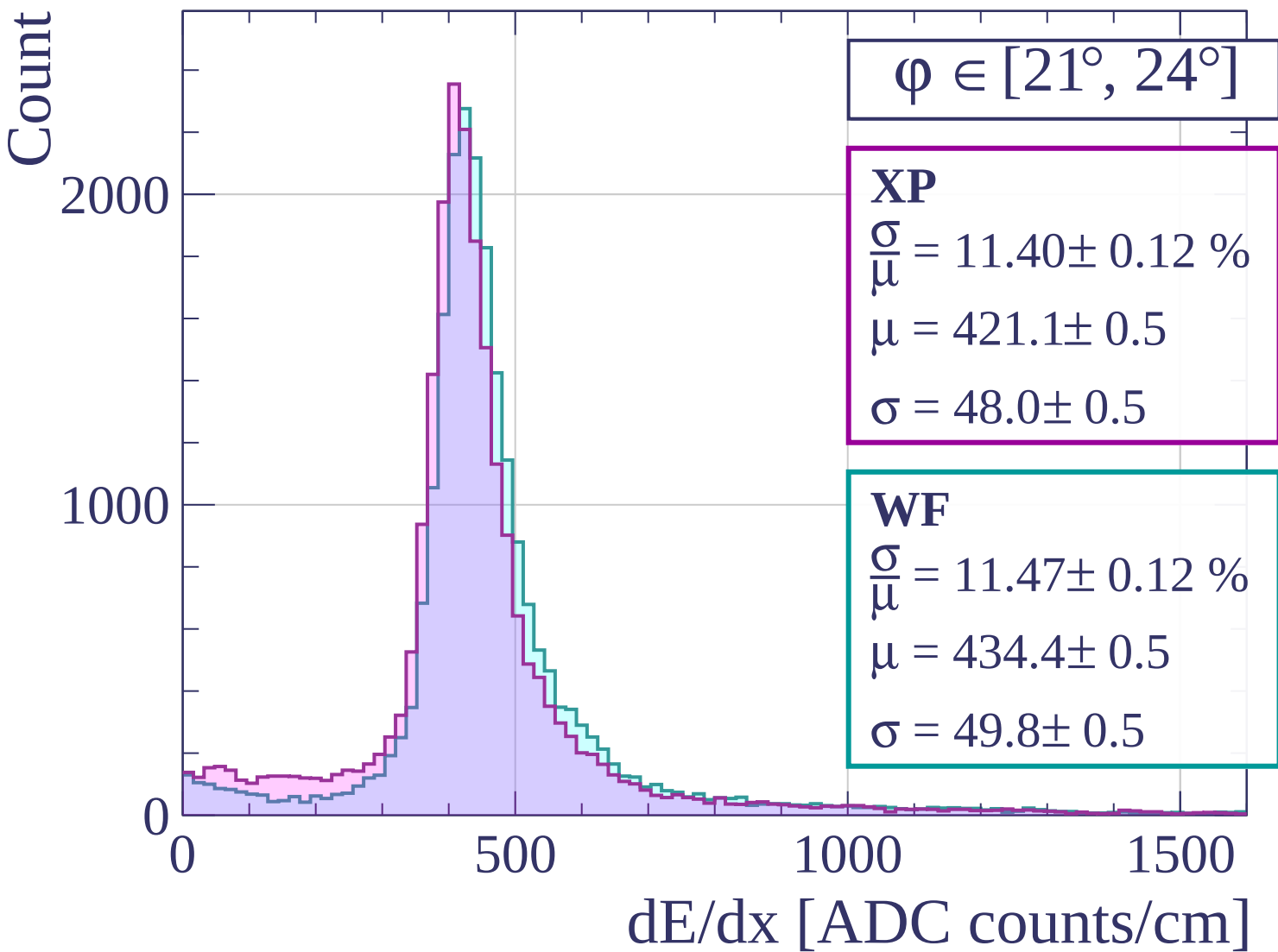


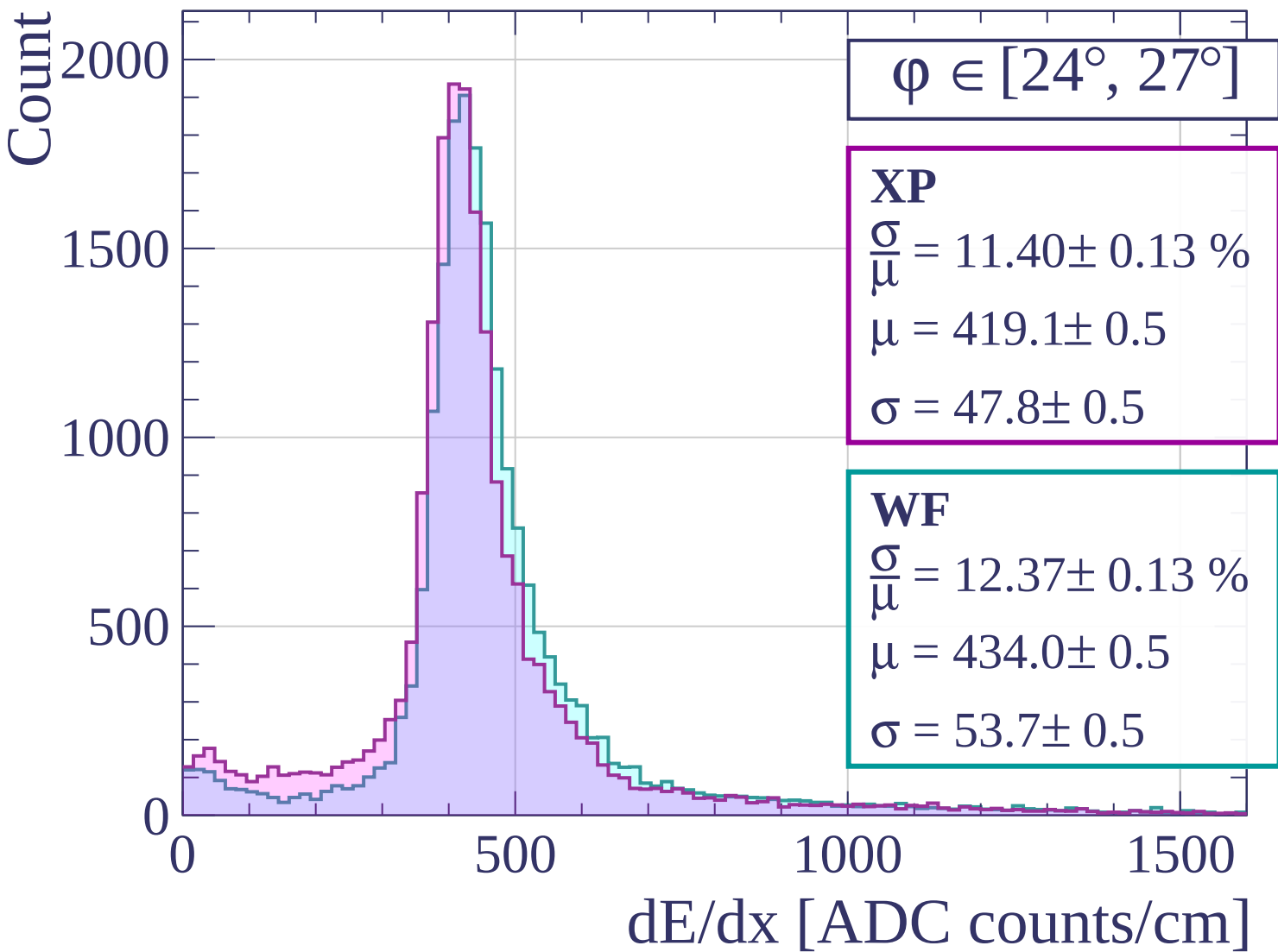


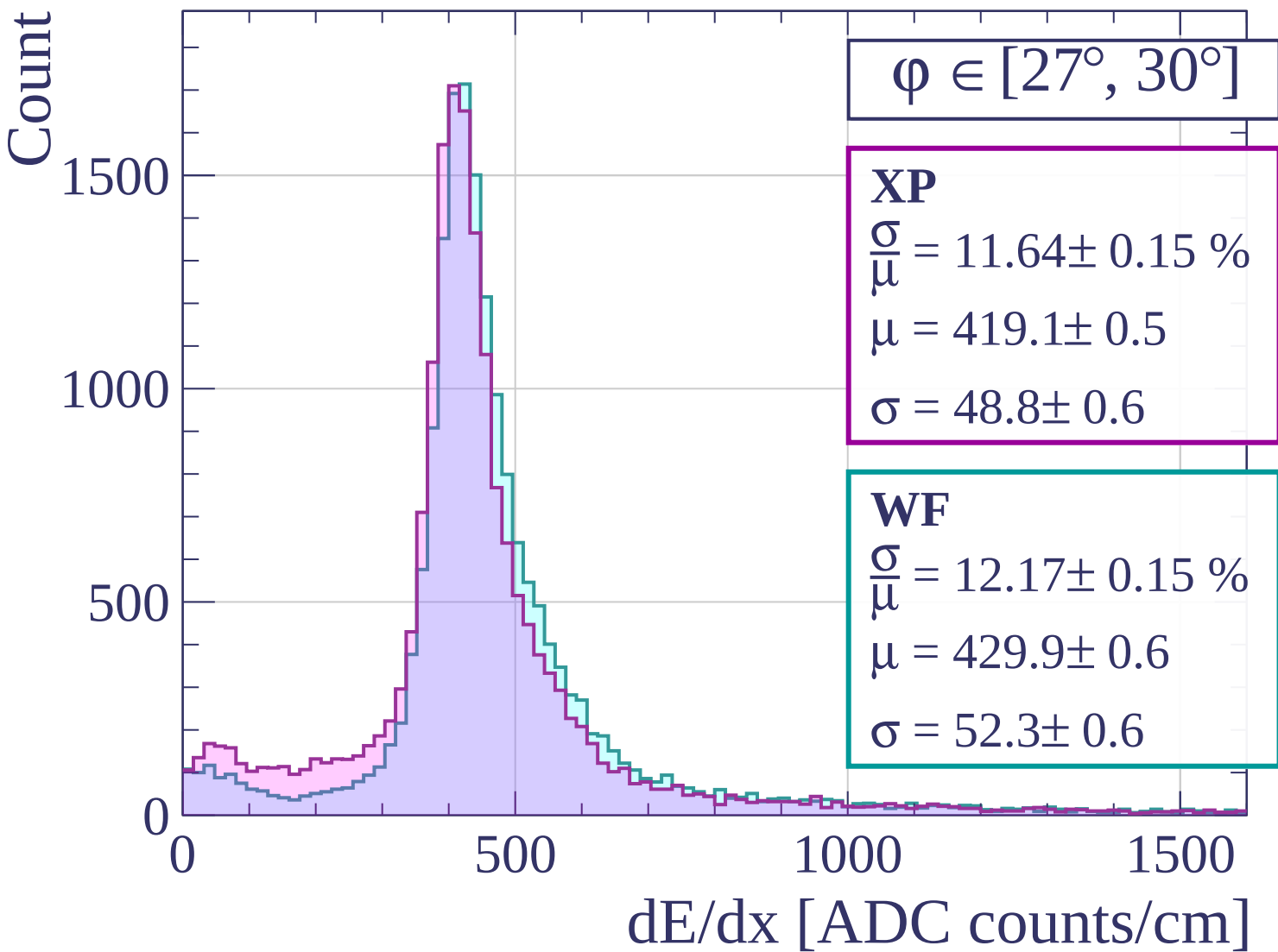


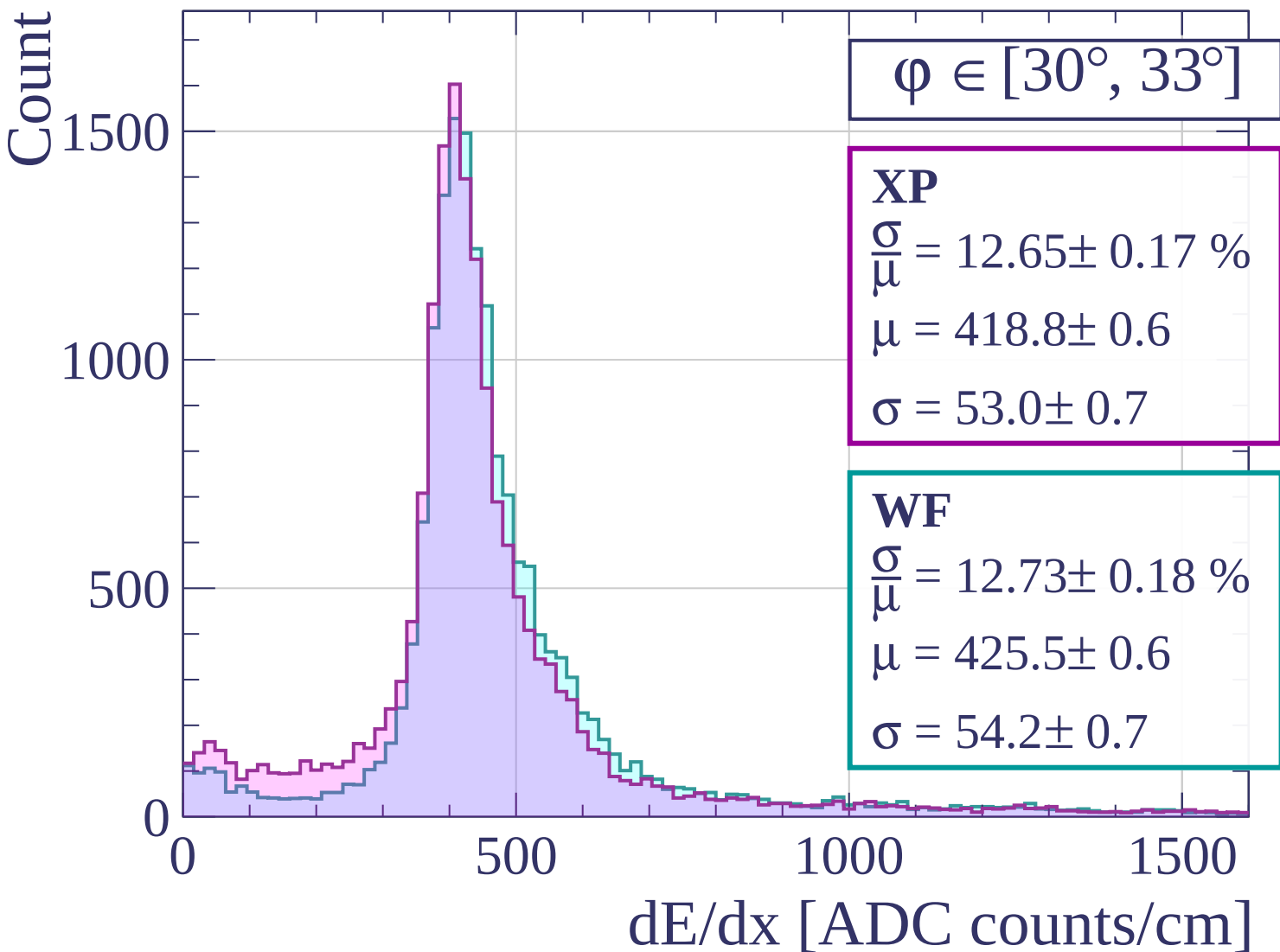




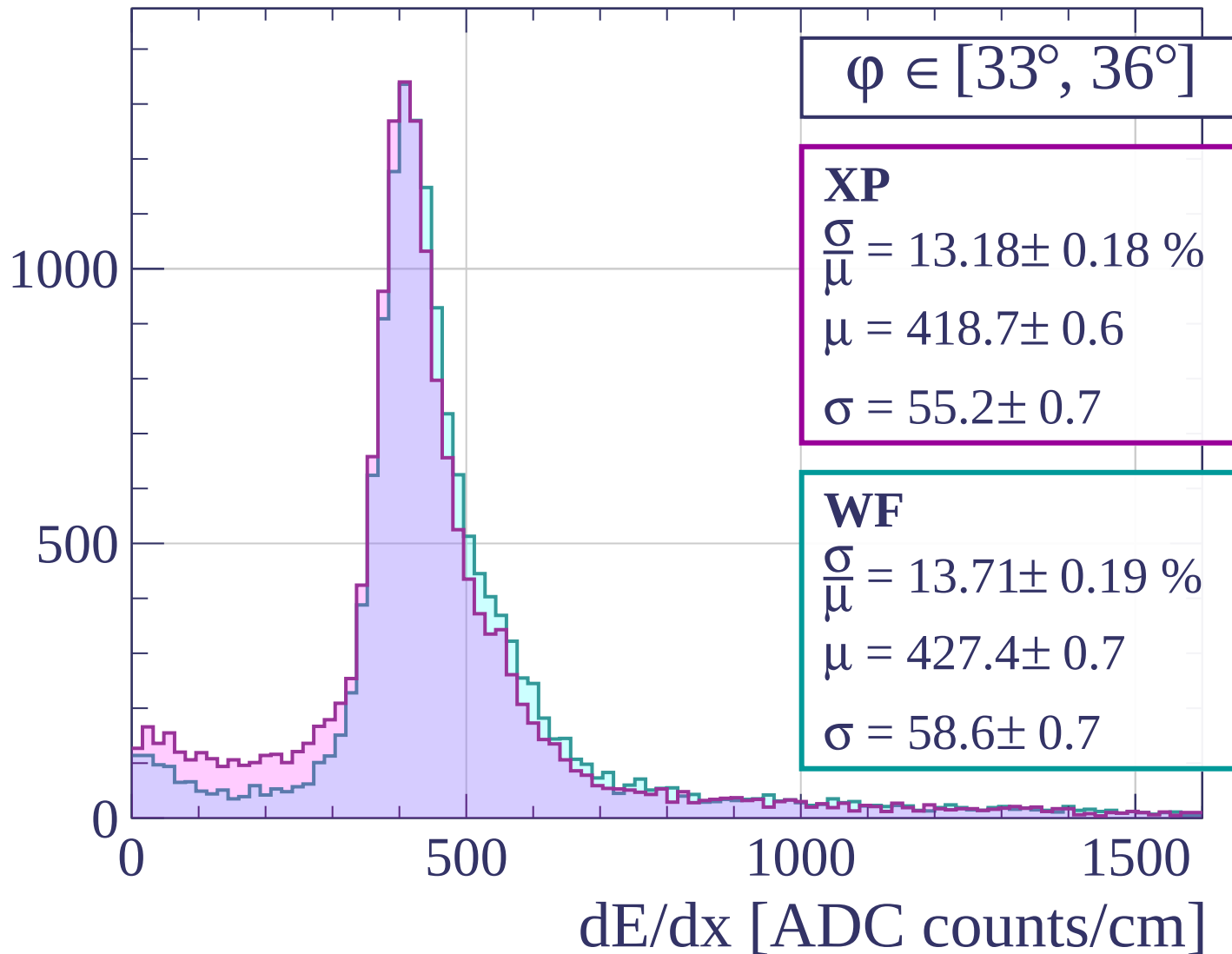


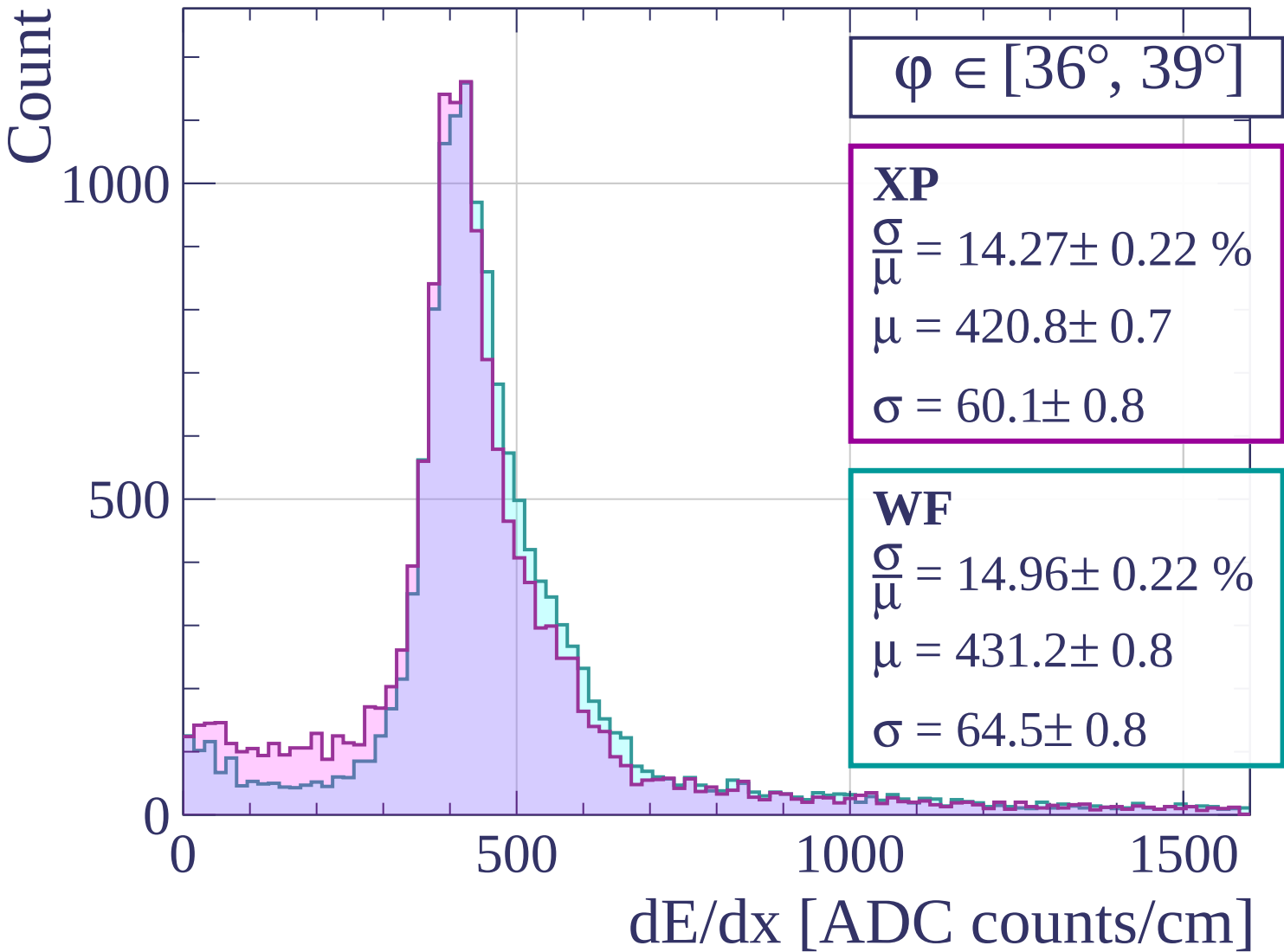


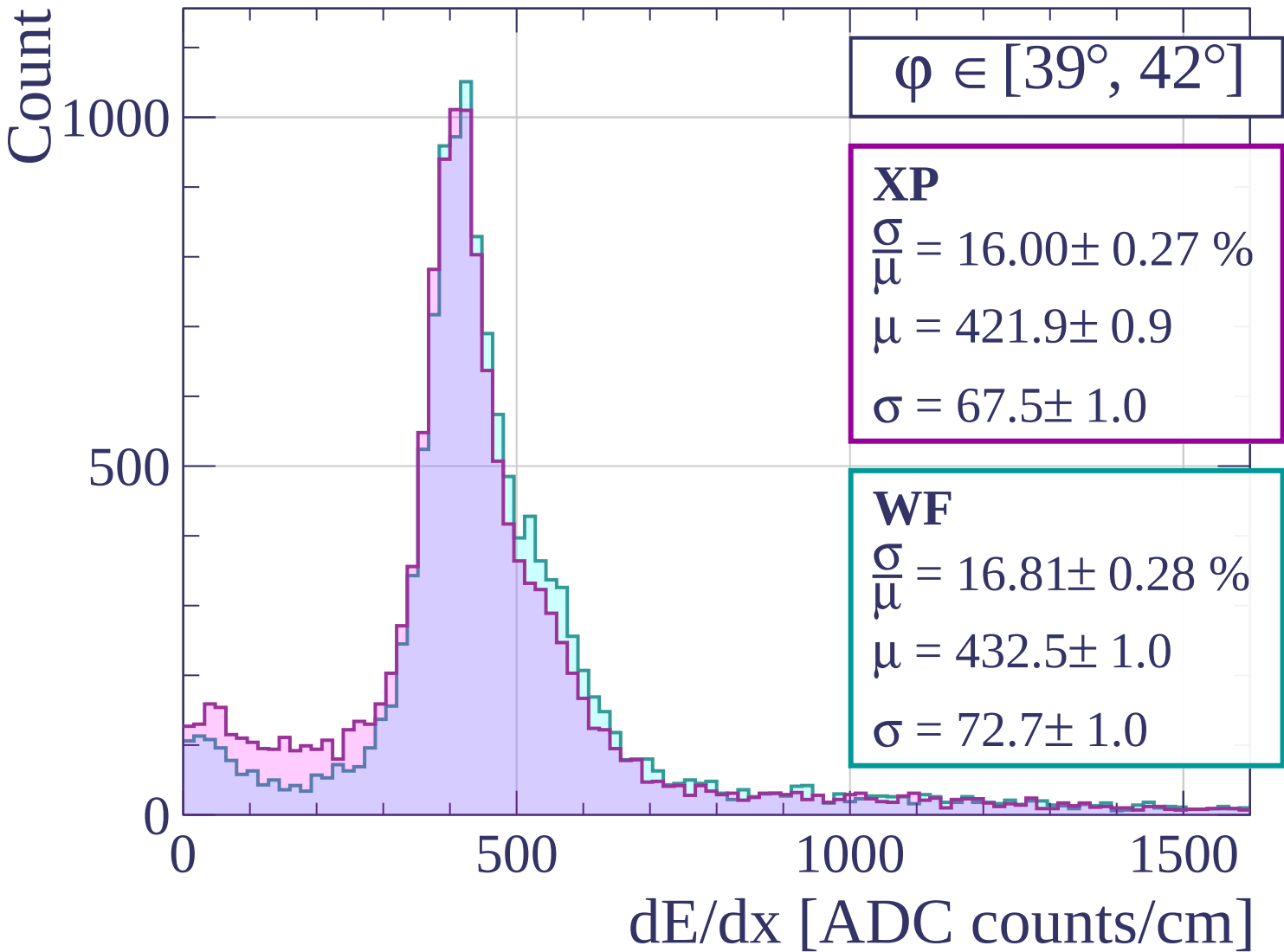


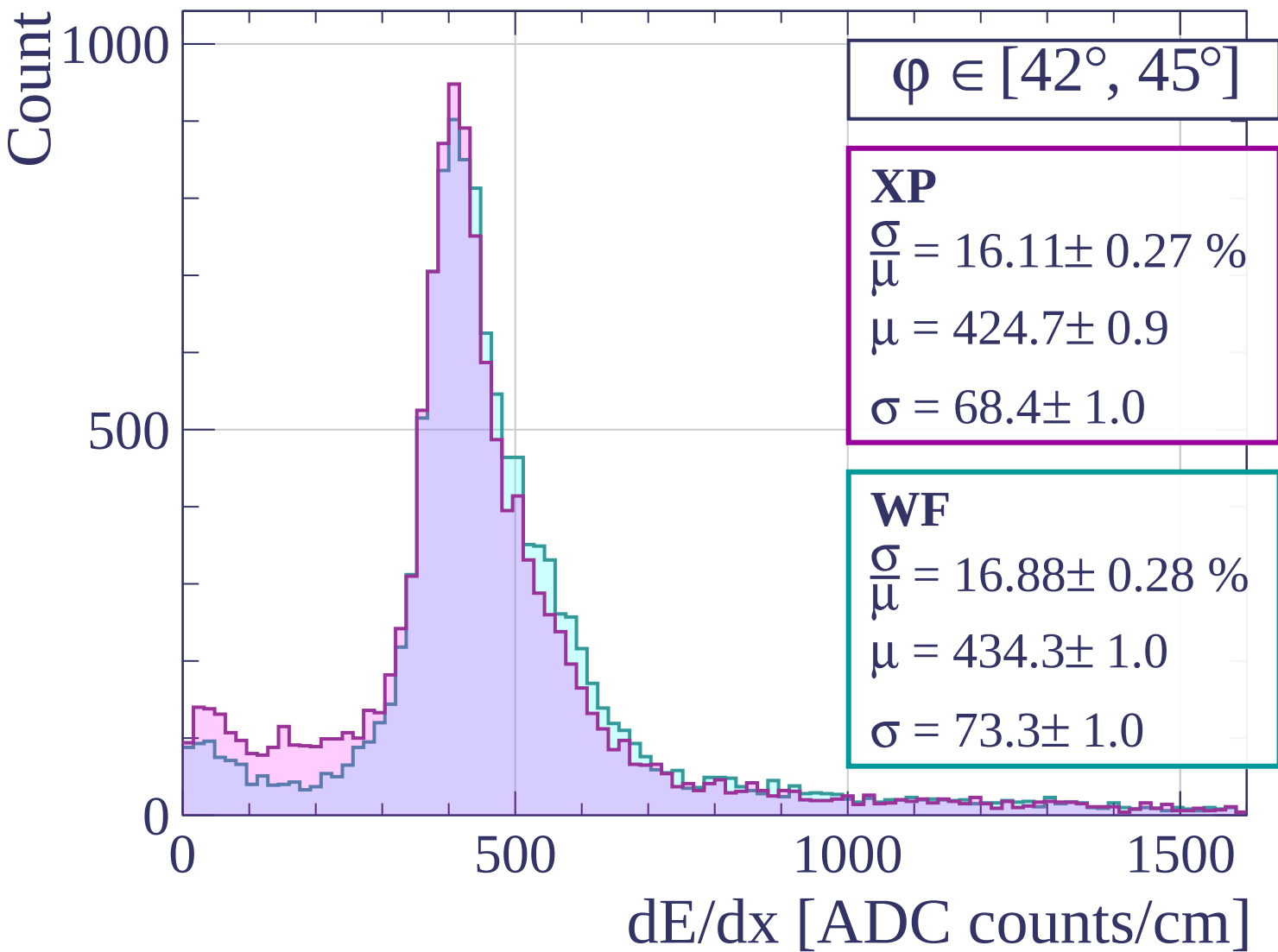


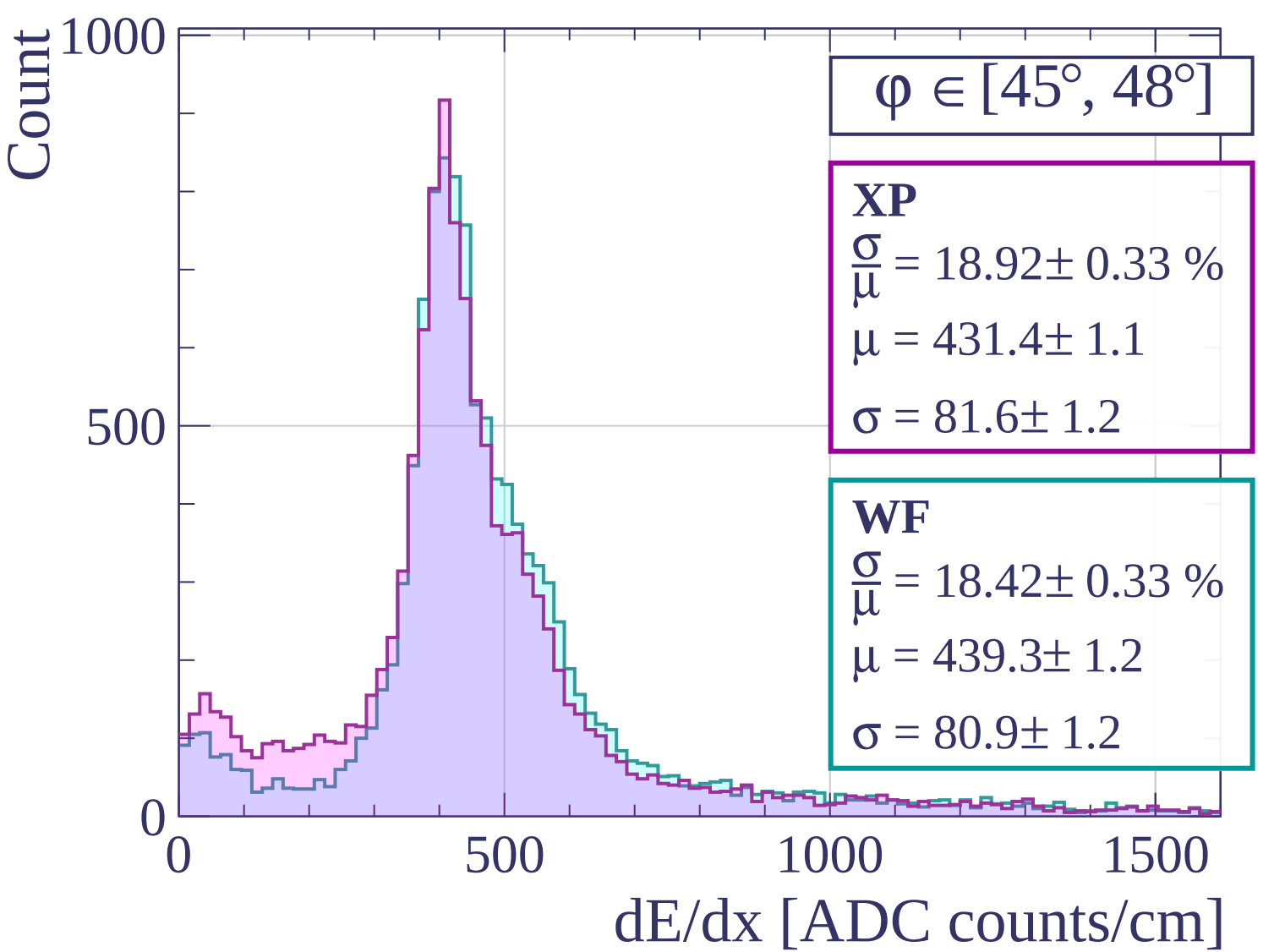
Count











Count

$\varphi \in [48^\circ, 51^\circ]$

XP

$$\frac{\sigma}{\mu} = 18.25 \pm 0.34 \%$$

$$\mu = 427.5 \pm 1.1$$

$$\sigma = 78.0 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 18.77 \pm 0.32 \%$$

$$\mu = 440.1 \pm 1.2$$

$$\sigma = 82.6 \pm 1.2$$

800

600

400

200

0

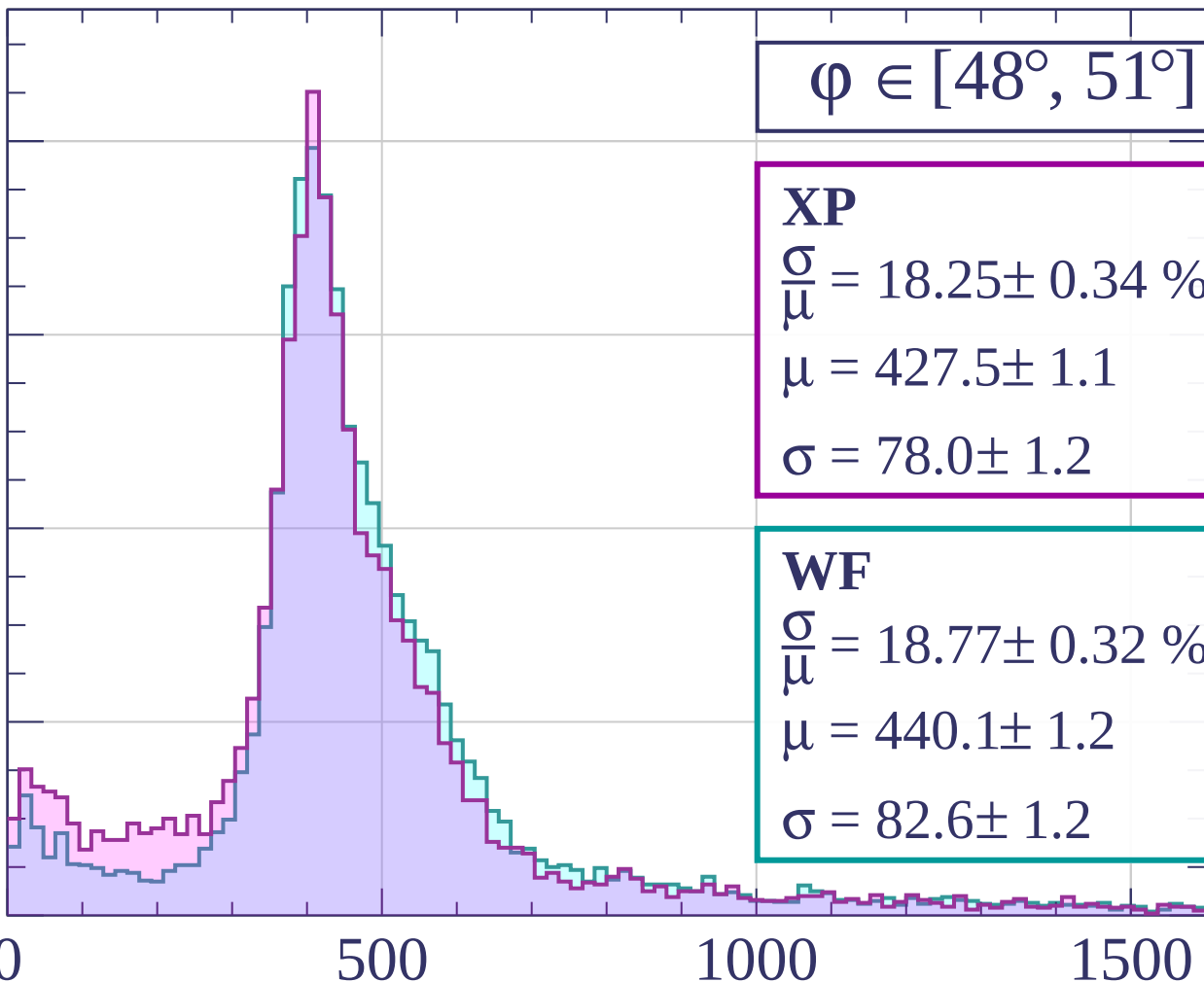
0

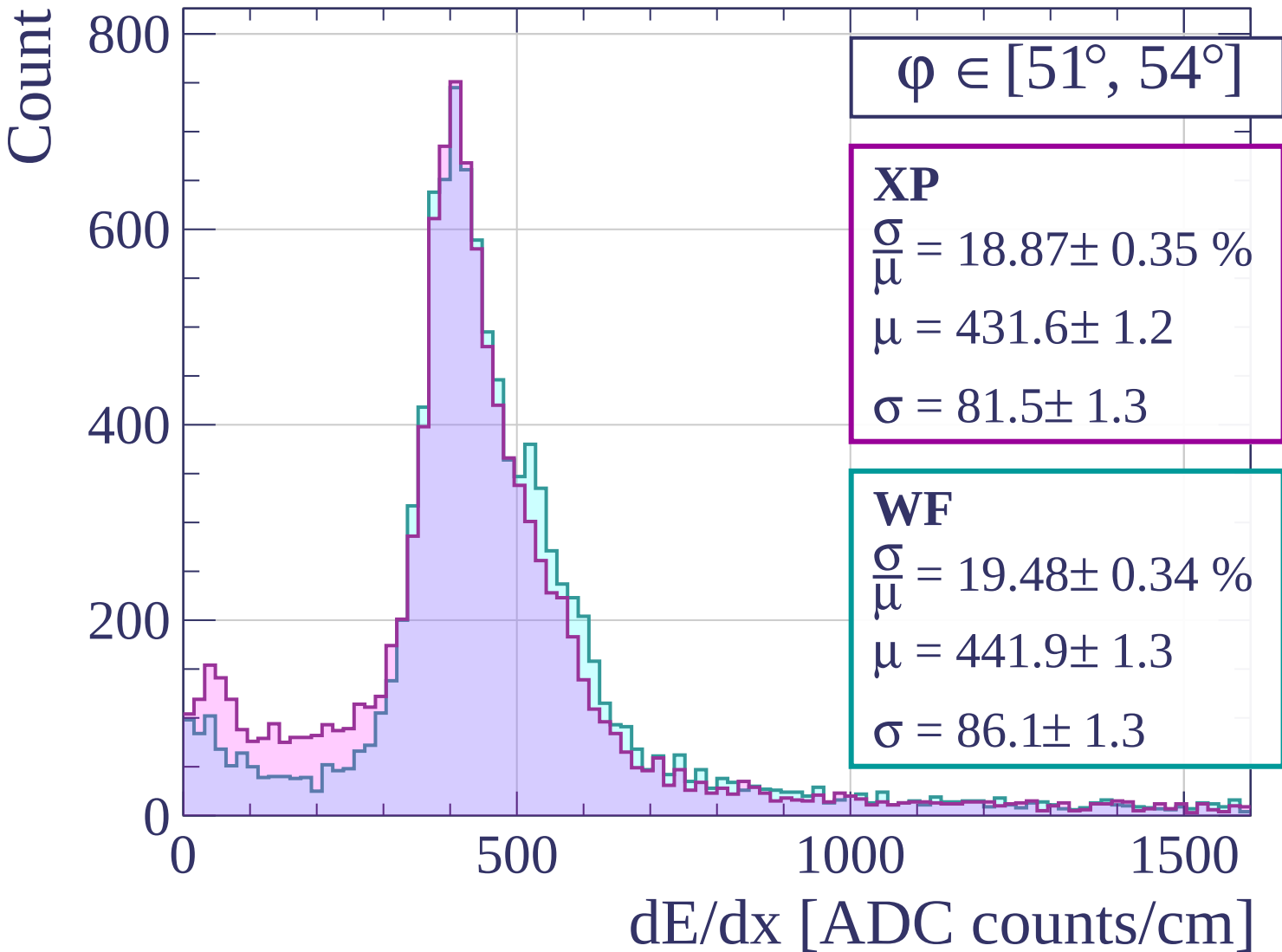
500

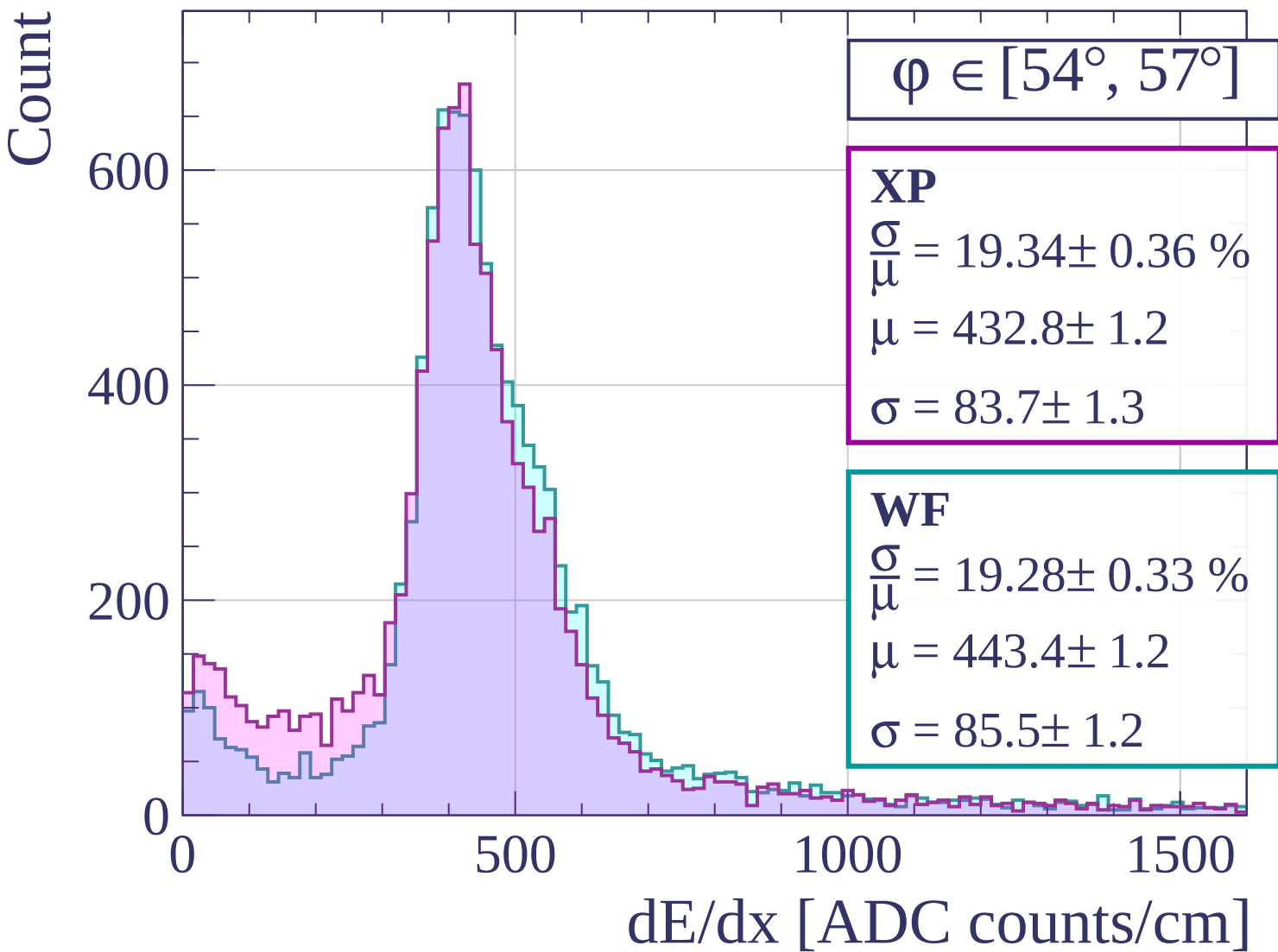
1000

1500

dE/dx [ADC counts/cm]







Count

$\phi \in [57^\circ, 60^\circ]$

XP

$$\frac{\sigma}{\mu} = 20.21 \pm 0.37 \%$$

$$\mu = 440.8 \pm 1.3$$

$$\sigma = 89.1 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 19.58 \pm 0.32 \%$$

$$\mu = 454.5 \pm 1.3$$

$$\sigma = 89.0 \pm 1.2$$

600

400

200

0

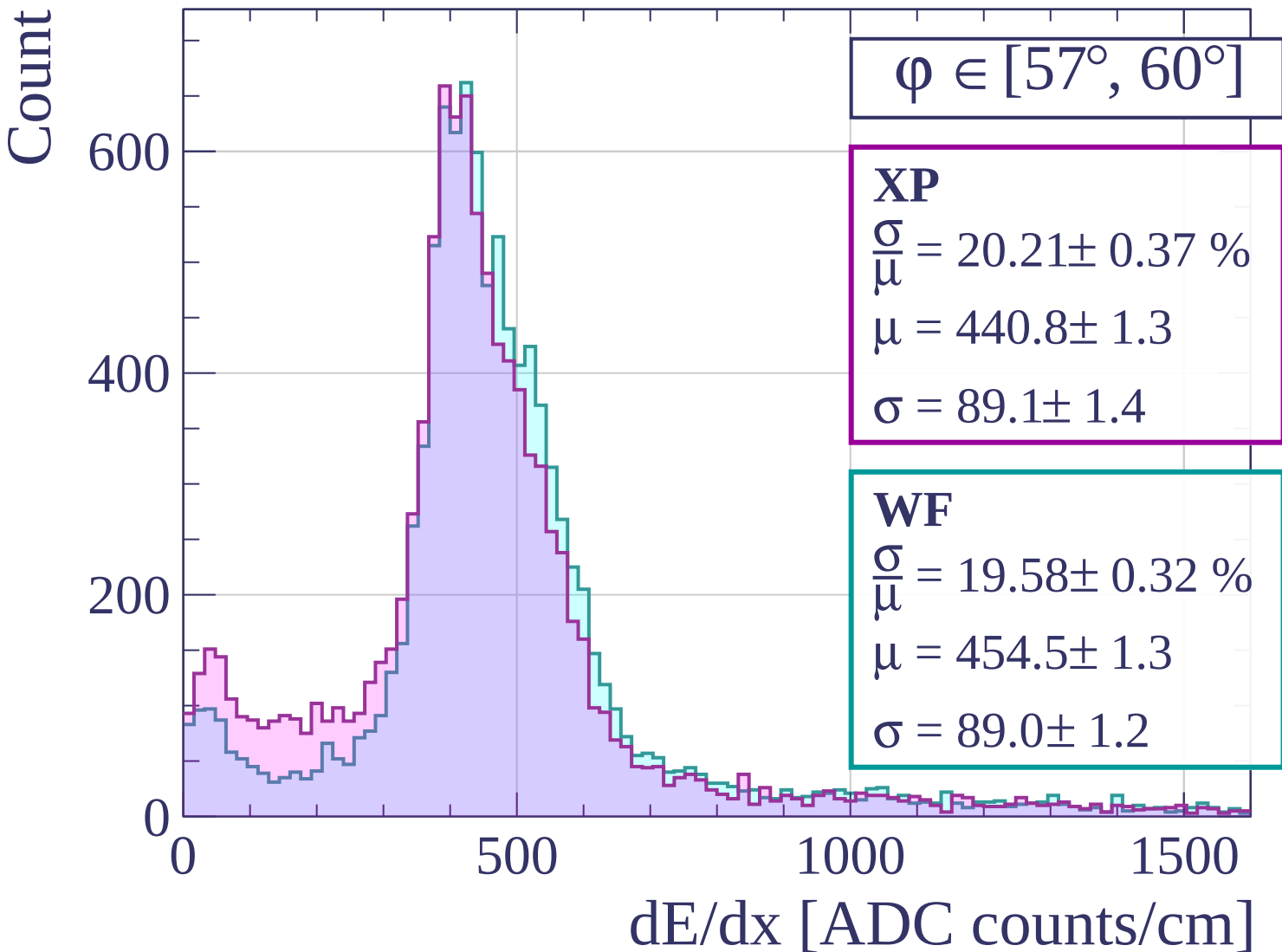
0

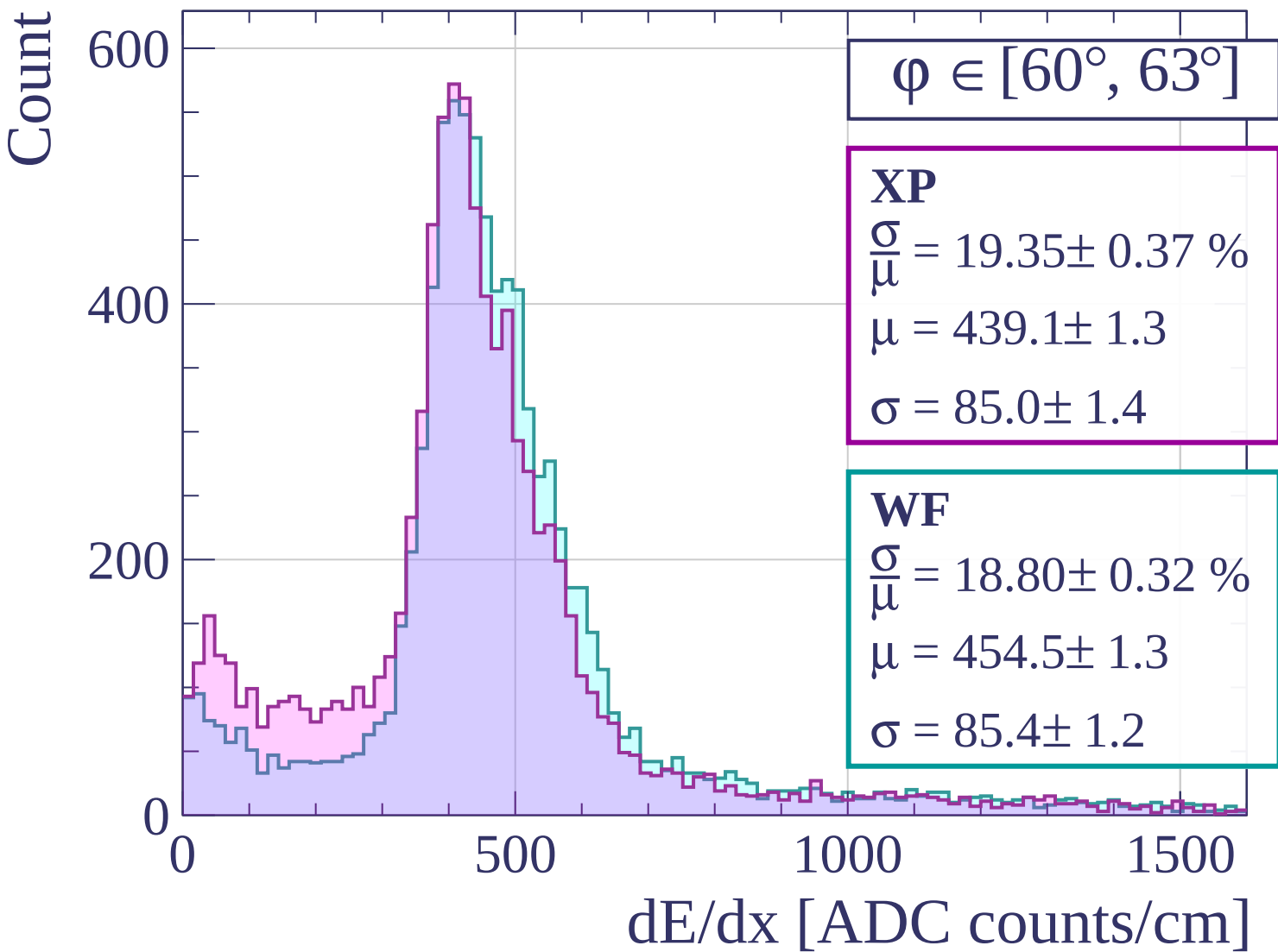
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\varphi \in [63^\circ, 66^\circ]$

XP

$$\frac{\sigma}{\mu} = 20.77 \pm 0.45 \%$$

$$\mu = 435.9 \pm 1.5$$

$$\sigma = 90.5 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 19.57 \pm 0.38 \%$$

$$\mu = 453.7 \pm 1.5$$

$$\sigma = 88.8 \pm 1.4$$

400

200

0

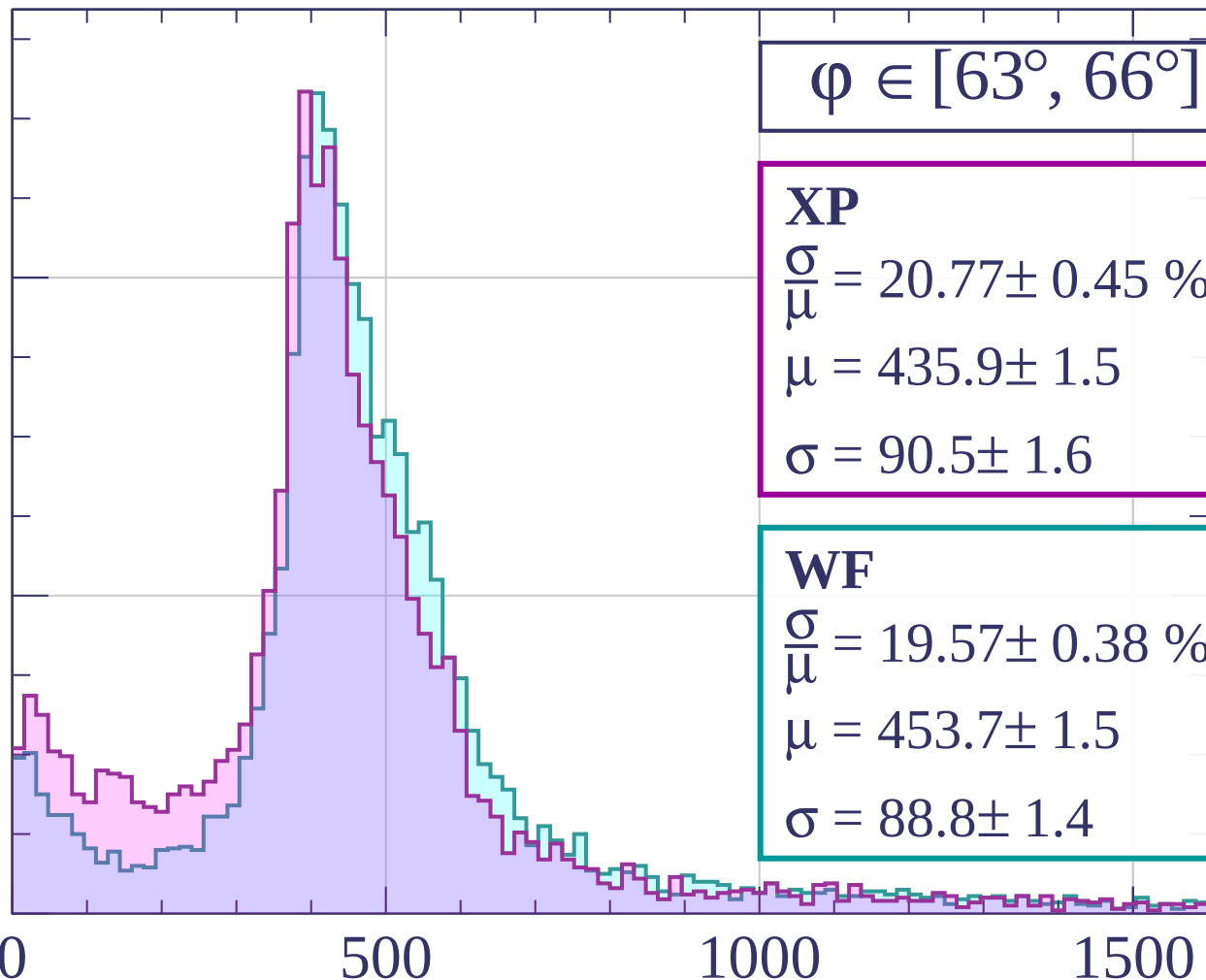
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [66^\circ, 69^\circ]$

XP

$$\frac{\sigma}{\mu} = 19.63 \pm 0.44 \%$$

$$\mu = 437.9 \pm 1.5$$

$$\sigma = 85.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 19.08 \pm 0.38 \%$$

$$\mu = 454.1 \pm 1.5$$

$$\sigma = 86.7 \pm 1.4$$

400

200

0

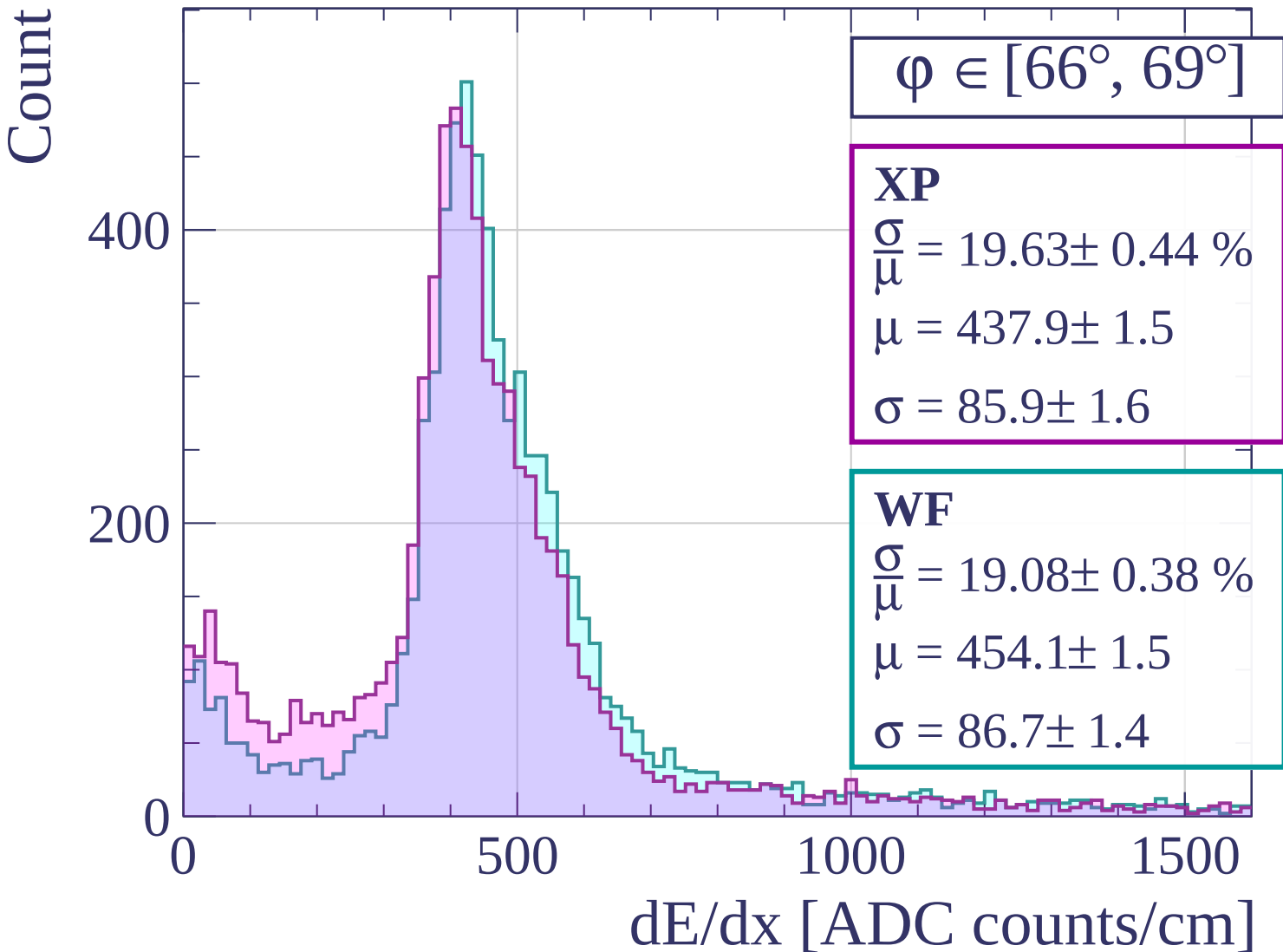
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [69^\circ, 72^\circ]$

XP

$$\frac{\sigma}{\mu} = 20.20 \pm 0.46 \%$$

$$\mu = 436.7 \pm 1.5$$

$$\sigma = 88.2 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 18.91 \pm 0.40 \%$$

$$\mu = 455.8 \pm 1.5$$

$$\sigma = 86.2 \pm 1.5$$

400

300

200

100

0

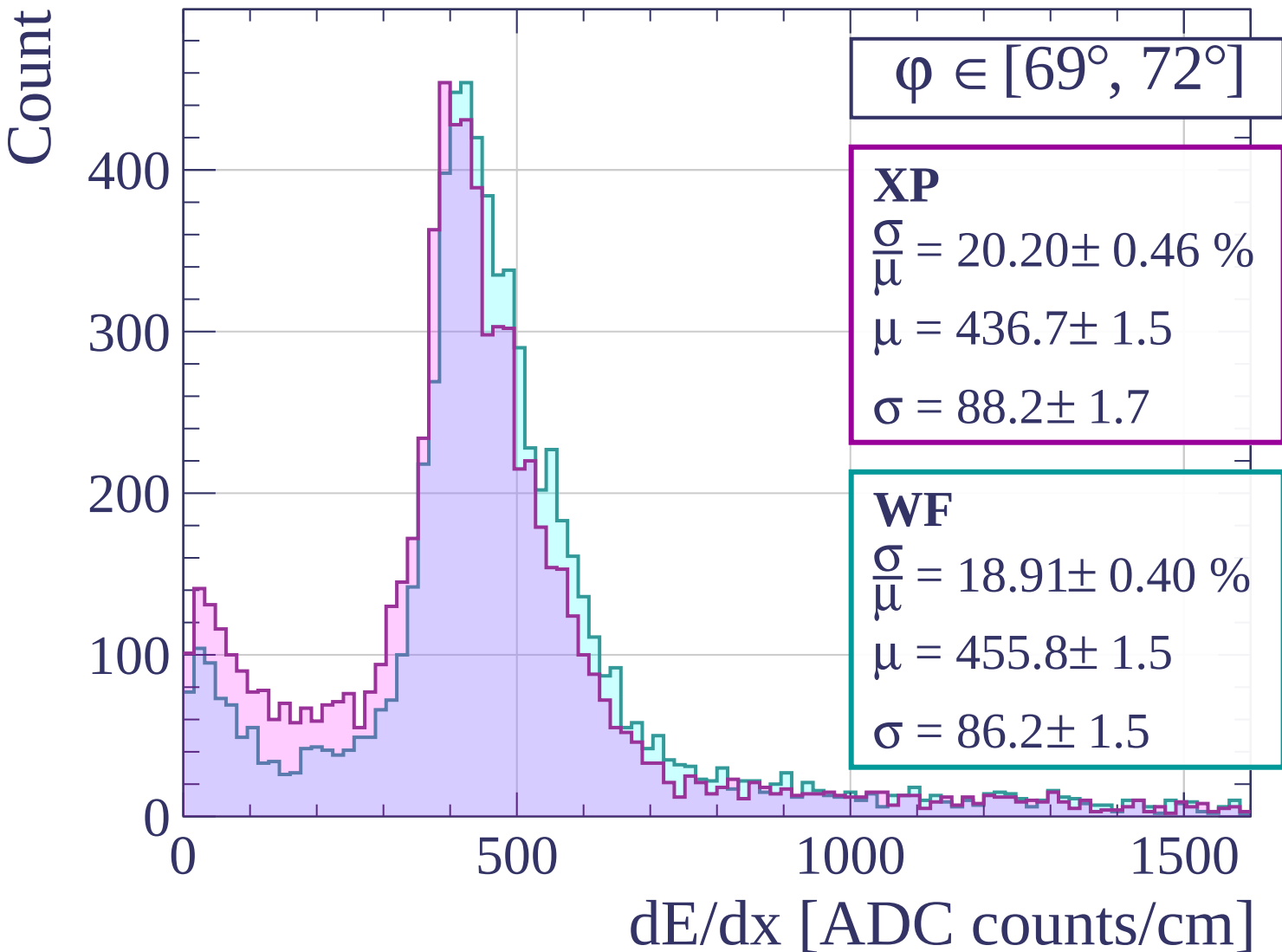
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [72^\circ, 75^\circ]$

XP

$$\frac{\sigma}{\mu} = 19.82 \pm 0.46 \%$$

$$\mu = 438.5 \pm 1.6$$

$$\sigma = 86.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 19.05 \pm 0.41 \%$$

$$\mu = 456.0 \pm 1.5$$

$$\sigma = 86.9 \pm 1.6$$

400

300

200

100

0

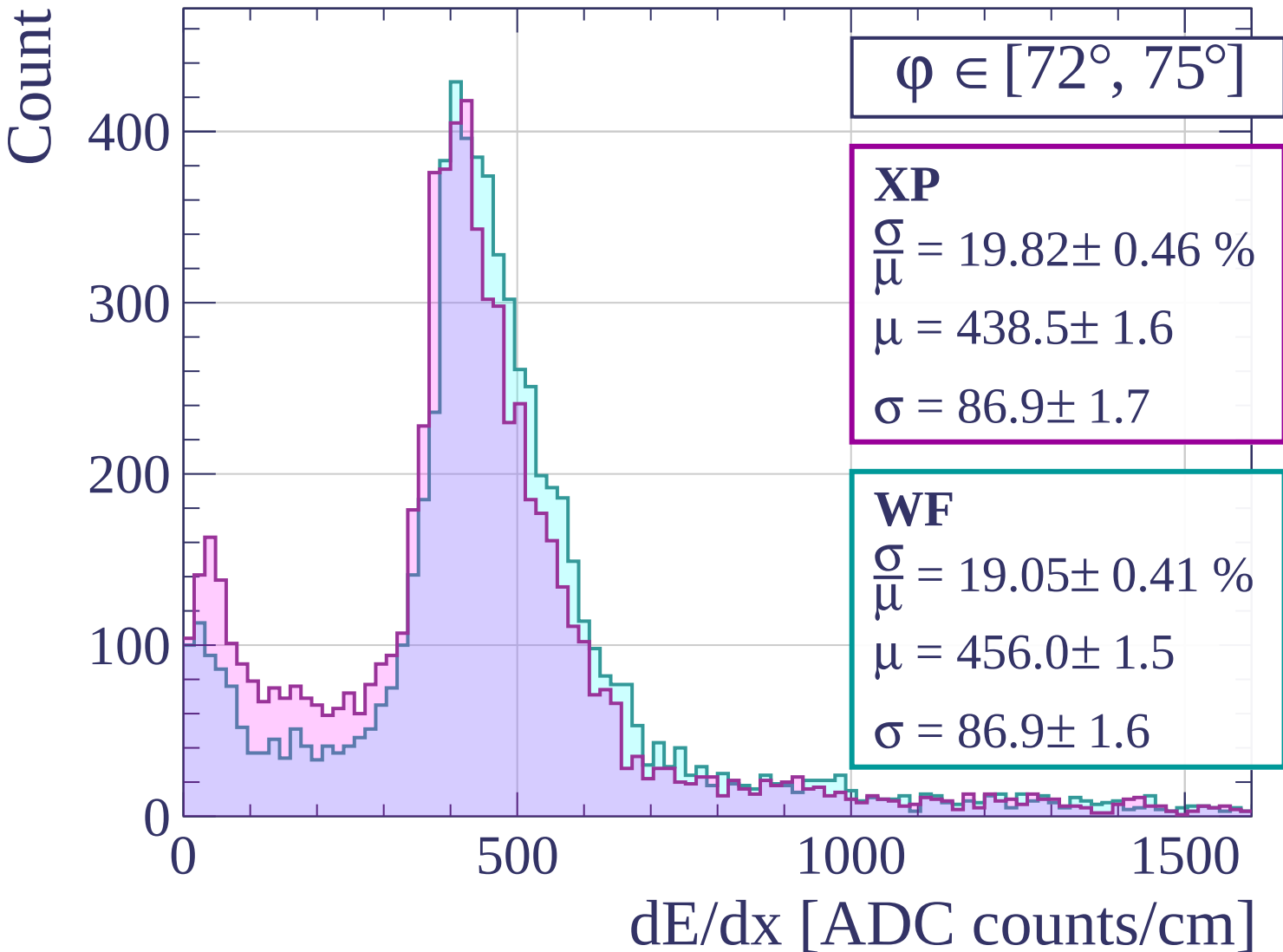
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\varphi \in [75^\circ, 78^\circ]$

XP

$$\frac{\sigma}{\mu} = 19.60 \pm 0.45 \%$$

$$\mu = 440.1 \pm 1.6$$

$$\sigma = 86.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 19.03 \pm 0.42 \%$$

$$\mu = 456.0 \pm 1.6$$

$$\sigma = 86.8 \pm 1.6$$

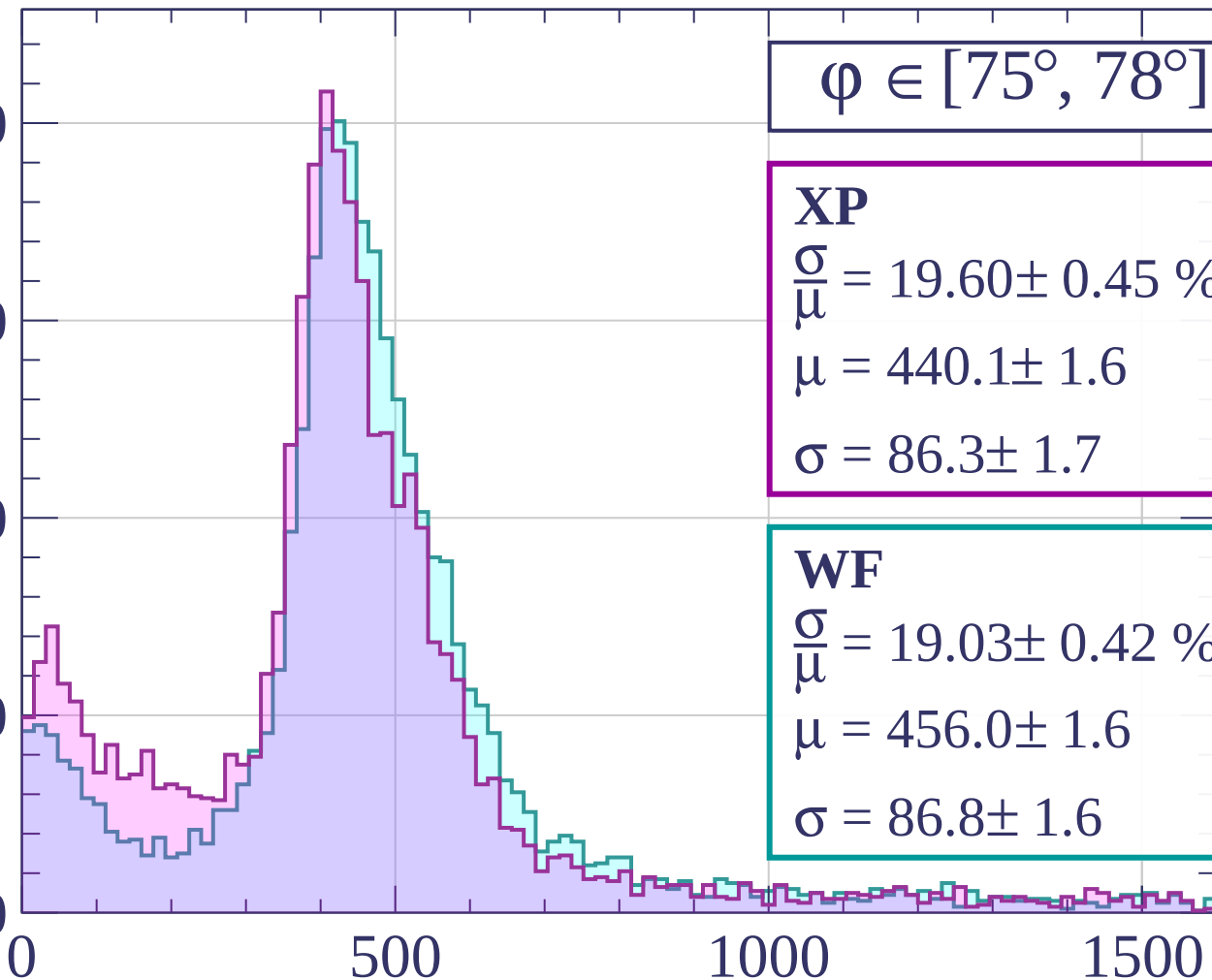
400

300

200

100

0



0

500

1000

1500

dE/dx [ADC counts/cm]

Count

$\varphi \in [78^\circ, 81^\circ]$

XP

$$\frac{\sigma}{\mu} = 18.77 \pm 0.42 \%$$

$$\mu = 436.5 \pm 1.5$$

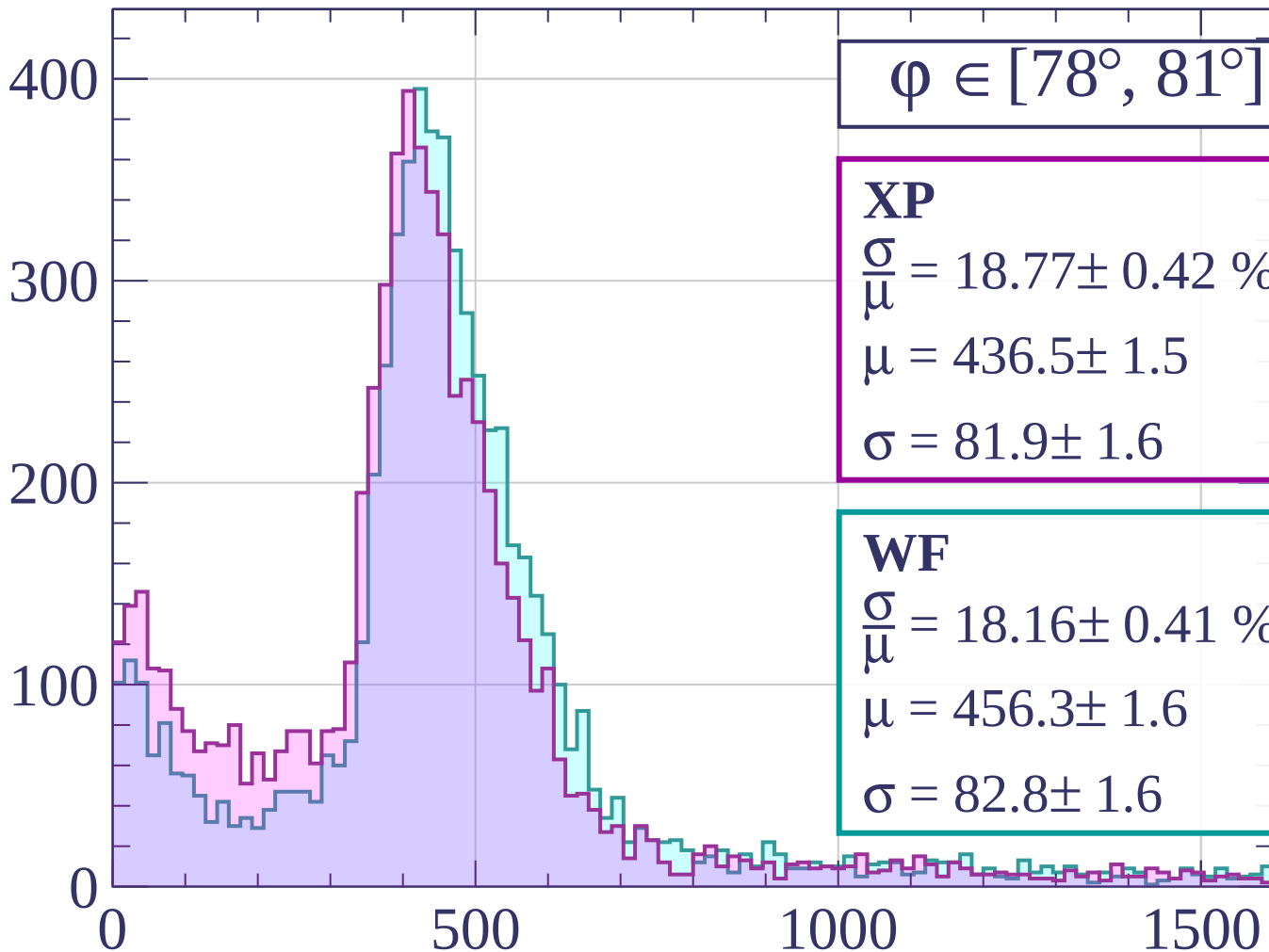
$$\sigma = 81.9 \pm 1.6$$

WF

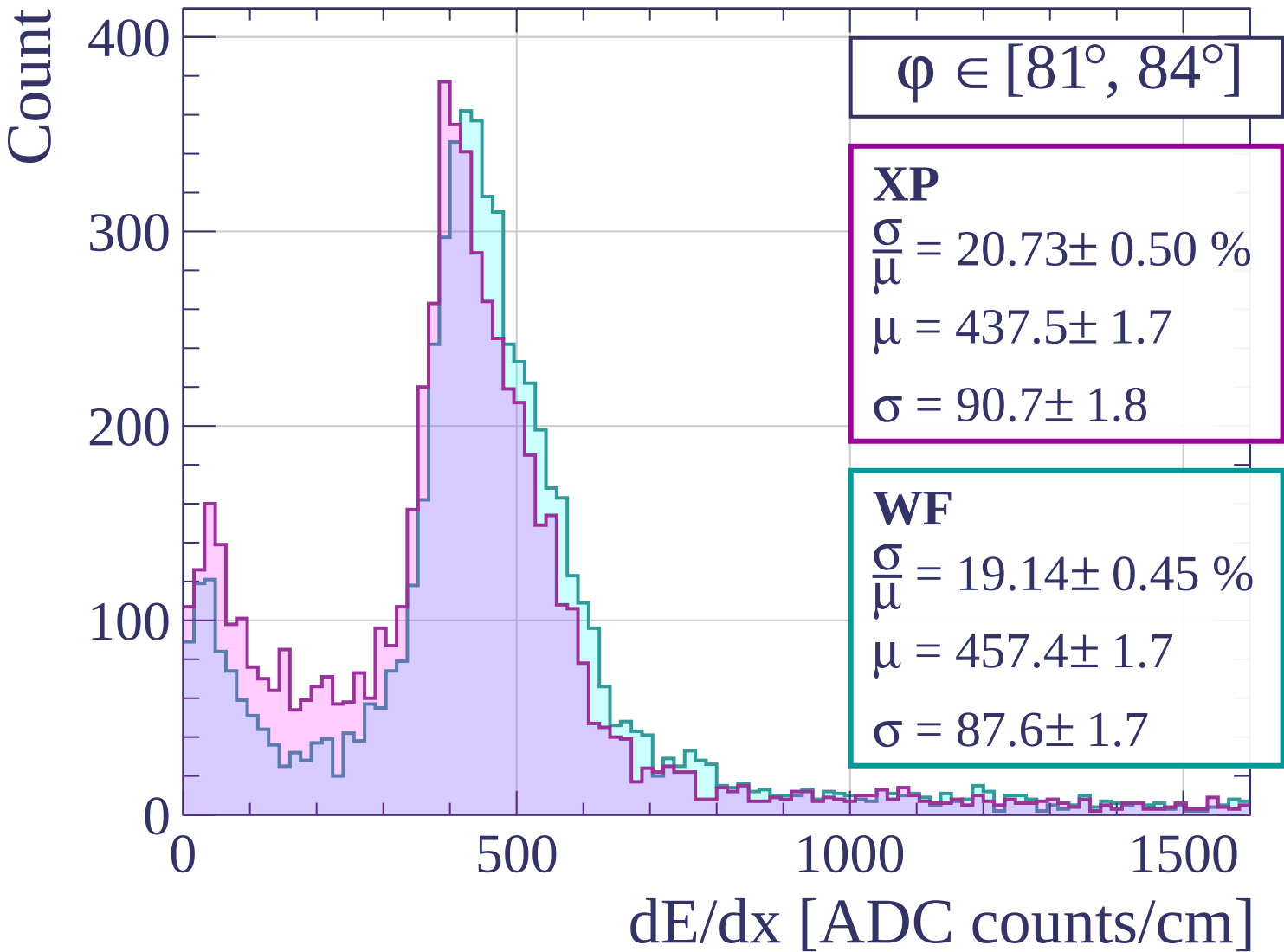
$$\frac{\sigma}{\mu} = 18.16 \pm 0.41 \%$$

$$\mu = 456.3 \pm 1.6$$

$$\sigma = 82.8 \pm 1.6$$



dE/dx [ADC counts/cm]



Count

$\phi \in [84^\circ, 87^\circ]$

XP

$$\frac{\sigma}{\mu} = 22.48 \pm 0.57 \%$$

$$\mu = 434.2 \pm 1.8$$

$$\sigma = 97.6 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 19.69 \pm 0.49 \%$$

$$\mu = 459.1 \pm 1.8$$

$$\sigma = 90.4 \pm 1.9$$

300

200

100

0

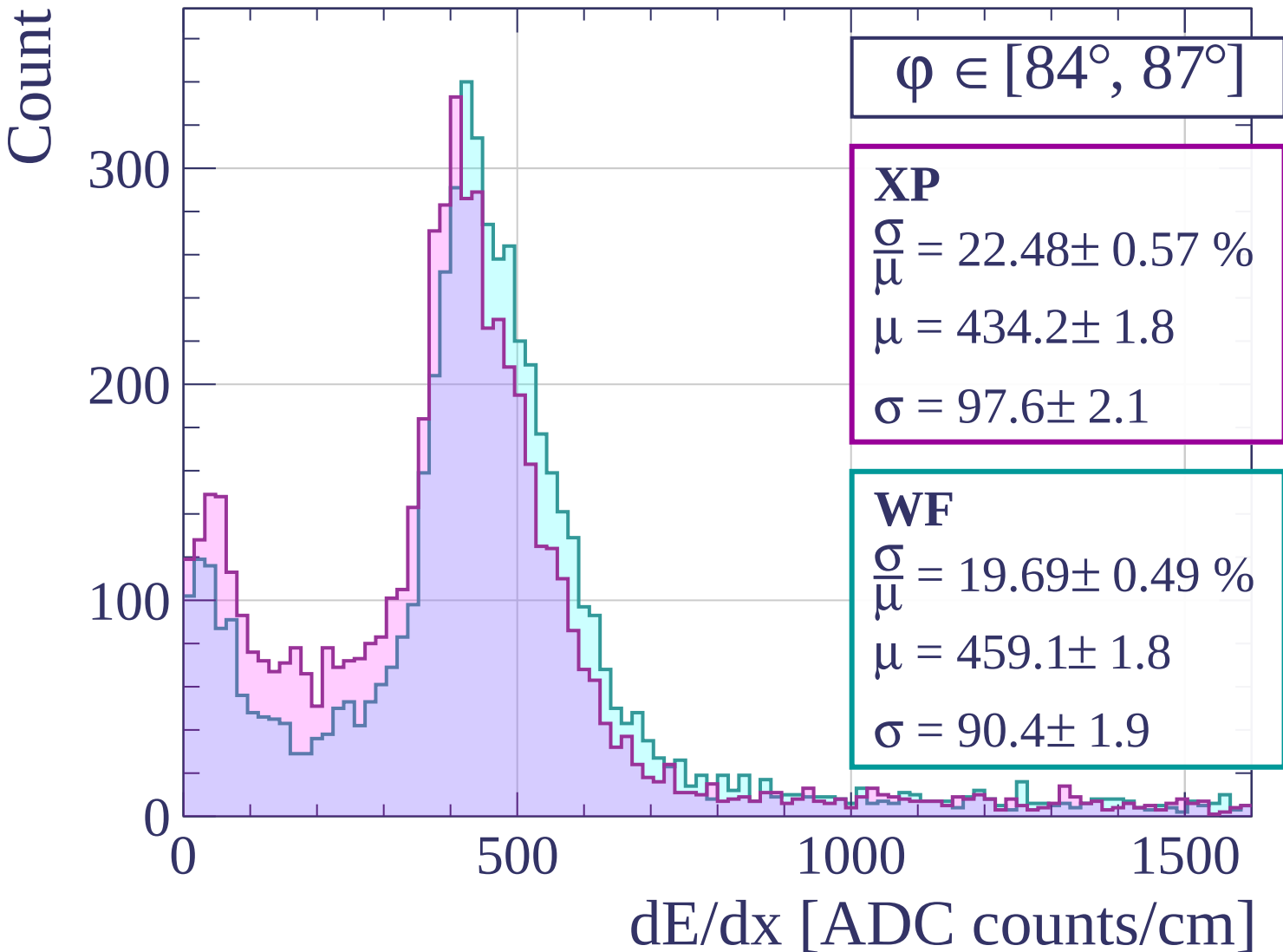
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [87^\circ, 90^\circ]$

XP

$$\frac{\sigma}{\mu} = 23.85 \pm 0.60 \%$$

$$\mu = 438.1 \pm 2.0$$

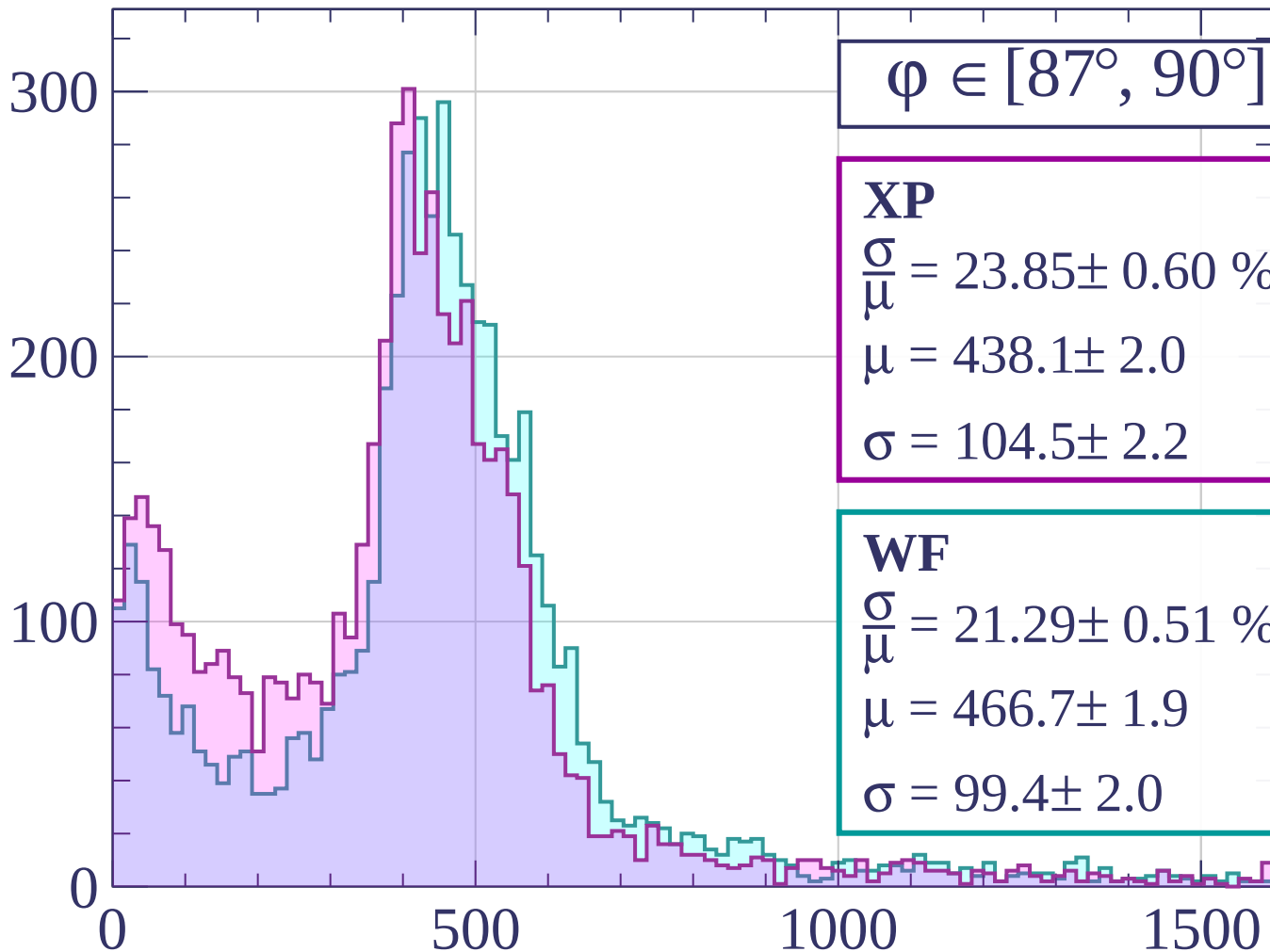
$$\sigma = 104.5 \pm 2.2$$

WF

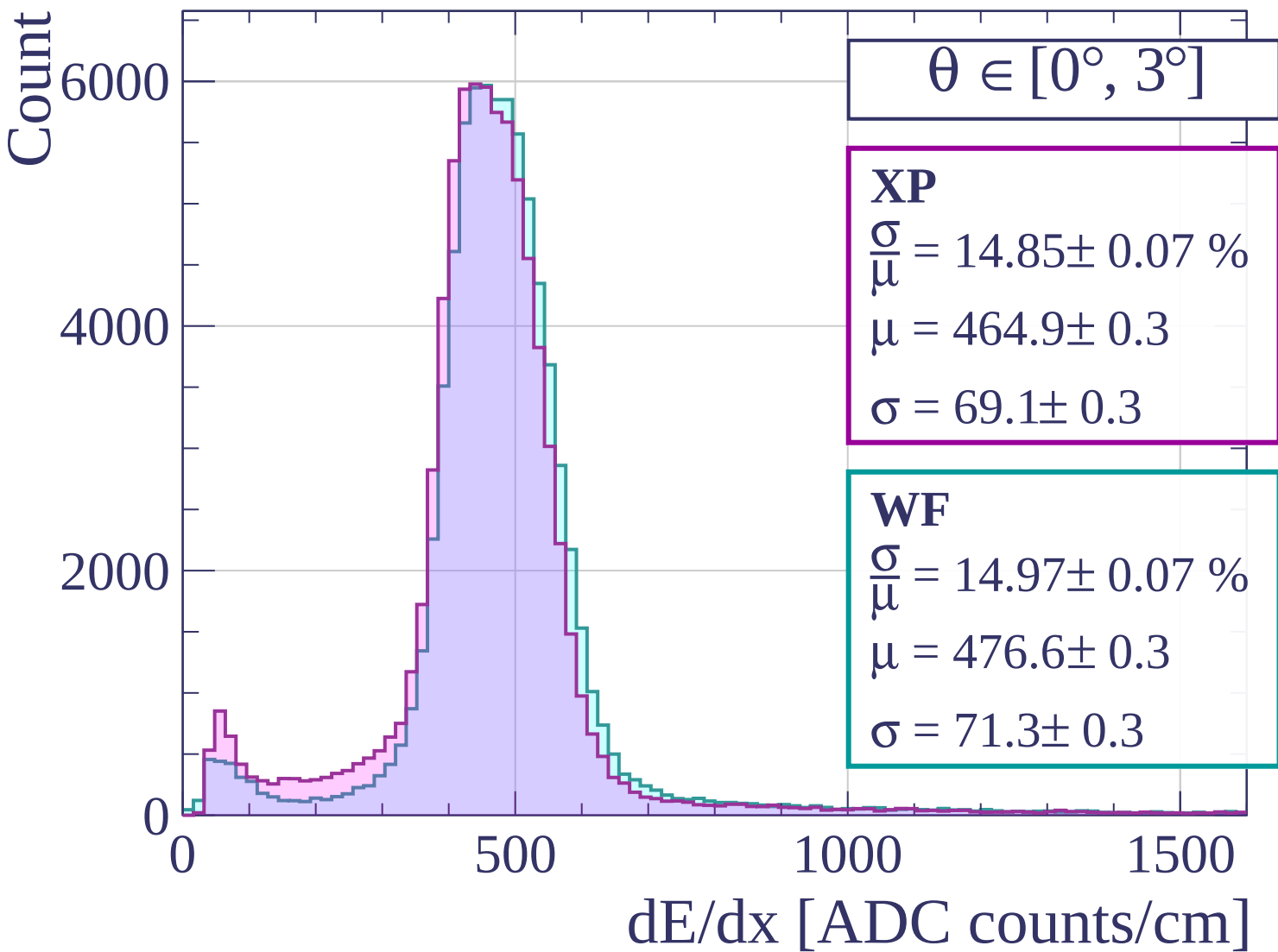
$$\frac{\sigma}{\mu} = 21.29 \pm 0.51 \%$$

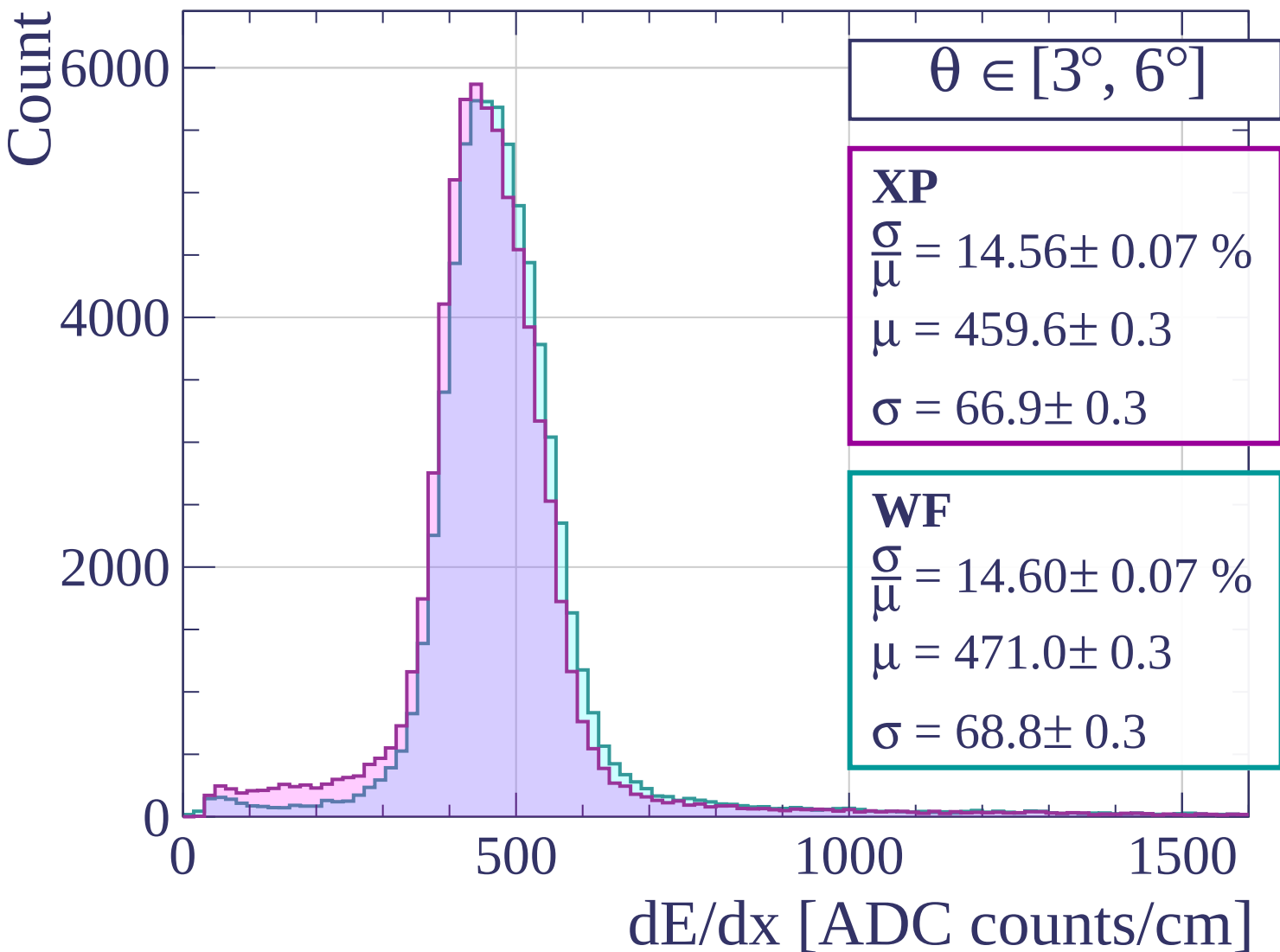
$$\mu = 466.7 \pm 1.9$$

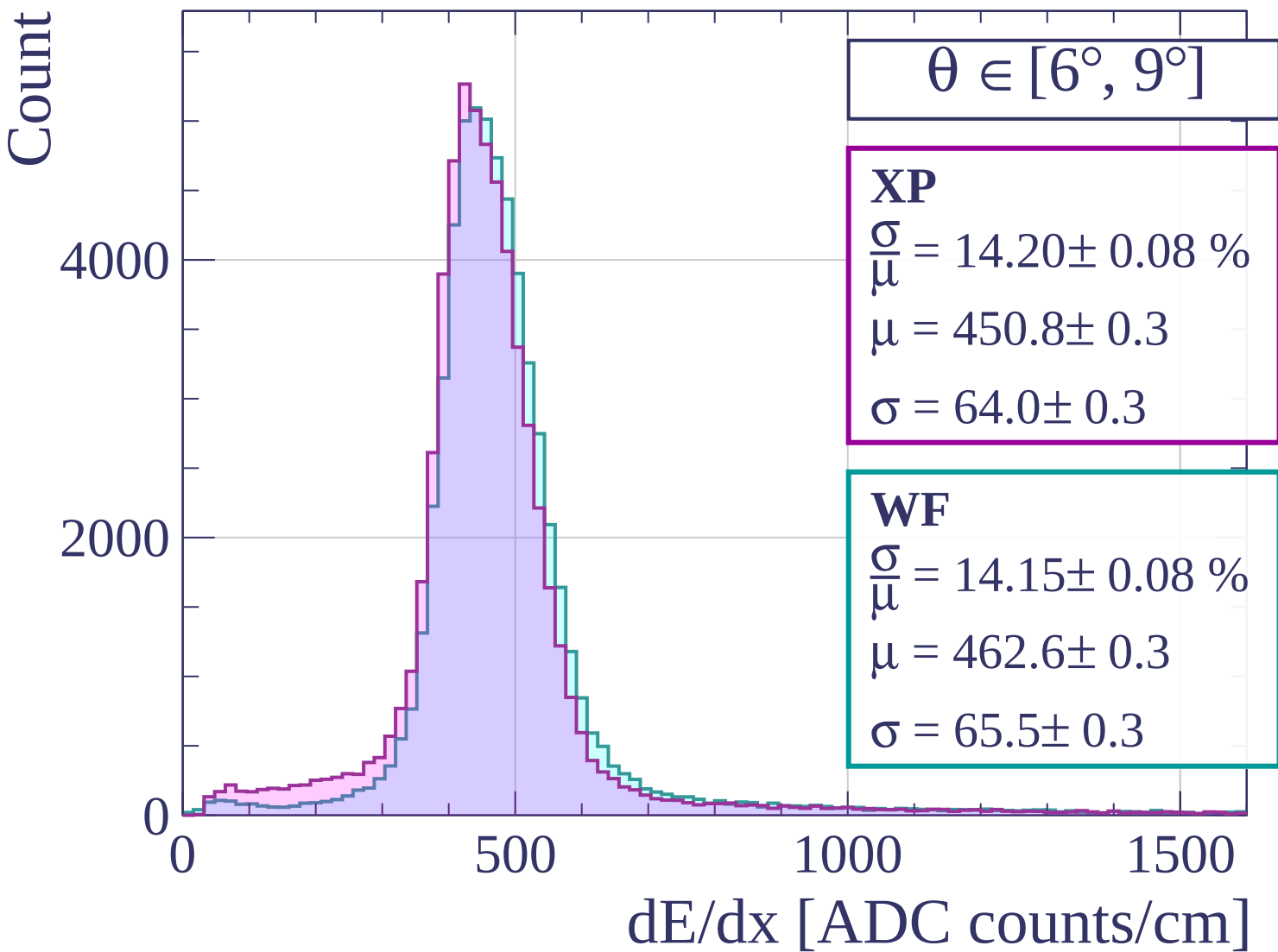
$$\sigma = 99.4 \pm 2.0$$

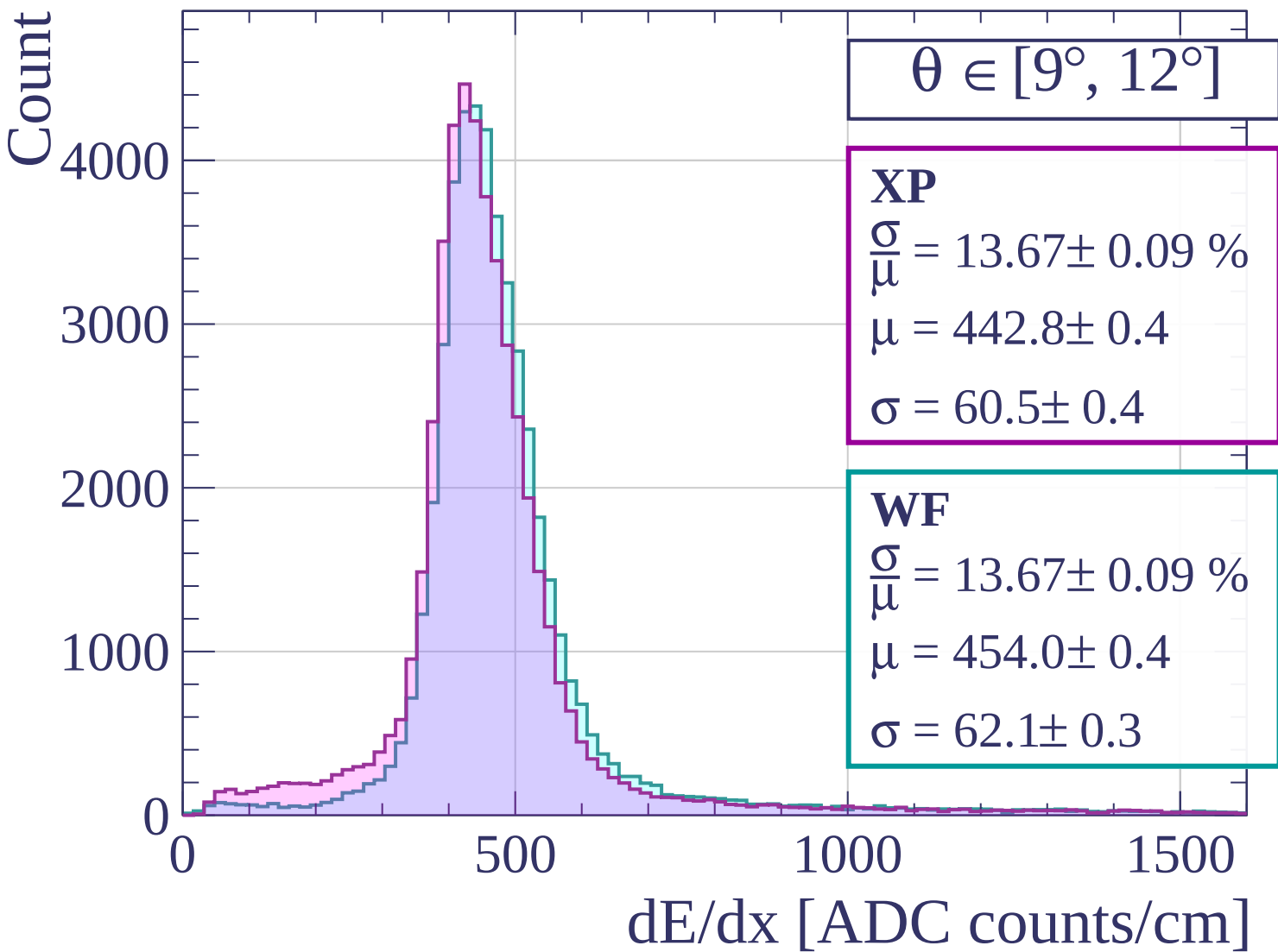


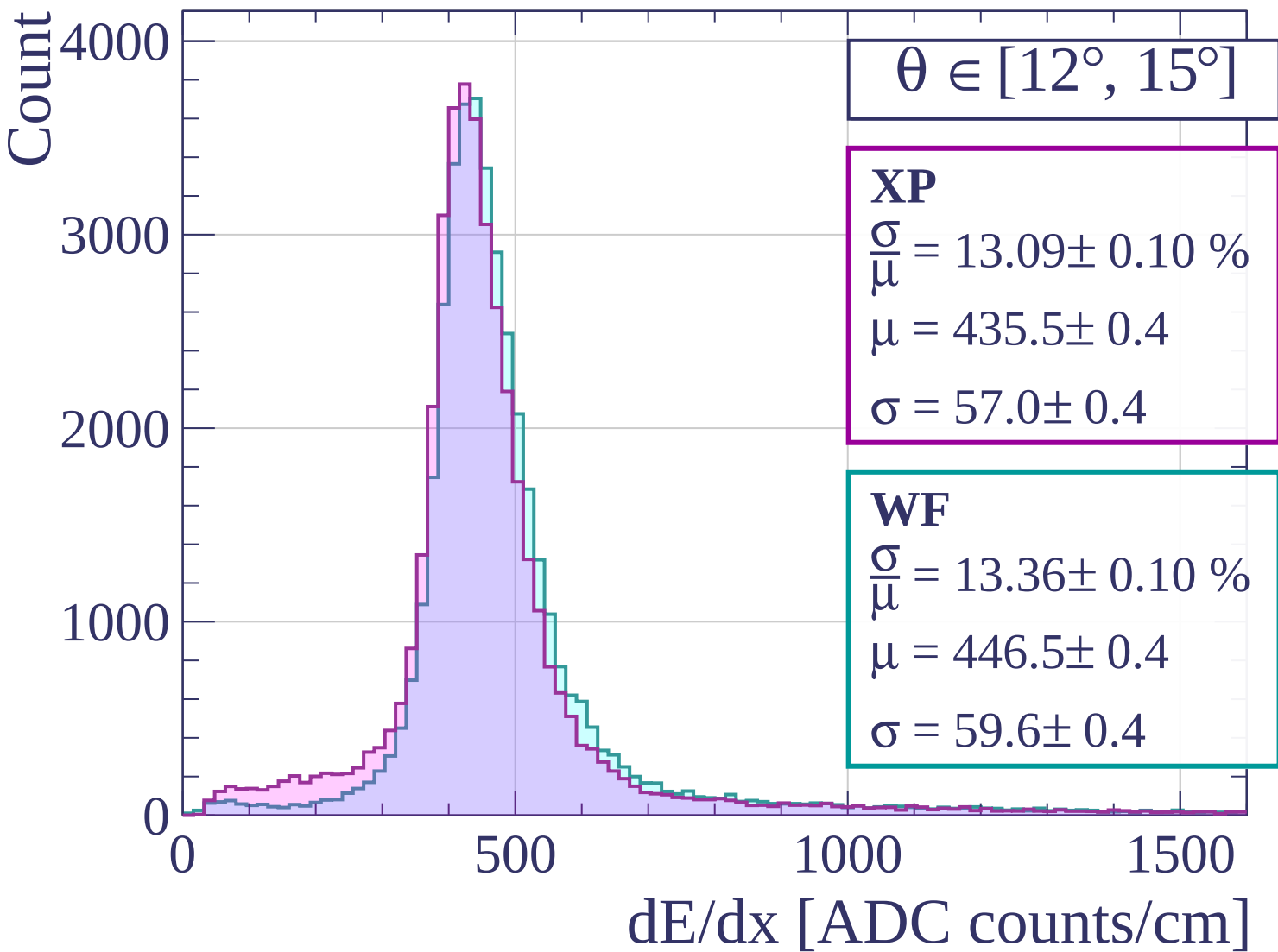
dE/dx [ADC counts/cm]

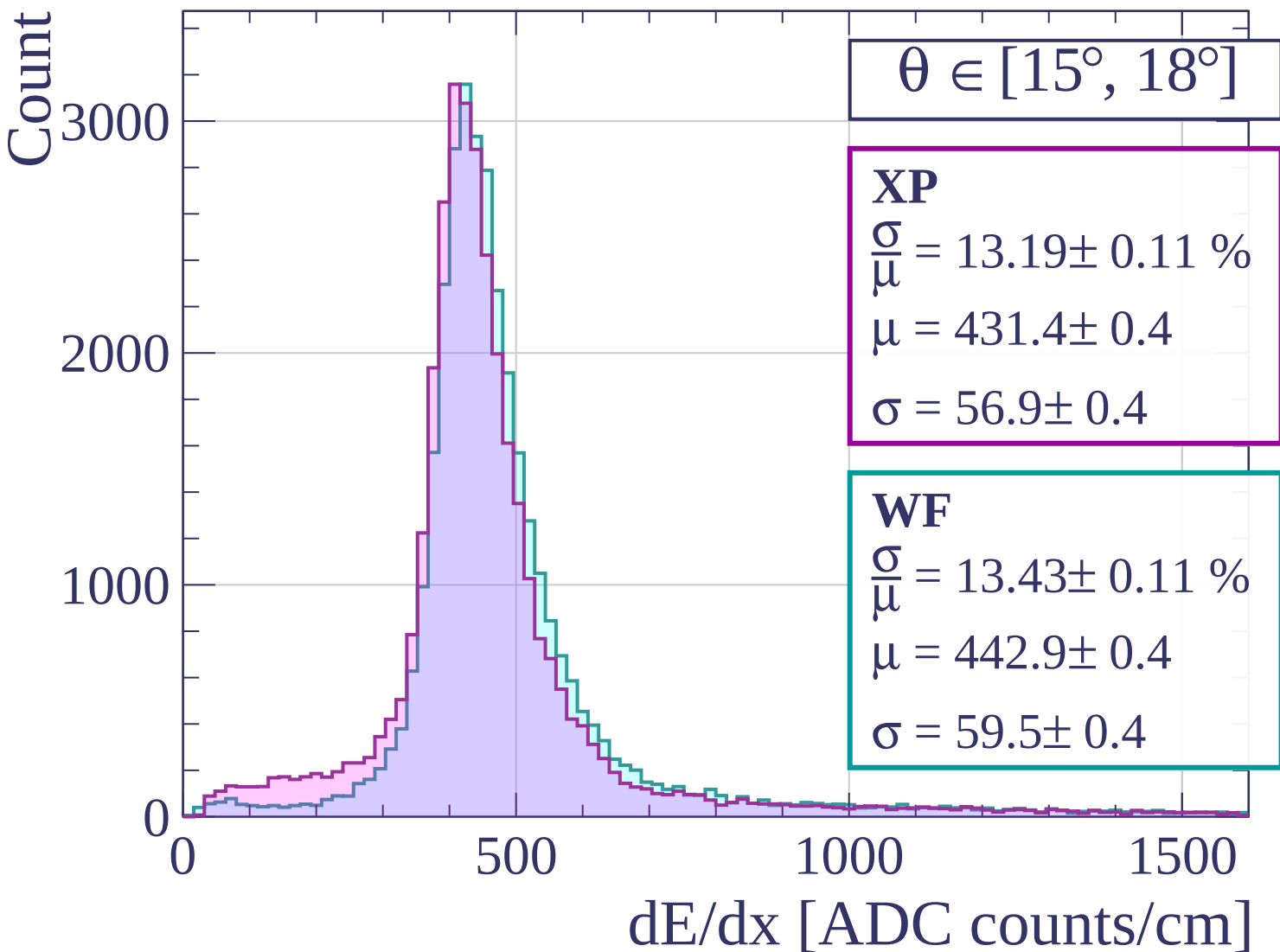


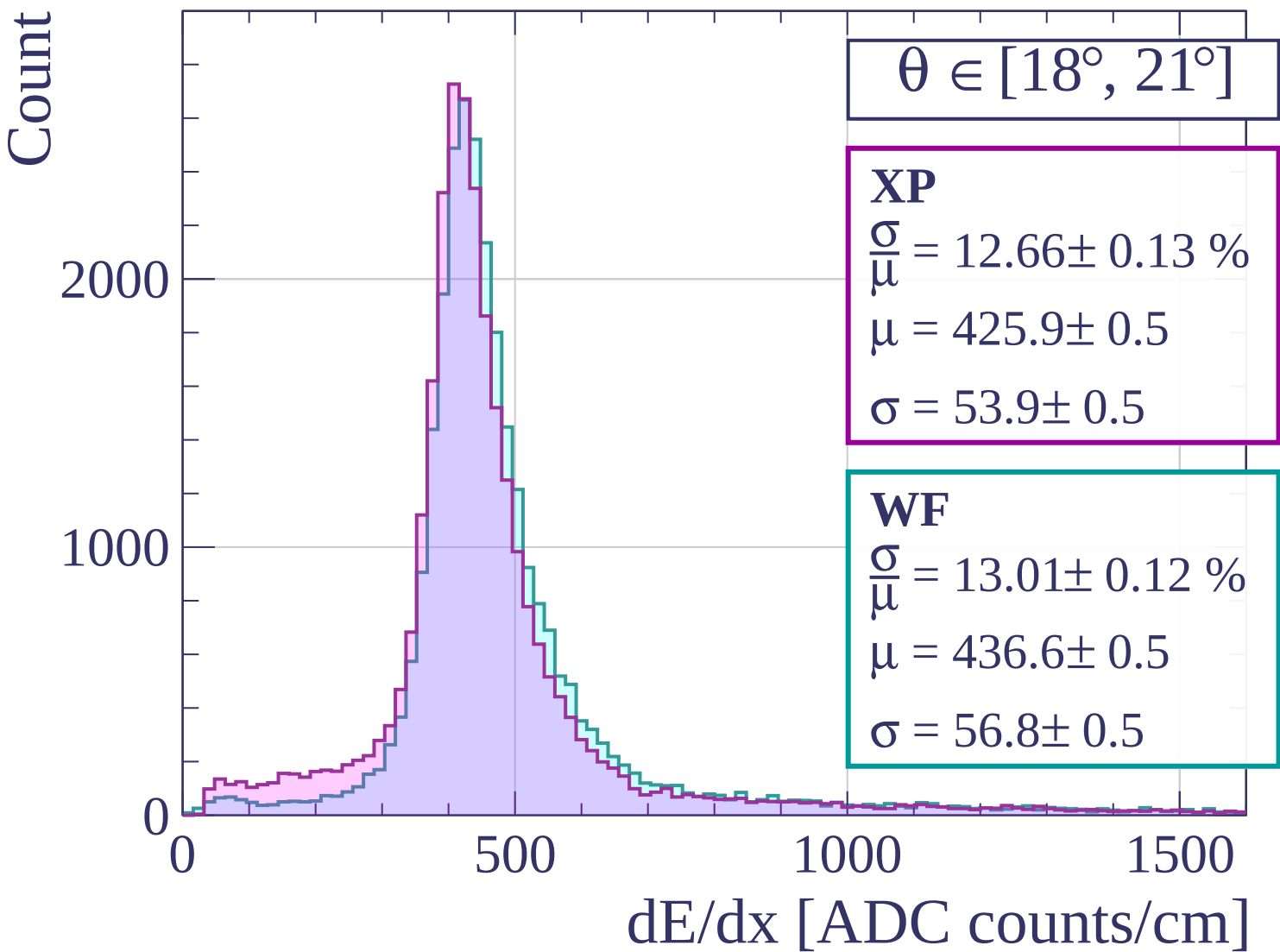


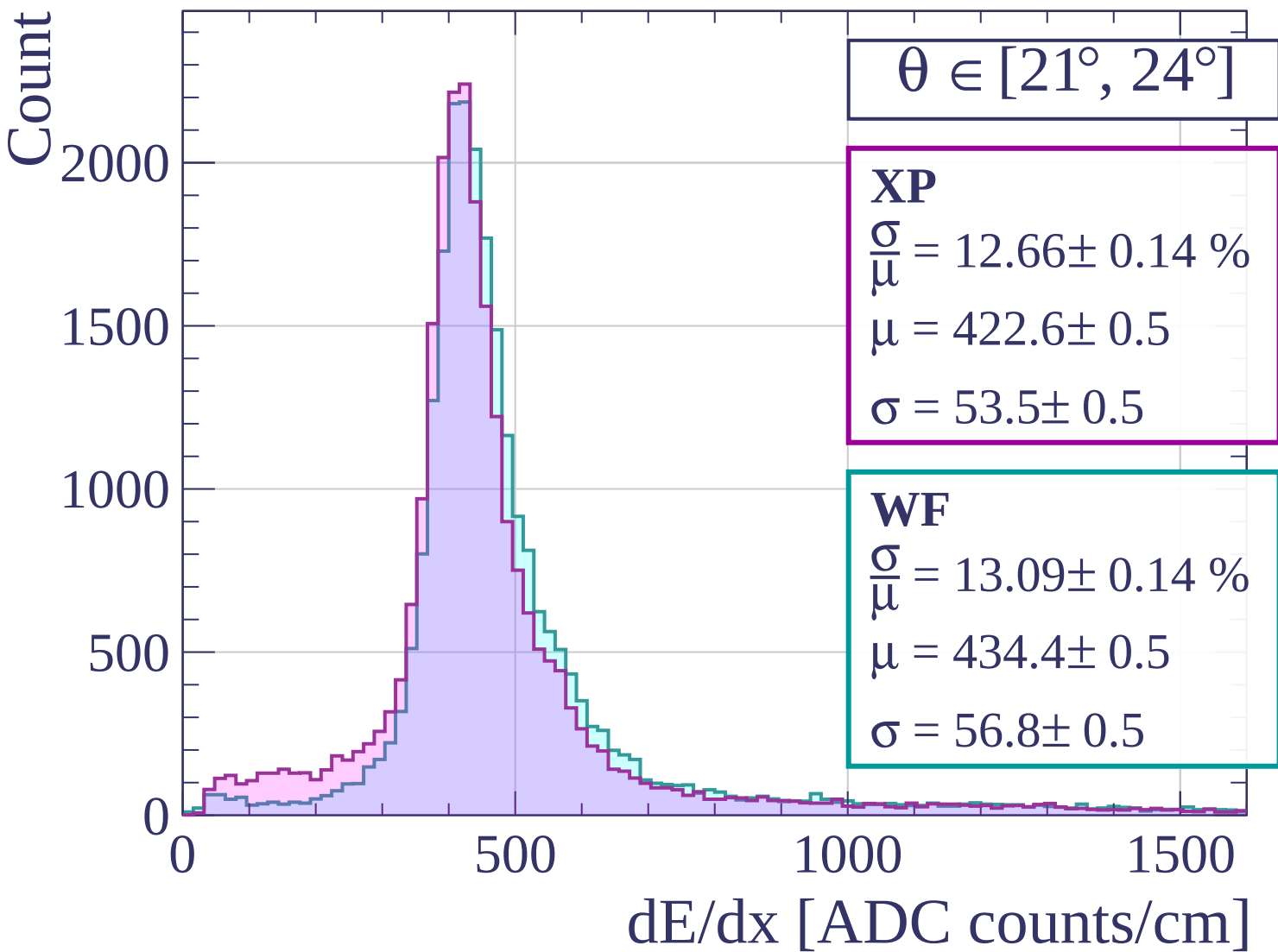


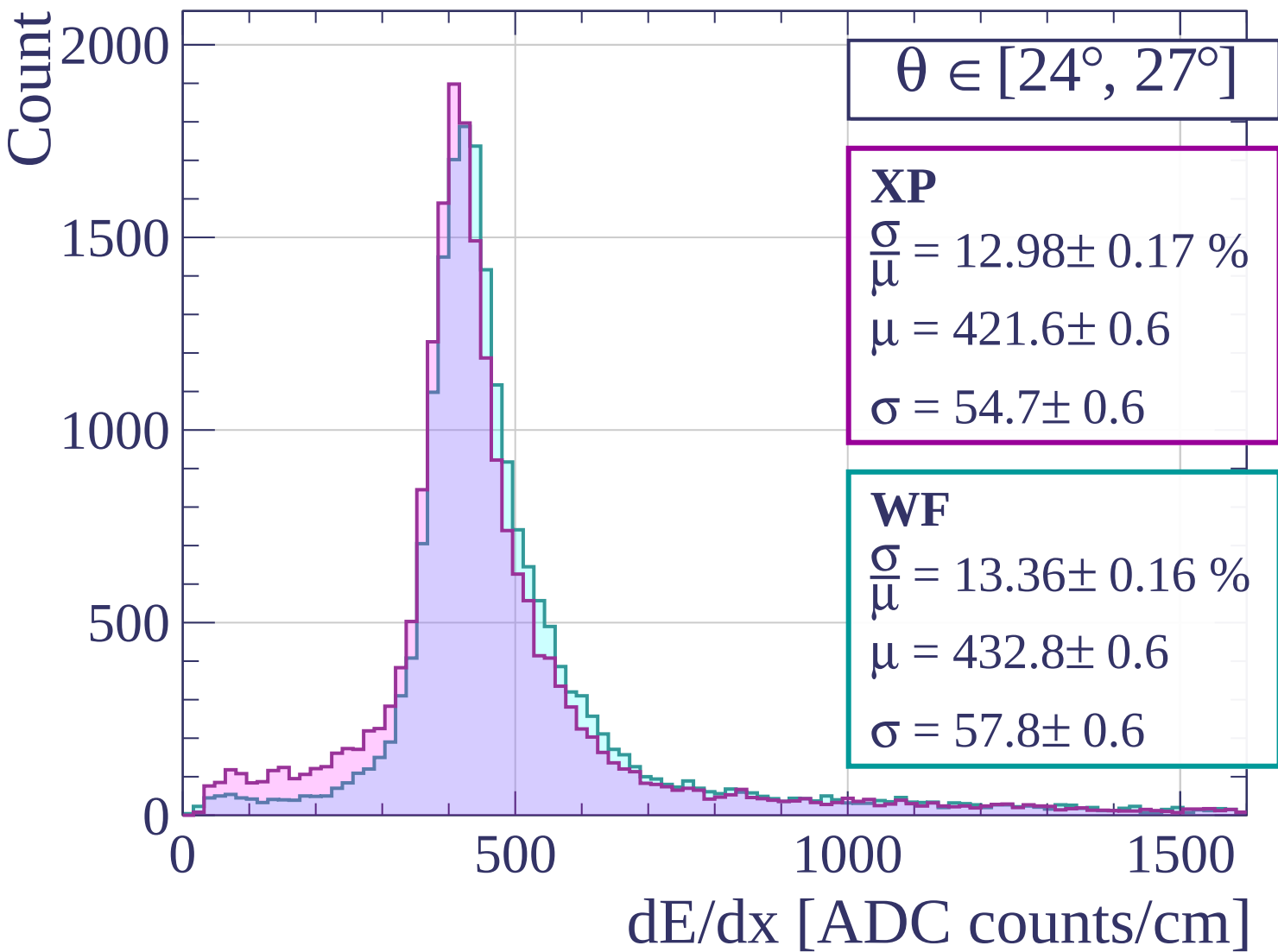


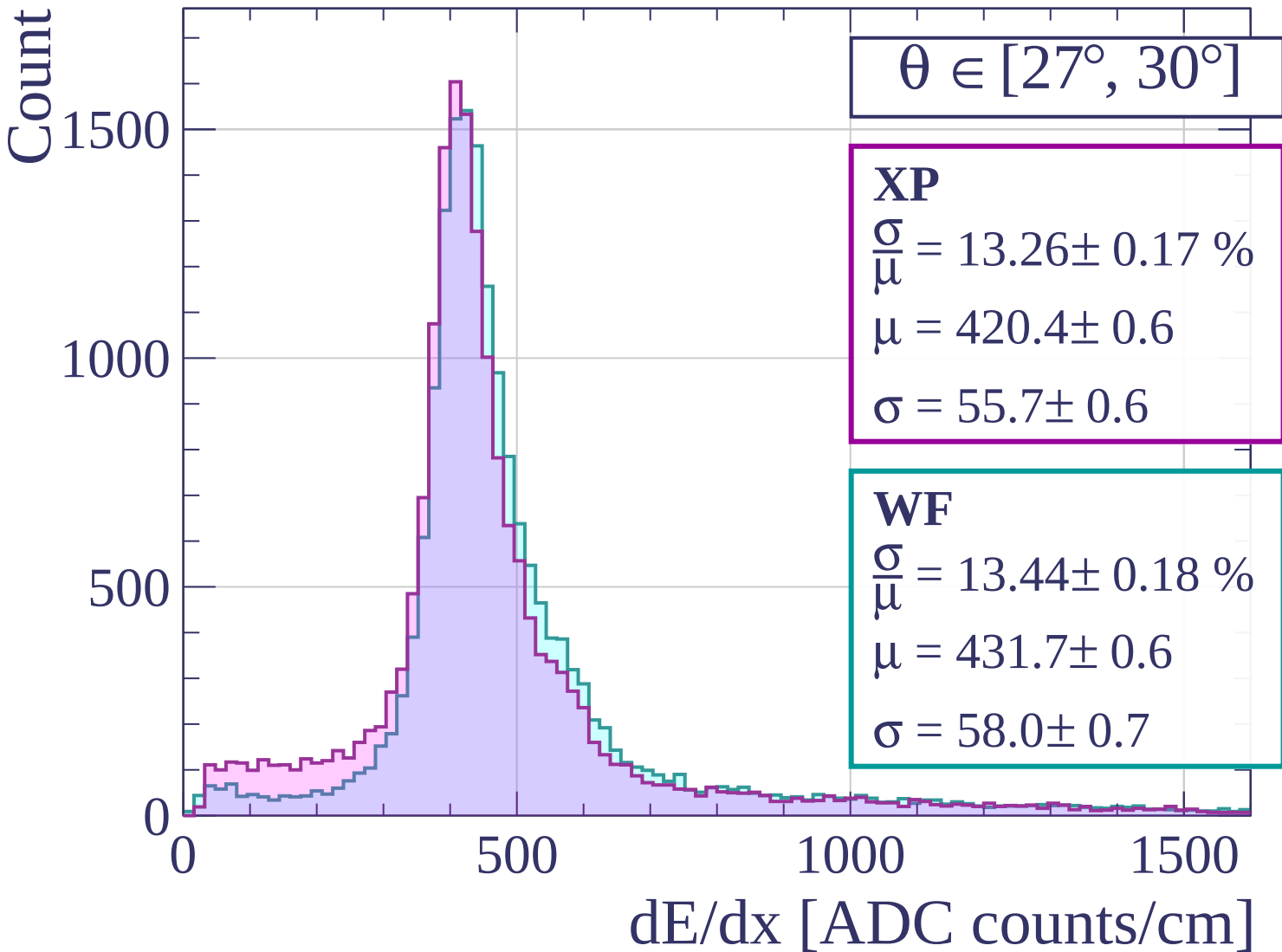


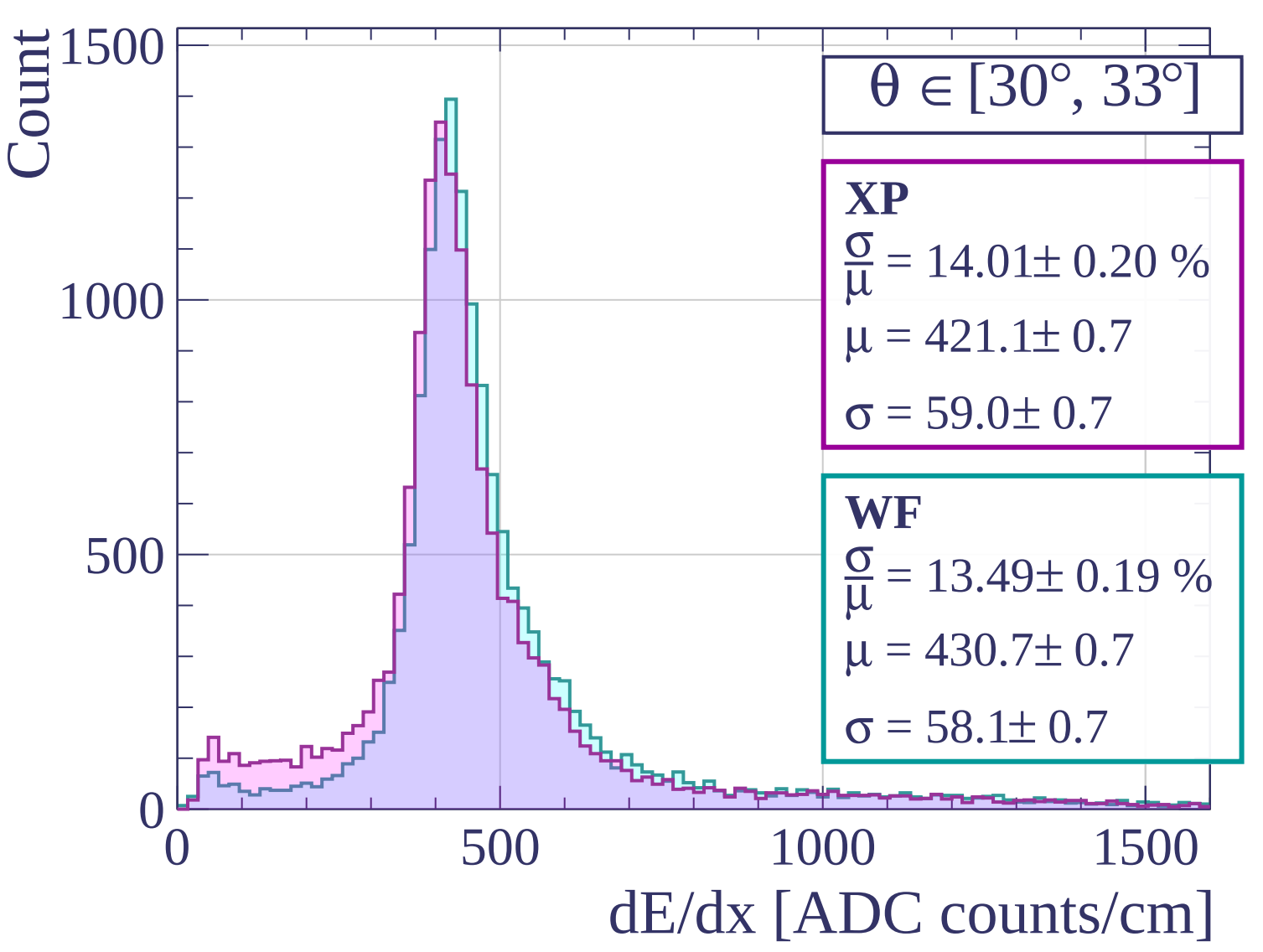


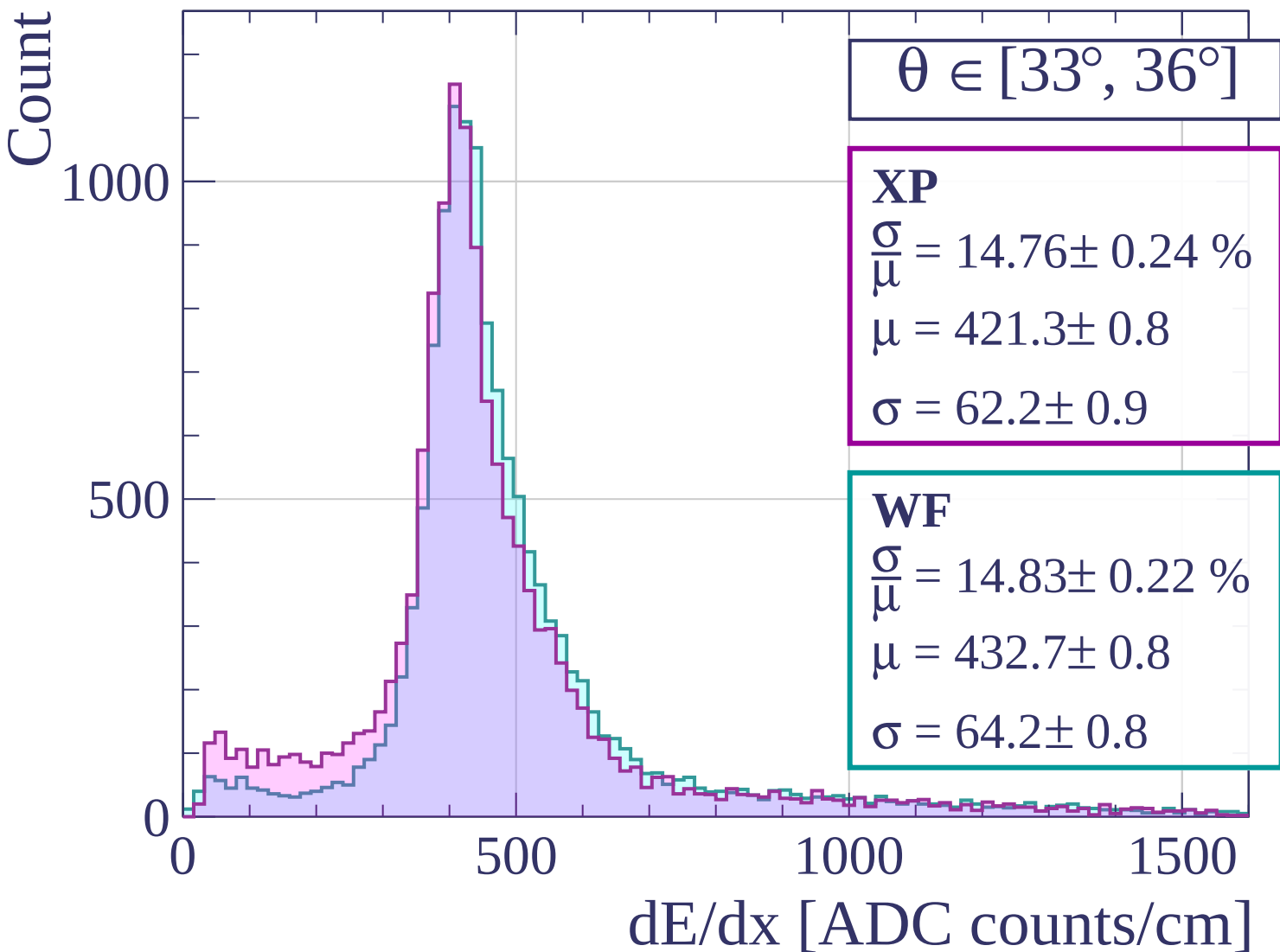


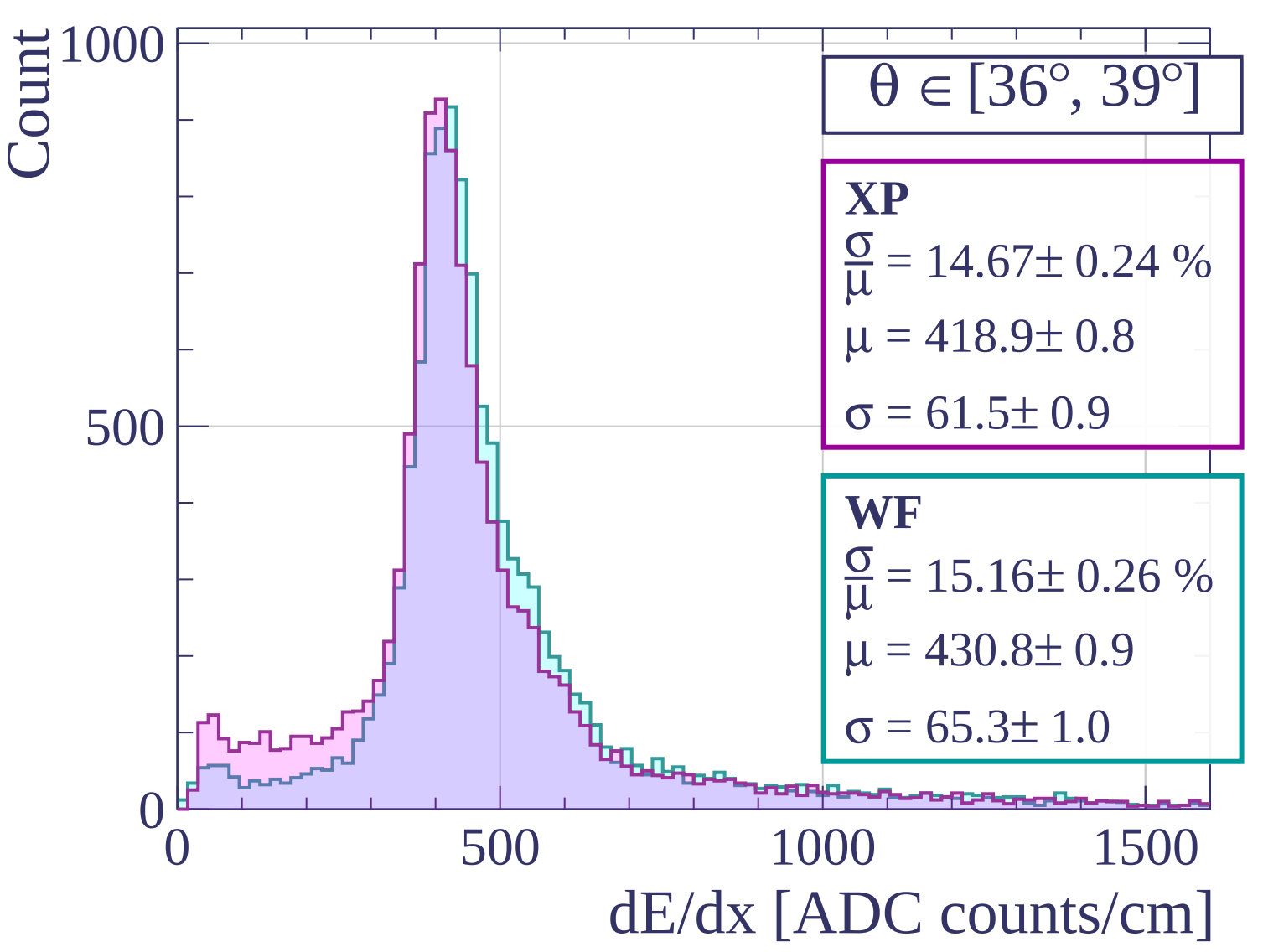












Count

$\theta \in [39^\circ, 42^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.03 \pm 0.28 \%$$

$$\mu = 418.9 \pm 0.9$$

$$\sigma = 62.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 14.94 \pm 0.27 \%$$

$$\mu = 430.8 \pm 0.9$$

$$\sigma = 64.4 \pm 1.0$$

800

600

400

200

0

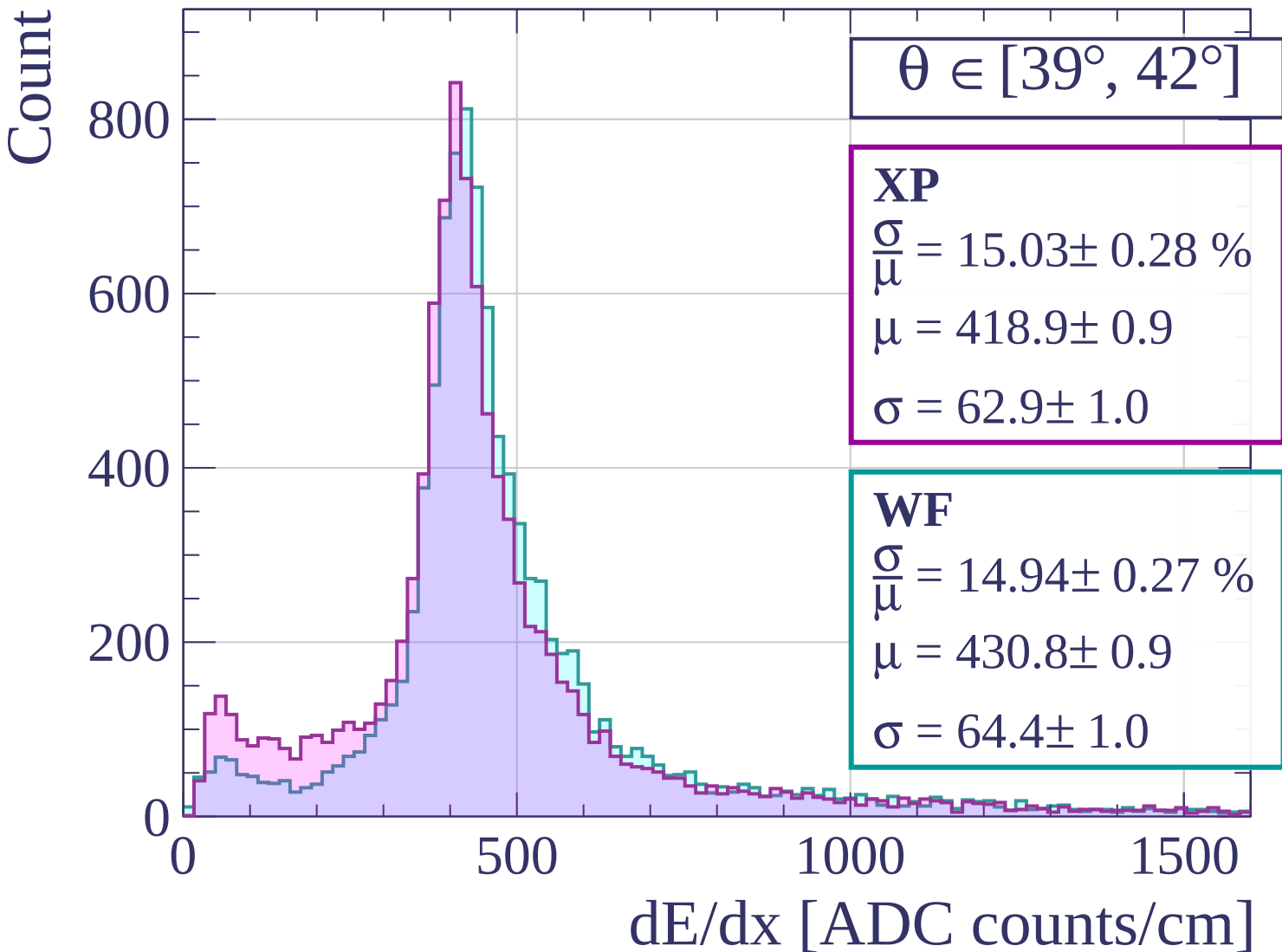
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [42^\circ, 45^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.69 \pm 0.34 \%$$

$$\mu = 420.3 \pm 1.1$$

$$\sigma = 70.1 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 16.16 \pm 0.32 \%$$

$$\mu = 429.1 \pm 1.1$$

$$\sigma = 69.4 \pm 1.2$$

600

400

200

0

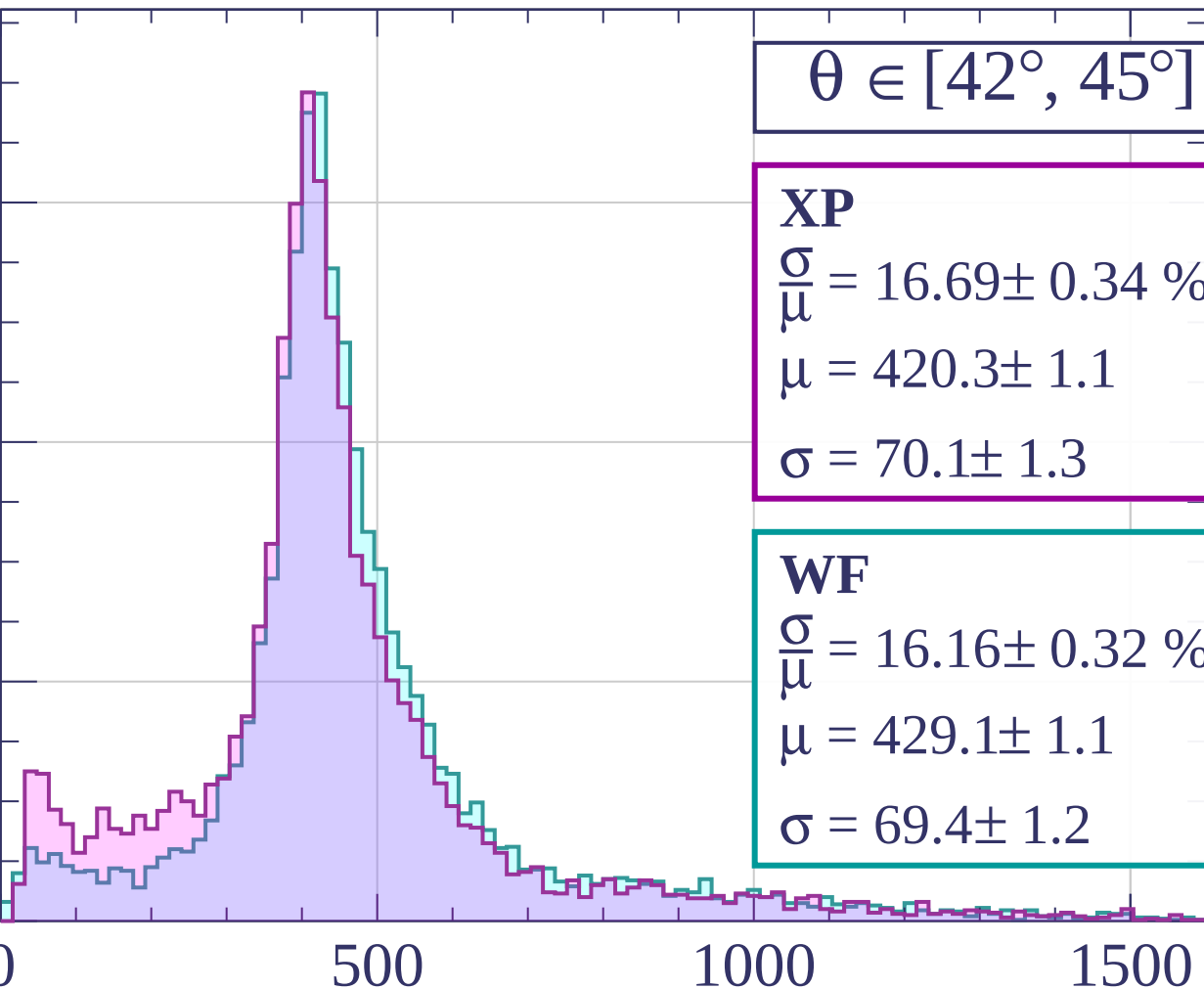
0

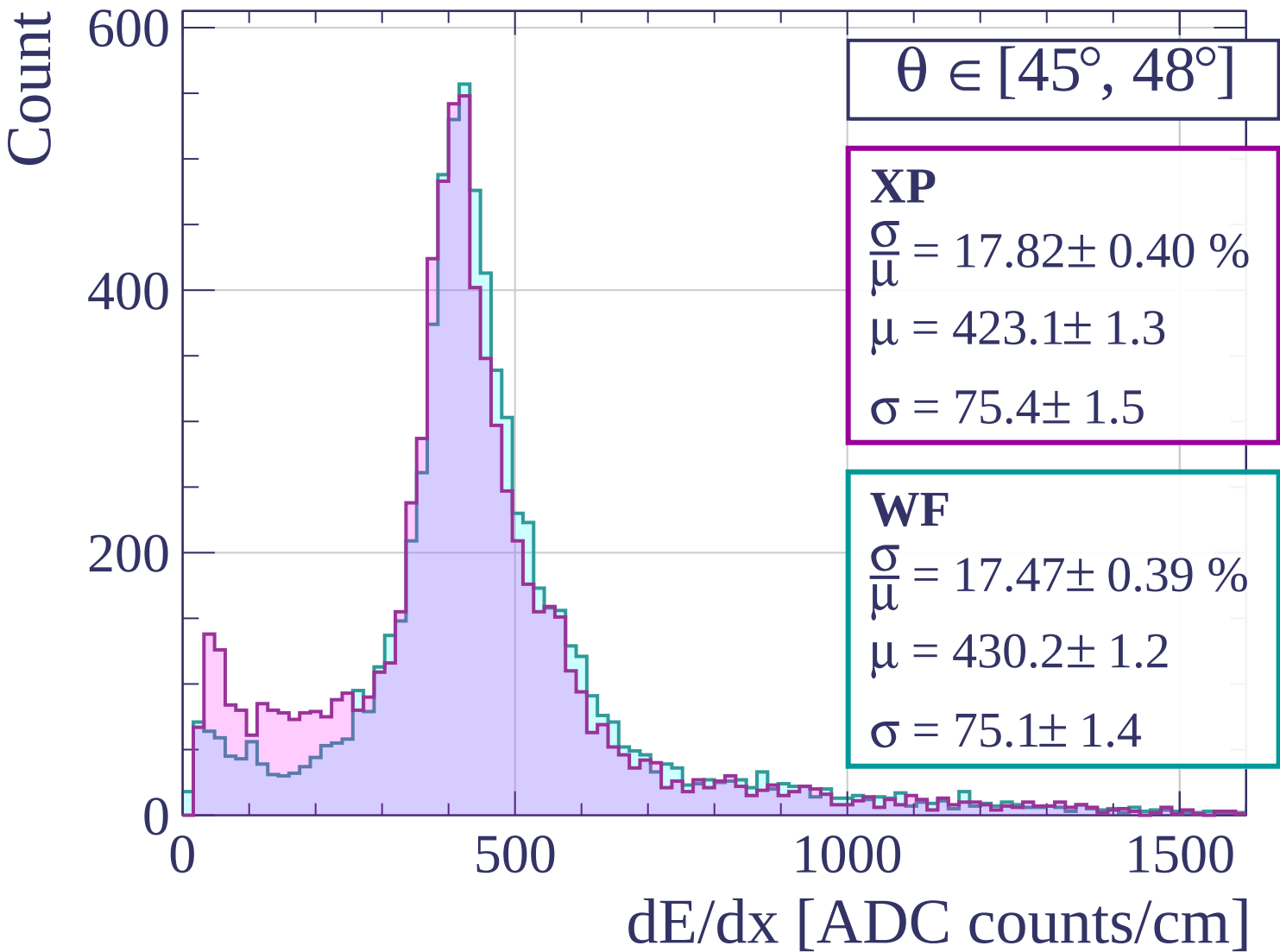
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\theta \in [48^\circ, 51^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.74 \pm 0.42 \%$$

$$\mu = 422.2 \pm 1.3$$

$$\sigma = 74.9 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 18.22 \pm 0.43 \%$$

$$\mu = 428.5 \pm 1.4$$

$$\sigma = 78.0 \pm 1.6$$

400

200

0

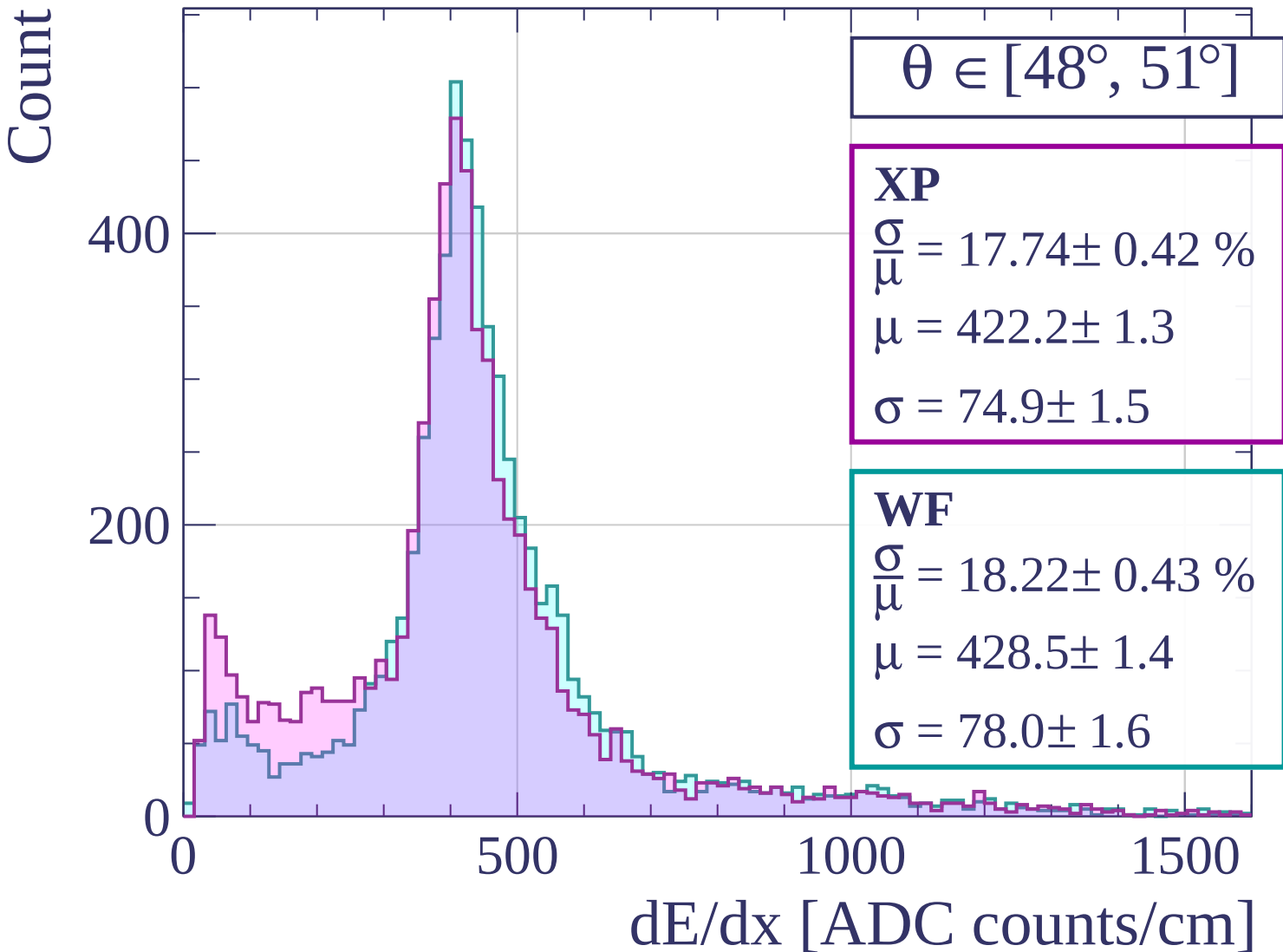
0

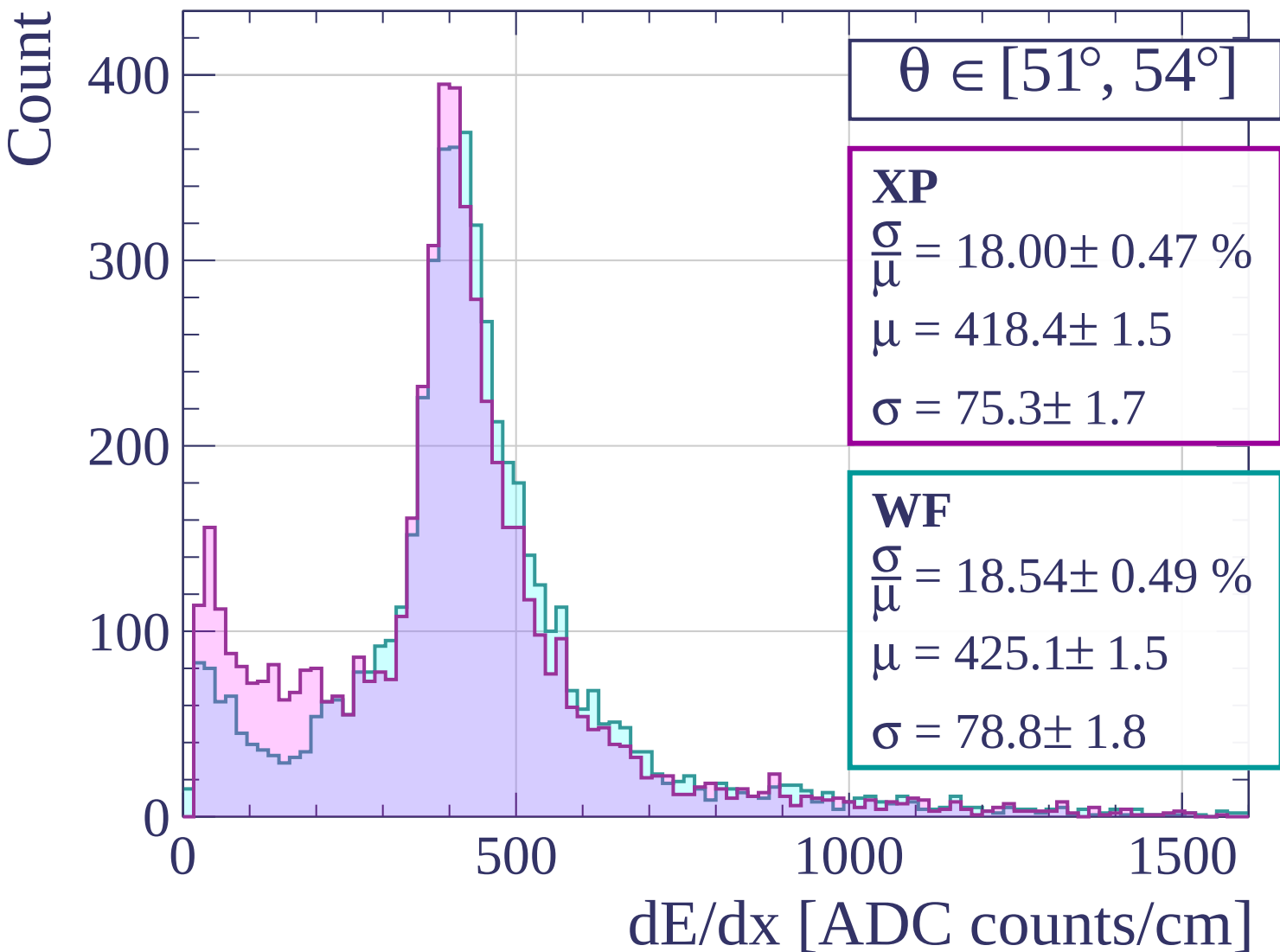
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\theta \in [54^\circ, 57^\circ]$

XP

$$\frac{\sigma}{\mu} = 23.19 \pm 0.64 \%$$

$$\mu = 423.5 \pm 1.9$$

$$\sigma = 98.2 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 20.09 \pm 0.55 \%$$

$$\mu = 426.8 \pm 1.7$$

$$\sigma = 85.7 \pm 2.0$$

300

200

100

0

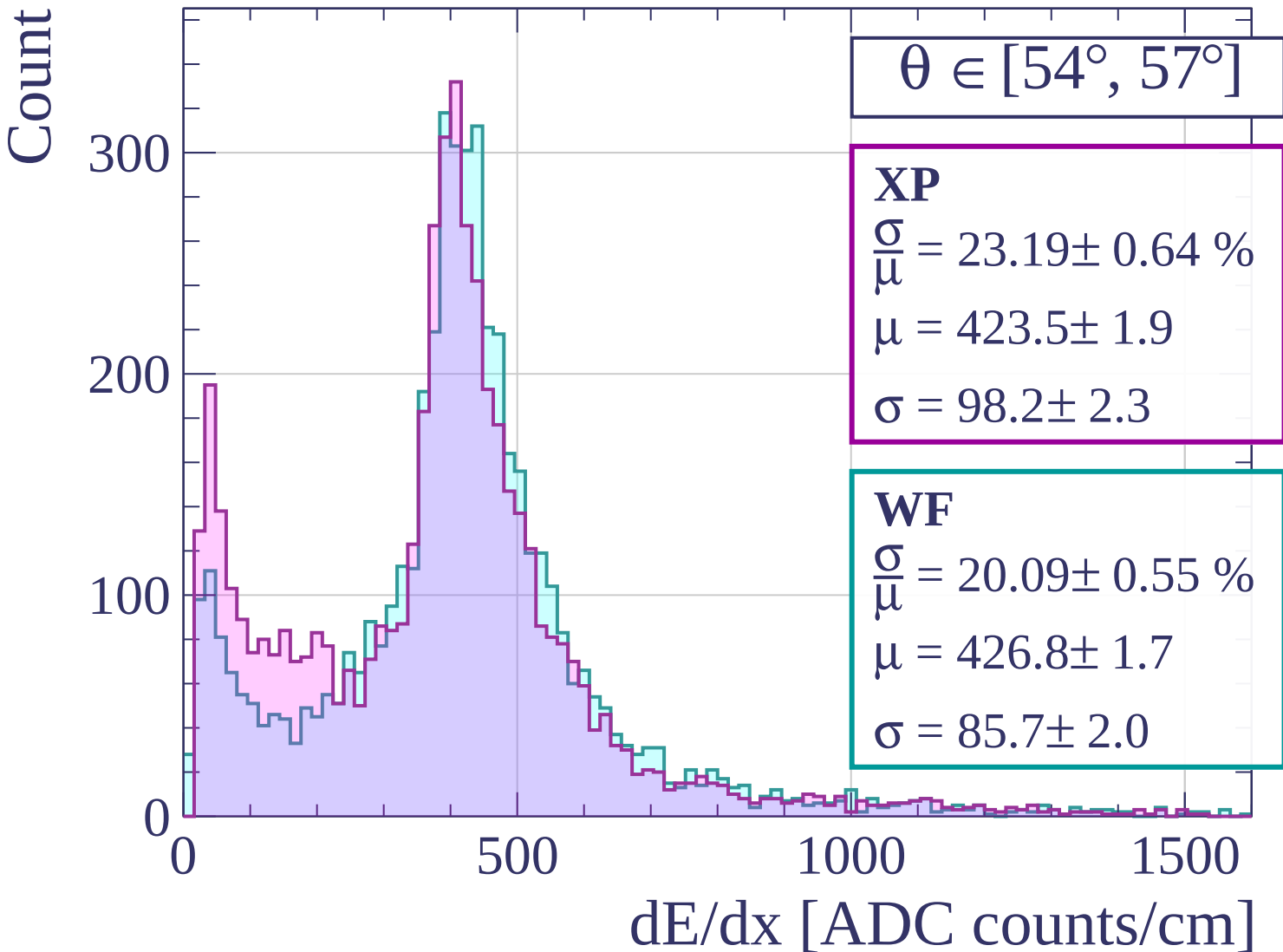
0

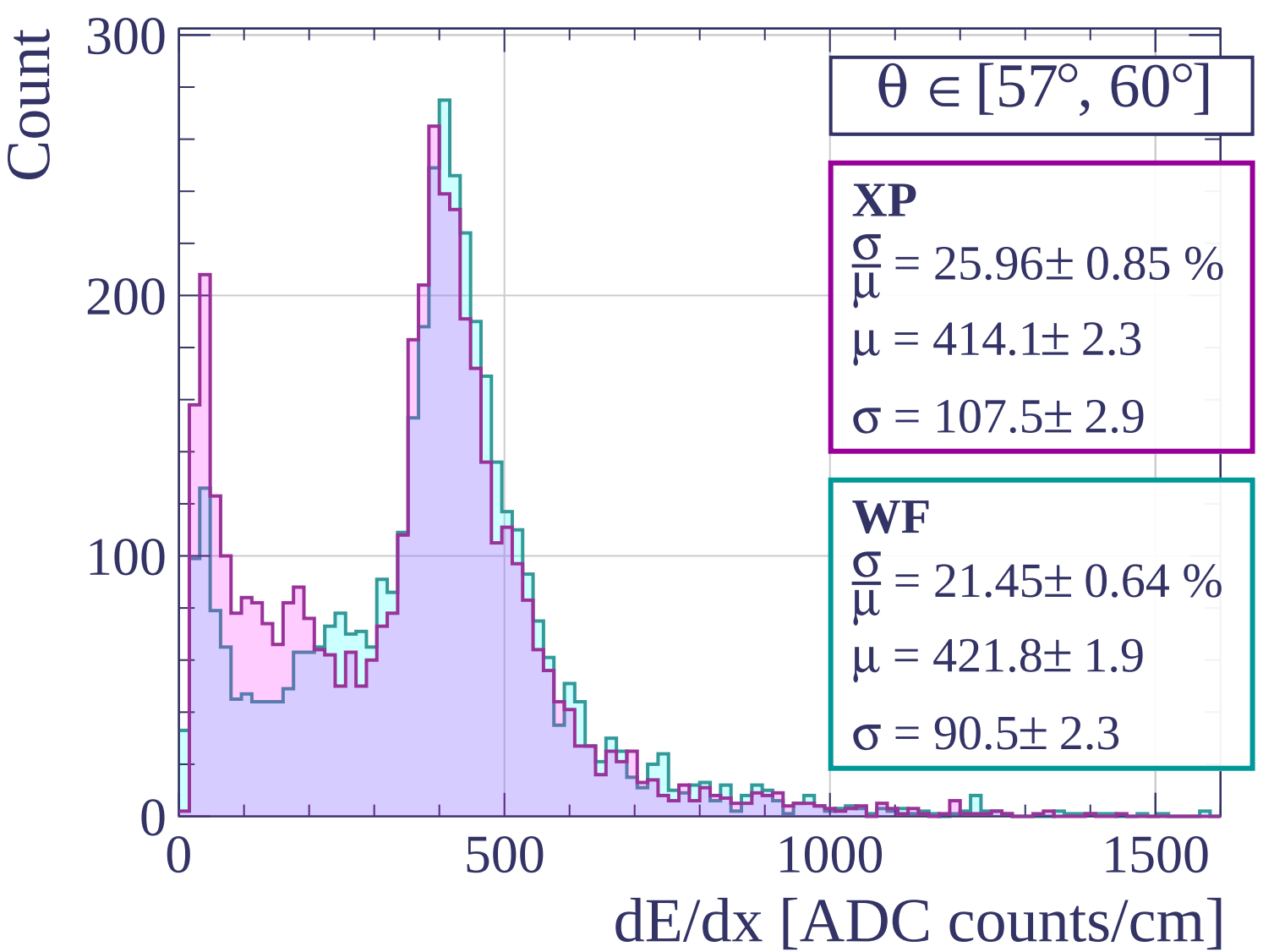
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\theta \in [60^\circ, 63^\circ]$

XP

$$\frac{\sigma}{\mu} = 25.42 \pm 3.96 \%$$

$$\mu = 4981.7 \pm 484.1$$

$$\sigma = 1266.2 \pm 74.1$$

WF

$$\frac{\sigma}{\mu} = 23.66 \pm 0.83 \%$$

$$\mu = 409.8 \pm 2.3$$

$$\sigma = 97.0 \pm 2.9$$

200

100

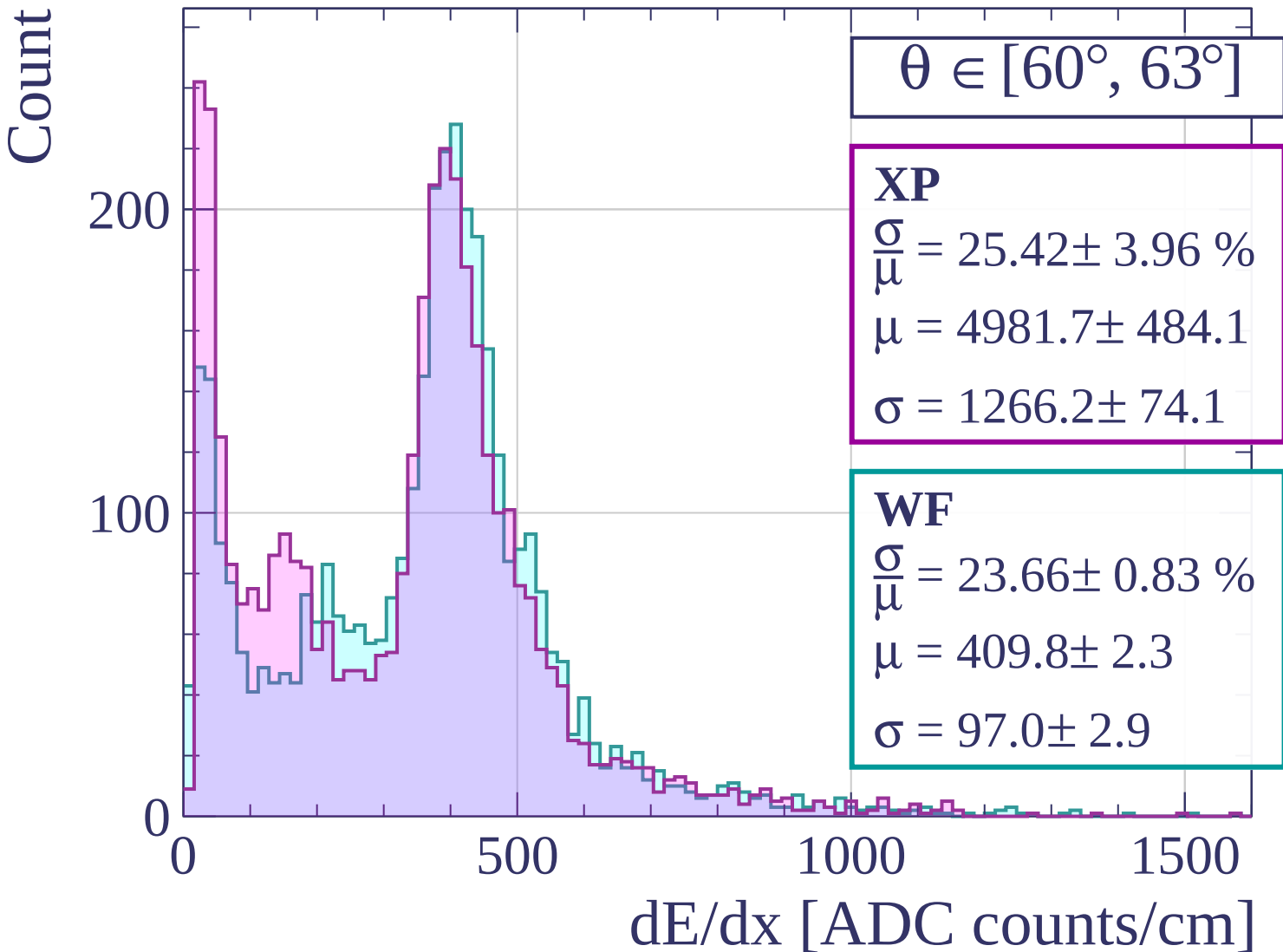
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [63^\circ, 66^\circ]$

XP

$$\frac{\sigma}{\mu} = 173.67 \pm 26.24 \%$$

$$\mu = 156.4 \pm 17.8$$

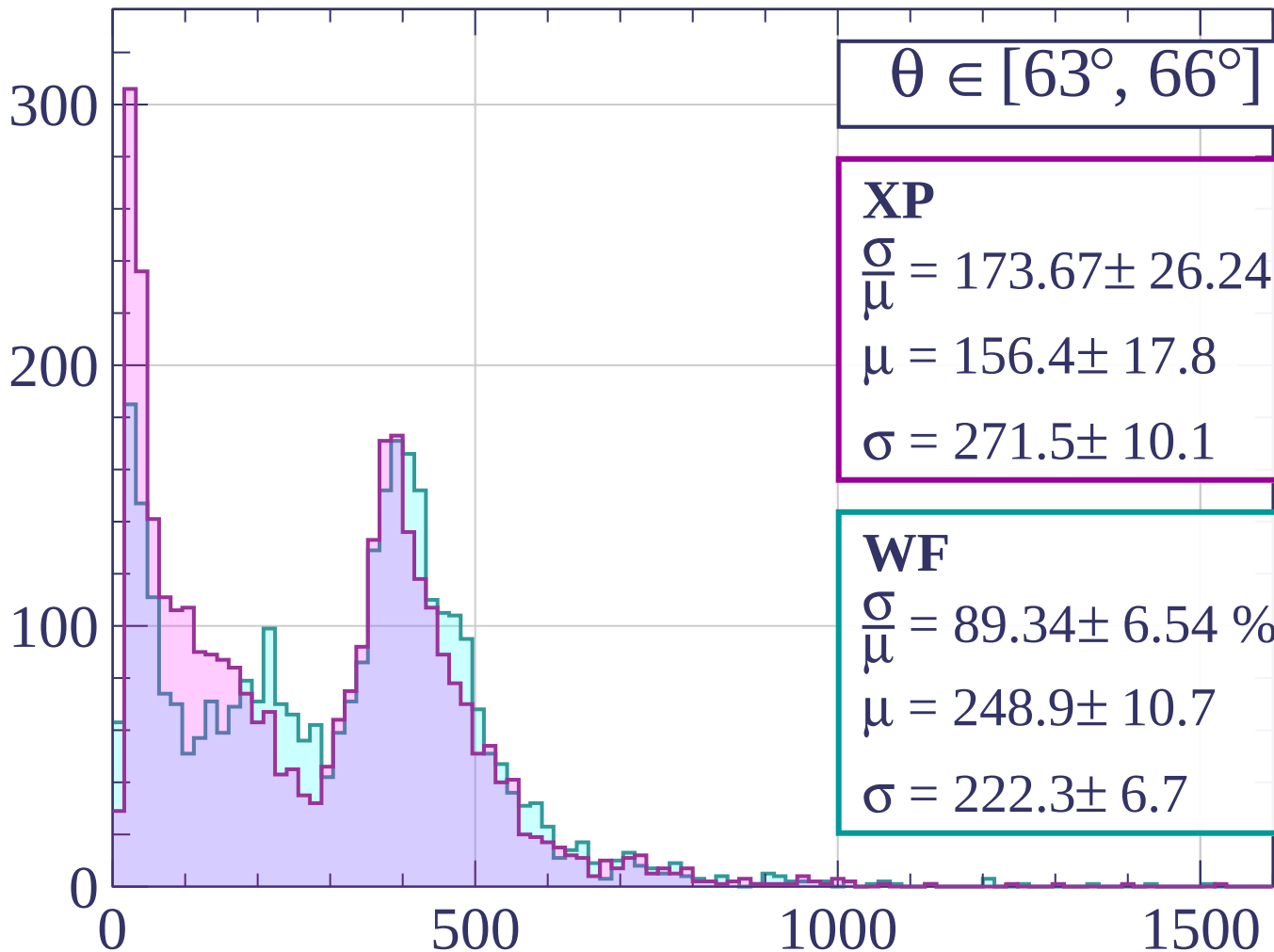
$$\sigma = 271.5 \pm 10.1$$

WF

$$\frac{\sigma}{\mu} = 89.34 \pm 6.54 \%$$

$$\mu = 248.9 \pm 10.7$$

$$\sigma = 222.3 \pm 6.7$$



dE/dx [ADC counts/cm]

Count

$\theta \in [66^\circ, 69^\circ]$

XP

$$\frac{\sigma}{\mu} = -29.62 \pm 2.75 \%$$

$$\mu = -2501.5 \pm 486.6$$

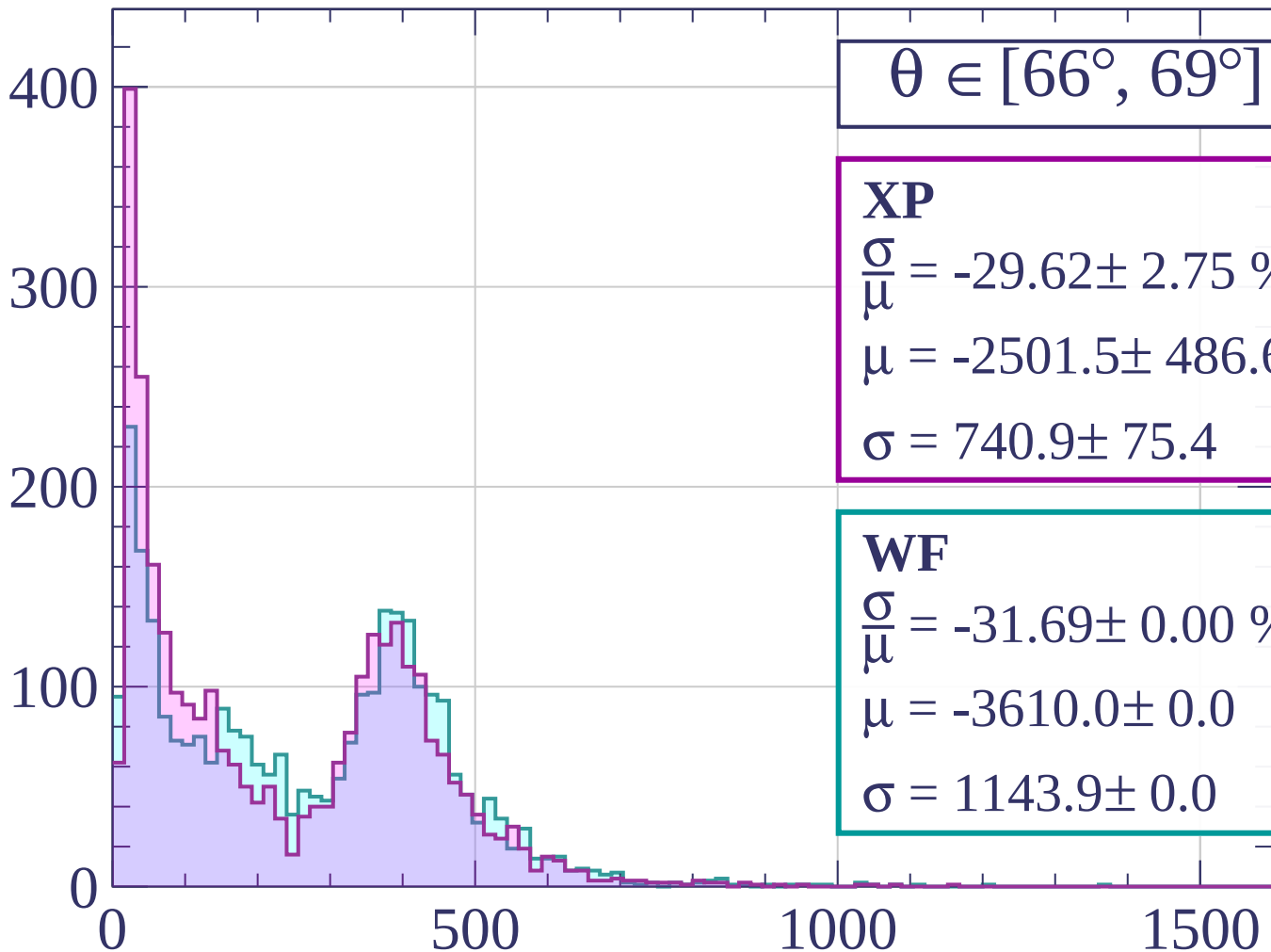
$$\sigma = 740.9 \pm 75.4$$

WF

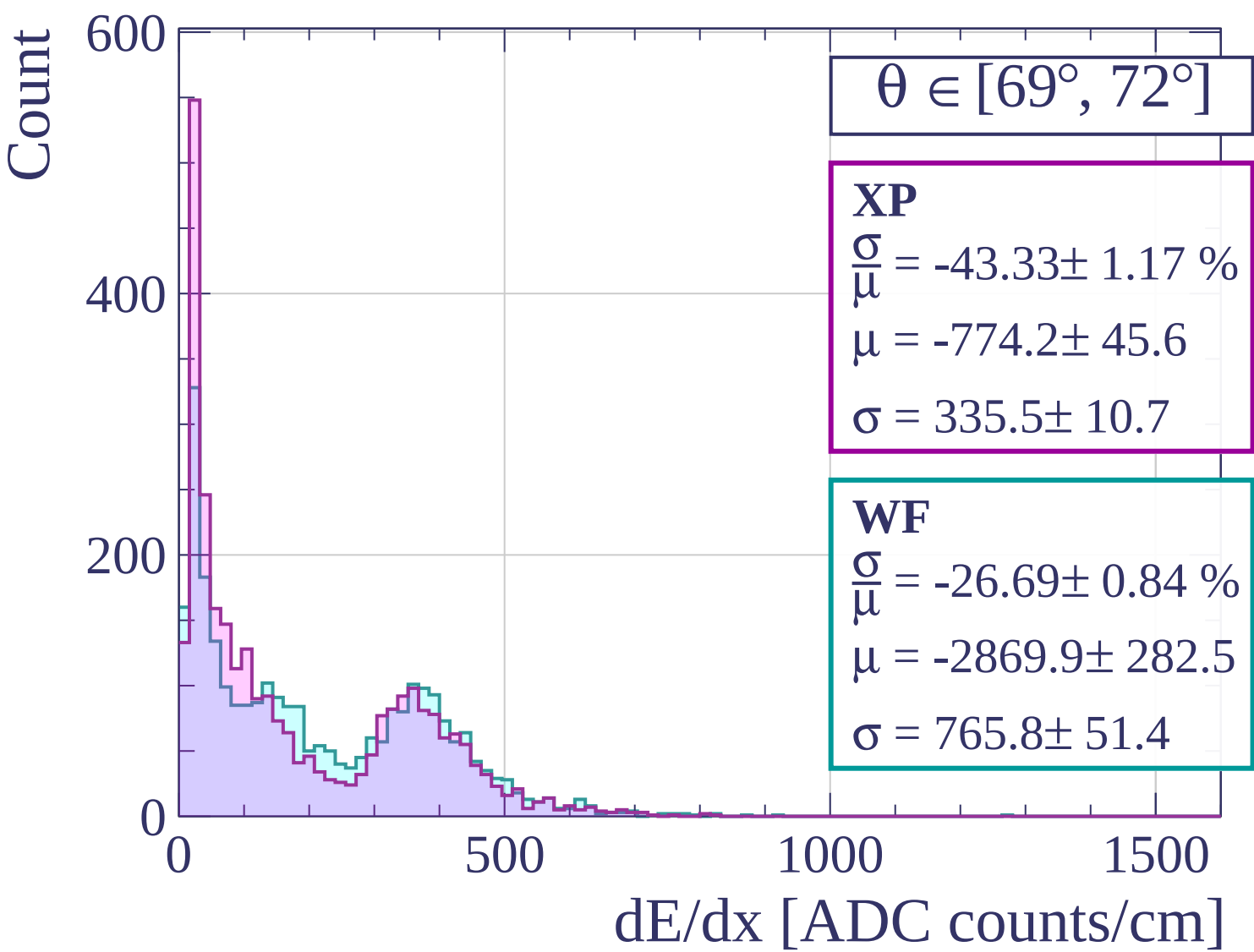
$$\frac{\sigma}{\mu} = -31.69 \pm 0.00 \%$$

$$\mu = -3610.0 \pm 0.0$$

$$\sigma = 1143.9 \pm 0.0$$



dE/dx [ADC counts/cm]



Count

$$\theta \in [72^\circ, 75^\circ]$$

XP

$$\frac{\sigma}{\mu} = -25.84 \pm 0.69 \%$$

$$\mu = -1286.1 \pm 87.6$$

$$\sigma = 332.3 \pm 13.7$$

WF

$$\frac{\sigma}{\mu} = -25.95 \pm 0.88 \%$$

$$\mu = -1777.8 \pm 162.7$$

$$\sigma = 461.3 \pm 26.5$$

400

200

0

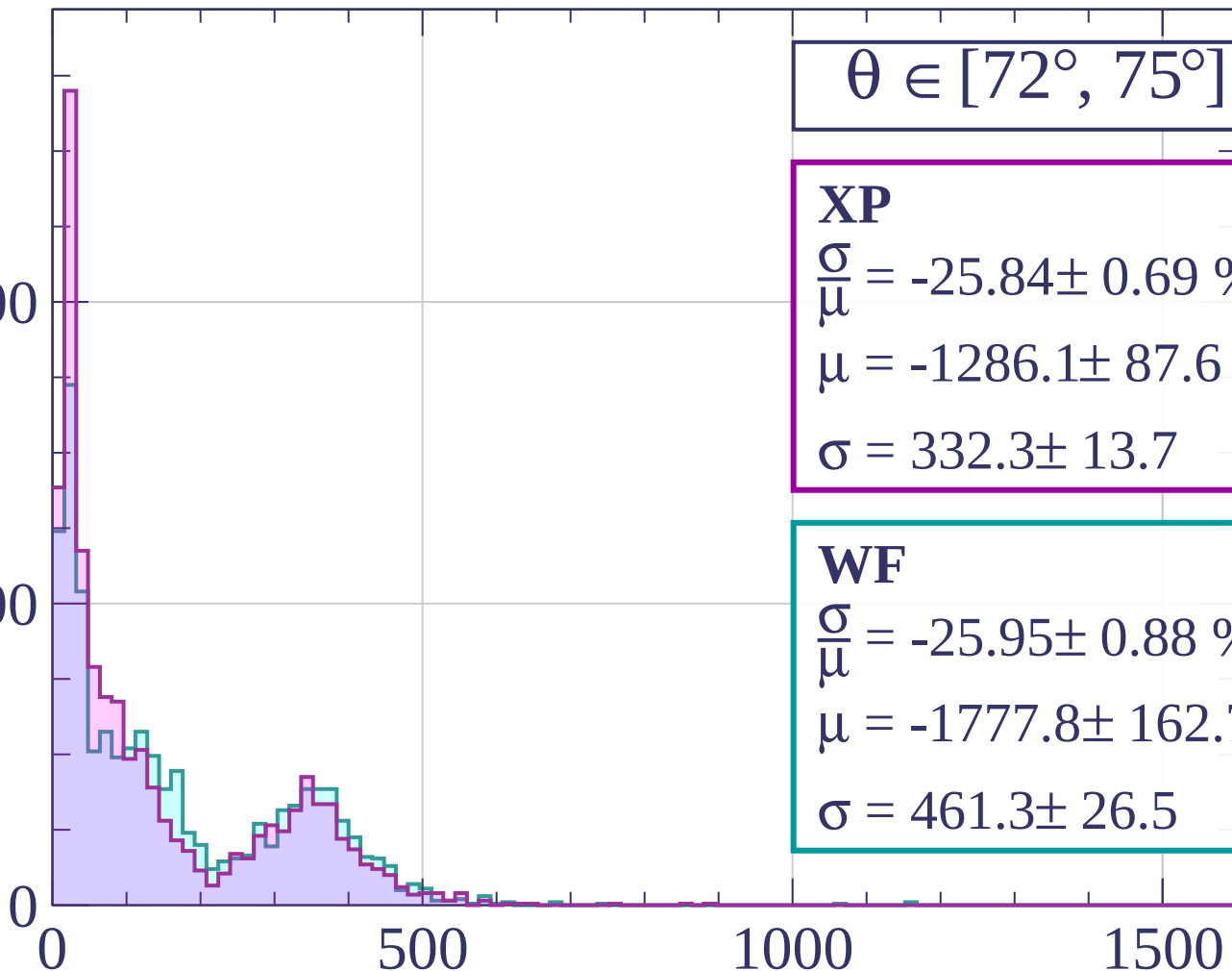
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [75^\circ, 78^\circ]$

XP

$$\frac{\sigma}{\mu} = -24.73 \pm 0.45 \%$$

$$\mu = -965.0 \pm 53.7$$

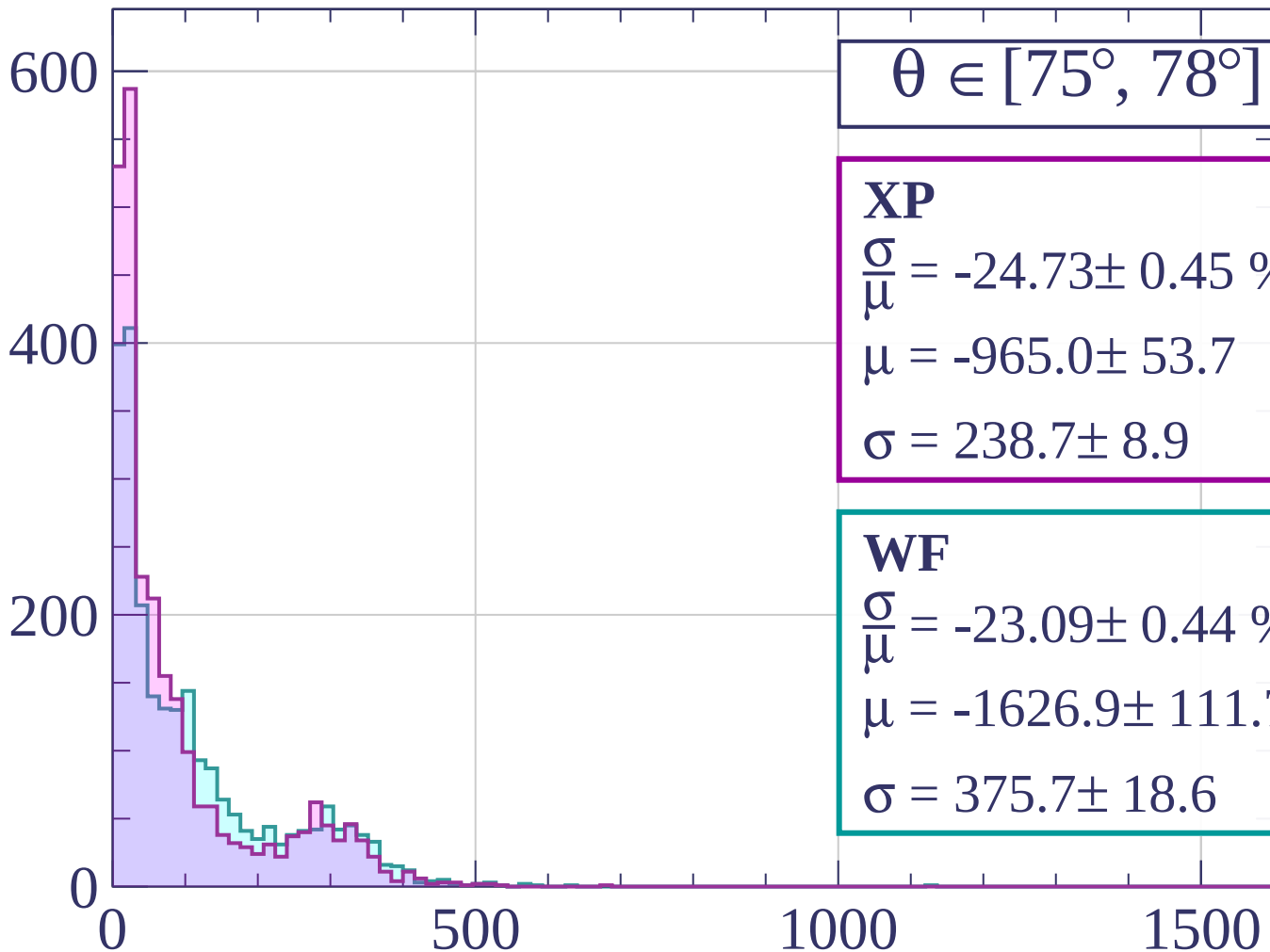
$$\sigma = 238.7 \pm 8.9$$

WF

$$\frac{\sigma}{\mu} = -23.09 \pm 0.44 \%$$

$$\mu = -1626.9 \pm 111.7$$

$$\sigma = 375.7 \pm 18.6$$



dE/dx [ADC counts/cm]

Count

$\theta \in [78^\circ, 81^\circ]$

XP

$$\frac{\sigma}{\mu} = -23.62 \pm 0.37 \%$$

$$\mu = -705.8 \pm 36.7$$

$$\sigma = 166.7 \pm 6.1$$

WF

$$\frac{\sigma}{\mu} = -24.09 \pm 0.42 \%$$

$$\mu = -970.7 \pm 61.0$$

$$\sigma = 233.8 \pm 10.6$$

800

600

400

200

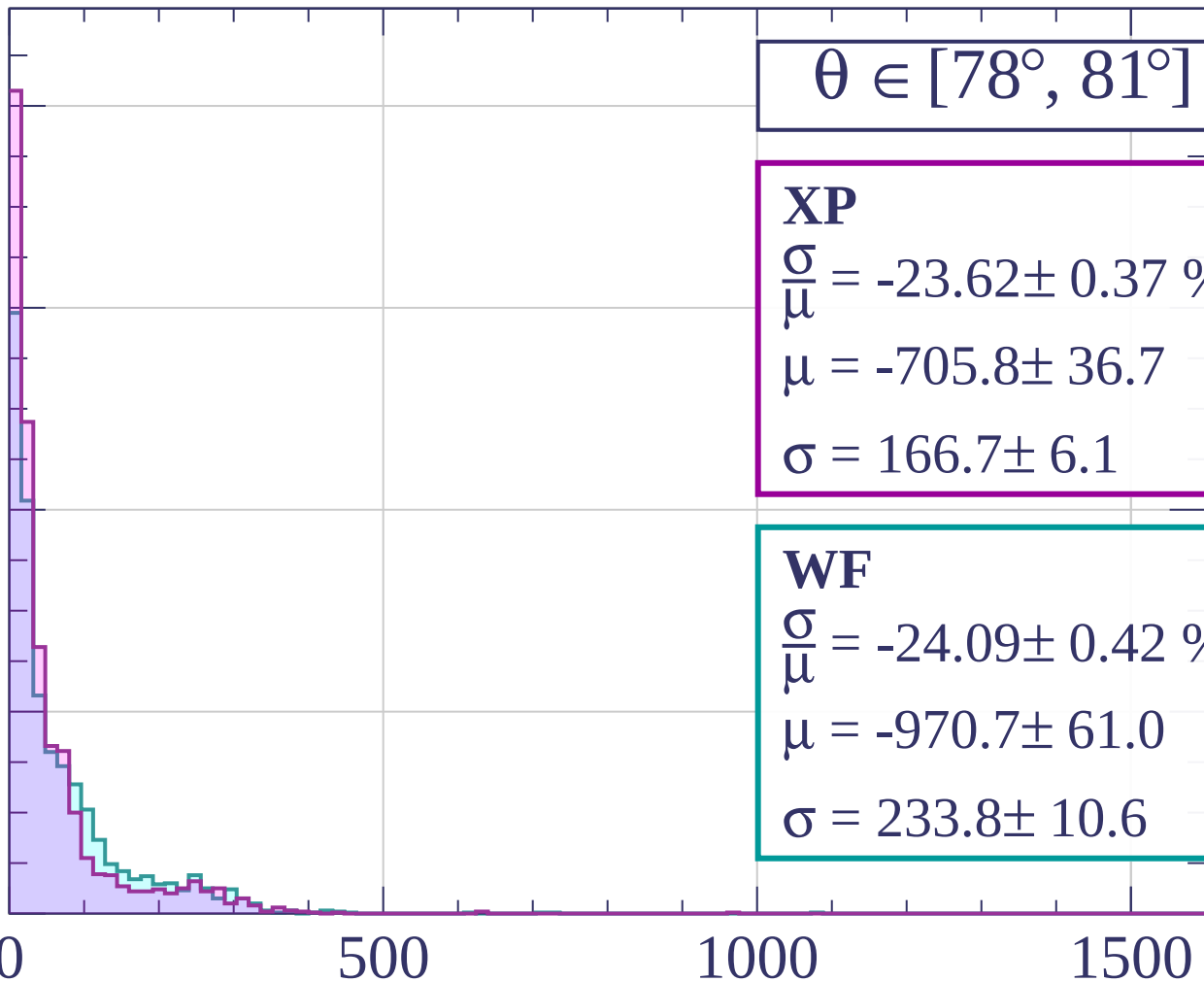
0

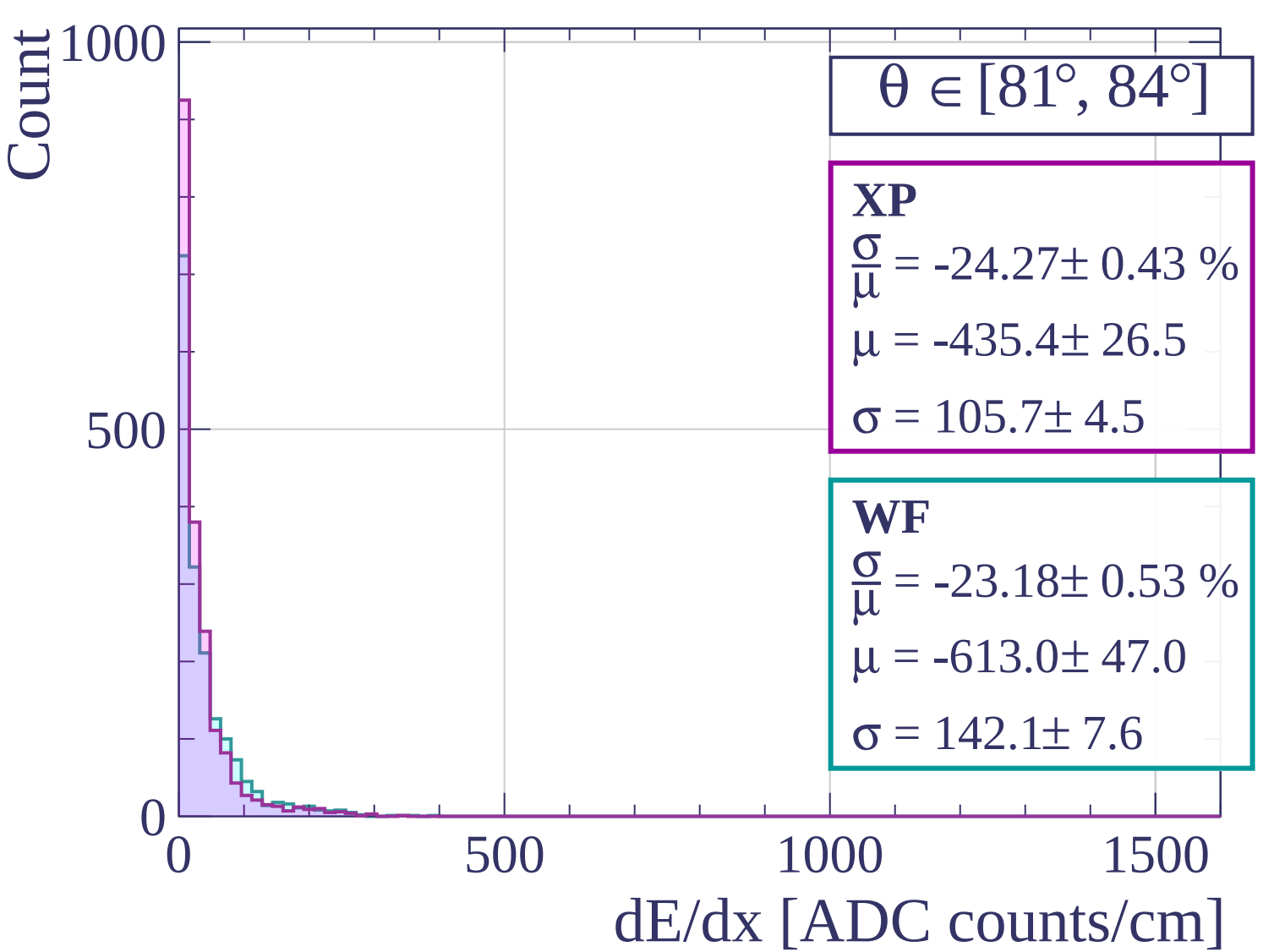
500

1000

1500

dE/dx [ADC counts/cm]





Count

$$\theta \in [84^\circ, 87^\circ]$$

XP

$$\frac{\sigma}{\mu} = -24.95 \pm 0.75 \%$$

$$\mu = -214.2 \pm 19.4$$

$$\sigma = 53.4 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = -24.93 \pm 0.79 \%$$

$$\mu = -254.3 \pm 25.5$$

$$\sigma = 63.4 \pm 4.4$$

600

400

200

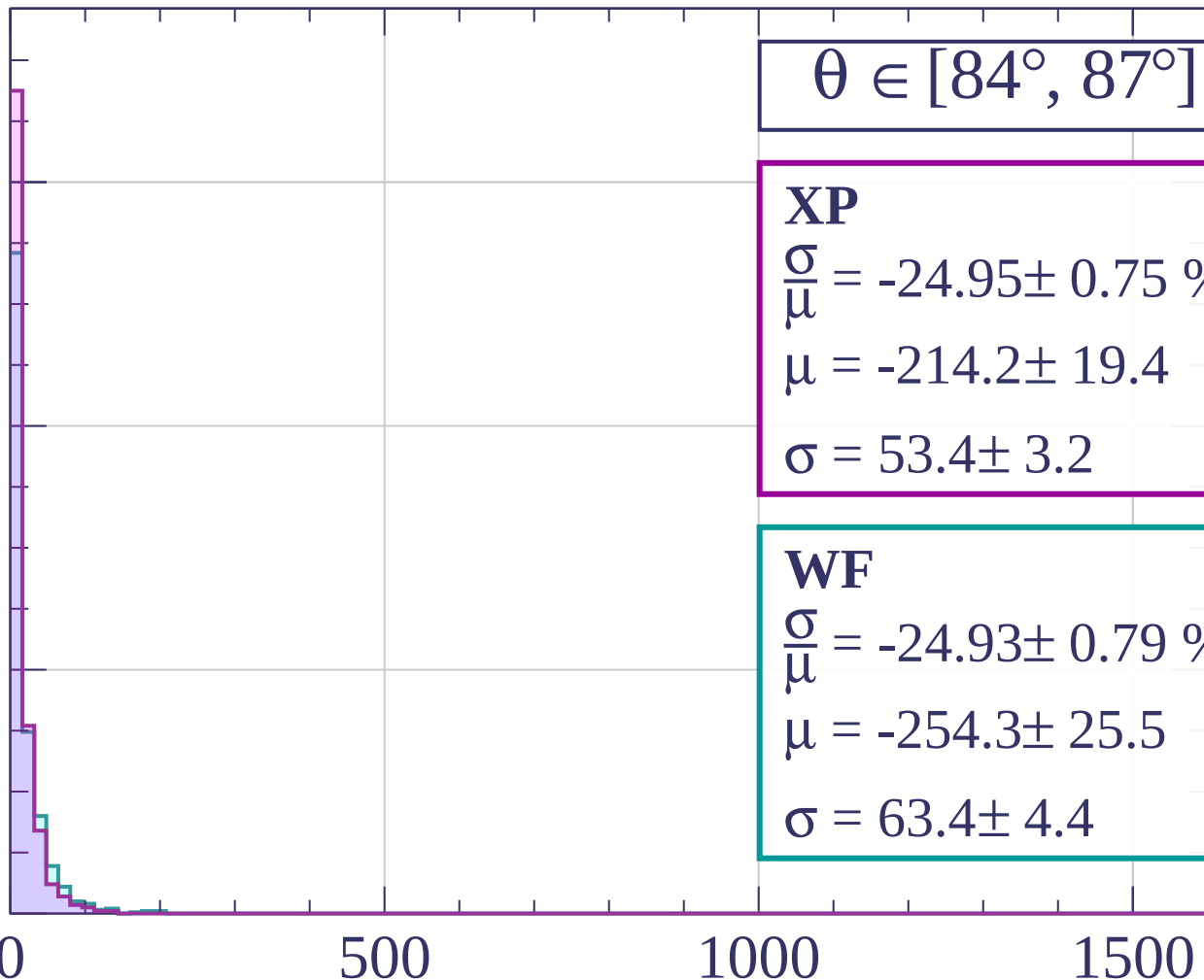
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [87^\circ, 90^\circ]$

XP

$$\frac{\sigma}{\mu} = 3.42 \pm 5.47 \%$$

$$\mu = 7.7 \pm 0.3$$

$$\sigma = 0.3 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 25.99 \pm 52.91 \%$$

$$\mu = 6.0 \pm 1.6$$

$$\sigma = 1.6 \pm 2.8$$

100

50

0

0

500

1000

1500

dE/dx [ADC counts/cm]

